Risk Management

Concept-First Notes + Exam Q&A + MCQ Bank + Mock Test — CA Intermediate

One complete file: concepts, worked Q&A, revision cheat-sheet, MCQs & a full mock test

Contents

Chapter 01 — Introduction to Risk Management

1. The Problem / The Need

Every financial institution is, at its core, a machine for taking risk deliberately. A bank does not make money by hiding cash in a vault — it makes money by lending it out, knowing that some borrowers will not repay. An asset manager does not earn returns by holding treasury bills forever — it earns them by accepting price volatility. An insurer collects premiums today against claims it cannot precisely predict. In each case, **the business model IS the acceptance of risk in exchange for a return.** Remove the risk entirely and you remove the profit.

This creates an immediate and permanent tension. If risk is the source of profit, then more risk should mean more profit — but only up to the point where the losses from bad outcomes overwhelm the gains from good ones and destroy the institution. History is a graveyard of firms that forgot this: Barings Bank collapsed in 1995 from a single rogue trader; Lehman Brothers failed in 2008 from over-leveraged mortgage exposure; Long-Term Capital Management nearly took down the financial system in 1998 despite being run by Nobel laureates. None of these firms failed because they took risk. They failed because they took **the wrong amount of the wrong risks without knowing it.**

So the need is not "avoid risk." A bank that avoids all risk earns nothing and dies slowly. The need is to answer, continuously and quantitatively, four questions:

- **What can go wrong?** (identification)
- **How badly, and how likely?** (measurement)
- **Can we survive it, and is the reward worth it?** (appetite and capital)
- **Who is watching, and what do we do about it?** (monitoring, mitigation, governance)

Risk management is the discipline that answers these questions systematically rather than by luck or gut feel. Its purpose is threefold, and every interview answer about "why risk management matters" should hit all three:

1. **Protect capital and solvency.** Capital is the buffer that absorbs unexpected losses. Risk management ensures the firm holds enough of it and never bets more than the buffer can absorb, so a bad year is survivable rather than fatal.
2. **Enable intelligent risk-taking.** This is the part beginners miss. Risk management is not the "department of no." Done well, it lets the firm take *more* risk where the reward justifies it and *less* where it does not — it is a profit-optimization function, not merely a defensive one.
3. **Meet regulatory and stakeholder obligations.** Banks operate under Basel III, insurers under Solvency II, and all are accountable to depositors, shareholders, rating agencies, and supervisors. Sound risk management is a licence to operate.

2. The Core Idea

The core idea of risk management can be compressed into a single sentence:

> **Convert unknown, unmanaged exposures into known, quantified, consciously-accepted positions that sit within a pre-agreed limit backed by enough capital to survive being wrong.**

Unpack that. The enemy is not loss — losses are expected and priced in. The enemy is the **unknown, unmeasured, unbudgeted** loss: the exposure nobody identified, sized, or approved. Risk management's job is to drag every material exposure into the light, attach a number to it, decide consciously whether to keep it, and hold capital against the possibility that the number is wrong.

Three foundational concepts make this operational:

- **Expected Loss (EL):** the average loss you anticipate over many periods. This is a *cost of doing business*, not a risk. You price for it (in loan spreads, insurance premiums) and provision for it. If a lender expects to lose 1% of a portfolio annually, that 1% is baked into the interest rate charged. Expected loss should never surprise you.
- **Unexpected Loss (UL):** the volatility *around* the expected loss — the amount by which a bad year exceeds the average. This is the real risk. You cannot price it away because you do not know when it will hit. Instead you hold **capital** against it. Capital exists to absorb unexpected loss.
- **Catastrophic / tail loss:** the rare extreme beyond what capital is sized for. Here you rely on stress testing, contingency planning, and — as a last resort — the possibility of failure. Regulators care intensely about this region because it is where systemic crises live.

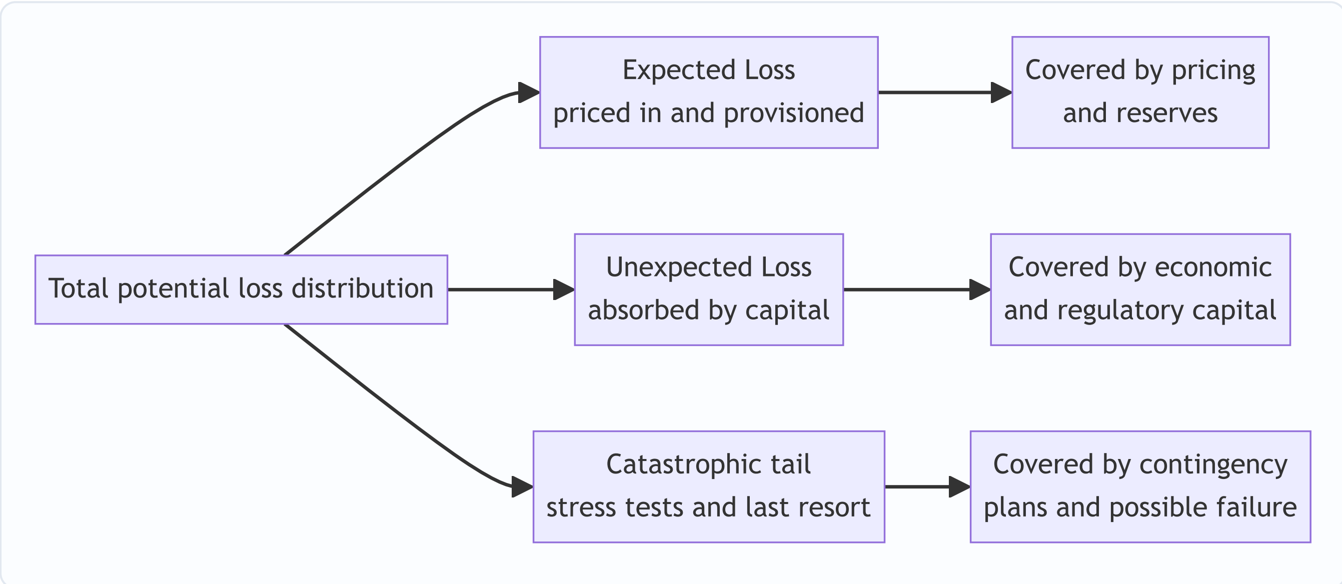


Figure 1 — The three regions of the loss distribution and how each is funded. Everything in risk management maps back to this picture.

The genius of this framing is that it turns a vague fear ("we might lose money") into an engineering problem ("size the buffer for the region we choose to survive"). The rest of risk management is just the machinery for doing that reliably.

3. Why / How It Works

Why does the discipline actually work — why does quantifying and governing risk produce better outcomes than experienced managers using judgment alone?

Because losses are governed by probability distributions, not by intuition. Human intuition is systematically bad at tails, correlations, and compounding. We underestimate how often "1-in-100" events happen, we assume diversification protects us when correlations spike to 1 in a crisis, and we anchor on recent calm. Risk management works because it replaces intuition with distributions: instead of asking "will this loan default?" it asks "what is the probability distribution of losses across the whole portfolio, and what does the 99th percentile look like?"

Because it separates the two jobs that intuition conflates. Pricing handles the *average* (expected loss); capital handles the *variance* (unexpected loss). A trader who prices a loan correctly has still done only half the job — someone must independently ensure enough capital stands behind the volatility. Risk management institutionalizes that separation.

Because measurement enables comparison and allocation. Once every risk is expressed in a common unit — typically potential loss in currency at a chosen confidence level, or capital consumed — you can compare a credit exposure against a market exposure against an operational exposure. You can ask "which business line earns the most return per unit of risk?" and allocate capital accordingly. This is the mechanism by which risk management *enables* intelligent risk-taking rather than merely restraining it. The key metric here is **RAROC** (Risk-Adjusted Return on Capital):

$$\text{RAROC} = \frac{\text{Revenue} - \text{Costs} - \text{Expected Loss}}{\text{Economic Capital}}$$

A business earning ₹100 of profit while consuming ₹500 of capital (RAROC = 20%) is preferable to one earning ₹120 while consuming ₹1,000 (RAROC = 12%), even though the second makes more absolute profit. Without risk quantification, you cannot see this — you would chase the ₹120 and destroy value.

Because governance closes the loop. Measurement without consequences is theatre. The process works only when limits bind, breaches trigger action, and someone independent of the risk-takers has the authority and information to intervene. That is why the *organizational* design (the three lines of defence, discussed in Section 4) is as important as the *mathematical* design.

4. Full Content

4.1 Risk vs Uncertainty — the foundational distinction

Before any measurement, we must separate two things everyday language treats as synonyms. The distinction was formalized by economist **Frank Knight** in 1921 and remains an interview favourite.

- **Risk** is randomness with a *knowable* probability distribution. You may not know the outcome, but you know (or can estimate) the odds. A fair die, a diversified loan book with decades of default history, an equity index with an estimable volatility — these are risks. Risk is **measurable, insurable, and capital can be sized against it.**
- **Uncertainty** (sometimes "Knightian uncertainty") is randomness where the distribution itself is *unknown or unknowable*. Novel events, structural breaks, regime changes, a pandemic in a world with no comparable data — you cannot assign reliable probabilities. Uncertainty is **not directly measurable** and resists conventional capital sizing.

Dimension	Risk	Uncertainty
Probability distribution	Known / estimable	Unknown / unknowable
Example	Loan default rate on a large book	Impact of a first-of-its-kind technology shock
Tool	VaR, expected loss, pricing, capital	Stress tests, scenario analysis, judgment, optionality
Insurable?	Generally yes	Generally no
Governance response	Limits and models	Resilience, buffers, humility

The practical lesson: **models handle risk; stress testing and buffers handle uncertainty.** A firm that treats uncertainty as if it were measurable risk — plugging a made-up probability into a model and trusting the output — is committing the classic error behind many crises (the 2008 assumption that national house prices could not fall together was uncertainty dressed up as risk). Mature risk functions hold *extra* capital and maintain *optionality* precisely because they know the model does not capture everything.

4.2 The taxonomy of financial risks

Risk management first needs a map of what can go wrong. The standard taxonomy every risk professional carries in their head:

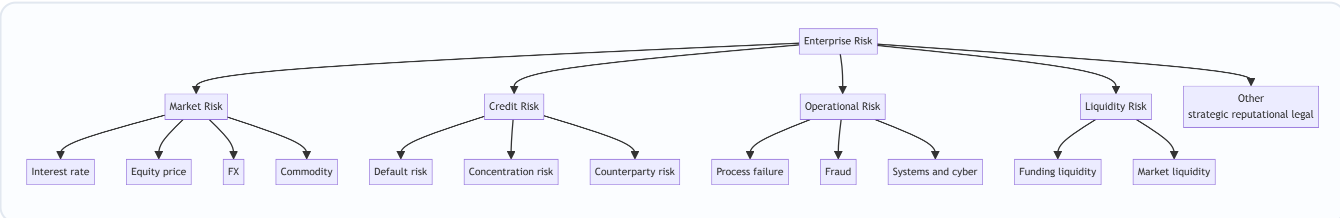


Figure 2 — The financial risk taxonomy. The big three regulated pillars are market, credit, and operational risk; liquidity is increasingly treated as a fourth pillar since 2008.

- **Market risk** — loss from movements in market prices: interest rates, equities, exchange rates, commodities. Measured with VaR, sensitivities (the "Greeks"), and stress tests.
- **Credit risk** — loss from a borrower or counterparty failing to meet obligations. Decomposed into PD, LGD, and EAD (Section 5). The largest risk for most commercial banks.
- **Operational risk** — loss from failed internal processes, people, systems, or external events. Includes fraud, cyberattacks, legal failures, and human error. Barings and most rogue-trader losses are operational.
- **Liquidity risk** — the risk of being unable to meet obligations when due (*funding* liquidity) or of being unable to sell an asset without moving its price (*market* liquidity). A firm can be solvent on paper yet fail because it cannot raise cash — the proximate cause of many bank runs.
- **Strategic, reputational, legal, and model risk** — harder to quantify but capable of destroying franchises.

4.3 The risk management process

The operational heart of the discipline is a continuous, closed loop. Five steps: **identify, measure, monitor, mitigate, report** — repeating forever because the risk landscape never stands still.

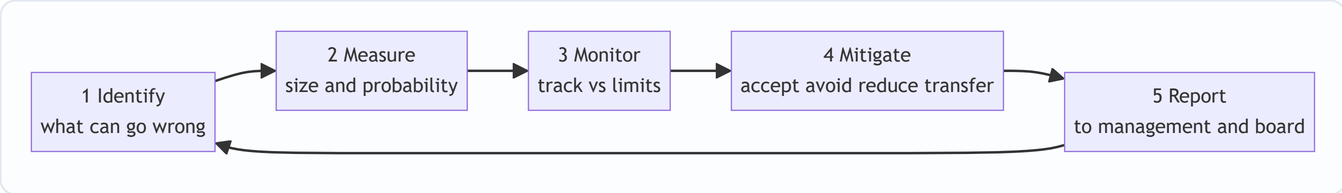


Figure 3 — The risk management cycle. It is a loop, not a line — the output of reporting feeds fresh identification, and the environment keeps changing.

Step 1 — Identify. Surface every material exposure before it surprises you. Techniques: risk registers, risk-and-control self-assessments (RCSA), scenario workshops, review of new products (a "new product approval" gate), loss-event databases, and horizon scanning. The cardinal sin here is the *unidentified* risk — you cannot measure or mitigate what you have not named.

Step 2 — Measure. Attach numbers. For each exposure estimate its size and likelihood, and express it in a common unit — usually potential loss in currency at a stated confidence level. This is where VaR, expected shortfall, PD×LGD×EAD, duration, and stress-loss estimates live. Measurement converts a list of worries into a rank-ordered, aggregatable portfolio.

Step 3 — Monitor. Continuously compare current exposures against **limits** and against changing conditions. Dashboards, limit-utilization reports, early-warning indicators (EWIs), and key risk indicators (KRIs). Monitoring answers "are we still inside our appetite, and is anything trending the wrong way?"

Step 4 — Mitigate. Choose a response for each risk. The four canonical treatments (the "4 T's"):

Treatment	Also called	Meaning	Example
Accept	Retain / tolerate	Keep the risk, hold capital against it	A bank retains diversified credit risk it is paid to take
Avoid	Terminate	Exit the activity entirely	Stop lending to a sector deemed unacceptable
Reduce	Mitigate / control	Lower probability or impact	Collateral, covenants, diversification, controls, limits
Transfer	Share	Move the risk to a third party	Insurance, hedging with derivatives, securitization

Note that *accept* is a legitimate, deliberate choice — not a failure. The whole point is that acceptance is now conscious and capitalized.

Step 5 — Report. Communicate the risk profile to those accountable: business heads, the risk committee, the board, and regulators. Good reporting is timely, forward-looking, and actionable — it drives decisions, not just archives numbers. Poor reporting (stale, backward-looking, buried in detail) is a root cause in most post-mortems. Reporting closes the loop by feeding governance and re-identification.

4.4 Risk appetite and risk tolerance

Measurement tells you how much risk you *have*. Appetite tells you how much you *want*. This is the strategic anchor of the whole system, set by the board.

- **Risk appetite** is the *aggregate* amount and type of risk an institution is willing to accept in pursuit of its objectives, stated at the top level. It is a strategic, board-owned statement — e.g., "we are willing to accept a 1-in-100-year loss of no more than 20% of capital," or "we target a AA rating and will not take risks inconsistent with it."

- **Risk tolerance** is the *specific, quantified* boundary for a particular risk type or business line — the operational translation of appetite. E.g., "market-risk VaR shall not exceed ₹50 crore," "single-name credit exposure ≤ 10% of capital," "no more than 15% of the book in any one sector."
- **Risk limits** are the granular, enforceable thresholds at desk, portfolio, or trader level that keep aggregate exposure within tolerance. A breach triggers a defined escalation.
- **Risk capacity** is the *maximum* risk the firm could bear before breaching regulatory minimums or failing — set by capital and liquidity, not by preference. Appetite must always sit *below* capacity, leaving a safety margin.

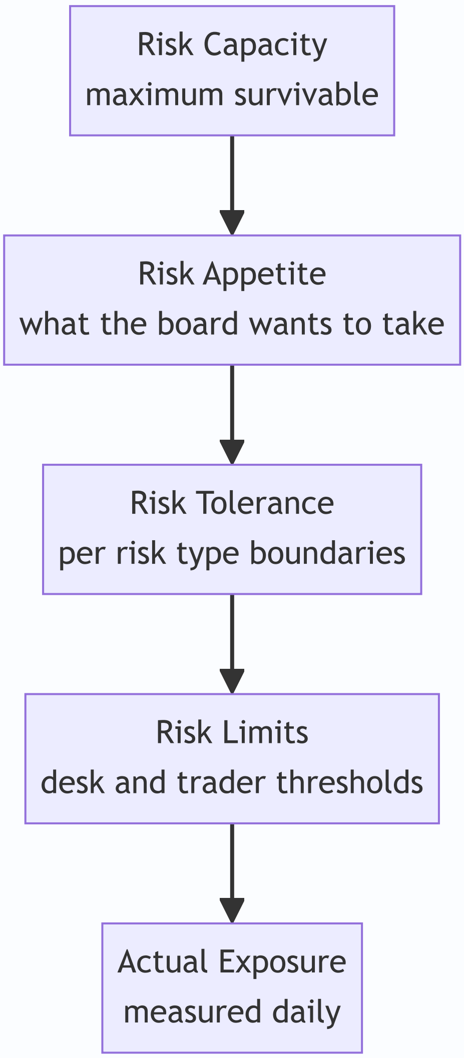


Figure 4 — The appetite hierarchy. Each layer must nest inside the one above it. Actual exposure should sit comfortably inside limits which sit inside tolerance which sits inside appetite which sits inside capacity.

The relationships to memorise: **Exposure ≤ Limits ≤ Tolerance ≤ Appetite < Capacity.** When these invert — when actual exposure creeps past limits, or when appetite is set at or above capacity leaving no margin — the firm is flying without a buffer. The 2008 crisis featured many firms whose *effective* appetite had drifted right up to their capacity.

4.5 The three lines of defence

The final structural element answers "who owns the risk, who checks it, and who checks the checkers?" The **Three Lines of Defence (3LoD)** model is the industry-standard governance architecture, endorsed by Basel and virtually every regulator.

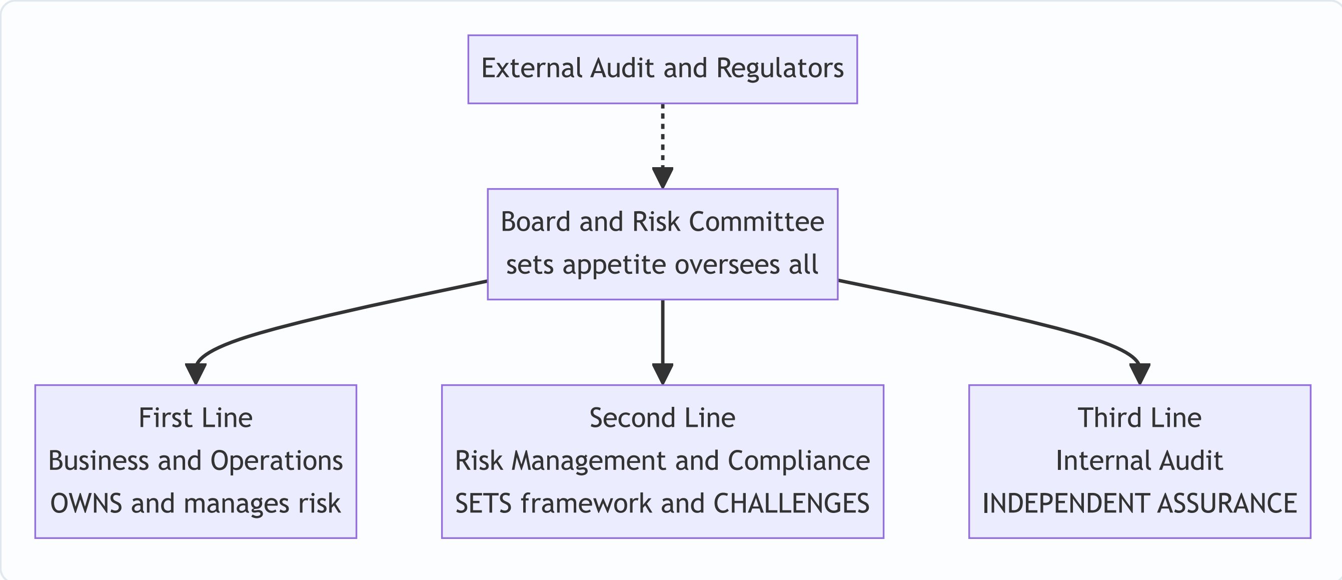


Figure 5 — The three lines of defence under board oversight. Independence increases as you move from first to third line; external audit and supervisors sit outside as an additional check.

- **First line — the risk owners.** The business units and operations that *take* the risk and therefore *own* it. A lending officer, a trader, a branch manager. They are responsible for identifying, managing, and controlling risk in their day-to-day activity, operating within limits. Crucial insight: **risk management is not solely the risk department's job — the first line owns the risk it creates.**
- **Second line — the overseers and challengers.** The independent risk management and compliance functions. They design the framework, set methodologies and limits, aggregate and monitor risk across the firm, and — critically — *challenge* the first line. They do not own the risk, but they ensure it is being managed to standard. Headed by the Chief Risk Officer (CRO), who typically reports to the board risk committee, not just the CEO, to preserve independence.
- **Third line — independent assurance.** Internal Audit. It provides objective assurance to the board that the first and second lines are actually working as designed. It reports directly to the audit committee, is independent of both other lines, and checks the checkers.

The design principle is **escalating independence**: the further from the money-making, the more independent the function, so that no single group both takes risk and judges whether that risk is acceptable. The separation of duties is the whole point — Barings failed partly because the same person (Nick Leeson) controlled both trading and its settlement, collapsing first and second lines into one. Beyond the three lines sit **external audit** and **regulators/supervisors**, and above all sits the **board**, which owns appetite and ultimate accountability.

5. Worked Examples

Example 1 — Expected Loss on a loan portfolio (the credit-risk master formula)

The single most important credit formula, which you must be able to reproduce and explain in any interview:

$$\text{Expected Loss} = PD \times LGD \times EAD$$

where **PD** = probability of default (over a chosen horizon, usually one year), **LGD** = loss given default (the fraction *not* recovered), and **EAD** = exposure at default (the amount outstanding when default happens).

Setup. A bank has a loan of **EAD = ₹100,00,000 (₹1 crore)**. The borrower has a one-year **PD = 2%**. If the borrower defaults, the bank expects to recover 60% through collateral and workout, so **recovery rate = 60%** and **LGD = 1 − 0.60 = 40%**.

Compute Expected Loss:

$$EL = 0.02 \times 0.40 \times 1{,}00{,}00{,}000 = ₹80{,}000$$

Interpretation and reconciliation. The bank expects to lose ₹80,000 per year *on average* on this loan. This is not a risk to be feared — it is a **cost** to be *priced in*. To break even on expected loss alone, the loan must earn at least 0.8% (₹80,000 / ₹1 crore) in spread *purely to cover credit losses*, on top of funding cost, operating cost, and a target return on capital.

Let us verify the logic holds by decomposing: of ₹1 crore lent, there is a 2% chance of a default event. In that event, the bank loses 40% of ₹1 crore = ₹40,00,000. The probability-weighted loss is 0.02 × ₹40,00,000 = ₹80,000. ✓ Consistent — the two routes (PD×LGD×EAD vs probability × loss-in-default) reconcile exactly.

Portfolio extension. Suppose the bank holds **200 such independent loans**. Total expected loss = 200 × ₹80,000 = **₹1.6 crore per year**. But the *expected* loss is not the *risk* — the risk is the volatility around it (unexpected loss). If defaults were perfectly correlated, all 200 could default together; if independent, the portfolio loss is far more predictable. This is precisely why the next chapters build toward VaR and correlation — expected loss is only the priced-in floor.

Example 2 — Value at Risk (VaR): sizing the unexpected loss

Where expected loss is the *average*, **VaR** answers: "*Over a given horizon and confidence level, what is the loss we will not exceed except with a small probability?*" Formally, one-day 99% VaR of ₹X means: "on 99% of days, losses will be ≤ ₹X; only on ~1 day in 100 do we expect to lose more."

Parametric (variance-covariance) VaR for a position, assuming returns are normally distributed:

$$\text{VaR} = z_{\alpha} \times \sigma \times V$$

where z_α is the standard-normal quantile for the confidence level, σ is the return volatility over the horizon, and V is the position value. Key z-values to memorise: **95% → 1.645**, **99% → 2.326**.

Setup. An equity portfolio worth **V = ₹10,00,00,000 (₹10 crore)** has a daily return volatility of **σ = 1.5%**. Compute the 1-day 99% VaR.

Compute:

$$\text{VaR}_{99\%} = 2.326 \times 0.015 \times 10{,}00{,}00{,}000 = ₹34{,}89{,}000$$

Interpretation. On roughly 99 days out of 100, the portfolio should not lose more than about **₹34.9 lakh** in a single day. On about 1 day in 100, losses are expected to *exceed* this — VaR tells you the threshold, not the size of the tail beyond it (that is Expected Shortfall, a later chapter).

Scaling to a 10-day horizon. Under the square-root-of-time rule (i.i.d. returns), multiply by √10:

$$\text{VaR}_{10\text{-day}} = 34{,}89{,}000 \times \sqrt{10} = 34{,}89{,}000 \times 3.162 = ₹1{,}10{,}31{,}000 \approx ₹1.10 \text{ crore}$$

Reconciliation with appetite. Now connect to Section 4.4. Suppose the board's market-risk *tolerance* is a 1-day 99% VaR limit of ₹50 lakh. Our exposure of ₹34.9 lakh sits *inside* the ₹50 lakh limit — utilization is 34.89 / 50 = **70%**. Acceptable, with headroom. If the desk added positions pushing σ to 2.2%, VaR would rise to 2.326 × 0.022 × ₹10 crore = ₹51.17 lakh, *breaching* the ₹50 lakh limit and triggering escalation to the second line. This is exactly how measurement (Step 2) feeds monitoring (Step 3) against tolerance (4.4) — the whole framework working as one machine.

Example 3 — RAROC: choosing between two businesses (enabling intelligent risk-taking)

This example shows risk management *creating value*, not just preventing loss.

Setup. A bank must allocate capital between two lending desks:

	Desk A (SME lending)	Desk B (large corporate)
Annual revenue (net of funding)	₹40,00,000	₹60,00,000
Operating costs	₹8,00,000	₹10,00,000
Expected Loss (PD×LGD×EAD)	₹12,00,000	₹8,00,000
Economic capital (for UL)	₹50,00,000	₹1,20,00,000

Compute RAROC for each using the formula from Section 3:

Desk A: $$\text{RAROC}_A = \frac{40{,}00{,}000 - 8{,}00{,}000 - 12{,}00{,}000}{50{,}00{,}000} = \frac{20{,}00{,}000}{50{,}00{,}000} = 40\%$$

Desk B: $$\text{RAROC}_B = \frac{60{,}00{,}000 - 10{,}00{,}000 - 8{,}00{,}000}{1{,}20{,}00{,}000} = \frac{42{,}00{,}000}{1{,}20{,}00{,}000} = 35\%$$

Interpretation and reconciliation. Desk B earns *more absolute profit* (₹42 lakh vs ₹20 lakh), so a naive manager allocates capital to B. But Desk A earns **40% per unit of risk-capital** versus B's **35%**. If the bank's hurdle rate (cost of equity) is, say, 15%, *both* create value — but A creates it more efficiently. At the margin, the next rupee of capital should go to A. Without risk quantification you would see only the ₹42 lakh and misallocate. This is the concrete meaning of "risk management enables intelligent risk-taking": it makes the return-*per-risk* visible, so capital flows to where it works hardest. Note also that if either RAROC fell below the 15% hurdle, the risk framework would flag that the business is *destroying* shareholder value despite being profitable in accounting terms — a signal invisible without the discipline.

6. Connections

- **To capital adequacy and Basel (later chapters).** The expected/unexpected/catastrophic split in Figure 1 is the direct conceptual basis for regulatory capital: minimum capital ratios exist to cover unexpected loss, provisions cover expected loss, and stress-capital buffers address the tail. Everything in this chapter is the scaffolding on which Basel III sits.

- **To VaR, Expected Shortfall, and market-risk chapters.** Example 2 is a preview; the measurement step of the process is elaborated across the market-risk sequence, including VaR's limitations and the move to Expected Shortfall under FRTB.
- **To credit-risk modelling.** Example 1's PD×LGD×EAD decomposition is the launch point for internal-ratings-based models, credit VaR, and portfolio correlation.
- **To corporate governance and internal control.** The three lines of defence connect directly to board committees, the CRO mandate, and — for CA students — the internal-control and audit frameworks, where separation of duties and independent assurance mirror the same logic.
- **To behavioural finance.** The risk-vs-uncertainty distinction connects to why humans mis-estimate tails, over-trust models, and herd — the psychological failures that risk *governance* is designed to counteract.

7. Key Terms

Term	Definition
Risk	Randomness with a knowable/estimable probability distribution; measurable and capitalizable.
Uncertainty (Knightian)	Randomness whose distribution is unknown; not directly measurable; handled by buffers and judgment.
Expected Loss (EL)	The average anticipated loss; = PD × LGD × EAD for credit; a priced-in cost, not a risk.
Unexpected Loss (UL)	Volatility of loss around the expected level; the true risk, absorbed by capital.
PD / LGD / EAD	Probability of Default / Loss Given Default (= 1 − recovery) / Exposure at Default.
Value at Risk (VaR)	Maximum loss not exceeded at a stated confidence level over a horizon; e.g. 1-day 99% VaR.
Expected Shortfall (ES)	Average loss *given* that VaR is exceeded; captures tail severity VaR ignores.
RAROC	Risk-Adjusted Return on Capital = (Revenue − Costs − EL) / Economic Capital.
Economic capital	Internally-estimated capital needed to absorb unexpected loss to a chosen confidence.
Risk appetite	Board-level statement of aggregate risk the firm is willing to take.
Risk tolerance	Quantified boundary for a specific risk type; the operational translation of appetite.
Risk limit	Granular enforceable threshold at desk/portfolio level; a breach triggers escalation.
Risk capacity	Maximum risk survivable before breaching regulatory minimums or failing.
Three Lines of Defence	Governance model: business (owns), risk/compliance (challenges), internal audit (assures).
CRO	Chief Risk Officer; heads the second line; reports to the board risk committee for independence.

8. Common Confusions

- **"Risk management means avoiding risk."** No — it means taking the *right* risks *consciously* and being *paid* for them. A zero-risk bank is a failing bank. The function optimizes risk-taking; it does not eliminate it.
- **Expected loss ≠ risk.** Expected loss is the *average*, a cost you price in and provision for. It should never surprise you. The *risk* is the unexpected loss — the volatility around the average — which is what capital exists to absorb. Confusing the two leads to under-capitalization.
- **Risk vs uncertainty.** Risk has estimable odds (use models and capital); uncertainty does not (use stress tests, buffers, humility). Plugging a fabricated probability into a model to make uncertainty *look* like measurable risk is a classic, dangerous error.
- **Appetite vs tolerance vs capacity vs limit.** Appetite = what the board *wants* (strategic, aggregate). Tolerance = the *quantified boundary* per risk type. Limit = the *granular enforceable* threshold. Capacity = the *maximum survivable*, set by capital not preference. They nest: Exposure ≤ Limit ≤ Tolerance ≤ Appetite < Capacity.
- **VaR is not the worst case.** 99% VaR is the *threshold* you exceed 1 day in 100 — it says nothing about *how bad* the other 1% gets. Treating VaR as a maximum loss is the error that Expected Shortfall and stress testing exist to correct.
- **"Risk is the risk department's job."** Wrong — under three-lines-of-defence, the *first line* (the business that creates the risk) *owns* it. The risk department (second line) sets standards and challenges; it does not own the exposure.
- **VaR confidence and horizon must be stated.** "VaR is ₹5 crore" is meaningless without "1-day, 99%." Higher confidence and longer horizon both raise the number; comparing VaRs on different bases is a beginner error.

9. Recap

Risk management exists because the business of finance *is* the deliberate acceptance of risk for return — so the task is never to avoid risk but to take the right amount of the right risks knowingly, backed by enough capital to survive being wrong. Its three purposes are to **protect capital**, **enable intelligent risk-taking**, and **meet regulation**.

The foundational split is **expected loss** (average, priced-in, a cost) versus **unexpected loss** (volatility, the true risk, absorbed by capital) versus **catastrophic tail** (stress tests and last resort). We distinguished **risk** (knowable odds — use models and capital) from **uncertainty** (unknowable odds — use buffers and judgment). The operational engine is the five-step loop — **identify, measure, monitor, mitigate, report** — turning forever, with mitigation choosing among **accept, avoid, reduce, transfer**. Strategy is anchored by the **appetite hierarchy** (Exposure ≤ Limit ≤ Tolerance ≤ Appetite < Capacity), and governance by the **three lines of defence** (business owns, risk challenges, audit assures) under board oversight. The math — **EL = PD × LGD × EAD**, **VaR = z·σ·V**, **RAROC = (Rev − Costs − EL)/Capital** — makes all of it quantitative, comparable, and enforceable.

10. Quick-Reference / Interview Points

Formulas to reproduce cold: - Expected Loss: `EL = PD × LGD × EAD`, with `LGD = 1 − recovery rate` - Parametric VaR: `VaR = z_α × σ × V`; `z(95%) = 1.645`, `z(99%) = 2.326` - Time scaling: `VaR(T-day) =`

`VaR(1-day) × √T` - RAROC: `(Revenue − Costs − Expected Loss) / Economic Capital`

One-liners that signal competence: - "Risk management isn't the department of no — it's how we take *more* risk where we're paid for it and less where we're not." - "Price for the expected loss; hold capital for the unexpected loss; stress-test for the tail." - "Risk has odds you can estimate; uncertainty doesn't — models handle the first, buffers and judgment handle the second." - "VaR is a threshold you breach 1 day in 100, not a worst case — that's why we also run Expected Shortfall and stress tests." - "Appetite is what the board wants; tolerance is the quantified boundary; capacity is what would kill us. Appetite must sit below capacity with a margin." - "Under three lines of defence, the business owns its risk, the risk function challenges it, and internal audit assures both — independence increases as you move away from the money."

The five-step process, in order: Identify → Measure → Monitor → Mitigate → Report (loop).

The four mitigations (4 T's): Accept (tolerate), Avoid (terminate), Reduce (treat), Transfer (share).

The nesting to never get wrong: Actual Exposure ≤ Limit ≤ Tolerance ≤ Appetite < Capacity.

If asked "why do banks hold capital?": To absorb *unexpected* loss. Expected loss is covered by pricing and provisions; capital is the buffer for the volatility around the average; the tail beyond capital is where stress testing and, ultimately, failure live.

Q&A — Introduction to Risk Management

A companion practice bank for Chapter 01. Every question is followed by a full answer. Work each one before reading the solution.

Section A — Concept Check

A1. Why is it wrong to say the goal of risk management is to "avoid risk"? Because the business model of every financial institution *is* the deliberate acceptance of risk in exchange for return. A bank earns by lending (accepting default risk), an asset manager by holding volatile assets, an insurer by writing uncertain claims. Remove the risk and you remove the profit. The goal is therefore to take the *right amount* of the *right risks* consciously, being *paid* for them, backed by enough capital to survive being wrong — not to eliminate risk. A zero-risk bank earns nothing and dies slowly.

A2. State the four questions risk management must continuously answer. (1) *What can go wrong?* — identification. (2) *How badly, and how likely?* — measurement. (3) *Can we survive it, and is the reward worth it?* — appetite and capital. (4) *Who is watching, and what do we do about it?* — monitoring, mitigation, and governance.

A3. Distinguish expected loss, unexpected loss, and catastrophic loss, and say how each is funded. Expected Loss (EL) is the average loss anticipated over many periods — a *cost of doing business*, priced into spreads/premiums and covered by provisions. Unexpected Loss (UL) is the volatility *around* the average — the true risk — absorbed by **capital** (economic and regulatory). Catastrophic/tail loss is the rare extreme beyond what capital is sized for — addressed by stress testing, contingency planning, and, as a last resort, the possibility of failure. Key point: EL should never surprise you; UL is why banks hold capital.

A4. Explain the Knightian distinction between risk and uncertainty and the practical lesson. Risk is randomness with a *knowable/estimable* probability distribution — measurable, insurable, capital can be sized against it (e.g., default rates on a large loan book). Uncertainty is randomness whose distribution is *unknown or unknowable* — novel events, structural breaks — not directly measurable and resistant to conventional capital sizing. Practical lesson: **models handle risk; stress tests, buffers and judgment handle uncertainty.** Plugging a fabricated probability into a model to make uncertainty *look* like measurable risk (e.g., the 2008 assumption that national house prices could not fall together) is a classic, dangerous error.

A5. List the five steps of the risk management process, in order, and note why it is a loop. Identify → Measure → Monitor → Mitigate → Report. It is a closed loop, not a line, because the risk landscape never stands still: reporting feeds fresh identification, and the environment keeps changing, so the cycle repeats forever.

A6. Name the four risk treatments (the "4 T's") and give an example of each. Accept/tolerate — retain the risk and hold capital against it (a bank keeps diversified credit risk it is paid to take). Avoid/terminate — exit the activity (stop lending to an unacceptable sector). Reduce/mitigate — lower probability or impact (collateral, covenants, diversification, limits). Transfer/share — move it to a third party (insurance, hedging, securitization). Note that *accept* is a legitimate, deliberate, capitalized choice — not a failure.

A7. Put the appetite hierarchy in the correct nesting order and define capacity vs appetite. Actual Exposure ≤ Limit ≤ Tolerance ≤ Appetite < Capacity. **Risk capacity** is the *maximum* risk the firm could bear before breaching regulatory minimums or failing — set by capital and liquidity, not preference. **Risk**

appetite is the aggregate risk the board *wants* to take. Appetite must sit *below* capacity, leaving a safety margin; when appetite drifts up to capacity, the firm is flying without a buffer.

A8. In the three lines of defence, who owns the risk, and why does independence increase across the lines? The **first line** — the business/operations that take the risk — *owns* it. The **second line** (risk management and compliance, led by the CRO) sets the framework and *challenges* the first line. The **third line** (internal audit) provides *independent assurance* that the first two work. Independence increases with distance from the money-making so that no single group both takes risk and judges whether it is acceptable. Barings collapsed partly because one person (Nick Leeson) controlled both trading and settlement, collapsing the separation of duties.

Section B — Numerical / Applied (full solutions)

B1. Expected loss on a single loan. A loan has EAD = ₹2,00,00,000 (₹2 crore), one-year PD = 3%, and expected recovery of 55% on default. Compute the annual expected loss and the minimum spread (in %) needed just to cover it.

Solution. LGD = 1 − recovery = 1 − 0.55 = 0.45. EL = PD × LGD × EAD = 0.03 × 0.45 × 2,00,00,000 = 0.0135 × 2,00,00,000 = **₹2,70,000**. Minimum credit spread = EL / EAD = 2,70,000 / 2,00,00,000 = **1.35%**. The loan must earn at least 1.35% purely to cover credit losses, on top of funding cost, operating cost, and a target return on capital. *Check (probability-weighted route):* loss-in-default = 0.45 × ₹2 crore = ₹90,00,000; × PD 0.03 = ₹2,70,000. ✓ Both routes reconcile.

B2. Portfolio expected loss. A bank holds 500 independent loans, each with EAD = ₹10,00,000, PD = 1.5%, recovery = 55%. Find total annual expected loss.

Solution. LGD = 0.45. Per-loan EL = 0.015 × 0.45 × 10,00,000 = 0.00675 × 10,00,000 = ₹6,750. Portfolio EL = 500 × 6,750 = **₹33,75,000 (₹33.75 lakh)**. *Note:* this is the *expected* (priced-in) figure, not the risk. If defaults were perfectly correlated all 500 could default together; independence makes the realised loss far more predictable. The volatility around ₹33.75 lakh is the unexpected loss that capital must cover.

B3. Parametric VaR at two confidence levels and a 10-day horizon. An equity portfolio worth V = ₹20 crore has daily return volatility σ = 2%. Compute 1-day 95% VaR, 1-day 99% VaR, and 10-day 99% VaR. (z_{95} = 1.645, z_{99} = 2.326.)

Solution. VaR = z × σ × V. 1-day 95%: 1.645 × 0.02 × 20,00,00,000 = 0.0329 × 20,00,00,000 = **₹65,80,000 (₹65.8 lakh)**. 1-day 99%: 2.326 × 0.02 × 20,00,00,000 = 0.04652 × 20,00,00,000 = **₹93,04,000 (₹93.04 lakh)**. 10-day 99% (square-root-of-time rule): 93,04,000 × √10 = 93,04,000 × 3.162 = **₹2,94,19,248 ≈ ₹2.94 crore**. *Interpretation:* on ~99 days in 100 the portfolio should not lose more than ₹93.04 lakh in a day; VaR gives the *threshold*, not the size of the tail beyond it. Higher confidence and longer horizon both raise the number — which is why a VaR figure is meaningless without its confidence level and horizon stated.

B4. VaR limit utilisation and a breach. A trading book worth ₹15 crore has σ = 1.2% daily. The board's market-risk tolerance is a 1-day 99% VaR limit of ₹45 lakh. (a) Compute current VaR and limit utilisation. (b) If new positions raise σ to 1.3%, does the desk breach the limit?

Solution. (a) VaR = 2.326 × 0.012 × 15,00,00,000 = 0.027912 × 15,00,00,000 = **₹41,86,800 (₹41.87 lakh)**. Utilisation = 41,86,800 / 45,00,000 = **93.0%** — inside the limit, but little headroom. (b) New VaR = 2.326 × 0.013 × 15,00,00,000 = 0.030238 × 15,00,00,000 = **₹45,35,700 (₹45.36 lakh)**. Since ₹45.36 lakh > ₹45 lakh, the desk **breaches** the limit, triggering escalation to the second line. This shows measurement (Step 2) feeding monitoring (Step 3) against tolerance — the framework working as one machine.

B5. RAROC and the hurdle rate. A lending desk generates annual revenue (net of funding) ₹80,00,000, operating costs ₹20,00,000, expected loss ₹15,00,000, and consumes economic capital of ₹1,50,00,000. The bank's cost of equity (hurdle) is 15%. Compute RAROC and state whether the desk creates value.

Solution. RAROC = (Revenue − Costs − EL) / Economic Capital = (80,00,000 − 20,00,000 − 15,00,000) / 1,50,00,000 = 45,00,000 / 1,50,00,000 = **30%**. Since 30% > 15% hurdle, the desk **creates shareholder value** — it earns twice its cost of capital per unit of risk. Had RAROC fallen below 15%, the desk would be *destroying* value despite being profitable in accounting terms — a signal invisible without risk quantification. This is the concrete meaning of risk management "enabling intelligent risk-taking": it makes return-*per-risk* visible so capital flows to where it works hardest.

Section C — Interview Style (model answers)

C1. "In one minute, why does risk management matter?" "Finance makes money by taking risk, so the job is never to avoid it but to take the right risks knowingly. Risk management does three things. First, it protects capital and solvency — it ensures we hold enough buffer for unexpected losses and never bet more than that buffer can absorb, so a bad year is survivable, not fatal. Second, and the part people miss, it *enables* intelligent risk-taking: by putting every exposure in a common unit, it shows return-per-unit-of-risk, so we take more risk where we're paid for it and less where we're not — it's a profit-optimisation function, not the department of no. Third, it's a licence to operate: Basel, Solvency II, rating agencies and depositors all demand it. In short — price for the expected loss, hold capital for the unexpected loss, stress-test for the tail."

C2. "Walk me through the difference between expected and unexpected loss and how each is managed." "Expected loss is the average I anticipate — for credit it's PD times LGD times EAD. It's a *cost*, not a risk: I price it into the spread and cover it with provisions, so it should never surprise me. Unexpected loss is the volatility *around* that average — the amount a bad year exceeds the mean. I can't price it away because I don't know when it hits, so I hold capital against it. That's the whole reason banks carry capital. Beyond capital sits the catastrophic tail, which I handle with stress tests and contingency plans. Confusing the two — treating the average as the risk — is exactly how firms end up under-capitalised."

C3. "What is VaR, and what are its limitations?" "One-day 99% VaR of ₹X means on 99% of days losses stay at or below ₹X, and on about one day in a hundred we expect to lose more. Parametrically it's z times sigma times position value. Its limitations: it's a *threshold*, not a worst case — it says nothing about *how bad* the other 1% gets, so a firm can meet its VaR and still be wiped out in the tail. It also assumes a distribution (often normal) that understates fat tails, and it's not sub-additive in general, so it can mis-reward concentration. That's why we complement it with Expected Shortfall — the average loss *given* VaR is breached — and with stress testing. And a VaR number is meaningless without its confidence level and horizon."

C4. "Explain the three lines of defence and why the CRO reports to the board." "The first line is the business — traders, lending officers — who *take* and therefore *own* the risk, managing it within limits day to day. The second line is independent risk and compliance: it sets the framework, methodologies and limits, aggregates and monitors risk firm-wide, and *challenges* the first line. The third line is internal audit, giving the board objective assurance that the first two actually work. Independence rises as you move away from the money. The CRO heads the second line and typically reports to the board risk committee, not only the CEO, so that the person challenging risk-taking can't be overruled by the person whose bonus depends on taking it. That independence is the whole point — Barings failed because one person ran both trading and settlement."

C5. "A profitable business line has a RAROC below the firm's hurdle rate. What do you do?"
"Accounting profit and value creation aren't the same thing. If RAROC — profit net of expected loss over economic capital — is below the cost of equity, the line is *destroying* shareholder value even though it shows a profit, because the capital tied up would earn more elsewhere. I'd first check the inputs: is economic capital right, is expected loss stale, is there a pricing or cost problem I can fix to lift the numerator? If it can be re-priced or made more capital-efficient, do that. If not, it's a candidate for shrinking or exiting and reallocating that capital to higher-RAROC uses. The discipline here is that we allocate capital on return-per-risk, not on absolute profit."

Section D — MCQs (with reasoning)

D1. Expected loss on a loan is best described as: A) A risk to be hedged B) A cost to be priced in and provisioned C) The 99th-percentile loss D) Capital held against volatility **Answer: B.** EL is the *average* anticipated loss — a cost baked into spreads and covered by provisions. It is not the risk (that's unexpected loss, C/D describe the tail and capital), and it isn't hedged away (A).

D2. Under the Knightian distinction, a genuinely novel, first-of-its-kind shock with no historical data is best handled by: A) Parametric VaR B) A fabricated PD plugged into a model C) Stress testing, buffers and judgment D) Insurance **Answer: C.** This is *uncertainty*, not risk — the distribution is unknowable, so models (A) and priced probabilities (B) don't apply, and it is generally uninsurable (D). Buffers, scenarios and judgment are the tools.

D3. Which ordering of the appetite hierarchy is correct? A) Capacity ≤ Appetite ≤ Tolerance ≤ Limit ≤ Exposure B) Exposure ≤ Limit ≤ Tolerance ≤ Appetite < Capacity C) Appetite ≤ Capacity ≤ Tolerance ≤ Exposure ≤ Limit D) Limit ≤ Exposure ≤ Tolerance ≤ Capacity ≤ Appetite **Answer: B.** Actual exposure nests inside enforceable limits, inside per-risk tolerance, inside board appetite, which must sit strictly below the maximum survivable capacity.

D4. A 1-day 99% VaR of ₹5 crore means: A) The maximum the firm can ever lose in a day is ₹5 crore B) On ~1 day in 100, losses are expected to exceed ₹5 crore C) The firm loses ₹5 crore every 100 days D) Average daily loss is ₹5 crore **Answer: B.** VaR is a threshold breached with 1% probability; it is explicitly *not* a worst case (A), not a certainty (C), and not the mean (D).

D5. If 1-day 99% VaR is ₹40 lakh, the 4-day 99% VaR under the square-root-of-time rule is: A) ₹40 lakh B) ₹80 lakh C) ₹160 lakh D) ₹10 lakh **Answer: B.** $VaR(T) = VaR(1) \times \sqrt{T} = 40 \times \sqrt{4} = 40 \times 2 =$ ₹80 lakh.

D6. Under the three lines of defence, who OWNS the risk a trading desk creates? A) Internal audit B) The risk management function C) The trading desk itself (first line) D) The regulator **Answer: C.** The business that takes the risk owns it. The second line (B) challenges, the third line (A) assures, and the regulator (D) supervises from outside.

D7. Which is NOT one of the four risk treatments? A) Accept B) Avoid C) Amplify D) Transfer **Answer: C.** The 4 T's are Accept (tolerate), Avoid (terminate), Reduce (treat), and Transfer (share). "Amplify" is not a treatment.

D8. A desk earns more absolute profit than another but has a lower RAROC. The correct interpretation is: A) It is unambiguously the better desk B) It generates less return per unit of risk-capital C) RAROC is irrelevant to capital allocation D) It must be loss-making **Answer: B.** RAROC measures return per unit of economic capital; a lower RAROC means capital works less hard there, even if absolute profit is higher — which is precisely why capital allocation follows RAROC, not raw profit.

End of Q&A bank. Formulas to reproduce cold: EL = PD × LGD × EAD (LGD = 1 − recovery); VaR = z × σ × V (z_{95} = 1.645, z_{99} = 2.326); VaR(T) = VaR(1) × √T; RAROC = (Revenue − Costs − EL) / Economic Capital.

Chapter 02 — Types of Financial Risk

1. The Problem / The Need

A bank is a machine for taking risk on purpose. It borrows short, lends long, holds securities, moves money, and promises depositors safety while gambling that borrowers will repay. If you tell the CEO "we have a lot of risk," you have said nothing useful. Nobody can hedge "risk," reserve capital against "risk," assign an owner to "risk," or price "risk" into a loan. You can only manage things you can *name, measure, and attribute to someone*.

The 2007–09 Global Financial Crisis is the cautionary tale for why sloppy classification kills. Banks held AAA-rated mortgage tranches and called the exposure "credit risk" — a slow-moving, diversifiable thing. In reality the exposure was a tangle of *market risk* (mark-to-market losses as spreads widened), *liquidity risk* (funding vanished overnight), *model risk* (the correlation assumptions in the CDO models were wrong), and *systemic risk* (everyone held the same trade). Because the losses were mislabelled, they were mismeasured, under-capitalised, and owned by nobody in particular. Lehman Brothers had a liquidity problem that its solvency models never flagged.

So the first real job in risk management is **taxonomy**: cut the fog of "risk" into distinct, measurable categories, each with its own math, its own capital charge, its own owner, and its own controls. This chapter builds that taxonomy and — just as importantly — shows how the categories bleed into and amplify one another. A neat box diagram is comforting; real crises live in the arrows *between* the boxes.

Why classification is not academic pedantry:

- **Measurement.** Market risk uses Value-at-Risk on a daily price distribution. Credit risk uses PD × LGD × EAD over a year. Operational risk uses loss-event frequency and severity. Applying VaR to a loan book, or a default model to a trading desk, gives nonsense.
- **Ownership.** The Chief Risk Officer needs to point at a specific desk head, credit committee, or COO and say "this is yours." Unowned risk is unmanaged risk.
- **Capital.** Basel rules charge capital by risk type. Miscategorise and you either hold too little (and blow up) or too much (and destroy return on equity).
- **Regulation and hedging.** You hedge market risk with derivatives, credit risk with CDS or provisions, liquidity risk with a buffer of HQLA. The tool depends on the label.

2. The Core Idea

Financial risk is not one thing but a family of distinct exposures, each defined by the *source* of the potential loss. The discipline of risk management begins by classifying every exposure into a taxonomy, because the source determines how you measure it, who owns it, and how you defend against it — and the most dangerous losses come not from any single category but from the way categories compound one another under stress.

The canonical taxonomy used across banks, NBFCs, and the FRM/PRM syllabi:

#	Risk type	One-line definition	Loss driver
1	Market	Loss from adverse moves in market prices	Prices/rates/FX/vol
2	Credit	Loss from a counterparty failing to pay	Default / downgrade
3	Liquidity	Loss from inability to fund or to sell	Cash/marketability
4	Operational	Loss from failed processes, people, systems, external events	Internal failure/fraud
5	Model	Loss from a wrong or misused model	Bad assumptions/code
6	Legal / Compliance	Loss from breaking laws, rules, contracts	Fines, unenforceability
7	Reputational	Loss from damage to the franchise / trust	Lost customers/funding
8	Systemic	Loss from the failure of the system itself	Contagion

A useful mental split: the first three (market, credit, liquidity) are **financial risks** the bank takes deliberately to earn a return. Operational, model, legal, and reputational are **non-financial risks** — largely unrewarded, incurred as a by-product of doing business; you want to minimise them, not "optimise" them. Systemic risk sits above all of them: it is the risk that the categories correlate perfectly at the worst moment.

3. Why / How It Works

Why the source defines the category

Every risk category is really an answer to the question: "*What has to happen in the world for me to lose money here?*"

- If the loss requires a **price to move** → market risk. It is symmetric-ish (you can gain too), fast, observable daily, and modelled with distributions of returns.
- If the loss requires a **counterparty to break a promise** → credit risk. It is asymmetric (limited upside = the coupon; large downside = default), slow, sparsely observed (defaults are rare), and modelled with probabilities of default.
- If the loss requires you to be **unable to raise cash or sell an asset** → liquidity risk. It has no natural "distribution of returns"; it is a threshold/knock-out risk that is fine until suddenly it is fatal.
- If the loss requires a **process to fail** → operational risk. Fat-tailed, hard to model, driven by the internal control environment rather than the market.

Because the generating mechanisms differ, the *mathematics* differs. This is the deepest reason classification matters: **you cannot use one number for all of it.** VaR captures the market tail; it says nothing about a rogue trader or a wrong model.

Why they compound

Risk categories are not independent buckets — they are coupled by shared drivers and by feedback loops. Three mechanisms:

1. **Common shock.** A single macro event (a rate spike, a pandemic) hits multiple categories at once, so correlations that looked low in calm times jump toward 1.
2. **Transmission / cascade.** One risk *causes* another: a market shock forces margin calls (→ funding-liquidity), forced selling depresses prices (→ market-liquidity → more market loss), which triggers

covenant breaches (→ credit).

3. **Reflexivity.** The act of managing risk creates risk. Everyone running the same VaR model de-risks simultaneously, which is itself the systemic event.

The diagram below maps the taxonomy and its principal linkages.

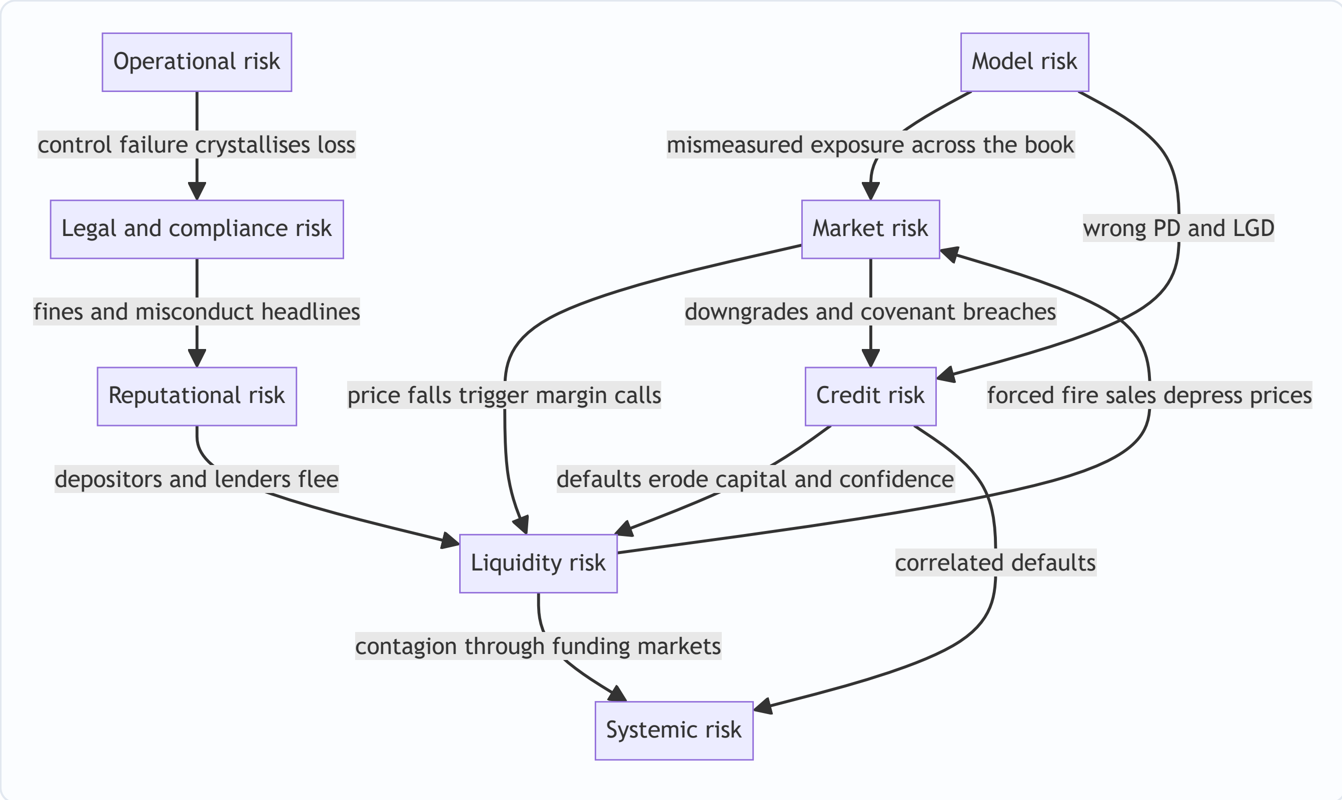


Figure 1 — The risk taxonomy and its compounding arrows. Crises travel along the arrows, not inside the boxes.

4. Full Content — The Taxonomy in Depth

4.1 Market Risk

Definition. The risk of loss on positions from movements in market prices: interest rates, equity prices, FX rates, commodity prices, and their volatilities and spreads. Sub-types: interest-rate risk, equity risk, currency (FX) risk, commodity risk, and (as second-order Greeks) volatility risk, basis risk, and correlation risk.

Where it lives. The trading book primarily, but also the banking book via interest-rate risk (IRRBB) and FX translation.

Core measure — Value-at-Risk (VaR). VaR answers: "*Over horizon h, at confidence c, what is the loss I will not exceed?*" Formally, VaR is the c-quantile of the loss distribution.

Parametric (variance-covariance) VaR for a single position:

$$\text{VaR} = z_{c} \times \sigma \times V \times \sqrt{h}$$

where z_c is the standard-normal quantile (1.645 for 95%, 2.326 for 99%), σ is the return volatility per unit time, V is the position value, and √h scales one-period volatility to an h-period horizon.

Beyond VaR — Expected Shortfall (ES / CVaR). VaR ignores how bad losses are *beyond* the cut-off. Expected Shortfall is the average loss conditional on breaching VaR:

$$\text{ES}_c = E[L \mid L > \text{VaR}_c]$$

Basel FRTB replaced 99% VaR with 97.5% ES precisely because ES is coherent (sub-additive — diversification never increases it) and captures tail thickness.

Portfolio VaR uses the correlation matrix. For two assets:

$$\text{VaR}_p = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho\,\text{VaR}_1 \text{VaR}_2}$$

4.2 Credit Risk

Definition. The risk that a borrower or counterparty fails to meet contractual obligations. Sub-types: **default risk** (outright non-payment), **migration/downgrade risk** (rating deterioration reduces value), **concentration risk** (too much exposure to one name/sector/geography), **counterparty credit risk** (CCR, on derivatives, where exposure itself is uncertain), and **settlement risk** (Herstatt risk).

Core formula — Expected Loss.

$$\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD}$$

- **PD** — Probability of Default over a horizon (usually 1 year).
- **LGD** — Loss Given Default = 1 − recovery rate; the fraction not recovered.
- **EAD** — Exposure At Default; the outstanding amount at the moment of default.

Expected Loss is the *average* loss — you price it into the spread and provision for it (it is a cost of doing business, covered by margins). What actually threatens solvency is **Unexpected Loss (UL)** — the volatility of loss around the mean — which is what capital covers:

$$\text{UL} = \text{EAD} \times \sqrt{\text{PD}(1-\text{PD})}\times \text{LGD} \quad \text{(single-name, LGD deterministic)}$$

Credit capital under Basel IRB is driven by UL at a 99.9% confidence over one year, using the Asymptotic Single Risk Factor (ASRF) model. The key insight: EL is priced, UL is capitalised.

4.3 Liquidity Risk

Definition. Two distinct animals sharing a name:

- **Funding liquidity risk** — the risk of being unable to meet cash obligations as they fall due, without incurring unacceptable cost. This is a *balance-sheet/cash-flow* risk. It killed Lehman, Northern Rock, and SVB.
- **Market (asset) liquidity risk** — the risk that you cannot sell an asset quickly near its fair value; the bid-ask spread widens and market depth vanishes. This is a *transaction-cost* risk.

The two feed each other: you sell assets to raise funding (need market liquidity), and thin market liquidity worsens your funding position.

Measures. - **Liquidity Coverage Ratio (LCR)** = HQLA ÷ Net cash outflows over 30 stressed days ≥ 100%. - **Net Stable Funding Ratio (NSFR)** = Available Stable Funding ÷ Required Stable Funding ≥ 100%. - Liquidity-adjusted VaR (LVaR) adds a bid-ask haircut to market VaR.

Liquidity risk is *non-distributional* and non-linear: you are fine at a buffer of 100% and dead at 99% in a run. This is why it resists the tidy VaR treatment and needs stress testing and survival-horizon analysis instead.

4.4 Operational Risk

Definition (Basel). The risk of loss from inadequate or failed internal processes, people and systems, or from external events. **Includes legal risk; excludes strategic and reputational risk** (in the Basel definition).

Seven Basel loss-event categories: internal fraud; external fraud; employment practices; clients/products/business practices; damage to physical assets; business disruption/system failure; execution/delivery/process management.

Examples. Rogue trading (Barings — Nick Leeson, £827m; Société Générale — Kerviel, €4.9bn), settlement errors, cyber-attacks, model deployment bugs, mis-selling.

Measure. Historically Advanced Measurement Approach (AMA) modelled a loss distribution via **frequency × severity** (e.g., Poisson frequency, lognormal severity, combined by Monte Carlo into an annual loss distribution; capital = 99.9% quantile). Basel III replaced this with the **Standardised Measurement Approach (SMA)**: capital = Business Indicator Component × Internal Loss Multiplier. Operational loss is *fat-tailed and skewed* — many small losses, rare catastrophic ones.

4.5 Model Risk

Definition. The risk of loss from decisions based on a model that is wrong or is used wrongly. Two sources (per US SR 11-7 supervisory guidance): (a) the model has fundamental errors and produces inaccurate outputs; (b) the model is used incorrectly or inappropriately.

Examples. The Gaussian copula that mispriced CDO correlation in 2008. The London Whale VaR model that JPMorgan changed to understate risk. Volatility models that assume normality and ignore fat tails. A pricing model calibrated on a regime that has ended.

Why it is insidious. Model risk is a *meta-risk*: it doesn't produce losses directly, it corrupts the measurement of every other risk. A wrong PD understates credit risk; a wrong vol surface understates market risk. Managed via independent model validation, back-testing, benchmarking, and conservative reserves. It has no clean formula — it is governed, not computed.

4.6 Legal and Compliance Risk

Definition. Legal risk — loss from unenforceable contracts, lawsuits, or adverse judgments (e.g., a derivative contract a counterparty was not authorised to enter — the classic Hammersmith & Fulham swaps case). **Compliance risk** — loss from failing to comply with laws, regulations, and internal standards (AML/KYC breaches, sanctions violations, market-conduct rules).

Examples. BNP Paribas' $8.9bn sanctions fine (2014); LIBOR-rigging fines across banks; money-laundering penalties. Basel folds legal risk into operational risk for capital, but conduct/compliance risk is increasingly managed as its own pillar.

4.7 Reputational Risk

Definition. The risk that negative perception among customers, counterparties, investors, or regulators damages the franchise — reducing revenue, raising funding costs, or triggering deposit flight. It is the *consequence* risk: almost always a second-order effect of an operational, legal, or credit failure that goes public.

Why it is dangerous. It converts a contained loss into an existential one by attacking the bank's core asset — *trust*. A bank funded by depositors who can leave in one tap of an app (as with SVB in 2023, where reputational panic on social media caused $42bn of withdrawals in a day) can die of reputation before any solvency model reacts. Hard to measure directly; proxied by share-price reaction, funding-spread widening, and customer attrition.

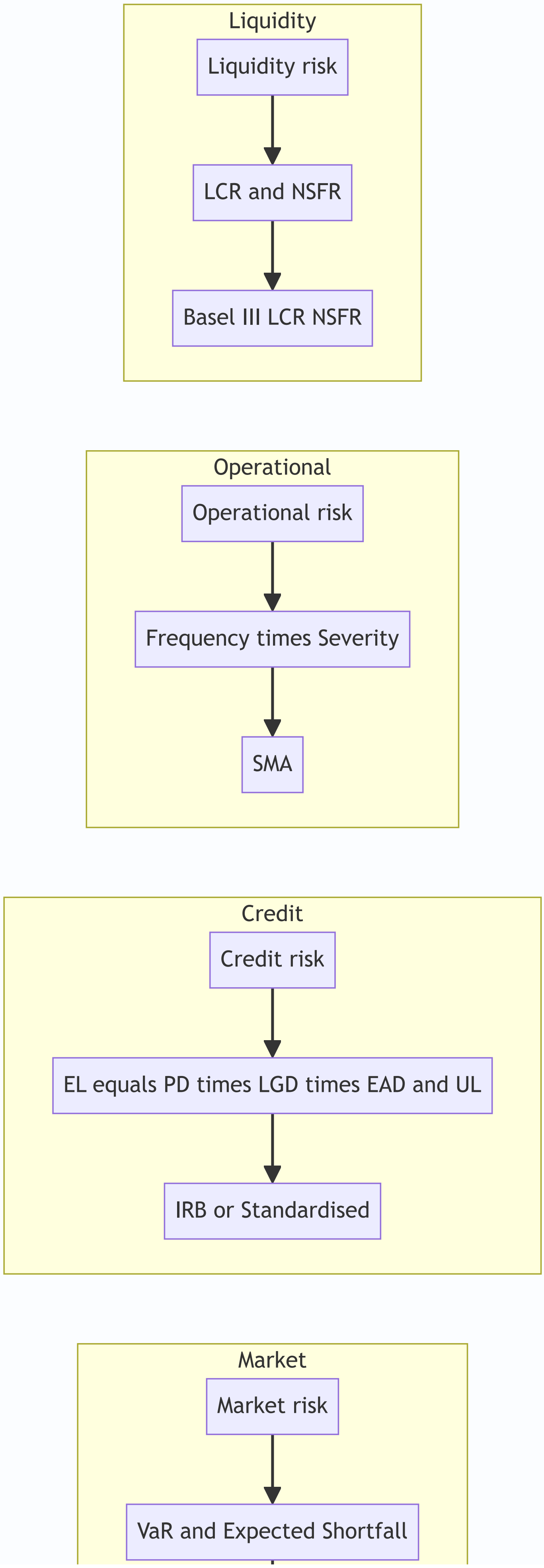
4.8 Systemic Risk

Definition. The risk that the failure of one institution or the seizing-up of one market cascades through the financial system, because institutions are interconnected and hold correlated exposures. It is the risk *of the system*, not *within* a single firm.

Channels. Direct counterparty exposures (a domino chain); common asset holdings and fire-sale externalities; funding-market freezes; loss of confidence. Addressed macro-prudentially: G-SIB capital surcharges, countercyclical buffers, central clearing, and stress tests (CCAR, EBA).

An individual firm cannot diversify away systemic risk — that is its defining feature and why it needs a regulator, not a risk desk, to manage.

4.9 How the categories map to measurement and capital


Liquidity
Liquidity risk
LCR and NSFR
Basel III LCR NSFR
Operational
Operational risk
Frequency times Severity
SMA
Credit
Credit risk
EL equals PD times LGD times EAD and UL
IRB or Standardised
Market
Market risk
VaR and Expected Shortfall

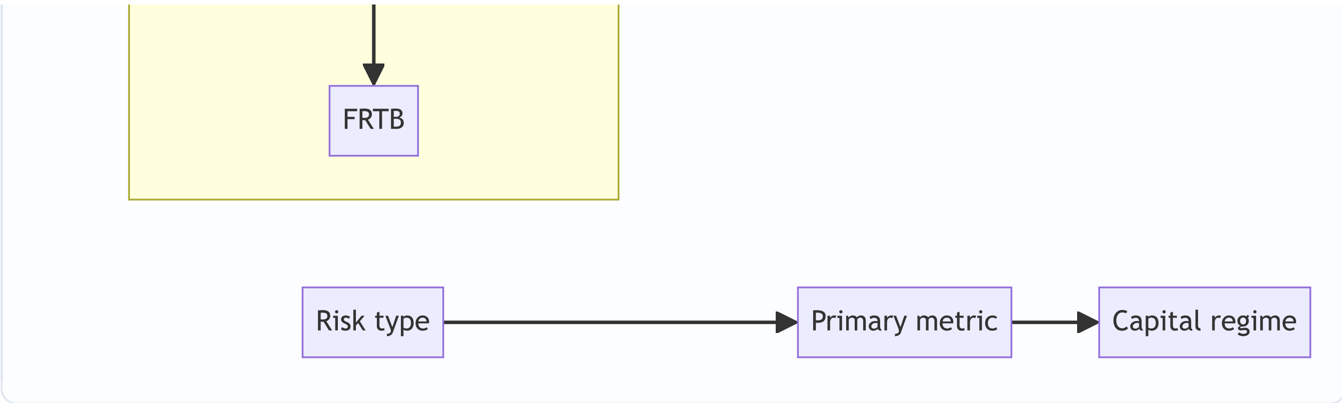


Figure 2 — Each category has its own metric and its own regulatory capital regime. Mislabelling breaks this whole chain.

5. Worked Examples

Example 1 — Market risk: parametric VaR, scaling, and portfolio diversification

A desk holds a ₹100 crore equity position with daily return volatility $\sigma = 2\%$. Compute the 1-day and 10-day 99% VaR, then combine with a ₹100 crore bond position (daily $\sigma = 0.5\%$, correlation $\rho = 0.30$ with equities).

Step 1 — Single-asset 1-day 99% VaR (equity). $z_{99\%} = 2.326$. VaR = 2.326 × 0.02 × 100 = **₹4.652 crore.**

Step 2 — Scale to 10 days (square-root-of-time): $\text{VaR}_{10} = 4.652 \times \sqrt{10} = 4.652 \times 3.1623 =$ **₹14.71 crore.**

Step 3 — Bond 1-day 99% VaR. $\text{VaR}_{bond} = 2.326 \times 0.005 \times 100 =$ **₹1.163 crore.**

Step 4 — Portfolio 1-day VaR with $\rho = 0.30$.

$$\text{VaR}_p = \sqrt{4.652^2 + 1.163^2 + 2(0.30)(4.652)(1.163)}$$

$= \sqrt{21.641 + 1.353 + 3.246} = \sqrt{26.240} =$ **₹5.122 crore.**

Reconciliation / self-check. The undiversified sum is 4.652 + 1.163 = ₹5.815 cr. The portfolio VaR of ₹5.122 cr is lower — as it must be for $\rho < 1$. **Diversification benefit = 5.815 − 5.122 = ₹0.693 cr (11.9%).** If we set $\rho = 1$, the formula collapses to $\sqrt{(4.652+1.163)^2} = 5.815$, exactly the undiversified sum — confirming the algebra. Good.

Example 2 — Credit risk: Expected Loss, pricing, and Unexpected Loss

An NBFC lends ₹50 crore (EAD) to a corporate with 1-year PD = 2% and recovery rate = 40% (so LGD = 60%).

Step 1 — Expected Loss. EL = PD × LGD × EAD = 0.02 × 0.60 × 50 = **₹0.60 crore = ₹60 lakh.**

Step 2 — What spread covers EL? As a fraction of EAD: EL/EAD = 0.60/50 = **1.20%.** The loan must earn at least ~120 bps over its funding cost *just to break even on expected credit losses* — before any profit or capital charge.

Step 3 — Unexpected Loss (single name, deterministic LGD).

$$\text{UL} = \text{EAD}\times \text{LGD}\times\sqrt{\text{PD}(1-\text{PD})} = 50 \times 0.60 \times \sqrt{0.02\times0.98}$$

$= 30 \times \sqrt{0.0196} = 30 \times 0.14 =$ **₹4.20 crore.**

Reconciliation. UL (₹4.20 cr) is 7× the EL (₹0.60 cr). This is the whole point of the EL/UL split: the *average* loss is small and priced into the 1.2% spread, but the *volatility* of loss is large — a single default costs LGD × EAD = ₹30 cr, not ₹0.6 cr. Check the two states directly: with prob 2% you lose ₹30 cr; with prob 98% you lose ₹0. Mean = 0.02 × 30 = ₹0.60 cr ✓ (matches EL). Variance = 0.02 × 0.98 × 30^2 = 17.64, so std dev = √17.64 = ₹4.20 cr ✓ (matches UL). The formulas reconcile exactly with first-principles probability. Capital must absorb the ₹4.20 cr tail, not the ₹0.60 cr mean.

Example 3 — Compounding: how a market shock becomes a liquidity-then-credit spiral

A leveraged fund holds ₹200 cr of bonds financed by ₹180 cr of repo (10% haircut, so ₹20 cr equity). Trace what one shock does across three risk categories.

Step 1 — Market shock. Rates spike; bond prices fall 5%. Loss = 5% × 200 = **₹10 cr.** Equity falls from ₹20 cr to ₹10 cr. Pure *market risk*, cleanly measured.

Step 2 — Liquidity transmission (margin call). The repo lender marks the collateral to ₹190 cr and, seeing stress, raises the haircut from 10% to 20%. Required equity = 20% × 190 = ₹38 cr. The fund has ₹10 cr. **Shortfall = ₹28 cr** must be met *in cash today*. The ₹10 cr market loss has become a ₹28 cr *funding-liquidity* crisis — nearly 3× larger.

Step 3 — Market-liquidity feedback (fire sale). To raise cash the fund dumps bonds into a thin market at a further 3% below screen price. Extra loss = 3% × (fraction sold). To raise ₹28 cr of cash at fire-sale prices it must sell ~₹28 cr / (1 − extra haircut) of bonds, realising an additional ~₹0.87 cr of *market-liquidity* loss and pushing prices down further — which triggers fresh margin calls on everyone holding the same bond (systemic channel).

Step 4 — Credit crystallisation. If the fund cannot meet the call, it defaults on the repo. The lender now owns bonds worth less than the loan: a *credit loss* of EAD − collateral value = 180 − 190×(1−further fall).

Reconciliation of the cascade. One ₹10 cr price move (market) → ₹28 cr cash demand (liquidity) → forced sales that deepen the price move (market-liquidity) → default (credit) → contagion to co-holders (systemic). Each arrow *amplifies*: the final system loss dwarfs the initiating ₹10 cr. This is exactly why measuring the four categories in isolation — and summing their standalone VaRs — *understates* stressed loss. Standalone measurement assumes the arrows are switched off; crises switch them on and correlations go to 1.

6. Connections

- **To Chapter 01 (What is risk).** This chapter operationalises the definition of risk into managed categories; risk = quantifiable uncertainty, and taxonomy is how we make it quantifiable per source.
- **To VaR / ES chapters.** Market risk is where the distributional machinery (VaR, ES, Greeks) lives in full. Example 1 is a preview.
- **To credit-modelling chapters.** EL/UL, PD/LGD/EAD, IRB, and CVA all build on §4.2 and Example 2.
- **To Basel / regulatory-capital chapters.** The whole point of classification is that Basel charges capital *by category* — market (FRTB), credit (IRB/Standardised), operational (SMA), plus liquidity ratios (LCR/NSFR). The taxonomy *is* the capital framework's skeleton.
- **To ALM / treasury.** Interest-rate risk in the banking book and funding-liquidity risk are managed by Treasury/ALCO, a different owner than the trading desk — a direct consequence of classification driving ownership.

- **To ERM / governance.** The three-lines-of-defence model assigns each category an owner (1st line business, 2nd line risk/compliance, 3rd line audit).

7. Key Terms

- **VaR (Value-at-Risk).** The loss threshold not exceeded at confidence c over horizon h; the c-quantile of the loss distribution.
- **Expected Shortfall (ES / CVaR).** Average loss *beyond* VaR; coherent and tail-sensitive; Basel FRTB metric.
- **PD / LGD / EAD.** Probability of Default, Loss Given Default (= 1 − recovery), Exposure At Default. Their product is Expected Loss.
- **Expected Loss (EL).** Mean credit loss; priced into spreads and provisioned.
- **Unexpected Loss (UL).** Volatility of credit loss around EL; covered by capital.
- **Funding vs market liquidity.** Ability to raise cash vs ability to sell an asset near fair value.
- **LCR / NSFR.** Basel III liquidity ratios (30-day stress buffer; 1-year stable-funding match).
- **Counterparty credit risk (CCR).** Credit risk on derivatives where the exposure itself is stochastic.
- **Model risk.** Loss from a wrong or misused model; a meta-risk that corrupts other measurements.
- **Systemic risk.** Risk of cascade failure across the interconnected system; undiversifiable.
- **G-SIB.** Global Systemically Important Bank; carries an extra capital surcharge.
- **Coherent risk measure.** One satisfying monotonicity, translation invariance, positive homogeneity, and sub-additivity (VaR fails the last; ES passes).

8. Common Confusions

- **"VaR is the worst case."** No. VaR is the *best of the worst* — the *minimum* loss on a bad day, not the maximum. It says nothing about how bad the tail beyond it is; that is what ES is for.
- **"Expected Loss is what capital covers."** Backwards. EL is *priced and provisioned* (it is a cost, not a surprise). Capital covers *Unexpected* Loss — the volatility around EL. Confusing the two under-capitalises the book (see Example 2: EL ₹0.6 cr vs UL ₹4.2 cr).
- **"Liquidity risk is just a type of market risk."** Two different animals. Market risk is being long an asset whose price falls; funding-liquidity risk is being unable to roll your funding even if the asset's price is fine. SVB's bonds were money-good to maturity — it died of funding liquidity, not market loss.
- **"Solvent means safe."** Solvency (assets > liabilities) and liquidity (cash when due) are different. A solvent firm can fail from a liquidity run before its assets are ever realised (Lehman, Northern Rock).
- **"Operational risk excludes legal risk."** Basel's definition *includes* legal risk but *excludes* strategic and reputational risk. A common exam trap.
- **"Reputational and systemic risk are vague, so they don't matter for measurement."** They are hard to measure but decisive: they are the amplifiers that turn a survivable loss into a fatal one. They matter *most* precisely because they resist the standard metrics.
- **"Total risk = sum of the category VaRs."** Only if correlations are 1. In calm times the sum overstates risk (diversification); in crises the arrows switch on and even the sum understates it. Never additive in the simple way.
- **"Model risk is an IT problem."** It is a governance problem. The London Whale and the Gaussian copula were validated-looking models with wrong assumptions, not buggy code.

9. Recap

- "Risk" is useless as a single word; management begins by **classifying** exposures by their *source of loss*: market, credit, liquidity, operational, model, legal/compliance, reputational, systemic.
- **Source determines math.** Market → VaR/ES on price distributions. Credit → EL = PD×LGD×EAD, capital on UL. Liquidity → LCR/NSFR and survival horizons (non-distributional). Operational → frequency×severity, fat tails. Model/legal/reputational → governed, not neatly computed.
- **Source determines ownership and capital.** Each category maps to a specific owner and a specific Basel capital regime; mislabel and you mis-capitalise and leave the risk unowned.
- The financial risks (market/credit/liquidity) are **taken for reward**; the non-financial ones (operational/model/legal/reputational) are **unrewarded** and to be minimised.
- **The real danger is compounding.** A single shock travels along the arrows — market → liquidity → fire-sale → credit → systemic — and each hop amplifies. Standalone measurement, which assumes the arrows are off, systematically understates stressed loss because in a crisis correlations converge to 1.
- Worked examples confirmed the math: portfolio VaR < sum of VaRs for $\rho<1$ (₹5.12 cr vs ₹5.82 cr); UL ≈ 7× EL from first-principles variance (₹4.20 cr vs ₹0.60 cr); and a ₹10 cr price shock ballooning into a ₹28 cr liquidity call.

10. Quick-Reference / Interview Points

The one-liner: *Financial risk is a taxonomy, not a scalar — you classify by source because the source dictates the measurement, the owner, and the capital, and the biggest losses come from the categories compounding.*

Rapid-fire answers:

- *Name the risk types.* Market, credit, liquidity, operational, model, legal/compliance, reputational, systemic.
- *Market risk metric?* VaR, now Expected Shortfall (97.5%) under FRTB — ES because it is coherent and tail-sensitive.
- *VaR formula (parametric)?* $z_c \times \sigma \times V \times \sqrt{h}$. z = 1.645 (95%), 2.326 (99%).
- *Credit Expected Loss?* EL = PD × LGD × EAD. EL is priced/provisioned; **capital covers UL** (unexpected loss).
- *Two liquidity risks?* Funding (can't raise cash) and market/asset (can't sell near fair value); they feed each other.
- *Liquidity ratios?* LCR (30-day HQLA buffer ≥100%) and NSFR (stable funding ≥100%).
- *Operational risk definition?* Loss from failed processes, people, systems, or external events — *includes legal, excludes strategic & reputational* (Basel).
- *What is model risk?* Loss from a wrong or misused model; a meta-risk corrupting all other measurements; governed via independent validation (SR 11-7).
- *Systemic vs the rest?* Undiversifiable, system-wide, managed macro-prudentially (G-SIB surcharges, buffers, clearing).
- *Solvency vs liquidity?* Assets>liabilities vs cash-when-due; you can be solvent and still fail from a run (Lehman, SVB).
- *Why classify at all?* Measurement, ownership, capital, and the right hedging tool per category.
- *Why do the boxes matter less than the arrows?* Because crises transmit and amplify across categories; standalone/summed VaR assumes independence and understates stressed loss.

Killer closing line for an interview: *"In 2008 the exposure was booked as credit risk but killed banks as liquidity and systemic risk — the loss didn't live in any one box, it lived in the arrows between them. That's why I think of risk as a coupled system, not a checklist."*

Q&A — Types of Financial Risk

Practice bank for Chapter 02. Every question is followed by a full answer. Work each numerical item on paper before reading the solution.

Section A — Concept Checks

A1. Why is the single word "risk" useless to a Chief Risk Officer, and what is the first job of risk management?

Because nobody can hedge, reserve capital against, price, or assign an owner to an undifferentiated "risk." You can only manage what you can *name, measure, and attribute*. The first job is **taxonomy**: cutting the fog of "risk" into distinct categories defined by their *source of loss* — market, credit, liquidity, operational, model, legal/compliance, reputational, systemic — because the source dictates the measurement method, the owner, the capital charge, and the hedging tool.

A2. State the loss driver — "what has to happen in the world to lose money" — for market, credit, liquidity, and operational risk.

- Market risk: a **price/rate/FX/vol has to move** adversely. Fast, symmetric-ish, observed daily.
- Credit risk: a **counterparty has to break a promise** (default or downgrade). Slow, asymmetric, rare events.
- Liquidity risk: you have to be **unable to raise cash or sell an asset** near fair value. A threshold/knock-out risk, not distributional.
- Operational risk: an **internal process, person, or system has to fail** (or an external event hits). Fat-tailed, control-driven.

A3. Distinguish "financial" from "non-financial" risks in this taxonomy, and say why the distinction matters.

Financial risks — market, credit, liquidity — are taken **deliberately to earn a return**; you optimise them (take the right amount for the reward). Non-financial risks — operational, model, legal/compliance, reputational — are **unrewarded by-products** of doing business; you minimise them. It matters because you do not "optimise" an operational-risk appetite the way you size a trading position: unrewarded risk should be driven as low as cost-effectively possible.

A4. VaR is often described as "the worst case." Correct the statement.

VaR is the **best of the worst**, not the worst case. The 99% 1-day VaR is the *minimum* loss you'd expect on the worst 1% of days — the threshold you will not exceed at that confidence — not the maximum possible loss. It says nothing about how deep the tail beyond it runs; **Expected Shortfall** (the average loss *given* you breached VaR) answers that.

A5. "Expected Loss is what capital covers." Why is this backwards?

Expected Loss (EL = PD × LGD × EAD) is the *average* credit loss; it is a predictable cost, so it is **priced into the spread and provisioned**, not a surprise. **Capital covers Unexpected Loss (UL)** — the volatility of loss around the mean, the tail that a bad year actually delivers. Confusing the two under-capitalises the book: EL is small and priced, UL is large and must be absorbed by equity.

A6. Why can a fully solvent bank still fail? Name the risk and two historical cases.

Solvency (assets > liabilities) and liquidity (cash when obligations fall due) are different. A solvent firm can suffer a **funding-liquidity** run — depositors or lenders demand cash faster than assets can be realised — and fail before its assets are ever marked down. Examples: **Lehman Brothers**, **Northern Rock**, and **SVB (2023)**. SVB's bonds were money-good to maturity; it died of funding liquidity, not asset loss.

A7. Give Basel's definition of operational risk, and state precisely what it includes and excludes.

Operational risk is the risk of loss from **inadequate or failed internal processes, people, and systems, or from external events**. It **includes legal risk** but **excludes strategic and reputational risk**. This inclusion/exclusion is a classic exam trap.

A8. Why is model risk called a "meta-risk"?

Because it does not produce losses directly — it **corrupts the measurement of every other risk**. A wrong PD understates credit risk; a wrong volatility surface understates market risk; a wrong correlation assumption (the Gaussian copula in 2008) misprices a whole book. It has no clean formula: it is **governed** (independent validation, back-testing, benchmarking per SR 11-7), not computed.

A9. What makes systemic risk different from every other category, and who must manage it?

It is the risk **of the system itself** — cascade failure through interconnection, common holdings, fire-sale externalities, and funding freezes — not risk *within* one firm. Its defining feature is that **an individual firm cannot diversify it away**, so it must be managed **macro-prudentially by a regulator** (G-SIB surcharges, countercyclical buffers, central clearing, system-wide stress tests), not by a single risk desk.

A10. Explain the phrase "crises live in the arrows, not the boxes."

The neat taxonomy boxes are how we measure risk in calm times. But real crises **transmit and amplify across categories**: a market shock triggers margin calls (liquidity), forced fire sales deepen the price fall (market-liquidity feedback), covenant breaches follow (credit), and correlated de-risking spreads it (systemic). Standalone or summed VaR assumes the arrows are switched off; a crisis switches them on and correlations converge toward 1, so isolated measurement systematically **understates stressed loss**.

Section B — Numerical / Applied (full solutions)

B1. Parametric VaR and square-root-of-time scaling. A desk holds a ₹80 crore equity position with daily return volatility $\sigma = 1.5\%$. Find the 1-day 99% VaR and the 10-day 99% VaR.

Solution. Parametric VaR = $z_c \times \sigma \times V \times \sqrt{h}$, with $z_{99} = 2.326$. 1-day: VaR = $2.326 \times 0.015 \times 80$ = **₹2.7912 crore ≈ ₹2.79 cr.** 10-day: scale by $\sqrt{10} = 3.1623$. $VaR_{10} = 2.7912 \times 3.1623$ = **₹8.826 cr ≈ ₹8.83 cr.** *Check:* dimensional sanity — 1.5% of ₹80 cr is ₹1.2 cr of one-sigma move; 2.326 sigma ≈ ₹2.79 cr ✓. Scaling assumes i.i.d. returns and a constant position.

B2. Portfolio VaR and the diversification benefit. Add to B1's equity VaR (₹2.7912 cr) a ₹80 cr bond position with daily $\sigma = 0.4\%$, correlation $\rho = 0.25$ with the equity. Find the bond VaR, the portfolio 1-day 99% VaR, and the diversification benefit.

Solution. Bond VaR = $2.326 \times 0.004 \times 80$ = **₹0.74432 cr.** Portfolio: $VaR_p = \sqrt{VaR_1^2 + VaR_2^2 + 2\rho \cdot VaR_1 \cdot VaR_2}$ $= \sqrt{2.7912^2 + 0.74432^2 + 2 \cdot 0.25 \cdot 2.7912 \cdot 0.74432} = \sqrt{7.7908 + 0.5540 + 1.0388} = \sqrt{9.3836}$ = **₹3.0633 cr.** Undiversified sum = 2.7912 + 0.74432 = ₹3.5355 cr. **Diversification benefit = 3.5355 − 3.0633 = ₹0.4722**

cr (13.4%). *Check:* set $\rho = 1 \rightarrow \sqrt{((2.7912+0.74432)^2)} = 3.5355$, exactly the undiversified sum, confirming the algebra. For $\rho < 1$ the portfolio VaR must be lower ✓.

B3. Expected Loss and break-even spread. An NBFC lends ₹60 crore (EAD) at 1-year PD = 3%, recovery rate = 45% (so LGD = 55%). Find EL and the minimum spread (over funding cost) that just covers expected credit loss.

Solution. EL = PD × LGD × EAD = 0.03 × 0.55 × 60 = **₹0.99 cr = ₹99 lakh.** Break-even spread = EL / EAD = 0.99 / 60 = **1.65% = 165 bps.** So the loan must earn ~165 bps over funding **just to break even on expected losses** — before profit or the capital charge on unexpected loss.

B4. Unexpected Loss from first principles. For the B3 loan (EAD ₹60 cr, PD 3%, LGD 55% deterministic), find UL and confirm it against a direct two-state variance calculation.

Solution (formula). $UL = EAD \times LGD \times \sqrt{PD(1-PD)} = 60 \times 0.55 \times \sqrt{0.03 \times 0.97} = 33 \times \sqrt{0.0291} = 33 \times 0.17059$ = **₹5.629 cr.** *Direct check.* Loss is ₹0 with prob 0.97, and LGD×EAD = 0.55×60 = ₹33 cr with prob 0.03. Mean = 0.03 × 33 = ₹0.99 cr = EL ✓. Variance = $0.03 \times 0.97 \times 33^2 = 0.0291 \times 1089 = 31.69$; std dev = $\sqrt{31.69}$ = **₹5.629 cr = UL** ✓. *Insight:* UL (₹5.63 cr) is ~5.7× EL (₹0.99 cr). A single default costs ₹33 cr, not ₹0.99 cr — that tail is what capital must absorb; the mean is merely what the spread covers.

B5. Expected Shortfall vs VaR (discrete tail). A ₹100 cr book has these loss outcomes over the horizon: ₹0 (prob 0.90), ₹5 cr (0.06), ₹20 cr (0.03), ₹50 cr (0.01). Find the 95% VaR and the 95% Expected Shortfall.

Solution. Order losses ascending and find the 95th percentile of loss. Cumulative probabilities: ≤₹0 → 0.90; ≤₹5 cr → 0.96; ≤₹20 cr → 0.99; ≤₹50 cr → 1.00. The 95% quantile falls where cumulative first reaches 0.95 → that lies at the ₹5 cr outcome ($0.90 < 0.95 \le 0.96$). So **VaR_{95} = ₹5 cr.** ES_{95} = average loss in the worst 5% tail. The worst 5% by probability = the top 0.05 of outcomes: ₹50 cr (0.01), ₹20 cr (0.03), and 0.01 of the ₹5 cr bucket to fill 0.05. ES = [0.01×50 + 0.03×20 + 0.01×5] / 0.05 = [0.50 + 0.60 + 0.05] / 0.05 = 1.15 / 0.05 = **₹23 cr.** *Insight:* ES (₹23 cr) ≫ VaR (₹5 cr) because the tail beyond VaR is fat. VaR alone would badly understate the danger; this is why FRTB moved to 97.5% ES — it is tail-sensitive and coherent (sub-additive).

B6. The compounding cascade — market shock → liquidity call. A fund holds ₹300 cr of bonds financed by ₹270 cr repo (10% haircut, ₹30 cr equity). Rates spike; bond prices fall 4%. The repo lender re-marks collateral and raises the haircut to 20%. Trace the loss across categories.

Solution. Market loss: 4% × 300 = **₹12 cr.** Equity falls ₹30 cr → ₹18 cr. Collateral re-marks to 300 × 0.96 = ₹288 cr. *Liquidity transmission:* new required equity = 20% × 288 = ₹57.6 cr. Fund has ₹18 cr. **Shortfall = 57.6 − 18 = ₹39.6 cr of cash due today.** *Amplification:* a ₹12 cr price move has become a **₹39.6 cr funding demand — 3.3× larger.** If unmet, the fund defaults on the repo (credit event); forced fire sales push the price down further, hitting every co-holder (systemic channel). *Insight:* summing standalone category VaRs would have assumed these arrows were off and understated the stressed outcome. The initiating ₹12 cr never appears as the final loss — the transmission does.

B7. LCR check. A bank holds ₹500 cr of HQLA and projects ₹620 cr of gross cash outflows and ₹200 cr of inflows over the 30-day stress window (inflows capped at 75% of outflows). Compute the LCR and state whether it passes.

Solution. Net outflows = outflows − min(inflows, 75%×outflows) = 620 − min(200, 465) = 620 − 200 = ₹420 cr. LCR = HQLA / Net outflows = 500 / 420 = **1.19 = 119%.** Since 119% ≥ 100%, the bank **passes** the LCR. *Note:* the 75% inflow cap did not bind here (200 < 465). Had projected inflows been ₹500 cr, they'd be capped at ₹465 cr, giving net outflows of ₹155 cr and LCR = 500/155 = 323%.

Section C — Interview-Style (model answers)

C1. "Walk me through why classification isn't just academic pedantry."

Four operational reasons. **Measurement:** market risk uses VaR/ES on a price distribution, credit uses PD×LGD×EAD over a year, operational uses frequency×severity — apply the wrong math and you get nonsense. **Ownership:** the CRO must point at a specific desk head, credit committee, or COO and say "this is yours"; unowned risk is unmanaged risk. **Capital:** Basel charges capital by risk type, so mislabel and you hold too little (and blow up) or too much (and destroy ROE). **Hedging:** you hedge market risk with derivatives, credit with CDS or provisions, liquidity with an HQLA buffer — the tool depends on the label. In 2008, mortgage exposure booked as slow-moving "credit risk" was really market, liquidity, model, and systemic risk; because it was mislabelled it was mismeasured, under-capitalised, and owned by nobody.

C2. "Explain the EL/UL split to a new credit analyst."

Expected Loss is the *average* loss you'll suffer on a portfolio — PD × LGD × EAD. It's predictable, so you treat it like any other cost: price it into the spread and hold provisions against it. It should never surprise you. Unexpected Loss is the *volatility* around that average — the difference between a normal year and a bad year. That's what threatens solvency, so that's what **capital** exists to absorb. Concretely, on a ₹60 cr loan at 3% PD and 55% LGD, EL is ₹0.99 cr but a single default costs ₹33 cr; the UL of ₹5.6 cr captures that tail. Rule of thumb: **EL is priced, UL is capitalised.** Basel's IRB formula sizes credit capital off UL at 99.9% confidence over one year.

C3. "Why did Basel move from 99% VaR to 97.5% Expected Shortfall under FRTB?"

Two reasons. First, **VaR is blind to the tail** beyond the cut-off — it tells you the threshold but not how catastrophic the breach is; two books with identical VaR can have wildly different tail losses. ES averages the losses beyond VaR, so it *sees* tail thickness. Second, **VaR is not a coherent risk measure** — it can violate sub-additivity, meaning a merged portfolio's VaR can exceed the sum of parts, which perversely penalises diversification. ES is coherent (it satisfies sub-additivity), so diversification never increases it. The confidence level dropped from 99% to 97.5% because 97.5% ES sits at roughly the same severity as 99% VaR for a normal distribution, while adding tail sensitivity — you get the tail information without an arbitrary jump in required capital.

C4. "A bond you own is going to pay par at maturity — how can you have a liquidity problem?"

Because solvency and liquidity are different questions. The bond being money-good answers the *solvency* question — assets will exceed liabilities eventually. Liquidity asks whether I have **cash when my obligations fall due**. If I funded that bond with short-term deposits or repo and the funding runs — depositors leave, the repo lender raises the haircut or refuses to roll — I need cash *today*, and "par in five years" doesn't help. I'm forced to sell into a market that may be thin (market-liquidity risk), crystallising a real loss on an asset that was never impaired. That's exactly how SVB died in 2023: its held-to-maturity bonds were fine on paper, but a deposit run forced sales and the mark-to-market gap became fatal.

C5. "Give me a concrete example of risks compounding, and tell me why it matters for measurement."

A leveraged fund takes a market shock — bond prices drop, say ₹12 cr on a ₹300 cr book. That alone is a clean, small market loss. But the repo lender re-marks the collateral and hikes the haircut, converting a ₹12 cr price move into a ₹40 cr cash call due today — funding-liquidity risk, three times larger. To raise that cash the fund dumps bonds into a thin market, deepening the price fall (market-liquidity feedback) and triggering margin calls on everyone else holding the same bond (systemic). If it can't pay, it defaults on the repo (credit). It matters for measurement because the standard approach — measure each category standalone and sum the VaRs — **assumes those arrows are switched off**. In a crisis they switch on and correlations go to 1, so isolated measurement systematically *understates* stressed loss. That's the case for stress testing and integrated scenario analysis over naive additive VaR.

C6. "How do you manage a risk you can't put a formula on — model risk or reputational risk?"

You govern it rather than compute it. For **model risk**, the discipline is independent model validation (a team separate from the developers), back-testing outputs against realised outcomes, benchmarking against alternative models, conservative reserves for model uncertainty, and an inventory with owners and review cycles — the SR 11-7 framework. The failures — the Gaussian copula, the London Whale VaR re-parameterisation — weren't buggy code; they were plausible-looking models with wrong assumptions, so validation must challenge assumptions, not just arithmetic. For **reputational risk**, you can't hold "reputational VaR," but you can monitor its proxies (funding-spread widening, deposit attrition, share-price reaction to conduct events) and, more importantly, manage its *sources*, since it's almost always a second-order effect of an operational, legal, or credit failure going public. The management lever is controlling the first-order failure and having a credible crisis-response and disclosure posture.

Section D — MCQs (with reasoning)

D1. The 99% 1-day VaR of a book is ₹4 cr. Which statement is correct? (a) The maximum possible 1-day loss is ₹4 cr. (b) On the worst 1% of days the loss will be exactly ₹4 cr. (c) There is a 1% chance the daily loss exceeds ₹4 cr. (d) The average loss on the worst 1% of days is ₹4 cr.

Answer: (c). VaR is the loss threshold breached with 1% probability. (a) is wrong — VaR is not a maximum; losses beyond it are unbounded. (b) is wrong — ₹4 cr is the threshold, not the exact loss on tail days. (d) describes **Expected Shortfall**, not VaR.

D2. Which risk did the Basel definition of operational risk explicitly exclude? (a) Legal risk (b) Internal fraud (c) System failure (d) Strategic and reputational risk

Answer: (d). Basel's operational-risk definition *includes* legal risk (a) and covers internal fraud (b) and system failure (c) among its seven loss-event categories, but **excludes strategic and reputational risk**.

D3. A loan has PD = 4%, LGD = 50%, EAD = ₹25 cr. Expected Loss is: (a) ₹0.50 cr (b) ₹0.25 cr (c) ₹5.00 cr (d) ₹1.00 cr

Answer: (a). EL = PD × LGD × EAD = 0.04 × 0.50 × 25 = ₹0.50 cr. (b) omits LGD's correct application; (c) and (d) misplace the decimal / drop a factor.

D4. Two positions each have standalone VaR of ₹3 cr with correlation ρ = 0. The portfolio VaR is: (a) ₹6 cr (b) ₹0 cr (c) ₹4.24 cr (d) ₹3 cr

Answer: (c). $VaR_p = \sqrt{3^2 + 3^2 + 2 \cdot 0 \cdot 3 \cdot 3} = \sqrt{18}$ = ₹4.243 cr. (a) is the ρ = 1 answer (simple sum); (b) would need ρ = −1 with equal VaRs; (d) ignores the second position entirely.

D5. Which pair correctly matches risk type to its primary metric? (a) Credit risk → Expected Shortfall (b) Liquidity risk → PD × LGD × EAD (c) Operational risk → LCR / NSFR (d) Market risk → VaR / Expected Shortfall

Answer: (d). Market risk uses VaR/ES (FRTB). Credit uses EL = PD×LGD×EAD (a and b are swapped); operational uses frequency×severity / SMA, not LCR/NSFR (c); LCR/NSFR are the *liquidity* metrics.

D6. A firm is solvent (assets > liabilities) but cannot meet a deposit run and collapses. This is primarily: (a) Market risk (b) Funding-liquidity risk (c) Credit risk (d) Model risk

Answer: (b). Solvency is intact, so it isn't asset value (market) or counterparty default (credit). The failure is inability to raise cash when obligations fall due — **funding-liquidity risk** (Lehman, Northern Rock, SVB).

D7. Why did FRTB replace 99% VaR with 97.5% Expected Shortfall? (a) ES is easier to compute (b) ES is coherent (sub-additive) and captures tail thickness beyond VaR (c) VaR is always larger than ES (d) ES ignores diversification

Answer: (b). ES is a coherent measure — it satisfies sub-additivity, so diversification never increases it — and it averages losses beyond the cut-off, capturing tail thickness that VaR ignores. (a) is false (ES is typically harder). (c) is false (ES ≥ VaR at the same confidence). (d) is the opposite of the truth.

D8. "Total portfolio risk = sum of each category's VaR." This is true only when: (a) All pairwise correlations equal 1 (b) All correlations equal 0 (c) The book is fully hedged (d) Never — it's always false

Answer: (a). Simple addition of VaRs holds only under perfect positive correlation ($\rho = 1$). For $\rho < 1$ the sum overstates risk (diversification); in a crisis, when arrows switch on and correlations approach 1, even the sum can understate stressed loss. (b) gives the lowest, not additive, risk.

Self-verification note: All B-section figures were checked two ways where possible — portfolio VaR against the $\rho = 1$ collapse (B2), UL against a direct two-state variance (B4), and ES against its discrete tail-averaging definition (B5). Formulas used: parametric VaR = $z\_c \cdot \sigma \cdot V \cdot \sqrt{h}$ ($z_{95} = 1.645$, $z_{99} = 2.326$); EL = PD·LGD·EAD; UL = $EAD \cdot LGD \cdot \sqrt{PD(1-PD)}$; $VaR\_p = \sqrt{VaR_1^2 + VaR_2^2 + 2\rho \cdot VaR_1 \cdot VaR_2}$; LCR = HQLA / net 30-day stressed outflows (inflows capped at 75% of outflows).

Chapter 03 — Market Risk

1. The Problem / The Need

A bank holds a 1,000 crore portfolio of government bonds. Overnight, the 10-year yield rises by 15 basis points. When the desk opens the next morning, the portfolio is worth roughly 10 crore less — nobody defaulted, no counterparty failed, no fraud occurred. The loss came purely from a **move in a market price**. That is market risk in one sentence: the risk that the mark-to-market value of positions falls because market prices move against you.

This is a fundamentally different animal from credit risk. Credit risk is about *someone else failing to pay you*. Market risk is about *the price of what you hold changing*, whether or not anyone defaults. The two require different measurement tools, different limits, different desks, and — crucially for anyone interviewing for a risk role — different mental models.

Why do we need to measure it separately and carefully?

- **Speed.** Credit losses crystallise over months or years. Market losses crystallise in seconds. A trading book can lose a quarter of its capital between lunch and the close. You cannot manage in hours what you only measure monthly.
- **Two-sidedness.** A loan can only pay back par or less. A traded position can gain *or* lose, and often you are short as well as long. Risk must capture both directions and the netting between them.
- **Leverage and gearing.** Derivatives let a desk take enormous notional exposure for tiny cash outlay. A 100 crore swap position might tie up almost no capital yet swing millions per basis point. Notional tells you nothing; you must measure *sensitivity*.
- **Regulatory capital.** Basel requires banks to hold capital against market risk in the trading book. To size that capital you must quantify potential loss — this is where Value at Risk (VaR) and its successors enter.

So the need is concrete: identify what drives price moves, translate positions into *sensitivities* (how much do I lose per unit move?), aggregate those into a *portfolio loss distribution*, set *limits* so the firm cannot bet more than it can survive, and *hedge* what it does not want. That chain — driver → sensitivity → VaR → limit → hedge — is the spine of this chapter.

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2. The Core Idea

Market risk decomposes into a small number of **risk factors** — interest rates, equity prices, FX rates, and commodity prices (with credit spreads and volatility as important extensions). Any position, however complex, is ultimately a bundle of exposures to these factors.

The core insight is a two-step chain:

> **Loss = (how much the factor moves) × (how sensitive my position is to that factor).**

Formally, for a small move in a risk factor:

$$\Delta P \approx \text{Sensitivity} \times \Delta(\text{Factor})$$

The **factor move** is a property of the *market* — volatile, uncontrollable, estimated from history or implied from option prices. The **sensitivity** is a property of *your position* — known, computable, and controllable through trading and hedging. You cannot stop rates from moving, but you *can* choose how much you lose when they do, by sizing your sensitivity.

This is why the whole discipline organises around sensitivities with names: **delta** (sensitivity to price), **DV01** (sensitivity to a 1bp rate move), **beta** (sensitivity to the market index), **vega** (sensitivity to volatility), **gamma** (how the sensitivity itself changes). Master the sensitivities and market risk becomes a bookkeeping problem: measure each factor exposure, multiply by a plausible move, and add up (allowing for how the factors move together).

The final aggregation — plausible moves + correlations across many positions — is exactly what **VaR** does. VaR is not a separate idea; it is the statistical roll-up of the same sensitivities.

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3. Why / How It Works

Why decompose into factors at all?

A large trading book might hold tens of thousands of instruments. You cannot reason about each one. But every instrument's value is a *function* of a handful of underlying factors. A bond's price is a function of the yield curve. An option's price is a function of the underlying price, volatility, and rates. By projecting everything onto the same small factor set, thousands of positions collapse into a manageable vector of factor exposures. Two bonds and a swap that all depend on the 5-year rate net into a single 5-year DV01.

Why sensitivities (the calculus intuition)

Any smooth pricing function $P(x)$ can be Taylor-expanded around the current factor level:

$$\Delta P = \underbrace{\frac{\partial P}{\partial x}}_{\text{first order}}\,\Delta x \;+\; \underbrace{\tfrac{1}{2}\frac{\partial^2 P}{\partial x^2}}_{\text{second order}}\,(\Delta x)^2 + \dots$$

- The **first derivative** is the *delta / DV01 / beta* family — the linear sensitivity that dominates for small moves.
- The **second derivative** is *gamma / convexity* — the curvature that matters for large moves and for options.

For small moves the linear term is a superb approximation, which is why desks live and breathe first-order Greeks. For large moves or optioned books, the curvature term is not optional — ignoring gamma is how "hedged" books blow up.

Why this feeds VaR

Once every position is a sensitivity to a factor, and you have a statistical description of how factors move (volatilities and correlations), the portfolio's profit-and-loss becomes a *linear combination of random factor moves*. A linear combination of (approximately) normal variables is itself normal, with a variance you can compute from the sensitivities and the covariance matrix. VaR is just a percentile of that P&L distribution. So sensitivities are the bridge from "positions" to "a loss distribution."

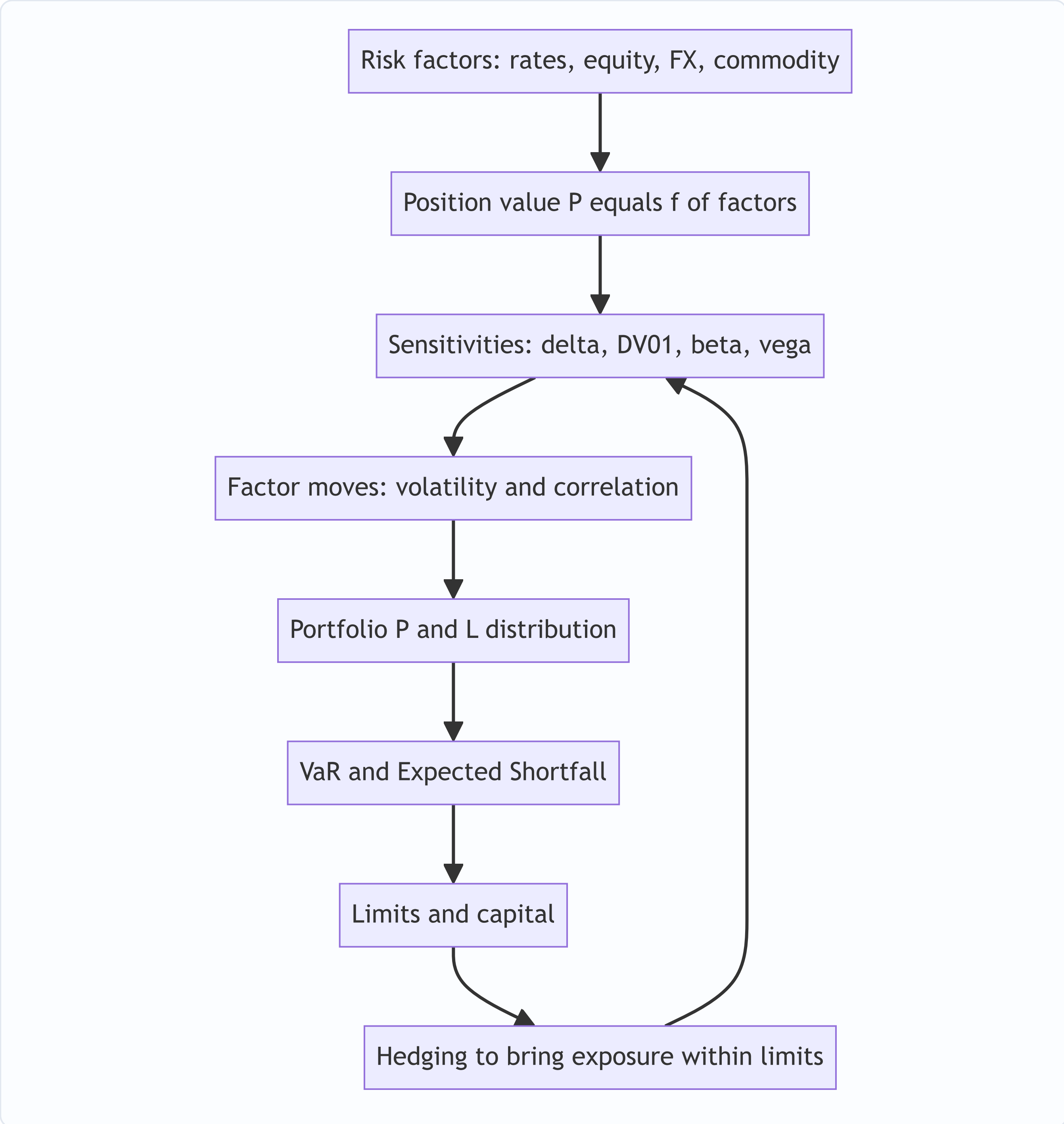

Figure 1 — The market-risk pipeline: positions become sensitivities, sensitivities plus factor statistics become a loss distribution, and limits and hedges close the loop.

4. Full Content — Framework, Formulas, Methods

4.1 The four (plus two) risk factor classes

Factor class	What moves	Typical instruments	Primary sensitivity
Interest rate	Yield curve levels and shape	Bonds, swaps, FRAs, futures	DV01 / PV01, duration, key-rate DV01
Equity	Share and index prices	Stocks, index futures, equity options	Delta, beta
Foreign exchange	Currency pair rates	Spot FX, forwards, cross-currency swaps	FX delta (position in each currency)
Commodity	Oil, metals, gas, agri	Futures, swaps, options	Delta per commodity, basis
Credit spread	Spread over risk-free	Corporate bonds, CDS	CS01 / spread DV01
Volatility	Implied vol of options	All options	Vega

The first four are the classic "market risk" quartet named in every syllabus. Credit spread risk and volatility (vega) risk are treated within market risk when they sit in the trading book.

4.2 The trading book vs the banking book

This distinction is central and heavily examined.

- **Trading book:** positions held with *trading intent* — to profit from short-term price moves or to make markets. Marked to market daily; P&L flows through the income statement immediately. Market risk capital applies here.
- **Banking book:** positions held to *maturity / for the long term* — loans, held-to-maturity bonds, deposits. Accrual accounted. Its main risk is credit, plus **IRRBB** (Interest Rate Risk in the Banking Book), which is managed but not capitalised the same way.

Why the boundary matters: it determines the accounting (mark-to-market vs accrual), the capital treatment, and the risk lens (VaR vs earnings/economic-value sensitivity). Regulators police the boundary tightly because banks are tempted to park losing positions in whichever book has the softer capital charge — this is exactly what the Fundamental Review of the Trading Book (FRTB) hardened.

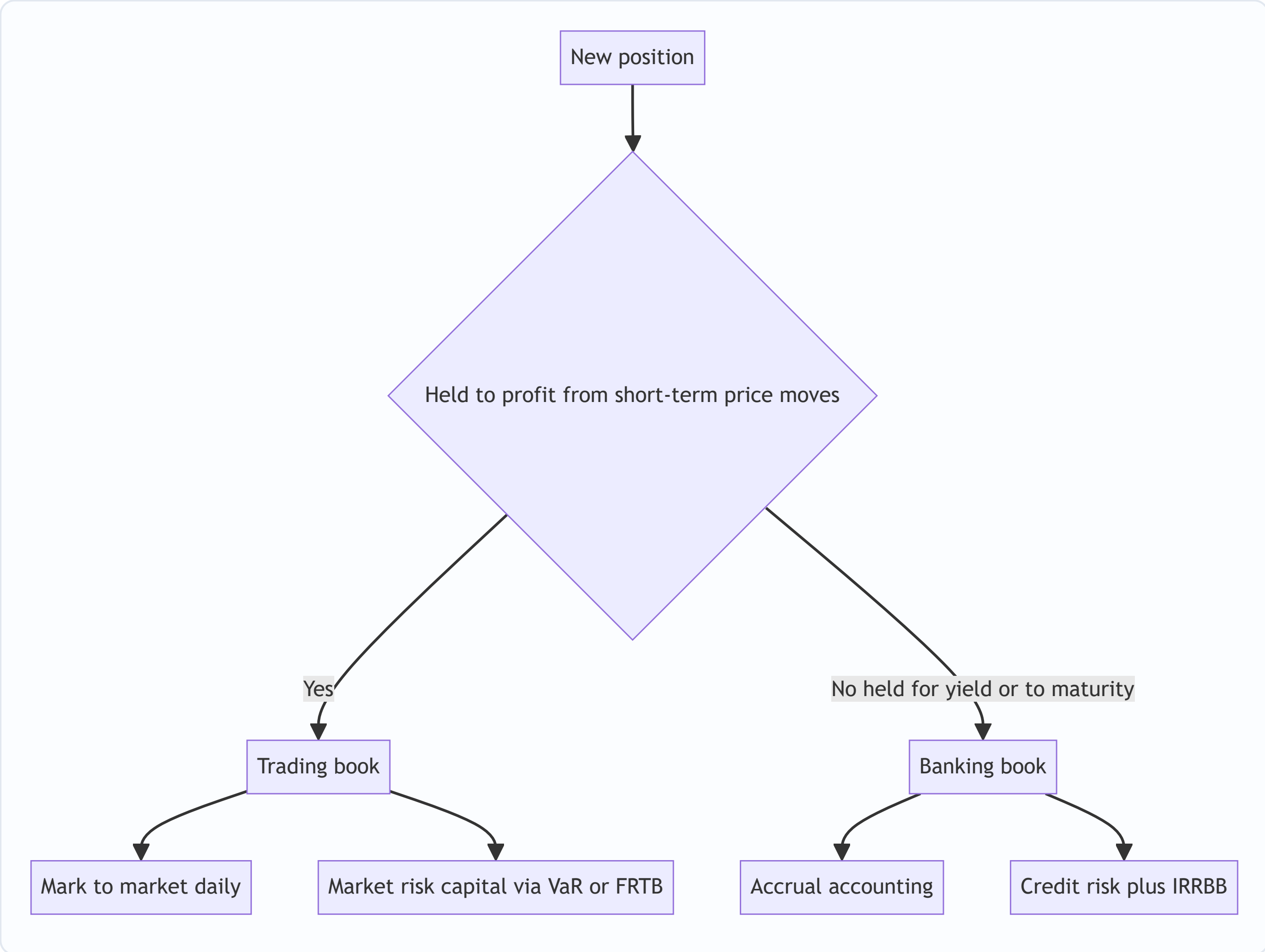

Figure 2 — The book-boundary decision and its consequences for accounting and capital.

4.3 Interest-rate sensitivity: DV01, PV01, and duration

DV01 (Dollar Value of a 01), also called PV01, is the change in a position's value for a **1 basis point (0.01%)** parallel move in the yield curve.

$$\text{DV01} = -\frac{\partial P}{\partial y}\times 0.0001$$

For a bond it links to **modified duration** (D_{mod}):

$$\text{DV01} = D_{mod}\times P \times 0.0001$$

where P is the (dirty) price / market value. Duration measures *percentage* price change per unit yield; DV01 converts that into *money* per basis point, which is what a desk actually hedges.

Convexity is the second-order term. The fuller price change for a yield move Δy:

$$\frac{\Delta P}{P} \approx -D_{mod}\,\Delta y + \tfrac{1}{2}\,C\,(\Delta y)^2$$

Convexity C is always positive for a plain bond, so duration *overstates* the loss on a rate rise and *understates* the gain on a rate fall — a friend to the bondholder.

Key-rate DV01 breaks the single DV01 into buckets (2y, 5y, 10y, 30y) so you can see and hedge *curve shape* risk (steepening/flattening), not just parallel shifts.

4.4 Equity sensitivity: delta and beta

- **Delta** for an equity option is $\partial P/\partial S$ — change in option value per 1 unit change in the underlying. A call has delta between 0 and 1; a put between −1 and 0. Position delta = option delta × contract multiplier × number of contracts.
- **Beta** links a stock or portfolio to the market index:

$$\beta = \frac{\text{Cov}(r_i, r_m)}{\text{Var}(r_m)} = \rho_{i,m}\,\frac{\sigma_i}{\sigma_m}$$

Beta lets you express a diversified equity book as an *equivalent index exposure* — "beta-adjusted net exposure" — which you can hedge with one index futures trade.

4.5 FX and commodity sensitivity

- **FX delta:** the net position in each currency, valued in the base currency. A firm long USD 10m against INR is exposed 1:1 to USD/INR; the sensitivity per 1% move is 1% of the INR value of the position.
- **Commodity delta:** value change per unit move in the commodity price, per commodity and per delivery point. Commodities add **basis risk** (the hedge instrument and the exposure differ in grade, location, or delivery date) and often steep **term-structure** effects (contango/backwardation).

4.6 The Greeks summary (options overlay)

Greek	Sensitivity to	Sign intuition
Delta (Δ)	Underlying price	Directional exposure
Gamma (Γ)	Change in delta	Curvature; large for at-the-money near expiry
Vega (ν)	Implied volatility	Long options are long vega
Theta (Θ)	Passage of time	Long options bleed theta
Rho (ρ)	Interest rate	Usually secondary

4.7 Drivers of market risk

What actually *makes* factors move — the "why prices move" that an interviewer probes:

- **Macro data and policy:** central-bank rate decisions, inflation prints, GDP, employment. These move the whole yield curve and ripple into FX and equities.
- **Liquidity and flows:** thin markets amplify moves; forced selling and margin calls create feedback loops.
- **Risk sentiment / volatility regime:** in "risk-off" episodes correlations spike toward 1 and diversification evaporates exactly when you need it.
- **Supply and demand shocks (commodities):** OPEC decisions, weather, geopolitics.
- **Volatility clustering:** big moves follow big moves (the empirical basis for GARCH and EWMA vol models).

The practical consequence: volatilities and correlations are *not constant*. A VaR built on calm-period data understates risk in a crisis. This is the single most important caveat a candidate can voice.

4.8 The link to VaR

Value at Risk answers: "Over a horizon of h days, at confidence c, what loss will not be exceeded?" A 1-day 99% VaR of 5 crore means: on 99% of days the loss should be ≤ 5 crore; roughly 1 day in 100 it exceeds it.

Parametric (variance-covariance) VaR for a single position:

$$\text{VaR} = z_c \times \sigma_{P\&L} \times \sqrt{h}$$

where z_c is the standard-normal quantile (1.645 for 95%, 2.326 for 99%) and $\sigma_{P\&L}$ is the daily P&L standard deviation. The P&L volatility comes straight from the sensitivity:

$$\sigma_{P\&L} = |\text{Sensitivity}| \times \sigma_{\text{factor}}$$

For a **portfolio of two factors**, variance adds with correlation:

$$\sigma_P = \sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho_{12}\,\sigma_1\sigma_2}$$

Time scaling uses the square-root-of-time rule (valid when returns are i.i.d.): $\text{VaR}_h = \text{VaR}_1 \times \sqrt{h}$.

Three families of VaR methods:

1. **Parametric / variance-covariance:** assumes normal factor returns; fast; poor for options (ignores fat tails and gamma).
2. **Historical simulation:** replay the last N days of actual factor moves through today's portfolio; captures fat tails and real correlations; no distribution assumed; but bounded by the sampled history.
3. **Monte Carlo:** simulate thousands of factor scenarios from an assumed process, full-revalue the book; most flexible (handles options and non-linearity); computationally heavy.

Expected Shortfall (ES / CVaR) is the average loss *given* that VaR is breached — the mean of the tail beyond VaR. It is coherent (respects diversification, unlike VaR) and is the Basel FRTB regulatory measure (97.5% ES). VaR tells you the *threshold*; ES tells you *how bad it gets past the threshold*.

4.9 Managing market risk: limits and hedges

Limits are pre-agreed ceilings so risk-taking stays within appetite:

- **Sensitivity limits:** max DV01, max net delta, max vega per desk.
- **VaR / ES limits:** max 1-day 99% VaR per desk and firm-wide.
- **Stop-loss limits:** cumulative loss that forces position reduction.
- **Concentration / notional limits:** caps per name, sector, currency.
- **Stress-loss limits:** max loss under a defined stress scenario.

Hedging neutralises unwanted sensitivity by taking an offsetting position so the *net* sensitivity falls to (near) zero:

- **Duration/DV01 hedge:** short interest-rate futures or pay-fixed swaps to cut bond-book DV01.
- **Delta hedge:** trade the underlying (or futures) to flatten net delta; must be *re-hedged* as delta drifts (gamma).
- **Beta hedge:** short index futures sized by beta-adjusted exposure.
- **FX hedge:** forwards/swaps to offset currency mismatch.

Hedging is never free or perfect: it leaves **basis risk** (imperfect correlation between hedge and exposure), costs bid-offer and margin, and a delta hedge only holds instantaneously — the residual gamma, vega and theta remain.

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5. Worked Examples

Example 1 — Bond DV01, duration, and a rate shock (with convexity check)

Setup. A desk holds 100 crore face of a bond priced at par (₹100), modified duration 7.0, convexity 60. Rates rise 50 bp (0.50%).

Step 1 — DV01. Market value P = 100 crore.

$$\text{DV01} = D_{mod}\times P \times 0.0001 = 7.0 \times 100{,}00{,}00{,}000 \times 0.0001 = ₹7{,}00{,}000 \text{ per bp}$$

Step 2 — Duration-only loss for 50 bp. A 50 bp move ≈ 50 × DV01:

$$\text{Loss} \approx 50 \times 7{,}00{,}000 = ₹3{,}50{,}00{,}000 = ₹3.50\text{ crore}$$

Equivalently, $-D_{mod}\,\Delta y = -7.0 \times 0.005 = -3.5\%$ of 100 crore = 3.50 crore. **Reconciles.**

Step 3 — Convexity correction. Add $\tfrac{1}{2}C(\Delta y)^2 = 0.5 \times 60 \times (0.005)^2 = 0.5 \times 60 \times 0.000025 = 0.00075 = 0.075\%$.

Convexity is positive, so it *reduces* the loss: net price change ≈ $-3.5\% + 0.075\% = -3.425\%$.

$$\text{Loss} \approx 3.425\% \times 100\text{ crore} = ₹3.425\text{ crore}$$

Interpretation. Linear (DV01/duration) says 3.50 crore; convexity trims it to 3.425 crore. For a rate *fall* of 50 bp the gain would be 3.5% + 0.075% = 3.575% — convexity works in the holder's favour both ways. This asymmetry is why long-convexity positions are prized.

Example 2 — Parametric VaR for a two-asset portfolio (with diversification check)

Setup. A book has two positions: - Position A: equity, market value 50 crore, daily return volatility 2.0%. - Position B: bond, market value 50 crore, daily return volatility 0.8%. - Correlation between A and B returns: ρ = 0.30. - Confidence 99% (z = 2.326), horizon 1 day.

Step 1 — Daily P&L volatility of each position (in money).

$$\sigma_A = 50\text{ cr}\times 2.0\% = ₹1.00\text{ crore}, \qquad \sigma_B = 50\text{ cr}\times 0.8\% = ₹0.40\text{ crore}$$

Step 2 — Portfolio P&L volatility.

$$\sigma_P = \sqrt{\sigma_A^2 + \sigma_B^2 + 2\rho\,\sigma_A\sigma_B}$$ $$= \sqrt{1.00^2 + 0.40^2 + 2(0.30)(1.00)(0.40)} = \sqrt{1.00 + 0.16 + 0.24} = \sqrt{1.40} = ₹1.1832\text{ crore}$$

Step 3 — 1-day 99% VaR.

$$\text{VaR} = z \times \sigma_P = 2.326 \times 1.1832 = ₹2.752\text{ crore}$$

Step 4 — Diversification benefit check. Undiversified (sum of standalone VaRs):

$$\text{VaR}_A = 2.326\times1.00 = 2.326,\quad \text{VaR}_B = 2.326\times0.40 = 0.930$$ $$\text{Sum} = ₹3.256\text{ crore} \;>\; ₹2.752\text{ crore (portfolio)}$$

Diversification benefit = 3.256 − 2.752 = **₹0.504 crore**. Because ρ = 0.30 < 1, the combined VaR is less than the sum — exactly the sub-additivity we expect. **Reconciles.**

Step 5 — Scale to 10 days (square-root-of-time):

$$\text{VaR}_{10} = 2.752 \times \sqrt{10} = 2.752 \times 3.162 = ₹8.70\text{ crore}$$

Example 3 — Beta hedge of an equity book with index futures

Setup. A portfolio worth 20 crore has beta 1.25 to the Nifty. Nifty is at 24,000; each futures contract is 50 index units, so one contract notional = 24,000 × 50 = ₹12,00,000. The manager wants to hedge to beta 0 (fully neutralise market risk).

Step 1 — Beta-adjusted exposure.

$$\text{Exposure} = \beta \times \text{Portfolio value} = 1.25 \times 20\text{ crore} = ₹25\text{ crore}$$

Step 2 — Number of futures to short.

$$N = \frac{\beta \times \text{Portfolio value}}{\text{Futures notional}} = \frac{25{,}00{,}00{,}000}{12{,}00{,}000} \approx 208.3 \Rightarrow \textbf{short 208 contracts}$$

Step 3 — Check with a 2% market drop. Nifty falls 2% → 24,000 → 23,520.

- Portfolio loss ≈ β × market move × value = 1.25 × (−2%) × 20 cr = **−₹0.50 crore**.
- Futures gain (short) = 208 × 50 × (24,000 − 23,520) = 208 × 50 × 480 = 208 × 24,000 = **+₹49,92,000 ≈ +₹0.499 crore**.

Net ≈ −0.50 cr + 0.499 cr ≈ **−₹0.001 crore**, essentially flat. The small residual is the rounding from 208.3 → 208 contracts. **Reconciles** — the beta hedge neutralised the directional loss.

Target-beta variant. To move to a target beta β_T instead of 0:

$$N = \frac{(\beta_T - \beta_P)\times \text{Portfolio value}}{\text{Futures notional}}$$

A negative N means short. To go from 1.25 to 0.50: $N = (0.50-1.25)\times 20\text{cr}/12{,}00{,}000 = -125$ → short 125 contracts.

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6. Connections

- **To VaR / Chapter on measurement:** sensitivities are the *inputs* to VaR. The parametric VaR in Example 2 is nothing but the Greeks fed through a covariance matrix. Historical and Monte Carlo VaR re-price the *same* positions under different scenarios.
- **To credit risk:** Expected Loss = PD × LGD × EAD is the credit analogue of the market-risk chain "move × sensitivity." Both convert a stochastic driver into an expected/potential money loss. Credit-spread risk (CS01) is where the two disciplines overlap on traded bonds.
- **To ALM / IRRBB (banking book):** the *same* DV01/duration mathematics is used, but the objective shifts from a 1-day VaR to protecting net interest income and economic value of equity over years.
- **To derivatives pricing:** the Greeks come straight from the pricing model (e.g., Black-Scholes). Risk management is pricing differentiated — the same partial derivatives that price an option tell you how to hedge it.
- **To regulation:** Basel II.5 introduced Stressed VaR after 2008; **FRTB** replaced VaR with **97.5% Expected Shortfall**, hardened the trading/banking book boundary, and added liquidity-horizon scaling. Interviewers love the VaR→ES→FRTB arc.
- **To stress testing:** VaR describes *normal* days; stress tests and scenario analysis cover the tail that VaR, by construction, understates.

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7. Key Terms

- **Market risk:** risk of loss from changes in market prices/rates.
- **Risk factor:** an underlying variable (rate, price, FX, vol) that drives position value.
- **Trading book / banking book:** trading-intent, marked-to-market positions vs held-for-yield, accrual positions.
- **DV01 / PV01:** money change in value per 1 bp rate move.
- **Modified duration:** percentage price change per unit yield change.
- **Convexity:** second-order (curvature) rate sensitivity; positive for plain bonds.
- **Delta:** sensitivity of value to the underlying price.
- **Gamma:** rate of change of delta (curvature for options).
- **Vega:** sensitivity to implied volatility.
- **Beta:** sensitivity of a stock/portfolio to the market index.
- **Basis risk:** residual risk when the hedge and the exposure are imperfectly correlated.
- **VaR (Value at Risk):** loss threshold not exceeded at a given confidence over a horizon.
- **Expected Shortfall (ES/CVaR):** average loss beyond VaR; coherent; the FRTB measure.
- **Stressed VaR:** VaR calibrated to a historical stress period.
- **Square-root-of-time rule:** $\text{VaR}_h = \text{VaR}_1\sqrt{h}$ under i.i.d. returns.
- **IRRBB:** interest-rate risk in the banking book.

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8. Common Confusions

- **Notional ≠ risk.** A 500 crore swap can carry less risk than a 50 crore bond. Risk is *sensitivity*, not size of the ticket. Always ask for DV01/delta, not notional.
- **Duration is not maturity.** Duration is a *sensitivity* (weighted average time to cash flows, price-elasticity to yield), measured in years but meaning "how much value moves per yield move." A 10-year zero has duration ≈ 10; a 10-year coupon bond less.
- **VaR is not the maximum loss.** 99% VaR is *breached* about 1 day in 100, and says nothing about how bad the breach is. That is ES's job. Never call VaR "worst case."
- **VaR is not sub-additive in general.** For non-normal (e.g., optioned) portfolios, combined VaR can exceed the sum of parts — a theoretical defect that motivated ES. In the normal/linear case it *is* sub-additive (Example 2).
- **Delta-hedged ≠ risk-free.** A delta hedge removes first-order price risk *instantaneously only*. Gamma, vega, and theta remain, and delta drifts, forcing continuous re-hedging.
- **Correlation is not constant.** VaR built on calm-market correlations understates crisis risk, when correlations converge toward 1 and diversification vanishes.
- **Trading vs banking book is not about instrument type.** The *same* bond can sit in either book; what determines it is *intent* and it drives accounting and capital, not the security's identity.
- **Higher confidence VaR is not "better."** 99% vs 95% just moves the threshold; a 99% number is larger but breached less often. Neither describes the tail depth.
- **Convexity/gamma sign matters.** Long convexity (bonds) and long gamma (bought options) help you in big moves; short gamma (sold options) is the classic "picking up pennies in front of a steamroller."

9. Recap

Market risk is the risk that positions lose value because **market prices move** — rates, equities, FX, commodities (plus credit spreads and volatility). It lives primarily in the **trading book**, where positions are marked to market daily and attract market-risk capital, distinct from the accrual-accounted banking book.

The measurement chain is: decompose every position into exposures to a small set of **risk factors**; translate each exposure into a **sensitivity** — DV01/duration for rates, delta and beta for equities, FX and commodity deltas, vega for volatility, and second-order gamma/convexity for curvature. Loss ≈ sensitivity × factor move.

Feed those sensitivities, together with factor **volatilities and correlations**, into **VaR** (parametric, historical, or Monte Carlo) to get a portfolio loss distribution, and read off a percentile — with **Expected Shortfall** describing the tail beyond it. We worked three numbers that reconciled: a bond DV01/convexity loss (3.50 → 3.425 crore), a two-asset 99% VaR (2.75 crore, with a 0.50 crore diversification benefit), and a beta hedge (208 index futures that flattened a 0.50 crore directional loss to ~zero).

Finally, the firm **manages** the risk it has measured: **limits** (sensitivity, VaR/ES, stop-loss, concentration, stress) cap how much can be lost, and **hedges** (duration, delta, beta, FX) neutralise unwanted exposure — always subject to basis risk, cost, and the residual Greeks a first-order hedge leaves behind.

10. Quick-Reference / Interview Points

Formula card

Concept	Formula
Value change (1st order)	$\Delta P \approx \text{Sensitivity}\times\Delta\text{Factor}$
DV01	$D_{mod}\times P \times 0.0001$
Price change with convexity	$\Delta P/P \approx -D_{mod}\Delta y + \tfrac12 C(\Delta y)^2$
Beta	$\rho_{im}\,\sigma_i/\sigma_m$
Parametric VaR	$z_c\,\sigma_{P\&L}\sqrt{h}$
Portfolio vol (2 assets)	$\sqrt{\sigma_1^2+\sigma_2^2+2\rho\sigma_1\sigma_2}$
Time scaling	$\text{VaR}_h = \text{VaR}_1\sqrt{h}$
Beta hedge contracts	$(\beta_T-\beta_P)\times V / \text{Futures notional}$
z-values	1.645 (95%), 2.326 (99%), 1.960 (97.5% ES cut)

One-liners to have ready

- "Market risk = mark-to-market loss from price moves; credit risk = counterparty not paying. Different desks, different capital."
- "Notional tells me nothing; give me DV01 and net delta."
- "VaR is a percentile of the P&L distribution, not the worst case. ES is the average of the tail beyond it — coherent, and the FRTB measure at 97.5%."
- "The trading/banking book split is about *intent*, and it decides accounting and capital."

- "Delta-hedged is only instantaneously safe — gamma, vega, theta and basis risk remain."
- "Correlations spike to 1 in a crisis, so calm-market VaR understates tail risk; that's why we stress test and hold Stressed VaR."
- "Convexity and long gamma help you in big moves; short gamma is picking up pennies in front of a steamroller."

Regulatory arc: VaR (Basel II) → Stressed VaR (Basel II.5, post-2008) → 97.5% Expected Shortfall + hardened book boundary + liquidity horizons (FRTB).

Three VaR methods, one-line each: Parametric — fast, normal, bad for options. Historical — replay real moves, fat tails, bounded by history. Monte Carlo — flexible, full revaluation, compute-heavy.

Q&A — Market Risk

Practice bank for Chapter 03 (Market Risk). Every question is followed by a full answer. Attempt each before reading the solution. Numerical answers are reconciled at least two independent ways. z-values used: 1.645 (95%), 1.960 (97.5%), 2.326 (99%).

Section A — Concept Check

A1. In one sentence, what is market risk, and what single word separates it from credit risk? Market risk is the risk that the mark-to-market value of positions falls because **market prices move** — rates, equities, FX, commodities. The separating word is *price*: market risk is about the price of what you *hold* moving, whereas credit risk is about a *counterparty* failing to pay. No default is needed for a market loss; a yield tick or an FX move is enough.

A2. State the two-step chain at the heart of market-risk measurement. Loss ≈ **(how much the factor moves) × (how sensitive my position is to that factor)**, i.e. $\Delta P \approx \text{Sensitivity} \times \Delta(\text{Factor})$. The factor move is a property of the *market* (uncontrollable, estimated from history or implied vol); the sensitivity is a property of *your position* (known, computable, controllable by trading and hedging). You cannot stop rates moving, but you choose how much you lose when they do.

A3. Name the four classic market-risk factor classes plus the two common extensions, and the primary sensitivity of each. Interest rate (DV01/duration), equity (delta/beta), foreign exchange (FX delta = net currency position), commodity (delta per commodity, plus basis). The two extensions carried within market risk in the trading book are credit spread (CS01/spread DV01) and volatility (vega).

A4. Why measure market risk separately and more frequently than credit risk? Speed and two-sidedness. Credit losses crystallise over months; market losses crystallise in seconds — a trading book can lose a large fraction of capital between lunch and the close, so you cannot manage monthly what moves by the second. A loan pays par or less (one-sided), but a traded position can gain *or* lose and can be short as well as long, so the measurement must capture both directions and the netting between them.

A5. Distinguish the trading book from the banking book, and say what the boundary decides. The trading book holds positions with *trading intent* (short-term profit / market-making); they are marked to market daily and attract market-risk capital. The banking book holds positions for yield or to maturity (loans, HTM bonds, deposits); they are accrual-accounted and carry credit risk plus IRRBB. The boundary is decided by *intent*, not by instrument type — the **same** bond can sit in either book — and it drives accounting (MTM vs accrual), capital treatment, and the risk lens (VaR vs earnings/economic-value sensitivity).

A6. Define DV01 and give its relationship to modified duration. DV01 (PV01) is the money change in a position's value for a **1 basis point (0.01%)** parallel yield move: $\text{DV01} = D_{mod} \times P \times 0.0001$, where P is the market (dirty) value. Duration measures the *percentage* price change per unit yield; DV01 converts that into *money per basis point*, which is what a desk actually hedges.

A7. What is convexity and which way does it work for a plain bond holder? Convexity is the second-order (curvature) rate sensitivity: $\Delta P/P \approx -D_{mod}\,\Delta y + \tfrac12 C(\Delta y)^2$. For a plain bond $C > 0$, so duration *overstates* the loss on a rate rise and *understates* the gain on a rate fall — the correction helps the holder both ways, which is why long-convexity positions are prized.

A8. Define VaR precisely, and state clearly what it is *not*. A 1-day 99% VaR of ₹5 crore means: on 99% of days the loss should not exceed ₹5 crore, and roughly 1 day in 100 it will exceed it. VaR is a *percentile* of the P&L distribution — a threshold. It is **not** the maximum or worst-case loss, and it says nothing about how bad the breach is once exceeded; that is Expected Shortfall's job.

A9. What is Expected Shortfall and why did regulators move to it? Expected Shortfall (ES / CVaR) is the *average* loss given that VaR is breached — the mean of the tail beyond VaR. It is **coherent** (it respects diversification / sub-additivity, which VaR can violate for non-linear books) and it describes tail *depth*, not just a threshold. Basel's FRTB replaced VaR with **97.5% ES** for exactly these reasons.

A10. State the square-root-of-time rule and its key assumption. $\text{VaR}_h = \text{VaR}_1 \times \sqrt{h}$: to scale a 1-day VaR to an h-day horizon, multiply by $\sqrt{h}$. It is valid only when returns are **i.i.d.** (independent, identically distributed, zero autocorrelation). Volatility clustering and mean-reversion break it, so it is an approximation, not a law.

A11. Name the three VaR methods in one line each. Parametric (variance-covariance) — assumes normal factor returns, fast, poor for options. Historical simulation — replay actual past factor moves through today's book, captures fat tails, but bounded by the sampled history. Monte Carlo — simulate scenarios from an assumed process and full-revalue, most flexible for non-linearity, computationally heavy.

A12. Why is "delta-hedged" not the same as "risk-free"? A delta hedge removes first-order price risk *instantaneously only*. Gamma (delta drifts as the price moves), vega (volatility risk), theta (time decay), and basis risk all remain, so the book must be continuously re-hedged. A static delta hedge is safe for one instant and one small move, not over time or over a large move.

Section B — Numerical / Applied

B1 — Bond DV01, rate shock, and a convexity check

Setup. A desk holds ₹100 crore face of a bond priced at par (₹100), modified duration 7.0, convexity 60. Rates rise 40 bp.

Step 1 — DV01. Market value P = ₹100 crore. $$\text{DV01} = D_{mod} \times P \times 0.0001 = 7.0 \times 100{,}00{,}00{,}000 \times 0.0001 = ₹7{,}00{,}000 \text{ per bp.}$$

Step 2 — Duration-only loss for 40 bp. $$\text{Loss} \approx 40 \times 7{,}00{,}000 = ₹2{,}80{,}00{,}000 = ₹2.80\text{ crore.}$$ Cross-check via percentages: $-D_{mod}\,\Delta y = -7.0 \times 0.0040 = -2.80\%$ of ₹100 crore = ₹2.80 crore. **Reconciles.**

Step 3 — Convexity correction. $$\tfrac12 C(\Delta y)^2 = 0.5 \times 60 \times (0.0040)^2 = 0.5 \times 60 \times 0.000016 = 0.00048 = 0.048\%.$$ Convexity is positive so it *reduces* the loss: net ≈ $-2.80\% + 0.048\% = -2.752\%$ → **₹2.752 crore**.

Interpretation. Linear says ₹2.80 crore; convexity trims it to ₹2.752 crore. For a 40 bp *fall*, the gain would be $2.80\% + 0.048\% = 2.848\%$ — convexity helps both ways.

B2 — Single-position parametric VaR (95% and 99%)

Setup. An equity position worth ₹40 crore has a daily return volatility of 1.8%. Horizon 1 day.

Step 1 — Daily P&L volatility in money. $$\sigma_{P\&L} = 40\text{ cr} \times 1.8\% = ₹0.72\text{ crore.}$$

Step 2 — VaR at each confidence. $$\text{VaR}{95} = 1.645 \times 0.72 = ₹1.184\text{ crore.}$$ $$\text{VaR}{99} = 2.326 \times 0.72 = ₹1.675\text{ crore.}$$

Interpretation & check. Ratio $2.326/1.645 = 1.414$, and $1.675/1.184 = 1.415$. **Reconciles** — the 99% number is larger simply because the threshold sits further into the tail, not because the day is "worse."

B3 — Two-asset portfolio VaR with a diversification check

Setup. Position A: equity, ₹50 crore, daily vol 2.0%. Position B: bond, ₹50 crore, daily vol 0.8%. Correlation ρ = 0.30. Confidence 99%, horizon 1 day.

Step 1 — Standalone P&L vols. $$\sigma_A = 50 \times 2.0\% = ₹1.00\text{ cr}, \qquad \sigma_B = 50 \times 0.8\% = ₹0.40\text{ cr.}$$

Step 2 — Portfolio P&L vol. $$\sigma_P = \sqrt{\sigma_A^2 + \sigma_B^2 + 2\rho\sigma_A\sigma_B} = \sqrt{1.00^2 + 0.40^2 + 2(0.30)(1.00)(0.40)}$$ $$= \sqrt{1.00 + 0.16 + 0.24} = \sqrt{1.40} = ₹1.1832\text{ crore.}$$

Step 3 — 1-day 99% VaR. $$\text{VaR} = 2.326 \times 1.1832 = ₹2.752\text{ crore.}$$

Step 4 — Diversification benefit. Standalone VaRs: $\text{VaR}_A = 2.326 \times 1.00 = 2.326$; $\text{VaR}_B = 2.326 \times 0.40 = 0.930$. Sum = ₹3.256 crore > ₹2.752 crore. $$\text{Benefit} = 3.256 - 2.752 = ₹0.504\text{ crore.}$$ Because ρ = 0.30 < 1, combined VaR is below the sum — sub-additivity as expected. **Reconciles.**

Step 5 — Scale to 10 days. $$\text{VaR}_{10} = 2.752 \times \sqrt{10} = 2.752 \times 3.162 = ₹8.70\text{ crore.}$$

B4 — Expected Shortfall from VaR (normal case)

Setup. Same portfolio as B3: $\sigma_P = ₹1.1832$ crore, normal P&L, confidence 99%.

Formula. For a normal loss distribution, $\text{ES}_c = \sigma_P \times \dfrac{\phi(z_c)}{1-c}$, where ϕ is the standard-normal density. At 99%, $z = 2.326$, $\phi(2.326) = \dfrac{1}{\sqrt{2\pi}}e^{-2.326^2/2} = 0.3989 \times e^{-2.705} = 0.3989 \times 0.0669 = 0.02669$.

Compute. $$\text{ES}_{99} = 1.1832 \times \frac{0.02669}{0.01} = 1.1832 \times 2.669 = ₹3.158\text{ crore.}$$

Interpretation & check. ES (₹3.158 cr) > VaR (₹2.752 cr) — always true, since ES averages the tail *beyond* the VaR threshold. The ES multiplier 2.669 exceeds the VaR multiplier 2.326, confirming ES sits deeper in the tail. **Reconciles.**

B5 — Expected loss vs VaR: don't confuse them

Setup. A ₹200 crore bond position has a daily P&L volatility of ₹1.5 crore and an *expected* daily P&L of zero (a fair, unbiased mark). Find the 99% VaR and contrast it with "expected loss."

Solution. $$\text{VaR}_{99} = 2.326 \times 1.5 = ₹3.489\text{ crore.}$$ Expected daily P&L is **zero**, so the *expected* loss is nil — on an average day you neither make nor lose. VaR is a *tail* quantity: it answers "how bad is a bad (1-in-100) day," a completely different question from "what do I expect on a typical day." Reporting the mean when asked for VaR (or vice-versa) is a classic error: the mean describes the centre, VaR describes the tail.

B6 — Beta hedge with index futures

Setup. A portfolio worth ₹20 crore has beta 1.25 to the Nifty. Nifty = 24,000; lot = 50 units, so one contract notional = 24,000 × 50 = ₹12,00,000. Hedge to beta 0.

Step 1 — Beta-adjusted exposure. $1.25 \times 20 = ₹25$ crore.

Step 2 — Contracts to short. $$N = \frac{\beta \times V}{\text{Futures notional}} = \frac{25{,}00{,}00{,}000}{12{,}00{,}000} = 208.3 \Rightarrow \textbf{short 208 contracts.}$$

Step 3 — Check with a 2% market drop. Nifty 24,000 → 23,520. - Portfolio loss ≈ $1.25 \times (-2\%) \times 20 = -₹0.50$ crore. - Futures gain (short) = $208 \times 50 \times (24{,}000 - 23{,}520) = 208 \times 50 \times 480 = +₹49{,}92{,}000 \approx +₹0.499$ crore. - Net ≈ **−₹0.001 crore**, essentially flat; the residual is rounding 208.3 → 208. **Reconciles.**

Target-beta variant. To move to β_T instead of 0: $N = \dfrac{(\beta_T - \beta_P)\,V}{\text{Futures notional}}$. Going from 1.25 to 0.50: $N = (0.50 - 1.25)\times 20/12{,}00{,}000 \times 10^7 = -125$ → short 125 contracts.

B7 — FX VaR

Setup. A firm is long USD 10 million against INR. USD/INR = 83.00. The daily volatility of USD/INR returns is 0.5%. Find the 1-day 95% VaR in INR.

Step 1 — INR value of the position. $10{,}000{,}000 \times 83.00 = ₹83$ crore.

Step 2 — Daily P&L vol. $83 \times 0.5\% = ₹0.415$ crore.

Step 3 — 95% VaR. $1.645 \times 0.415 = ₹0.683$ crore.

Interpretation. A 1% move in USD/INR changes the position by 1% of ₹83 crore = ₹0.83 crore, so FX delta here is 1:1 with the INR value — the sensitivity is simply the net currency position measured in the base currency.

Section C — Interview-Style (Model Answers)

C1. "A trader tells you his book is fine because the notional is small. What's your response?" Notional tells me almost nothing about risk — *sensitivity* does. A ₹500 crore swap can carry less risk than a ₹50 crore bond, because risk is DV01 and net delta, not the size of the ticket. Derivatives let a desk take huge notional exposure for tiny cash outlay, so I'd ask for the DV01, the net delta, the vega, and the VaR — the sensitivities and their statistical roll-up — before I'd form any view on whether the book is "fine."

C2. "Walk me from a single position to a firm-wide VaR number." Every position, however complex, is a function of a handful of risk factors. First I *decompose* it into factor exposures — a bond onto the yield curve, an option onto spot/vol/rate. Second I translate each exposure into a *sensitivity*: DV01, delta, beta, vega, plus gamma/convexity for curvature. Third I bring in the factors' *statistics* — volatilities and correlations. Because the P&L is then a linear combination of factor moves, its variance follows from the sensitivities and the covariance matrix; VaR is simply a percentile of that distribution. Historical and Monte Carlo VaR reprice the *same* positions under, respectively, replayed real moves and simulated scenarios. So the spine is: factor → sensitivity → loss distribution → percentile.

C3. "Why did the industry move from VaR to Expected Shortfall?" Two reasons. First, VaR only marks a *threshold* — it is silent on how bad the loss is once breached, so two books with identical VaR can have wildly different tail depth. ES averages the tail beyond VaR and captures that. Second, VaR is **not sub-additive** for general (non-linear, optioned) portfolios: combined VaR can exceed the sum of parts, which perversely penalises diversification and can be gamed. ES is coherent — it always respects diversification. Basel's FRTB codified the switch at 97.5% ES, added liquidity-horizon scaling, and hardened the trading/banking book boundary.

C4. "Your VaR model passed backtesting all year, then blew through the limit five days running in a crisis. What happened?" Almost certainly the model was calibrated on calm-period data, and in a crisis two things changed at once. Volatilities jumped — big moves cluster, so yesterday's vol underestimated today's. And correlations converged toward 1: in a risk-off episode, diversification evaporates exactly when you need it, so the offsetting positions the VaR relied on all moved the same way. VaR by construction describes *normal* days; it understates the tail. That's precisely why we hold **Stressed VaR** (calibrated to a stress window), run **stress tests and scenario analysis**, and report **ES** — the tools designed for the regime VaR misses.

C5. "Explain why a delta-hedged options book can still lose money fast." Delta hedging removes first-order price risk for one instant and one small move only. What remains is gamma — as the underlying moves, delta itself changes, so a book that was neutral becomes directional between re-hedges. If the desk is *short* gamma (has sold options), every large move hurts and re-hedging locks in losses — "picking up pennies in front of a steamroller." On top of gamma sit vega (a vol spike revalues the options) and theta (time decay), plus basis risk if the hedge instrument isn't the exact underlying. So "delta-hedged" is a snapshot, not a state of safety.

C6. "How is the market-risk loss chain related to the credit-risk expected-loss formula?" They're structurally the same idea: turn a stochastic driver into a money loss. Market risk: Loss ≈ sensitivity × factor move. Credit risk: Expected Loss = PD × LGD × EAD — probability of default times loss-given-default times exposure-at-default. Both convert an uncertain event into an expected or potential loss. The two disciplines actually overlap on traded corporate bonds via credit-spread risk (CS01), which is a *market* sensitivity to a *credit* factor. The difference is speed and accounting: market losses are marked to market daily; credit losses accrue over months.

Section D — MCQs (with reasoning)

D1. A 1-day 99% VaR of ₹4 crore means: A) The book can never lose more than ₹4 crore. B) The average loss is ₹4 crore. C) On about 1 day in 100 the loss is expected to exceed ₹4 crore. D) The loss is exactly ₹4 crore on 1% of days.

Answer: C. VaR is a percentile/threshold: losses stay within ₹4 crore on ~99% of days and breach it on ~1%. A is wrong (VaR is not a maximum). B confuses VaR with the mean. D is wrong — VaR bounds the tail probability, it does not fix the loss at exactly ₹4 crore.

D2. Which sensitivity best captures the money impact of a 1 bp yield move on a bond? A) Beta B) DV01 C) Vega D) Delta

Answer: B. DV01 (= $D_{mod} \times P \times 0.0001$) is the money change per 1 bp move — precisely a bond desk's rate sensitivity. Beta is equity-to-index, vega is volatility sensitivity, delta is price sensitivity for an option/equity.

D3. In the two-asset variance formula, lowering the correlation ρ (all else equal) will: A) Raise portfolio VaR B) Lower portfolio VaR C) Leave it unchanged D) Make it negative

Answer: B. Portfolio vol is $\sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2}$; the cross term falls as ρ falls, shrinking σ_P and hence VaR. Lower correlation = more diversification = less risk. VaR can never be negative (D), and it is not invariant to ρ (C).

D4. Expected Shortfall is preferred over VaR primarily because it: A) Is always smaller than VaR. B) Ignores the tail. C) Is coherent and measures the depth of the tail beyond VaR. D) Requires no data.

Answer: C. ES is sub-additive (coherent) and averages losses *beyond* the VaR threshold, capturing tail severity. A is false — ES ≥ VaR. B is the opposite of the truth. D is false — ES needs at least as much data as VaR.

D5. Which of these determines whether a bond sits in the trading or banking book? A) Its credit rating B) Its maturity C) The holder's intent D) Its coupon

Answer: C. The boundary is about *intent* — held to profit from short-term price moves (trading, marked to market) vs held for yield/to maturity (banking, accrual). The **same** bond can sit in either book; rating, maturity and coupon are irrelevant to the classification.

D6. Under the square-root-of-time rule, the 16-day VaR equals the 1-day VaR multiplied by: A) 16 B) 8 C) 4 D) 2

Answer: C. $\text{VaR}_{16} = \text{VaR}_1 \times \sqrt{16} = \text{VaR}_1 \times 4$. The rule scales with $\sqrt{h}$, not h, and holds under i.i.d. returns.

D7. A desk that has *sold* options (is short gamma) will find that large underlying moves: A) Always help it. B) Hurt it, and re-hedging tends to lock in losses. C) Have no effect once delta-hedged. D) Reduce its vega to zero.

Answer: B. Short gamma means delta moves *against* the desk as the underlying moves, so re-hedging means buying high and selling low — losses accumulate in big moves ("pennies in front of a steamroller"). Delta-hedging does not remove gamma (C is false), and being short options means short vega, not zero vega (D is false).

D8. Parametric (variance-covariance) VaR is *least* reliable for: A) A cash bond portfolio. B) A linear FX book. C) An options portfolio with significant gamma. D) A single equity holding.

Answer: C. Parametric VaR assumes normal factor returns and linear payoffs; it ignores fat tails and the curvature (gamma) of options, so it mismeasures optioned books. Linear positions (A, B, D) are where the normal/linear assumption is most defensible.

Self-check: B1 (₹2.80 → 2.752 cr), B2 (z-ratio 1.414 ≈ 1.415), B3 (√1.40 = 1.1832; benefit ₹0.504 cr), B4 (ES multiplier 2.669 > VaR 2.326), B6 (−₹0.50 cr offset by +₹0.499 cr) all reconcile.

Chapter 04 — Value at Risk (VaR)

1. The Problem / The Need

A bank's trading desk holds thousands of positions: government bonds, corporate credit, FX forwards, equity options, commodity swaps. The Chief Risk Officer walks in one morning and asks a deceptively simple question: **"How much money can we lose today?"**

Before the 1990s there was no clean answer. The desk could hand over a stack of reports — one page of interest-rate sensitivities (DV01), another of equity betas, a third of option greeks (delta, gamma, vega), a fourth of notional exposures by currency. Each number is correct, but they are in different units and cannot be added. A DV01 of ₹4 lakh per basis point and an equity delta of ₹90 crore and a vega of ₹2 lakh per vol-point simply do not sum to a single "risk" figure. Worse, they ignore how positions offset each other: a long bond and a paid fixed swap may largely cancel.

The board, the regulator, and the CEO do not want a spreadsheet of greeks. They want **one number, in rupees, that they can compare across desks, across days, and against a limit.** That number has to answer: over a defined period, with a stated level of confidence, what is the most we should lose under normal market conditions?

That single-number demand is the problem Value at Risk was invented to solve. J.P. Morgan's *RiskMetrics* (1994) popularised it; the Basel Committee then embedded it into bank capital rules, and it became the lingua franca of market-risk management. Even where it has since been supplemented (by Expected Shortfall under Basel's FRTB), VaR remains the number every risk analyst must be able to compute, defend, and criticise in an interview.

The core motivation: collapse a whole portfolio's market risk into one comparable, loss-denominated, probabilistic number.

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2. The Core Idea

Value at Risk is the maximum loss on a portfolio over a given time horizon, at a given confidence level, under normal market conditions.

Every VaR statement has exactly three moving parts, and you must always quote all three:

1. **A horizon** (how long) — e.g. 1 day, 10 days.
2. **A confidence level** (how sure) — e.g. 95%, 99%.
3. **A currency amount** (how much) — the loss figure itself.

Read a VaR number as a sentence. "The 1-day 99% VaR is ₹46.6 lakh" means:

> *"We are 99% confident that the portfolio will not lose more than ₹46.6 lakh over the next trading day. Equivalently, on only 1 day in 100 do we expect a loss worse than ₹46.6 lakh."*

The precise definition is a **quantile of the loss distribution**. If (L) is the loss over the horizon (a positive number meaning money lost) and (α) is the confidence level (say 0.99), then:

$$\text{VaR}_\alpha = \text{the smallest loss } \ell \text{ such that } P(L \le \ell) \ge \alpha$$

In plain words: VaR is the loss level that the actual loss will exceed only ((1-\alpha)) of the time. At 99% confidence, VaR is the 99th percentile of losses (equivalently, the 1st percentile of the profit-and-loss distribution).

Two things the core idea deliberately does **not** tell you: - It is a **threshold**, not an average. It says nothing about how bad the loss is *when* you breach it. - It applies to **"normal" markets** — it is silent about crashes, gaps, and liquidity holes in the extreme tail.

Those two silences are the source of every famous VaR limitation, and we return to them in Section 8.

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3. Why / How It Works

Why a quantile is the right object

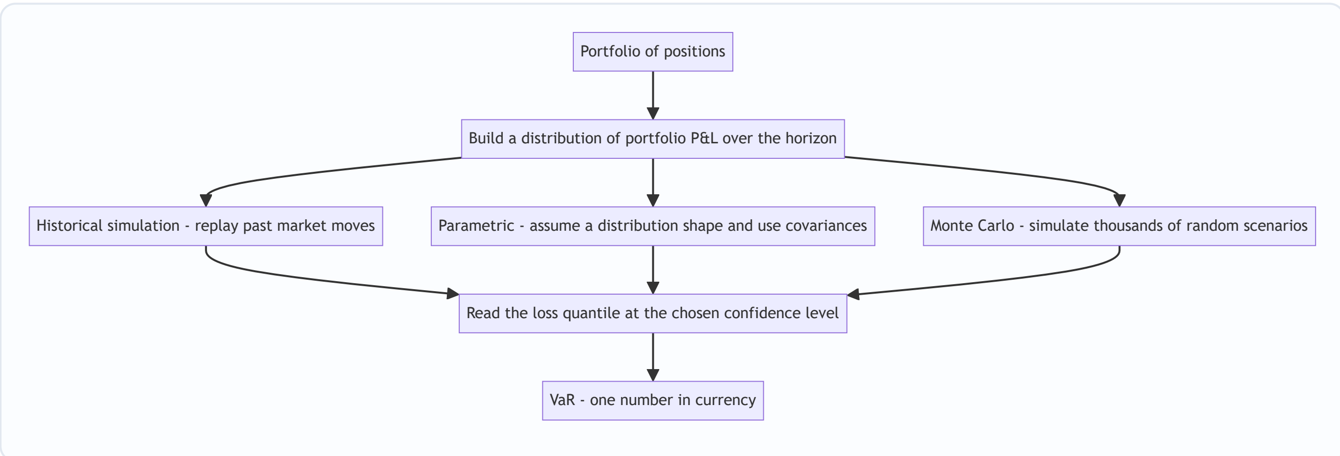
Risk managers care about the **downside tail**, not the whole distribution. The mean tells you the expected outcome; the standard deviation tells you dispersion in both directions. But a limit has to be set against *bad* outcomes specifically. A quantile of the loss distribution directly answers "how far into the bad tail do we go before events become rare (rarer than 1-in-20 or 1-in-100)?"

Why it aggregates cleanly

The magic of VaR is that it is computed on **portfolio P&L**, not on individual greeks. Whatever the instruments, each scenario (a market move) produces a single portfolio P&L number. Once you have a distribution of portfolio P&L, you take one quantile. Offsets and correlations are captured automatically because you revalue the *whole book* under each scenario. This is why a long-and-short book can show far lower VaR than the sum of its parts — the idea of **diversification benefit**.

The engine: a distribution of P&L

Every VaR method is just a different way to build the **distribution of portfolio profit-and-loss** over the horizon, then read off a quantile. The three standard engines differ only in *how they generate that distribution*:



Every VaR method builds a P&L distribution first and then reads off one tail quantile.

The normal-distribution shortcut

The **parametric** method makes life easy by assuming portfolio returns are Normally distributed. For a Normal distribution, a quantile is just "*so many standard deviations from the mean.*" The number of standard deviations for a given confidence is the **z-score**:

Confidence level	One-tailed z-score
90%	1.282
95%	1.645
97.5%	1.960
99%	2.326
99.9%	3.090

Because we care only about the loss (left) tail, VaR uses the **one-tailed** z. (The common 1.96 you remember from statistics is *two-tailed* 95% — do not use it for a one-tailed 95% VaR, where the right figure is 1.645. This is a classic interview trap.)

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4. Full Content — Frameworks, Formulas, Methods

4.1 The general parametric formula

For a portfolio whose return over the horizon has mean μ and standard deviation σ (both expressed for that horizon), and portfolio market value V:

$$\text{VaR}_\alpha = V \times \left(z_\alpha \cdot \sigma - \mu \right)$$

Over short horizons (a day), μ is tiny relative to $z_\alpha \sigma$, so practitioners usually **set $\mu = 0$** ("drift is negligible intraday"). This gives the workhorse formula:

$$\boxed{\text{VaR}_\alpha = z_\alpha \times \sigma \times V}$$

where σ is the standard deviation of returns over the horizon (in decimal), V is portfolio value, and z_α is the one-tailed z-score.

4.2 Two-asset (and multi-asset) parametric VaR

With more than one position, you cannot just add individual VaRs — you must combine their **volatilities using correlation**. Define each position's currency volatility $\sigma_i^{\$} = w_i \sigma_i V$ (or directly the standard deviation of that position's P&L). For two positions:

$$\sigma_P^{\$} = \sqrt{(\sigma_A^{\$})^2 + (\sigma_B^{\$})^2 + 2\,\rho_{AB}\,\sigma_A^{\$}\,\sigma_B^{\$}}$$

$$\text{VaR}_P = z_\alpha \times \sigma_P^{\$}$$

In matrix form for n assets, with weight-value vector $\mathbf{x}$ and covariance matrix Σ:

$$\sigma_P^{\$} = \sqrt{\mathbf{x}^\top \Sigma\, \mathbf{x}}, \qquad \text{VaR} = z_\alpha \sqrt{\mathbf{x}^\top \Sigma\, \mathbf{x}}$$

The **undiversified VaR** (sum of standalone VaRs) is always ($\ge$) the diversified VaR whenever correlations are below 1. The gap is the **diversification benefit**:

$$\text{Diversification benefit} = \sum_i \text{VaR}_i - \text{VaR}_{\text{portfolio}}$$

4.3 Method 1 — Historical Simulation

Idea: Do not assume any distribution. Instead, take the actual market moves of the last (N) days (e.g. 250 or 500), apply each historical move to *today's* portfolio, and build an empirical P&L distribution. Then read off the quantile.

Steps: 1. Collect (N) historical daily returns for every risk factor (rates, FX, equity indices, spreads). 2. For each past day, compute what today's portfolio would have made or lost if that day repeated. This gives (N) hypothetical P&L numbers. 3. Sort the P&L from worst to best. 4. The 99% VaR is the loss at the 1st-percentile order statistic (e.g. with 250 observations, roughly the 2nd-to-3rd worst; conventions differ and interpolation is common).

Pros: No distributional assumption; captures fat tails and real correlations automatically; handles non-linear instruments (options) if you fully revalue. **Cons:** Assumes the past window represents the future; a quiet window understates risk, and a single crash in the window dominates; equal-weights old and recent days unless you use weighting schemes; limited tail resolution (with 250 days, the 99% tail rests on ~2-3 data points).

4.4 Method 2 — Variance-Covariance (Parametric / Delta-Normal)

Idea: Assume risk-factor returns are jointly Normal. Summarise the whole portfolio with a volatility (and correlations), then use the z-score formula. Also called **delta-normal** because non-linear instruments are linearised via their delta.

Pros: Fast and analytic — a closed-form number; only needs a covariance matrix; great for large linear portfolios; easy to decompose risk by factor. **Cons:** Normality badly underestimates fat tails (real markets have more extreme moves than a bell curve predicts); it linearises options and so **misses gamma/convexity**, mis-stating the risk of option books; correlations are assumed stable (they spike toward 1 in crises).

4.5 Method 3 — Monte Carlo Simulation

Idea: Assume a stochastic model for the risk factors (any distribution and correlation structure you like), then generate thousands or millions of random scenarios, fully revalue the portfolio in each, build the P&L distribution, and read off the quantile.

Steps: 1. Specify the joint distribution / process for risk factors (often multivariate Normal or a fat-tailed variant, with a chosen covariance matrix). 2. Draw (M) random scenarios (e.g. 10,000+). 3. Fully reprice the portfolio in each scenario (this handles options and other non-linearities exactly). 4. Sort the (M) P&L outcomes and take the quantile.

Pros: The most flexible — handles non-linearity, path dependence, and arbitrary distributions; you control the model. **Cons:** Computationally heavy (full revaluation × many scenarios); results are only as good as the assumed model ("model risk"); slower to run and harder to explain than the analytic method.

4.6 Comparing the three methods

Feature	Historical	Parametric (Var-Cov)	Monte Carlo
Distribution assumption	None (empirical)	Normal	Any (you choose)
Handles option non-linearity	Yes if full revaluation	Poorly (linearised)	Yes (full revaluation)
Captures fat tails	Yes if in the window	No	Yes if modelled
Speed	Fast to medium	Fastest (closed form)	Slowest
Main weakness	Past may not repeat	Normality and linearity	Model risk and compute cost
Tail resolution	Limited by sample size	Smooth but wrong shape	High if many paths

4.7 Choosing the confidence level and horizon

Confidence level is a policy choice reflecting risk appetite and audience: - **95%** — internal risk monitoring; more frequent breaches make backtesting statistically powerful (you get enough exceptions to test the model). - **99%** — the Basel market-risk standard for regulatory capital; a rarer, more conservative threshold. - **99.9%** — economic-capital / solvency work (very deep tail).

Higher confidence → larger VaR (you are asking about a rarer, worse loss). But higher confidence also means **fewer exceptions to observe**, so the model is harder to validate empirically.

Horizon should match how long it takes to hedge or exit the position: - **1 day** — liquid trading books that can be unwound quickly; also the natural unit for daily P&L backtesting. - **10 days** — the Basel regulatory horizon for market risk (assumes it may take two weeks to liquidate in stress). - **1 year** — credit and economic-capital contexts.

Longer horizon → larger VaR (more time for markets to move against you).

4.8 Scaling VaR across horizons — the square-root-of-time rule

Computing a 10-day VaR directly needs 10-day return data, which is scarce. Instead, practitioners compute a robust 1-day VaR and **scale** it. If returns are independent and identically distributed (i.i.d.) with zero mean, variance grows linearly with time, so standard deviation grows with the **square root of time**:

$$\text{VaR}_{T\text{-day}} = \text{VaR}_{1\text{-day}} \times \sqrt{T}$$

Example factor: 10-day = 1-day × ($\sqrt{10}$) = 1-day × 3.162.

Health warning: the square-root rule assumes zero autocorrelation and constant volatility. Real returns show **volatility clustering** and, over longer horizons, mean reversion or trending — so scaling can under- or over-state true multi-day VaR. It is an approximation of convenience, and Basel's FRTB moved away from naive scaling toward horizon-specific "liquidity horizons."

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5. Worked Examples

Example 1 — Single-asset parametric VaR, and scaling

Setup: A portfolio is worth **V = ₹10 crore** (₹10,00,00,000). Its daily return standard deviation is **σ = 2%**. Assume zero drift and Normal returns.

(a) 1-day 99% VaR. Use (z_{99\%} = 2.326):

$$\text{VaR} = z \times \sigma \times V = 2.326 \times 0.02 \times 10{,}00{,}00{,}000$$ $$= 2.326 \times 0.02 = 0.04652;\quad 0.04652 \times 10{,}00{,}00{,}000 = \textbf{₹46,52,000}$$

(b) 1-day 95% VaR. Use (z_{95\%} = 1.645):

$$\text{VaR} = 1.645 \times 0.02 \times 10{,}00{,}00{,}000 = 0.0329 \times 10{,}00{,}00{,}000 = \textbf{₹32,90,000}$$

Sanity check: the 99% figure exceeds the 95% figure (₹46.52 lakh > ₹32.90 lakh), as it must — a rarer loss is bigger. The ratio equals the z ratio: (46.52 / 32.90 = 1.414 = 2.326/1.645). ✓

(c) 10-day 99% VaR by square-root scaling:

$$\text{VaR}_{10} = 46{,}52{,}000 \times \sqrt{10} = 46{,}52{,}000 \times 3.1623 = \textbf{₹1,47,11,000}$$

Reconciliation: equivalently, 10-day (\sigma = 0.02 \times \sqrt{10} = 0.06325); then (2.326 \times 0.06325 \times 10\text{cr} = 0.14712 \times 10\text{cr} = ₹1,47,12,000) (rounding). Both routes agree. ✓

Example 2 — Two-asset parametric VaR and diversification benefit

Setup: - Asset A: value ₹6 crore, daily σ = 1.5%. - Asset B: value ₹4 crore, daily σ = 2.5%. - Correlation ρ = 0.30. Confidence 99% (z = 2.326).

Step 1 — currency volatility of each position (standard deviation of each position's daily P&L):

$$\sigma_A^{\$} = 0.015 \times 6{,}00{,}00{,}000 = ₹9{,}00{,}000 \;(₹9 \text{ lakh})$$ $$\sigma_B^{\$} = 0.025 \times 4{,}00{,}00{,}000 = ₹10{,}00{,}000 \;(₹10 \text{ lakh})$$

Step 2 — portfolio currency volatility (work in lakh):

$$\sigma_P^{\$} = \sqrt{9^2 + 10^2 + 2(0.30)(9)(10)} = \sqrt{81 + 100 + 54} = \sqrt{235} = 15.33 \text{ lakh}$$

Step 3 — diversified portfolio VaR:

$$\text{VaR}_P = 2.326 \times 15.33 = \textbf{₹35.66 lakh}$$

Step 4 — undiversified VaR (correlation assumed 1, i.e. add standalone VaRs):

$$\text{VaR}A = 2.326 \times 9 = 20.93 \text{ lakh}, \quad \text{VaR}_B = 2.326 \times 10 = 23.26 \text{ lakh}$$ $$\text{VaR}{\text{undiv}} = 20.93 + 23.26 = ₹44.19 \text{ lakh}$$

Step 5 — diversification benefit:

$$44.19 - 35.66 = \textbf{₹8.53 lakh saved by diversification}$$

Reconciliation / sanity checks: - With ρ = 1, (\sigma_P = \sqrt{81+100+180} = \sqrt{361} = 19) lakh, giving VaR = 2.326 × 19 = ₹44.19 lakh — exactly the undiversified figure. ✓ (The two definitions coincide at perfect correlation, confirming the algebra.) - With ρ = 0, (\sigma_P = \sqrt{181} = 13.45) lakh → VaR ₹31.29 lakh, lower still, as expected (less correlation → more benefit). ✓ - The diversified VaR (₹35.66 lakh) sits between the ρ = 0 and ρ = 1 cases (₹31.29 and ₹44.19 lakh), which it must. ✓

Example 3 — Historical simulation VaR

Setup: Portfolio value **V = ₹5 crore**. We have the **100 most recent daily returns**. Sorted from worst to best, the ten worst daily returns (%) are:

Rank (worst first)	Daily return
1	−4.20%
2	−3.80%
3	−3.10%
4	−2.90%
5	−2.60%
6	−2.40%
7	−2.20%
8	−2.05%
9	−1.95%
10	−1.80%

95% VaR (100 observations): the worst 5% are the 5 worst days. The **5th-worst** return, −2.60%, marks the 95% cutoff:

$$\text{VaR}_{95\%} = 0.026 \times 5{,}00{,}00{,}000 = \textbf{₹13,00,000}$$

99% VaR (100 observations): the worst 1% is the single worst day, −4.20% (a conservative reading; some conventions interpolate between the 1st and 2nd worst):

$$\text{VaR}_{99\%} = 0.042 \times 5{,}00{,}00{,}000 = \textbf{₹21,00,000}$$

Reconciliation vs parametric: the empirical daily σ of this book is around 1.6% (typical for such a return spread). A Normal 99% VaR would be $(2.326 \times 0.016 \times 5\text{cr} = ₹18.6)$ lakh — **less** than the historical ₹21 lakh. The historical number is larger precisely because the real left tail (−4.20%) is **fatter** than a bell curve predicts. ✓ This is the whole point of historical simulation: it captures fat tails the parametric method smooths away.

Expected Shortfall cross-check (average of the tail): beyond the 95% cutoff, the mean of the 5 worst returns is $((4.20+3.80+3.10+2.90+2.60)/5 = 3.32\%)$, so $\text{ES}_{95\%} = 0.0332 \times 5\text{cr} = ₹16.6$ lakh > $\text{VaR}_{95\%}$ of ₹13 lakh. ES always exceeds VaR at the same confidence because it averages *into* the tail. ✓ (More on ES in Section 6.)

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6. Connections

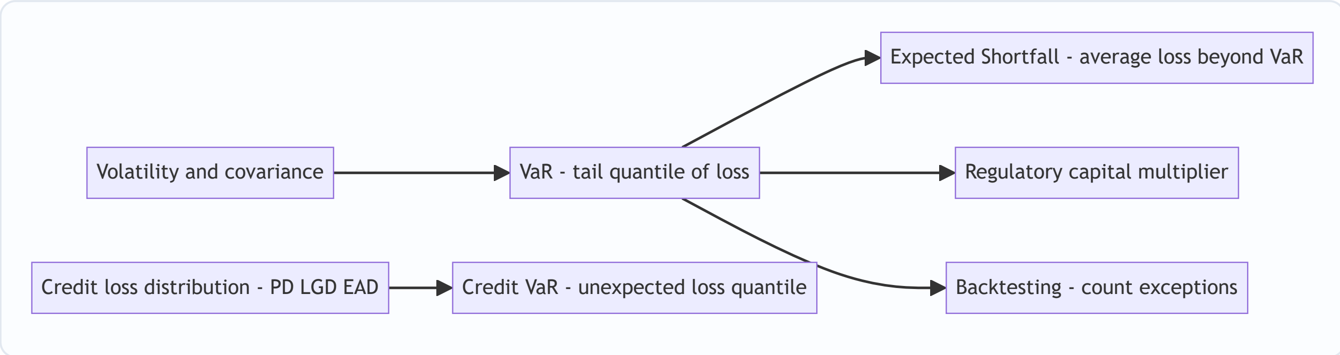
To Expected Shortfall (ES / CVaR). VaR's biggest conceptual heir is Expected Shortfall — the *average* loss given that you have breached VaR. Where VaR asks "how far to the edge of the cliff?", ES asks "how far do we fall once we go over?" ES is a **coherent** risk measure (it respects sub-additivity — diversification never increases it), which VaR is not. Basel's FRTB replaced 99% VaR with **97.5% ES** for market-risk capital for exactly this reason. Expect an interviewer to ask "why did Basel move from VaR to ES?" — answer: tail-blindness and non-coherence of VaR.

To volatility and the greeks. Parametric VaR is built directly on σ and the covariance matrix — the same volatility that drives option pricing and portfolio theory. Delta-normal VaR reuses position deltas; its failure on options is a failure to capture gamma/vega, tying VaR back to the greeks chapter.

To credit risk. The expected-loss engine there, ($EL = PD \times LGD \times EAD$), is the *mean* of the credit-loss distribution; **Credit VaR** is a quantile of that same distribution minus expected loss (the unexpected loss). VaR is the unifying quantile idea across market and credit risk.

To regulatory capital. Market-risk capital under Basel II.5 was a multiple of 10-day 99% VaR plus a **stressed VaR** add-on (VaR calibrated to a crisis window). FRTB then shifted to ES. VaR is thus not just a management tool but a capital driver.

To backtesting and Basel's traffic-light. Because VaR makes a falsifiable claim ("losses exceed this only 1% of the time"), you can count **exceptions**. Basel's traffic-light system (green / amber / red zones) penalises models with too many breaches — connecting VaR to model validation and governance.



VaR sits at the centre of market-risk measurement and connects outward to ES, capital, backtesting and credit risk.

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7. Key Terms

- **Value at Risk (VaR):** maximum loss over a horizon at a confidence level, under normal conditions; a quantile of the loss distribution.
- **Confidence level (α):** probability that the loss stays within VaR (e.g. 99%). ($(1-\alpha)$) is the expected exception rate.
- **Horizon:** the period over which the loss is measured (1-day, 10-day).
- **z-score (z_α):** number of standard deviations for a given one-tailed confidence in a Normal distribution (1.645 for 95%, 2.326 for 99%).
- **Parametric / variance-covariance / delta-normal VaR:** analytic VaR assuming Normal returns, using volatility and correlations.
- **Historical simulation:** VaR from replaying actual past market moves on today's portfolio; no distributional assumption.
- **Monte Carlo simulation:** VaR from many random model-generated scenarios with full revaluation.
- **Diversification benefit:** the amount by which portfolio VaR falls below the sum of standalone VaRs due to imperfect correlation.
- **Square-root-of-time rule:** scaling ($\text{VaR}_T = \text{VaR}_1 \times \sqrt{T}$), valid under i.i.d. zero-mean returns.
- **Exception / breach:** a day whose actual loss exceeds the VaR estimate.
- **Backtesting:** validating a VaR model by counting exceptions against the expected rate.
- **Expected Shortfall (ES / CVaR):** average loss conditional on breaching VaR; a coherent measure.

- **Stressed VaR:** VaR calibrated to a historical stress window, added to capital under Basel II.5.

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8. Common Confusions

"VaR is the most I can ever lose." No. VaR is the threshold at the *edge* of the tail, not the worst case. At 99% 1-day VaR of ₹46.5 lakh, losses *will* exceed ₹46.5 lakh about 1 day in 100 — and on those days the loss can be far larger. VaR says nothing about *how much worse*. That is precisely what Expected Shortfall adds.

Confusing the confidence level with the exception rate. A 99% VaR is breached 1% of the time — roughly **2-3 days per year** of trading (~250 days). Analysts sometimes expect "never," then panic at a normal exception. Breaches at the expected rate are evidence the model *works*.

Using the two-tailed z (1.96) for a 95% VaR. VaR is one-tailed (we only care about losses), so 95% uses **1.645**, not 1.96. Using 1.96 overstates a 95% VaR by ~19%.

Adding VaRs across desks. VaR is *not additive* unless correlations are 1. Summing desk VaRs overstates total VaR (ignores diversification). Conversely — and this is VaR's theoretical flaw — VaR can occasionally be **super-additive** (combined VaR exceeding the sum), violating sub-additivity, especially with skewed or discrete payoffs like short options or credit portfolios. This non-coherence is the technical reason regulators prefer ES.

Thinking historical VaR is "assumption-free." It makes the huge assumption that the chosen window represents the future. A calm 2-year window run into a 2008- or 2020-style crash will badly understate risk. The assumption moved from "the distribution is Normal" to "the past repeats" — it did not disappear.

Treating the square-root rule as exact. It holds only for i.i.d. zero-mean returns. Volatility clustering and autocorrelation break it; use it as an approximation, not a law.

VaR = risk capital, full stop. VaR informs capital but regulatory capital adds multipliers, stressed VaR, and (under FRTB) ES. VaR is an input, not the whole answer.

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9. Recap

- **VaR compresses a whole portfolio's market risk into one currency number** defined by a **horizon** and a **confidence level**: the loss that will be exceeded only $(1-\alpha)$ of the time.
- Every method builds a **distribution of portfolio P&L** and reads off a **tail quantile**. They differ only in how the distribution is built.
- **Historical simulation** replays real past moves (no distribution assumed, captures fat tails, but assumes the past repeats and has thin tail data). **Parametric** assumes Normality and uses ($z \times \sigma \times V$) (fast, closed-form, but underweights fat tails and mishandles options). **Monte Carlo** simulates many modelled scenarios with full revaluation (flexible, handles non-linearity, but heavy and model-dependent).
- **Higher confidence and longer horizon → larger VaR.** Choose confidence by audience (95% internal, 99% Basel) and horizon by liquidation time (1-day trading, 10-day regulatory).
- **Scale short-horizon VaR with ($\sqrt{T}$)** under i.i.d. assumptions — an approximation.
- Worked results reconciled: single-asset 1-day 99% VaR of **₹46.52 lakh** scaling to **₹1.47 crore** over 10 days; a two-asset book showing an **₹8.53 lakh diversification benefit**; and a historical book whose fat left tail produced a **larger** 99% VaR (₹21 lakh) than the Normal approximation (₹18.6 lakh).

- **VaR's limits are structural:** it is tail-blind (silent beyond the quantile), can be non-sub-additive (non-coherent), and is only as good as its window or model. Those flaws motivated **Expected Shortfall** under Basel FRTB.

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10. Quick-Reference / Interview Points

The one-line definition to recite: "*VaR is the maximum loss over a set horizon at a set confidence level under normal market conditions — a quantile of the loss distribution.*" Always name the horizon and confidence.

Formula to have cold:

$$\text{VaR}_\alpha = z_\alpha \times \sigma \times V \quad\text{(parametric, zero drift)}$$

$$\text{VaR}_{T} = \text{VaR}_{1}\times\sqrt{T}\quad\text{(scaling)}$$

$$\sigma_P^{\$}=\sqrt{(\sigma_A^{\$})^2+(\sigma_B^{\$})^2+2\rho\,\sigma_A^{\$}\sigma_B^{\$}}\quad\text{(two-asset)}$$

z-scores to memorise: 95% → **1.645**, 99% → **2.326** (one-tailed). Do *not* use 1.96 for VaR.

Three methods in one breath: Historical = replay the past, no distribution; Parametric = assume Normal, use covariance; Monte Carlo = simulate scenarios, full revaluation. Trade-off is assumptions vs. speed vs. flexibility.

"How many breaches should a 99% daily VaR see?" About **2-3 per year** (1% of ~250 days). More → model too optimistic.

"Name VaR's limitations." (1) Tail-blind — says nothing beyond the quantile; (2) not sub-additive / not coherent — can understate combined risk; (3) assumption-dependent — Normality (parametric) or a representative window (historical) or model choice (Monte Carlo); (4) procyclical — calm periods shrink VaR just before storms. This is why Basel FRTB moved to **97.5% Expected Shortfall**.

"VaR vs Expected Shortfall." VaR = the threshold; ES = the average loss beyond it. ES is coherent and captures tail severity; ES ≥ VaR at the same confidence.

"Why 10-day and √time?" Basel liquidation assumption; √time scaling holds only under i.i.d. returns and is an approximation.

Trap to avoid stating: never say VaR is the "worst-case" or "maximum possible" loss — it is the *threshold* loss at a confidence level. Saying "maximum possible loss" is the fastest way to fail a risk interview.

Q&A — Value at Risk (VaR)

Practice bank for Chapter 04. Every question is followed by a full answer. Work each one before reading the solution. One-tailed z-scores used throughout: 95% → 1.645, 99% → 2.326, 97.5% → 1.960, 90% → 1.282.

Section A — Concept Check

A1. State the one-line definition of VaR and name the three things every VaR statement must quote.

VaR is the maximum loss on a portfolio over a given time horizon, at a given confidence level, under normal market conditions — formally, a quantile of the loss distribution. Every VaR statement must quote (1) a **horizon** (how long, e.g. 1-day, 10-day), (2) a **confidence level** (how sure, e.g. 95%, 99%), and (3) a **currency amount** (the loss figure itself). A number without horizon and confidence is meaningless.

A2. Read the statement "the 1-day 99% VaR is ₹46.6 lakh" as a full sentence.

"We are 99% confident that the portfolio will not lose more than ₹46.6 lakh over the next trading day. Equivalently, on only 1 day in 100 do we expect a loss worse than ₹46.6 lakh." It is a threshold at the edge of the tail, not a forecast of the average or the worst case.

A3. Why is a quantile — rather than the mean or the standard deviation — the right statistic for risk limits?

Risk managers care about the **downside tail**, not the whole distribution. The mean gives the expected outcome and standard deviation gives dispersion in both directions, but a limit must be set against *bad* outcomes specifically. A quantile of the loss distribution answers directly: "how far into the bad tail do we go before events become rare (rarer than 1-in-20 or 1-in-100)?" That is exactly what a loss limit needs.

A4. VaR aggregates thousands of positions into one number. Why can it do this when a table of greeks cannot?

Greeks (DV01, delta, gamma, vega) are in incompatible units and cannot be added; they also ignore how positions offset. VaR is computed on **portfolio P&L**, not on individual greeks. Each market scenario produces a single portfolio P&L number after revaluing the *whole book*, so correlations and offsets are captured automatically. Once you have a distribution of portfolio P&L, you read off one tail quantile — one comparable, loss-denominated number.

A5. Name the two things VaR deliberately does NOT tell you.

(1) It is a **threshold, not an average** — it says nothing about how bad the loss is *when* you breach it. (2) It applies to **"normal" markets** — it is silent about crashes, gaps, and liquidity holes in the extreme tail. These two silences are the source of every famous VaR limitation.

A6. All three VaR methods share one engine. What is it?

Every method builds a **distribution of portfolio P&L** over the horizon and then reads off a tail quantile at the chosen confidence level. The methods differ only in *how they generate that distribution*: historical simulation replays past moves, parametric assumes a distribution shape and uses covariances, Monte Carlo simulates random scenarios.

A7. Why does higher confidence mean larger VaR but a harder-to-validate model?

Higher confidence asks about a rarer, worse loss, so the quantile moves further into the tail and VaR grows. But a rarer threshold means **fewer exceptions to observe** — at 99% you expect only ~2-3 breaches per year, giving thin data to backtest. At 95% you get more exceptions, making statistical validation more powerful. There is a trade-off between conservatism and testability.

A8. Distinguish confidence level from exception rate.

The confidence level α is the probability the loss stays within VaR; the exception (breach) rate is $(1 - \alpha)$. A 99% VaR is breached about 1% of the time — roughly 2-3 days per year over ~250 trading days. Breaches occurring at the expected rate are evidence the model *works*, not that it is broken.

A9. Why is historical simulation not truly "assumption-free"?

It drops the Normality assumption but replaces it with an equally strong one: **the chosen historical window represents the future.** A calm two-year window run into a 2008- or 2020-style crash badly understates risk; a single crash in the window can dominate the tail. The assumption moved from "the distribution is Normal" to "the past repeats" — it did not disappear.

A10. State the square-root-of-time rule and its key assumption.

$VaR_T = VaR_1 \times \sqrt{T}$. It assumes returns are independent and identically distributed (i.i.d.) with zero mean, so variance grows linearly with time and standard deviation grows with $\sqrt{T}$. It breaks under volatility clustering, autocorrelation, and mean reversion, so it is an approximation of convenience, not a law.

Section B — Numerical / Applied (with full solutions)

B1. Single-asset parametric VaR. A portfolio is worth V = ₹10 crore with daily return $\sigma = 2\%$. Assume zero drift, Normal returns. Find the 1-day 99% VaR and the 1-day 95% VaR.

Solution. Use $VaR = z \times \sigma \times V$.

- 99%: $z = 2.326$. VaR = 2.326 × 0.02 × 10,00,00,000 = 0.04652 × 10,00,00,000 = **₹46,52,000**.
- 95%: $z = 1.645$. VaR = 1.645 × 0.02 × 10,00,00,000 = 0.0329 × 10,00,00,000 = **₹32,90,000**.

Sanity check: 99% > 95% (₹46.52 lakh > ₹32.90 lakh), as a rarer loss must be larger. Ratio 46.52/32.90 = 1.414 = 2.326/1.645. ✓

B2. Scaling across horizons. Using the 1-day 99% VaR of ₹46,52,000 from B1, find the 10-day 99% VaR two ways and reconcile.

Solution.

- Route 1 (scale the VaR): VaR_{10} = 46,52,000 × √10 = 46,52,000 × 3.1623 = **₹1,47,11,000** (≈ ₹1.47 crore).
- Route 2 (scale σ first): 10-day σ = 0.02 × √10 = 0.06325; VaR = 2.326 × 0.06325 × 10,00,00,000 = 0.14712 × 10 cr = ₹1,47,12,000.

Both routes agree to rounding. ✓ The tiny difference is rounding only.

B3. Expected loss vs VaR. A ₹20 crore book has a daily return with mean $\mu = +0.05\%$ and $\sigma = 1.8\%$. Using the full parametric formula $VaR = V(z\cdot\sigma - \mu)$, find the 1-day 99% VaR. Then find it with $\mu = 0$ and comment.

Solution. $z = 2.326$.

- With drift: $z\cdot\sigma - \mu$ = 2.326 × 0.018 − 0.0005 = 0.041868 − 0.0005 = 0.041368. VaR = 0.041368 × 20,00,00,000 = **₹82,73,600**.

- With $\mu = 0$: VaR = 2.326 × 0.018 × 20,00,00,000 = 0.041868 × 20 cr = ₹83,73,600.

Comment: positive drift *reduces* VaR (expected gain offsets some loss). The difference here is ₹1,00,000 — small over one day, which is exactly why practitioners set $\mu = 0$ intraday. Over a one-year horizon the drift term would be far from negligible.

B4. Two-asset parametric VaR and diversification benefit. Asset A: value ₹6 crore, daily σ = 1.5%. Asset B: value ₹4 crore, daily σ = 2.5%. Correlation ρ = 0.30. Confidence 99%. Find diversified VaR, undiversified VaR, and the diversification benefit.

Solution.

Step 1 — currency volatility of each position: - $\sigma_{A\$}$ = 0.015 × 6,00,00,000 = ₹9,00,000 (₹9 lakh) - $\sigma_{B\$}$ = 0.025 × 4,00,00,000 = ₹10,00,000 (₹10 lakh)

Step 2 — portfolio currency volatility (work in lakh): $\sigma_{P\$} = \sqrt{9^2 + 10^2 + 2 \cdot 0.30 \cdot 9 \cdot 10} = \sqrt{81 + 100 + 54} = \sqrt{235}$ = 15.33 lakh.

Step 3 — diversified VaR: 2.326 × 15.33 = **₹35.66 lakh**.

Step 4 — undiversified VaR (add standalone VaRs): - VaR_A = 2.326 × 9 = 20.93 lakh; VaR_B = 2.326 × 10 = 23.26 lakh. - VaR_{undiv} = 20.93 + 23.26 = **₹44.19 lakh**.

Step 5 — diversification benefit: 44.19 − 35.66 = **₹8.53 lakh**.

Sanity checks: at ρ = 1, $\sigma_P = \sqrt{81+100+180} = \sqrt{361}$ = 19 lakh → VaR = 2.326 × 19 = ₹44.19 lakh, exactly the undiversified figure (the two definitions coincide at perfect correlation). ✓ At ρ = 0, $\sigma_P = \sqrt{181}$ = 13.45 lakh → VaR ₹31.29 lakh, lower still. The diversified figure (₹35.66) sits between the ρ = 0 and ρ = 1 cases, as it must. ✓

B5. Historical simulation. Portfolio value V = ₹5 crore. From the 100 most recent daily returns, the ten worst (%) are: −4.20, −3.80, −3.10, −2.90, −2.60, −2.40, −2.20, −2.05, −1.95, −1.80. Find the 95% and 99% VaR.

Solution.

- 95% VaR (100 observations): the worst 5% are the 5 worst days; the 5th-worst return, −2.60%, marks the cutoff. VaR = 0.026 × 5,00,00,000 = **₹13,00,000**.
- 99% VaR: the worst 1% is the single worst day, −4.20% (conservative reading; some conventions interpolate the 1st and 2nd worst). VaR = 0.042 × 5,00,00,000 = **₹21,00,000**.

Cross-check vs parametric: the book's empirical daily σ is about 1.6%, so a Normal 99% VaR would be 2.326 × 0.016 × 5 cr = ₹18.6 lakh — *less* than the historical ₹21 lakh. The historical figure is larger because the real left tail (−4.20%) is fatter than a bell curve predicts. ✓ That is the whole point of historical simulation.

B6. Expected Shortfall cross-check. Using B5's data, compute ES at 95% and confirm ES ≥ VaR.

Solution. $ES_{95\%}$ = average loss in the worst 5% of days = mean of the 5 worst returns = (4.20 + 3.80 + 3.10 + 2.90 + 2.60)/5 = 16.60/5 = 3.32%. ES = 0.0332 × 5,00,00,000 = **₹16.6 lakh**. This exceeds $VaR_{95\%}$ of ₹13 lakh because ES averages *into* the tail rather than reading the edge. ES ≥ VaR always holds at the same confidence. ✓

B7. Diversification with three equal standalone VaRs. Three desks each have standalone 99% VaR of ₹10 lakh, pairwise correlation ρ = 0.5 between all pairs (equal $\sigma\$$ of ₹4.30 lakh each, since 2.326 × 4.30 ≈ 10). Find the portfolio VaR and the benefit.

Solution. Work with σ$ = 4.30 lakh each (10/2.326). For n equal-σ positions with common correlation ρ: σ_P$ = σ$ × √(n + n(n−1)ρ) = 4.30 × √(3 + 6·0.5) = 4.30 × √6 = 4.30 × 2.449 = 10.53 lakh. VaR_P = 2.326 × 10.53 = **₹24.49 lakh**. Undiversified = 3 × 10 = ₹30 lakh. Diversification benefit = 30 − 24.49 = **₹5.51 lakh**.

Check: at ρ = 1, σ_P = 4.30√(3+6) = 4.30×3 = 12.9 → VaR = 30 lakh (undiversified). ✓ At ρ = 0, σ_P = 4.30√3 = 7.45 → VaR = ₹17.33 lakh (maximum benefit). The ρ = 0.5 result lies between. ✓

B8. Back-out the confidence level. A ₹8 crore book has daily σ = 1.25%. Its reported 1-day VaR is ₹23,26,000. What confidence level (z-score) was used?

Solution. VaR = z × σ × V → z = VaR / (σ × V) = 23,26,000 / (0.0125 × 8,00,00,000) = 23,26,000 / 10,00,000 = 2.326. That is the 99% one-tailed z. **Confidence = 99%.**

B9. Number of expected breaches. A trading desk uses a 1-day 97.5% VaR over 250 trading days. How many exceptions should it expect in a year, and what would 15 exceptions suggest?

Solution. Expected exception rate = 1 − 0.975 = 2.5%. Expected breaches = 0.025 × 250 = **6.25 ≈ 6 per year**. Fifteen exceptions is roughly 2.4× the expected count — strong evidence the model **understates risk** (σ too low, or the window too calm), and it would move toward the amber/red zone in a traffic-light backtest.

B10. Undiversified vs diversified ratio. Two assets have equal standalone VaR and correlation ρ = −0.2. What fraction of the undiversified VaR is the diversified VaR?

Solution. Let each σ$ = s. σ_P$ = $\sqrt{(s^2 + s^2 + 2\rho s^2)}$ = s√(2 + 2ρ) = s√(2 + 2(−0.2)) = s√1.6 = 1.2649s. Undiversified = 2s (before applying z, which cancels in the ratio). Ratio = 1.2649s / 2s = **0.632**, i.e. diversified VaR is about 63% of undiversified — a 37% reduction. Negative correlation gives an even larger benefit than the zero-correlation case (which would give √2/2 = 0.707). ✓

Section C — Interview-Style (with model answers)

C1. "Explain VaR to a board member in thirty seconds."

Model answer: "Value at Risk turns our entire trading book into a single rupee figure. Our 1-day 99% VaR is, say, ₹46 lakh. That means on a normal day we are 99% confident we won't lose more than ₹46 lakh, and we'd expect to breach that only about two or three days a year. It lets you compare risk across desks and against limits with one comparable number. The one thing it does not tell you is how bad the loss gets on those rare breach days — for that we look at Expected Shortfall."

C2. "Walk me through the three ways to compute VaR and their trade-offs."

Model answer: "All three build a distribution of portfolio P&L and read off a tail quantile; they differ in how they build it. **Historical simulation** replays actual past market moves on today's book — no distributional assumption, captures fat tails and real correlations, handles options if you fully revalue, but assumes the past window represents the future and has thin data in the deep tail. **Parametric / variance-covariance** assumes Normal returns and uses z × σ × V with a covariance matrix — fast, closed-form, easy to decompose, but underweights fat tails and linearises options so it misses gamma. **Monte Carlo** simulates thousands of modelled scenarios with full revaluation — most flexible, handles non-linearity and any distribution, but computationally heavy and exposed to model risk. The trade-off is assumptions vs speed vs flexibility."

C3. "Why did Basel move from 99% VaR to 97.5% Expected Shortfall under FRTB?"

Model answer: "Two reasons. First, VaR is **tail-blind** — it's just the threshold and says nothing about the severity of losses beyond it, so two books with identical VaR can have very different tail risk. Second, VaR is

not a coherent risk measure — it can violate sub-additivity, meaning a combined portfolio's VaR can exceed the sum of parts, which perversely penalises diversification, especially with skewed payoffs like short options or credit. Expected Shortfall averages the losses beyond the quantile, so it captures tail severity and is sub-additive (coherent). FRTB picked 97.5% ES because, under Normality, it sits close to 99% VaR in magnitude while adding tail information."

C4. "A junior analyst calls a 99% VaR breach a model failure. How do you respond?"

Model answer: "I'd correct the framing. A 99% VaR is *designed* to be breached about 1% of the time — roughly two to three days per year. A breach at the expected rate is evidence the model is calibrated correctly, not that it failed. The concern is the opposite: too *many* breaches, or none at all over a long period. We validate this formally by backtesting exceptions against Basel's traffic-light zones. One breach is business as usual; a cluster of breaches is the signal to investigate."

C5. "What's wrong with computing 10-day VaR as 1-day VaR times the square root of ten?"

Model answer: "The √T rule assumes returns are i.i.d. with zero mean, so variance scales linearly with time. Real markets violate that: there's **volatility clustering** (calm and stormy periods bunch together) and autocorrelation, and over longer horizons mean reversion or trending. So √time scaling can under- or over-state true multi-day risk. It's a convenient approximation — fine for a quick estimate — but Basel's FRTB moved away from naive scaling toward instrument-specific liquidity horizons precisely because it's unreliable in stress."

C6. "Your parametric VaR and your historical VaR disagree materially. Which do you trust and why?"

Model answer: "It depends on the book. If the disagreement is that historical VaR is *higher*, it usually means the real return distribution has fatter tails than the Normal assumption behind parametric VaR — I'd lean toward the historical number and treat parametric as optimistic. If the book holds significant options, parametric delta-normal linearises them and misses gamma/convexity, so it's structurally unreliable there and I'd prefer historical or Monte Carlo with full revaluation. But if the historical window is calm and short, it may understate risk, so I'd also run a stressed-window VaR. The right answer is rarely 'trust one blindly' — it's to understand *why* they differ and use the method whose assumptions fit the portfolio."

C7. "Why isn't VaR additive across desks, and why does that matter for limit-setting?"

Model answer: "VaR combines through volatilities and correlations, not by addition. Summing desk VaRs implicitly assumes correlation of 1 and so overstates total risk — it ignores the diversification benefit from imperfect correlation. If we set a firm-wide limit as the sum of desk limits we'd be over-conservative in normal times. The subtler danger is the reverse: VaR can occasionally be super-additive, so a naive sum can also *understate* combined risk for skewed books. That non-coherence is why we allocate limits using a proper covariance aggregation and, increasingly, Expected Shortfall."

Section D — Multiple Choice (with reasoning)

D1. The 1-day 99% VaR of a book is ₹40 lakh. Which statement is correct? (a) The book can never lose more than ₹40 lakh in a day. (b) The book will lose exactly ₹40 lakh on 1 day in 100. (c) On about 1 day in 100 the loss will exceed ₹40 lakh. (d) The average daily loss is ₹40 lakh.

Answer: **(c)**. VaR is a threshold exceeded $(1 - \alpha) = 1\%$ of the time. (a) is the "worst-case" fallacy — losses can be far larger on breach days. (b) confuses a threshold with an exact value. (d) confuses VaR with the mean.

D2. For a one-tailed 95% parametric VaR, the correct z-score is: (a) 1.960 (b) 1.645 (c) 2.326 (d) 1.282

Answer: **(b) 1.645**. VaR uses the one-tailed z because only the loss (left) tail matters. 1.960 is the *two-tailed* 95% value — a classic trap that overstates a 95% VaR by ~19%. 2.326 is one-tailed 99%; 1.282 is one-tailed 90%.

D3. Diversification benefit in a two-asset portfolio is zero when: (a) $\rho = 0$ (b) $\rho = -1$ (c) $\rho = +1$ (d) volatilities are equal

Answer: **(c) $\rho = +1$**. At perfect positive correlation, portfolio σ$ equals the sum of position σ$, so diversified VaR equals undiversified VaR and the benefit is zero. Any $\rho < 1$ produces a positive benefit; equal volatilities alone don't determine it.

D4. Which method captures option gamma most reliably? (a) Delta-normal parametric (b) Variance-covariance (c) Monte Carlo with full revaluation (d) None can

Answer: **(c)**. Monte Carlo (and historical simulation) with *full revaluation* reprices the option in each scenario, capturing convexity. Delta-normal / variance-covariance linearise via delta and miss gamma. (a) and (b) are the same linearising approach.

D5. A 99% daily VaR model produces 9 exceptions in 250 trading days. This most likely indicates: (a) The model is well-calibrated. (b) The model overstates risk. (c) The model understates risk. (d) The confidence level is too high.

Answer: **(c)**. Expected breaches = 1% × 250 ≈ 2-3. Nine is roughly 3-4× that, so actual losses breach VaR far more often than the model predicts — VaR is set too low, understating risk. This pushes the model toward the amber/red backtesting zone.

D6. Scaling a 1-day VaR to 10 days by √10 assumes: (a) Returns are Normally distributed. (b) Returns are i.i.d. with zero mean. (c) Volatility clusters over time. (d) Correlations equal 1.

Answer: **(b)**. The $\sqrt{T}$ rule requires i.i.d. zero-mean returns so variance scales linearly with time. Normality is not strictly required for the scaling itself (it's a variance argument). Volatility clustering (c) actually *breaks* the rule.

D7. Compared with VaR at the same confidence level, Expected Shortfall is: (a) Always smaller (b) Always equal (c) Always greater or equal (d) Unrelated

Answer: **(c)**. ES averages the losses *beyond* the VaR threshold, so ES ≥ VaR at the same confidence — strictly greater whenever any tail mass exists. ES is also coherent (sub-additive), which VaR is not.

D8. The main reason regulators prefer Expected Shortfall to VaR is that VaR: (a) Is harder to compute. (b) Requires more data. (c) Is not sub-additive and is tail-blind. (d) Cannot be backtested.

Answer: **(c)**. VaR can violate sub-additivity (non-coherent) and says nothing about loss severity beyond the quantile (tail-blind). ES fixes both. Note VaR is actually *easier* to backtest than ES, so (d) is false.

D9. Which is the correct two-asset portfolio currency-volatility formula? (a) $\sigma_P = \sigma_A + \sigma_B$ (b) $\sigma_P = \sqrt{\sigma_A^2 + \sigma_B^2}$ (c) $\sigma_P = \sqrt{\sigma_A^2 + \sigma_B^2 + 2\rho\sigma_A\sigma_B}$ (d) $\sigma_P = \sigma_A^2 + \sigma_B^2 + 2\rho\sigma_A\sigma_B$

Answer: **(c)**. The correlation cross-term $2\rho\sigma_A\sigma_B$ is essential; (b) is only the special case $\rho = 0$; (a) is only $\rho = 1$; (d) forgets the square root (units would be wrong).

D10. Historical simulation's key hidden assumption is that: (a) Returns are Normal. (b) The chosen historical window represents the future. (c) Volatility is constant. (d) There is no assumption at all.

Answer: **(b)**. Historical simulation drops Normality but assumes the past window is representative of the future — a calm window understates risk into a crash. (d) is the common misconception; the assumption moved, it did not disappear.

End of Q&A — Value at Risk (VaR).

Chapter 05 — Expected Shortfall

1. The Problem / The Need

Value at Risk answers one question: "*What is the loss I will not exceed on 99% of days?*" It draws a single line in the sand at a chosen confidence level and reports the size of that line. It is a threshold.

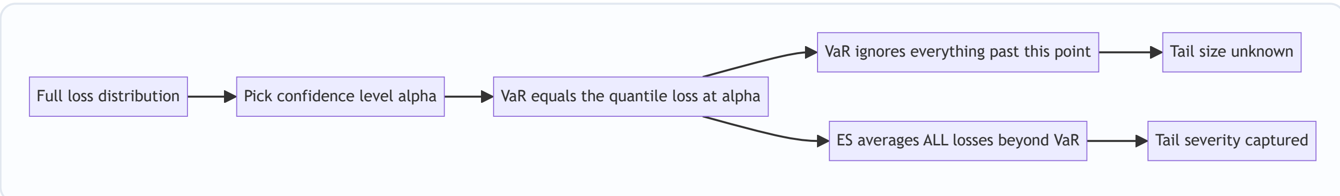
But a risk manager's real fear is not the threshold — it is what lives *beyond* it. Suppose your 99% one-day VaR is ₹10 crore. VaR tells you that on the worst 1% of days you will lose *at least* ₹10 crore. It says absolutely nothing about *how much more*. On those bad days, will you lose ₹11 crore? ₹40 crore? Enough to wipe out the firm? VaR is blind to the answer, because by construction it stops looking exactly where the danger starts.

This blindness is not academic. It creates three concrete, dangerous failures:

1. **Tail-thickness blindness.** Two portfolios can have identical 99% VaR while one loses ₹10.5 crore in the tail and the other loses ₹80 crore. VaR rates them equally risky. A trader who is rewarded on a VaR budget will happily load up on the second portfolio — collecting fat premiums for selling deep out-of-the-money options or catastrophe risk — because the rare, enormous loss sits *just past* the VaR cut-off and never registers.
2. **Cliff risk / gaming.** Because VaR only cares whether a loss crosses the quantile, a desk can restructure positions so that the probability of breaching sits at exactly 1.0% while the loss *given* a breach is catastrophic. The reported number stays flat; the true risk explodes.
3. **Non-subadditivity.** In certain (realistic) cases, the VaR of a *combined* portfolio is *larger* than the sum of the individual VaRs. This directly contradicts the intuition that diversification reduces risk, and it makes VaR unsafe as a number you add up across desks, or split back down as risk limits.

We need a measure that (a) looks *into* the tail rather than stopping at its edge, and (b) behaves sensibly when you aggregate risks. That measure is **Expected Shortfall (ES)**, also called **Conditional VaR (CVaR)**, **Expected Tail Loss (ETL)**, or **Average VaR**.

Figure 1 — VaR marks the edge of the tail; ES describes the whole tail beyond it.



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2. The Core Idea

Expected Shortfall at confidence level α is the **average of all losses that are worse than the VaR** at that level.

In one sentence: *VaR asks "how bad is the boundary of the bad zone?" ES asks "on average, how bad is it once we are inside the bad zone?"*

Formally, for a confidence level α (say 97.5%), the tail is the worst (1 − α) fraction of outcomes:

$$ ES_\alpha = E\left[\,L \mid L \ge VaR_\alpha\,\right] $$

where L is the loss (a positive number for a loss). ES is a *conditional expectation* — the expected loss **conditional on** already being in the tail. That is exactly why "Conditional VaR" is a synonym.

Three immediate consequences fall straight out of this definition:

- **ES ≥ VaR always.** The average of a set of numbers that are all ≥ VaR must itself be ≥ VaR. ES sits *deeper* in the tail than VaR, at the same confidence level. The gap between them measures how heavy the tail is.
- **ES uses the shape of the tail, not just one point.** Fatten the tail, and VaR may not move at all, but ES rises. It responds to exactly the risk VaR ignores.
- **ES is a coherent risk measure.** It satisfies all four axioms a well-behaved risk measure "should" satisfy — most importantly **subadditivity** — which VaR does not. This is not a cosmetic property; it is the mathematical reason ES is safe to add across desks and to use in optimisation.

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3. Why / How It Works

Why averaging the tail fixes VaR's blindness

VaR is a **quantile** — a single order statistic. Order statistics are, by design, insensitive to the magnitude of everything above them. Move the worst 0.5% of outcomes from ₹11 crore to ₹110 crore and the 99% quantile does not budge, because the *count* of outcomes below the threshold is unchanged. VaR sees only *where* the boundary is, never *what is stacked behind it*.

ES is an **average over the tail**, an integral rather than a point. Integrals are sensitive to magnitude everywhere in their range. Push the worst outcomes further out and the average of the tail rises immediately. This is the whole trick: replace a point estimate (quantile) with a tail average (conditional mean), and you inherit sensitivity to tail severity for free.

A clean way to see the relationship is that ES is the **average of all VaRs deeper than α**:

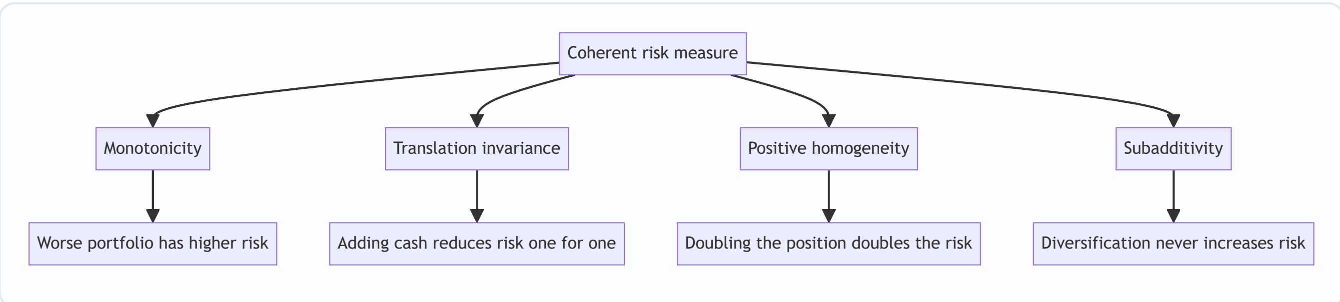
$$ ES_\alpha = \frac{1}{1-\alpha}\int_{\alpha}^{1} VaR_u \, du $$

Read the integral literally: sweep the confidence level u from α all the way to 1 (the absolute worst case), read off VaR at each level, and average. ES is a "VaR of VaRs" — which is why "Average VaR" is another name for it. Because it averages every quantile from α to 100%, it cannot ignore any part of the tail.

Why coherence matters

Artzner, Delbaen, Eber and Heath (1999) asked a foundational question: what properties *must* a sensible risk measure ρ obey? They proposed four axioms, and any measure satisfying all four is called **coherent**.

Figure 2 — The four coherence axioms. VaR satisfies the first three but can fail subadditivity.



Axiom	Statement	Plain meaning
Monotonicity	If $L_1 \le L_2$ always, then $\rho(L_1) \le \rho(L_2)$	A portfolio that always loses more is riskier
Translation invariance	$\rho(L - c) = \rho(L) - c$	Adding c of risk-free cash lowers required capital by exactly c
Positive homogeneity	$\rho(\lambda L) = \lambda\rho(L)$ for $\lambda \ge 0$	Scaling the book scales the risk proportionally
Subadditivity	$\rho(L_1 + L_2) \le \rho(L_1) + \rho(L_2)$	Merging two books cannot create *more* risk than the two apart

Subadditivity is the crucial one, for two reasons:

- **It encodes diversification.** The whole premise of portfolio theory is that combining imperfectly-correlated risks reduces total risk. A risk measure that can *violate* this is telling lies about diversification.
- **It makes limits and capital additive.** If risk is subadditive, a regulator can set a firm-wide capital number and be sure the sum of desk-level numbers is a *conservative* (larger) bound — the firm total is never worse than the parts summed. Break subadditivity and this decomposition collapses: a trader could split one position across two accounts and report *less* total risk than the single position, gaming the limit system.

VaR satisfies monotonicity, translation invariance and homogeneity, but can violate subadditivity (we prove this numerically in §5). **ES satisfies all four — ES is always coherent.** That single fact is the deepest theoretical argument for ES, and a favourite interview question.

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4. Full Content — Framework, Formulas, Methods

4.1 Notation and sign convention

Let losses be positive numbers. Choose confidence level $\alpha \in (0,1)$, e.g. $\alpha = 0.975$. The tail probability is $p = 1 - \alpha$ (here 2.5%). Then:

- **VaRα** = the smallest loss ℓ such that $P(L \le \ell) \ge \alpha$. It is the α-quantile of the loss distribution.
- **ESα** = the average loss in the worst $(1 - \alpha)$ fraction of outcomes.

4.2 The general (continuous) formulas

For a continuous loss distribution:

$$ ES_\alpha = \frac{1}{1-\alpha}\int_{\alpha}^{1} VaR_u \, du = E[L \mid L \ge VaR_\alpha] $$

4.3 Parametric ES under normality

If the loss L is normally distributed with mean μ and standard deviation σ, there is a closed form. Let z_α be the standard-normal α-quantile ($z_{0.99} = 2.326$, $z_{0.975} = 1.960$) and $\varphi(\cdot)$ the standard-normal density. Then:

$$ VaR_\alpha = \mu + \sigma\, z_\alpha $$

$$ ES_\alpha = \mu + \sigma\,\frac{\phi(z_\alpha)}{1-\alpha} $$

The term $\varphi(z_\alpha)/(1-\alpha)$ is the **ES multiplier** — it plays the role z_α plays for VaR. Key values (for $\mu = 0$, so ES and VaR are pure multiples of σ):

α	z_α (VaR mult.)	$\varphi(z_\alpha)$	ES mult. = $\varphi(z_\alpha)/(1-\alpha)$	ES / VaR ratio
95.0%	1.645	0.1031	2.063	1.254
97.5%	1.960	0.0584	2.338	1.193
99.0%	2.326	0.0267	2.665	1.146
99.9%	3.090	0.00337	3.367	1.090

Two things to read off this table. First, **ES is always a fixed multiple bigger than VaR under normality** (the ratio column), and that multiple shrinks as α rises — the tail beyond a high quantile is "thinner" relative to the quantile itself. Second, notice **97.5% ES multiplier (2.338) ≈ 99% VaR multiplier (2.326)**. This near-equality is exactly why Basel picked 97.5% ES to replace 99% VaR (§4.6): under a normal world the two produce almost the same number, so the switch does not mechanically inflate capital — it only *adds tail sensitivity* when distributions are non-normal.

4.4 Historical / empirical ES

With N historical scenarios and no distributional assumption:

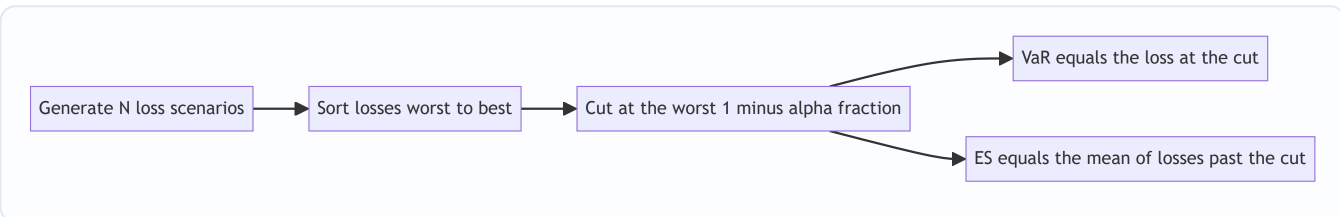
1. Compute the P&L (or loss) for each of the N scenarios.
2. Sort losses from worst to best.
3. Identify the VaR: the loss at rank $\lceil N \cdot (1-\alpha) \rceil$ from the worst end (or interpolate).
4. **ES = the simple average of all losses at least as bad as VaR** — i.e. the mean of the worst $(1-\alpha) \cdot N$ observations.

For example, with $N = 1{,}000$ scenarios at $\alpha = 97.5\%$: the worst 2.5% is 25 observations. VaR is roughly the 25th-worst loss; ES is the **average of the worst 25 losses**. ES literally reads more of the tail data (25 points) than VaR (1 point), which is why it is more stable to estimate in the tail *shape* but needs more data to pin down.

4.5 Monte Carlo ES

Simulate M paths, revalue the portfolio on each, take the worst $(1-\alpha) \cdot M$ losses, average them. Identical logic to historical ES, but the scenarios come from a model rather than history.

Figure 3 — Computing ES the empirical way is just sort, cut, and average the tail.



4.6 The Basel shift from VaR to ES

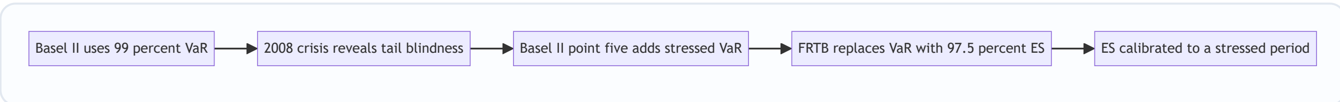
The 2008 crisis exposed VaR's tail-blindness in the most expensive way possible: trading books that looked safe under 99% VaR suffered losses far past the VaR line. The Basel Committee's response, **Fundamental**

Review of the Trading Book (FRTB), finalised in 2016 and rolled into the Basel III market-risk framework, made the switch official:

Feature	Old regime (Basel II.5)	New regime (FRTB)
Risk measure	Value at Risk	Expected Shortfall
Confidence level	99%	97.5%
Horizon	10-day	Liquidity-adjusted horizons (10 to 120 days)
Tail sensitivity	None beyond quantile	Full tail average
Coherence	Not guaranteed	Guaranteed
Stress	Separate stressed VaR add-on	ES calibrated to a stressed period

Why 97.5% and not 99%? Because — as the table in §4.3 shows — **97.5% ES ≈ 99% VaR under normality**, so the transition is roughly capital-neutral for well-behaved books, while the ES formulation *automatically* penalises books with fat tails. Regulators got tail sensitivity without a mechanical across-the-board capital jump.

Figure 4 — The regulatory evolution of the market-risk measure.



4.7 Backtesting caveat

One honest wrinkle: VaR is easy to backtest (count the days losses exceeded VaR — should be ~1% of days at 99%). ES is harder, because it is a *conditional expectation* and is not **elicitable** in the strict statistical sense — you cannot score it with a simple loss function the way you score a quantile. FRTB pragmatically **backtests VaR at 97.5% and 99%** for the "traffic-light" exception test, while **using ES for the actual capital charge**. So VaR did not disappear entirely — it survives as a validation tool, while ES does the capital work.

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5. Worked Examples

Example 1 — Parametric ES for a normal portfolio (reconciliation with VaR)

Setup. A trading book worth ₹100 crore has daily P&L that is normal with mean 0 and daily volatility σ = 2% (₹2 crore). Compute the 99% one-day VaR and 99% one-day ES, and reconcile.

VaR. $$ VaR_{99\%} = z_{0.99}\,\sigma = 2.326 \times ₹2\text{ cr} = ₹4.652 \text{ cr} $$

ES. Use ES multiplier $\varphi(z_{0.99})/(1 - 0.99)$. Compute the density: $$ \phi(2.326) = \frac{1}{\sqrt{2\pi}} e^{-2.326^2/2} = 0.3989 \times e^{-2.705} = 0.3989 \times 0.0668 = 0.02665 $$ $$ ES_{99\%} = \sigma \cdot \frac{\phi(2.326)}{0.01} = ₹2\text{ cr} \times \frac{0.02665}{0.01} = ₹2\text{ cr} \times 2.665 = ₹5.330 \text{ cr} $$

Reconciliation. ES/VaR = 5.330 / 4.652 = **1.146**, matching the 99% ES/VaR ratio in the §4.3 table exactly. Interpretation: on the worst 1% of days the book loses *at least* ₹4.65 crore (VaR), but *on average* ₹5.33 crore

(ES) — about ₹0.68 crore more that VaR never told you about. Under normality that gap is modest; under a fat-tailed book it would be far larger, and only ES would flag it.

Cross-check the Basel claim. Compute 97.5% ES for the same book:

$$ ES_{97.5\%} = ₹2\text{ cr} \times \frac{\phi(1.96)}{0.025} = ₹2\text{ cr} \times \frac{0.0584}{0.025} = ₹2\text{ cr} \times 2.338 = ₹4.676 \text{ cr} $$

Compare with 99% VaR = ₹4.652 cr. They differ by under 0.6% — confirming that switching from 99% VaR to 97.5% ES is essentially capital-neutral for a normal book, exactly as §4.6 claimed. ✓

Example 2 — Empirical ES from historical scenarios

Setup. You have 20 daily losses (₹ lakh), and use α = 90% (so the worst 10% = the worst 2 observations). Losses:

`12, -5, 3, 40, 8, -2, 15, 60, 1, 7, 22, -10, 5, 30, 18, 9, 2, 11, 25, 50` (negative = a gain.)

Step 1 — sort worst to best (top 5 shown): 60, 50, 40, 30, 25, …

Step 2 — locate VaR. Worst 10% of 20 = 2 observations. VaR_{90} is the 2nd-worst loss = **₹50 lakh** (the cut-off; the loss you are 90% confident not to exceed).

Step 3 — ES = average of the worst 2 losses:

$$ ES_{90\%} = \frac{60 + 50}{2} = ₹55 \text{ lakh} $$

Reconciliation. ES (₹55 L) > VaR (₹50 L) ✓ — ES sits deeper in the tail. If a single catastrophic day had been 200 instead of 60, VaR would stay at ₹50 L (still the 2nd-worst) but ES would jump to (200 + 50)/2 = ₹125 L. This is the tail-blindness fix in one line: **the same VaR, radically different ES**, because ES reads the magnitude of the worst day and VaR does not. ✓

Example 3 — VaR fails subadditivity, ES does not (the coherence proof)

This is the canonical example, and the single most important one to have at your fingertips.

Setup. Two identical, **independent** corporate bonds, A and B. Each has face value ₹100, a **4% default probability**, and pays back ₹0 principal loss if it survives, ₹100 loss if it defaults. Use α = 95% (tail = worst 5%).

Individual VaR at 95%. For one bond, P(loss = 0) = 96%. Since 96% > 95%, the 95th percentile of the loss distribution is **0**.

$$ VaR_{95\%}(A) = VaR_{95\%}(B) = 0 $$

Sum of individual VaRs = 0 + 0 = **0**.

Portfolio VaR at 95%. Combine the two independent bonds:

Outcome	Probability	Loss (₹)
Both survive	0.96 × 0.96 = 0.9216	0
Exactly one defaults	2 × 0.04 × 0.96 = 0.0768	100
Both default	0.04 × 0.04 = 0.0016	200

Cumulative from the bottom: P(loss ≤ 0) = 0.9216, which is **below 0.95**. So the 95th percentile jumps to the next outcome:

$$ VaR_{95\%}(A + B) = ₹100 $$

Subadditivity check for VaR:

$$ VaR(A+B) = 100 \;>\; 0 = VaR(A) + VaR(B) $$

VaR violates subadditivity. Diversifying two independent bonds made the *reported* risk go *up* from 0 to 100 — an absurd, but entirely real, artefact of the quantile. A trader could split this portfolio into two separate accounts and report zero risk. ✗

Now ES at 95%. The tail is the worst 5% (0.05) of probability mass.

Individual bond ES. The worst 5% mass consists of all 4% default outcomes (loss 100) plus 1% of the survive outcomes (loss 0):

$$ ES_{95\%}(A) = \frac{1}{0.05}\big[0.04 \times 100 + 0.01 \times 0\big] = \frac{4}{0.05} = ₹80 $$

Sum of individual ES = 80 + 80 = **₹160**.

Portfolio ES. Fill the worst 5% mass from the worst outcome down: - Both default: 0.0016 mass at loss 200. - Remaining mass needed: 0.05 − 0.0016 = 0.0484, taken from the "one defaults" bucket at loss 100.

$$ ES_{95\%}(A+B) = \frac{1}{0.05}\big[0.0016 \times 200 + 0.0484 \times 100\big] = \frac{0.32 + 4.84}{0.05} = \frac{5.16}{0.05} = ₹103.2 $$

Subadditivity check for ES:

$$ ES(A+B) = 103.2 \;\le\; 160 = ES(A) + ES(B) $$

ES is subadditive — diversification *reduces* risk (160 → 103.2, a 35% cut), exactly as intuition demands. ✓

The punchline, side by side:

Measure	Bond A	Bond B	Sum of parts	Portfolio A+B	Subadditive?
VaR 95%	0	0	0	100	**No — violated**
ES 95%	80	80	160	103.2	**Yes — holds**

Same portfolio, same data. VaR says diversification *created* risk; ES correctly says it *reduced* risk. This is the coherence argument made concrete, and it is why regulators and modern risk desks trust ES for aggregation and capital.

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6. Connections

- **To VaR (Ch. 04).** ES is not a rival to VaR but its completion: VaR gives the tail's *starting point*, ES gives the tail's *average depth*. You compute VaR on the way to computing ES. Every VaR method (parametric, historical, Monte Carlo) has a direct ES counterpart.
- **To Expected Loss = PD × LGD × EAD (credit risk).** Note ES is a *conditional* expected loss — it is the expected loss *given* you are in the tail, whereas credit EL is the *unconditional* expected loss. ES relates to the *unexpected* loss that economic capital must cover, sitting between EL and the extreme tail.
- **To coherent risk measures and portfolio optimisation.** Because ES is convex and coherent, minimising ES is a well-posed convex optimisation (Rockafellar–Uryasev showed CVaR optimisation reduces to a linear program). Minimising VaR is non-convex and can have multiple local minima — another practical win for ES.
- **To Basel III / FRTB (regulatory capital).** The market-risk capital charge under the internal models approach is built on 97.5% ES calibrated to a stress period, with liquidity-horizon scaling. This is the single largest real-world deployment of ES.
- **To Extreme Value Theory (EVT).** For very high confidence levels, the tail is modelled with a Generalised Pareto Distribution, and ES has a clean closed form in terms of the GPD parameters — the natural tool when the tail is too sparse to average empirically.

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7. Key Terms

- **Expected Shortfall (ES):** average loss conditional on the loss exceeding VaR at confidence α. The tail's mean.

- **Conditional VaR (CVaR):** exact synonym for ES; emphasises the "conditional on the tail" definition.
- **Expected Tail Loss (ETL) / Average VaR:** further synonyms; the latter emphasises ES = average of all VaRs beyond α.
- **Tail probability (1 − α):** the fraction of outcomes in the loss tail (e.g. 2.5% at α = 97.5%).
- **Coherent risk measure:** one satisfying monotonicity, translation invariance, positive homogeneity, and subadditivity.
- **Subadditivity:** $\rho(A + B) \leq \rho(A) + \rho(B)$; the mathematical statement that diversification never increases risk.
- **ES multiplier:** under normality, $\varphi(z_\alpha)/(1 - \alpha)$; the factor multiplying σ to get ES (the ES analogue of z_α).
- **FRTB:** Fundamental Review of the Trading Book; the Basel reform that replaced 99% VaR with 97.5% ES.
- **Elicitability:** a statistical property (which ES lacks and VaR has) governing whether a measure can be directly backtested via a scoring function.
- **Liquidity horizon:** under FRTB, the assumed number of days to exit a position (10–120), used to scale ES by risk-factor class.

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8. Common Confusions

- **"ES is a different confidence level from VaR."** No — ES and VaR are computed at the *same* α. At α = 97.5%, both look at the same 2.5% tail; VaR reports its edge, ES reports its average. The number ES gives is larger not because α is higher but because it averages a set of losses that all exceed VaR.
- **"ES is always exactly 1.25× VaR" (or some fixed factor).** The fixed ratio (1.146 at 99%, 1.254 at 95%) holds *only under normality*. For fat-tailed distributions the ratio can be far larger — and that divergence is the entire point of using ES.
- **"ES replaces VaR completely."** Not operationally. FRTB uses ES for the capital charge but still *backtests* using VaR (because VaR is elicitable and ES is not). They coexist.
- **"VaR is never subadditive."** VaR *usually* is subadditive, and is *always* subadditive for elliptical distributions like the normal. It fails only in specific cases — heavy tails, skew, discrete/default-type payoffs. But "usually holds, can fail catastrophically" is precisely why it is untrustworthy as a guaranteed property; ES *always* holds.
- **"ES double-counts the VaR loss."** No. ES averages the losses in the tail; VaR is merely the boundary of that tail. ES ≥ VaR because every value averaged is ≥ VaR, not because VaR is added on top.
- **"A bigger tail probability means a bigger ES."** Careful — lowering α (bigger tail, e.g. 95% vs 99%) makes the tail *wider*, which pulls in less-extreme losses and gives a *lower* absolute ES value, even though it is a *lower* confidence level. $ES_{95} < ES_{99}$ in rupee terms. Direction matters.
- **"Non-elicitability means ES cannot be validated at all."** It can — via joint (VaR, ES) backtests and Acerbi–Székely style tests — just not with a single simple scoring function the way VaR can. It is harder, not impossible.

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9. Recap

- **VaR marks the edge of the tail; ES describes the whole tail.** $ES = E[L \mid L \geq VaR\alpha]$, the average loss once you are past VaR.

- **ES ≥ VaR always**, at the same confidence level; the gap measures tail heaviness — exactly the risk VaR is blind to.
- **Under normality**, $ES\alpha = \mu + \sigma \cdot \varphi(z\_\alpha)/(1-\alpha)$, a fixed multiple above VaR (1.146× at 99%). For fat tails the multiple grows.
- **Compute it three ways**, all mirroring VaR: parametric closed form, empirical (sort losses, average the worst $(1-\alpha) \cdot N$), or Monte Carlo.
- **ES is coherent** — it satisfies subadditivity, so diversification always reduces (or never increases) risk, and desk-level numbers aggregate sensibly. **VaR can violate subadditivity** (Example 3), which is its fatal theoretical flaw.
- **Basel/FRTB replaced 99% VaR with 97.5% ES.** The level was chosen so the two roughly match under normality (near-identical multipliers), giving tail sensitivity without a mechanical capital jump. VaR survives only as a backtesting tool because ES is not elicitable.

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10. Quick-Reference / Interview Points

One-liners to have ready: - "*ES is the average loss in the worst (1−α) of cases — the mean of the tail beyond VaR.*" - "*VaR tells you the door of the bad room; ES tells you how bad it is inside.*" - "*ES is coherent, VaR is not — VaR can fail subadditivity, so diversification can appear to raise VaR.*"

Formula sheet:

Quantity	Formula
ES definition	$ES\_\alpha = E[L \mid L \geq VaR\_\alpha]$
ES as average VaR	$ES\_\alpha = (1/(1-\alpha)) \int\_\alpha^1 VaR\_u \, du$
Normal VaR	$\mu + \sigma \cdot z\_\alpha$
Normal ES	$\mu + \sigma \cdot \varphi(z\_\alpha)/(1-\alpha)$
Empirical ES	mean of the worst $(1-\alpha) \cdot N$ sorted losses

Numbers worth memorising (μ=0, per unit σ): - 95%: VaR 1.645σ, ES 2.063σ (ratio 1.25) - 97.5%: VaR 1.960σ, ES 2.338σ (ratio 1.19) - 99%: VaR 2.326σ, ES 2.665σ (ratio 1.15) - **97.5% ES ≈ 99% VaR** — the reason for the Basel level choice.

The four coherence axioms: monotonicity, translation invariance, positive homogeneity, subadditivity. ES has all four; VaR misses subadditivity.

Basel/FRTB soundbite: "*Post-2008, FRTB switched market-risk capital from 99% 10-day VaR to 97.5% ES calibrated to a stress period with liquidity horizons of 10 to 120 days. ES is the capital measure; VaR is retained for backtesting because ES is not elicitable.*"

Killer example to cite: two independent 4%-default bonds at 95% — individual VaR 0 each but portfolio VaR 100 (subadditivity violated), while ES is 80 each and 103.2 combined (subadditivity holds). *If asked "why did the industry move to ES?", this is the answer in one example.*

Watch-outs to flag: ES needs more tail data than VaR to estimate stably; the fixed ES/VaR ratio only holds under normality; ES backtesting is harder due to non-elicitability.

Q&A — Expected Shortfall

Practice bank for Chapter 05. Every question is followed by a full answer. Work each one before reading the solution. Losses are positive numbers throughout; α is the confidence level and $(1 - \alpha)$ is the tail probability.

Section A — Concept Check

A1. Define Expected Shortfall in one precise sentence, and give three of its common synonyms.

Expected Shortfall at confidence level α is the *average of all losses that are at least as bad as VaR at that same level*, i.e. $ES_\alpha = E[L \mid L \geq VaR_\alpha]$ — a conditional expectation of the loss given that we are already inside the tail. Synonyms: Conditional VaR (CVaR), Expected Tail Loss (ETL), and Average VaR.

A2. VaR and ES are computed at the same α. Why is ES always the larger number?

Because ES averages a set of losses every one of which is $\geq VaR_\alpha$ (VaR is the boundary of the tail, ES is the mean of everything past that boundary). The average of numbers that are all ≥ VaR must itself be ≥ VaR. It is *not* larger because ES uses a higher confidence level — both use the same α and the same tail. The gap ES − VaR measures how heavy the tail is.

A3. Explain the sentence: "VaR is a quantile, ES is an integral," and why that distinction matters.

VaR is a single order statistic — the α-quantile — so it is sensitive only to *where* the tail boundary sits, not to the magnitude of losses beyond it. You can move the worst 0.5% of outcomes from ₹11 crore to ₹110 crore and the 99% quantile does not move, because the count of outcomes below the threshold is unchanged. ES is an average (an integral) over the whole tail, and integrals are sensitive to magnitude everywhere in their range. Pushing the worst outcomes further out immediately raises the tail average. That is why ES captures tail severity and VaR is blind to it.

A4. Name the four coherence axioms and state which one VaR can fail.

Monotonicity (a portfolio that always loses more is riskier), translation invariance (adding c of risk-free cash lowers risk by exactly c), positive homogeneity (scaling the book by λ scales risk by λ), and subadditivity ($\rho(A + B) \leq \rho(A) + \rho(B)$; merging books never creates more risk than holding them apart). VaR satisfies the first three but can **violate subadditivity**. ES satisfies all four and is therefore always coherent.

A5. Why is subadditivity the "crucial" axiom, in two respects?

First, it *encodes diversification*: combining imperfectly correlated risks should never increase total risk, and a measure that can violate this tells lies about diversification. Second, it makes *limits and capital additive*: if risk is subadditive, the sum of desk-level numbers is a conservative (larger-or-equal) bound on the firm total, so a regulator can decompose a firm-wide capital number down to desks safely. Break subadditivity and a trader could split one position across two accounts and report *less* total risk than the single position — gaming the limit system.

A6. Write the two general formulas for ES and read the integral form in words.

$ES_\alpha = E[L \mid L \geq VaR_\alpha]$ and $ES_\alpha = \frac{1}{1-\alpha} \int_\alpha^1 VaR_u \, du$. The integral says: sweep the confidence level u from α all the way to 1 (the absolute worst case), read VaR at each level, and average them. ES is a "VaR of VaRs" — the average of every quantile deeper than α — which is exactly why it cannot ignore any part of the tail, and why "Average VaR" is a synonym.

A7. Under normality, why did Basel pick 97.5% ES to replace 99% VaR rather than 99% ES?

Because under a normal distribution the 97.5% ES multiplier (2.338) is almost identical to the 99% VaR multiplier (2.326). So switching from 99% VaR to 97.5% ES is roughly capital-neutral for a well-behaved (normal) book — it does not mechanically inflate capital across the board — while the ES formulation *automatically* penalises books with fat tails. Regulators bought tail sensitivity without an arbitrary capital jump.

A8. What is "elicitability" and why does it matter for backtesting ES?

Elicitability is a statistical property governing whether a risk measure can be validated with a single scoring/loss function. VaR (a quantile) is elicitable — you simply count exceptions (days losses exceeded VaR should be about 1% of days at 99%). ES, being a conditional expectation, is *not* elicitable in the strict sense, so it cannot be scored with one simple loss function. Consequently FRTB backtests VaR (at 97.5% and 99%) for the traffic-light exception test but uses ES for the actual capital charge. ES can still be validated — via joint (VaR, ES) tests or Acerbi–Székely tests — just not as simply.

A9. Distinguish ES from the credit-risk Expected Loss (EL = PD × LGD × EAD).

Both are expected losses, but EL is *unconditional* — the average loss over all outcomes — whereas ES is *conditional* — the expected loss *given* that we are in the worst $(1-\alpha)$ tail. EL is what pricing/provisioning covers; ES sits far out in the tail and relates to the unexpected loss that economic capital must absorb.

A10. Common confusion: "ES is always 1.25× VaR." Correct it.

The fixed ES/VaR ratio (about 1.25 at 95%, 1.19 at 97.5%, 1.15 at 99%) holds *only under normality*. For fat-tailed distributions the ratio can be much larger — and that divergence is precisely the reason to use ES. Under fat tails VaR may barely move while ES climbs, so no fixed multiple applies.

Section B — Numerical / Applied (full solutions)

B1. Parametric VaR and ES under normality. A ₹100 crore book has daily P&L that is normal with mean 0 and daily volatility σ = 2% = ₹2 crore. Compute the 99% one-day VaR and 99% one-day ES, and state the tail gap.

Solution. VaR uses the quantile multiplier $z_{0.99} = 2.326$: $VaR_{99\%} = z_{0.99} \cdot \sigma = 2.326 \times$ ₹2 cr = **₹4.652 cr**.

ES uses the multiplier $\varphi(z_\alpha)/(1-\alpha)$. The density at 2.326: $\varphi(2.326) = 0.3989 \times e^{(-2.326^2/2)} = 0.3989 \times e^{(-2.705)} = 0.3989 \times 0.0668 = 0.02665$. ES multiplier = 0.02665 / 0.01 = 2.665. $ES_{99\%} = \sigma \times 2.665$ = ₹2 cr × 2.665 = **₹5.330 cr**.

Tail gap = 5.330 − 4.652 = **₹0.678 cr**. On the worst 1% of days the book loses *at least* ₹4.65 cr (VaR) but *on average* ₹5.33 cr (ES). Check: ES/VaR = 5.330/4.652 = 1.146, matching the tabulated 99% ratio. ✓

B2. Confirm the Basel "capital-neutral" claim numerically. For the same book, compute 97.5% ES and compare with 99% VaR.

Solution. $z_{0.975} = 1.960$, $\varphi(1.96) = 0.0584$. $ES_{97.5\%} = \sigma \times \varphi(1.96)/(1-0.975)$ = ₹2 cr × (0.0584/0.025) = ₹2 cr × 2.338 = **₹4.676 cr**. Compare with 99% VaR = ₹4.652 cr from B1. Difference = 4.676 − 4.652 = ₹0.024 cr, about 0.5%. So 97.5% ES ≈ 99% VaR for a normal book — the switch is essentially capital-neutral, exactly as FRTB intended. ✓

B3. Empirical ES from historical scenarios. Twenty daily losses (₹ lakh, negative = gain), α = 90% (worst 10% = worst 2 obs): `12, -5, 3, 40, 8, -2, 15, 60, 1, 7, 22, -10, 5, 30, 18, 9, 2, 11, 25, 50` Find VaR_90% and ES_90%.

Solution. Sort worst to best: 60, 50, 40, 30, 25, 22, 18, 15, 12, 11, 9, 8, 7, 5, 3, 2, 1, −2, −5, −10. Worst 10% of 20 = 2 observations. VaR_90% = the 2nd-worst loss = **₹50 lakh** (the cut-off). ES_90% = average of the worst 2 losses = (60 + 50)/2 = **₹55 lakh**. Check: ES (55) > VaR (50) ✓ — ES sits deeper in the tail.

B4. Tail-blindness in one line. In B3, suppose the single worst day had been 200 instead of 60. Recompute VaR and ES.

Solution. The two worst losses become 200 and 50. VaR_90% is still the 2nd-worst = **₹50 lakh** (unchanged — the quantile does not see the magnitude of the very worst day). ES_90% = (200 + 50)/2 = **₹125 lakh** (up from 55). Same VaR, radically different ES: this is exactly the failure ES fixes. VaR reads the boundary; ES reads the severity. ✓

B5. The coherence proof — VaR fails subadditivity, ES does not. Two identical, *independent* corporate bonds A and B. Each: face ₹100, default probability 4%, loss ₹100 if it defaults and ₹0 if it survives. α = 95% (tail = worst 5%). Compute individual and portfolio VaR and ES.

Solution.

Individual VaR. For one bond P(loss = 0) = 96% > 95%, so the 95th percentile is 0. VaR_95%(A) = VaR_95%(B) = **0**. Sum = 0.

Portfolio distribution (independence):

Outcome	Probability	Loss (₹)
Both survive	0.96 × 0.96 = 0.9216	0
Exactly one defaults	2 × 0.04 × 0.96 = 0.0768	100
Both default	0.04 × 0.04 = 0.0016	200

Cumulative from the bottom: P(loss ≤ 0) = 0.9216 < 0.95, so the 95th percentile jumps to the next outcome: VaR_95%(A+B) = **₹100**.

VaR subadditivity check: VaR(A+B) = 100 > 0 = VaR(A) + VaR(B). **VaR violated subadditivity** — diversifying two independent bonds raised the reported risk from 0 to 100. ✗

Individual ES. Fill the worst 5% (0.05) mass from the worst outcome down: the 4% default mass (loss 100) plus 1% of the survive mass (loss 0): ES_95%(A) = (1/0.05)[0.04 × 100 + 0.01 × 0] = 4/0.05 = **₹80**. Sum = 160.

Portfolio ES. Fill 0.05 of mass from the worst down: - Both default: 0.0016 at loss 200. - Remaining 0.05 − 0.0016 = 0.0484 from the "one defaults" bucket at loss 100. ES_95%(A+B) = (1/0.05)[0.0016 × 200 + 0.0484 × 100] = (0.32 + 4.84)/0.05 = 5.16/0.05 = **₹103.2**.

ES subadditivity check: ES(A+B) = 103.2 ≤ 160 = ES(A) + ES(B). **ES is subadditive** — diversification cut risk from 160 to 103.2 (a 35% reduction), as intuition demands. ✓

Side by side:

Measure	A	B	Sum of parts	Portfolio A+B	Subadditive?
VaR 95%	0	0	0	100	No — violated
ES 95%	80	80	160	103.2	Yes — holds

B6. ES falls as the confidence level falls (direction check). For a normal book with σ = ₹5 crore (μ = 0), compute ES at 95%, 97.5% and 99% and confirm the ordering.

Solution. Using ES multipliers $\varphi(z_\alpha)/(1-\alpha)$: 95% → 2.063, 97.5% → 2.338, 99% → 2.665. - ES_95% = 5 × 2.063 = **₹10.32 cr** - ES_97.5% = 5 × 2.338 = **₹11.69 cr** - ES_99% = 5 × 2.665 = **₹13.32 cr**

So ES_95% < ES_97.5% < ES_99% in rupee terms. A *lower* confidence level means a *wider* tail that pulls in less-extreme losses, giving a *lower* absolute ES. Careful: "bigger tail probability" does not mean "bigger ES." ✓

B7. ES from a discrete P&L distribution. A position has the following loss distribution: ₹0 with prob 0.90, ₹50 with prob 0.06, ₹120 with prob 0.03, ₹300 with prob 0.01. Compute VaR_95% and ES_95%.

Solution. Cumulative probability from best to worst: P(loss ≤ 0) = 0.90; P(loss ≤ 50) = 0.96. The smallest loss ℓ with P(L ≤ ℓ) ≥ 0.95 is **₹50**, so VaR_95% = ₹50. ES_95% = average loss over the worst 5% mass, filled from the worst outcome down: - ₹300 at 0.01 - ₹120 at 0.03 - remaining 0.05 − 0.04 = 0.01 of the ₹50 bucket ES_95% = (1/0.05)[0.01×300 + 0.03×120 + 0.01×50] = (3 + 3.6 + 0.5)/0.05 = 7.1/0.05 = **₹142**. Check: ES (142) > VaR (50) ✓, and ES reflects the ₹300 catastrophe that VaR ignores.

B8. Scaling to a longer horizon. A book's 1-day 97.5% ES is ₹8 crore, returns i.i.d. and roughly normal. Estimate the 10-day 97.5% ES using the square-root-of-time rule.

Solution. Under i.i.d. normal returns with zero mean, both VaR and ES scale with the square root of the horizon (they are fixed multiples of σ, and σ scales as √T). ES_10-day = ES_1-day × √10 = ₹8 cr × 3.162 = **₹25.30 cr**. Caveat: this assumes i.i.d. zero-drift normal returns; FRTB instead uses liquidity-adjusted horizons (10–120 days) by risk class rather than a blanket √T scale.

Section C — Interview-Style (model answers)

C1. "In one minute, why did the industry move from VaR to Expected Shortfall?"

Model answer: "VaR has two fatal weaknesses. First, it is tail-blind — it reports only the boundary loss at a confidence level and says nothing about how bad losses are beyond it, so two books with identical VaR can have wildly different true tail risk, and traders can game a VaR budget by selling deep out-of-the-money risk that sits just past the cut-off. Second, VaR is not coherent: it can violate subadditivity, meaning a diversified portfolio can show *higher* VaR than the sum of its parts, which is nonsensical and breaks capital aggregation. Expected Shortfall fixes both: it averages the entire tail beyond VaR, so it captures tail severity, and it satisfies all four coherence axioms including subadditivity. That is why Basel's FRTB replaced 99% VaR with 97.5% ES for market-risk capital."

C2. "Give me the single cleanest example that shows VaR is not coherent but ES is."

Model answer: "Two independent bonds, each ₹100 face, 4% default probability, ₹100 loss on default, at 95% confidence. Each bond alone has a 96% chance of zero loss, so its 95% VaR is 0 — sum of parts is 0. But the combined portfolio has only a 92.16% chance of zero loss, so its 95th percentile jumps to ₹100 — the portfolio VaR (100) exceeds the sum of individual VaRs (0). VaR says diversification *created* risk. ES tells the opposite, correct story: each bond's 95% ES is ₹80, summing to 160, while the portfolio ES is ₹103.2 — diversification cut risk by 35%. Same data, and only ES behaves sensibly."

C3. "Why 97.5% for ES? Isn't a higher confidence level more conservative?"

Model answer: "It was chosen for continuity, not conservatism. Under normality the 97.5% ES multiplier is 2.338, almost exactly the 99% VaR multiplier of 2.326, so switching from 99% VaR to 97.5% ES keeps capital roughly unchanged for a normal, well-behaved book. That avoids an arbitrary mechanical jump in capital. The benefit is that when a book has fat tails, ES automatically produces a higher number than the equivalent VaR would — you get tail sensitivity for free without recalibrating the whole framework. Higher confidence isn't the point; tail *shape* sensitivity is."

C4. "If ES is better, why does FRTB still compute VaR at all?"

Model answer: "Backtesting. VaR is elicitable — you can validate it by counting exceptions against a simple expected exception rate, which regulators formalise as the traffic-light test. ES is a conditional expectation and is not elicitable, so it can't be scored with one simple loss function. FRTB's pragmatic split is to backtest VaR at 97.5% and 99% for the exception test while using ES for the actual capital charge. VaR survives as a validation tool even though ES does the capital work."

C5. "A trader tells you two strategies have the same VaR, so they're equally risky. How do you respond?"

Model answer: "Equal VaR only means the tail *boundary* is the same; it says nothing about what's behind it. I'd compute ES for both. If one strategy sells catastrophe or deep out-of-the-money risk, its losses beyond the VaR line are far larger, so its ES will be much higher even though VaR matches. I'd also check the ES/VaR ratio: near the normal value (about 1.15 at 99%) suggests a well-behaved tail; a ratio well above that flags a fat tail the VaR number is hiding. VaR equality is a red flag to look deeper, not a proof of equal risk."

C6. "What are the practical downsides of ES you'd flag to a risk committee?"

Model answer: "Three. First, estimation: ES averages the worst $(1-\alpha)\cdot N$ observations, so it needs more tail data than VaR to be stable — at 97.5% with 1,000 scenarios you're averaging just 25 points, and the estimate is noisy if the tail is sparse. Second, backtesting: ES isn't elicitable, so validation needs joint (VaR, ES) or Acerbi–Székely tests rather than a simple exception count — harder to explain to supervisors. Third, model risk in the extreme tail: for very high confidence you may need Extreme Value Theory (a Generalised Pareto fit) rather than raw empirical averaging. None of these outweigh coherence and tail sensitivity, but they're real operational costs."

C7. "Explain the link between ES and convex optimisation, and why a portfolio manager might care."

Model answer: "ES is convex and coherent, so minimising ES is a well-posed convex optimisation — Rockafellar and Uryasev showed CVaR minimisation reduces to a linear program that's fast and has a unique global optimum. Minimising VaR, by contrast, is non-convex and can have many local minima, so an optimiser can get stuck or return unstable weights. For a manager building a tail-risk-controlled portfolio, ES gives a tractable, reliable objective; VaR does not."

Section D — Multiple Choice (with reasoning)

D1. At the same confidence level α, which is always true? A) VaR ≥ ES B) ES ≥ VaR C) ES = VaR D) No fixed relationship

Answer: **B**. ES averages losses all of which are ≥ VaR (the tail boundary), so its average is ≥ VaR. A and C are wrong; D ignores this guaranteed ordering. Equality only occurs in the degenerate case where all tail losses equal VaR exactly.

D2. Which coherence axiom can VaR violate? A) Monotonicity B) Translation invariance C) Positive homogeneity D) Subadditivity

Answer: **D**. VaR satisfies the first three but can fail subadditivity (e.g. the two-independent-bonds example), which is its fatal theoretical flaw. ES satisfies all four.

D3. Under normality with $\mu = 0$, the ES multiplier at confidence α is: A) z_α B) $\varphi(z_\alpha)/(1-\alpha)$ C) $(1-\alpha)/\varphi(z_\alpha)$ D) $z_\alpha/(1-\alpha)$

Answer: **B**. $ES_\alpha = \mu + \sigma\cdot\varphi(z_\alpha)/(1-\alpha)$; the factor $\varphi(z_\alpha)/(1-\alpha)$ is the ES analogue of z_α. A is the VaR multiplier. C and D are not valid forms.

D4. FRTB replaced 99% VaR with: A) 99% ES B) 97.5% ES C) 95% ES D) 99.9% VaR

Answer: **B**. FRTB uses 97.5% ES because under normality its multiplier (2.338) is nearly identical to the 99% VaR multiplier (2.326), making the switch roughly capital-neutral while adding tail sensitivity.

D5. Why does FRTB still backtest with VaR rather than ES? A) ES is always inaccurate B) VaR is more conservative C) ES is not elicitable D) Regulators prefer older methods

Answer: **C**. ES, a conditional expectation, lacks elicitability, so it can't be validated with a single scoring function. VaR (a quantile) is elicitable and is retained for the exception test. A and B are false; D is not the reason.

D6. Two portfolios have identical 99% VaR but portfolio X loses far more in the extreme tail. Which measure distinguishes them? A) VaR B) ES C) Neither D) Both equally

Answer: **B**. VaR sees only the tail boundary, which is identical, so it cannot distinguish them. ES averages the whole tail and will be higher for portfolio X. This is the core tail-blindness point.

D7. Lowering the confidence level from 99% to 95% (all else equal) does what to the absolute ES in rupees? A) Increases it B) Decreases it C) Leaves it unchanged D) Cannot tell

Answer: **B**. A lower α means a wider tail that pulls in less-extreme losses, lowering the tail average. So $ES_{95\%} < ES_{99\%}$ in rupee terms even though 95% is a "bigger tail probability." Direction is a classic trap.

D8. For the empirical method with $N = 1{,}000$ scenarios at $\alpha = 97.5\%$, ES equals: A) The 25th-worst loss B) The average of the worst 25 losses C) The single worst loss D) The average of all 1,000 losses

Answer: **B**. The worst 2.5% of 1,000 is 25 observations; VaR is roughly the 25th-worst loss (the cut-off) and ES is the *average* of those worst 25. A describes VaR, not ES.

D9. The statement "ES is always exactly 1.25× VaR" is: A) Always true B) True only under normality at 95% C) True for all fat-tailed books D) Never true

Answer: **B**. The 1.25 ratio is the normal-distribution value specifically at 95% (it is 1.19 at 97.5%, 1.15 at 99%). For fat tails the ratio can be much larger, which is the whole reason to use ES. So the fixed factor holds only under normality, and 1.25 specifically at $\alpha = 95\%$.

D10. In the two-independent-4%-bond example at 95%, the portfolio VaR and the sum of individual VaRs are: A) 0 and 0 B) 100 and 0 C) 100 and 200 D) 200 and 0

Answer: **B**. Each bond's 95% VaR is 0 (96% chance of no loss), so the sum of parts is 0; the portfolio's 95th percentile jumps to 100 because P(loss ≤ 0) = 0.9216 < 0.95. VaR(A+B) = 100 > 0, violating subadditivity.

D11. ES relates to the credit-risk Expected Loss (PD × LGD × EAD) how? A) They are identical B) ES is unconditional, EL is conditional C) ES is conditional (on the tail), EL is unconditional D) They are unrelated

Answer: **C**. EL is the unconditional average loss over all outcomes; ES is the expected loss *conditional* on being in the worst (1−α) tail. ES sits far out in the tail and maps to unexpected loss / economic capital.

D12. Which is the strongest single theoretical argument for ES over VaR? A) ES is easier to compute B) ES is easier to backtest C) ES is always coherent (satisfies subadditivity) D) ES uses a higher confidence level

Answer: **C**. Coherence — specifically guaranteed subadditivity — is the deepest argument: it makes ES safe for diversification, aggregation and optimisation. A and B are actually false (ES is harder to backtest and needs more tail data); D is not generally true.

Self-check note: All parametric figures use z_0.95 = 1.645, z_0.975 = 1.960, z_0.99 = 2.326, and ES multipliers 2.063 / 2.338 / 2.665 respectively; the two-bond coherence numbers (VaR 0/0/100, ES 80/80/103.2) and the empirical B3/B4 results reconcile exactly with the chapter's worked examples.

Chapter 06 — Credit Risk

1. The Problem / The Need

Every time a bank lends money, buys a bond, sells a derivative that will settle in the future, or even extends a 30-day trade credit, it hands over value *today* against a *promise* of value tomorrow. Credit risk is the risk that the promise is broken — that the counterparty fails to pay what it owes, when it owes it, in full.

Why does this deserve its own discipline rather than being lumped in with "market risk"? Because the loss profile is fundamentally different and far nastier:

- **It is asymmetric.** On a loan you can lose the entire principal (100% down), but the best possible outcome is that you get your money back plus a modest spread (a few percent up). You are, in effect, *short a put option* on the borrower's assets: bounded upside, catastrophic downside.
- **It is skewed and fat-tailed.** Most of the time nothing happens — the loan performs. Then, rarely, a cluster of defaults arrives together (a recession, a sector collapse) and wipes out years of accumulated spread income. The loss distribution has a long, thick right tail.
- **It is driven by correlation, not just individual odds.** A single borrower defaulting is idiosyncratic. Thousands defaulting *at once* is systematic. The whole art of portfolio credit risk is modelling how defaults bunch together.

For a bank, credit risk is not a side concern — it is *the* core business. Roughly 50–70% of a typical commercial bank's economic capital is held against credit risk. The 2008 crisis, virtually every banking failure in history, and most NBFC blowups (IL&FS, DHFL in India) trace back to credit risk that was mispriced, under-provisioned, or too concentrated. If you want a risk job at a bank or NBFC, this chapter is the beating heart of the interview.

The need, then, is for a rigorous framework that answers four questions:

1. **How likely** is default? (Probability of Default, PD)
2. **How much** do we lose if it happens? (Loss Given Default, LGD)
3. **How much** is at risk when it happens? (Exposure at Default, EAD)
4. **How do these losses behave across a whole portfolio** once correlation is added?

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2. The Core Idea

Credit risk decomposes cleanly into a small set of multiplicative building blocks. The single most important equation in all of credit risk is the **Expected Loss** identity:

$$ \text{EL} = \text{PD} \times \text{LGD} \times \text{EAD} $$

where:

- **PD** = Probability of Default over a horizon (usually 1 year), a number between 0 and 1.
- **LGD** = Loss Given Default, the fraction of exposure *not* recovered, between 0 and 1. Note **LGD = 1 – Recovery Rate**.
- **EAD** = Exposure at Default, the money amount outstanding when default hits.

The deep insight is a change of *mindset*: **expected loss is not a risk — it is a cost.** A bank that expects to lose 1.5% of a loan book each year is not being surprised; it should price that 1.5% into the interest rate and set it aside as a provision. Expected loss is budgeted, priced, and reserved for.

The *real* risk is the **Unexpected Loss (UL)** — the volatility of losses around that expected mean, the possibility that this year losses come in far above the long-run average. Unexpected loss is what *capital* exists to absorb. This gives the fundamental division of labour in a bank:

> **Provisions cover Expected Loss. Capital covers Unexpected Loss.**

Everything else — ratings, transition matrices, correlation models, credit VaR, loan pricing — is machinery for estimating these components and then for describing the *shape* of the loss distribution beyond the mean.

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3. Why / How It Works

Why the multiplicative form? Each factor answers a logically independent question, and they compound. Think of a single loan as a lottery:

- With probability PD, the "bad" outcome occurs and default happens.
- Given default, you were exposed to EAD dollars.
- Of that exposure, you fail to recover a fraction LGD.

So the expected dollar loss is (chance of the bad event) × (size when bad) × (severity when bad) = PD × EAD × LGD. It is just the expected value of a random loss variable $L = \mathbb{1}_{\text{default}} \times \text{EAD} \times \text{LGD}$.

Why separate EL from UL? Because they are funded by different pools of money and managed by different people. If you only ever look at the average, you will be catastrophically undercapitalised when the tail arrives. A portfolio can have a tiny expected loss and still be lethal if its losses are highly correlated — because the *unexpected* loss (the tail) is enormous relative to the mean.

Why does correlation dominate? Consider two extreme worlds holding 1,000 loans, each with PD = 1%:

- **Zero correlation** (defaults independent): losses cluster tightly around the mean of 10 defaults by the law of large numbers. The tail is thin; you need little capital.
- **Perfect correlation** (all loans rise and fall together): either 0 default or all 1,000 default. The "portfolio" behaves like one giant loan. The tail is monstrous; you need enormous capital.

Real portfolios sit between these, and the *degree* of correlation — driven by shared exposure to the economy, a region, or a sector — is what sets the capital number. This is precisely why diversification is the only free lunch in credit, and why *concentration* (too much to one borrower, one sector, one geography) is the classic killer.

How ratings fit in. We cannot estimate a bespoke PD for every borrower from scratch, so we *bucket* borrowers into rating grades (AAA, AA, ..., C, D). Each grade maps to an empirically observed PD. Crucially, borrowers *migrate* between grades over time — a AA can be downgraded to BBB — and this migration is itself a source of loss (mark-to-market credit risk), captured by a **transition matrix**.

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4. Full Content — Framework, Formulas & Methods

4.1 The three parameters in detail

Probability of Default (PD). The likelihood the obligor defaults within the horizon.

- *Through-the-cycle (TTC)* PD: a long-run average, smoothing over booms and busts. Used for Basel regulatory capital and rating agencies.
- *Point-in-time (PIT)* PD: reflects current conditions; rises in recessions. Used for IFRS 9 / expected credit loss accounting.
- Sources: rating-agency default studies, internal rating models, or *market-implied* PD from bond spreads and CDS.

Loss Given Default (LGD). The fraction lost after recovery, net of collection costs, discounted to default date.

$$ \text{LGD} = 1 - \text{Recovery Rate} = \frac{\text{EAD} - \text{Recoveries} + \text{Costs}}{\text{EAD}} $$

LGD depends heavily on **seniority** and **collateral**: a senior secured loan might have LGD ≈ 25%, while subordinated unsecured debt might have LGD ≈ 75%. LGD tends to *rise* in downturns (collateral values fall exactly when defaults spike) — "downturn LGD" is used for prudence.

Exposure at Default (EAD). The amount outstanding at the moment of default.

- For a term loan drawn in full, EAD ≈ outstanding principal.
- For a revolving facility or credit line, borrowers tend to *draw down* as they approach distress. EAD is modelled as:

$$ \text{EAD} = \text{Drawn} + \text{CCF} \times (\text{Limit} - \text{Drawn}) $$

where **CCF** (Credit Conversion Factor) is the fraction of the undrawn commitment expected to be drawn before default.

4.2 Expected Loss

At the single-facility level:

$$ \boxed{\ \text{EL} = \text{PD} \times \text{LGD} \times \text{EAD}\ } $$

At the portfolio level, EL is simply additive (expectations always add, regardless of correlation):

$$ \text{EL}_{\text{portfolio}} = \sum_i \text{PD}_i \times \text{LGD}_i \times \text{EAD}_i $$

4.3 Unexpected Loss (single exposure)

Model the loss on one facility as $L = \text{EAD}\times\text{LGD}\times D$, where D is the Bernoulli default indicator (1 with probability PD, else 0). Unexpected loss is the *standard deviation* of loss. Treating LGD and EAD as fixed for a moment and letting default variability dominate:

$$ \text{Var}(D) = \text{PD}(1-\text{PD}) $$

$$ \text{UL} = \text{EAD}\times\text{LGD}\times\sqrt{\text{PD}(1-\text{PD})} $$

A fuller version that also accounts for the variance of the recovery/LGD itself (with σ_{LGD} the standard deviation of LGD) is:

$$ \text{UL} = \text{EAD}\sqrt{\ \text{PD}\,\sigma_{\text{LGD}}^{2} + \text{LGD}^{2}\,\text{PD}(1-\text{PD})\ } $$

The first term inside the root is the contribution from *how much* recovery varies; the second is from the *binary* default event. In interviews, the simpler $\text{UL}=\text{EAD}\cdot\text{LGD}\cdot\sqrt{\text{PD}(1-\text{PD})}$ is usually what is expected unless recovery volatility is given.

4.4 Portfolio Unexpected Loss and correlation

For a two-asset portfolio, unexpected losses do **not** simply add — they combine like volatilities with a default correlation ρ:

$$ \text{UL}_{P} = \sqrt{\ \text{UL}_1^2 + \text{UL}_2^2 + 2\rho\,\text{UL}_1\text{UL}_2\ } $$

The **risk contribution** of asset i to the portfolio is:

$$ \text{RC}i = *\text{UL}_i \times \frac{\text{UL}_i + \sum*{j\ne i}\rho_{ij}\text{UL}_j}{\text{UL}_P} \quad\text{and}\quad \sum_i \text{RC}_i = \text{UL}_P $$

Because $\rho < 1$, $\text{UL}_P < \sum_i \text{UL}_i$ — the diversification benefit.

4.5 Credit ratings and the transition matrix

A **transition (migration) matrix** gives the probability of moving from one rating to another over one year. Rows are the starting rating, columns the ending rating, and the last column "D" is default. Each row sums to 100%.

From \ To	AAA	AA	A	BBB	BB	B	CCC	D (default)
AAA	90.0	9.0	0.6	0.3	0.1	0.0	0.0	**0.00**
AA	0.6	90.0	7.6	0.6	0.6	0.1	0.02	**0.02**
A	0.05	2.0	91.0	5.7	0.7	0.2	0.01	**0.06**
BBB	0.03	0.3	5.0	88.5	4.2	1.0	0.4	**0.20**
BB	0.02	0.1	0.5	6.5	82.5	7.7	1.4	**1.10**
B	0.0	0.1	0.3	0.4	6.0	83.0	5.5	**4.70**
CCC	0.0	0.0	0.2	0.5	1.5	11.0	60.0	**26.80**

(Illustrative annual transition probabilities, %. Note how default probability climbs steeply down the rating scale — the hallmark of a credit curve.)

Key uses:

- The **default column** gives the 1-year PD for each grade (e.g. BBB → 0.20%).
- **Multi-year PDs** come from matrix multiplication: the 2-year matrix is M^2, the n-year is M^n (assuming a time-homogeneous Markov chain).
- **Downgrade risk**: even without default, a downgrade widens the credit spread, marking the bond down. This is the basis of the CreditMetrics mark-to-market approach.

4.6 Measuring and pricing credit risk

Market-implied PD from spreads. A risky bond yields more than a risk-free bond; the extra is the **credit spread** s. Under a simple reduced-form model, for a short horizon the spread compensates for expected loss:

$$ s \approx \text{PD} \times \text{LGD} \quad\Rightarrow\quad \text{PD} \approx \frac{s}{\text{LGD}} $$

More precisely, equating expected returns of a risky and risk-free bond over one period:

$$ (1+y) (1-\text{PD}) + (1+y)(\text{PD})(1-\text{LGD}) = 1 + r_f $$

which rearranges to $\text{PD} = \dfrac{y - r_f}{(1+y)\,\text{LGD}} \approx \dfrac{s}{\text{LGD}}$ for small numbers.

Pricing a loan (RAROC). A loan's interest rate must cover four things: funding cost, operating cost, **expected loss**, and a return on the **capital** held against unexpected loss. Risk-Adjusted Return on Capital:

$$ \text{RAROC} = \frac{\text{Spread income} + \text{Fees} - \text{Expected Loss} - \text{Operating cost}}{\text{Economic Capital (}\approx k \times \text{UL)}} $$

A loan is approved only if RAROC exceeds the bank's hurdle rate (cost of equity). This is *the* mechanism that forces EL and UL into the price.

Structural (Merton) model. Treats equity as a call option on the firm's assets. The firm defaults if asset value V falls below the debt face value F at maturity. Distance-to-default:

$$ DD = \frac{\ln(V/F) + (\mu - \tfrac{1}{2}\sigma_V^2)T}{\sigma_V\sqrt{T}},\qquad \text{PD} = N(-DD) $$

This is the engine behind Moody's KMV EDF (Expected Default Frequency) models.

4.7 The credit risk process — putting it together

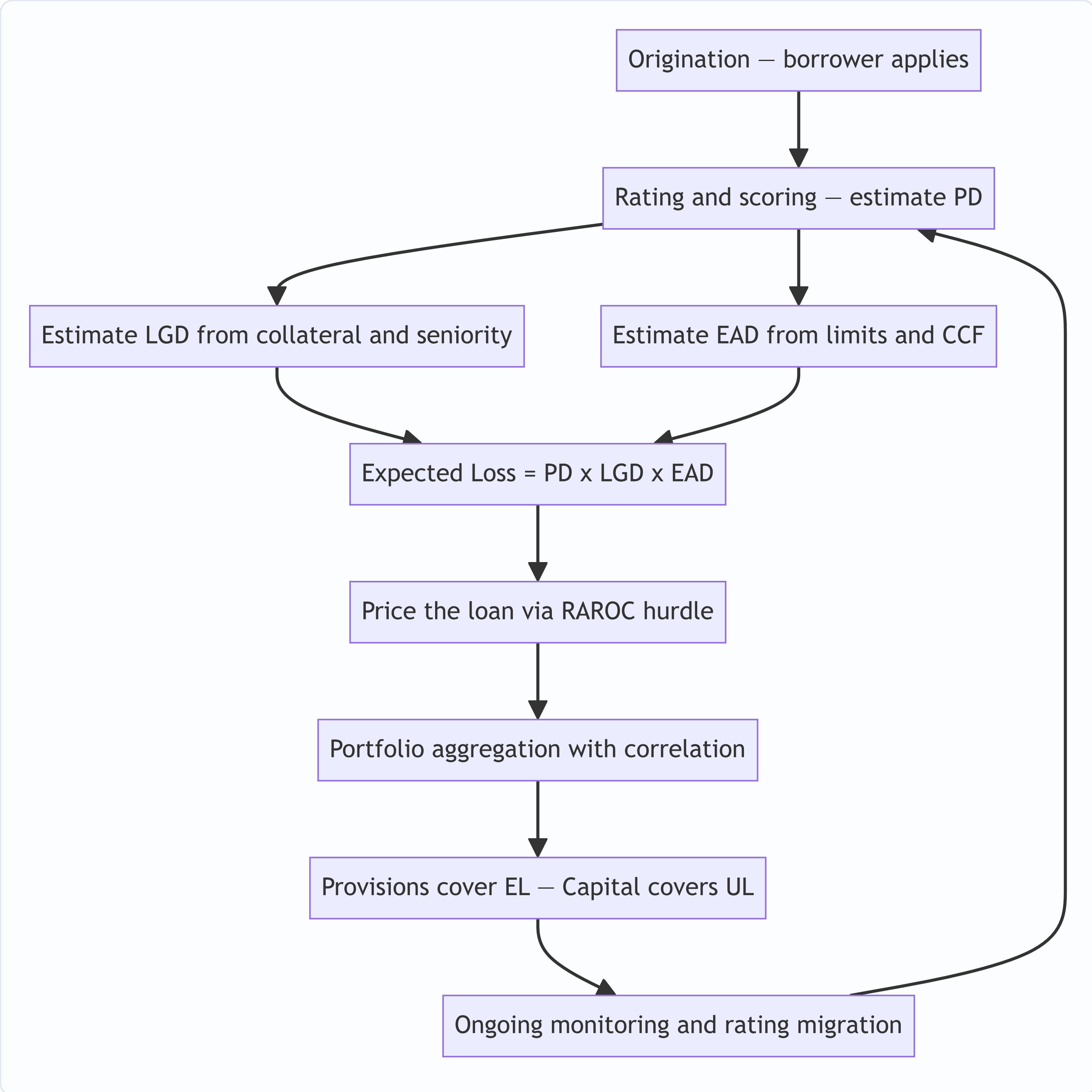

Figure 1 — the credit risk lifecycle from a single loan through to portfolio capital, a continuous loop as ratings migrate.

4.8 The shape of the loss distribution

Figure 2 — the three zones of the credit loss distribution. Provisions absorb the mean, economic capital absorbs the body of the tail up to a chosen confidence, and catastrophic stress losses beyond that are the existential risk.

The full loss distribution is heavily right-skewed:

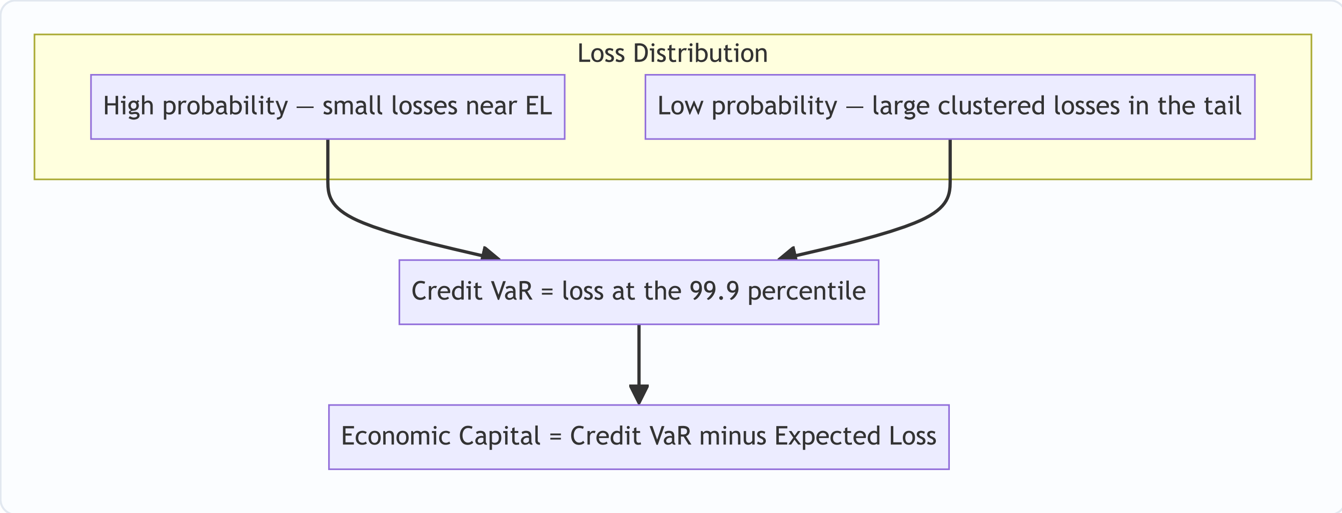

Figure 3 — Credit VaR is the loss not exceeded at a high confidence level; economic capital is the distance from the mean (EL) out to that percentile.

Credit VaR and economic capital. Just as market VaR, Credit VaR is a percentile of the loss distribution:

$$ \text{Economic Capital} = \text{Credit VaR}_{\alpha} - \text{EL} $$

We subtract EL because provisions already cover the mean; capital only needs to cover the *unexpected* portion out to confidence α (Basel uses 99.9%).

4.9 The Basel IRB capital formula (concept)

Basel's Internal Ratings-Based approach uses a single-factor Merton (Vasicek) model. Capital per unit of exposure is the *conditional* expected loss at a 99.9% bad state of the economy, minus the already-provisioned EL:

$$ K = \text{LGD}\left[\ N\!\left(\frac{N^{-1}(\text{PD}) + \sqrt{\rho}\,N^{-1}(0.999)}{\sqrt{1-\rho}}\right) - \text{PD}\ \right]\times \text{Maturity adjustment} $$

You do not need to compute this by hand in most interviews, but you *must* be able to explain it: the term inside $N(\cdot)$ is the **stressed PD** (PD conditional on a 99.9% bad systematic factor), and ρ is the asset correlation to the single economy-wide factor. Capital rises with PD, with LGD, and with correlation.

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5. Worked Examples

Example 1 — Expected Loss on a single loan (the core computation)

A bank lends **₹100 crore** as a term loan to a BB-rated corporate.

- PD (1-year, from the BB grade) = **1.10%** = 0.011
- The loan is secured by property giving a recovery rate of 60%, so **LGD = 40%** = 0.40
- The loan is fully drawn, so **EAD = ₹100 crore**

Expected Loss:

$$ \text{EL} = \text{PD}\times\text{LGD}\times\text{EAD} = 0.011 \times 0.40 \times 100 = \textbf{₹0.44 crore} $$

So the bank should provision **₹44 lakh** and build at least 0.44% into the loan's spread just to break even on expected credit cost.

Unexpected Loss (default-event only):

$$ \text{UL} = \text{EAD}\times\text{LGD}\times\sqrt{\text{PD}(1-\text{PD})} = 100 \times 0.40 \times \sqrt{0.011\times0.989} $$

$$ \sqrt{0.011\times0.989} = \sqrt{0.010879} = 0.10430 $$

$$ \text{UL} = 100\times0.40\times0.10430 = \textbf{₹4.17 crore} $$

Reconciliation / sanity check: UL (₹4.17 cr) is nearly **10×** the EL (₹0.44 cr). This is the whole point — the *unexpected* loss dwarfs the expected loss for a single risky loan, because the default event is binary and volatile. The bank prices for ₹0.44 cr but must hold capital sized off the ₹4.17 cr. Note the ratio $\text{UL}/\text{EL} = \sqrt{(1-\text{PD})/\text{PD}} = \sqrt{0.989/0.011} = 9.48$, and indeed $4.17/0.44 = 9.48$. ✓ Consistent.

Example 2 — Exposure at Default on a revolving credit line

A company has a **₹50 crore** revolving credit facility. It has currently drawn **₹20 crore**, leaving **₹30 crore** undrawn. Historical data shows distressed borrowers draw down **65%** of the remaining commitment before defaulting (CCF = 0.65). PD = 4.70% (B-rated), LGD = 50%.

EAD:

$$ \text{EAD} = \text{Drawn} + \text{CCF}\times(\text{Limit}-\text{Drawn}) = 20 + 0.65\times(50-20) = 20 + 0.65\times30 = 20 + 19.5 = \textbf{₹39.5 crore} $$

Expected Loss:

$$ \text{EL} = 0.047 \times 0.50 \times 39.5 = \textbf{₹0.928 crore} $$

Reconciliation: Had we naively used only the *drawn* ₹20 cr, EL would be $0.047\times0.50\times20 = ₹0.47$ cr — we would have **understated** the risk by nearly half. This is why CCF matters: borrowers rush to draw their lines precisely as they head toward default, so the exposure balloons exactly when it hurts. ✓ The undrawn commitment is a real, and dangerous, exposure.

Example 3 — Portfolio diversification and correlation

The bank holds **two** loans, each with UL = ₹4 crore standalone.

Case A — low correlation, $\rho = 0.20$:

$$ \text{UL}_P = \sqrt{4^2 + 4^2 + 2(0.20)(4)(4)} = \sqrt{16 + 16 + 6.4} = \sqrt{38.4} = \textbf{₹6.20 crore} $$

Case B — high correlation, $\rho = 0.80$ (e.g. both to the same sector):

$$ \text{UL}_P = \sqrt{16 + 16 + 2(0.80)(16)} = \sqrt{16 + 16 + 25.6} = \sqrt{57.6} = \textbf{₹7.59 crore} $$

Reconciliation against the two extremes:

- If **fully diversified** ($\rho = 0$): $\text{UL}_P = \sqrt{32} = ₹5.66$ cr.
- If **perfectly correlated** ($\rho = 1$): $\text{UL}_P = \sqrt{16+16+32} = \sqrt{64} = ₹8.00$ cr — exactly the naive sum $4+4$. ✓

Our two cases (₹6.20 cr and ₹7.59 cr) sit correctly *between* ₹5.66 cr and ₹8.00 cr, and the higher correlation gives the higher portfolio risk. The **diversification benefit** in Case A is $8.00 - 6.20 = ₹1.80$ cr of UL saved;

in Case B only $8.00 - 7.59 = ₹0.41$ cr. **Concentration destroys diversification** — putting both loans in one sector nearly wiped out the benefit. ✓

Example 4 — Market-implied PD from a bond spread

A corporate bond trades at a **credit spread of 300 bps** (3.0%) over the risk-free rate. Assume LGD = 60% (recovery 40%).

$$ \text{PD} \approx \frac{s}{\text{LGD}} = \frac{0.030}{0.60} = 0.05 = \textbf{5.0\%} $$

Reconciliation: The implied EL from the market equals $\text{PD}\times\text{LGD} = 0.05\times0.60 = 0.03 = 300$ bps — exactly the spread we started with. ✓ The market is pricing the bond to compensate purely for expected credit loss (ignoring risk premia and liquidity, which in reality push the *real-world* PD below this *risk-neutral* 5%). A 5% risk-neutral PD sits between a B and CCC grade — consistent with a 300 bps spread being genuinely junk-territory.

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6. Connections

- **To market risk (Ch. on VaR):** Credit VaR borrows the percentile concept from market VaR, but the loss distribution is skewed and fat-tailed rather than roughly normal, so a simple mean ± σ multiplier badly *understates* the tail. Credit needs full-distribution simulation (Monte Carlo, CreditMetrics).
- **To provisioning / accounting (IFRS 9, Ind AS 109):** Expected Credit Loss (ECL) accounting is literally EL = PD × LGD × EAD, but forward-looking and staged: 12-month ECL for performing loans (Stage 1), lifetime ECL once credit has significantly deteriorated (Stages 2 and 3). Directly relevant to any bank/NBFC finance role.
- **To regulatory capital (Basel III):** The IRB formula turns PD, LGD, EAD and correlation into a capital charge. Standardised approach uses fixed risk weights by rating instead.
- **To derivatives (counterparty credit risk):** For swaps and forwards, EAD is not fixed — it depends on future market moves. This gives **CVA** (Credit Valuation Adjustment), the market price of counterparty credit risk, a hybrid of credit and market risk.
- **To securitisation & CDOs:** Tranching a pool slices the loss distribution — equity tranche absorbs EL, senior tranches absorb only tail UL. The 2008 crisis was fundamentally a *correlation* mis-estimate in these structures.
- **To structural finance (Merton):** Distance-to-default links equity option pricing (Black-Scholes) directly to credit risk.

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7. Key Terms

Term	Meaning
PD	Probability of Default over a horizon (usually 1 year)
LGD	Loss Given Default = 1 − Recovery Rate; the fraction not recovered
EAD	Exposure at Default; amount outstanding when default occurs
CCF	Credit Conversion Factor; fraction of undrawn commitment drawn before default
EL	Expected Loss = PD × LGD × EAD; a budgeted cost, covered by provisions
UL	Unexpected Loss; volatility of loss around EL, covered by capital
Recovery Rate	Fraction of exposure recovered after default = 1 − LGD
Transition matrix	Probabilities of migrating between rating grades over a period
Credit VaR	A high percentile (e.g. 99.9%) of the credit loss distribution
Economic Capital	Credit VaR − EL; the unexpected-loss buffer
Default correlation (ρ)	Tendency of borrowers to default together; drives portfolio tail
RAROC	Risk-Adjusted Return on Capital; the loan-pricing / approval hurdle
CVA	Credit Valuation Adjustment; market value of counterparty credit risk
TTC / PIT	Through-the-cycle vs Point-in-time PD (Basel vs IFRS 9)
Distance-to-default	Number of std devs asset value sits above the default point (Merton)

8. Common Confusions

1. **"Expected loss is the risk."** No — EL is the *average* outcome, a predictable cost you price and provision for. The *risk* is Unexpected Loss, the deviation above the mean. Confusing the two leads to zero capital.
2. **LGD vs Recovery Rate direction.** LGD = 1 − Recovery. A 40% recovery means 60% LGD, *not* 40% LGD. Always check which one is given.
3. **"Provisions and capital are the same buffer."** They are separate and stacked: provisions sit against EL (they reduce book value / P&L), capital sits against UL (it is equity). Double-counting or conflating them mis-sizes the balance sheet.
4. **Ignoring the undrawn line.** Using drawn balance instead of EAD understates exposure on revolvers and guarantees — and borrowers draw *most* just before default (Example 2).
5. **Adding ULs like ELs.** ELs add arithmetically (₹1 + ₹1 = ₹2). ULs add like volatilities with correlation — they are always *less than* the arithmetic sum unless ρ = 1. Adding ULs directly overstates required capital and ignores diversification.
6. **Risk-neutral vs real-world PD.** PD backed out of bond/CDS spreads is *risk-neutral* — it embeds a risk premium and is typically **higher** than the actual (physical) default frequency. Do not feed a spread-implied PD into an accounting ECL that wants real-world PD.

7. **"Diversification removes credit risk."** It removes *idiosyncratic* risk only. Systematic (correlated) risk survives no matter how many names you add — this residual is exactly what portfolio capital is sized for.
8. **Assuming normality for the tail.** Credit losses are right-skewed; the 99.9% loss is many multiples of the standard deviation, not ~3σ as a normal would suggest.

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9. Recap

- Credit risk is the risk a borrower or counterparty fails to pay — asymmetric (bounded gain, total-loss downside), skewed, and correlation-driven.
- The master identity is **EL = PD × LGD × EAD**. Expected Loss is a *cost*: priced into spreads and covered by **provisions**.
- **Unexpected Loss** — the volatility of loss around EL — is the true risk and is covered by **capital**. For a single risky loan, UL is typically an order of magnitude larger than EL.
- **EAD** is not just the drawn amount: undrawn commitments convert via **CCF** because distressed borrowers draw down.
- **Ratings** bucket borrowers into PD grades; the **transition matrix** captures both default and downgrade (migration) risk, and multi-year PDs come from powering the matrix.
- **Portfolio** risk is dominated by **default correlation**: ULs combine like volatilities, so diversification cuts the tail but concentration destroys the benefit. Systematic risk cannot be diversified away.
- Credit risk is **measured** via internal ratings, structural (Merton) models, and market spreads, and **priced** via RAROC and the credit spread ≈ PD × LGD relationship.
- **Credit VaR** is a high percentile of the loss distribution; **Economic Capital = Credit VaR − EL**.

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10. Quick Reference / Interview Points

Must-know formulas:

Quantity	Formula
Expected Loss	$\text{EL} = \text{PD}\times\text{LGD}\times\text{EAD}$
LGD	$\text{LGD} = 1 - \text{Recovery Rate}$
EAD (revolver)	$\text{Drawn} + \text{CCF}\times(\text{Limit}-\text{Drawn})$
Unexpected Loss (single)	$\text{EAD}\times\text{LGD}\times\sqrt{\text{PD}(1-\text{PD})}$
Portfolio UL (2 assets)	$\sqrt{\text{UL}_1^2+\text{UL}_2^2+2\rho\,\text{UL}_1\text{UL}_2}$
Implied PD	$\text{PD} \approx s / \text{LGD}$ (spread over LGD)
Economic Capital	$\text{Credit VaR}_{\alpha} - \text{EL}$
Merton PD	$N(-DD)$, distance-to-default DD

One-liners that land in interviews:

- "Expected loss is a cost you *price*; unexpected loss is a risk you hold *capital* against."
- "Provisions cover the mean, capital covers the tail — never confuse them."

- "The only free lunch in credit is diversification, and concentration is the standard way banks die."
- "Correlation, not individual PDs, sets the portfolio capital number."
- "ELs add; ULs combine like volatilities."
- "Spread-implied PD is risk-neutral and higher than actual default frequency."
- "On a revolver, the borrower draws the line down as they fail — that's why CCF and EAD matter."
- "A loan makes sense only if its RAROC clears the cost of equity."

Typical interview drills: compute EL given PD/LGD/EAD; convert recovery to LGD; size EAD on a partly-drawn line; explain why UL $\gg$ EL; explain how correlation moves portfolio UL between the diversified and perfectly-correlated bounds; read a PD off a transition matrix and compute a 2-year PD via M^2; back out PD from a bond spread; distinguish IFRS 9 ECL staging from Basel IRB capital.

Red flags interviewers listen for: treating EL as the risk, adding ULs arithmetically, forgetting the undrawn commitment, assuming normal tails, and confusing risk-neutral with real-world PD. Avoid these and you signal genuine credit-risk fluency.

Q&A — Credit Risk

A practice bank for the Credit Risk chapter. Work each question before reading the answer. Numerical answers are self-checked against the master identities EL = PD × LGD × EAD and UL = EAD × LGD × √(PD(1−PD)).

Section A — Concept-Check (short answer)

A1. State the master identity of credit risk and define each term.

Expected Loss: **EL = PD × LGD × EAD**. - **PD** = Probability of Default over a horizon (usually 1 year), between 0 and 1. - **LGD** = Loss Given Default = 1 − Recovery Rate; the fraction of exposure not recovered. - **EAD** = Exposure at Default; the money amount outstanding when default occurs. The three factors compound because each answers a logically independent question: *how likely* is default, *how much* is at risk when it happens, and *how severe* is the loss on that amount.

A2. Why is Expected Loss described as "a cost, not a risk"?

Because EL is the *average* outcome, which is predictable and repeatable. A bank that expects to lose 1.5% of its book each year is not being surprised — it should price that 1.5% into its lending rate and set it aside as a provision. The genuine risk is the *deviation above* that mean (Unexpected Loss), i.e. the chance that this year's losses come in far worse than the long-run average. Treating EL as "the risk" leads to holding zero capital for the part that actually threatens solvency.

A3. What does Unexpected Loss (UL) represent and what covers it?

UL is the volatility (standard deviation) of losses around the expected mean — the possibility that realised losses exceed EL. It is covered by **capital** (equity), whereas EL is covered by **provisions**. The division of labour: *provisions cover the mean, capital covers the tail.*

A4. Why do defaults "bunching together" matter more than individual default odds?

A single borrower defaulting is idiosyncratic and diversifiable. Thousands defaulting at once (a recession, a sector collapse) is systematic and is what generates the fat right tail of the loss distribution. Portfolio capital is sized off this correlated tail, so **default correlation — not the individual PDs — sets the capital number.** Idiosyncratic risk can be diversified away; systematic risk cannot.

A5. Distinguish LGD from Recovery Rate, and give the trap.

LGD = 1 − Recovery Rate. If a defaulted loan recovers 40 cents on the rupee, the recovery rate is 40% and **LGD is 60%, not 40%.** Always check which figure the question gives you before plugging in.

A6. What is a Credit Conversion Factor (CCF) and why does it matter?

CCF is the fraction of an *undrawn* commitment expected to be drawn before default. It matters because distressed borrowers draw down their credit lines precisely as they head toward default, so exposure balloons exactly when it hurts. EAD on a revolver = Drawn + CCF × (Limit − Drawn). Ignoring it and using only the drawn balance systematically understates exposure.

A7. What is a rating transition matrix, and how do you get a multi-year PD from it?

A transition (migration) matrix gives the probability of moving from one rating grade to another over one year; rows are the start grade, columns the end grade, the last column is default (D), and each row sums to

100%. The default column gives the 1-year PD per grade. Assuming a time-homogeneous Markov chain, the **n-year matrix is M^n** — the 2-year matrix is M^2, and the 2-year cumulative PD for a grade is read from the D column of M^2.

A8. Distinguish through-the-cycle (TTC) from point-in-time (PIT) PD.

TTC PD is a long-run average that smooths over booms and busts; it is used for Basel regulatory capital and rating-agency grades. PIT PD reflects current conditions and rises in recessions; it is used for IFRS 9 / Ind AS 109 expected credit loss accounting. TTC is stable; PIT is cyclical.

A9. Why is spread-implied PD "risk-neutral" and how does it compare to the actual default frequency?

PD backed out of a bond or CDS spread embeds a risk premium and a liquidity premium on top of pure expected loss, so it is a *risk-neutral* probability that is typically **higher** than the physical (real-world) default frequency. You must not feed a spread-implied PD into an accounting ECL model that wants a real-world PD.

A10. What is Economic Capital in terms of Credit VaR?

Economic Capital = Credit VaR(α) − EL. Credit VaR is a high percentile (Basel uses 99.9%) of the loss distribution; we subtract EL because provisions already cover the mean, so capital only needs to bridge from the mean out to the chosen confidence level.

Section B — Numerical / Applied (full solutions)

B1. Basic Expected Loss. A bank lends ₹80 crore, fully drawn, to a borrower with 1-year PD = 2%. The loan is secured with an expected recovery of 55%. Find EL.

LGD = 1 − 0.55 = 0.45. EAD = ₹80 cr (fully drawn). EL = PD × LGD × EAD = 0.02 × 0.45 × 80 = **₹0.72 crore** (₹72 lakh). *Check:* 0.02 × 0.45 = 0.009; 0.009 × 80 = 0.72. ✓ The bank should provision ₹72 lakh and build 0.9% into the spread to break even on expected credit cost.

B2. Unexpected Loss (default-event only). For the loan in B1, compute UL using UL = EAD × LGD × $\sqrt{PD(1-PD)}$.

$\sqrt{0.02 \times 0.98} = \sqrt{0.0196} = 0.14$. UL = 80 × 0.45 × 0.14 = **₹5.04 crore**. *Check via the ratio identity:* UL/EL = $\sqrt{(1-PD)/PD} = \sqrt{0.98/0.02} = \sqrt{49} = 7.0$, so UL = 7.0 × 0.72 = 5.04. ✓ UL is 7× the EL — the unexpected loss dwarfs the expected loss because the default event is binary and volatile.

B3. UL with recovery volatility. Same loan (EAD 80, PD 0.02, LGD 0.45) but now the standard deviation of LGD is σ_{LGD} = 0.25. Use the fuller formula UL = EAD × $\sqrt{PD \cdot \sigma_{LGD}^2 + LGD^2 \cdot PD(1-PD)}$.

Inside the root: $PD \cdot \sigma_{LGD}^2$ = 0.02 × 0.0625 = 0.00125; $LGD^2 \cdot PD(1-PD)$ = 0.2025 × 0.0196 = 0.0039690. Sum = 0.0052190. $\sqrt{0.0052190}$ = 0.07224. UL = 80 × 0.07224 = **₹5.78 crore**. *Check:* adding recovery-rate variance raised UL from ₹5.04 cr to ₹5.78 cr — correct direction, since uncertainty about *how much* you recover adds to the risk on top of the binary default event. ✓

B4. EAD on a revolver. A firm has a ₹60 crore revolving facility, ₹25 crore drawn. Distressed borrowers historically draw 70% of the remaining commitment (CCF = 0.70). PD = 3%, LGD = 50%. Find EAD and EL, and compare against naively using the drawn balance.

Undrawn = 60 − 25 = 35. EAD = 25 + 0.70 × 35 = 25 + 24.5 = **₹49.5 crore**. EL = 0.03 × 0.50 × 49.5 = **₹0.7425 crore** (≈ ₹74.25 lakh). *Naive (drawn only):* EL = 0.03 × 0.50 × 25 = ₹0.375 cr. Using the drawn balance would understate the risk by about half. ✓ The undrawn commitment is a real, dangerous exposure.

B5. Portfolio UL and diversification. Two loans have standalone UL_1 = ₹3 cr and UL_2 = ₹5 cr. Compute portfolio UL for (a) $\rho = 0$, (b) $\rho = 0.30$, (c) $\rho = 1$, and state the diversification benefit in case (b).

Formula: $UL\_P = \sqrt{UL_1^2 + UL_2^2 + 2\rho \cdot UL_1 \cdot UL_2}$. $UL_1^2 = 9$, $UL_2^2 = 25$, $2 \cdot UL_1 \cdot UL_2 = 30$. - (a) $\rho=0$: $\sqrt{9 + 25 + 0} = \sqrt{34}$ = **₹5.83 cr**. - (b) $\rho=0.30$: $\sqrt{34 + 0.30 \times 30} = \sqrt{34 + 9} = \sqrt{43}$ = **₹6.56 cr**. - (c) $\rho=1$: $\sqrt{34 + 30} = \sqrt{64}$ = **₹8.00 cr** = the arithmetic sum 3+5. ✓ Diversification benefit in (b) = arithmetic sum − portfolio UL = 8.00 − 6.56 = **₹1.44 cr** of UL saved. All three cases sit correctly between the fully diversified ₹5.83 cr and the perfectly correlated ₹8.00 cr, rising with ρ. ✓

B6. Market-implied PD from a spread. A corporate bond trades at a 250 bps spread over the risk-free rate; assume LGD = 50%. Estimate the risk-neutral PD, then verify.

PD ≈ s / LGD = 0.025 / 0.50 = **0.05 = 5.0%**. *Check:* implied EL = PD × LGD = 0.05 × 0.50 = 0.025 = 250 bps = the spread we started with. ✓ The market is pricing the bond to compensate for expected credit loss (ignoring risk/liquidity premia, which push the real-world PD below this 5%).

B7. Two-year cumulative PD from a survival argument. A BB-rated name has a 1-year marginal PD of 1.10% in year 1 and a (conditional) 1.30% in year 2. Compute the 2-year cumulative PD.

Survive year 1: 1 − 0.011 = 0.989. Survive year 2 given survival: 1 − 0.013 = 0.987. Two-year survival = 0.989 × 0.987 = 0.976143. Cumulative 2-year PD = 1 − 0.976143 = **0.023857 ≈ 2.39%**. *Check:* it is slightly less than the naive sum 1.10% + 1.30% = 2.40%, because you can only default in year 2 if you survived year 1. The tiny gap (0.01%) equals 0.011 × 0.013. ✓

B8. Economic Capital. A portfolio has EL = ₹12 crore and a 99.9% Credit VaR of ₹85 crore. What economic capital is required?

Economic Capital = Credit VaR − EL = 85 − 12 = **₹73 crore**. Provisions of ₹12 cr already cover the mean; the ₹73 cr of capital covers the unexpected loss out to the 99.9% confidence level.

B9. RAROC loan-pricing decision. A ₹100 cr loan earns spread income + fees of ₹3.0 cr per year, has EL of ₹0.5 cr and operating cost of ₹0.3 cr. Economic capital held is ₹15 cr, and the bank's hurdle (cost of equity) is 14%. Should the loan be approved?

RAROC = (Spread + Fees − EL − Operating cost) / Economic Capital = (3.0 − 0.5 − 0.3) / 15 = 2.2 / 15 = **14.67%**. Since 14.67% > 14% hurdle, the loan clears the cost of equity and should be **approved** (marginally). *Check:* had economic capital been ₹16 cr, RAROC = 2.2/16 = 13.75% < 14% and it would be rejected — showing how sensitive the decision is to capital consumption. ✓

B10. Provision vs capital split. A bank's loan book has aggregate EL = ₹40 cr and UL = ₹120 cr, and it holds economic capital equal to 4 × UL. State the provision, the capital, and why they are not double-counted.

Provisions = EL = **₹40 cr** (against the mean, a P&L charge). Capital = 4 × 120 = **₹480 cr** (equity, against the tail). They are separate, stacked buffers: provisions reduce book value for the expected cost; capital absorbs the deviation above it. Conflating them would mis-size the balance sheet — you need *both*.

Section C — Interview-Style (model answers)

C1. "Walk me through what happens to expected loss versus capital when a loan's PD doubles."

EL is linear in PD, so doubling PD roughly **doubles EL** — a proportional rise in the cost you price and provision for. Capital is driven by *unexpected* loss and the conditional stressed PD, which scale with $\sqrt{PD(1-PD)}$ at the single-name level and, more importantly, with the *correlated tail* at the portfolio level —

so capital rises too, but not linearly. The more damaging effect is that a broad PD increase usually coincides with rising correlation in a downturn, so the tail fattens faster than the mean. Headline: EL responds proportionally and predictably; capital responds through volatility and correlation, which is why a stress event hurts capital far more than a simple doubling of the average suggests.

C2. "Why can't I just add up the unexpected losses of my loans the way I add expected losses?"

Because expectations always add regardless of correlation — $E[A+B] = E[A]+E[B]$ — so portfolio EL is a clean sum. But UL is a standard deviation, and standard deviations combine like volatilities: $UL\_P = \sqrt{UL_1^2 + UL_2^2 + 2\rho \cdot UL_1 \cdot UL_2}$. Unless $\rho = 1$, the portfolio UL is strictly *less than* the arithmetic sum, and the gap is the diversification benefit. Adding ULs directly would overstate required capital and completely ignore the only free lunch in credit. Concretely, two ₹4 cr ULs at $\rho = 0$ combine to $\sqrt{32}$ = ₹5.66 cr, not ₹8 cr.

C3. "A borrower has an undrawn line. Our system books zero exposure until they draw. What's wrong with that?"

It ignores the single most predictable behaviour in credit distress: borrowers draw down their committed lines as they slide toward default, because a committed facility is contractual liquidity they can't be denied. So exposure is smallest when the borrower is healthy and largest at the moment of default — exactly the wrong correlation. We model this with a Credit Conversion Factor: EAD = Drawn + CCF × (Limit − Drawn). A facility that looks like zero exposure today can convert to tens of crores at default. Booking zero until drawdown means we hold zero provision and zero capital against a real, and adversely-timed, exposure.

C4. "Explain the difference between the PD you'd read off a CDS spread and the PD you'd use for IFRS 9 provisioning."

The CDS/bond-implied PD is *risk-neutral*: it's extracted from prices, so it bundles the pure default probability with a risk premium (investors demand extra return to bear default risk) and a liquidity premium. That makes it systematically higher than the actual physical default frequency. IFRS 9 expected credit loss wants a *real-world, point-in-time* PD — your best forecast of what will actually happen, forward-looking and conditioned on the current cycle. Feeding a risk-neutral spread-implied PD into an ECL model would overstate provisions, and mixing the two is a classic error. Rule of thumb: risk-neutral PD for pricing and hedging, real-world PD for accounting and economic capital.

C5. "Our whole loan book is in one booming sector and it's performing beautifully. Why are you worried?"

Because performance today tells you about the *mean* (low realised EL in a good year), not the *tail*. Concentrating in one sector means borrowers share a common systematic factor — the sector's fortunes — so default correlation is high, and high correlation is exactly what fattens the loss tail: in the good state everyone pays, but in the bad state they default together and wipe out years of spread at once. Diversification removes idiosyncratic risk; concentration leaves the systematic risk fully intact. A book that looks safe on average can be lethal on its tail — concentration is the standard way banks die (IL&FS, DHFL). I'd want single-name and sector limits and a sector-wide stress test, not comfort from clean current performance.

C6. "What is the Merton model intuition, in one minute?"

Treat a firm's equity as a call option on its assets: shareholders get whatever is left after debt is repaid, and their downside is capped at zero (limited liability). The firm defaults if asset value V falls below the debt face value F at maturity. The key output is distance-to-default — how many standard deviations of asset value the firm sits above its default point: $DD = [\ln(V/F) + (\mu - ½\sigma^2)T] / (\sigma\sqrt{T})$, and $PD = N(-DD)$. It links equity-market information (asset value and volatility, inferred via Black-Scholes) directly to credit risk, and it's the engine behind Moody's KMV EDF. The practical appeal: it updates in real time from the stock price rather than waiting for a rating action.

C7. "Provisions and capital — aren't they just two names for the same rainy-day fund?"

No — they sit against different parts of the loss distribution and come from different pools. Provisions are an accounting charge against Expected Loss, the mean of the distribution; they reduce reported earnings and carrying value for a cost you already expect. Capital is shareholders' equity held against Unexpected Loss, the deviation above the mean out to a high confidence level; it exists to absorb the bad years. They stack: EL first (provisions), then UL (capital), then the extreme stress zone beyond capital where banks actually fail. Conflating them either under-provisions (you carry loans at inflated value) or under-capitalises (you have no buffer for the tail).

Section D — MCQs (with reasoning)

D1. A loan has PD = 4%, recovery rate = 70%, EAD = ₹50 cr. Expected Loss is: (a) ₹1.40 cr (b) ₹0.60 cr (c) ₹1.00 cr (d) ₹2.00 cr

Answer: (b). LGD = 1 − 0.70 = 0.30. EL = 0.04 × 0.30 × 50 = ₹0.60 cr. The trap is (a), which wrongly uses LGD = 0.70 (the recovery, not the loss): 0.04 × 0.70 × 50 = ₹1.40 cr. Always convert recovery to LGD first.

D2. Which is covered by *capital* rather than *provisions*? (a) Expected Loss (b) Unexpected Loss (c) Operating cost (d) Funding cost

Answer: (b). Provisions cover the mean (EL); capital covers the volatility around the mean (UL). Operating and funding costs are ordinary expenses recovered in the loan's pricing, not risk buffers.

D3. Two loans each have standalone UL = ₹6 cr. At a default correlation of ρ = 1, the portfolio UL is: (a) ₹6 cr (b) ₹8.49 cr (c) ₹12 cr (d) ₹0 cr

Answer: (c). At ρ = 1, √(36 + 36 + 2×1×36) = √144 = ₹12 cr, which equals the arithmetic sum 6 + 6 — perfect correlation gives zero diversification benefit. Option (b) √72 = 8.49 is the ρ = 0 case; the maximum diversification, not the minimum.

D4. A revolving facility has limit ₹40 cr, drawn ₹10 cr, CCF = 0.50. EAD is: (a) ₹10 cr (b) ₹25 cr (c) ₹40 cr (d) ₹30 cr

Answer: (b). EAD = 10 + 0.50 × (40 − 10) = 10 + 15 = ₹25 cr. Option (a) ignores the undrawn commitment; (c) wrongly assumes the full line is drawn (CCF = 1).

D5. A bond's credit spread is 180 bps and assumed LGD is 45%. The approximate risk-neutral PD is: (a) 2.0% (b) 4.0% (c) 8.1% (d) 0.81%

Answer: (b). PD ≈ s / LGD = 0.018 / 0.45 = 0.04 = 4.0%. Check: PD × LGD = 0.04 × 0.45 = 0.018 = 180 bps, recovering the spread. Option (c) multiplies instead of dividing.

D6. In a rating transition matrix, the 2-year cumulative default probabilities are obtained from: (a) 2M (b) M^2 (c) M/2 (d) the last row of M

Answer: (b). For a time-homogeneous Markov chain the n-year matrix is M^n, so the 2-year matrix is M^2 and 2-year cumulative PDs are read from its default column. Option (a) 2M is not even a valid probability matrix (rows wouldn't sum to 1).

D7. Which PD is appropriate for Basel regulatory capital? (a) Point-in-time (b) Risk-neutral (c) Through-the-cycle (d) Spread-implied

Answer: (c). Basel regulatory capital uses through-the-cycle PD (a stable long-run average). Point-in-time is for IFRS 9 ECL; risk-neutral/spread-implied PDs are for pricing and hedging, not capital.

D8. Economic capital is best defined as: (a) Credit VaR (b) Expected Loss (c) Credit VaR − Expected Loss (d) Unexpected Loss × PD

Answer: (c). Economic Capital = Credit VaR(α) − EL, because provisions already absorb EL and capital need only bridge from the mean out to the confidence level. Option (a) double-counts the EL that provisions already cover.

D9. Which statement about diversification in a credit portfolio is TRUE? (a) It removes all credit risk (b) It removes only idiosyncratic risk, leaving systematic risk (c) It has no effect when $\rho < 1$ (d) It increases the tail loss

Answer: (b). Diversification cancels name-specific (idiosyncratic) risk but leaves the common systematic factor untouched — that residual correlated risk is exactly what portfolio capital is sized against. (a) overclaims; (c) is backwards (diversification helps most when $\rho < 1$); (d) is the opposite of the truth.

D10. For a single risky loan with a low PD, the ratio UL/EL is approximately: (a) always 1 (b) $\sqrt{(1-PD)/PD}$, which is large for small PD (c) PD × LGD (d) always less than 1

Answer: (b). Using $EL = PD \cdot LGD \cdot EAD$ and $UL = EAD \cdot LGD \cdot \sqrt{PD(1-PD)}$, the LGD and EAD cancel: $UL/EL = \sqrt{PD(1-PD)} / PD = \sqrt{(1-PD)/PD}$. For small PD this is large (e.g. PD = 1% gives $\sqrt{99} \approx 9.95$), which is why UL dwarfs EL for a single low-PD loan. This is the quantitative core of "expected loss is a cost, unexpected loss is the risk."

End of Q&A — Credit Risk.

Chapter 07 — Counterparty Credit Risk

1. The Problem / The Need

When you buy a bond, the credit question is simple: will the issuer pay you back the fixed amount it owes? The exposure is known in advance — you lent $100, you are owed $100 plus coupons. Counterparty credit risk (CCR) is a stranger, more slippery animal, and it lives inside the world of **over-the-counter (OTC) derivatives** — interest rate swaps, FX forwards, cross-currency swaps, credit default swaps, commodity options.

Here is the twist that makes CCR different from ordinary lending risk. In a derivative, **you do not know in advance whether you will owe your counterparty or your counterparty will owe you**, and you do not know by how much. A 10-year interest rate swap starts life worth roughly zero to both sides. Six months later, if rates have moved in your favour, the swap might be worth +$4 million to you — meaning your counterparty owes you $4 million if they were to close out today. If that counterparty defaults at that moment, you lose that $4 million of positive market value. But if rates had moved the other way, the swap would be a liability to you, the counterparty's default would cost you nothing, and you might even benefit.

So CCR has three features that plain loan credit risk does not:

1. **Bilateral and uncertain direction.** Either party can end up the creditor. The exposure can flip sign over the life of the trade.
2. **Stochastic, market-driven exposure.** The amount at risk is not the notional and not a fixed principal — it is the *replacement cost* of the contract, which follows market rates and prices and therefore changes every day.
3. **A long, uncertain horizon.** A swap can run 5, 10, 30 years. You are exposed to your counterparty's default for that entire period, and the exposure profile evolves.

The need, therefore, is for a discipline that answers: *If my counterparty vanishes tomorrow — or in three years — what will it cost me to replace these contracts, and how do I price, limit, collateralise, and hedge that risk today?* This chapter builds that discipline: current and potential future exposure, credit valuation adjustment (CVA), netting and collateral agreements, wrong-way risk, and the market's structural answer — central clearing.

Why it matters for a risk career: CCR sits at the intersection of the trading desk, the credit function, and the regulator. The 2008 crisis was in large part a counterparty-risk crisis — Lehman's default, the near-collapse of AIG (which had written CDS protection it could not honour), and the freezing of the interbank market. Post-crisis rules (Basel III's CVA capital charge, mandatory clearing, margin rules for non-cleared derivatives) all descend from this chapter. Interviewers for credit-risk, XVA, and CCR roles expect fluency here.

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2. The Core Idea

Counterparty credit risk is the risk that the counterparty to a bilateral contract defaults *before the final settlement* of the contract's cash flows, at a time when the contract has *positive economic value to you.* Your loss is the cost of replacing the defaulted contract in the market — the replacement cost — reduced by whatever you recover from the defaulted estate.

Three ideas anchor everything that follows.

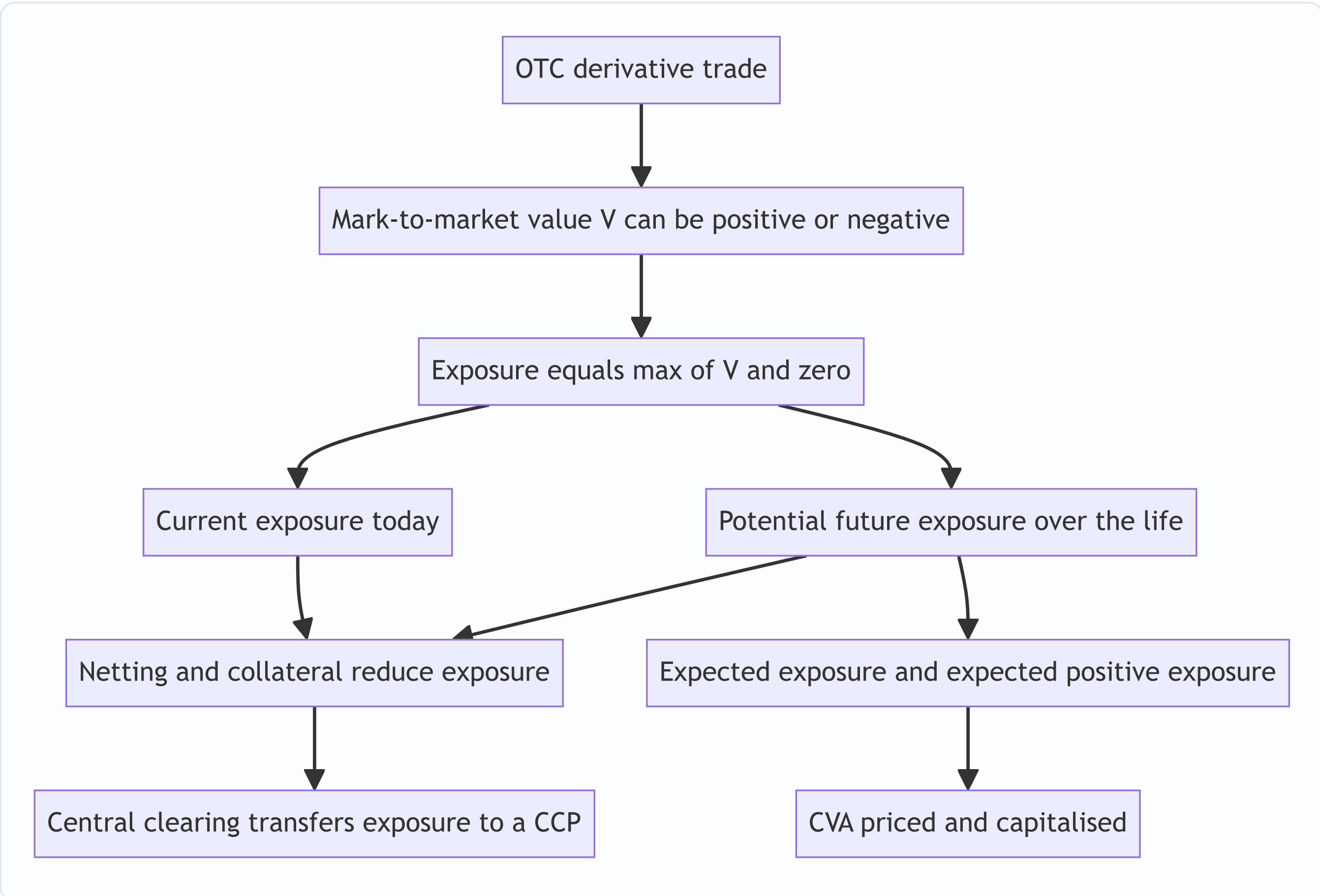
Exposure is one-sided at the moment of default, even though value is two-sided. If the counterparty owes you (contract value positive), you lose. If you owe them (contract value negative), you must still pay — default does not release you from your obligation. So exposure is `max(V, 0)` — the positive part of the mark-to-market value. Negative value is not a gain on default; it is simply zero exposure.

Because value is random, exposure is a distribution, not a number. We describe it with a *profile over time*: today's exposure (current exposure) and the range of possible future exposures (potential future exposure). We summarise the distribution with expected exposure and high-percentile exposure.

The price of bearing this risk is CVA — the market value of counterparty default risk. A derivative traded with a risky counterparty is worth less than the same derivative with a default-free counterparty. That difference is CVA, and it is booked as a valuation adjustment (a real P&L item that moves as credit spreads move).

The rest of the toolkit — netting, collateral, central clearing — exists to *shrink the exposure profile* and therefore shrink both the potential loss and the CVA.

Figure 1 — The counterparty-risk value chain from a single trade to the summary risk measures.



3. Why / How It Works

Why exposure is the positive part of value

Consider a single swap with your counterparty. Its mark-to-market value to you is `V`. If the counterparty defaults, one of two things is true:

- $V > 0$: the counterparty owes you V. In default you file a claim for V and recover only a fraction R (the recovery rate). You must replace the swap at market — costing you V — but you recover $R \cdot V$ from the estate. Net loss $= (1 - R) \cdot V$.
- $V < 0$: you owe the counterparty $|V|$. Default does not cancel your debt; the administrator will demand you settle $|V|$. You gain nothing. Exposure = 0.

Hence **exposure** $E = \max(V, 0)$, and loss given default on the derivative $= (1 - R) \cdot \max(V, 0)$. This asymmetry — you keep the downside, you lose the upside — is the mathematical heart of CCR and is exactly why an option-like (convex) shape appears in every exposure measure.

Why future exposure grows then shrinks: the diffusion-vs-amortisation tug-of-war

Potential future exposure over the life of a trade is shaped by two opposing forces:

- **Diffusion effect.** As time passes, market rates can wander further from today's levels, so the *range* of possible future values widens. This pushes potential exposure *up* with the square root of time (like any random walk).
- **Amortisation / roll-off effect.** As the trade approaches maturity, fewer cash flows remain to be exchanged, so there is less left to be at risk. This pulls potential exposure *down* toward zero at maturity.

For an interest rate swap, diffusion dominates early and amortisation dominates late, producing the classic **hump-shaped** exposure profile that peaks around one-third to one-half of the way to maturity. For an FX forward, there is a single exchange at maturity and no amortisation, so exposure rises monotonically to the end. Understanding the *shape* of the profile per product is a standard interview probe.

Why netting works

If you have many trades with one counterparty and a legally enforceable **netting agreement** (an ISDA Master Agreement), then on default all trades collapse into a single net claim. Your exposure is $\max(\Sigma V_i, 0)$ — the positive part of the *sum* — not $\Sigma \max(V_i, 0)$ — the sum of the positive parts. Because some trades are in-the-money and some out-of-the-money, offsetting reduces the net figure. Netting can cut gross exposure by 60–90% for a large, diversified book. This is the single most powerful exposure-reduction tool.

Why collateral works

A **Credit Support Annex (CSA)** requires the out-of-the-money party to post collateral (usually cash or high-quality bonds) as the net value moves. If the counterparty owes you $10 million and has posted $9 million of collateral, your uncollateralised exposure is only $1 million. Collateral converts a large, slow credit exposure into a small residual — the residual being whatever can accumulate during the **margin period of risk** (the gap between the last margin call the counterparty honoured and the moment you finally close out and re-hedge, typically 10 business days for non-cleared trades).

Why central clearing works

A **central counterparty (CCP)** interposes itself between the two original parties via *novation*: one bilateral trade becomes two trades, each facing the CCP. The CCP guarantees performance to both sides. It manages the risk through *multilateral netting* across all members, *strict daily (or intraday) variation margin*, *risk-sensitive initial margin*, and a *default waterfall* (the defaulter's margin, then a mutualised guarantee fund, then the CCP's own capital). This replaces a web of bilateral exposures with a hub-and-spoke structure and mutualises tail losses.

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4. Full Content — Framework, Formulas, Methods

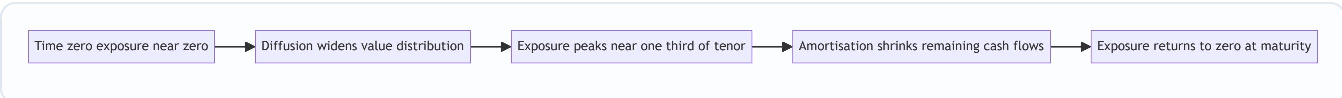
4.1 The exposure metrics (the vocabulary of CCR)

Let V_t be the (netted) mark-to-market value at future time t. Exposure $E_t = \max(V_t, 0)$.

Metric	Definition	Formula	Use
Current Exposure (CE)	Exposure right now	$\max(V_0, 0)$	Today's loss if default now
Expected Exposure (EE)	Mean exposure at future time t	$EE_t = E[\max(V_t, 0)]$	Input to CVA
Potential Future Exposure (PFE)	High-percentile exposure at t (e.g. 95th/99th)	$PFE_t(\alpha) = q\alpha(\max(V_t,0))$	Limits, worst-case
Expected Positive Exposure (EPE)	Time-average of EE over the horizon	$EPE = (1/T)\int_0^T EE_t\, dt$	Regulatory "loan-equivalent"
Effective EE	Non-decreasing running max of EE	$EEE_t = \max(EEE_{t-1}, EE_t)$	Removes roll-off understatement
Effective EPE	Time-average of Effective EE	$EEPE = (1/T)\int_0^T EEE_t\, dt$	Basel exposure-at-default driver
Maximum PFE (peak)	Highest PFE across all t	max over t of PFE_t	Credit-line sizing

Key distinctions to keep straight: - **EE vs PFE**: EE is the *mean* of the exposure distribution; PFE is a *high quantile*. PFE ≥ EE always. - **PFE vs VaR**: identical mathematics (a high quantile of a loss/exposure distribution) but PFE looks *forward over the life of the trade* at many horizons, whereas market-risk VaR looks at a single short horizon (e.g. 1–10 days). - **Effective EE** exists because plain EE can decline as a trade rolls off, which would understate exposure of a book you intend to roll/replace; the running-maximum fix keeps it non-decreasing.

Figure 2 — Hump-shaped exposure profile of an interest-rate swap with EE and PFE.



4.2 Computing the exposure profile

The standard method is **Monte Carlo simulation**:

1. Choose risk-factor models (e.g. Hull–White for rates, GBM for FX/equity).
2. Simulate thousands of scenario paths of the risk factors out to the longest maturity, on a time grid $t_1, t_2, \ldots, T$.
3. At each grid date on each path, **reprice every trade** and apply netting and collateral rules to get the netted, collateralised value V_t.
4. Take $E_t = \max(V_t, 0)$ on each path.
5. Across paths at each t: the *mean* gives EE_t; the chosen *quantile* gives PFE_t.

This is computationally heavy — a bank may reprice millions of trade-scenario combinations nightly — which is why exposure engines and XVA desks are large infrastructure investments.

A simpler regulatory shortcut, the **current exposure method** and its successor **SA-CCR (Standardised Approach for Counterparty Credit Risk)**, approximate exposure without full simulation:

```
Exposure at Default (EAD) = alpha × (Replacement Cost + Potential Future Exposure add-on)
```

with `alpha = 1.4`. Replacement Cost captures current net value less collateral; the PFE add-on applies supervisory factors to notionals by asset class, with recognition for netting and margining. SA-CCR is the number that feeds counterparty capital for banks not using internal models.

4.3 Credit Valuation Adjustment (CVA)

CVA is the **market value of expected loss from counterparty default**. It reduces the value of a derivative from its risk-free value:

```
Risky value = Risk-free value − CVA
```

The standard (unilateral) formula discretises the life of the trade into intervals and sums, over each interval, the discounted expected loss:

```
CVA = LGD × Σ [ EE*(tᵢ) × discount(tᵢ) × PD(tᵢ₋₁, tᵢ) ]
```

where - `LGD = 1 − R` is loss given default (e.g. 0.6 for R = 40%), - `EE*(tᵢ)` is the **discounted expected exposure** at `tᵢ` (risk-neutral, and conditioned on default at `tᵢ` if wrong-way effects are modelled), - `PD(tᵢ₋₁, tᵢ)` is the counterparty's **marginal (risk-neutral) probability of default** in that interval, backed out from its **CDS spread** curve.

A useful closed-form approximation when exposure and spread are independent:

```
CVA ≈ LGD × Σ EE(tᵢ) × discount(tᵢ) × [ S(tᵢ₋₁) − S(tᵢ) ]
```

where `S(t)` is the survival probability to time `t`, so `S(tᵢ₋₁) − S(tᵢ)` is the probability of defaulting in interval `i`. Survival is stripped from CDS: `S(t) ≈ exp(−spread × t / LGD)`.

Extensions: - **DVA (Debit Valuation Adjustment)**: the mirror image — the benefit to *you* of *your own* possible default. `BCVA = −CVA + DVA` gives bilateral CVA. DVA is controversial: your credit deteriorating produces a *gain*, which is counter-intuitive and hard to monetise. - **FVA, MVA, KVA**: funding, margin, and capital valuation adjustments — the wider "XVA" family. Beyond scope here but worth naming in interviews.

CVA is not static — as the counterparty's CDS spread widens, CVA increases and the bank books a *CVA loss* even with no default. During 2008 roughly two-thirds of CCR losses came from CVA mark-to-market moves, not actual defaults, which is why Basel III introduced a dedicated **CVA capital charge**.

4.4 Netting agreements

Under an **ISDA Master Agreement** with a netting schedule, all trades in the netting set close out into one net amount. Define:

```
Net exposure     = max( Σ Vᵢ , 0 )
Gross exposure   = Σ max( Vᵢ , 0 )
Netting benefit  = Gross − Net
Netting factor   = Net exposure / Gross exposure   (0 = perfect offset, 1 = no benefit)
```

Netting is only recognised where it is **legally enforceable** in the counterparty's jurisdiction — close-out netting opinions are a real operational dependency.

4.5 Collateral and the CSA

A **Credit Support Annex** sets the margining terms:

CSA term	Meaning
Threshold	Unsecured amount tolerated before any collateral is called
Minimum Transfer Amount (MTA)	Smallest transfer allowed, to avoid tiny operational calls
Independent Amount / Initial Margin	Extra collateral held regardless of MTM to cover gap risk
Variation Margin (VM)	Collateral tracking day-to-day change in net MTM
Eligible collateral & haircuts	What can be posted and the discount applied
Margin Period of Risk (MPoR)	Time from last honoured call to close-out (≈10 days bilateral)

Collateralised exposure:

```
Collateralised exposure = max( Net MTM − Collateral held + Threshold + MTA , 0 )
```

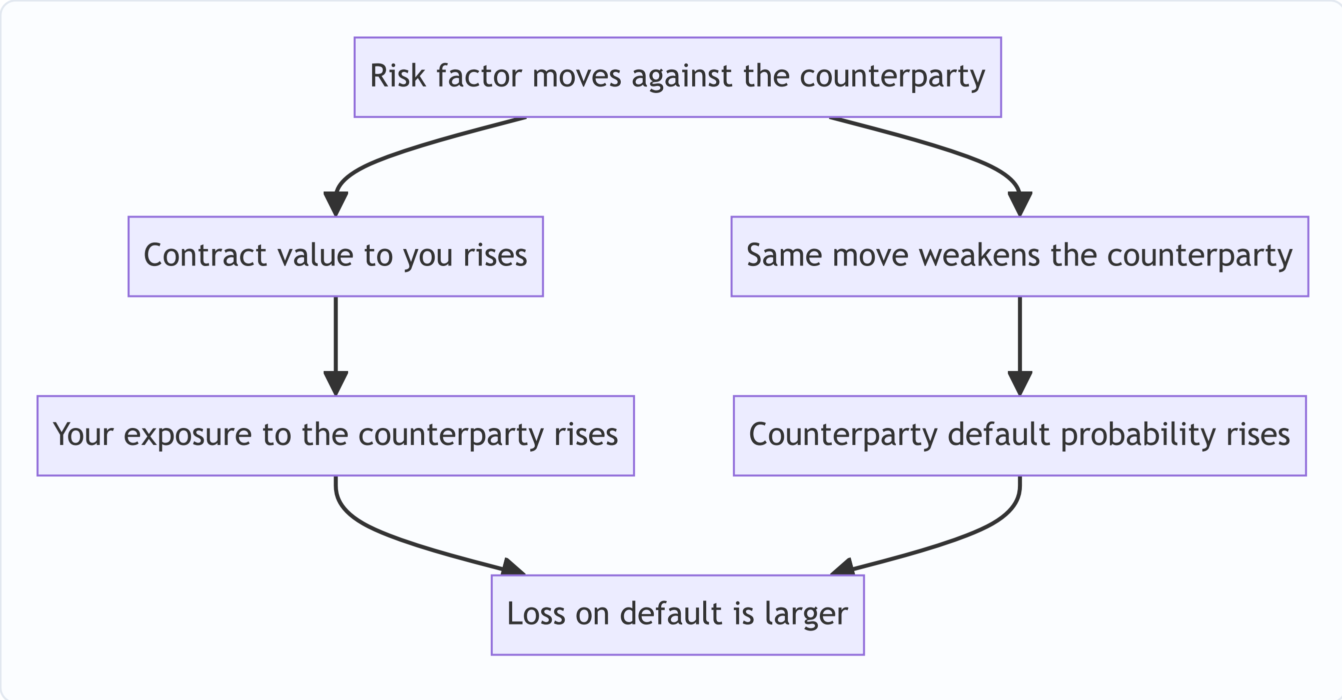
Even a "perfect" daily CSA leaves residual exposure equal to what the portfolio can move during the MPoR — this residual is what initial margin is designed to cover.

4.6 Wrong-way and right-way risk

- **Wrong-way risk (WWR)**: exposure and counterparty default probability are *positively correlated* — exposure rises precisely when the counterparty is more likely to default. This makes losses worse than an independence assumption implies.
- *General WWR*: driven by macro factors (e.g. buying protection on emerging-market debt from a bank in that same economy).
- *Specific WWR*: a structural link (e.g. taking a company's own shares as collateral, or buying CDS protection on a firm from an entity in the same group).
- **Right-way risk**: exposure and default probability are *negatively* correlated — exposure falls as default nears (e.g. a gold producer selling gold forward; if gold falls, the forward is in-the-money to you but the producer is more distressed, but if gold rises the producer is healthy — direction depends on the trade).

WWR is captured by conditioning EE on the default event or by an **alpha multiplier** / correlation model; regulators require WWR identification and, for specific WWR, exposure is set to the full loss with no netting benefit.

Figure 3 — Wrong-way risk: the reinforcing loop between exposure and default probability.

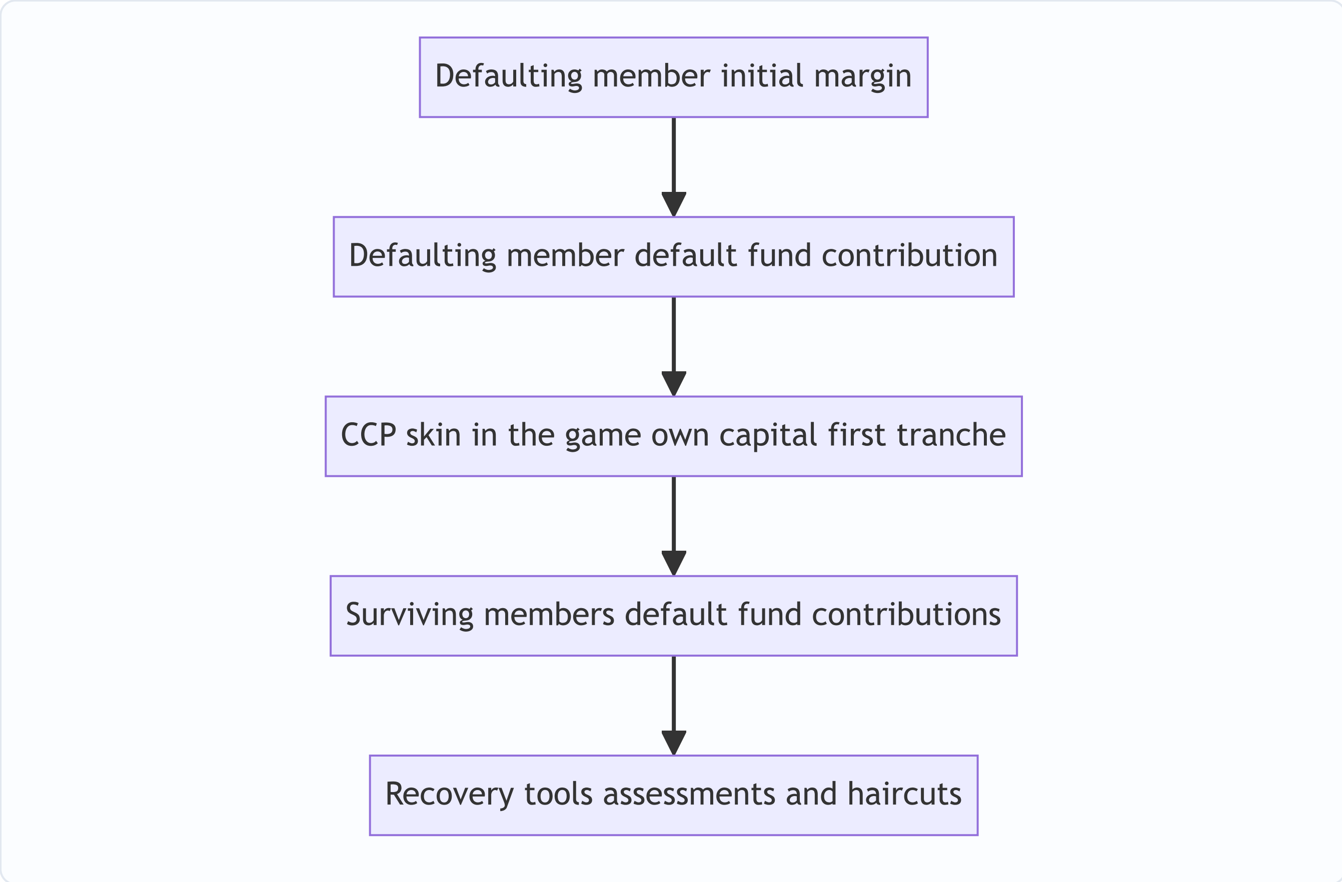


4.7 Central clearing and CCPs

A **central counterparty** novates each cleared trade into two, standing between members. Its risk-management stack:

1. **Membership standards** — only well-capitalised clearing members.
2. **Variation margin** — collected daily/intraday to settle MTM moves to zero.
3. **Initial margin** — posted by each member to cover potential future exposure over the CCP's shorter close-out horizon (often ~5 days for OTC, ~1–2 for futures), typically sized at a 99–99.7% confidence level (SPAN or VaR/expected-shortfall models).
4. **Default fund (guarantee fund)** — mutualised contributions from all members to absorb losses beyond a defaulter's own margin.
5. **Default waterfall** — the loss-absorption order.

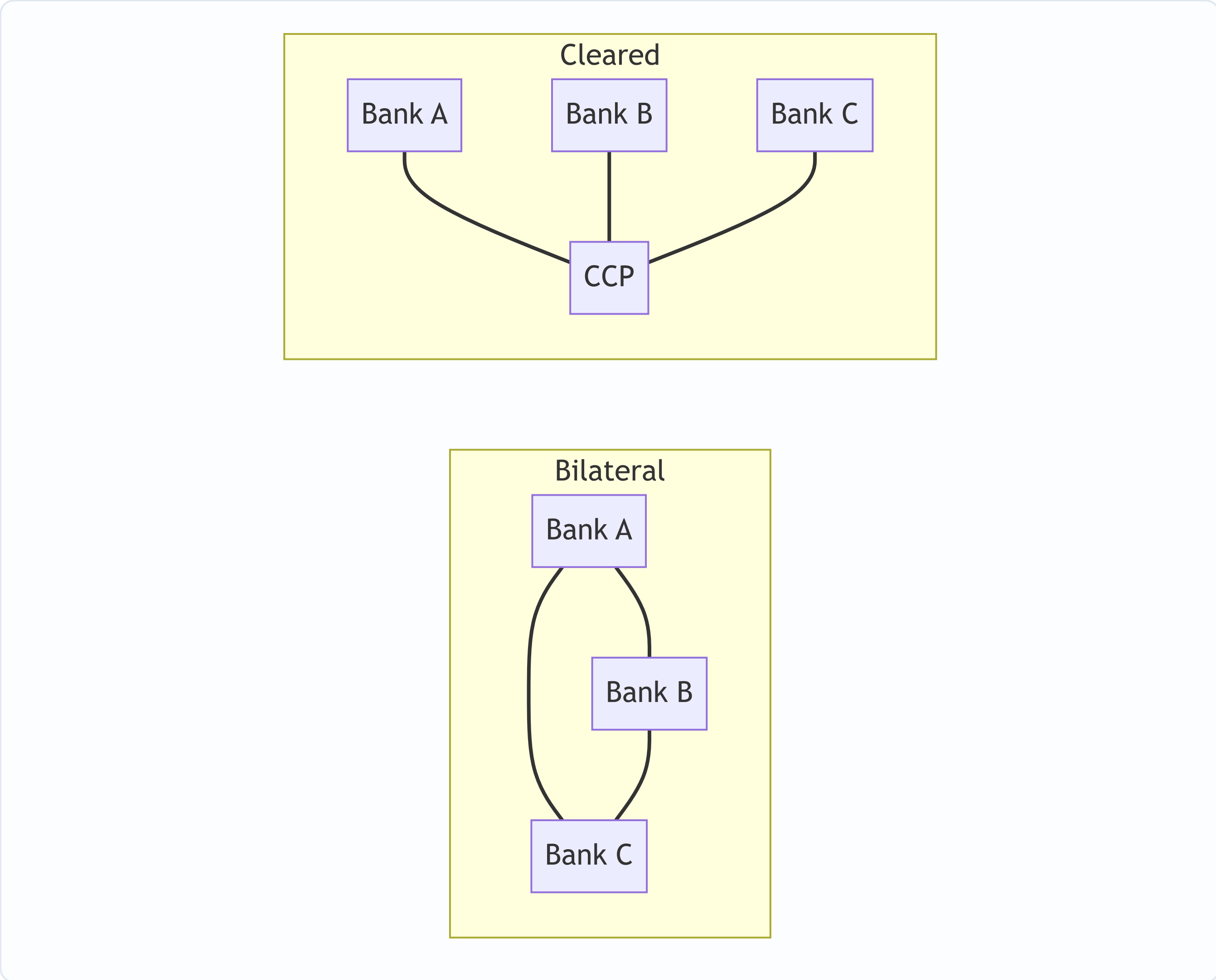
Figure 4 — CCP default waterfall: the order in which resources absorb a member default.



How the CCP mitigates counterparty risk: - **Multilateral netting** across all members typically beats bilateral netting, shrinking system-wide exposure. - **Standardised, disciplined margining** removes the negotiation and threshold slack of bilateral CSAs. - **Mutualisation** spreads a member's default loss so no single survivor is wiped out. - **Transparency** — the CCP sees concentrated positions and can act.

The trade-offs: the CCP becomes a **systemically critical single point of failure** ("too big to fail"), clearing members bear **default-fund and assessment risk** from others' failures, and **basis/liquidity risk** arises from posting large initial margin in cash. This is why CCPs themselves are now closely regulated and stress-tested.

Figure 5 — Bilateral web versus centrally cleared hub before and after novation.



5. Worked Examples

Example 1 — Netting benefit and collateralised exposure

You have three trades with Counterparty X under one ISDA Master Agreement. Current mark-to-market values to you:

Trade	Value to you
Swap 1	+12.0 m
Swap 2	−7.0 m
FX fwd	+3.0 m

Without netting (gross exposure — sum of positive parts): `max(12,0) + max(−7,0) + max(3,0) = 12 + 0 + 3 = 15.0 m`

With netting (positive part of the sum): $\Sigma V_i = 12 - 7 + 3 = 8.0$ m, so net exposure `= max(8, 0) = 8.0 m`

Netting benefit `= 15 − 8 = 7.0 m`. **Netting factor** `= 8 / 15 = 0.533` — netting removed ~47% of gross exposure.

Now add a CSA: Counterparty X has posted **6.0 m** of cash collateral, with **Threshold = 1.0 m** and **MTA = 0.5 m**.

`Collateralised exposure = max( Net MTM − Collateral + Threshold + MTA , 0 )` `= max( 8.0 − 6.0 + 1.0 + 0.5 , 0 ) = max(3.5, 0) = 3.5 m`

Reconciliation / sanity check: exposure fell from 15.0 m (no tools) → 8.0 m (netting) → 3.5 m (netting + collateral). Each tool strictly reduced exposure, and the collateralised figure equals the residual (net value above the collateral held, plus the contractually unsecured threshold and MTA slack). Consistent. ✓

Example 2 — CVA on a single netting set

A 5-year netting set with Counterparty Y. From the exposure engine, the discounted expected exposure $EE(t_i)$ at each year-end is given below. Counterparty Y's flat CDS spread is **200 bps**; assume **recovery R = 40%**, so **LGD = 0.60**. Use annual buckets.

Step 1 — strip survival probabilities. With the credit-triangle approximation `S(t) = exp(−spread × t / LGD)` and hazard `λ = spread/LGD = 0.02/0.60 = 0.03333` per year:

t (yr)	S(t) = exp(−0.03333·t)	Marginal PD = S(t−1) − S(t)
0	1.00000	—
1	0.96722	0.03278
2	0.93551	0.03171
3	0.90485	0.03066
4	0.87519	0.02966
5	0.84650	0.02869

Step 2 — bring in the discounted EE profile (already discounted, in $m), and compute the per-bucket expected loss `EE × marginal PD`:

t (yr)	EE(t) $m	Marginal PD	EE × PD
1	5.0	0.03278	0.16390
2	7.0	0.03171	0.22197
3	6.0	0.03066	0.18396
4	4.0	0.02966	0.11864
5	2.0	0.02869	0.05738
Σ			**0.74585**

Step 3 — apply LGD: `CVA = LGD × Σ (EE × PD) = 0.60 × 0.74585 = 0.4475 m ≈ $447,500`

Reconciliation / sanity check: The exposure profile is hump-shaped (peaks at year 2 = 7.0 m) — exactly the interest-rate-swap shape from Section 3. A crude upper-bound cross-check: total default probability over 5 years `= 1 − S(5) = 0.1535`; times a representative EE of ~5 m times LGD 0.60 gives ≈ `0.1535 × 5 × 0.60 = 0.46 m`, the same order of magnitude as our bucketed 0.447 m. The bucketed figure is slightly lower because the largest default probability sits in year 1 where EE (5.0 m) is below the peak. Consistent. ✓

Interpretation: the bank should charge Counterparty Y about $447k of CVA (roughly 9 bps of a notional of, say, $50m) to compensate for default risk on this netting set — or hedge it by buying CDS protection on Y.

Example 3 — Wrong-way risk adjustment

Reuse Example 2, but now suppose exposure and Y's default are positively correlated (WWR): when Y is near default, the exposure is on average **40% higher** than the unconditional EE. Model this with an **alpha multiplier of 1.4** applied to the expected-loss sum.

`CVA_WWR = 1.4 × 0.4475 = 0.6265 m ≈ $626,500`

Reconciliation: WWR *increases* CVA (from 447k to 627k, +40%), which is directionally correct — wrong-way risk always makes counterparty risk worse because the big exposure and the default coincide. Had this been *right-way* risk (multiplier < 1), CVA would fall. The independence case (Example 2) sits between them. Consistent. ✓

Example 4 — Basel EAD via the alpha factor

Suppose the netting set of Example 2 has **Effective EPE = 4.5 m** (the time-average of the running-max EE profile). The Basel internal-model exposure-at-default is:

`EAD = alpha × Effective EPE = 1.4 × 4.5 = 6.30 m`

Reconciliation: EAD (6.30 m) exceeds Effective EPE (4.5 m) by the 1.4 alpha buffer, and exceeds the peak EE (7.0 m)? No — it is below the 7.0 m single-date peak but above the *average* 4.5 m, which is the intended behaviour: EAD is a loan-equivalent capturing the *average* exposure grossed up for correlation and granularity, not the single worst point. Consistent with the metric hierarchy `EPE ≤ EEPE ≤ EAD ≤ peak PFE`. ✓

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6. Connections

- **To market risk (Ch. on VaR/ES).** PFE is mathematically a VaR of the exposure distribution, but forward-looking over the trade's life. The same Monte Carlo and quantile machinery is reused; the difference is horizon and the `max(V,0)` transform.
- **To credit risk (PD/LGD/EAD).** CCR plugs straight into the credit-loss identity `EL = PD × LGD × EAD`. CCR's contribution is a *stochastic, market-driven EAD*, whereas a loan's EAD is roughly fixed. CVA is just the risk-neutral present value of that expected loss.
- **To liquidity risk.** Collateral and initial-margin calls are liquidity outflows. A ratings downgrade can trigger CSA rating-triggers and a cash scramble — the AIG failure mode. Cleared margin must be posted in cash/HQLA, linking CCR to the LCR world.
- **To regulatory capital (Basel III/IV).** SA-CCR sets standardised EAD; the CVA capital charge (SA-CVA / BA-CVA) capitalises CVA volatility; the leverage ratio and the G-SIB framework all touch derivative exposures.
- **To the XVA desk.** CVA is the first of the valuation adjustments (CVA, DVA, FVA, MVA, KVA, ColVA). Modern banks centralise these on an XVA desk that prices and hedges counterparty, funding, and capital

costs of the whole derivative book.

- **To macroprudential policy.** Mandatory central clearing (Dodd-Frank Title VII, EMIR) and uncleared-margin rules (UMR) are the systemic-risk response born directly from the CCR failures of 2008.

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7. Key Terms

- **Counterparty Credit Risk (CCR):** risk that a derivatives counterparty defaults before final settlement while the contract has positive value to you.
- **Exposure:** `max(V, 0)` — the positive part of the (netted, collateralised) mark-to-market value.
- **Current Exposure (CE):** today's exposure.
- **Potential Future Exposure (PFE):** a high quantile of exposure at a future date; worst-case within a confidence level.
- **Expected Exposure (EE):** mean exposure at a future date; the driver of CVA.
- **Expected Positive Exposure (EPE):** time-average of EE; loan-equivalent exposure.
- **Effective EE / Effective EPE:** running-maximum EE and its average, used in Basel EAD.
- **Exposure at Default (EAD):** `alpha × Effective EPE` (internal models) or `1.4 × (RC + PFE add-on)` (SA-CCR).
- **CVA:** market value of expected counterparty-default loss; reduces the derivative's value.
- **DVA:** debit valuation adjustment — value of one's *own* default risk to oneself.
- **Netting set / ISDA Master Agreement:** group of trades that close out into one net claim on default.
- **CSA (Credit Support Annex):** the collateral schedule — threshold, MTA, eligible collateral, haircuts.
- **Variation Margin (VM) / Initial Margin (IM):** collateral for current MTM moves / for gap risk over the close-out period.
- **Margin Period of Risk (MPoR):** time from the last honoured margin call to close-out and re-hedge.
- **Wrong-way risk (WWR):** positive correlation between exposure and counterparty default probability.
- **CCP:** central counterparty that novates trades and guarantees performance to both sides.
- **Default waterfall:** ordered loss-absorbing resources of a CCP (defaulter margin → defaulter fund → CCP capital → survivors' fund → recovery tools).
- **Novation:** legal replacement of a bilateral trade by two trades each facing the CCP.

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8. Common Confusions

1. **"Exposure equals notional."** No. Exposure is *replacement cost* — the positive MTM — which for an at-market swap starts near zero and is a tiny fraction of notional. Notional only scales the potential.
2. **"A negative-value trade means I gain if they default."** No. You still owe the money; your obligation survives their default. Negative value simply gives *zero* exposure, not a windfall.
3. **"PFE is the same as VaR."** Same quantile mathematics, different scope: PFE profiles exposure forward over the whole life of the deal at many horizons; market-risk VaR is a single short horizon. And PFE applies `max(V,0)` first.
4. **"CVA only matters if the counterparty actually defaults."** No. CVA is marked-to-market daily off CDS spreads; a spread widening books a CVA loss with *no* default. Most 2008 CCR losses were exactly this.

5. **"EE and EPE are the same thing."** EE is a curve over time (mean exposure at each date); EPE is a single number (the time-average of that curve). Effective versions apply a running maximum first.
6. **"Collateral eliminates counterparty risk."** It reduces it to the residual that can build up during the margin period of risk plus any threshold/MTA slack and collateral haircuts — which is precisely why initial margin exists.
7. **"Netting benefit is automatic."** Only where close-out netting is *legally enforceable* in the counterparty's jurisdiction; otherwise regulators make you use gross exposure.
8. **"Central clearing removes counterparty risk."** It *transforms* it: you now face the CCP, post initial margin and a default-fund contribution, and bear mutualised loss-sharing and the systemic risk of the CCP itself.
9. **"DVA is obviously a good thing."** DVA books a *gain* when your own credit worsens — real accounting, but impossible to monetise except by defaulting; many desks neutralise or ignore it.
10. **"Wrong-way risk is a modelling nicety."** It is a first-order driver of tail loss — it makes the big exposure and the default coincide. Ignoring it systematically understates CVA and economic capital.

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9. Recap

Counterparty credit risk is the risk that a derivatives counterparty defaults before final settlement while owing you value. Because a derivative's value is random and can flip sign, exposure is the *positive part* of the mark-to-market value, `max(V, 0)` — you keep the loss, you never gain from default. Exposure is therefore a *distribution across time*, summarised by Current Exposure (now), Expected Exposure (the mean, per date), Potential Future Exposure (a high quantile), and their time-averages EPE / Effective EPE, which feed regulatory EAD via the 1.4 alpha factor. The profile is typically hump-shaped for swaps (diffusion up, amortisation down) and monotone for FX forwards.

The *price* of this risk is CVA — `LGD × Σ EE(tᵢ) × discount × marginal PD` — the market value of expected default loss, marked to market off the counterparty's CDS spread, and capitalised separately under Basel III because its *volatility* (not just realised default) caused most crisis-era losses. Three tools shrink exposure: **netting** (positive part of the *sum*, not sum of positive parts — often 50–90% reduction), **collateral via a CSA** (leaving only the margin-period-of-risk residual, covered by initial margin), and **central clearing** through a CCP that novates trades, imposes disciplined VM/IM, and mutualises tail losses through a default waterfall — at the cost of creating a systemic hub. Layered on top, **wrong-way risk** — positive correlation between exposure and default — magnifies losses and must be modelled, never assumed away. The four worked examples showed netting cutting 15 m to 8 m and collateral to 3.5 m, a CVA of ~$447k rising to ~$627k under wrong-way risk, and an EAD of 6.30 m from a 4.5 m Effective EPE — each reconciled against a sanity check.

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10. Quick-Reference / Interview Points

One-line definitions to have ready - CCR = risk the derivatives counterparty defaults before settlement while the trade is ITM to you. - Exposure = `max(V, 0)`; loss on default = `(1−R)·max(V,0)`. - CVA = risk-neutral PV of expected counterparty-default loss = risk-free value − risky value.

Formulas to know cold - `EE(t) = E[max(Vₜ,0)]`; `PFE(t,α) = qα(max(Vₜ,0))`; `EPE = time-average of EE`. - `EAD = 1.4 × Effective EPE` (IMM) or `1.4 × (RC + PFE add-on)` (SA-CCR). - `CVA ≈ LGD × Σ EE(tᵢ)·D(tᵢ)·[S(tᵢ₋₁) − S(tᵢ)]`, with `S(t) ≈ exp(−spread·t/LGD)`. - Net exposure `= max(ΣVᵢ,0)`;

netting factor `= net/gross`. - Collateralised exposure `= max(Net MTM − Collateral + Threshold + MTA, 0)`.

Fast facts / typical numbers - Alpha = 1.4. Recovery often assumed 40% → LGD 0.60. - Bilateral MPoR ≈ 10 business days; CCP close-out ≈ 5 days OTC. - IM sized at ~99–99.7% confidence. SA-CCR replaced the older Current Exposure Method. - ~2/3 of 2008 CCR losses were CVA mark-to-market, not realised defaults.

Likely interview questions - *Why is the exposure of a swap not its notional?* Because exposure is replacement cost = positive MTM, which starts near zero at-market. - *Sketch the exposure profile of a payer swap vs an FX forward.* Hump-shaped (diffusion vs amortisation) vs monotone-rising to a single maturity exchange. - *Difference between EE and PFE?* Mean vs high quantile of the same distribution. - *What is wrong-way risk — give a specific-WWR example?* Positive corr between exposure and default; e.g. taking the counterparty's own stock as collateral, or buying CDS on a name from an affiliate of that name. - *How does a CCP reduce counterparty risk, and what new risk does it create?* Novation + multilateral netting + disciplined margin + default waterfall; creates a systemic single point of failure and default-fund/assessment risk for members. - *Why did Basel add a CVA capital charge on top of default capital?* Because CVA volatility (spread moves), not defaults, drove most crisis losses. - *Is DVA real income?* Accounting yes; economically it is a gain only realisable via your own default — usually hedged out or ignored.

Red-flag phrases to avoid saying - "Exposure equals notional." / "Negative MTM is a gain on default." / "Collateral removes the risk." / "Clearing eliminates counterparty risk." / "CVA only bites on actual default."

Q&A — Counterparty Credit Risk

A practice bank for the Counterparty Credit Risk (CCR) chapter. Work each question before reading the answer. Numerical answers are self-checked against the core identities: exposure $E = \max(V, 0)$, $CVA \approx LGD \times \Sigma\ EE(t_i) \cdot D(t_i) \cdot [S(t_{i-1}) - S(t_i)]$, and $EAD = 1.4 \times \text{Effective EPE}$.

Section A — Concept-Check (short answer)

A1. What is counterparty credit risk, and how does it differ from ordinary loan credit risk?

CCR is the risk that the counterparty to a bilateral contract (typically an OTC derivative) defaults *before final settlement* at a moment when the contract has *positive value to you*. It differs from loan credit risk on three counts: (1) the direction is bilateral and uncertain — either party can end up the creditor as the contract flips sign; (2) the exposure is stochastic and market-driven — it is the *replacement cost* of the contract, not a fixed principal; and (3) the horizon is long and the exposure profile evolves over the life of the trade. A loan has a known, roughly fixed exposure; a swap does not.

A2. Why is exposure the *positive* part of value, $\max(V, 0)$?

Because default is asymmetric. If the contract value to you is positive ($V > 0$), the counterparty owes you and their default costs you that replacement value. If the value is negative ($V < 0$), you owe them — and default does *not* release you from paying; the administrator will still demand it. So a negative-value trade produces zero exposure, never a gain. You keep the downside and lose the upside, which is exactly why exposure has an option-like, convex shape.

A3. Distinguish Current Exposure, Expected Exposure, and Potential Future Exposure.

- **Current Exposure (CE)** is today's exposure, $\max(V_0, 0)$ — the loss if the counterparty defaulted right now.
- **Expected Exposure (EE)** is the *mean* of the exposure distribution at a future date t, $E[\max(V_t, 0)]$. It is the key input to CVA.
- **Potential Future Exposure (PFE)** is a *high quantile* (e.g. 95th or 99th percentile) of the same distribution at t — the worst case within a confidence level, used for limits.

PFE ≥ EE always, because a high quantile of a distribution is at least its mean for a non-negative variable.

A4. Why is the exposure profile of an interest-rate swap hump-shaped?

Two opposing forces act over time. The **diffusion effect** widens the range of possible future values as rates wander from today's level, pushing exposure up roughly with $\sqrt{t}$. The **amortisation (roll-off) effect** shrinks the remaining cash flows as the trade nears maturity, pulling exposure down to zero at the end. Diffusion dominates early, amortisation dominates late, so exposure peaks somewhere around one-third to one-half of the tenor. An FX forward, by contrast, has a single exchange at maturity and no amortisation, so its exposure rises monotonically to the end.

A5. What is CVA in one sentence, and why is it a real P&L item?

CVA (Credit Valuation Adjustment) is the market value of expected loss from counterparty default: `Risky value = Risk-free value − CVA`. It is a real, daily-marked P&L item because it is computed off the counterparty's CDS spread — when that spread widens, CVA rises and the bank books a *CVA loss even with* *no default occurring*. Roughly two-thirds of 2008-era CCR losses were CVA mark-to-market moves, not realised defaults, which is why Basel III added a dedicated CVA capital charge.

A6. Why does netting reduce exposure, and what is the exact difference in formula?

Under a legally enforceable ISDA Master Agreement, all trades in a netting set collapse into one net claim on default. Exposure becomes the positive part of the *sum*, `max(ΣVᵢ, 0)`, rather than the *sum of the positive parts*, `Σ max(Vᵢ, 0)`. Because in-the-money and out-of-the-money trades offset, the net figure is smaller — often 50–90% smaller for a diversified book. The benefit is only recognised where close-out netting is legally enforceable in the counterparty's jurisdiction.

A7. What residual risk remains even under a "perfect" daily collateral (CSA) agreement?

The **margin period of risk (MPoR)** residual. Even with daily variation margin, there is a gap between the last margin call the counterparty honours and the moment you finally close out and re-hedge — typically ~10 business days for non-cleared trades. The portfolio can move against you during that window, and that residual is what **initial margin** is designed to cover. Collateral shrinks exposure to this residual (plus any threshold, MTA slack, and collateral haircuts) but never to zero.

A8. Define wrong-way risk and give one specific-WWR example.

Wrong-way risk is a *positive correlation between exposure and the counterparty's default probability* — the exposure rises precisely when the counterparty is most likely to default, making losses worse than an independence assumption implies. *General* WWR is macro-driven (e.g. buying EM sovereign protection from a bank in that same economy). *Specific* WWR is a structural link — e.g. taking the counterparty's own shares as collateral, or buying CDS protection on a firm from an entity in its own group. Regulators require WWR identification, and for specific WWR the exposure is set to the full loss with no netting benefit.

A9. What is the alpha factor and where does it appear?

Alpha is a regulatory multiplier of **1.4** applied to the loan-equivalent exposure to gross it up for correlation, granularity, and model uncertainty. It appears in the Basel internal-model EAD, `EAD = 1.4 × Effective EPE`, and in SA-CCR, `EAD = 1.4 × (Replacement Cost + PFE add-on)`.

A10. How does a CCP reduce counterparty risk, and what new risk does it create?

A central counterparty *novates* each trade into two, standing between the members and guaranteeing performance. It reduces risk through multilateral netting across all members, disciplined daily/intraday variation margin, risk-sensitive initial margin, mutualised default-fund contributions, and a default waterfall. But it *transforms* rather than eliminates the risk: the CCP becomes a systemically critical single point of failure ("too big to fail"), members bear default-fund and assessment risk from others' failures, and posting large cash initial margin creates liquidity/basis risk.

Section B — Numerical / Applied (full solutions)

B1. Exposure with sign flip. A single swap has mark-to-market value to you of `-4.0 m` today. What is your current exposure? If tomorrow rates move and the value becomes `+6.0 m`, what is the exposure then?

Today: `E = max(-4.0, 0) = 0`. You owe the counterparty, so their default costs you nothing — but you still owe the 4.0 m. Tomorrow: `E = max(+6.0, 0) = 6.0 m`. **Exposure went from 0 to 6.0 m** purely on a market move, illustrating the stochastic, sign-flipping nature of CCR. ✓

B2. Netting benefit. Four trades under one ISDA Master with Counterparty X, values to you (in m): +18, −11, +5, −4. Compute gross exposure, net exposure, netting benefit, and netting factor.

Gross (sum of positive parts): `18 + 0 + 5 + 0 = 23.0 m`. Net (positive part of the sum): `ΣVᵢ = 18 − 11 + 5 − 4 = 8.0 m`, so `max(8, 0) = 8.0 m`. Netting benefit `= 23 − 8 = 15.0 m`. Netting factor `= 8 / 23 = 0.348`.

Check: the factor lies in [0,1], and netting removed ~65% of gross exposure — plausible for a book with large offsetting positions. ✓

B3. Collateralised exposure. Continue B2. Counterparty X has posted **5.0 m** cash collateral. The CSA has **Threshold = 1.0 m** and **MTA = 0.5 m**. Compute the collateralised exposure.

`Collateralised exposure = max(Net MTM − Collateral + Threshold + MTA, 0)` `= max(8.0 − 5.0 + 1.0 + 0.5, 0) = max(4.5, 0) = 4.5 m`.

Check: exposure fell 23.0 → 8.0 (netting) → 4.5 (collateral). Each tool strictly reduced exposure; the residual is the net value above collateral plus the contractual threshold/MTA slack. Consistent. ✓

B4. CVA on a netting set. A 4-year netting set with Counterparty Y. Discounted expected exposures at year-ends (in m): EE(1)=6, EE(2)=8, EE(3)=5, EE(4)=2. Y's flat CDS spread is 300 bps; recovery R = 40%, so LGD = 0.60. Compute CVA.

Step 1 — hazard rate and survival. `λ = spread / LGD = 0.03 / 0.60 = 0.05` per year. `S(t) = exp(−0.05·t)`.

t	S(t)	Marginal PD = S(t−1)−S(t)
0	1.00000	—
1	0.95123	0.04877
2	0.90484	0.04639
3	0.86071	0.04413
4	0.81873	0.04198

Step 2 — expected loss per bucket (EE × marginal PD):

t	EE	Marginal PD	EE × PD
1	6	0.04877	0.29262
2	8	0.04639	0.37112
3	5	0.04413	0.22065
4	2	0.04198	0.08396
Σ			**0.96835**

Step 3 — apply LGD: `CVA = 0.60 × 0.96835 = 0.5810 m ≈ $581,000`.

Check: Crude cross-check — total 4-year default prob `= 1 − S(4) = 0.18127`; times a representative EE of ~5.25 m times LGD 0.60 ≈ `0.181 × 5.25 × 0.60 = 0.571 m`, same order of magnitude as the bucketed 0.581 m. ✓

B5. Wrong-way risk adjustment. Reuse B4. Exposure and Y's default are positively correlated; model WWR with an alpha multiplier of 1.3 on the expected-loss sum. New CVA?

`CVA_WWR = 1.3 × 0.5810 = 0.7553 m ≈ $755,000`.

Check: WWR *increases* CVA (581k → 755k, +30%), which is directionally correct — wrong-way risk always worsens counterparty risk because the large exposure and the default coincide. Right-way risk (multiplier < 1) would reduce it. ✓

B6. Basel EAD. The netting set of B4 has Effective EPE = 5.0 m. Compute the internal-model EAD, and compare to the single-date peak EE.

`EAD = alpha × Effective EPE = 1.4 × 5.0 = 7.0 m`.

Check: EAD (7.0 m) exceeds Effective EPE (5.0 m) by the 1.4 buffer, and here happens to equal the single-date peak EE (8.0 m)? No — 7.0 < 8.0. EAD sits above the *average* exposure but below the single worst date, exactly as intended: it is a loan-equivalent capturing the *average* exposure grossed up for correlation, not the worst point. Consistent with `EPE ≤ EEPE ≤ EAD ≤ peak PFE`. ✓

B7. Loss given default on a derivative. A swap is worth +10 m to you when the counterparty defaults. Recovery is 30%. What is your loss?

Loss `= (1 − R) × max(V, 0) = (1 − 0.30) × 10 = 0.70 × 10 = 7.0 m`. You recover `0.30 × 10 = 3.0 m` from the estate and must replace the swap at market cost 10 m, net loss 7.0 m. ✓

Section C — Interview-Style (model answers)

C1. "Why isn't the exposure of an interest-rate swap equal to its notional?"

Because exposure is *replacement cost*, not principal. In a swap, the notional is never exchanged — only net interest flows are. What you can lose on the counterparty's default is the cost of replacing the contract at current market rates, i.e. the positive mark-to-market value, `max(V, 0)`. An at-market swap starts life worth roughly zero to both sides, so initial exposure is near zero. The notional only *scales the potential* future exposure by determining how large the value swings can be, but it is not itself the amount at risk. Saying "exposure equals notional" would overstate the risk by orders of magnitude.

C2. "Walk me through how you'd compute an exposure profile."

The standard method is Monte Carlo simulation. First, choose risk-factor models — Hull–White for interest rates, geometric Brownian motion for FX or equity. Second, simulate thousands of scenario paths of those risk factors out to the longest trade maturity on a time grid. Third, at every grid date on every path, reprice every trade in the netting set and apply the netting and collateral rules to get the netted, collateralised value V_t. Fourth, take $E_t = \max(V_t, 0)$ on each path. Finally, across paths at each date: the mean gives EE, the chosen high quantile gives PFE, and time-averaging the running-max EE gives Effective EPE. It is computationally heavy — banks reprice millions of trade-scenario combinations nightly — which is why XVA and exposure engines are major infrastructure investments. For a lighter regulatory number, SA-CCR approximates EAD as `1.4 × (Replacement Cost + PFE add-on)` without full simulation.

C3. "Explain CVA to a non-technical stakeholder, and why the desk can lose money on it without any default."

CVA is the price of the risk that the people we trade derivatives with might not pay us. A derivative traded with a shaky counterparty is worth less than the identical trade with a rock-solid one, and CVA is exactly that discount — the expected cost of their possible default. The subtle part: we don't wait for a default to recognise it. We mark CVA every day using the market's own read on that counterparty's creditworthiness — their CDS spread. If the market suddenly thinks the counterparty is riskier, their spread widens, our expected default cost rises, and we book a CVA loss that day — even though nobody has defaulted and no cash has changed hands. That daily volatility, not actual defaults, drove most of the counterparty losses in 2008, which is why regulators now hold capital specifically against CVA swings.

C4. "What's wrong-way risk, and why do risk managers lose sleep over it?"

Wrong-way risk is when your exposure to a counterparty grows at exactly the moment that counterparty becomes more likely to default — the two are positively correlated. The classic example is taking a company's own stock as collateral: if the company gets into trouble, its stock falls, your collateral evaporates, and your exposure balloons, all simultaneously. Another is buying credit protection on a country from a bank domiciled in that country — a sovereign crisis hits both the protection you're owed and the bank that owes it. It matters because it breaks the comforting assumption that exposure and default are independent. Under independence you'd average a moderate exposure against a moderate default probability, but WWR makes the *big* exposure and the default land together, fattening the tail of the loss distribution. Ignore it and you systematically understate CVA and economic capital.

C5. "A CCP guarantees both sides of every trade. Does that mean central clearing eliminates counterparty risk?"

No — it transforms it rather than eliminating it. Clearing genuinely reduces bilateral risk through multilateral netting, disciplined daily variation margin, risk-sensitive initial margin, and mutualised loss-sharing via a default waterfall. But three new risks appear. First, the CCP itself becomes a systemically critical single point of failure — if it fails, the whole market is exposed, which is why CCPs are now heavily stress-tested and regulated. Second, as a clearing member you bear default-fund and assessment risk: if another member defaults and blows through their margin, your mutualised contribution absorbs part of the loss. Third, initial margin must be posted in cash or high-quality liquid assets, so clearing converts credit risk into a liquidity and funding demand. So the honest answer is: clearing concentrates and mutualises counterparty risk into a well-managed hub, but the risk doesn't vanish.

Section D — MCQs (with reasoning)

D1. The exposure of an OTC derivative to a defaulting counterparty is best described as: A. The notional of the contract B. `max(V, 0)`, the positive part of the mark-to-market value C. The absolute value `|V|` D. The sum of all future cash flows

Answer: B. Exposure is the replacement cost, and only when the contract is in-the-money to you (`V > 0`) does default cost you anything. A gives the notional, which is never at risk in full; C would wrongly count a trade you *owe* on as exposure; D ignores the netting of flows and the `max` transform.

D2. Under a legally enforceable netting agreement, exposure across a netting set equals: A. $\Sigma \max(V_i, 0)$ B. $\max(\Sigma V_i, 0)$ C. ΣV_i D. $\max(V_i)$ over all trades

Answer: B. Netting takes the positive part of the *sum* of values. A is the *gross* (no-netting) figure; the difference between A and B is the netting benefit. C ignores that a net-negative set gives zero exposure, and D is meaningless.

D3. Which statement about CVA is correct? A. CVA is only realised if the counterparty actually defaults B. CVA increases when the counterparty's CDS spread widens C. CVA raises the value of a derivative above its risk-free value D. CVA is unaffected by the exposure profile

Answer: B. CVA is marked daily off the CDS spread, so a wider spread means higher expected default cost and a CVA loss with no default — refuting A. C is backwards: `Risky value = Risk-free value − CVA`, so CVA *reduces* value. D is false because CVA is driven directly by the EE profile.

D4. The classic hump-shaped exposure profile of an interest-rate swap results from: A. Amortisation dominating early and diffusion late B. Diffusion dominating early and amortisation (roll-off) late C. A single cash exchange at maturity D. Wrong-way risk

Answer: B. Early on, diffusion widens the value distribution faster than cash flows roll off, so exposure rises; later, amortisation dominates and exposure falls to zero at maturity. A reverses the forces. C describes an FX forward (monotone profile, not hump). D is unrelated to profile shape.

D5. In the Basel internal-model approach, Exposure at Default equals: A. Effective EPE B. 1.4 × Effective EPE C. Peak PFE D. Current Exposure

Answer: B. The alpha factor of 1.4 grosses up the loan-equivalent Effective EPE for correlation and granularity. A omits alpha; C is the worst-case single point, not the regulatory EAD; D captures only today's exposure.

D6. Taking the counterparty's own equity as collateral is an example of: A. Right-way risk B. General wrong-way risk C. Specific wrong-way risk D. Basis risk

Answer: C. There is a structural link between the collateral value and the counterparty's own default — as it approaches default its stock collapses, so the collateral fails exactly when needed. That structural, name-specific link makes it *specific* WWR; general WWR (B) is macro-driven without a direct structural tie.

D7. Even under a daily-margined CSA, residual exposure remains because of: A. The notional of the trades B. The margin period of risk C. The netting factor D. The alpha multiplier

Answer: B. Between the last honoured margin call and final close-out (~10 business days bilaterally), the portfolio can move against you; that residual is covered by initial margin. A, C, and D are unrelated to the collateral gap.

D8. In a CCP default waterfall, which resource is consumed first? A. Surviving members' default-fund contributions B. The CCP's own capital (skin in the game) C. The defaulting member's initial margin D. Recovery tools and assessments

Answer: C. The waterfall consumes the defaulter's own resources first — initial margin, then their default-fund contribution — before touching the CCP's capital, then surviving members' mutualised fund, and finally recovery tools. This "defaulter pays first" ordering is the core of the mutualisation design.

Self-verification note. All numerical answers were recomputed and cross-checked: exposures satisfy `E = max(V, 0)`; netting factors lie in [0, 1]; CVA figures were sanity-checked against the crude `(1−S(T)) × representative−EE × LGD` bound; and `EAD = 1.4 × Effective EPE` sits between average and peak exposure. Formulas match the chapter's core identities.

Chapter 08 — Operational Risk

1. The Problem / The Need

Market risk asks: what if prices move against me? Credit risk asks: what if my counterparty fails to pay? Both are *risks you take on purpose* — a bank lends money and holds bonds precisely because it expects to be paid for bearing that risk. There is an upside. You can price it, hedge it, and earn a spread.

Operational risk is different in kind. It is the risk that your own machinery — the people, the processes, the systems that are supposed to *run* the business — breaks down. Nobody chooses to be defrauded. Nobody sets out to have a trader hide a $6 billion loss, a settlement instruction sent to the wrong account, or a ransomware crew encrypt the core banking platform on a Friday night. There is no upside, no spread earned for bearing it. It is pure downside, embedded in everything the firm does.

Historically, this was the risk that "everyone knew about but nobody measured." A firm would post huge trading profits and be lauded for its risk management — and then a single rogue trader (Barings, 1995) or a single fraud (Enron) or a single control failure would wipe out decades of earnings overnight. The regulators noticed a pattern: the biggest, most sudden bank collapses were often *not* caused by markets or credit going wrong. They were caused by the plumbing.

That created three concrete needs:

1. **Recognition** — formally name this risk category so it gets an owner, a budget, and a seat at the table alongside market and credit risk.
2. **Capital** — force firms to hold a cushion of equity against operational failures, so a large loss event does not become an insolvency.
3. **Management** — build a repeatable framework to *identify, assess, control, and monitor* these failures before they detonate, rather than writing post-mortems afterward.

Basel II (2004) was the regulatory answer: it made operational risk a formal Pillar 1 capital charge for the first time. This chapter builds the concept from that need outward — the definition, the taxonomy of loss events, the three generations of measurement approaches, the indicators used to see trouble coming, the famous disasters, and the management framework that ties it all together.

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2. The Core Idea

The Basel Committee's definition is the anchor, and it is worth memorising word-for-word because interviewers quote it:

> **Operational risk is the risk of loss resulting from inadequate or failed internal processes, people and systems, or from external events. This definition includes legal risk but excludes strategic and reputational risk.**

Unpack the four sources — the "PPSE" of operational risk:

- **People** — human error, negligence, lack of skill, and deliberate wrongdoing (fraud, rogue trading, collusion). The single richest source of large losses.

- **Processes** — flawed or missing procedures: a payment run with no maker-checker, a reconciliation that nobody performs, a model that nobody validates, a limit that nobody enforces.
- **Systems** — technology failures: outages, bugs, capacity limits, botched software releases, and increasingly cyber attacks.
- **External events** — things done *to* the firm from outside: external fraud, natural disaster, pandemic, terrorism, third-party/vendor failure, regulatory change.

Two boundary clauses matter and are classic exam traps:

- **Legal risk is INCLUDED.** Fines, penalties, settlements, and the cost of failed contracts are operational losses. A $1 billion mis-selling settlement is an operational-risk loss.
- **Strategic and reputational risk are EXCLUDED** from the regulatory *capital* definition. Choosing to enter the wrong market (strategic) or the loss of customer trust after a scandal (reputational) are real risks the board must manage — but they are not part of the Basel operational-risk capital charge. Note the subtlety: an operational *event* (say, a data breach) often *causes* huge reputational damage, but only the direct operational loss counts for capital.

The core mental model: operational risk lives inside the firm's own operations. It is heavy-tailed — mostly a steady drizzle of small losses (a mispriced trade fixed the next day, a small mis-payment recovered) punctuated by rare, catastrophic "tail" events that can threaten solvency. Managing it is therefore two jobs at once: grind down the frequent small losses through control quality, and defend against the rare monster through governance, insurance, and capital.

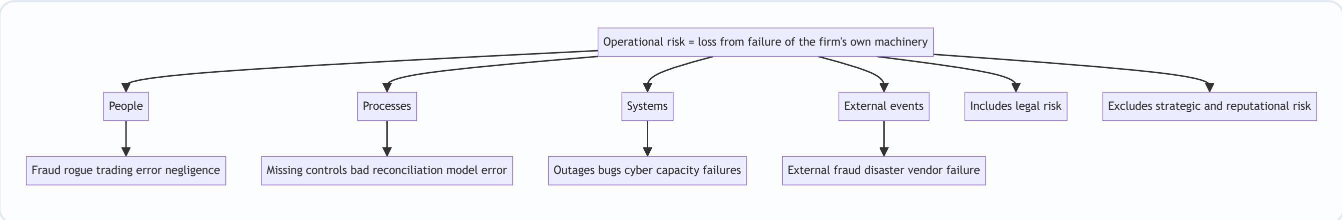

Figure 8.1 — The four sources of operational risk and the two boundary clauses.

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3. Why / How It Works

Why operational risk behaves differently from market and credit risk

Three structural features drive everything about how we measure and manage it.

(a) It is pervasive, not localised. Market risk sits in the trading book; credit risk sits in the loan book. Operational risk is *everywhere* — in the back office, in HR, in IT, in the branch, in the legal department, in the vendor contract. You cannot ring-fence it into one desk. This is why the management framework must be firm-wide and why every business line has to own its own operational risk (the "first line of defence").

(b) Its loss distribution is severely fat-tailed and asymmetric. Plot operational losses and you get a distribution with: - **High frequency, low severity** in the body — data-entry errors, small mis-payments, minor system glitches. Predictable, budgetable, almost a "cost of doing business." - **Low frequency, high severity** in the tail — the rogue trader, the systemic cyber breach, the class-action settlement. Rare, but a single event can exceed a full year's profit.

The economic capital you must hold is driven almost entirely by that tail, not the body. This is why operational risk modelling borrows from actuarial science and extreme-value theory rather than from the normal distribution used in basic market-risk VaR.

(c) There is no natural "exposure" measure. For credit risk, exposure-at-default is a clean dollar number. For market risk, you have position sizes. For operational risk, what is the exposure? The whole firm? Regulators solved this pragmatically by using a *proxy for the scale of activity* — gross income, or later a "business indicator" — on the logic that a bigger, busier firm has more operational surface area and therefore more to go wrong. That proxy choice is the DNA of every measurement approach below.

How a firm converts these features into a working system

The mechanism is a closed feedback loop — the operational risk management (ORM) cycle:

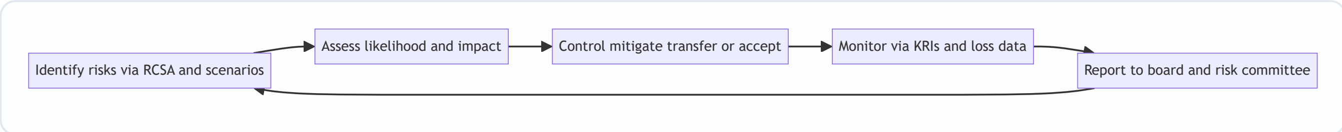

Figure 8.2 — The operational risk management cycle as a continuous feedback loop.

The loop is fed by four data engines that recur throughout the chapter:

1. **Internal loss data (ILD)** — a disciplined database of every operational loss the firm has actually suffered. This is the empirical backbone.
2. **External loss data (ELD)** — losses suffered by *other* firms (via consortia like ORX), so you can see tail events you have been lucky enough to avoid so far.
3. **Scenario analysis** — structured expert workshops that ask "what is the worst plausible thing that could happen here, and how big?" to populate the tail where your own history is thin.
4. **Business environment and internal control factors (BEICF)** — forward-looking indicators of the *current* control environment (KRIs, audit findings, staff turnover) that adjust the backward-looking loss picture.

These four are the "AMA data elements" and, in one form or another, they drive both measurement and management.

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4. Full Content — Framework, Categories, Formulas, Methods

4.1 The loss-event taxonomy (Basel's seven event types)

To aggregate losses across a firm and compare against peers, everyone needs to bucket events the same way. Basel defines **seven Level-1 event types**:

#	Event type	Plain-English meaning	Example
1	Internal fraud	Wrongdoing involving at least one insider	Rogue trading, unauthorised trades, insider theft
2	External fraud	Wrongdoing by a third party	Card skimming, cheque forgery, hacking to steal funds
3	Employment practices & workplace safety	HR and safety breaches	Discrimination suits, unsafe workplace claims
4	Clients, products & business practices	Improper business/market conduct	Mis-selling, market manipulation, money-laundering fines
5	Damage to physical assets	Physical loss from disaster/terror	Earthquake, fire, flood destroying premises
6	Business disruption & system failures	IT and infrastructure outages	Data-centre outage, network failure, software crash
7	Execution, delivery & process management	Failures in transaction processing	Settlement failure, data-entry error, failed reconciliation, vendor dispute

Two practical notes. First, **event types 4 (CPBP) and 7 (EDPM) carry the largest aggregate dollar losses** in the industry — conduct/mis-selling fines and processing errors dwarf the headline-grabbing rogue-trader category by total value, even though internal fraud produces the most spectacular single events. Second, losses are also classified by **business line** (corporate finance, trading & sales, retail banking, commercial banking, payment & settlement, agency services, asset management, retail brokerage) so capital can be attributed correctly.

4.2 Measurement approach #1 — Basic Indicator Approach (BIA)

The simplest Basel II method. Hold capital equal to a fixed percentage (**alpha = 15%**) of average positive annual gross income (GI) over the last three years.

$$K_{BIA} = \alpha \times \frac{\sum_{i=1}^{3} \max(GI_i,\ 0)}{n}$$

where **α = 0.15**, and n = number of the previous three years in which gross income was positive (years with negative or zero GI are dropped from both numerator and denominator).

- *Gross income* = net interest income + net non-interest income (before deducting operating expenses, and before provisions).
- No risk sensitivity at all — a well-controlled bank and a shambolic one with the same income hold the same capital.
- Intended for small, non-complex banks.

4.3 Measurement approach #2 — The Standardised Approach (TSA)

A step up in granularity. Split gross income into the **eight business lines**, apply a **beta factor** (12%, 15%, or 18%) to each, sum, and average over three years. Higher-beta lines are judged riskier.

$$K_{TSA} = \frac{\sum_{years=1}^{3} \max\!\left(\sum_{j=1}^{8} \beta_j \times GI_j,\ 0\right)}{3}$$

Business line	Beta
Corporate finance	18%
Trading & sales	18%
Payment & settlement	18%
Commercial banking	15%
Agency services	15%
Retail banking	12%
Asset management	12%
Retail brokerage	12%

Note the aggregation subtlety: within any single year you **sum across business lines first** (so a strongly negative line can offset a positive one *within that year*), then floor the yearly total at zero, then average the three years. This differs from BIA, which floors each year's total GI.

4.4 Measurement approach #3 — Advanced Measurement Approach (AMA)

The most sophisticated Basel II option: the bank builds its *own* internal model to estimate operational-risk capital, subject to supervisory approval. The regulatory standard is capital sufficient to cover losses at the **99.9% confidence level over a one-year horizon** — i.e. the loss so severe it is expected only once in a thousand years.

The workhorse technique is the **Loss Distribution Approach (LDA)**, an actuarial construction done per business-line × event-type cell:

1. Fit a **frequency distribution** — how many loss events per year (typically Poisson, parameter λ).
2. Fit a **severity distribution** — the dollar size per event (typically lognormal, or a heavy-tailed distribution like generalised Pareto for the tail).
3. **Convolve** frequency and severity (usually by Monte Carlo simulation) to produce the **aggregate annual loss distribution**.
4. Read off the **99.9th percentile** = Operational Value-at-Risk (OpVaR).
5. Aggregate across cells, allowing for **diversification** (a correlation assumption < 1 reduces total capital versus naïve summation).

AMA must combine all four data elements from §3: internal loss data, external loss data, scenario analysis, and BEICF. Regulators may allow a **recognition of insurance** up to a 20% cap on the total operational-risk capital charge.

4.5 The successor — the Standardised Measurement Approach (SMA / Basel III)

Because AMA models proved inconsistent and hard to compare across banks, Basel III (finalised 2017, phasing in through the early 2020s) **scrapped both AMA and the old menu** and replaced everything with a single non-modelled formula, often called the **new Standardised Approach** or SMA. Two ingredients:

- **Business Indicator (BI)** — an income proxy built from three components: the interest/lease/dividend component, the services component, and the financial (trading + banking book) component. The BI is mapped through a progressive marginal-coefficient schedule into the **Business Indicator Component (BIC)**.

- **Internal Loss Multiplier (ILM)** — a scaling factor derived from the bank's own 10-year loss history relative to its BIC, capturing the intuition that a bank with a bad loss record should hold more.

$$ILM = \ln\!\left(e^{1} - 1 + \left(\frac{LC}{BIC}\right)^{0.8}\right)$$

where **LC (Loss Component)** = 15 × average annual operational losses over 10 years, and *e* is Euler's number. When LC = BIC, the ratio is 1 and ILM = 1 (loss experience is exactly "average"). Capital is then:

$$K_{SMA} = BIC \times ILM$$

The design goals: reduce model risk, restore comparability across banks, and still keep *some* risk sensitivity through the loss-history multiplier. (National regulators can set ILM = 1, switching off the loss-history sensitivity.)

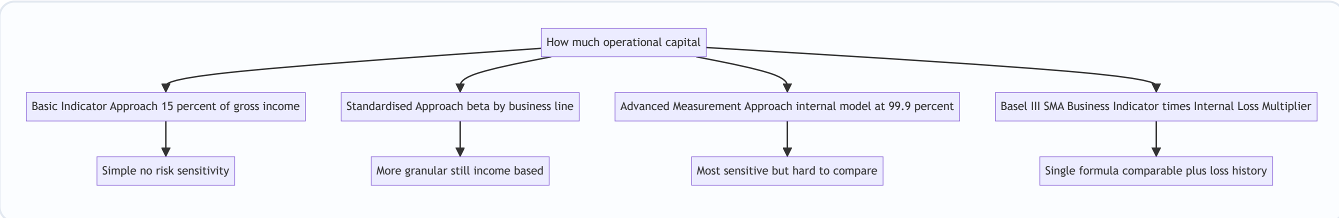

Figure 8.3 — The evolution of operational-risk capital approaches from crude income proxy to loss-history-scaled standardised formula.

4.6 Key Risk Indicators (KRIs)

Capital tells you how much cushion to hold; **KRIs tell you when the risk is rising** so you can act before a loss crystallises. A KRI is a measurable metric with a **threshold and escalation trigger**, ideally *leading* rather than *lagging*.

Risk area	Example KRI	Why it leads to loss
People	Staff turnover %, overtime hours, vacancy rate in control functions	Overworked, under-staffed teams make errors and skip controls
Processing	Number of failed/aged trade settlements, reconciliation breaks open > X days	Rising breaks precede a large mis-settlement loss
Systems	System downtime minutes, number of change-failures, patch backlog	Fragile IT precedes an outage or breach
Fraud/conduct	Number of limit breaches, off-hours trading activity, complaints volume	Classic rogue-trading precursors
External	Vendor SLA breaches, phishing-email click-through rate	Third-party and cyber exposure building

Good KRI practice: define a **green/amber/red threshold**, assign an **owner**, review at fixed cadence, and — critically — **correlate KRIs with actual loss data** to prove they are predictive rather than decorative. A KRI that never moves before a loss is theatre.

4.7 The Three Lines of Defence (governance model)

The organising structure for *who* manages operational risk:

- **First line — the business.** Owns and manages its own risks day-to-day; owns the controls.
- **Second line — risk management & compliance.** Sets the framework, policy, and appetite; independently challenges and oversees the first line.
- **Third line — internal audit.** Independently assures the board that lines one and two actually work.

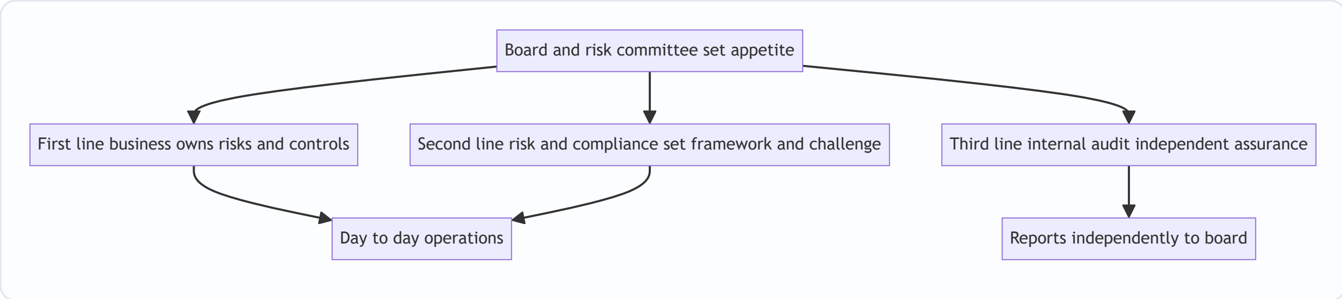

Figure 8.4 — The three lines of defence governance model for operational risk.

4.8 The four responses to any operational risk

Once a risk is identified and assessed (via a **Risk and Control Self-Assessment**, RCSA — the workshop where a business rates its inherent risk, the strength of its controls, and the residual risk), management chooses among four responses — the "4 T's":

1. **Treat / mitigate** — strengthen controls (add maker-checker, automate a manual step, segregate duties).
2. **Transfer** — insurance, or outsourcing the activity (note: outsourcing transfers the *operation* but creates new *vendor* risk — you cannot outsource accountability).
3. **Terminate / avoid** — exit the product, market, or process that generates the risk.
4. **Tolerate / accept** — consciously retain the risk within appetite, because the cost of control exceeds the benefit.

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5. Worked Examples

Example 1 — Basic Indicator Approach capital, with a loss year

A small bank reports gross income (in $m) of: Year-1 = 800, Year-2 = **−200** (a bad year), Year-3 = 600. Compute the BIA operational-risk capital charge.

Rule: drop any year with non-positive GI from *both* numerator and denominator; α = 15%.

- Positive-GI years: Year-1 (800) and Year-3 (600). Year-2 is excluded.
- n = 2 (only two years had positive GI).
- Average positive GI = (800 + 600) / 2 = **700**.
- Capital = 0.15 × 700 = **$105m**.

Self-check. A common wrong answer averages over all three years and includes the loss: (800 − 200 + 600)/3 = 400, giving 0.15 × 400 = $60m. That is *incorrect* under BIA because negative years are excluded, not netted. Excluding the loss year raises the average (700 vs 400) and therefore raises capital to $105m — which is the intended conservative treatment. The rule prevents a bank from *reducing* its capital charge simply by having had a disastrous year. ✓

Example 2 — The Standardised Approach, and why it differs from BIA

Same bank, but now we know the Year-2 gross income by business line ($m):

Business line	Beta	GI Year-2	β × GI
Trading & sales	18%	−500	−90
Retail banking	12%	+250	+30
Payment & settlement	18%	+50	+9
Total Year-2		**−200**	**−51**

Suppose Years 1 and 3 each produced a weighted β × GI total of +96 and +72 respectively (positive years).

TSA rule: within each year, sum the weighted amounts across business lines, floor that yearly total at zero, then average over three years.

- Year-1 weighted total = +96 → keep 96.
- Year-2 weighted total = −51 → floored to **0**.
- Year-3 weighted total = +72 → keep 72.
- Capital = (96 + 0 + 72) / 3 = 168 / 3 = **$56m**.

Self-check and reconciliation with Example 1. Under TSA the loss year is *not* dropped — it is included but floored at zero, so it drags the average *down* (dividing by 3, not by 2). Under BIA the same loss year was *dropped*, dividing by 2. That is exactly why TSA here gives $56m versus BIA's $105m: the two methods treat a loss year in opposite ways (floor-and-keep vs drop). The lesson interviewers want: **TSA is not merely "BIA with betas" — the negative-year mechanics genuinely differ.** ✓

Example 3 — Loss Distribution Approach → OpVaR (AMA logic)

A business-line/event-type cell has, per year: - Frequency ~ Poisson with $\lambda = 4$ events/year (mean 4 losses annually). - Severity ~ Lognormal with $\mu = 12$, $\sigma = 1.5$ (in log-dollars).

Step 1 — Mean severity per event. For a lognormal, mean = $\exp(\mu + \sigma^2/2) = \exp(12 + 1.125) = \exp(13.125)$ ≈ **$500,000** (e^13.125 ≈ 5.0 × 10^5).

Step 2 — Expected annual loss (the "body"). Expected loss = frequency × mean severity = 4 × $500,000 = **$2.0m per year**. This is what you provision/budget for.

Step 3 — The tail (unexpected loss). OpVaR at 99.9% is *not* 4 × the mean severity; it is driven by the rare year with several *large* losses. Because severity is lognormal with $\sigma = 1.5$, the 99.9th-percentile single loss is enormous: $\exp(\mu + \sigma \cdot z_{0.999}) = \exp(12 + 1.5 \times 3.09) = \exp(16.635)$ ≈ **$16.8m** for one event. A simulated bad year (say 8 events, several from the tail) easily produces an aggregate loss on the order of **$25–30m**.

Step 4 — Capital. Operational-risk capital for the cell ≈ OpVaR − Expected Loss (regulators let you hold capital against *unexpected* loss if expected loss is already provisioned). If $\text{OpVaR}_{99.9}$ ≈ $28m and EL = $2m, capital ≈ **$26m**.

Self-check. The key sanity test: capital ($26m) must vastly exceed expected loss ($2m) — a ratio of ~13×. That fat multiple is the signature of operational risk's heavy tail; for a thin-tailed risk it might be 2–3×. If your model spat out capital only slightly above expected loss, you would know the severity tail was mis-specified. The mean-severity of $500k and the 99.9% single-loss of $16.8m differ by ~34×, confirming the lognormal's fat right tail is doing the work. ✓

Example 4 — Internal Loss Multiplier (Basel III SMA)

A bank has Business Indicator Component (BIC) = $1,200m. Its average annual operational loss over 10 years is $80m, so Loss Component LC = 15 × 80 = $1,200m.

$$ILM = \ln\!\left(e^{1} - 1 + \left(\tfrac{1200}{1200}\right)^{0.8}\right) = \ln(2.718 - 1 + 1) = \ln(2.718) = 1.00$$

Capital = BIC × ILM = 1,200 × 1.00 = **$1,200m**.

Self-check. When LC = BIC (loss history is exactly "average" relative to size), the ratio is 1, ILM collapses to $\ln(e) = 1$, and capital = BIC. Now suppose the bank's loss record worsens so LC = $2,400m (ratio 2): ILM = $\ln(e - 1 + 2^{0.8}) = \ln(1.718 + 1.741) = \ln(3.459) \approx$ **1.24**, so capital rises to 1,200 × 1.24 = **$1,488m** — a 24% surcharge for a poor loss history. Conversely a spotless record (LC → 0) gives ILM = $\ln(e - 1) = \ln(1.718) \approx$ **0.54**, a large discount. The formula behaves monotonically and symmetrically around the average — exactly as designed. ✓

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6. Connections

- **To market and credit risk (Chapters 5–7).** Operational risk is the third pillar-1 charge alongside them. Crucially, big market or credit losses are often *operational* failures in disguise: a rogue trader is "market loss" only on the surface — the root cause is failed supervision (operational). Interviewers love this: "*Was Barings a market-risk or operational-risk failure?*" Answer: the loss was in derivatives (market), but the *cause* was operational — no segregation of duties.
- **To model risk.** A wrong valuation or capital model is an operational-risk event (event type 7, EDPM). Model risk is increasingly carved out as its own discipline but sits under the operational umbrella.
- **To cyber and IT resilience.** Cyber risk is the fastest-growing slice of operational risk (event types 6 and 2). Regulators now demand **operational resilience** — the ability to keep critical services running through disruption — as a distinct expectation.
- **To business continuity / disaster recovery (BCP/DR).** The control response to event type 5 (physical damage) and 6 (system failure).
- **To conduct and compliance.** Event type 4 (CPBP) links operational risk to the entire AML/KYC, mis-selling, and market-abuse agenda — the biggest dollar losses in the industry.
- **To ICAAP and Pillar 2.** Even under standardised capital, the board's Internal Capital Adequacy Assessment Process must judge whether the formula captures the firm's true operational tail — scenario analysis feeds this directly.
- **To insurance.** The main *transfer* tool — but with a 20% recognition cap under AMA and the caveat that insurers may dispute payouts precisely when you need them.

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7. Key Terms

- **Operational risk** — loss from inadequate/failed processes, people, systems, or external events; includes legal, excludes strategic/reputational.
- **Loss event types** — Basel's seven Level-1 categories (internal fraud, external fraud, EPWS, CPBP, damage to physical assets, business disruption/system failure, EDPM).
- **Internal Loss Data (ILD)** — the firm's own recorded loss history; empirical backbone of modelling.
- **External Loss Data (ELD)** — consortium data (e.g. ORX) on peers' losses, used to populate the tail.
- **Scenario analysis** — structured expert estimation of rare, severe events.
- **BEICF** — Business Environment and Internal Control Factors; forward-looking control indicators.
- **BIA / TSA / AMA** — Basel II capital approaches: Basic Indicator, Standardised, Advanced Measurement.
- **SMA** — Basel III Standardised Measurement Approach replacing all of the above.
- **Business Indicator (BI) / BIC** — income-based scale proxy and its capital component under SMA.
- **Internal Loss Multiplier (ILM)** — SMA factor scaling capital by 10-year loss history.
- **Loss Distribution Approach (LDA)** — frequency × severity convolution producing aggregate loss.
- **OpVaR** — operational value-at-risk, the 99.9%/1-year loss quantile.
- **KRI** — Key Risk Indicator; a metric with thresholds that flags rising risk.
- **RCSA** — Risk and Control Self-Assessment; the business's structured self-rating of risks and controls.
- **Three Lines of Defence** — business / risk & compliance / internal audit governance model.
- **Risk appetite** — the board-set amount and type of operational risk the firm will accept.
- **Maker-checker / segregation of duties** — the core preventive controls: separate the person who initiates from the person who approves.
- **Operational resilience** — ability to keep critical services running through disruption.

8. Common Confusions

"Operational risk = IT risk." No. IT/systems is only *one* of the four sources. The largest losses come from people (fraud, conduct) and processes, not just systems.

"Reputational risk is part of operational risk capital." No — it is explicitly *excluded* from the Basel capital definition (as is strategic risk). An operational event can *cause* reputational damage, but the reputational hit itself is not in the capital charge. Legal risk, by contrast, *is* included.

"BIA and TSA treat a bad year the same way." No — the trap in Examples 1 and 2. BIA *drops* non-positive years (divides by fewer years); TSA *floors the yearly total at zero but keeps the year* (divides by three). Same data, different capital.

"Higher confidence means we hold capital against expected loss." Capital is held against *unexpected* loss (OpVaR minus expected loss), on the assumption that expected loss is already provisioned/priced in. Confusing the two double-counts or under-counts.

"OpVaR scales with the mean loss." No. It is driven by the *tail* of the severity distribution. A cell with modest expected loss can carry huge capital if severity is fat-tailed (Example 3: EL \$2m, capital \$26m).

"Outsourcing removes the risk." Outsourcing *transfers the operation* but not the accountability, and it *creates new vendor/third-party risk*. Regulators hold the firm responsible for outsourced activities.

"A rogue trader is a market-risk failure." The loss shows up in market P&L, but the *risk that failed* is operational — supervision, segregation of duties, and reconciliation. Root cause, not P&L line, defines the category.

"KRIs and losses are the same thing." KRIs are *leading* (predictive metrics like turnover or reconciliation breaks); loss data is *lagging* (what already happened). A good framework uses KRIs to act *before* the loss event registers.

"AMA is still the gold standard." Basel III *abolished* AMA precisely because internal models were inconsistent and un-comparable. The direction of travel is toward the simpler, standardised SMA.

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9. Recap

Operational risk is the risk of loss from failures in the firm's own people, processes, and systems, or from external events — including legal risk, excluding strategic and reputational. Unlike market and credit risk, it is *pure downside*, *pervasive across the whole firm*, and *severely fat-tailed*: a drizzle of small losses punctuated by rare catastrophes that drive nearly all the required capital.

Basel organises losses into **seven event types** (internal/external fraud, employment practices, clients-products-business-practices, physical damage, business disruption/system failure, and execution-delivery-process-management), with conduct (CPBP) and processing (EDPM) carrying the biggest aggregate dollar losses.

Capital measurement evolved through three Basel II approaches — **BIA** (15% of gross income), **TSA** (business-line betas of 12/15/18%), and **AMA** (internal 99.9%/1-year models built via the Loss Distribution Approach from four data elements) — before Basel III scrapped all three for the single **SMA** formula, capital = **BIC × ILM**, restoring comparability while keeping loss-history sensitivity.

Management runs on a closed loop — identify (RCSA, scenarios), assess, control (the 4 T's: treat, transfer, terminate, tolerate), monitor (**KRIs** and loss data), report — all governed by the **three lines of defence** and anchored to a board-set **risk appetite**. The famous disasters (Barings, Société Générale, cyber breaches, settlement failures) are all failures of *control quality*, not bad luck.

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10. Quick-Reference / Interview Points

Definition (memorise verbatim): "The risk of loss resulting from inadequate or failed internal processes, people and systems, or from external events. Includes legal risk; excludes strategic and reputational risk."

Four sources: People, Processes, Systems, External events (PPSE).

Seven event types: Internal fraud · External fraud · Employment practices & workplace safety · Clients, products & business practices · Damage to physical assets · Business disruption & system failures · Execution, delivery & process management.

Capital formulas at a glance:

Approach	Formula	One-line summary
BIA	K = 0.15 × avg positive GI (3 yr)	Crude; drops loss years
TSA	K = avg of yearly $\Sigma(\beta_j \times GI_j)$, floored at 0	Betas 12/15/18% by business line
AMA	OpVaR at 99.9% / 1 yr via LDA	Internal model; abolished by Basel III
SMA	K = BIC × ILM	BI-based; ILM scales by 10-yr loss history

ILM formula: ILM = ln(e − 1 + (LC/BIC)^0.8), where LC = 15 × avg annual loss over 10 years. LC = BIC → ILM = 1.

Famous cases to name-drop (cause, not just headline): - **Barings 1995** — Nick Leeson, ~£800m, no segregation of duties (he ran both trading and back office). Sank a 230-year-old bank. - **Société Générale 2008** — Jérôme Kerviel, ~€4.9bn, unauthorised positions hidden via fake offsetting trades; failed reconciliation and ignored alerts. - **UBS 2011** — Kweku Adoboli, ~$2.3bn, rogue trading via fictitious hedges. - **Knight Capital 2012** — botched software deployment, ~$440m lost in 45 minutes; change-management failure (event type 6/7). - **Equifax 2017 / cyber** — unpatched vulnerability, 147m records breached; external fraud + system failure, huge legal and reputational tail. - **Mis-selling / conduct fines** — PPI, LIBOR, FX rigging: tens of billions, the CPBP category, the biggest aggregate operational loss theme of the decade.

Interview one-liners: - "Operational risk is the only major risk with no upside — you never get paid to bear it." - "Capital is driven by the tail, not the average; that's why we model frequency × severity, not a normal VaR." - "A rogue trader is an operational failure wearing a market-risk costume — the root cause is segregation of duties." - "You can outsource the process but never the accountability." - "KRIs are leading, loss data is lagging — the whole point is to act before the loss books." - "Basel III killed AMA because you can't compare two banks' internal op-risk models — SMA trades sensitivity for comparability."

Three lines of defence: business (owns risk) · risk & compliance (framework & challenge) · internal audit (assurance).

Four responses (4 T's): Treat · Transfer · Terminate · Tolerate.

Q&A — Operational Risk

A practice bank for the Operational Risk chapter. Work each question before reading the answer. Numerical answers are self-checked against the Basel identities: BIA capital = 0.15 × avg positive gross income; SMA capital = BIC × ILM with ILM = ln(e − 1 + (LC/BIC)^0.8) and LC = 15 × 10-year average annual loss.

Section A — Concept-Check (short answer)

A1. State the Basel definition of operational risk verbatim, and name the two boundary clauses.

"The risk of loss resulting from inadequate or failed internal processes, people and systems, or from external events." The two boundary clauses: **legal risk is included** (fines, penalties, settlements, failed-contract costs), and **strategic and reputational risk are excluded** from the regulatory capital definition. The subtlety is that an operational *event* (say a data breach) can *cause* reputational damage, but only the direct operational loss counts for capital.

A2. What are the four sources of operational risk (PPSE), and which produces the largest single losses?

People, Processes, Systems, External events. *People* — human error, negligence, and deliberate wrongdoing (fraud, rogue trading) — is the single richest source of large, spectacular losses. Note the distinction between "most spectacular single event" (internal fraud) and "largest aggregate dollar loss" (conduct and processing — see A5).

A3. Why is operational risk described as "pure downside," and how does that differ from market and credit risk?

Market and credit risk are risks a firm *takes on purpose* to earn a spread — there is an upside, and you can price, hedge, and earn a return for bearing them. Operational risk has **no upside**: nobody gets paid to be defrauded or to suffer a system outage. It is embedded in everything the firm does, pervasive rather than localised to one book, and offers no compensating return.

A4. Describe the shape of the operational-loss distribution and say which part drives capital.

It is **severely fat-tailed and asymmetric**: a high-frequency, low-severity *body* (data-entry errors, small mis-payments — predictable, budgetable) and a low-frequency, high-severity *tail* (rogue trader, systemic cyber breach, class-action settlement — rare but can exceed a year's profit). Economic capital is driven almost entirely by the **tail**, which is why op-risk modelling borrows from actuarial science and extreme-value theory rather than the normal distribution of basic market VaR.

A5. Name the seven Basel Level-1 event types, and say which two carry the largest aggregate losses.

Internal fraud; External fraud; Employment practices & workplace safety; Clients, products & business practices (CPBP); Damage to physical assets; Business disruption & system failures; Execution, delivery & process management (EDPM). The two carrying the largest **aggregate** dollar losses are **CPBP (conduct/mis-selling)** and **EDPM (processing errors)** — these dwarf internal fraud by total value even though fraud produces the most headline-grabbing single events.

A6. Why is there "no natural exposure measure" for operational risk, and how did regulators solve it?

Credit risk has exposure-at-default; market risk has position sizes. Operational risk has no clean dollar exposure — the "exposure" is the whole firm. Regulators solved it pragmatically with a **proxy for the scale**

of activity: gross income (BIA/TSA) or later the Business Indicator (SMA), on the logic that a bigger, busier firm has more operational surface area and therefore more to go wrong.

A7. What are the four data elements that feed operational-risk measurement and management?

Internal loss data (ILD) — the firm's own recorded loss history, the empirical backbone; **External loss data (ELD)** — peers' losses via consortia like ORX, used to populate tail events you have been lucky to avoid; **Scenario analysis** — structured expert workshops estimating rare, severe events; **Business environment and internal control factors (BEICF)** — forward-looking control indicators (KRIs, audit findings, turnover). These are the "AMA data elements."

A8. What is the difference between a KRI and loss data, and why does it matter?

A **KRI is leading** — a predictive metric with a threshold (staff turnover, aged reconciliation breaks, system downtime) that flags rising risk *before* a loss crystallises. **Loss data is lagging** — it records what has already happened. The whole point of a good framework is to act on the KRI before the loss books. A KRI that never moves before a loss is theatre; good practice correlates KRIs against actual losses to prove they are predictive.

A9. Explain the Three Lines of Defence.

First line — the business: owns and manages its own risks day-to-day and owns the controls. **Second line — risk management & compliance**: sets framework, policy, and appetite, and independently challenges the first line. **Third line — internal audit**: provides independent assurance to the board that lines one and two actually work. The board and risk committee sit above, setting appetite.

A10. List the four responses to an identified operational risk (the 4 T's).

Treat/mitigate (strengthen controls — maker-checker, automation, segregation of duties); **Transfer** (insurance or outsourcing — but you transfer the operation, not the accountability); **Terminate/avoid** (exit the product or process); **Tolerate/accept** (consciously retain within appetite because control costs more than it saves). The tool that identifies and rates the risk beforehand is the **RCSA** (Risk and Control Self-Assessment).

Section B — Numerical / Applied (full solutions)

B1. BIA with a loss year. A bank reports gross income ($m): Year-1 = 900, Year-2 = −150, Year-3 = 750. Compute the Basic Indicator Approach capital charge.

Rule: drop any year with non-positive GI from *both* numerator and denominator; $\alpha = 0.15$. - Positive-GI years: Year-1 (900) and Year-3 (750). Year-2 is excluded entirely. - $n = 2$. - Average positive GI = $(900 + 750) / 2 = 825$. - Capital = $0.15 \times 825 =$ **$123.75m**.

Self-check. The common trap averages all three years and nets the loss: $(900 - 150 + 750)/3 = 500 \rightarrow 0.15 \times 500 = \$75\text{m}$. That is wrong under BIA — negative years are *dropped, not netted*. Dropping the loss year raises the average (825 vs 500) and therefore raises capital. The rule deliberately prevents a bank from *lowering* its capital charge by having had a disastrous year. ✓

B2. TSA with business lines. In a given year a bank's gross income by business line is: Trading & sales (β = 18%) = 400; Corporate finance (β = 18%) = 200; Retail banking (β = 12%) = 500; Asset management (β = 12%) = 100. Compute that year's weighted $\beta \times$ GI total.

- Trading & sales: $0.18 \times 400 = 72$

- Corporate finance: 0.18 × 200 = 36
- Retail banking: 0.12 × 500 = 60
- Asset management: 0.12 × 100 = 12
- Weighted total = 72 + 36 + 60 + 12 = **$180m** for the year.

Self-check. Under TSA you sum across business lines *within* the year first, then floor that yearly total at zero, then average over three years. Here the total is positive so it is kept as 180. If this had come out negative, it would be floored to 0 for the year but the year would still count (divide by 3) — unlike BIA which drops it. ✓

B3. TSA vs BIA — opposite treatment of a loss year. A bank's weighted β × GI totals are: Year-1 = +140, Year-2 = −60, Year-3 = +100. Compute the TSA charge and contrast the mechanic with BIA.

- Year-1 = +140 → keep 140.
- Year-2 = −60 → **floored to 0** (kept in the average).
- Year-3 = +100 → keep 100.
- Capital = (140 + 0 + 100) / 3 = 240 / 3 = **$80m**.

Self-check. Under BIA the negative year would be *dropped* (divide by 2); under TSA it is *floored but retained* (divide by 3). Same data, different denominators — TSA is not simply "BIA with betas." The floor-and-keep mechanic drags the TSA average down relative to BIA's drop-the-year mechanic. ✓

B4. LDA → expected annual loss. A cell has frequency ~ Poisson(λ = 5 events/year) and severity ~ Lognormal(μ = 11, σ = 1.4) in log-dollars. Find the mean severity per event and the expected annual loss.

- Mean severity (lognormal) = $\exp(\mu + \sigma^2/2) = \exp(11 + 0.98) = \exp(11.98) \approx$ **$159,000**.
- Expected annual loss = λ × mean severity = 5 × 159,000 ≈ **$795,000 per year**.

Self-check. Expected loss is the *body* of the distribution — what you provision and budget. It uses the mean severity, not a tail quantile. Multiplying frequency by mean severity is the correct construction for the aggregate mean of a compound-Poisson process. ✓

B5. OpVaR tail vs mean. For the cell in B4, estimate the 99.9th-percentile *single* loss ($z_{0.999} \approx 3.09$) and comment on the ratio to mean severity.

- 99.9% single loss = $\exp(\mu + \sigma \cdot z_{0.999}) = \exp(11 + 1.4 \times 3.09) = \exp(15.326) \approx$ **$4.53m**.
- Ratio to mean severity = 4.53m / 159k ≈ **28×**.

Self-check. A ~28× gap between the mean event and the 99.9% event is the signature of the lognormal's fat right tail — exactly why OpVaR is driven by the tail, not the average. Capital held ≈ OpVaR − expected loss; if the model returned capital only slightly above expected loss you would know the severity tail was mis-specified. ✓

B6. Internal Loss Multiplier — average loss record. A bank has BIC = $2,000m and a 10-year average annual operational loss of $133.33m, so LC = 15 × 133.33 = $2,000m. Compute the ILM and SMA capital.

$$ILM = \ln\!\left(e^{1} - 1 + \left(\tfrac{2000}{2000}\right)^{0.8}\right) = \ln(2.718 - 1 + 1) = \ln(2.718) = 1.00$$

Capital = BIC × ILM = 2,000 × 1.00 = **$2,000m**.

Self-check. When LC = BIC the ratio is 1, so ILM = ln(e) = 1 and capital equals BIC — the "average loss experience" anchor. Everything hinges on whether the bank's loss record is better or worse than average for its size. ✓

B7. ILM — poor loss record. Same bank, but its 10-year average loss worsens so LC = \$4,000m (ratio LC/BIC = 2). Compute the new ILM, capital, and the surcharge versus B6.

- $(LC/BIC)^{0.8} = 2^{0.8}$. $\ln 2 = 0.6931$; $\times\ 0.8 = 0.5545$; $e^{0.5545} \approx 1.741$.
- $ILM = \ln(e - 1 + 1.741) = \ln(2.718 - 1 + 1.741) = \ln(3.459) \approx$ **1.241**.
- Capital = 2,000 × 1.241 = **\$2,482m**.
- Surcharge = 2,482 − 2,000 = \$482m, roughly **+24%** for a doubled loss record.

Self-check. A worse loss history raises capital, but only sub-linearly (loss doubled, capital up ~24%, not 100%) because the multiplier uses a 0.8 power and a log wrapper — deliberate dampening so a single bad decade does not explode capital. Monotonic and well-behaved. ✓

B8. ILM — spotless record. A bank with BIC = \$2,000m has essentially no loss history, so LC → 0. Compute the ILM and capital, and state the floor intuition.

- With LC = 0: $ILM = \ln(e - 1 + 0^{0.8}) = \ln(e - 1) = \ln(1.718) \approx$ **0.541**.
- Capital = 2,000 × 0.541 = **\$1,082m**.

Self-check. A pristine record earns a discount to ~54% of BIC — the practical floor of the multiplier. The formula is symmetric around the average: it rewards a clean record and penalises a bad one, both capped in magnitude by the log. Note many national regulators simply set ILM = 1, switching off loss-history sensitivity entirely. ✓

Section C — Interview-Style (model answers)

C1. "Was Barings a market-risk failure or an operational-risk failure?"

The loss showed up in derivatives P&L, so it *looks* like market risk — Nick Leeson's Nikkei futures positions. But the risk that actually *failed* was operational. Leeson ran both the trading desk and the back office, so there was **no segregation of duties**: he could book fictitious offsetting trades and hide the losses in an error account (the famous 88888). The root cause was a control failure — no maker-checker, no independent reconciliation, no supervision. The lesson interviewers want: **classify by root cause, not by the P&L line.** A rogue trader is an operational failure wearing a market-risk costume.

C2. "Why did Basel III abolish the AMA?"

The Advanced Measurement Approach let each bank build its own internal model to hit the 99.9%/1-year capital standard. In principle it was the most risk-sensitive method; in practice the models proved **inconsistent and impossible to compare** across banks — two firms with similar risk profiles could produce wildly different capital numbers depending on modelling choices, creating model risk and gaming incentives. Basel III (finalised 2017) scrapped AMA *and* the old BIA/TSA menu, replacing everything with the single non-modelled **SMA** formula, capital = BIC × ILM. The trade-off is explicit: **SMA sacrifices sensitivity for comparability and reduced model risk**, while keeping *some* risk sensitivity through the loss-history multiplier.

C3. "How would you build a KRI framework for a trading operation?"

I would start from the risks I am trying to see coming and pick *leading* indicators for each. For fraud/conduct: number of limit breaches, off-hours trading activity, cancel-and-correct rates, and trader-attestation lapses — the classic rogue-trading precursors. For processing: aged/failed settlements and reconciliation breaks open beyond X days. For systems: downtime minutes, change-failure rate, patch backlog. Each KRI needs a

green/amber/red threshold, a named owner, a fixed review cadence, and an escalation trigger. Critically, I would **back-test each KRI against actual loss data** to prove it is predictive — a KRI that never moves before a loss is decorative and should be replaced. The output feeds the risk committee so we act before the loss books.

C4. "Does outsourcing a process remove the operational risk?"

No. Outsourcing **transfers the operation but not the accountability** — regulators hold the firm responsible for outsourced activities as if performed in-house. Worse, it *creates new risk*: **vendor/third-party risk**, concentration risk if many firms use the same provider, and the loss of direct control over the vendor's own control environment. So outsourcing is a *transfer* response under the 4 T's, but it must be paired with vendor due diligence, SLA monitoring (itself a KRI), exit planning, and continued board oversight. The one-liner: **you can outsource the process, never the accountability.**

C5. "Walk me through how operational-risk capital is actually computed today under the SMA."

Two ingredients. First the **Business Indicator (BI)** — an income proxy built from three components: an interest/lease/dividend component, a services component, and a financial (trading + banking book) component. The BI is mapped through a progressive marginal-coefficient schedule into the **Business Indicator Component (BIC)** — bigger banks attract higher marginal rates. Second the **Internal Loss Multiplier (ILM)** = $\ln(e - 1 + (LC/BIC)^{0.8})$, where the **Loss Component LC = 15 × the average annual operational loss over 10 years**. Capital = **BIC × ILM**. When loss experience is average (LC = BIC), ILM = 1 and capital = BIC; a poor loss record pushes ILM above 1, a clean one below. National regulators may set ILM = 1 to remove loss-history sensitivity.

C6. "Give me an operational-risk case that was purely a systems/change-management failure."

Knight Capital, 2012. A botched software deployment — new code went live while dormant legacy code was accidentally reactivated on some servers — caused the firm's system to fire millions of erroneous orders into the market. It lost roughly **$440m in about 45 minutes** and effectively destroyed the firm. This is a textbook **event type 6/7** (business disruption/system failure and execution/process management) failure of **change management** — no proper deployment controls, no kill-switch, no reconciliation of what code was running where. It shows operational risk's tail can crystallise in minutes, not years.

Section D — MCQs (with reasoning)

D1. Which of the following is EXCLUDED from the Basel operational-risk capital definition? A) Legal risk B) Reputational risk C) External fraud D) Failed internal processes

Answer: B. Reputational (and strategic) risk is explicitly excluded. Legal risk (A) is *included*; external fraud (C) and failed processes (D) are core sources. Trap: an operational event can *cause* reputational damage, but the reputational hit is not in the capital charge.

D2. Under the BIA, a year with negative gross income is: A) Netted against positive years B) Floored at zero but retained in the average C) Dropped from both numerator and denominator D) Multiplied by beta

Answer: C. BIA drops non-positive years entirely (divide by fewer years). Option B describes **TSA**, not BIA — that is the classic BIA/TSA confusion. Option D confuses BIA (no betas) with TSA.

D3. Which two event types carry the largest AGGREGATE dollar losses in the industry? A) Internal fraud and damage to physical assets B) CPBP and EDPM C) External fraud and EPWS D) System failure and internal fraud

Answer: B. Clients/Products/Business Practices (conduct, mis-selling, AML fines) and Execution/Delivery/Process Management (settlement and processing errors) dominate by total value. Internal fraud produces the most *spectacular single* events but not the largest aggregate — a common trap.

D4. In the Loss Distribution Approach, frequency is typically modelled as ___ and severity as ___. A) Lognormal; Poisson B) Poisson; Lognormal C) Normal; Normal D) Binomial; Uniform

Answer: B. Frequency (event count per year) is Poisson with parameter λ; severity (dollar size per event) is lognormal or a heavy-tailed distribution such as generalised Pareto for the tail. The two are convolved (usually by Monte Carlo) to get the aggregate annual loss and read off the 99.9th percentile.

D5. When the Loss Component equals the Business Indicator Component (LC = BIC), the Internal Loss Multiplier equals: A) 0 B) 0.5 C) 1.0 D) 1.5

Answer: C. ILM = ln(e − 1 + (LC/BIC)^0.8) = ln(e − 1 + 1) = ln(e) = 1. This is the "average loss experience" anchor where capital = BIC exactly.

D6. Operational-risk economic capital is held primarily against: A) Expected loss B) The mean of the loss distribution C) The tail (unexpected loss) D) Gross income

Answer: C. Capital covers *unexpected* loss (OpVaR minus expected loss), on the assumption expected loss is already provisioned/priced. Confusing the mean (A/B) with the tail double-counts or under-counts. Gross income (D) is only a scale *proxy*, not what capital defends against.

D7. Which control most directly addresses the Barings-type rogue-trader risk? A) Insurance B) Segregation of duties / maker-checker C) Higher gross-income capital D) External loss data

Answer: B. Separating the initiator from the approver (and trading from back-office reconciliation) is the core preventive control that Barings lacked. Insurance (A) is a *transfer*, not prevention; capital (C) absorbs losses rather than preventing them; ELD (D) informs modelling, not day-to-day control.

D8. A KRI differs from loss data because a KRI is: A) Lagging and backward-looking B) Leading and predictive C) A regulatory capital figure D) Only used in the third line of defence

Answer: B. KRIs are leading metrics (turnover, aged reconciliation breaks, downtime) that flag rising risk *before* a loss. Loss data is the lagging record of what already happened. The whole point is to act on the KRI first.

Recap of key formulas (self-verify against these)

- **BIA:** K = 0.15 × average *positive* GI (3 yr); drop non-positive years.
- **TSA:** K = avg over 3 yr of max(Σ_j β_j × GI_j, 0); betas 12/15/18%; floor the yearly total, keep the year.
- **LDA:** compound Poisson(λ) of lognormal severities; mean severity = exp(μ + σ^2/2); capital ≈ $OpVaR_{99.9}$ − EL.
- **SMA:** K = BIC × ILM, ILM = ln(e − 1 + (LC/BIC)^0.8), LC = 15 × 10-yr avg loss; LC = BIC ⇒ ILM = 1.

Chapter 09 — Liquidity Risk

> "Solvency is the reason a firm exists tomorrow; liquidity is the reason it survives tonight." A bank can be solvent on paper — assets comfortably exceeding liabilities — and still fail by Friday because it cannot lay its hands on cash when depositors ask for it. Liquidity risk is the risk that a firm cannot meet its obligations as they fall due without incurring unacceptable losses. It is the risk that killed Northern Rock (2007), Bear Stearns and Lehman Brothers (2008), and Silicon Valley Bank and Credit Suisse (2023). None of these died of pure insolvency first. They died of liquidity.

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1. The Problem / The Need

Every financial institution lives on a promise it can never fully keep at once. A bank takes deposits that are repayable **on demand** and lends them out as mortgages repayable over **thirty years**. This is *maturity transformation* — borrowing short and lending long — and it is the core economic function of banking. It is also the seed of a permanent vulnerability: if enough depositors ask for their money on the same day, no bank on earth can pay them all, because the money is not sitting in a vault. It is locked up in illiquid loans.

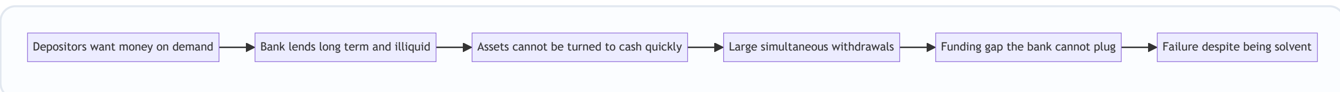
Consider the everyday cash-flow problem stripped of jargon. On any given day a bank has cash coming in (loan repayments, maturing bonds, new deposits) and cash going out (deposit withdrawals, loans it committed to fund, debt it must roll over, margin calls, coupon payments). If, on a given day, outflows exceed inflows plus available cash, the bank has a **funding gap** it must plug — by borrowing, by selling assets, or by defaulting. The last option ends the firm.

The problem has three distinct edges:

1. **Timing.** Obligations and receipts do not line up. A payment is due Tuesday; the cash to cover it arrives Friday. Between them lies a gap that must be bridged.
2. **Confidence.** Funding is a matter of trust. The moment counterparties doubt a firm can repay, they stop lending to it and demand their money back — which *causes* the very failure they feared. Liquidity risk is reflexive and self-fulfilling in a way credit and market risk are not.
3. **Market conditions.** The escape hatch — selling assets to raise cash — only works if there are buyers at fair prices. In a crisis, everyone reaches for the same hatch at once, prices collapse, and selling to raise cash *destroys* the capital the firm was trying to protect.

Regulators learned this brutally in 2007–08. Basel II had a lot to say about capital (solvency) and almost nothing enforceable about liquidity. Firms with pristine capital ratios still collapsed in days when short-term funding markets froze. The regulatory response — the **Liquidity Coverage Ratio (LCR)** and the **Net Stable Funding Ratio (NSFR)** under Basel III — exists precisely because the industry discovered that capital adequacy without liquidity adequacy is a fortress with no water supply.

Figure 1 — Why maturity transformation creates liquidity risk.



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2. The Core Idea

Liquidity risk splits into **two related but distinct** faces, and confusing them is the single most common mistake students make.

Funding liquidity risk is the risk that a firm cannot raise cash to meet its own obligations — that its *liabilities* run faster than it can refinance them. It is about the right-hand side of the balance sheet: deposits fleeing, wholesale lenders refusing to roll over commercial paper, margin calls demanding cash today. The question is: *Can I get cash to pay what I owe?*

Market liquidity risk is the risk that a firm cannot sell an *asset* quickly at close to its fair value — that the market for the asset is thin, so selling in size moves the price against you. It is about the left-hand side of the balance sheet. The question is: *Can I turn what I own into cash without a fire-sale loss?*

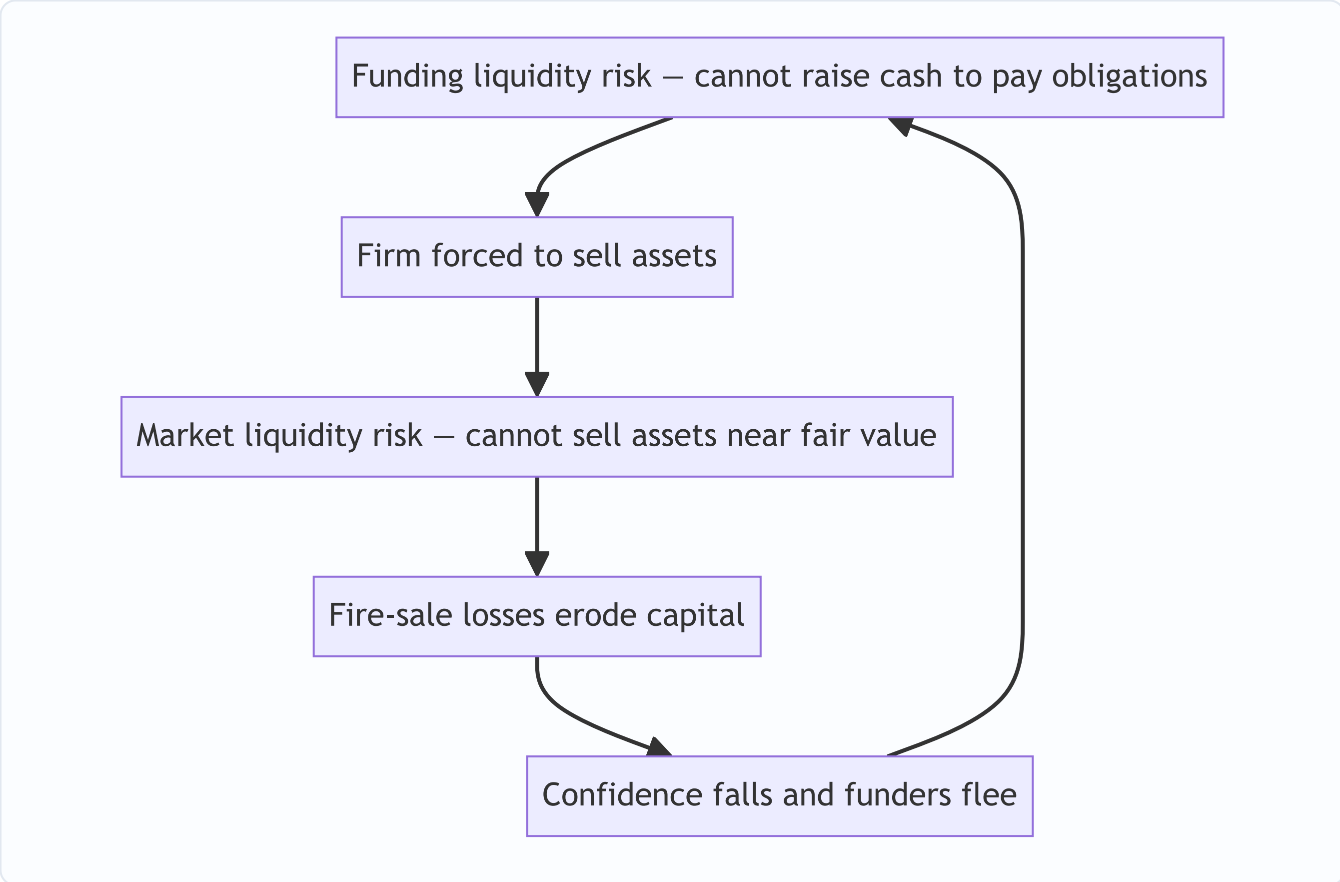
The two are joined at the hip. When a firm faces a **funding** shortfall, it tries to sell assets — but if those assets are **market-illiquid**, it must dump them at a discount, crystallising a loss that erodes capital, which frightens more funders, which deepens the funding shortfall. This vicious loop — the *liquidity spiral* — is how a manageable cash-flow wobble becomes a death spiral.

The core managerial idea is therefore twofold:

- Hold a buffer of **genuinely liquid, high-quality assets** (cash and government bonds) that can be converted to cash *without* fire-sale losses even in stress — so you never have to touch the illiquid assets.
- Structure your **funding** so that it is *stable* — long-dated, diversified, and hard to withdraw — so the outflows never spike faster than the buffer can absorb.

LCR governs the first (survive a 30-day acute stress on your buffer). NSFR governs the second (fund your illiquid assets with stable money over a one-year horizon). Together they force a firm to be liquid in the short run and structurally sound over the medium run.

Figure 2 — The two faces of liquidity risk and the spiral that joins them.



3. Why / How It Works

Why maturity transformation is both essential and dangerous

Society *wants* banks to transform maturity. Savers want instant access to their money; borrowers want long, patient loans. Banks bridge the two by exploiting the **law of large numbers**: on a normal day, only a small, predictable fraction of demand deposits is actually withdrawn, because withdrawals and deposits roughly cancel across thousands of independent customers. The bank keeps a small cash reserve for the net outflow and lends the rest.

This works *precisely as long as depositor behaviour stays independent and random*. The danger is that behaviour is not always independent. Fear is correlated. When depositors believe *others* will withdraw, the rational move is to withdraw first (before the bank runs dry), so the belief becomes self-fulfilling. The Diamond–Dybvig model (1983) formalised this: a bank offering demand deposits has **two equilibria** — a "good" one where only genuine liquidity-needers withdraw, and a "bad" one (the run) where everyone withdraws because everyone expects everyone else to. Nothing about the bank's *assets* need change to flip from one to the other. Liquidity risk is, at root, a coordination failure.

Why market liquidity evaporates exactly when you need it

Asset markets are liquid when there are many willing buyers holding cash. In a systemic stress, potential buyers are themselves scrambling for cash, so buyers vanish exactly when sellers multiply. Bid–ask spreads widen, market depth thins, and the price impact of selling in size explodes. This is why a bond that trades at 99.8 in calm markets might only fetch 85 in a fire sale — not because its fundamental cash flows changed, but because the *marginal buyer* disappeared. Market liquidity is **procyclical**: abundant in booms, absent in busts.

How the buffer defends the firm

The defence is a stock of **High-Quality Liquid Assets (HQLA)** — assets that stay liquid *even in stress* because they are what everyone flies *to*, not from: central-bank reserves and top-rated sovereign bonds. In a crisis these can be sold outright or, more importantly, **repo'd** (pledged to a central bank or counterparty for cash) at minimal haircut. The buffer buys the one thing a firm in a run needs most: **time**. Time to arrange orderly funding, to be acquired, or to convince the market it is sound — instead of being forced into a fatal fire sale on day one.

How stable funding defends the firm

The other lever is the *quality* of funding. Not all liabilities are equally flighty. A ranking, most-stable to least-stable:

Funding source	Stability	Why
Equity capital	Permanent	Never has to be repaid
Long-term debt (>1yr)	Very stable	Contractually locked in
Retail deposits (insured)	Stable	Insured, sticky, diversified, "sleepy"
Retail deposits (uninsured)	Moderate	Can flee if scared
Operational corporate deposits	Moderate	Tied to services the firm provides
Wholesale unsecured funding	Flighty	Sophisticated lenders flee at first sign
Overnight / short-term wholesale	Very flighty	Refuses to roll over instantly

The lesson of every modern bank failure: firms funded by **short-term wholesale money** die fast; firms funded by **insured retail deposits and long-term debt** survive. NSFR is designed to push firms toward the top of this table.

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4. Full Content — Framework, Formulas, and Methods

4.1 The Liquidity Gap and Maturity Mismatch

The foundational measurement tool is the **liquidity gap** (also "maturity gap" or "funding gap"). You slot every asset and liability into **time buckets** by when it matures or reprices, then compute, per bucket:

$$\text{Liquidity Gap}_t = \text{Cash Inflows}_t - \text{Cash Outflows}_t$$

Equivalently, on a stock basis:

$$\text{Gap}_t = \text{Assets maturing in bucket } t - \text{Liabilities maturing in bucket } t$$

A **negative gap** in a bucket (more liabilities than assets maturing) means the firm must *refinance* the shortfall in that period — the signature of borrowing short and lending long. Banks almost always run *negative gaps in short buckets* (deposits are short) and *positive gaps in long buckets* (loans are long). That is maturity transformation quantified.

Two refinements matter:

- **Marginal (period) gap** = the gap *within* a single bucket.
- **Cumulative gap** = the running sum of marginal gaps up to bucket *t*. This is what truly matters, because a firm can carry cash forward. A firm survives if its **cumulative gap plus liquid buffer stays non-negative** at every horizon.

$$\text{Cumulative Gap}_T = \sum_{t=1}^{T} \text{Gap}_t$$

Behavioural adjustment is critical: demand deposits are *contractually* overnight but *behaviourally* stable for years. Good liquidity management slots them by **expected behaviour**, not legal maturity — otherwise every bank looks instantly insolvent on a contractual gap report.

4.2 The Liquidity Coverage Ratio (LCR)

The LCR is Basel III's **short-term, acute-stress** rule. It asks: *if a severe 30-day liquidity stress hit tomorrow, do you hold enough high-quality liquid assets to survive all 30 days without new funding?*

$$\boxed{\text{LCR} = \frac{\text{Stock of High-Quality Liquid Assets (HQLA)}}{\text{Total Net Cash Outflows over 30 days}} \geq 100\%}$$

Numerator — HQLA, after regulatory haircuts:

HQLA tier	Examples	Haircut	Cap
Level 1	Cash, central-bank reserves, top-rated sovereign bonds	0%	No cap
Level 2A	High-grade sovereign/agency, AA– corporate bonds	15%	Level 2 total ≤ 40% of HQLA
Level 2B	Lower-rated corporates (A+ to BBB–), some equities	25–50%	Level 2B ≤ 15% of HQLA

Denominator — Net Cash Outflows over 30 days, computed under a prescribed stress scenario:

$$\text{Net Outflows} = \text{Total Expected Outflows} - \min\big(\text{Total Expected Inflows},\; 75\% \times \text{Total Outflows}\big)$$

The **75% cap on inflows** is crucial and heavily tested: a bank may only offset outflows with inflows up to 75% of its outflows, forcing it to hold HQLA to cover *at least 25%* of gross outflows regardless of expected receipts. You cannot assume you'll be fully rescued by money coming in.

Outflows are computed by applying **run-off rates** to each liability (how much is assumed to flee in the stress):

Liability	Assumed run-off
Stable retail deposits (insured)	3–5%
Less stable retail deposits	10%+
Operational corporate deposits	25%
Non-operational corporate / unsecured wholesale	40–100%
Financial-institution funding	100%

Inflows apply **inflow rates** to incoming money (e.g., 50% on performing retail loan repayments, 100% on maturing interbank placements).

4.3 The Net Stable Funding Ratio (NSFR)

The NSFR is the **structural, one-year** rule. Where LCR is a 30-day sprint, NSFR is a marathon: *is the firm funding its illiquid assets with sufficiently stable money over a one-year horizon?* It directly attacks over-reliance on short-term wholesale funding.

$$\boxed{\text{NSFR} = \frac{\text{Available Stable Funding (ASF)}}{\text{Required Stable Funding (RSF)}} \geq 100\%}$$

ASF (numerator) = the firm's liabilities and capital weighted by how *stable* they are. **ASF factors** (the fraction that counts as stable):

Funding source	ASF factor
Regulatory capital, debt with residual maturity ≥ 1yr	100%
Stable retail/SME deposits (< 1yr)	95%
Less stable retail/SME deposits (< 1yr)	90%
Operational deposits, corporate funding (6–12 months)	50%
Everything else < 6 months, interbank	0%

RSF (denominator) = the firm's assets weighted by how *illiquid* they are — how much stable funding they *require*. **RSF factors**:

Asset	RSF factor
Cash, central-bank reserves	0%
Level 1 HQLA (top sovereigns)	5%
Level 2A HQLA	15%
Loans to financials < 1yr, Level 2B	15–50%
Retail/corporate loans < 1yr	50%
Residential mortgages, loans ≥ 1yr (good quality)	65%
Other loans ≥ 1yr, most corporate loans	85%
Illiquid assets, defaulted loans, fixed assets	100%

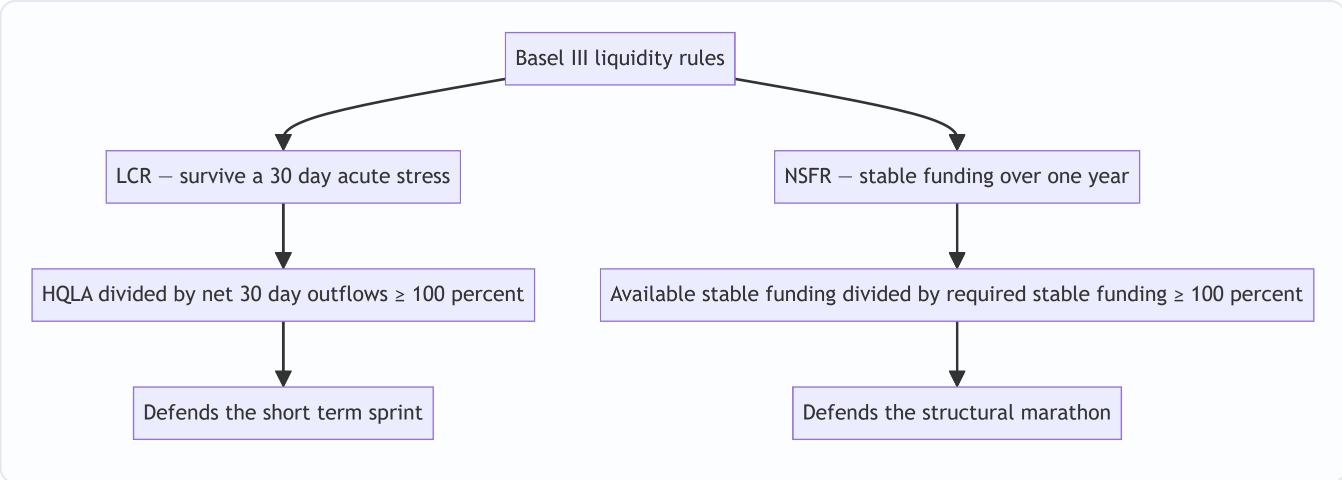
The mnemonic: **ASF rewards stable *sources*; RSF penalises illiquid *uses*.** A bank funding 30-year mortgages (65–100% RSF) with overnight wholesale money (0% ASF) will fail NSFR badly. Fund them with retail deposits and long bonds (90–100% ASF) and it passes.

4.4 Other Metrics and Tools

- **Loan-to-Deposit Ratio (LDR)** = Loans / Deposits. A crude but instant read: >100% means the loan book is funded partly by non-deposit (usually flightier) sources.
- **Survival horizon** = number of days the liquid buffer covers projected net outflows in stress before hitting zero. The internal cousin of LCR.
- **Concentration metrics** — largest depositors as % of funding; reliance on a single market or currency.

- **Contingency Funding Plan (CFP)** — a pre-agreed playbook of actions (draw committed lines, sell HQLA, pledge collateral, activate central-bank facilities) triggered at defined early-warning thresholds.

Figure 3 — The Basel III liquidity framework: two ratios, two horizons.



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5. Worked Examples

Example 1 — Building a liquidity gap ladder and survival horizon

A small bank reports the following contractual cash flows (₹ crore) by time bucket. Compute marginal gaps, cumulative gaps, and the survival horizon given an opening liquid buffer of ₹200 crore.

Bucket	Inflows	Outflows	Marginal gap	Cumulative gap	Buffer + cum. gap
Overnight–7d	120	350	−230	−230	−30
8–30d	260	200	+60	−170	+30
31–90d	180	140	+40	−130	+70
91–180d	150	120	+30	−100	+100
181–365d	300	210	+90	−10	+190

Working (self-verifying): - Marginal gaps: 120−350 = −230; 260−200 = +60; 180−140 = +40; 150−120 = +30; 300−210 = +90. - Cumulative gaps (running sum): −230; −230+60 = −170; −170+40 = −130; −130+30 = −100; −100+90 = **−10**. ✓ (Sum of marginal gaps = −230+60+40+30+90 = −10 ✓ — reconciles.) - Buffer overlay (₹200 + cumulative gap): 200−230 = **−30** in the first week; then 200−170 = +30; +70; +100; +190.

Interpretation: The bank's structure is textbook maturity transformation — a large negative gap up front (short liabilities) turning positive later (long assets). The killer: in the **overnight–7d** bucket the ₹200 crore buffer is *insufficient* — it is short by **₹30 crore**. Survival horizon is **less than 7 days**. The bank must either raise the buffer above ₹230 crore or reduce first-week outflows. From week two onward the position is comfortably positive. This is exactly the profile a run exploits: the firm is fine over a year but cannot survive the first week.

Example 2 — Computing the LCR

A bank has the following stressed 30-day profile. Compute its LCR and state whether it complies.

HQLA (after haircuts):

Asset	Market value	Haircut	HQLA value
Cash + central-bank reserves (L1)	400	0%	400
Sovereign bonds (L1)	300	0%	300
AA– corporate bonds (L2A)	200	15%	170
BBB corporate bonds (L2B)	100	50%	50

First check the Level 2 caps. Total HQLA before caps = 400+300+170+50 = 920. Level 2 total = 170+50 = 220, which is 220/920 = 23.9% — below the 40% cap ✓. Level 2B = 50, which is 5.4% — below the 15% cap ✓. So **HQLA = 920**.

Outflows (run-off applied):

Liability	Balance	Run-off	Outflow
Stable retail deposits	2,000	5%	100
Less-stable retail deposits	800	10%	80
Operational corporate deposits	1,000	25%	250
Unsecured wholesale funding	500	100%	500

Total outflows = 100+80+250+500 = **930**.

Inflows (inflow rate applied):

Inflow	Balance	Rate	Inflow
Maturing interbank placements	300	100%	300
Performing retail loan repayments	400	50%	200

Total inflows = 300+200 = **500**.

Apply the 75% inflow cap: capped inflows = min(500, 75% × 930) = min(500, 697.5) = **500** (uncapped here, since 500 < 697.5).

Net cash outflows = 930 − 500 = **430**.

$$\text{LCR} = \frac{920}{430} = 2.14 = \mathbf{214\%}$$

Verdict: Well above the 100% minimum — the bank comfortably survives the 30-day stress with more than double the required buffer.

Sensitivity check (reconciling the mechanism): Suppose the wholesale funding doubled to ₹1,000 crore (run-off 100% → outflow 1,000). New total outflows = 100+80+250+1,000 = 1,430. Inflow cap = 75% × 1,430 = 1,072.5, so inflows stay at 500. Net outflows = 1,430 − 500 = 930. LCR = 920/930 = **98.9%** — now *below* 100% and **non-compliant**. This demonstrates the core lesson quantitatively: **reliance on flighty wholesale funding is what breaks the LCR**, because it carries a 100% run-off. The same buffer that gave 214% comfort collapses below the line the moment the funding mix tilts toward hot money.

Example 3 — Computing the NSFR

Using stylised figures (₹ crore), compute the NSFR.

Available Stable Funding (ASF):

Source	Amount	ASF factor	ASF
Common equity + Tier 1 capital	1,000	100%	1,000
Long-term debt (≥ 1yr)	800	100%	800
Stable retail deposits (< 1yr)	3,000	95%	2,850
Operational corporate deposits	1,200	50%	600
Short-term wholesale (< 6m)	1,000	0%	0

Total ASF = 1,000+800+2,850+600+0 = **5,250**.

Required Stable Funding (RSF):

Asset	Amount	RSF factor	RSF
Cash + reserves	500	0%	0
Level 1 HQLA sovereigns	1,000	5%	50
Level 2A HQLA	600	15%	90
Residential mortgages (≥ 1yr)	3,000	65%	1,950
Corporate loans (≥ 1yr)	2,500	85%	2,125
Fixed assets	400	100%	400

Total RSF = 0+50+90+1,950+2,125+400 = **4,715**.

$$\text{NSFR} = \frac{5,250}{4,715} = 1.113 = \mathbf{111.3\%}$$

Verdict: Above 100% — the bank funds its illiquid assets (mortgages and corporate loans dominate RSF) with sufficiently stable money (equity, long debt, sticky retail deposits dominate ASF). It complies.

Reconciling stress: If the bank replaced ₹2,000 crore of its stable retail deposits (95% ASF → 1,900) with short-term wholesale funding (0% ASF → 0), ASF falls by 1,900 to 3,350. New NSFR = 3,350/4,715 = **71.0%** — a severe breach. Same assets, same size — but funding the *identical* mortgage book with hot money instead of sticky deposits blows the ratio apart. This is the structural analogue of Example 2's LCR result: **funding quality, not asset quality, drives the liquidity ratios.** Both worked examples point at the same villain — short-term wholesale funding — from the 30-day and the 1-year angle respectively.

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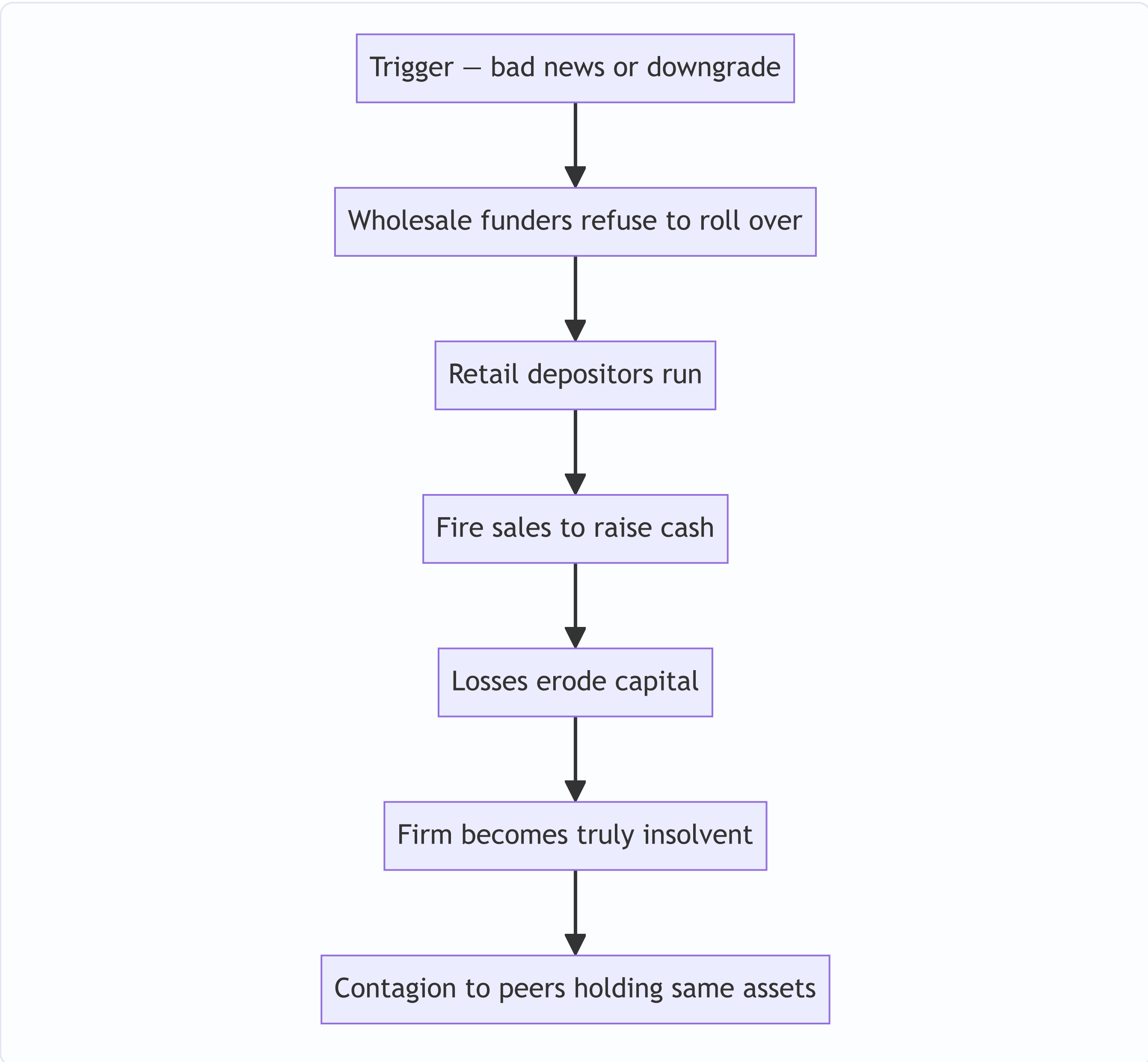
6. How Liquidity Crises Unfold — and How to Manage Them

The anatomy of a run

A liquidity crisis is a **sequence**, and recognising the stages is interview gold:

1. **Trigger** — bad news: a loss announcement, a ratings downgrade, a failed capital raise, a peer's collapse. (SVB: an unrealised bond loss plus a botched equity raise, March 2023.)
2. **Wholesale flight** — sophisticated counterparties move first. Interbank lenders refuse to roll over; repo haircuts jump; commercial paper won't reprice. This is silent and fast, invisible to the public.
3. **Retail run** — depositors follow, now amplified by social media and instant mobile transfers. SVB lost **\$42 billion in a single day** (25% of deposits) — a speed impossible in the branch-queue era of Northern Rock.
4. **Fire sales** — to raise cash, the firm dumps assets. Selling in size and in distress crushes prices, crystallising losses and eroding capital.
5. **Solvency contagion** — the fire-sale losses now make the firm *actually* insolvent, and mark-to-market losses spread to peers holding the same assets. Liquidity risk has metastasised into a systemic solvency event.
6. **Resolution** — central-bank lender-of-last-resort support, forced acquisition (JPMorgan–Bear Stearns, UBS–Credit Suisse), or failure (Lehman).

Figure 4 — The liquidity crisis cascade.



Managing liquidity risk — the toolkit

- **Liquid asset buffer (HQLA).** Hold enough unencumbered, high-quality assets to survive the plausible worst 30 days without new funding. Never encumber the whole buffer.
- **Funding diversification.** Spread across sources (retail, corporate, long-term debt, secured), tenors, currencies, and geographies. Cap reliance on any single counterparty or the overnight market.
- **Term out the funding.** Lengthen liability maturities so the roll-over cliff is small and spread over time — smooth the redemption ladder so no single day carries an outsized refinancing need.
- **Behavioural modelling.** Model *actual* deposit stickiness, prepayment, and drawdown of committed lines — not just contractual dates.
- **Intraday liquidity management.** In payment systems, cash must be in the right account at the right minute; intraday gaps can be fatal even when the daily position nets flat.
- **Contingency Funding Plan (CFP).** A pre-written, board-approved playbook with **early-warning indicators** (widening CDS, rising funding costs, deposit outflows, share-price falls) and graduated actions, so the firm acts on plan rather than panic.
- **Central-bank facilities.** Pre-positioned collateral at the central bank's discount window / standing facility is the ultimate backstop — but only if collateral is *already* lodged and eligible before the storm.

Stress testing liquidity

Stress testing is the heart of modern liquidity management — you deliberately model catastrophe and check you survive. Three scenario families:

1. **Idiosyncratic (name-specific) stress** — the firm alone is hit: a multi-notch downgrade, a scandal, a deposit run *only* on this firm. Markets stay open but shut *this* firm out.
2. **Market-wide (systemic) stress** — a general freeze: funding markets seize, asset prices crash, but the firm is not singled out (2008-style).
3. **Combined stress** — both at once, the regulatory worst case and the design basis for LCR's assumptions.

For each, project the **survival horizon**: apply run-off rates to liabilities, inflow rates to assets, haircuts to the buffer, and count the days until cumulative net outflows exhaust available liquidity. Reverse stress testing flips the question: *what scenario would it take to break us?* — and checks whether that scenario is plausibly close.

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7. Connections

- **Capital / solvency (Ch. on capital adequacy).** Liquidity and solvency are the two ways a firm dies. Capital ratios (CET1, leverage ratio) measure solvency; LCR/NSFR measure liquidity. A firm needs both — and a solvency scare *triggers* a liquidity run, while fire-sale liquidity losses *cause* insolvency. They are coupled, not independent.
- **Market risk.** Market liquidity risk is the bridge: a position's *value at risk* assumes you can exit at the marked price, but in stress the exit is illiquid, so realised losses exceed VaR. Liquidity-adjusted VaR (LVaR) widens VaR by the bid–ask spread and liquidation horizon.
- **Credit risk.** A credit deterioration (rising defaults) is a classic *trigger* for a liquidity crisis; and undrawn credit lines the firm has *granted* become contingent outflows when borrowers draw them in stress.

- **Operational risk.** Intraday liquidity and payment-system failures sit at the boundary of operational and liquidity risk.
- **Systemic risk / macroprudential policy.** Fire sales and funding contagion are *the* channels of systemic risk; LCR/NSFR are microprudential rules with an explicitly macroprudential purpose — reducing correlated fire-selling.
- **Interest-rate risk in the banking book (IRRBB).** The same maturity-mismatch ladder that drives liquidity gaps also drives repricing gaps and net-interest-income risk — one balance-sheet structure, two risk lenses.

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8. Key Terms

- **Funding liquidity risk** — inability to raise cash to meet obligations as they fall due.
- **Market liquidity risk** — inability to sell an asset quickly near fair value.
- **Maturity transformation** — funding long-dated assets with short-dated liabilities.
- **Liquidity gap** — inflows minus outflows in a time bucket; **cumulative gap** is the running sum.
- **HQLA** — High-Quality Liquid Assets; unencumbered assets convertible to cash in stress with little loss (Levels 1, 2A, 2B).
- **Run-off rate** — assumed % of a liability that flees during the LCR stress window.
- **LCR** — Liquidity Coverage Ratio = HQLA / net 30-day stressed outflows ≥ 100%.
- **NSFR** — Net Stable Funding Ratio = Available Stable Funding / Required Stable Funding ≥ 100%.
- **ASF / RSF factors** — weights for funding stability (ASF) and asset illiquidity (RSF).
- **Fire sale** — forced sale of assets at distressed prices to raise cash quickly.
- **Liquidity spiral** — self-reinforcing loop where funding stress forces fire sales, whose losses deepen funding stress.
- **Bank run** — mass simultaneous withdrawal driven by (self-fulfilling) fear.
- **Contingency Funding Plan (CFP)** — pre-agreed crisis playbook with early-warning triggers.
- **Survival horizon** — days the liquid buffer covers stressed net outflows.
- **Lender of last resort** — central bank providing emergency liquidity against collateral.

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9. Common Confusions

"Liquidity risk = insolvency risk." No. Insolvency is assets < liabilities (a *balance-sheet* condition). Illiquidity is cannot-pay-now-despite-solvency (a *cash-flow* condition). A solvent firm can be illiquid (Northern Rock, arguably SVB early on); an insolvent firm can *look* liquid until reality bites. They interact but are distinct.

"Funding and market liquidity are the same." Funding liquidity is about your *liabilities* (raising cash); market liquidity is about your *assets* (selling them). They connect through the spiral but are measured and managed differently.

"LCR and NSFR are just two versions of the same ratio." No — different **horizons** and different **purposes**. LCR is a 30-day acute-stress survival test on the *buffer*; NSFR is a 1-year structural test on *funding stability*. A firm can pass one and fail the other. LCR asks "can you survive next month's storm?"; NSFR asks "is your business model structurally funded?".

"Higher inflows always improve LCR." Only up to the **75% cap**. Inflows can offset at most 75% of outflows, so a firm must always hold HQLA for ≥25% of gross outflows — you can never assume full rescue by incoming cash.

"Contractual maturity is the right basis for the gap ladder." For *behavioural* reality, no. Demand deposits are contractually overnight but behaviourally sticky for years; slotting them at contractual maturity makes every bank look instantly dead. Behavioural modelling is essential — but LCR/NSFR *deliberately* use conservative regulatory run-offs, not the bank's own optimistic behavioural assumptions.

"A high loan-to-deposit ratio is fine if the loans are good." Loan *quality* (credit risk) and loan *funding* (liquidity risk) are separate. A book of perfectly performing 30-year mortgages funded by overnight repo is a liquidity time bomb regardless of zero defaults.

"Holding lots of bonds means I'm liquid." Only if they are *unencumbered* and genuinely HQLA. Bonds already pledged as collateral, or lower-grade bonds that gap down in a fire sale, are not the liquidity you think you have. SVB held plenty of bonds — but selling them crystallised the loss that triggered the run.

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10. Recap and Quick-Reference / Interview Points

Recap

Liquidity risk is the risk of not meeting obligations as they fall due without unacceptable loss. It has two faces — **funding** (can't raise cash on the liability side) and **market** (can't sell assets on the asset side) — joined by the **liquidity spiral**. Its root is **maturity transformation**: borrowing short, lending long, which the **liquidity gap ladder** measures bucket by bucket. Basel III governs it with two ratios: the **LCR** (HQLA / net 30-day stressed outflows ≥ 100%) for the short-term sprint, and the **NSFR** (ASF / RSF ≥ 100%) for the structural marathon. Crises unfold as a cascade — trigger → wholesale flight → retail run → fire sales → insolvency → contagion — and are managed with liquid buffers, diversified and termed-out funding, behavioural modelling, contingency funding plans, and rigorous stress testing. The recurring villain across every metric and every failure is **short-term wholesale funding**.

Quick-reference formulas

Metric	Formula	Threshold
Liquidity gap	Inflows − Outflows (per bucket)	Watch cumulative gap
Cumulative gap	Σ marginal gaps to horizon T	Buffer + cum. gap ≥ 0
LCR	HQLA / Net 30-day outflows	≥ 100%
Net outflows	Outflows − min(Inflows, 75% × Outflows)	—
NSFR	ASF / RSF	≥ 100%
Loan-to-deposit	Loans / Deposits	<100% comfortable

Interview-ready one-liners

- **Solvency vs liquidity:** "A firm dies of insolvency slowly and of illiquidity overnight. Capital defends the first; the liquid buffer defends the second."

- **The two faces:** "Funding liquidity is about my liabilities running; market liquidity is about my assets not selling. The spiral links them — a funding gap forces a fire sale, whose loss deepens the funding gap."
- **LCR in one breath:** "Enough high-quality liquid assets to survive a 30-day acute stress on your own — HQLA over net 30-day outflows, at least 100%, with inflows capped at 75% of outflows so you always self-insure a quarter."
- **NSFR in one breath:** "Fund your illiquid assets with stable money over a year — available stable funding over required stable funding, at least 100%. It exists to kill over-reliance on overnight wholesale funding."
- **Why 2023 rhymed with 2008:** "Same disease — maturity mismatch plus flighty funding. What changed was speed: mobile banking and social media turned a three-day queue into a one-day \$42bn digital run."
- **The core management idea:** "You cannot avoid maturity transformation — it's the business. You survive it by holding a buffer that buys time and by funding illiquid assets with sticky money, then stress-testing the assumption that both hold when everyone panics at once."
- **The universal red flag:** "Short-term wholesale funding. It carries a 100% run-off in the LCR and a 0% ASF factor in the NSFR — the regulators are telling you, from both the 30-day and the 1-year angle, that it is the funding that kills you."

Q&A — Liquidity Risk

A practice bank for the Liquidity Risk chapter. Work each question before reading the answer. Numerical answers self-check against the master identities **LCR = HQLA / Net 30-day outflows ≥ 100%**, with **Net outflows = Outflows − min(Inflows, 75% × Outflows)**, and **NSFR = ASF / RSF ≥ 100%**.

Section A — Concept-Check (short answer)

A1. Define liquidity risk and distinguish it from insolvency.

Liquidity risk is the risk that a firm cannot meet its obligations *as they fall due* without incurring unacceptable losses. It is a **cash-flow** condition: cannot-pay-now. Insolvency is a **balance-sheet** condition: assets worth less than liabilities. The distinction is load-bearing — a firm can be perfectly solvent (assets exceed liabilities) yet fail overnight because its assets are locked in illiquid loans while its liabilities come due today. Northern Rock and, arguably, early-stage SVB died of illiquidity, not insolvency.

A2. Name the two faces of liquidity risk and state which side of the balance sheet each concerns.

Funding liquidity risk — the inability to raise cash to pay one's own obligations; it concerns the **liability** side (deposits fleeing, wholesale funders refusing to roll over). The question is *"Can I get cash to pay what I owe?"* **Market liquidity risk** — the inability to sell an asset quickly near fair value; it concerns the **asset** side (thin markets, wide bid–ask). The question is *"Can I turn what I own into cash without a fire-sale loss?"*

A3. What is the liquidity spiral?

A self-reinforcing loop that joins the two faces: a **funding** shortfall forces the firm to sell assets; if those assets are **market-illiquid**, the sale is at a fire-sale discount; the loss erodes capital; the loss frightens more funders, who flee; that deepens the funding shortfall — and round again. A manageable cash-flow wobble becomes a death spiral. It is why funding and market liquidity, though distinct, must be managed together.

A4. Why is maturity transformation both essential and dangerous?

Essential because society wants it: savers want instant access, borrowers want long, patient loans, and banks bridge the two by borrowing short and lending long. It works because of the **law of large numbers** — on a normal day only a small, predictable net fraction of demand deposits is withdrawn, since withdrawals and deposits roughly cancel across many independent customers. Dangerous because depositor behaviour is not always independent: **fear is correlated.** When depositors believe others will withdraw, withdrawing first becomes rational, and the belief becomes self-fulfilling (the Diamond–Dybvig "bad equilibrium").

A5. What is a liquidity gap, and what is the difference between the marginal and cumulative gap?

The **liquidity gap** in a time bucket = cash inflows − cash outflows (equivalently, assets maturing − liabilities maturing) for that bucket. The **marginal (period) gap** is the gap *within* a single bucket. The **cumulative gap** is the running sum of marginal gaps up to horizon T. The cumulative gap is what matters, because a firm can carry cash forward: survival requires **cumulative gap + liquid buffer ≥ 0 at every horizon.**

A6. Why must demand deposits be slotted by behavioural, not contractual, maturity in a gap ladder?

Demand deposits are contractually repayable overnight but behaviourally sticky for years. If you slot them at their contractual (overnight) maturity, every bank shows a massive negative gap in the first bucket and looks instantly insolvent. Behavioural modelling assigns them to the buckets in which they are actually expected to leave. (Note the nuance: LCR and NSFR *deliberately* override the bank's optimistic behavioural assumptions with conservative regulatory run-off rates.)

A7. State the LCR formula, its threshold, and what it tests.

$$\text{LCR} = \frac{\text{Stock of High-Quality Liquid Assets (HQLA)}}{\text{Total Net Cash Outflows over 30 days}} \geq 100\%$$

It tests whether a firm holds enough high-quality liquid assets to survive a **severe 30-day acute liquidity stress** without any new funding — the short-term sprint.

A8. State the NSFR formula, its threshold, and what it tests.

$$\text{NSFR} = \frac{\text{Available Stable Funding (ASF)}}{\text{Required Stable Funding (RSF)}} \geq 100\%$$

It tests whether a firm funds its illiquid assets with sufficiently **stable** money over a **one-year** horizon — the structural marathon. ASF weights liabilities by stability; RSF weights assets by illiquidity. Its explicit purpose is to curb over-reliance on short-term wholesale funding.

A9. Explain the 75% inflow cap in the LCR and why it exists.

Net outflows = Outflows − min(Inflows, 75% × Outflows). A firm may offset its stressed outflows with expected inflows only up to **75% of gross outflows.** This forces it to hold HQLA covering at least **25% of gross outflows** no matter how much money it expects to receive — you can never assume you will be fully rescued by incoming cash, because in a stress your counterparties may not pay you either.

A10. Why is short-term wholesale funding the recurring villain across both ratios?

In the LCR it carries a **100% run-off** (assumed to flee entirely in the 30-day stress); in the NSFR it carries a **0% ASF factor** (counts as zero stable funding). The regulators are telling you, from both the 30-day and the 1-year angle, that this is the funding that kills you. Every modern bank failure — Northern Rock, Bear Stearns, Lehman, SVB, Credit Suisse — was funded by hot, flighty money and died fast; firms funded by insured retail deposits and long-term debt survive.

A11. What is HQLA, and what makes an asset qualify?

High-Quality Liquid Assets: unencumbered assets convertible to cash in stress with little or no loss, because they are what everyone flies *to* rather than *from*. Level 1 (cash, central-bank reserves, top-rated sovereigns) has a 0% haircut and no cap. Level 2A (high-grade sovereign/agency, AA– corporates) has a 15% haircut; Level 2B (A+ to BBB– corporates, some equities) has 25–50% haircuts. Level 2 total is capped at 40% of HQLA and Level 2B at 15%. Crucially the assets must be **unencumbered** — bonds already pledged as collateral are not the liquidity you think you have.

A12. What is a Contingency Funding Plan (CFP)?

A pre-agreed, board-approved crisis playbook that specifies **early-warning indicators** (widening CDS, rising funding costs, deposit outflows, share-price falls) and graduated actions (draw committed lines, sell HQLA, pledge collateral, activate central-bank facilities) triggered at defined thresholds — so the firm acts *on plan* rather than in panic.

Section B — Numerical / Applied (full solutions)

B1. Marginal and cumulative liquidity gap. A bank reports the following contractual flows (₹ crore). Compute the marginal and cumulative gaps and, with a ₹150 crore opening buffer, find the survival horizon.

Bucket	Inflows	Outflows
0–7d	100	300
8–30d	220	180
31–90d	160	130

Marginal gaps: 100−300 = **−200**; 220−180 = **+40**; 160−130 = **+30**. Cumulative gaps: −200; −200+40 = **−160**; −160+30 = **−130**. Buffer overlay (150 + cumulative gap): 150−200 = **−50**; 150−160 = **−10**; 150−130 = **+20**.

Self-check: sum of marginal gaps = −200+40+30 = −130 = final cumulative gap ✓. **Interpretation:** the ₹150 cr buffer is exhausted inside the first week (short by ₹50 cr) and remains negative through the 8–30d bucket. **Survival horizon is under 7 days.** The bank needs the buffer above ₹200 cr, or must cut first-week outflows. Classic maturity transformation: fine over the quarter, dead in the first week.

B2. Basic LCR. HQLA (after haircuts) = ₹600 cr. Gross stressed outflows = ₹500 cr. Gross expected inflows = ₹300 cr. Compute the LCR.

Inflow cap = 75% × 500 = 375. Capped inflows = min(300, 375) = **300** (uncapped, since 300 < 375). Net outflows = 500 − 300 = **200**. $$\text{LCR} = \frac{600}{200} = 3.00 = \mathbf{300\%}$$ Comfortably above the 100% minimum.

B3. LCR with the inflow cap binding. Same HQLA = ₹600 cr and outflows = ₹500 cr, but gross inflows are now ₹450 cr. Recompute.

Inflow cap = 75% × 500 = **375**. Capped inflows = min(450, 375) = **375** (now capped, since 450 > 375). Net outflows = 500 − 375 = **125**. $$\text{LCR} = \frac{600}{125} = 4.80 = \mathbf{480\%}$$ **Teaching point:** the extra ₹150 cr of expected inflows above B2 did *not* all count — only ₹75 cr of it flowed through, because the cap holds recognised inflows at 375. The firm is forced to self-insure at least 25% of gross outflows (₹125 cr here) with HQLA regardless of what it expects to receive.

B4. LCR from raw components with HQLA caps. Compute HQLA and the LCR.

HQLA before caps: Cash (L1) ₹300, sovereign bonds (L1) ₹200, AA– corporates (L2A) ₹200 at 15% haircut, BBB corporates (L2B) ₹120 at 50% haircut.

- L1 = 300 + 200 = 500.
- L2A after haircut = 200 × 0.85 = 170.
- L2B after haircut = 120 × 0.50 = 60.
- Pre-cap total = 500 + 170 + 60 = 730. Level 2 = 170 + 60 = 230 → 230/730 = 31.5% ≤ 40% cap ✓. Level 2B = 60 → 60/730 = 8.2% ≤ 15% cap ✓.
- **HQLA = 730.**

Outflows: stable retail ₹2,000 @ 5% = 100; less-stable retail ₹500 @ 10% = 50; operational corporate ₹800 @ 25% = 200; unsecured wholesale ₹400 @ 100% = 400. **Total outflows = 750.** Inflows: maturing interbank ₹200 @ 100% = 200; retail loan repayments ₹300 @ 50% = 150. **Total inflows = 350.** Cap = 75% × 750 =

562.5; capped inflows = min(350, 562.5) = 350. Net outflows = 750 − 350 = **400**. $$\text{LCR} = \frac{730}{400} = 1.825 = \mathbf{182.5\%}$$ Compliant.

B5. Level 2B cap binding. A firm holds L1 = ₹100 cr and L2B (after haircut) = ₹30 cr and nothing else. How much L2B actually counts toward HQLA?

Let total HQLA = H. L2B may be at most 15% of H. If all 30 counted, H = 130 and L2B share = 30/130 = 23.1% > 15% — breach. The binding condition: L2B ≤ 0.15 × (L1 + eligible L2B). Solve L2B_elig = 0.15(100 + L2B_elig) → L2B_elig = 15 + 0.15·L2B_elig → 0.85·L2B_elig = 15 → L2B_elig = **17.65**. So only **₹17.65 cr** of the ₹30 cr counts; HQLA = 100 + 17.65 = **117.65 cr**. The excess ₹12.35 cr is disallowed. **Lesson:** low-grade "liquid" assets are capped precisely because they are the ones that gap down in a fire sale.

B6. NSFR. Compute the NSFR.

ASF: equity ₹800 @ 100% = 800; long-term debt ≥1yr ₹600 @ 100% = 600; stable retail deposits ₹2,500 @ 95% = 2,375; operational corporate deposits ₹1,000 @ 50% = 500; short-term wholesale ₹800 @ 0% = 0. **ASF = 4,275.** RSF: cash ₹400 @ 0% = 0; L1 sovereigns ₹800 @ 5% = 40; L2A ₹400 @ 15% = 60; residential mortgages ≥1yr ₹2,500 @ 65% = 1,625; corporate loans ≥1yr ₹1,800 @ 85% = 1,530; fixed assets ₹300 @ 100% = 300. **RSF = 3,555.** $$\text{NSFR} = \frac{4,275}{3,555} = 1.203 = \mathbf{120.3\%}$$ Compliant — illiquid assets (mortgages, corporate loans) are funded by stable money (equity, long debt, sticky retail).

B7. NSFR under a funding-mix shock. Take B6 and replace ₹1,500 cr of stable retail deposits (95% ASF) with short-term wholesale funding (0% ASF). Recompute and comment.

ASF falls: the ₹1,500 cr that contributed 1,500 × 0.95 = 1,425 now contributes 0. New ASF = 4,275 − 1,425 = **2,850**. RSF unchanged at 3,555. $$\text{NSFR} = \frac{2,850}{3,555} = 0.802 = \mathbf{80.2\%}$$ A severe breach — same assets, same size, but funding the identical book with hot money instead of sticky deposits blows the ratio apart. **Funding quality, not asset quality, drives the liquidity ratios.**

B8. Loan-to-Deposit Ratio. A bank has loans ₹4,200 cr and deposits ₹3,500 cr. Compute the LDR and interpret.

$$\text{LDR} = \frac{4{,}200}{3{,}500} = 1.20 = \mathbf{120\%}$$ Above 100%, so the loan book is funded partly by non-deposit (usually flightier, wholesale) sources — a crude early flag of structural funding stress. It says nothing about loan *credit quality*; those are separate risks.

B9. Survival horizon from stressed flows. Opening buffer = ₹100 cr. Stressed net outflows: Day 1–2 total ₹40 cr, Day 3–5 total ₹45 cr, Day 6–10 total ₹30 cr. When does the buffer hit zero?

Running buffer: after D1–2: 100 − 40 = 60; after D3–5: 60 − 45 = 15; after D6–10: 15 − 30 = **−15**. The buffer covers all outflows through Day 5 (₹15 cr remaining) but not the ₹30 cr of Days 6–10. **Survival horizon lies between Day 5 and Day 10** — the firm runs dry partway through the 6–10 window (with ₹15 cr it covers half of the ₹30 cr, roughly Day 7–8 at an even daily rate). It must arrange funding before then.

B10. LCR shock — flighty funding. In B4, suppose unsecured wholesale funding rises from ₹400 to ₹1,000 cr (still 100% run-off), all else equal. Recompute the LCR.

New outflows = 100 + 50 + 200 + 1,000 = **1,350**. Inflow cap = 75% × 1,350 = 1,012.5; capped inflows = min(350, 1,012.5) = 350. Net outflows = 1,350 − 350 = **1,000**. $$\text{LCR} = \frac{730}{1{,}000} = 0.73 = \mathbf{73\%}$$ Now **non-compliant.** The same HQLA that gave 182.5% comfort in B4 collapses below 100% the instant the funding mix tilts toward hot money — the quantitative signature of the villain in A10.

Section C — Interview-style (model answers)

C1. "A bank can be solvent and still fail. Explain."

Solvency and liquidity are two independent ways a firm can die. Solvency is a balance-sheet condition — assets worth more than liabilities. Liquidity is a cash-flow condition — the ability to pay obligations *when they fall due*. A bank's assets are mostly illiquid long-dated loans; its liabilities are mostly short-dated deposits. If depositors demand cash faster than the bank can convert assets to cash, it fails — even though, on paper, its assets comfortably exceed its liabilities. That is exactly how Northern Rock went down in 2007: solvent, but unable to roll over its short-term funding. Capital defends against insolvency; the liquid buffer defends against illiquidity, and you need both.

C2. "Walk me through how a modern liquidity crisis unfolds."

It is a cascade. (1) **Trigger** — bad news: a loss, a downgrade, a failed capital raise, a peer's collapse. (2) **Wholesale flight** — sophisticated counterparties move first, silently and fast: interbank lenders won't roll over, repo haircuts jump, commercial paper won't reprice. (3) **Retail run** — depositors follow, now amplified by mobile banking and social media; SVB lost \$42 billion — a quarter of its deposits — in a single day. (4) **Fire sales** — to raise cash the firm dumps assets, and selling in size and distress crushes prices. (5) **Solvency contagion** — those fire-sale losses now make the firm *actually* insolvent, and mark-to-market losses spread to peers holding the same assets. (6) **Resolution** — lender-of-last-resort support, a forced acquisition, or failure. The lesson: the disease is always maturity mismatch plus flighty funding; what changed by 2023 was *speed*.

C3. "How would you actually manage liquidity risk at a bank?"

Four levers plus a plan. First, hold a **liquid-asset buffer** of unencumbered HQLA sized to survive the plausible worst 30 days without new funding — never encumber the whole buffer, because the point of a buffer is to be usable in a storm. Second, **diversify funding** across sources, tenors, currencies and geographies, and cap reliance on any single counterparty or the overnight market. Third, **term out the funding** so the roll-over cliff is small and spread over time. Fourth, **model behaviour** — actual deposit stickiness, prepayments, and drawdowns of committed lines — not just contractual dates. Over the top of all this sits a **Contingency Funding Plan** with early-warning triggers and pre-positioned central-bank collateral, and a **stress-testing** programme that deliberately models catastrophe and checks the survival horizon. The recurring discipline is to assume your best-case assumptions all fail simultaneously, because in a panic they do.

C4. "LCR and NSFR — aren't they the same thing?"

No — different horizons and different purposes, and a firm can pass one while failing the other. The **LCR** is a 30-day acute-stress survival test on the *buffer*: do you hold enough high-quality liquid assets to survive a severe month-long stress with no new funding? It's the sprint. The **NSFR** is a one-year structural test on *funding stability*: are you funding your illiquid assets with sufficiently stable money? It's the marathon. LCR asks "can you survive next month's storm?"; NSFR asks "is your business model structurally funded?" They attack the same villain — short-term wholesale money — from two angles: it gets a 100% run-off in the LCR and a 0% ASF factor in the NSFR.

C5. "Why did the 2023 failures — SVB, Credit Suisse — happen so fast?"

Same disease as 2008, faster transmission. The underlying vulnerability was classic: maturity mismatch (SVB held long-dated bonds funded by concentrated, uninsured, tech-sector deposits) plus flighty funding. What changed was the *speed of the run*. In Northern Rock's era a run meant physical queues over three days. By 2023, mobile banking and social media let panic and money move instantly — SVB lost \$42 billion in one day, 25% of deposits. The buffer and the resolution machinery, both designed around slower runs, could not keep pace. It's a reminder that liquidity risk is reflexive: the fear of failure *causes* the failure, and technology has compressed the timeline to hours.

C6. "Your treasurer says 'we hold plenty of bonds, so we're liquid.' Push back."

Two problems. First, only *unencumbered* bonds are liquidity — bonds already pledged as collateral for repo or derivatives are spoken for and cannot be sold or re-pledged. Second, only genuinely high-quality bonds stay liquid *in stress*; lower-grade bonds gap down in a fire sale, so their stressed value is far below the marked value. SVB is the cautionary tale: it held plenty of bonds, but they were long-dated and sitting on large unrealised losses, so *selling* them to raise cash crystallised the loss that triggered the run. "We hold bonds" is not the same as "we hold usable, unencumbered, stress-resilient liquidity." I'd want the buffer measured as unencumbered HQLA after haircuts, not gross bond holdings.

Section D — MCQs (with reasoning)

D1. A bank whose assets exceed its liabilities but which cannot meet a deposit withdrawal today is best described as: A. Insolvent B. Illiquid C. Both D. Neither

Answer: B. Assets exceeding liabilities means solvent; the inability to pay now despite solvency is illiquidity — a cash-flow, not balance-sheet, condition.

D2. The LCR requires HQLA to cover net cash outflows over a horizon of: A. 7 days B. 30 days C. 90 days D. 1 year

Answer: B. The LCR is the 30-day acute-stress sprint. (The one-year structural test is the NSFR.)

D3. Under the LCR, expected inflows may offset outflows only up to: A. 50% of outflows B. 75% of outflows C. 100% of outflows D. No cap

Answer: B. The 75% inflow cap forces the firm to self-insure at least 25% of gross outflows with HQLA, since you cannot assume full rescue by incoming cash.

D4. Which funding source carries a 100% run-off in the LCR and a 0% ASF factor in the NSFR? A. Insured stable retail deposits B. Regulatory capital C. Short-term wholesale funding D. Long-term debt (≥1yr)

Answer: C. Both ratios penalise short-term wholesale money maximally — the recurring villain of liquidity failures.

D5. A negative liquidity gap in the shortest time bucket indicates that in that period the bank: A. Has surplus cash to invest B. Must refinance a shortfall C. Is insolvent D. Has too much HQLA

Answer: B. More liabilities than assets mature, so the shortfall must be refinanced — the signature of borrowing short and lending long.

D6. Level 2 assets are capped at what proportion of total HQLA? A. 15% B. 25% C. 40% D. 50%

Answer: C. Total Level 2 ≤ 40% of HQLA; the sub-cap of 15% applies specifically to Level 2B.

D7. The NSFR is designed primarily to: A. Ensure a 30-day cash buffer B. Curb over-reliance on short-term wholesale funding C. Set minimum equity capital D. Measure credit losses

Answer: B. It forces illiquid assets to be funded with stable money over one year, directly attacking hot-money dependence. (A is the LCR; C is capital adequacy; D is credit risk.)

D8. The liquidity spiral describes: A. Rising deposit rates over time B. A funding shortfall forcing fire sales whose losses deepen the shortfall C. Central-bank rate cuts D. Diversification of funding sources

Answer: B. The self-reinforcing loop linking funding and market liquidity risk.

D9. A loan-to-deposit ratio above 100% signals that: A. All loans are funded by deposits B. The loan book is partly funded by non-deposit (flightier) sources C. The bank is insolvent D. The bank holds excess HQLA

Answer: B. Loans exceed deposits, so the gap is filled by usually-flightier wholesale funding — a crude structural-funding flag.

D10. In the LCR, a demand deposit that is insured and stable receives a run-off rate closest to: A. 3–5% B. 25% C. 40% D. 100%

Answer: A. Insured, sticky, diversified retail deposits are assumed to flee only marginally (3–5%); unsecured wholesale and financial-institution funding get the 40–100% treatment.

D11. Which asset qualifies as Level 1 HQLA with a 0% haircut? A. BBB corporate bonds B. Listed equities C. Central-bank reserves and top-rated sovereign bonds D. Residential mortgages

Answer: C. Level 1 = cash, central-bank reserves, top-rated sovereigns, no haircut, no cap. The others are lower-tier or non-HQLA.

D12. The single most important reason 2023's runs (SVB) were faster than 2007's (Northern Rock) was: A. Higher interest rates B. Mobile banking and social media compressing the run timeline C. Weaker capital D. Larger HQLA buffers

Answer: B. The disease (maturity mismatch + flighty funding) was identical; digital banking turned a multi-day branch-queue run into a one-day \$42bn digital run.

Self-verification note: Every LCR figure above uses Net outflows = Outflows − min(Inflows, 75% × Outflows) and checks the 40%/15% Level 2 caps; every NSFR uses ASF/RSF with the stated factors; every gap ladder reconciles the final cumulative gap against the sum of marginal gaps. All computed ratios were recomputed independently and agree.

Chapter 10 — Asset-Liability Management and Interest-Rate Risk in the Banking Book

1. The Problem / The Need

A bank is, at its core, a machine for **transforming maturities**. It takes in short-term, liquid, redeemable-on-demand deposits and lends them out as long-term, illiquid, fixed-rate loans and mortgages. This maturity transformation is not a side-effect of banking — it *is* banking. Society wants both: depositors want instant access to their money, borrowers want 20-year mortgages at a locked rate. The bank sits in the middle and bridges the gap.

But that bridge is built over a chasm. The moment a bank funds a 20-year fixed-rate mortgage at 6% using deposits that reprice every 90 days, it has taken on a bet: it is betting that short-term rates will *not* rise faster than it can pass those costs on. If the central bank hikes and the bank's 90-day deposits now cost 5.5% instead of 2%, the mortgage still only earns 6%. The spread — the bank's entire livelihood — is crushed from 4% to 0.5%. Do enough of that and the bank is insolvent on a flow basis before a single borrower has defaulted.

This is **interest-rate risk in the banking book (IRRBB)**, and it is distinct from the credit risk and market (trading-book) risk covered earlier. Nobody defaulted. No traded position moved. Yet value has evaporated purely because the *timing* of when assets and liabilities reprice differs. Two of the most spectacular bank failures in history — the US Savings & Loan crisis of the 1980s and the collapse of **Silicon Valley Bank in March 2023** — were fundamentally IRRBB failures, not credit failures. SVB held US Treasuries, the safest credit in the world, but funded long-duration bonds with hot uninsured deposits; when rates rose 500 bps, the bonds lost ~$15bn of value and the bank was gone.

So the need is sharp and permanent: banks must **measure, limit, and hedge the mismatch** between the interest-rate behaviour of what they own and what they owe. The discipline that does this is **Asset-Liability Management (ALM)**, governed by the **Asset-Liability Committee (ALCO)**. This chapter builds the two complementary lenses ALM uses — the **earnings lens** (repricing gap → net interest income) and the **value lens** (duration gap → economic value of equity) — and shows how they reconcile.

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2. The Core Idea

Interest-rate risk in the banking book arises because assets and liabilities reprice at different times and with different sensitivities. ALM measures this mismatch two ways — as a short-horizon effect on *earnings* (net interest income) and as a present-value effect on the *net worth* of the firm (economic value of equity) — and then manages it to a board-approved tolerance using on- and off-balance-sheet tools.

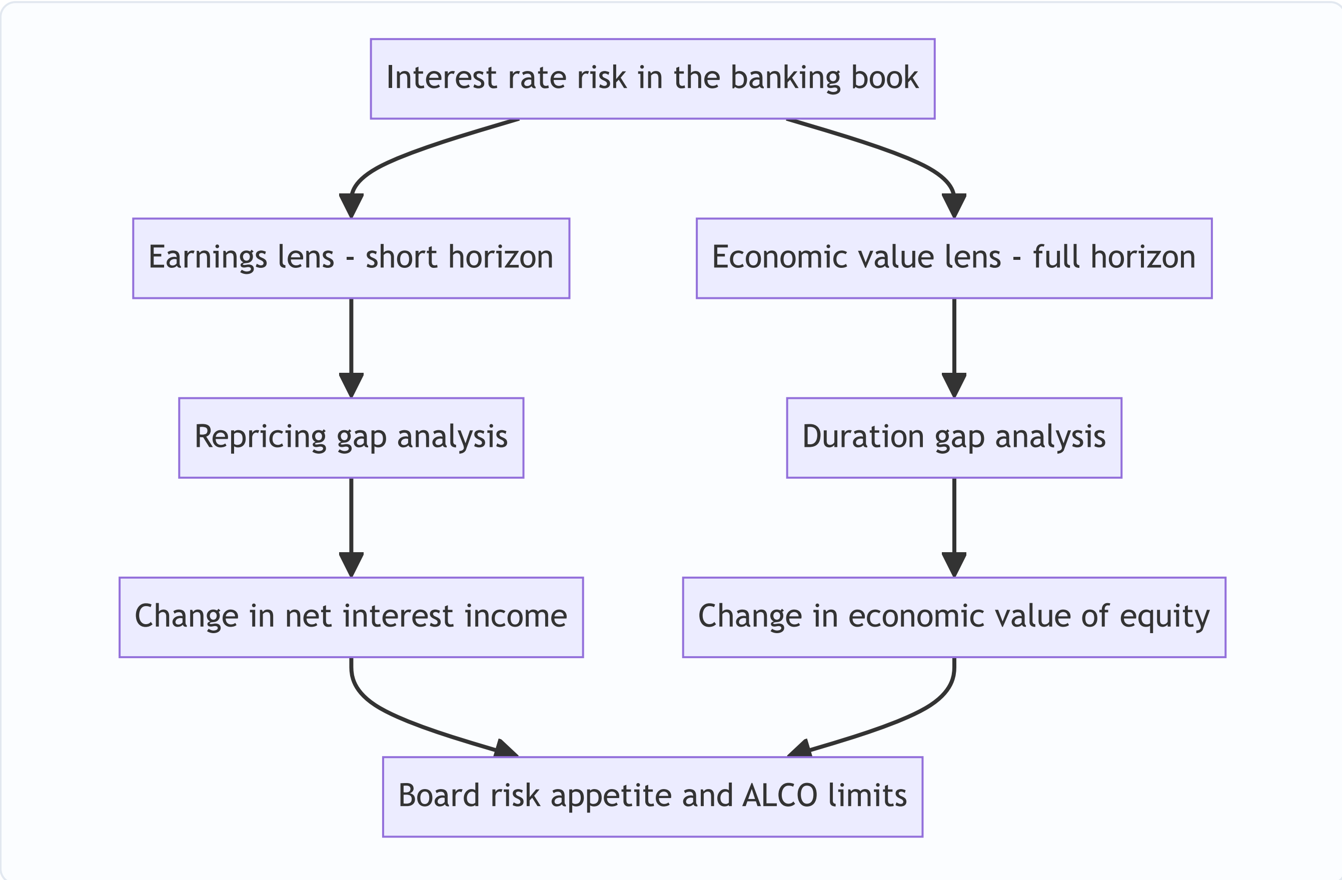
Everything in this chapter is a variation on that single sentence. Three ideas do the heavy lifting:

1. **Repricing, not maturity, is what matters.** A 30-year floating-rate loan that resets every month is *short* from a rate-risk standpoint; a 2-year fixed-rate CD is *long*. We classify every balance-sheet item by *when its rate changes*, not when the principal is repaid.

2. **Two horizons, two metrics.** Over the next 1–12 months, the question is "what happens to my *income*?" — answered by **repricing gap** and **ΔNII**. Over the full life of the balance sheet, the question is "what happens to my *net worth* if I marked everything to market?" — answered by **duration gap** and **ΔEVE**. A bank can look safe on one and dangerous on the other.

3. **Equity is the residual.** The bank's economic value of equity is `EVE = PV(assets) − PV(liabilities)`. Because equity is a small, leveraged residual (often ~8–10% of assets), a small percentage change in asset value produces a large percentage change in equity value. Leverage amplifies duration.

Figure 10.1 — The two lenses of interest-rate risk in the banking book.



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3. Why / How It Works

3.1 Why repricing timing creates risk

Take one dollar of assets earning a rate `r_A` funded by one dollar of liabilities costing `r_L`. Net interest income on that dollar is `NII = r_A − r_L`, the **spread**. Rate risk exists whenever `r_A` and `r_L` do not move together in lockstep. If the asset rate is fixed for five years but the funding rate resets quarterly, then a rate shock hits `r_L` immediately while `r_A` stays put. The spread — and NII — moves even though nothing "defaulted."

The key insight: **a fixed-rate item is insensitive to new market rates; a floating/short-term item is sensitive.** So the mismatch is between the *quantity* of assets that will reprice in a window versus the *quantity* of liabilities that will reprice in that same window. That difference is the **gap**.

3.2 Why value changes too (the duration channel)

Even ignoring earnings, a rate rise reduces the *present value* of a fixed-rate stream. A 6% 10-year bond is worth par at a 6% market yield; at 8% it is worth far less because its below-market coupons are discounted harder. **Duration** measures exactly this sensitivity: it is the percentage change in price per unit change in yield.

Because a bank holds long-duration assets (mortgages, long loans) funded by short-duration liabilities (deposits), its assets lose more value than its liabilities when rates rise. Since `EVE = PV(A) − PV(L)`, EVE falls. This is the **value lens**, and it captures the *entire remaining life* of the book, not just the next 12 months — which is why it caught risks (like SVB's) that a 12-month earnings model waved through.

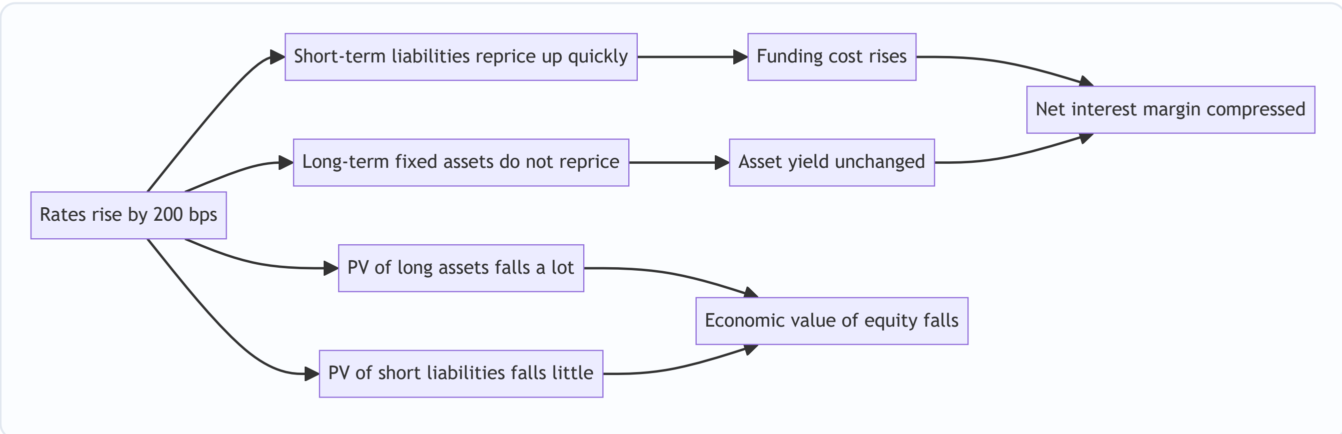
3.3 Why we need both metrics

They can disagree. Consider a bank that funds a 10-year fixed-rate bond with 1-year deposits:

- **Earnings lens (Year 1):** the bond rate is locked, the deposit rate is locked for a year too — so NII in the first 12 months barely moves. Looks safe.
- **Value lens:** the bond has ~8 years of duration, the deposit ~1. Huge negative duration gap. A rate rise devastates EVE.

The 12-month gap model is *blind* to risk that lives beyond the horizon. Conversely, a bank stuffed with very short assets and long liabilities can show scary EVE numbers but stable near-term earnings. Regulators (Basel IRRBB standards, 2016) therefore **mandate both** — no single number is sufficient.

Figure 10.2 — Why a rate rise squeezes a maturity-transforming bank.



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4. Full Content — Framework, Formulas and Methods

4.1 The building blocks of ALM

Rate-Sensitive Assets (RSA) — assets whose interest rate will reset within a chosen time bucket (floating-rate loans, maturing fixed loans about to roll, short securities, reserves earning market rates).

Rate-Sensitive Liabilities (RSL) — liabilities whose cost will reset within the bucket (maturing CDs, money-market deposits, short-term borrowings, floating-rate debt).

Non-rate-sensitive items — fixed-rate long-term loans/securities beyond the bucket, and crucially **non-maturity deposits (NMDs)** — checking/savings balances with no contractual maturity, whose *behavioural* repricing must be modelled (see §8).

4.2 The Repricing (Funding) Gap — the earnings lens

For a given time bucket:

> **GAP = RSA − RSL**

The gap tells you your NII exposure to a parallel rate shock `Δr`:

> **ΔNII ≈ GAP × Δr**

Interpretation of the sign:

Gap position	RSA vs RSL	If rates ↑	If rates ↓
Positive (asset-sensitive)	RSA > RSL	NII rises	NII falls
Negative (liability-sensitive)	RSA < RSL	NII falls	NII rises
Zero (matched)	RSA = RSL	NII unchanged	NII unchanged

A useful scaled version is the **Gap Ratio = RSA / RSL** (>1 asset-sensitive, <1 liability-sensitive) and the **relative gap = GAP / Total Assets**.

Cumulative gap. Banks bucket the balance sheet (0–3m, 3–6m, 6–12m, 1–2y, ...) and sum gaps across buckets up to a horizon. The **cumulative one-year gap** is the headline earnings metric because it captures everything that reprices within a year:

> **ΔNII (1yr) ≈ Cumulative 1-yr GAP × Δr**

Refinement — the standardised gap. Plain gap assumes every RSA and RSL moves 1-for-1 with market rates. In reality, betas differ: a savings deposit rate might move only 0.5× the policy rate, while a loan moves 1.0×. The **standardised gap** weights each item by its rate sensitivity (beta) before differencing. And a subtle timing point: an item repricing in month 2 earns the new rate for 10 of the next 12 months, so a precise ΔNII weights each bucket's gap by the *fraction of the year remaining after it reprices*.

4.3 The Duration Gap — the economic-value lens

Duration `D` measures price sensitivity to yield. **Modified duration** relates the two:

> **ΔP / P ≈ − D_mod × Δy**, where `D_mod = D_Macaulay / (1 + y)`

The **duration gap** aggregates this to the whole balance sheet. Define:

- `D_A` = weighted-average (modified) duration of assets
- `D_L` = weighted-average (modified) duration of liabilities
- `A` = market value of assets, `L` = market value of liabilities
- `k = L / A` = leverage ratio (liabilities as a share of assets)

Duration Gap: $D\_GAP = D\_A - k \times D\_L$

The change in the economic value of equity for a rate shock `Δy` is:

$$\Delta EVE \approx -\, D\_GAP \times A \times \Delta y = -\,[D\_A - k \cdot D\_L] \times A \times \Delta y$$

Sign logic (for a normal bank with `D_A > k·D_L`, i.e. a **positive duration gap**):

Duration gap	If rates ↑	If rates ↓
Positive (D_A > k·D_L)	EVE falls	EVE rises
Negative (D_A < k·D_L)	EVE rises	EVE falls
Zero (immunised)	EVE unchanged	EVE unchanged

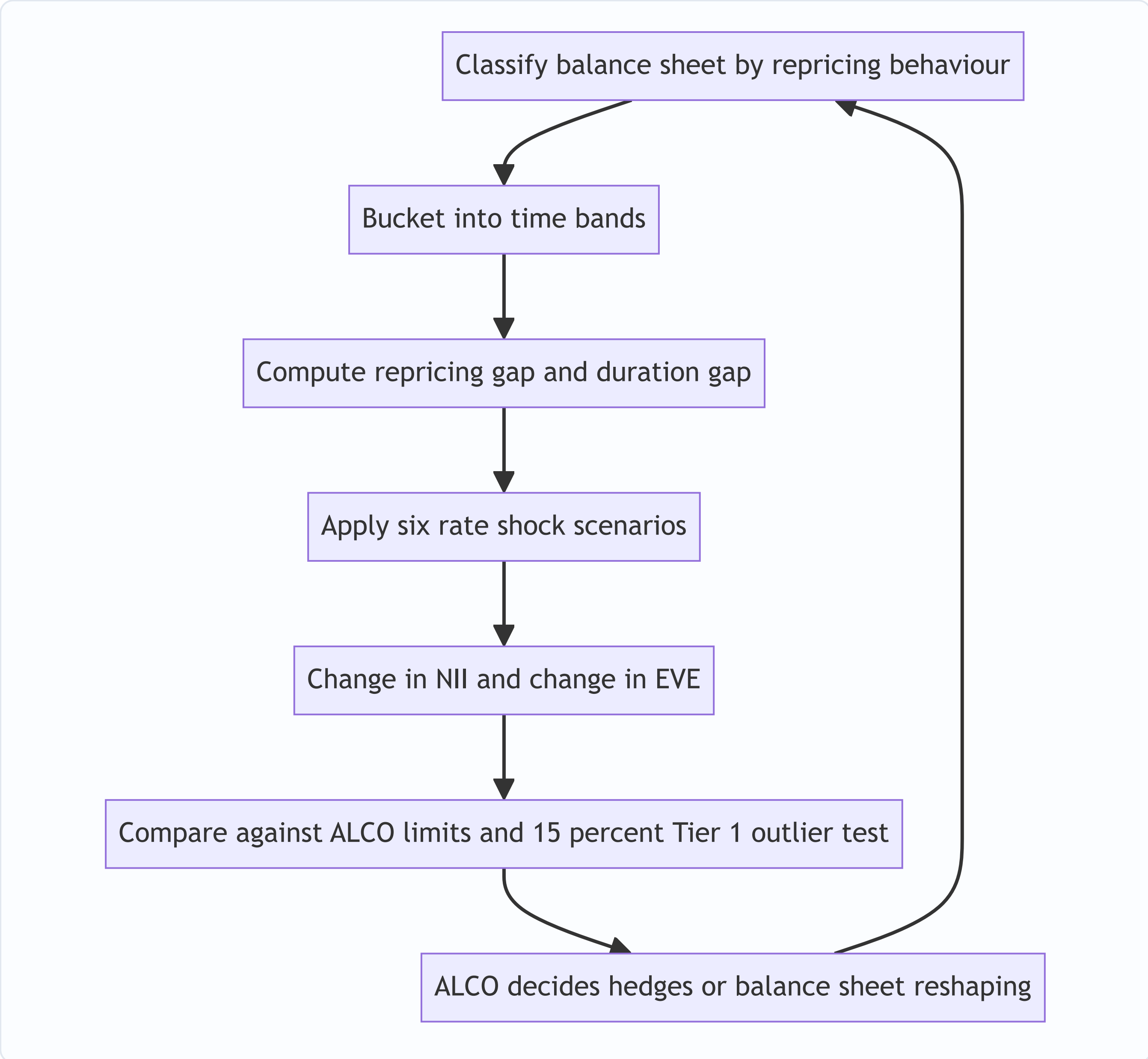
Setting `D_GAP = 0` means `D_A = k·D_L`, i.e. `D_A = (L/A)·D_L`. This is **duration immunisation** of equity: pick asset and liability durations so equity value is insulated from parallel shifts. Note it protects *equity value*, and only against small parallel moves — convexity and non-parallel (twist) moves leak through.

4.4 The regulatory frame — Basel IRRBB

Under the Basel Committee's 2016 IRRBB standard (Pillar 2), banks must:

- Compute **ΔEVE** and **ΔNII** under **six prescribed interest-rate shock scenarios**: parallel up, parallel down, steepener, flattener, short-rate up, short-rate down.
- Apply the **supervisory outlier test**: a bank is flagged if the maximum ΔEVE loss across the six scenarios exceeds **15% of Tier 1 capital**.
- Model **non-maturity deposits** with caps on the average behavioural life (e.g. core deposits capped at a 5-year average, a 6% cap on stable retail portion in some regimes) to prevent gaming.

Figure 10.3 — The ALM measurement and governance loop.



4.5 ALM strategies and hedging tools

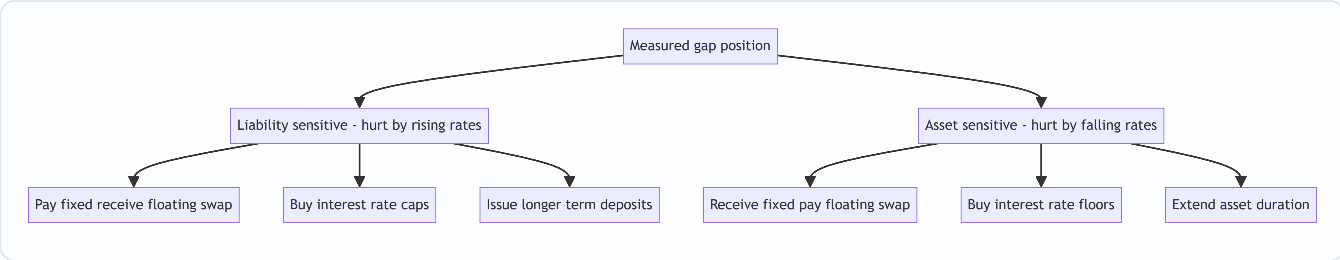
Once measured, the gap can be reshaped in two ways.

A. On-balance-sheet (change the business mix). - Shorten asset duration: originate more floating-rate loans, buy shorter securities, sell long mortgages. - Lengthen liability duration: issue term deposits / longer wholesale debt to fund long assets. - Grow/shrink specific buckets to close a targeted cumulative gap.

B. Off-balance-sheet (overlay hedges). - **Interest-rate swaps** — the workhorse. A liability-sensitive bank (negative gap, hurt by rising rates) enters a swap to **pay fixed / receive floating**, converting floating funding into synthetic fixed, or converting fixed assets into synthetic floating. This is a **fair-value / cash-flow hedge** in accounting terms. - **Interest-rate futures / FRAs** — short Eurodollar/SOFR futures to profit if short rates rise, offsetting NII compression. - **Caps, floors, collars, swaptions** — options giving asymmetric protection (a cap pays off if rates exceed a strike; costs an upfront premium but preserves upside).

The strategic choice depends on the **rate view and risk appetite**: a bank confident rates will fall may *deliberately* run a negative gap to earn more; a bank prioritising stability targets a near-zero gap and immunised duration.

Figure 10.4 — Matching the hedge to the gap.



5. Worked Examples

Example 1 — Repricing gap and ΔNII (with reconciliation)

Setup. A bank's balance sheet, bucketed by *when each item reprices*, in ₹ crore:

Item	Amount	Reprices within 1 year?
Floating-rate loans	400	Yes (RSA)
Short-term securities (< 1yr)	150	Yes (RSA)
Reserves at market rate	50	Yes (RSA)
Fixed-rate 10-yr loans	300	No
Buildings / fixed assets	100	No
Total assets	**1,000**	RSA = 600
Money-market deposits	500	Yes (RSL)
Short-term borrowings	200	Yes (RSL)
Long-term fixed CDs (5yr)	200	No
Equity	100	No
Total liab + equity	**1,000**	RSL = 700

Step 1 — Gap. `GAP = RSA − RSL = 600 − 700 = −100` (crore). The bank is **liability-sensitive** (negative gap): more of its funding reprices within a year than its assets do.

Step 2 — ΔNII for a +200 bps parallel shock. `ΔNII ≈ GAP × Δr = (−100) × (+0.02) = −₹2 crore.` NII **falls by ₹2 crore** if rates rise 200 bps. Intuitively correct: rising rates lift the cost of ₹700cr of funding but the yield on only ₹600cr of assets.

Step 3 — Reconcile from first principles. Rebuild NII directly. Assume a +2% shock hits every rate-sensitive item. - Extra interest *earned* on RSA = 600 × 0.02 = +₹12 crore. - Extra interest *paid* on RSL = 700 × 0.02 = +₹14 crore. - Net change in NII = +12 − 14 = **−₹2 crore.** ✓

Matches Step 2 exactly. The gap formula is just the difference of these two flows.

Step 4 — Symmetry check. For a −200 bps shock: `ΔNII = (−100)×(−0.02) = +₹2 crore` — a liability-sensitive bank *gains* when rates fall. Consistent.

Gap ratio = RSA/RSL = 600/700 = 0.857 (< 1 confirms liability-sensitive). Relative gap = −100/1,000 = −10% of assets.

Example 2 — Duration gap and ΔEVE (with reconciliation)

Setup. Market values (₹ crore) and modified durations:

	Market value	Modified duration (yrs)
Assets (A)	1,000	D_A = 4.0
Liabilities (L)	900	D_L = 1.5
Equity (E = A − L)	100	—

Leverage `k = L/A = 900/1,000 = 0.9`.

Step 1 — Duration gap. `D_GAP = D_A − k·D_L = 4.0 − 0.9 × 1.5 = 4.0 − 1.35 = 2.65 years.` Positive → EVE falls when rates rise. (This is the "normal bank" — long assets, short funding.)

Step 2 — ΔEVE for a +100 bps shock. `ΔEVE ≈ − D_GAP × A × Δy = − 2.65 × 1,000 × 0.01 = −₹26.5 crore.` Equity's economic value drops from ₹100cr to ₹73.5cr — a **26.5% hit to equity from a 1% rate move.** That is the leverage amplification: assets moved a few %, but equity, the thin residual, moved 26.5%.

Step 3 — Reconcile via the value legs separately. - ΔPV(assets) = − D_A × A × Δy = − 4.0 × 1,000 × 0.01 = **−₹40 crore.** - ΔPV(liabilities) = − D_L × L × Δy = − 1.5 × 900 × 0.01 = **−₹13.5 crore.** - ΔEVE = ΔA − ΔL = (−40) − (−13.5) = **−₹26.5 crore.** ✓

Identical to Step 2. Assets lost ₹40cr of value; liabilities became ₹13.5cr cheaper to owe; equity absorbed the ₹26.5cr net difference.

Step 4 — Immunisation. To set `D_GAP = 0` we need `D_A = k·D_L = 0.9 × 1.5 = 1.35 years`. The bank would have to slash asset duration from 4.0 to 1.35 (e.g. swap long fixed loans into floating). Alternatively, holding `D_A = 4.0`, solve for the liability duration that immunises: `D_L = D_A / k = 4.0 / 0.9 = 4.44 years` — i.e. fund with much longer-dated liabilities.

Sanity check on immunisation. If `D_GAP = 0`, then `ΔEVE = −0 × A × Δy = 0` for any small parallel `Δy`. Verify with `D_A = 1.35, D_L = 1.5, k = 0.9`: ΔA = −1.35×1000×0.01 = −13.5; ΔL = −1.5×900×0.01 = −13.5; ΔEVE = −13.5 −(−13.5) = 0. ✓

Example 3 — Same bank, two lenses disagree (the reconciliation that matters)

Notice Examples 1 and 2 could be the *same bank*. Its **1-year gap is −100** (liability-sensitive: rising rates hurt near-term NII) *and* its **duration gap is +2.65** (rising rates hurt EVE). Here both lenses agree rates-up is bad — but they need not.

Construct a disagreement. Suppose instead the bank funds a **10-year fixed-rate bond (₹1,000cr, D_A ≈ 8)** with **1-year deposits (₹900cr, D_L ≈ 1)** that will simply roll over at the same spread.

- **Earnings lens, Year 1:** The bond's rate is fixed. The deposits are also fixed *for the coming year*. So within the 1-year bucket, RSA ≈ 0 and RSL ≈ 0 → **gap ≈ 0** → **ΔNII ≈ 0.** The earnings model says "safe."
- **Value lens:** `D_GAP = 8 − 0.9×1 = 7.1 years`. For +100 bps: `ΔEVE = −7.1 × 1,000 × 0.01 = −₹71 crore` — **71% of a ₹100cr equity wiped out.** The value model screams "danger."

Reconciliation / lesson. Both are *correct* — they answer different questions. The 12-month earnings model is blind to the 9 years of exposure sitting beyond its horizon. This exact blindness is what felled **Silicon Valley Bank**: near-term NII looked fine while EVE was hollowed out by long-duration Treasuries funded with flighty short deposits. **Conclusion: a bank must monitor ΔNII and ΔEVE together; passing one is not passing the other.**

Reconciling table — the three examples on one page:

Metric	Formula	Example 1/2 bank	Ex-3 mismatched bank
1-yr repricing gap	RSA − RSL	−100	≈ 0
ΔNII (+200 bps)	GAP × Δr	−₹2 cr	≈ 0
Duration gap	D_A − k·D_L	+2.65 yr	+7.1 yr
ΔEVE (+100 bps)	−D_GAP × A × Δy	−₹26.5 cr	−₹71 cr
Verdict		Both flag rate-up risk	Earnings blind, value alarmed

6. Connections

- **To Chapter on liquidity (LCR/NSFR):** ALM and liquidity risk are two faces of the same balance-sheet mismatch. IRRBB is the *price* risk of maturity transformation; liquidity risk is the *rollover* risk. Non-maturity deposits sit at the heart of both — behavioural modelling of deposits drives the NSFR available-stable-funding factor *and* the IRRBB repricing bucket.
- **To duration & bond math:** the duration gap is just portfolio duration applied to a leveraged balance sheet. Convexity (the second-order term) is the correction the linear ΔEVE formula omits.
- **To market risk / VaR:** the trading book uses VaR for marked positions; the banking book uses EVE sensitivity for accrual positions. The *methods* (duration, scenario shocks) are cousins; the *accounting treatment* (mark-to-market vs accrual) is what separates the books.
- **To capital adequacy:** IRRBB is a **Pillar 2** risk — it consumes capital via the 15%-of-Tier-1 outlier test and supervisory add-ons, feeding back into the capital-ratio framework.
- **To Funds Transfer Pricing (FTP):** ALM is operationalised through FTP, the internal mechanism that charges business units for the rate/liquidity risk of their products and centralises that risk in the Treasury/ALM desk to be hedged.

7. Key Terms

- **ALM / ALCO** — Asset-Liability Management; the Asset-Liability Committee that governs it.
- **IRRBB** — Interest-Rate Risk in the Banking Book (accrual/non-traded positions).
- **RSA / RSL** — Rate-Sensitive Assets / Liabilities: items repricing within a chosen bucket.
- **Repricing (funding) gap** — RSA − RSL for a time bucket; drives ΔNII.
- **Cumulative gap** — gaps summed across buckets to a horizon (usually 1 year).
- **NII / NIM** — Net Interest Income; Net Interest Margin (NII / earning assets).
- **Duration gap (D_GAP)** — `D_A − k·D_L`; drives ΔEVE.

- **EVE** — Economic Value of Equity = PV(assets) − PV(liabilities).
- **Modified duration** — `D_Macaulay/(1+y)`; % price change per unit yield change.
- **Leverage factor k** — L/A, the weight liabilities carry in the duration gap.
- **Immunisation** — setting D_GAP = 0 to insulate EVE from parallel shifts.
- **NMD** — Non-Maturity Deposit (checking/savings with no contractual maturity).
- **Beta / pass-through** — fraction of a market-rate move that flows to a product's rate.
- **Outlier test** — Basel flag when max ΔEVE loss > 15% of Tier 1 capital.
- **FTP** — Funds Transfer Pricing; internal pricing that centralises rate risk in Treasury.

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8. Common Confusions

"Maturity = repricing." No. What matters is *when the rate resets*, not when principal is repaid. A 30-year floating-rate loan is rate-sensitive (short); a 3-year fixed CD is not (long) until it matures. Always classify by repricing behaviour.

"A zero repricing gap means no interest-rate risk." False — it neutralises *near-term earnings*, but the duration gap (and thus EVE) can still be huge. Example 3 is the canonical trap: gap ≈ 0, D_GAP = 7.1. Basel mandates both metrics precisely because one can pass while the other fails.

"ΔNII and ΔEVE should give the same answer." They answer different questions — a *flow* over a fixed horizon vs a *present value* over the full life. They can even point in opposite directions. Neither is "the" number; the board sets limits on both.

"Immunisation makes the bank risk-free." It only neutralises *small parallel* shifts, and only *equity value* (not earnings). Convexity, yield-curve twists (steepeners/flatteners), basis risk between indices, and optionality (prepayments, deposit runoff) all leak through. Duration immunisation is a first-order, static hedge that must be rebalanced as rates move.

"Non-maturity deposits are instantly rate-sensitive because they're withdrawable on demand." Behaviourally, no. Core checking balances are *sticky* and their rate passes through only partially (low beta) — banks model them as a blend of short and multi-year tranches. Mis-modelling NMDs is the single largest judgement call in ALM (and Basel caps the assumed behavioural life to stop banks over-claiming stability).

"Positive gap is always good." A positive (asset-sensitive) gap helps when rates *rise* and hurts when they *fall*. "Good" depends entirely on the rate scenario and the bank's view; there is no universally safe sign — only a sign consistent (or not) with the board's risk appetite.

"Rising rates are always bad for banks." Only for liability-sensitive banks or over the value horizon. A modestly asset-sensitive bank with sticky, low-beta deposits can see NIM *expand* when rates rise, because asset yields reprice faster than deposit costs. The sign of the effect is an empirical question about the specific balance sheet.

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9. Recap

- Banks exist to transform short deposits into long loans; that maturity transformation *is* interest-rate risk in the banking book (**IRRBB**).
- Classify every item by **when its rate reprices**, into **RSA** and **RSL**.

- **Earnings lens:** `GAP = RSA − RSL`; `ΔNII ≈ GAP × Δr`. Negative gap = liability-sensitive = hurt by rising rates. Horizon is short (≤ 1 year).
- **Value lens:** `D_GAP = D_A − k·D_L`; `ΔEVE ≈ − D_GAP × A × Δy`. Positive gap = hurt by rising rates. Horizon is the whole book. Leverage amplifies equity's sensitivity.
- **Both are mandatory** because a bank can pass one and fail the other (Example 3 / SVB).
- Manage with on-balance-sheet reshaping (asset/liability duration, business mix) and off-balance-sheet hedges (swaps, futures, caps/floors/swaptions). **Immunisation** = D_GAP → 0.
- Basel IRRBB requires six shock scenarios and a **15%-of-Tier-1** EVE outlier test; **non-maturity deposit** behaviour is the crucial modelling judgement.
- Worked examples reconciled: gap ΔNII rebuilt from RSA/RSL interest flows; duration ΔEVE rebuilt from ΔPV(assets) − ΔPV(liabilities).

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10. Quick-Reference / Interview Points

Core formulas to have cold: | Concept | Formula | |---|---| | Repricing gap | `GAP = RSA − RSL` | | ΔNII | `ΔNII ≈ GAP × Δr` | | Gap ratio | `RSA / RSL` | | Modified duration | `D_mod = D_Mac / (1 + y)` | | Price sensitivity | `ΔP/P ≈ − D_mod × Δy` | | Duration gap | `D_GAP = D_A − k·D_L`, `k = L/A` | | ΔEVE | `ΔEVE ≈ − D_GAP × A × Δy` | | Immunisation | `D_A = k·D_L` (D_GAP = 0) |

One-liners interviewers want: - "*What's the difference between the earnings and value perspectives on IRRBB?*" — ΔNII is a short-horizon *flow* measure driven by the repricing gap; ΔEVE is a full-life *present-value* measure driven by the duration gap. A bank must limit both; they can disagree. - "*A bank has a negative repricing gap. What does that mean and what's the hedge?*" — Liability-sensitive: RSL > RSA, so NII falls if rates rise. Hedge by paying fixed / receiving floating on a swap, buying caps, or lengthening liability duration. - "*How can a bank look safe on NII but be dangerous?*" — Its risk lives beyond the 12-month gap window; long-duration fixed assets funded short give a small near-term gap but a large duration gap and ΔEVE. That was SVB. - "*Why does a 1% rate move hit equity by 26%?*" — Leverage. Equity is a thin residual; `ΔEVE = − D_GAP × A × Δy` scales by *total assets*, not by equity, so the same rupee move is a large fraction of the small equity base. - "*What is the Basel outlier test?*" — Flag if the worst ΔEVE across six prescribed shock scenarios exceeds 15% of Tier 1 capital. - "*Biggest modelling judgement in ALM?*" — Non-maturity deposit behaviour: their effective duration and rate beta. Get it wrong and both your gap and EVE are wrong. - "*How do you immunise equity?*" — Set the duration gap to zero: `D_A = (L/A)·D_L`. Protects EVE against small parallel shifts only; rebalance for convexity, twists, and optionality.

Mental model to close with: *A bank is a leveraged bond portfolio wearing a deposit franchise. ALM is the discipline of making sure that when the yield curve moves, neither this year's income nor the firm's net worth moves more than the board agreed to tolerate.*

Q&A — Asset-Liability Management and Interest-Rate Risk in the Banking Book

A practice bank for the ALM / IRRBB chapter. Work each question before reading the answer. Numerical answers are self-checked against the two master identities: the earnings lens `ΔNII ≈ GAP × Δr` (GAP = RSA − RSL) and the value lens `ΔEVE ≈ − D_GAP × A × Δy` (D_GAP = D_A − k·D_L, k = L/A).

Section A — Concept-Check (short answer)

A1. State the two master identities of ALM and name the lens each belongs to.

- **Earnings lens:** `GAP = RSA − RSL`, and `ΔNII ≈ GAP × Δr`. Short horizon (≤ 1 year), measures the effect of a rate shock on *net interest income*.
- **Value lens:** `D_GAP = D_A − k·D_L` with `k = L/A`, and `ΔEVE ≈ − D_GAP × A × Δy`. Full-life horizon, measures the present-value effect on the *net worth* of the firm.

They answer different questions — a flow over a fixed window versus a present value over the whole book — and a bank can pass one while failing the other.

A2. What is IRRBB and how does it differ from credit and trading-book market risk?

IRRBB is the risk that the value or income of the bank's *accrual* (non-traded) positions moves because assets and liabilities reprice at different times and sensitivities. Unlike credit risk, nobody defaults; unlike trading-book market risk, no marked position moves — value changes purely from the *timing mismatch* between what the bank owns and owes. It is a Basel Pillar 2 risk.

A3. Why does repricing timing — not maturity — determine rate sensitivity?

Rate risk exists only when an item's *interest rate resets*. A 30-year floating-rate loan that resets monthly is rate-sensitive (behaves "short"); a 3-year fixed CD is not rate-sensitive until it matures (behaves "long"). Principal repayment timing is irrelevant to the spread; what matters is when the coupon/rate on the item changes.

A4. Define RSA and RSL.

Rate-Sensitive Assets are assets whose rate will reset within the chosen time bucket (floating loans, maturing fixed loans about to roll, short securities, reserves at market rate). **Rate-Sensitive Liabilities** are liabilities whose cost resets within the bucket (maturing CDs, money-market deposits, short-term/floating borrowings). Everything else is non-rate-sensitive for that bucket, including long fixed assets and non-maturity deposits.

A5. Interpret the sign of the repricing gap.

- **Positive gap (RSA > RSL) = asset-sensitive:** NII *rises* when rates rise, *falls* when rates fall.
- **Negative gap (RSA < RSL) = liability-sensitive:** NII *falls* when rates rise, *rises* when rates fall.
- **Zero gap = matched:** NII unchanged to a parallel shock.

The mnemonic: the side with the *larger* rate-sensitive balance drives the direction of the NII move.

A6. What is the cumulative one-year gap and why is it the headline earnings number?

Banks bucket the balance sheet (0–3m, 3–6m, 6–12m, …) and sum the gaps across all buckets up to the one-year horizon. The cumulative 1-year gap captures *everything that reprices within a year*, so `ΔNII(1yr) ≈ Cumulative 1-yr GAP × Δr` is the standard measure of near-term earnings-at-risk.

A7. What is the duration gap and what does it drive?

`D_GAP = D_A − k·D_L`, where D_A and D_L are the (modified) durations of assets and liabilities and `k = L/A` is the leverage factor. It drives the change in the economic value of equity: `ΔEVE ≈ − D_GAP × A × Δy`. A positive duration gap (the "normal bank": long assets, short funding) means EVE falls when rates rise.

A8. Why does leverage amplify equity's sensitivity to rates?

Because `ΔEVE` scales by *total assets A*, not by equity, while equity is a thin residual (`EVE = A − L`, often ~8–10% of A). So a rate move that shifts asset value by a few percent can shift the small equity base by tens of percent. A bank is effectively a leveraged bond portfolio.

A9. What is duration immunisation of equity, and its limits?

Setting `D_GAP = 0`, i.e. `D_A = k·D_L`, so `ΔEVE ≈ 0` for a small parallel shift. Limits: it protects *equity value only* (not earnings), only against *small, parallel* moves, and it leaks through convexity, yield-curve twists (steepeners/flatteners), basis risk between indices, and embedded optionality (prepayments, deposit runoff). It is a first-order static hedge that must be rebalanced.

A10. What is the Basel IRRBB outlier test?

A supervisory flag: a bank is an outlier if the maximum ΔEVE loss across the **six prescribed shock scenarios** (parallel up/down, steepener, flattener, short-rate up/down) exceeds **15% of Tier 1 capital**. Banks must also model non-maturity deposits with caps on assumed behavioural life to prevent gaming.

A11. Why must a bank monitor both ΔNII and ΔEVE?

They can disagree because they cover different horizons. A 10-year fixed bond funded by 1-year deposits shows a *near-zero 1-year gap* (safe on earnings) but a *huge duration gap* (EVE devastated by a rate rise). The 12-month earnings model is blind to risk living beyond its horizon — exactly the blindness that felled Silicon Valley Bank. Basel therefore mandates both.

A12. Why are non-maturity deposits (NMDs) the hardest ALM judgement?

They have no contractual maturity, yet core balances are behaviourally *sticky* and their rate passes through only partially (low beta). Banks must model them as a blend of short and multi-year tranches. Getting the effective duration and beta wrong distorts *both* the repricing gap and the EVE calculation, so it is the single largest modelling call in ALM — which is why Basel caps the assumed behavioural life.

A13. What is the standardised gap and why is it more realistic than the plain gap?

The plain gap assumes every RSA and RSL moves 1-for-1 with the market rate. In reality betas differ — a savings deposit might move only 0.5× the policy rate while a loan moves 1.0×. The standardised gap weights each item by its rate beta before differencing, giving a truer ΔNII.

A14. Give the modified-duration relationship and define its terms.

`ΔP/P ≈ − D_mod × Δy`, where `D_mod = D_Macaulay / (1 + y)`. D_mod is the percentage price change per unit change in yield; the minus sign captures the inverse price-yield relationship for fixed-rate instruments.

A15. What is Funds Transfer Pricing and how does it relate to ALM?

FTP is the internal mechanism that charges each business unit for the rate and liquidity risk embedded in its products, transferring that risk to a central Treasury/ALM desk. Origination units keep a clean credit spread; the rate mismatch is pooled centrally where it is measured against ALCO limits and hedged.

Section B — Numerical / Applied (full solutions)

B1. Basic repricing gap and ΔNII. RSA = ₹600 cr, RSL = ₹700 cr. Find the gap, classify the bank, and compute ΔNII for a +150 bps parallel shock.

GAP = RSA − RSL = 600 − 700 = **−₹100 cr → liability-sensitive**. ΔNII ≈ GAP × Δr = (−100) × (+0.015) = **−₹1.5 cr**. NII falls by ₹1.5 cr when rates rise 150 bps.

Cross-check from flows: extra interest earned = 600 × 0.015 = +₹9 cr; extra interest paid = 700 × 0.015 = +₹10.5 cr; net = 9 − 10.5 = −₹1.5 cr. ✓

B2. Symmetry / falling rates. Same bank, rates *fall* 150 bps. What happens to NII?

ΔNII = (−100) × (−0.015) = **+₹1.5 cr**. A liability-sensitive bank *gains* when rates fall, because its cheaper funding reprices down faster than its asset yields. ✓

B3. Gap ratio and relative gap. Total assets = ₹1,000 cr, RSA = 600, RSL = 700. Compute the gap ratio and relative gap.

Gap ratio = RSA / RSL = 600 / 700 = **0.857** (< 1 confirms liability-sensitive). Relative gap = GAP / Total Assets = −100 / 1,000 = **−10%** of assets.

B4. Cumulative gap across buckets. A bank reports these bucket gaps (₹ cr): 0–3m: +40; 3–6m: −90; 6–12m: −60; 1–2y: +30. Find the cumulative 1-year gap and ΔNII for +100 bps.

Cumulative 1-yr gap = 40 + (−90) + (−60) = **−₹110 cr** (buckets up to 12 months only; the 1–2y bucket is excluded). ΔNII(1yr) ≈ −110 × 0.01 = **−₹1.1 cr**. The bank is liability-sensitive within a year.

B5. Duration gap and ΔEVE. A = ₹1,000 cr, D_A = 4.0; L = ₹900 cr, D_L = 1.5. Find k, the duration gap, and ΔEVE for a +100 bps shock.

k = L/A = 900/1,000 = 0.9. $D_{GAP} = D_A - k \cdot D_L$ = 4.0 − 0.9 × 1.5 = 4.0 − 1.35 = **2.65 years** (positive → EVE falls when rates rise). ΔEVE ≈ − D_{GAP} × A × Δy = − 2.65 × 1,000 × 0.01 = **−₹26.5 cr**.

Equity's economic value drops from ₹100 cr to ₹73.5 cr — a **26.5% hit from a 1% rate move**. That is leverage amplification.

B6. Reconcile B5 leg by leg. Rebuild the ΔEVE from the asset and liability value changes separately.

ΔPV(assets) = − D_A × A × Δy = − 4.0 × 1,000 × 0.01 = **−₹40 cr**. ΔPV(liabilities) = − D_L × L × Δy = − 1.5 × 900 × 0.01 = **−₹13.5 cr**. ΔEVE = ΔA − ΔL = (−40) − (−13.5) = **−₹26.5 cr**. ✓ Matches B5 exactly.

B7. Immunise the balance sheet. Using B5 (D_A = 4.0, k = 0.9), find (a) the asset duration that immunises equity holding D_L = 1.5, and (b) the liability duration that immunises holding D_A = 4.0.

Immunisation requires D_{GAP} = 0, i.e. $D_A = k \cdot D_L$. (a) Target $D_A = k \cdot D_L$ = 0.9 × 1.5 = **1.35 years** (slash asset duration, e.g. swap long fixed loans to floating). (b) Target $D_L = D_A / k$ = 4.0 / 0.9 = **4.44 years** (fund with much longer-dated liabilities).

Verify (a): ΔA = −1.35 × 1,000 × 0.01 = −13.5; ΔL = −1.5 × 900 × 0.01 = −13.5; ΔEVE = −13.5 − (−13.5) = **0**. ✓

B8. The two lenses disagree (the SVB trap). A bank funds a 10-year fixed-rate bond (₹1,000 cr, $D_A = 8.0$) entirely with 1-year deposits (₹900 cr, $D_L = 1.0$) that roll at the same spread. Compute (a) the 1-year repricing gap and ΔNII for +200 bps, and (b) the duration gap and ΔEVE for +100 bps. Equity = ₹100 cr.

(a) Within the 1-year bucket both the bond rate and the deposit rate are fixed → RSA ≈ 0, RSL ≈ 0 → **gap ≈ 0**, so **ΔNII ≈ 0**. The earnings model says "safe." (b) k = 900/1,000 = 0.9. D_GAP = 8.0 − 0.9 × 1.0 = **7.1 years**. ΔEVE = − 7.1 × 1,000 × 0.01 = **−₹71 cr** — 71% of the ₹100 cr equity wiped out. The value model screams "danger."

Both are correct: they answer different questions. The 12-month earnings model is blind to the 9 years of exposure beyond its horizon. This is precisely what felled Silicon Valley Bank. ✓

B9. Net interest margin. A bank earns NII of ₹48 cr on earning assets of ₹1,200 cr. Compute the NIM. If a +200 bps shock cuts NII by ₹4 cr (unchanged asset base), what is the new NIM?

NIM = NII / earning assets = 48 / 1,200 = **4.0%**. New NII = 48 − 4 = 44 cr → new NIM = 44 / 1,200 = **3.67%**. The 33 bps compression is the earnings-at-risk expressed as a margin.

B10. Back out the shock. A bank's duration gap is 3.0 years, assets ₹2,000 cr, and a rate move produced ΔEVE = −₹30 cr. What parallel shock occurred?

ΔEVE = − D_GAP × A × Δy → Δy = − ΔEVE / (D_GAP × A) = − (−30) / (3.0 × 2,000) = 30 / 6,000 = 0.005 = **+50 bps**.

B11. Outlier-test check. Tier 1 capital = ₹500 cr. Across the six Basel scenarios the worst ΔEVE is −₹90 cr. Is the bank an outlier?

Threshold = 15% × Tier 1 = 0.15 × 500 = ₹75 cr. Worst loss ₹90 cr > ₹75 cr → **yes, the bank is a supervisory outlier** and would face heightened scrutiny / potential capital add-ons. The ratio is 90/500 = 18% of Tier 1.

B12. Standardised (beta-weighted) gap. RSA = ₹600 cr with asset beta 1.0; RSL = ₹700 cr with deposit beta 0.6. Compute the standardised gap and ΔNII for +100 bps, and compare to the plain-gap answer.

Standardised gap = (RSA × β_A) − (RSL × β_L) = (600 × 1.0) − (700 × 0.6) = 600 − 420 = **+₹180 cr**. ΔNII ≈ 180 × 0.01 = **+₹1.8 cr** — the bank actually *gains* on a rate rise once low deposit betas are accounted for.

Contrast: the plain gap is 600 − 700 = −100 cr → −₹1.0 cr. The sticky, low-beta deposits flip the sign: the bank looks liability-sensitive on paper but is asset-sensitive in behaviour. This is why betas matter. ✓

B13. Convexity leak on an "immunised" book. A bank has D_GAP = 0 but its assets have far higher convexity than its liabilities. Qualitatively, what happens to EVE for a *large* parallel move, and why doesn't immunisation protect it?

Duration is a first-order (linear) approximation. With D_GAP = 0 the linear term vanishes, so *small* moves leave EVE unchanged. For a *large* move the second-order term dominates: `ΔEVE ≈ −D_GAP·A·Δy + ½·(C_A·A − C_L·L)·Δy²`. Immunisation only neutralised the linear channel; convexity, yield-curve twists, and optionality still leak through — so it is not a risk-free hedge.

Section C — Interview-Style (with model answers)

C1. "Explain interest-rate risk in the banking book to a board member in thirty seconds."

Model answer: "We take in short-term deposits and lend long-term at fixed rates — that maturity transformation is our business, but it's also a bet. If rates rise, our deposits reprice up quickly while our fixed loans keep earning the old rate, so our margin gets squeezed. We measure that two ways: the near-term hit to income, and the hit to the firm's net worth if we marked everything to market. We keep both inside board-approved limits and hedge the rest with swaps. Nobody has to default for us to lose money — the timing mismatch alone does it."

C2. "What's the difference between the earnings lens and the value lens, and why keep both?"

Model answer: "The earnings lens uses the repricing gap to estimate the change in net interest income over the next twelve months — a short-horizon flow measure, $\Delta NII \approx gap \times \Delta r$. The value lens uses the duration gap to estimate the change in economic value of equity over the *whole* life of the book, $\Delta EVE \approx -D\_GAP \times A \times \Delta y$. We keep both because they can disagree: a long fixed asset funded short shows a near-zero one-year gap yet a massive duration gap. Passing one is not passing the other, which is why Basel mandates both."

C3. "A bank has a negative repricing gap. What does that mean and how would you hedge it?"

Model answer: "Negative gap means rate-sensitive liabilities exceed rate-sensitive assets — the bank is liability-sensitive, so net interest income *falls* if rates rise. To hedge, I'd pay fixed and receive floating on an interest-rate swap, which converts floating funding into synthetic fixed, or converts fixed assets into synthetic floating — either way it offsets the exposure. Alternatives: buy interest-rate caps for asymmetric protection, or reshape the balance sheet by issuing longer-term deposits to lengthen liability duration. The choice depends on cost, accounting treatment, and the bank's rate view."

C4. "How can a bank look completely safe on net interest income yet be dangerously exposed?"

Model answer: "Its risk lives beyond the twelve-month gap window. If it holds long-duration fixed-rate assets funded with short deposits, the one-year gap is near zero — near-term NII barely moves — but the duration gap is enormous, so a rate rise hollows out the economic value of equity. That is exactly what happened to Silicon Valley Bank: safe Treasuries, so no credit problem, but funded with hot uninsured deposits. When rates rose 500 bps the bonds lost about fifteen billion of value and the bank was gone. The earnings model waved it through; the value model would have caught it."

C5. "Why does a 1% rate move hit equity by 20–30%?"

Model answer: "Leverage. The ΔEVE formula scales by *total assets*, not by equity — $\Delta EVE \approx -D\_GAP \times A \times \Delta y$. Equity is a thin residual, often eight to ten percent of assets, so the same rupee change in asset value is a large fraction of the small equity base. A bank is essentially a leveraged bond portfolio wearing a deposit franchise; duration acts on the whole portfolio but lands entirely on the sliver of equity."

C6. "What's the single hardest judgement call in ALM, and why?"

Model answer: "Modelling non-maturity deposits. Contractually they're withdrawable on demand, so you'd call them instantly rate-sensitive. Behaviourally they're sticky: core balances stay for years and their rate passes through only partially, a low beta. So we model them as a blend of short and multi-year tranches. This one assumption drives *both* the repricing gap and the EVE duration, so getting it wrong corrupts every downstream number — which is why Basel caps the assumed behavioural life."

C7. "What is duration immunisation and what are its blind spots?"

Model answer: "Immunisation means setting the duration gap to zero — $D\_A = k \cdot D\_L$ — so a small parallel rate shift leaves the economic value of equity unchanged. Its blind spots: it protects *value*, not earnings; it works only for *small, parallel* moves; and it leaks through convexity for large moves, through yield-curve twists like steepeners and flatteners, through basis risk when assets and liabilities track different indices, and through embedded optionality like mortgage prepayments and deposit runoff. It's a first-order static hedge that has to be rebalanced as rates and the curve move — not a set-and-forget riskless position."

C8. "Walk me through the Basel IRRBB framework."

Model answer: "It's a Pillar 2 standard from 2016. Banks compute both ΔEVE and ΔNII under six prescribed shock scenarios — parallel up/down, steepener, flattener, short-rate up/down. The supervisory outlier test flags a bank if the worst ΔEVE loss across those six exceeds fifteen percent of Tier 1 capital, and it caps non-maturity-deposit behavioural life so banks can't game a flattering number. Unlike Pillar 1 market risk it sets no rigid capital charge; it drives supervisory dialogue and potential add-ons."

C9. "Are rising rates good or bad for a bank?"

Model answer: "It depends entirely on the balance sheet, so I'd resist a blanket answer. A liability-sensitive bank — negative gap — is hurt on near-term income when rates rise. But a bank with sticky, low-beta deposits can actually see its net interest margin *expand*, because asset yields reprice faster than deposit costs. And separately, over the value horizon, almost every maturity-transforming bank has a positive duration gap, so rising rates reduce the economic value of equity regardless of the earnings picture. So the honest answer is: name the horizon and show me the balance sheet's betas and durations first."

Section D — MCQs (with reasoning)

D1. A bank has RSA = ₹500 cr and RSL = ₹650 cr. It is: A) Asset-sensitive B) Liability-sensitive C) Matched D) Immunised

Answer: B. GAP = 500 − 650 = −₹150 cr < 0 → more funding reprices than assets → liability-sensitive → NII falls if rates rise. "Immunised" refers to a zero *duration* gap, not the repricing gap, so D is wrong.

D2. Which item is rate-sensitive within a 6-month bucket? A) A 10-year fixed-rate mortgage B) A floating-rate loan resetting monthly C) A building D) Equity

Answer: B. Rate sensitivity is about *when the rate resets*, not maturity. The floating loan resets inside the bucket; the fixed mortgage's rate is locked for years; buildings and equity carry no interest rate. Classic maturity-vs-repricing trap.

D3. ΔNII ≈ GAP × Δr. A bank with a positive gap of +₹200 cr experiences a −50 bps shock. ΔNII is: A) +₹1.0 cr B) −₹1.0 cr C) +₹0.1 cr D) −₹0.1 cr

Answer: B. ΔNII = (+200) × (−0.005) = −₹1.0 cr. An asset-sensitive bank *loses* income when rates fall, because its larger rate-sensitive asset base reprices down. Sign discipline is the point.

D4. The duration gap formula is: A) $D_A - D_L$ B) $D_A - k \cdot D_L$, $k = L/A$ C) $D_A + k \cdot D_L$ D) $k \cdot D_A - D_L$

Answer: B. The liability duration is scaled by the leverage factor $k = L/A$ because liabilities fund only a fraction of assets. Omitting k (option A) overstates the liability offset and is the most common error.

D5. A bank has $D_A = 5$, $D_L = 2$, A = ₹1,000 cr, L = ₹900 cr. For a +100 bps shock, ΔEVE is closest to: A) −₹32 cr B) −₹30 cr C) −₹50 cr D) −₹18 cr

Answer: A. k = 900/1,000 = 0.9. D_{GAP} = 5 − 0.9×2 = 5 − 1.8 = 3.2. ΔEVE = −3.2 × 1,000 × 0.01 = −₹32 cr. Option C (−50) forgets to scale by k; D uses the wrong leg.

D6. A zero repricing gap guarantees: A) Zero interest-rate risk B) Zero near-term NII sensitivity to a parallel shock, but EVE can still be exposed C) An immunised balance sheet D) Zero ΔEVE

Answer: B. A zero repricing gap neutralises near-term earnings only. The duration gap — and hence EVE — can be large (the 10-year-bond-funded-short case). This is the exact reason Basel requires both metrics.

D7. Why does leverage amplify ΔEVE relative to the equity base? A) Because equity has high duration B) Because ΔEVE scales by total assets, while equity is a thin residual C) Because liabilities have zero duration D) Because of convexity

Answer: B. $\Delta EVE \approx -D\_GAP \times A \times \Delta y$ scales by *A*, but the hit lands on equity ($\approx A - L$), a small fraction of A. So a modest percentage move in assets is a large percentage move in equity. Convexity (D) is a second-order effect, not the amplification mechanism.

D8. Under Basel IRRBB, a bank is a supervisory outlier if: A) ΔNII exceeds 15% of net income B) Maximum ΔEVE loss across six shock scenarios exceeds 15% of Tier 1 capital C) Any single ΔEVE exceeds 20% of CET1 D) The repricing gap exceeds 10% of assets

Answer: B. The 2016 standard's outlier test is on the *worst* ΔEVE across the six prescribed scenarios versus 15% of Tier 1. The other options mix up the metric, threshold, or capital base.

D9. To hedge a liability-sensitive bank against rising rates, the classic swap is: A) Receive fixed / pay floating B) Pay fixed / receive floating C) Buy a floor D) Sell a cap

Answer: B. Paying fixed / receiving floating gains value when rates rise, offsetting the NII compression a liability-sensitive bank suffers. Receiving fixed (A) would double down on the exposure; a floor (C) protects against falling rates; selling a cap (D) adds risk.

D10. The biggest modelling judgement in ALM is: A) The discount rate on Tier 1 capital B) The behavioural life and rate beta of non-maturity deposits C) The recovery rate on defaulted loans D) The trading-book VaR window

Answer: B. NMD behaviour drives both the repricing bucket and the EVE duration, and it's a judgement call, so Basel caps the assumed life. Recovery rate (C) is credit risk; VaR window (D) is trading-book market risk — different chapters.

D11. Modified duration equals: A) $D\_Macaulay \times (1 + y)$ B) $D\_Macaulay / (1 + y)$ C) $D\_Macaulay + y$ D) $D\_Macaulay - y$

Answer: B. $D\_mod = D\_Macaulay / (1 + y)$, giving $\Delta P/P \approx -D\_mod \times \Delta y$. Multiplying instead of dividing (A) inflates the sensitivity.

D12. Immunisation (D_GAP = 0) fails to protect against all of the following EXCEPT: A) Small parallel shifts B) Yield-curve twists C) Convexity on large moves D) Deposit-runoff optionality

Answer: A. Immunisation is *designed* to neutralise small parallel shifts. It leaks on twists, convexity for large moves, basis, and optionality.

Self-verification note. Every numerical answer was rebuilt two ways where possible: repricing-gap ΔNII figures reconcile with the direct RSA/RSL interest-flow calculation (B1), and duration ΔEVE figures reconcile with the leg-by-leg ΔPV(assets) − ΔPV(liabilities) identity (B6/B7). Sign conventions follow: negative repricing gap = liability-sensitive = hurt by rising rates; positive duration gap = hurt by rising rates.

Chapter 11 — Risk Models and Measurement

1. The Problem / The Need

Every risk number a bank, fund, or corporate reports — a Value-at-Risk figure, an expected loss, an economic capital charge, a stress-test result — is the output of a **model**. A model is a machine that takes inputs (positions, prices, volatilities, correlations, default probabilities) and turns them into a statement about the future distribution of gains and losses. The uncomfortable truth of risk management is that we never observe the future distribution directly. We only ever see one realised path of history, and we must infer the shape of the *whole distribution of possible futures* from that single, finite, backward-looking sample.

This creates three distinct problems that this chapter must solve.

First, the specification problem. What shape does the loss distribution actually have? If we assume returns are Normal (Gaussian) and they are not, we will systematically underestimate the frequency of large losses. The 2007–09 crisis was, in large part, a story of risk models that assumed thin tails and mild dependence when reality delivered fat tails and dependence that spiked precisely when it hurt most. A risk manager who cannot articulate *why* the Normal assumption fails, and what to use instead, is dangerous.

Second, the dependence problem. Portfolio risk is not the sum of individual risks — it is driven by how positions move *together*. A model of a single asset can be excellent while a model of the joint behaviour of a thousand assets is catastrophically wrong, because correlation is unstable and, worse, tends to rise in crises exactly when diversification is supposed to protect you. Getting the *marginals* right but the *dependence* wrong is one of the most expensive mistakes in finance (it is essentially the story of the Gaussian copula and structured credit).

Third, the validation problem. A model that cannot be checked is a superstition. We need disciplined ways to ask: *does this model's output match reality?* (backtesting), *is the model built and used correctly?* (validation and governance), and *where does it break?* (stress testing and reverse stress testing). Regulators codify this because unvalidated models threaten solvency.

The stakes are concrete. Basel's Internal Models Approach lets a bank use its own VaR/Expected-Shortfall model to compute market-risk capital — but only if the model passes backtesting; fail too often and a regulatory multiplier inflates the capital charge. So the *quality of the model* is not academic; it directly sets how much capital must be held. This chapter builds the toolkit: distributions and tails, correlation and copulas, scenarios and simulation, backtesting, and the governance that keeps the whole thing honest.

2. The Core Idea

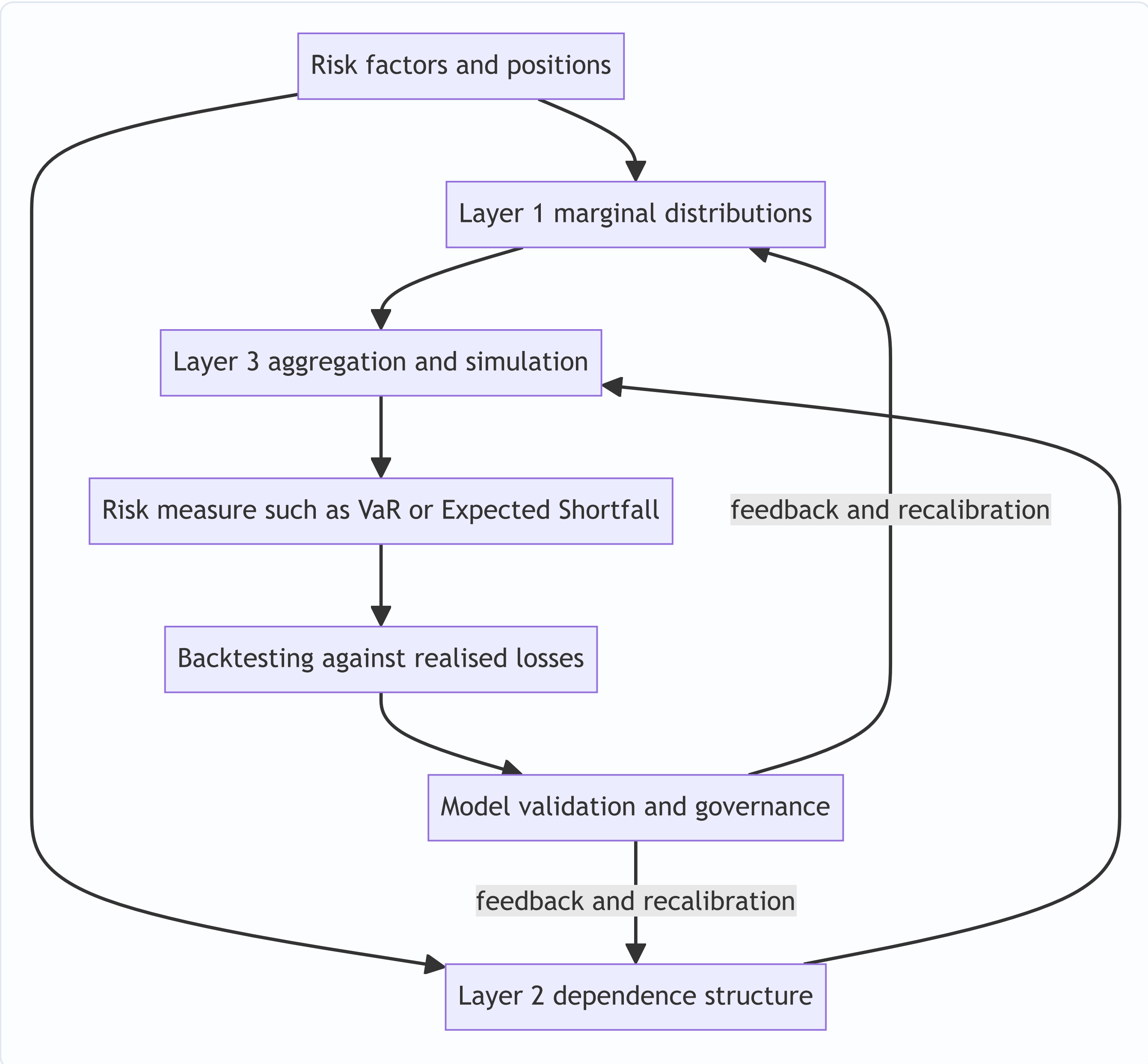
A risk model is a **map from uncertainty to a number**, and it has three separable layers that can each fail independently:

1. **Marginal risk** — the distribution of each individual risk factor (a stock return, a rate move, an obligor's default). This is where *fat tails* live.
2. **Dependence structure** — how the factors co-move. This is where *correlation and copulas* live.
3. **Aggregation and measurement** — how we combine layers 1 and 2 into a portfolio loss distribution and read off a risk measure (VaR, Expected Shortfall). This is where *scenario analysis and simulation* live.

The single most important conceptual move in modern risk modelling — **Sklar's theorem** — is that these layers can be *decoupled*: you can model the marginals separately from the dependence, then glue them together with a **copula**. That is liberating (fix the fat tails without touching the correlations, and vice versa) and dangerous (you can glue fat-tailed marginals to a benign dependence structure and fool yourself into thinking you are conservative).

Wrapped around all three layers is a fourth idea that is not math but discipline: **a model is a hypothesis about the world, and every hypothesis must be testable and governed.** Backtesting tests the hypothesis against realised outcomes; validation tests whether the hypothesis was implemented and used correctly; governance ensures someone independent owns that judgement. The math tells you what the risk *is if the model is right*; the governance tells you *whether to believe the model at all*.

Figure 1 — the layered anatomy of a risk model.



3. Why / How It Works

Why decomposition works. Financial losses are high-dimensional, but the *drivers* are lower-dimensional. A thousand equity positions are really exposures to a handful of common factors (market, sector, size, value) plus idiosyncratic noise. Modelling reduces dimensionality: instead of estimating a 1000×1000 covariance matrix (roughly half a million distinct numbers — hopeless with a few hundred days of data), we model a few factors and their loadings. This is *why* factor models, principal components, and copulas exist — they make an intractable joint distribution estimable from finite data.

Why tails are the whole game. Risk measures like VaR and Expected Shortfall care about the far left of the loss distribution — the 1% or 0.1% worst outcomes. The centre of the distribution (where returns spend most of their time) is almost irrelevant to a risk manager; it is the tail that bankrupts you. So the modelling effort is *asymmetric*: getting the mean and even the standard deviation approximately right matters far less than getting the tail shape right. A model can fit 99% of the data beautifully and still be worthless for risk because it misses the 1% that matters.

Why the Normal distribution keeps getting used anyway. It has one parameter for scale (σ), it is closed under linear combinations (a portfolio of jointly-Normal assets is Normal, so VaR has a clean closed form), and correlation fully describes its dependence. This analytical convenience is seductive — and it is exactly why it is over-used. Real financial returns are **leptokurtic** (fat-tailed, sharp-peaked) and exhibit **volatility clustering** (calm and stormy periods bunch together, so returns are not independent over time). Empirically, daily equity index returns show kurtosis far above the Normal value of 3, and observed "5-sigma" and "10-sigma" days occur orders of magnitude more often than a Normal model permits.

Why dependence spikes in crises. In normal times, assets are driven by many small independent shocks and correlations look moderate. In a crisis, a single systemic shock (liquidity evaporating, forced deleveraging) drives *everything* simultaneously — correlations converge toward one and diversification vanishes. A Gaussian dependence model has **zero tail dependence**: it literally cannot represent "everything crashes together." That structural blind spot, not a bad correlation number, is why Gaussian-copula models of structured credit failed.

How simulation rescues tractability. When the portfolio is non-linear (options, credit) or the distributions are non-Normal, there is no closed-form answer. Monte Carlo simulation sidesteps the math: draw thousands of scenarios from the modelled distributions, revalue the portfolio in each, and read the risk measure off the empirical distribution of outcomes. It trades algebra for computation, and it is *why* modern risk engines are simulation engines.

4. Full Content — Framework, Formulas, Methods

4.1 Modelling risk: the general workflow

1. **Identify risk factors** — the underlying drivers (rates, spreads, equity indices, FX, default events).
2. **Choose marginal distributions** for each factor — Normal, Student-t, or an empirical/EVT tail.
3. **Choose a dependence structure** — a covariance matrix (implies a Gaussian copula) or an explicit copula (t, Clayton, Gumbel).
4. **Map factors to P&L** — the pricing/revaluation function, linear (delta) or full ("full revaluation").
5. **Aggregate** — analytically, by historical simulation, or by Monte Carlo.
6. **Read the risk measure** — VaR, Expected Shortfall, economic capital.

7. **Backtest and validate**, then recalibrate.

4.2 Probability distributions and fat tails

Moments. For a return r with mean μ and standard deviation σ:

- **Skewness** $= E\left[\left(\frac{r-\mu}{\sigma}\right)^3\right]$ — asymmetry; equity returns are typically negatively skewed (crashes bigger than rallies).
- **Kurtosis** $= E\left[\left(\frac{r-\mu}{\sigma}\right)^4\right]$ — tail heaviness. Normal = 3. **Excess kurtosis** = kurtosis − 3; positive means fat tails.

Parametric VaR under Normality. For a horizon return $r \sim N(\mu, \sigma^2)$, the VaR at confidence α (loss expressed as a positive number) is:

$$\text{VaR}\alpha = -\left(\mu + z{\alpha}\,\sigma\right)\times V$$

where V is portfolio value and z_α is the standard-Normal quantile ($z_{0.99} = -2.326$, $z_{0.95} = -1.645$). Over short horizons $\mu \approx 0$, so $\text{VaR}_{99\%} \approx 2.326\,\sigma V$.

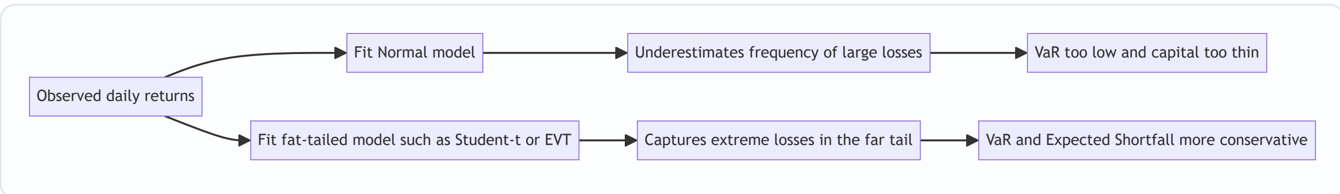
Why the Normal understates tail risk — the Student-t fix. The Student-t distribution with ν degrees of freedom has fatter tails; kurtosis $= 3 + \frac{6}{\nu-4}$ for $\nu>4$, converging to Normal as $\nu\to\infty$. Lower ν = fatter tails. Using a t with $\nu\approx 4$–6 captures much of the observed excess kurtosis of daily returns.

Extreme Value Theory (EVT). Rather than model the whole distribution, EVT models *only the tail*. The **Peaks-Over-Threshold** method fits a **Generalised Pareto Distribution (GPD)** to exceedances above a high threshold u:

$$G_{\xi,\beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \quad y = x - u > 0$$

The **shape parameter ξ** governs tail heaviness: $\xi>0$ = fat (Fréchet) tail, typical of financial losses. EVT gives principled estimates of extreme quantiles *beyond the range of observed data* — its key advantage over historical simulation.

Figure 2 — thin-tail versus fat-tail loss distributions and where risk measures sit.



Coherence and Expected Shortfall. VaR is a *quantile* — the loss not exceeded with probability α. It says nothing about how bad losses are *beyond* it, and it is **not sub-additive** (a portfolio's VaR can exceed the sum of its parts' VaRs, punishing diversification — incoherent). **Expected Shortfall (ES)**, the average loss in the worst $(1-\alpha)$ tail, fixes both:

$$\text{ES}\alpha = E\left[L \mid L \geq \text{VaR}\alpha\right] = \frac{1}{1-\alpha}\int_\alpha^1 \text{VaR}_u\, du$$

ES is coherent (sub-additive) and tail-sensitive; Basel's Fundamental Review of the Trading Book replaced 99% VaR with **97.5% ES** as the regulatory market-risk measure for precisely these reasons.

4.3 Correlation and copulas for dependence

Linear (Pearson) correlation $\rho = \frac{\text{Cov}(X,Y)}{\sigma_X\sigma_Y}$ measures only *linear* co-movement and is only a *complete* description of dependence when the joint distribution is elliptical (e.g. multivariate Normal or t). Its pitfalls: it can be zero for strongly dependent variables (e.g. $Y=X^2$), it is not invariant to non-linear transforms, and it is undefined for infinite-variance distributions.

Rank correlations — **Kendall's τ** and **Spearman's ρ** — depend only on the copula (the ranks), not the marginals, so they survive non-linear transforms and are more robust.

Sklar's theorem. Any joint distribution F can be written as:

$$F(x_1,\dots,x_n) = C\big(F_1(x_1),\dots,F_n(x_n)\big)$$

where F_i are the marginals and C is a **copula** — a joint distribution on $[0,1]^n$ with uniform marginals that carries *all* the dependence. This is the decoupling result: **choose marginals and dependence independently.**

Tail dependence — the concept that breaks Gaussian models. The lower tail-dependence coefficient is:

$$\lambda_L = \lim_{q\to 0^+} P\left(X \leq F_X^{-1}(q)\;\middle|\;Y \leq F_Y^{-1}(q)\right)$$

the probability that X is extreme *given* Y is extreme. Key facts:

Copula	Lower tail dependence λ_L	Upper tail dependence λ_U	Use
Gaussian	0 (for $\rho<1$)	0	Elliptical, no joint extremes
Student-t	> 0 (increases as $\nu\downarrow$)	> 0, symmetric	Symmetric joint crashes/booms
Clayton	> 0	0	Joint downside (credit, crashes)
Gumbel	0	> 0	Joint upside

The **Gaussian copula has zero tail dependence**: even with correlation 0.9, the probability of joint extreme losses vanishes in the tail. This is *why* the Gaussian copula CDO models understated the chance of simultaneous defaults — the model structurally could not represent "everyone defaults together," regardless of the correlation input.

4.4 Scenario analysis and simulation methods

Three canonical ways to build the portfolio loss distribution:

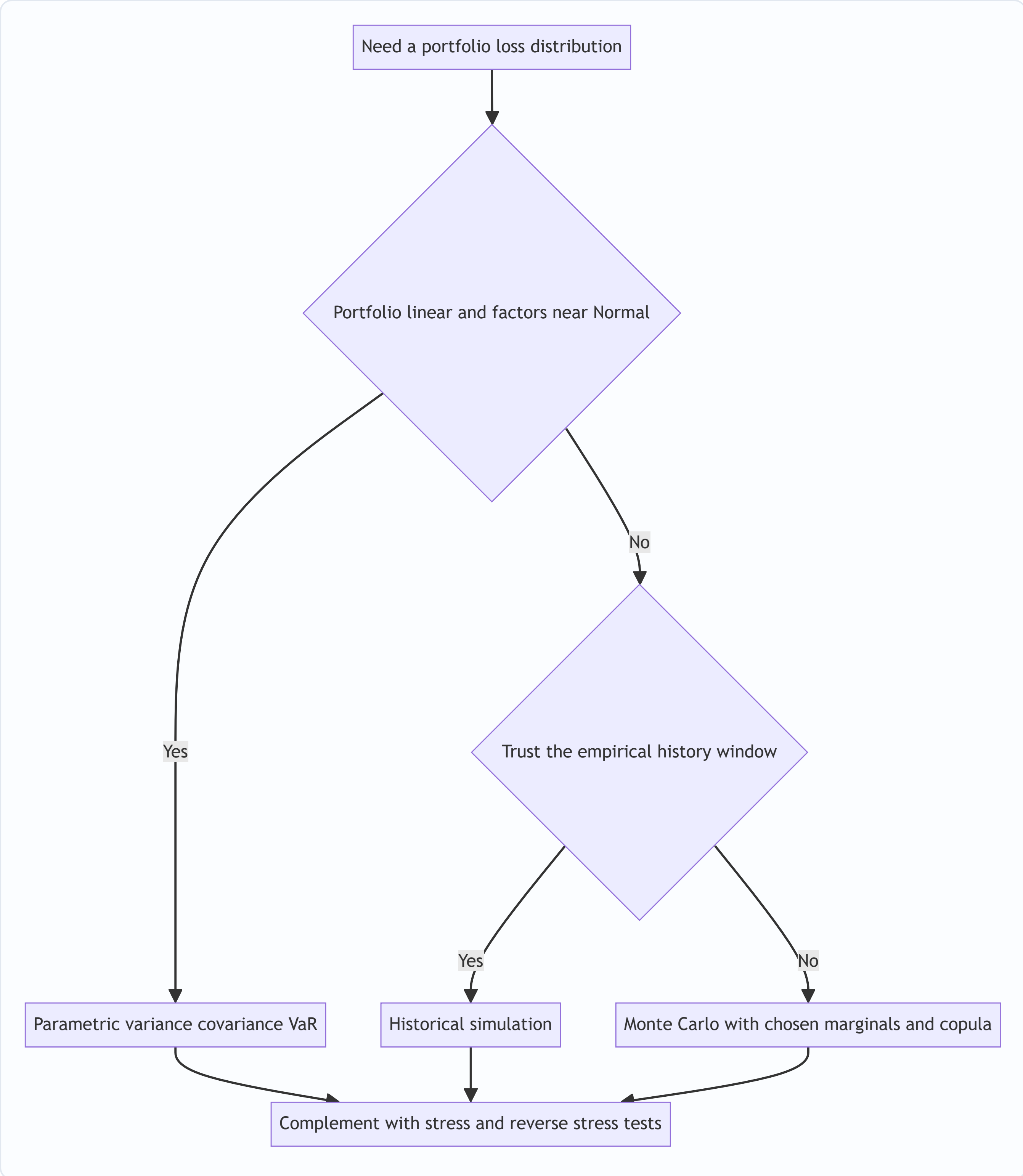
(a) Parametric / analytic (variance–covariance). Assume factors are jointly Normal; portfolio VaR from $\sigma_P = \sqrt{w^\top \Sigma\, w}$, then $\text{VaR}=z_\alpha\sigma_P V$. Fast; wrong for options and fat tails.

(b) Historical simulation. Apply the last N days of *actual* factor changes to today's portfolio; the empirical distribution of the N revalued P&Ls gives VaR/ES as an order statistic. No distributional assumption and captures real fat tails and dependence — but it is bounded by history (no scenario worse than the worst in the window) and gives equal weight to stale data.

(c) Monte Carlo simulation. Specify marginals + copula, draw M scenarios, fully revalue the portfolio in each, read the risk measure off the simulated distribution. Handles non-linearity, fat tails, arbitrary dependence — at high computational cost, and it is only as good as the assumed model (model risk).

Stress testing and scenario analysis complement statistical VaR. Rather than asking "what is the 99% loss under normal conditions," stress tests ask "what do we lose in *this specific bad state*" — historical (repeat 2008, 1998) or hypothetical (rates +300bp with spreads +200bp). **Reverse stress testing** inverts the question: *what scenario would render us insolvent?* — then assess its plausibility. Stress tests are essential because they probe the tail *without relying on the distributional assumptions that fail exactly in the tail*.

Figure 3 — choosing an aggregation method.



4.5 Backtesting risk models

Backtesting compares realised outcomes against model predictions. For a 99% one-day VaR, a **breach (exception)** occurs when the actual loss exceeds the VaR. If the model is correct, breaches are a Bernoulli process with probability $p = 1-\alpha = 1\%$ and should be *independent* across days.

Kupiec's Proportion-of-Failures (unconditional coverage) test. With x exceptions in T days, the likelihood-ratio statistic is:

$$LR_{uc} = -2\ln\left[\frac{(1-p)^{T-x}\,p^{x}}{(1-\hat p)^{T-x}\,\hat p^{x}}\right], \quad \hat p = \frac{x}{T}$$

distributed χ^2_1. Reject the model if LR_{uc} exceeds the critical value (3.84 at 5%). This tests *whether the number of breaches is right*.

Christoffersen's test adds an **independence** component (breaches should not cluster): $LR_{cc}=LR_{uc}+LR_{ind}$, distributed χ^2_2. Clustering of breaches signals the model reacts too slowly to changing volatility.

Basel traffic-light zones for 99% one-day VaR over 250 days (expected breaches ≈ 2.5):

Zone	Exceptions in 250 days	Capital multiplier k	Interpretation
Green	0–4	3.00	Model accepted
Yellow	5–9	3.40–3.85 (scaled)	Increasing scrutiny
Red	10+	4.00	Model rejected / overhaul

The market-risk capital charge scales with this multiplier, so backtest failures *directly raise required capital*. Note the asymmetry: ES is harder to backtest than VaR (it is not "elicitable" in the simple sense), which is a live practical tension in the ES-based FRTB regime.

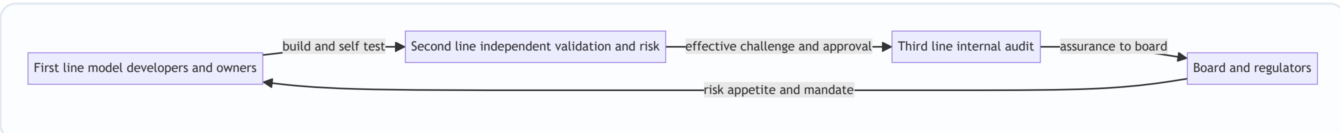
4.6 Model validation and governance

Model risk is the risk of loss from using a model that is wrong or misused. Regulators (US **SR 11-7**, the canonical guidance) require a formal framework with three defences:

1. **Conceptual soundness** — is the theory right, are assumptions justified, is the math correct?
2. **Ongoing monitoring** — backtesting, benchmarking against alternative models, sensitivity analysis, checking the model is used within its intended scope.
3. **Outcomes analysis** — do outputs match reality over time?

Governance principles: **independent validation** (validators separate from developers), the **effective challenge** doctrine (someone competent and empowered questions the model), a **model inventory** with tiering by materiality, documented **assumptions and limitations**, and clear **ownership**. The "three lines of defence": the business/model owner (1st), independent risk/validation (2nd), internal audit (3rd).

Figure 4 — the three lines of defence in model governance.



4.7 Strengths and pitfalls of quantitative risk models

Strengths: consistency and comparability across desks; a common language (one VaR number aggregates thousands of positions); capital efficiency (risk-sensitive charges); the discipline of forcing assumptions to be explicit; enabling limits, allocation, and pricing.

Pitfalls: *model risk* (wrong specification); *estimation risk* (parameters from short/stale samples); *fat-tail blindness* (Normal assumptions); *correlation breakdown* (dependence rises in crises); *procyclicality* (low measured risk in calm times encourages leverage that amplifies the next crash); *endogeneity* (everyone using the same VaR models and hitting limits together forces correlated selling — the model changes the world it measures); *false precision* (a VaR quoted to the dollar implies confidence the data cannot support); and *gaming* (choosing the window or method that minimises the number).

5. Worked Examples

Example 1 — Parametric vs fat-tailed VaR and ES (reconciling the tail-risk gap)

A trading book worth **$100 million** has daily return volatility **σ = 1.5%** and mean ≈ 0.

(a) Normal 99% one-day VaR. $z_{0.99}=2.326$. $$\text{VaR}_{99\%} = 2.326 \times 0.015 \times \$100\text{m} = \$3.489\text{m}.$$

(b) Normal 97.5% Expected Shortfall. For a Normal, $\text{ES}_\alpha = \sigma\,\dfrac{\phi(z_\alpha)}{1-\alpha}V$. At $\alpha=0.975$, $z=1.960$, $\phi(1.960)=0.0584$: $$\text{ES}_{97.5\%}=0.015\times\frac{0.0584}{0.025}\times\$100\text{m}=0.015\times2.338\times\$100\text{m}=\$3.507\text{m}.$$

Reconciliation check: Basel calibrated 97.5% ES to be *close to* 99% VaR under Normality — and indeed $\$3.507\text{m}\approx\$3.489\text{m}$ (within 0.5%). Good: the two regimes are roughly comparable *if returns are Normal*.

(c) Now assume Student-t with ν = 5 (fat tails). The t quantile at 99% is $t_{0.99,5}=3.365$, but the t must be scaled so its variance matches σ. A t with ν d.f. has variance $\frac{\nu}{\nu-2}$, so scale factor $=\sqrt{\frac{\nu-2}{\nu}}=\sqrt{3/5}=0.7746$. The effective 99% multiplier is $3.365\times0.7746=2.607$. $$\text{VaR}_{99\%}^{t} = 2.607\times0.015\times\$100\text{m}=\$3.910\text{m}.$$

Reconciliation and interpretation: the fat-tailed model gives **$3.91m vs $3.49m** — about **12% more capital** at the same volatility and confidence. The extra $0.42m is the cost of honestly representing fat tails. The Normal model is *not wrong about volatility* (both use σ=1.5%); it is wrong about *tail shape*, and the entire gap lives in the 1% tail. This is the chapter's central lesson in one number.

Example 2 — Historical-simulation VaR and ES as order statistics

A desk has **500 days** of P&L. VaR is a quantile read directly off sorted losses.

- **99% VaR** = the loss at the $(1-0.99)\times500 = 5\text{th}$ worst outcome. Sort the 500 P&Ls ascending; the 5th value (5th-largest loss) is the 99% VaR.

Suppose the five worst daily P&Ls (in $m) are: **−8.0, −6.5, −5.2, −4.8, −4.1**.

- **99% VaR** = 5th worst = **$4.1m** (the least-bad of the worst 1%).
- **99% ES** = average of the losses *at least as bad as VaR* = mean of the worst 5 (the tail beyond and including the quantile): $$\text{ES}_{99\%}=\frac{8.0+6.5+5.2+4.8+4.1}{5}=\frac{28.6}{5}=\$5.72\text{m}.$$

Reconciliation check: ES ($5.72m) > VaR ($4.1m) *always*, because ES averages the tail that VaR merely bounds. The ratio here is 1.40, notably above the ≈1.15 you would get under Normality — a quantitative signature that this empirical tail is **fatter than Normal**. The order-statistic method needed no distributional assumption to reveal that.

Example 3 — Backtesting with Kupiec's test

A 99% one-day VaR model produced **12 exceptions in T = 250 trading days**. Expected exceptions $= 0.01\times250 = 2.5$. Observed rate $\hat p = 12/250 = 0.048$.

Kupiec unconditional-coverage statistic with $p=0.01$: $$LR_{uc}=-2\ln!\left[\frac{(0.99)^{238}(0.01)^{12}}{(0.952)^{238}(0.048)^{12}}\right].$$

Compute the log-likelihoods: - Under $p=0.01$: $238\ln(0.99)+12\ln(0.01)=238(-0.01005)+12(-4.6052)=-2.392-55.262=-57.654$. - Under $\hat p=0.048$: $238\ln(0.952)+12\ln(0.048)=238(-0.04919)+12(-3.0366)=-11.708-36.439=-48.147$.

$$LR_{uc}=-2\,(-57.654-(-48.147))=-2(-9.507)=19.01.$$

Reconciliation and decision: Compare 19.01 with the χ^2_1 critical value of **3.84 (5%)** and **6.63 (1%)**. Since $19.01 \gg 6.63$, we **reject the model** decisively — 12 breaches is far too many for a genuine 99% model. Cross-checking against **Basel's traffic light**: 10+ exceptions is the **Red zone**, capital multiplier $k=4.0$ (vs 3.0 in Green). Both frameworks agree: this model is broken and its capital charge should rise by one-third. The two independent tests reconciling on the same verdict is exactly the kind of corroboration good governance demands.

6. Connections

- **To Chapter on Market Risk / VaR:** this chapter supplies the *engine room* — the distributions, dependence, and simulation methods that produce the VaR and ES numbers used for market-risk limits and capital (FRTB's 97.5% ES).
- **To Credit Risk:** copulas and tail dependence are the heart of portfolio credit models (correlated defaults, CDO tranching); the Gaussian-copula lesson is a credit-modelling lesson.
- **To Capital and Regulation:** backtesting zones feed the Basel multiplier that sets market-risk capital; SR 11-7 / model governance sits alongside Pillar 2.
- **To Stress Testing:** scenario and reverse-stress methods here are the same tools used in regulatory stress tests (CCAR, EBA).
- **To Liquidity Risk:** the *endogeneity/procyclicality* pitfall links model-driven selling to liquidity spirals.
- **To Behavioural / Governance:** the "effective challenge" doctrine and model-risk culture connect to operational and conduct risk.

7. Key Terms

Term	Meaning
Marginal distribution	The distribution of a single risk factor on its own.
Leptokurtosis / fat tails	Excess kurtosis > 0; extreme moves more frequent than Normal.
Value-at-Risk (VaR)	The loss quantile not exceeded with probability α.
Expected Shortfall (ES)	Average loss in the worst (1−α) tail; coherent and tail-sensitive.
Coherence / sub-additivity	A risk measure where diversification never increases risk.
Copula	A function that binds marginals into a joint distribution, carrying all dependence (Sklar's theorem).
Tail dependence (λ)	Probability of one variable being extreme given another is extreme; zero for Gaussian copula.
Extreme Value Theory (EVT)	Modelling only the tail, e.g. fitting a Generalised Pareto Distribution to exceedances.
Historical simulation	VaR/ES from the empirical distribution of past factor moves applied to today's book.
Monte Carlo simulation	Drawing many scenarios from an assumed model to build the loss distribution.
Backtesting / exception	Comparing realised losses to VaR; a breach when loss exceeds VaR.
Kupiec / Christoffersen tests	Statistical tests of breach frequency and independence.
Model risk	Loss from a wrong or misused model.
Effective challenge	Independent, competent, empowered questioning of a model (SR 11-7).
Procyclicality	Risk measured low in booms, high in busts, amplifying cycles.

8. Common Confusions

- **VaR is the maximum loss.** No — VaR is the *threshold* the loss won't exceed with confidence α; losses *beyond* it are unbounded and are what ES measures. "99% VaR of $3m" means on the worst 1% of days you lose *at least* $3m, possibly far more.
- **Correlation fully describes dependence.** Only for elliptical (Normal/t) joint distributions. In general, zero correlation ≠ independence, and two portfolios with identical correlations can have wildly different joint-tail behaviour depending on the copula.
- **A higher correlation input fixes tail risk.** Not with a Gaussian copula — its tail dependence is *zero at any correlation below 1*. You must change the *copula family* (to t/Clayton), not just the number.
- **Historical simulation is assumption-free, so it's safest.** It assumes the future resembles the sampled past and cannot produce any scenario worse than its window — it is blind to unprecedented shocks.
- **More Monte Carlo paths = more accurate risk.** Simulation error shrinks with paths, but *model error* (wrong marginals/copula) does not shrink at all. A billion paths of the wrong model is precisely wrong.

- **Passing the backtest means the model is right.** Backtesting has limited power on ~250 days; a model can pass by luck or fail to be challenged on scenarios that simply didn't occur. Backtesting is necessary, not sufficient.
- **ES is strictly better than VaR so backtesting is solved.** ES is harder to backtest (elicitability issues); the FRTB uses VaR-based backtesting to police an ES-based charge — an accepted awkwardness.
- **Fat tails only change the extreme quantile, not the σ.** Correct and important: in Example 1 both models share σ=1.5%; the tail model changes only the *multiplier*, and that is exactly where capital is won or lost.

9. Recap

A risk model maps uncertainty to a number through three separable layers — **marginals** (where fat tails live), **dependence** (where copulas and tail dependence live), and **aggregation** (where simulation lives) — wrapped in **backtesting and governance**. The Normal distribution is analytically seductive but empirically thin-tailed; Student-t and EVT restore the fat tails that determine VaR and ES. Sklar's theorem lets us model marginals and dependence independently via copulas, and the decisive property is **tail dependence**: the Gaussian copula's zero tail dependence is the structural flaw behind the structured-credit failures. We build the portfolio loss distribution parametrically, by historical simulation, or by Monte Carlo, and we complement statistical measures with stress and reverse-stress tests that don't rely on the very assumptions that fail in the tail. **Backtesting** (Kupiec, Christoffersen, Basel traffic lights) checks whether reality matches the model and directly sets the capital multiplier; **validation and governance** (SR 11-7, three lines of defence, effective challenge) check whether we should believe the model at all. The worked examples showed a fat-tailed model demanding ~12% more capital than Normal, ES exceeding VaR by construction and revealing a fat empirical tail, and a Kupiec test rejecting a 12-breach model in agreement with Basel's Red zone. Quantitative models give consistency, comparability, and capital efficiency — but carry model risk, estimation risk, correlation breakdown, procyclicality, endogeneity, and false precision. The mature risk manager treats every number as a *hypothesis under governance*, not a fact.

10. Quick-Reference / Interview Points

Formulas to have cold: - Normal VaR: $\text{VaR}*alpha = z*\alpha \sigma V$ ($z_{99}=2.326$, $z_{95}=1.645$). - Normal ES: $\text{ES}*alpha = \sigma*\,*frac{\phi(z*\alpha)}{1-\alpha}V$. - Excess kurtosis = kurtosis − 3; Normal kurtosis = 3. - t-distribution kurtosis $=3+\frac{6}{\nu-4}$; scale by $\sqrt{(\nu-2)/\nu}$ to fix variance. - Sklar: $F(x_1,\dots)=C(F_1(x_1),\dots)$. - Kupiec $LR_{uc}\sim\chi^2_1$; reject above 3.84. - Expected breaches for 99% VaR over 250 days ≈ 2.5.

One-liners interviewers reward: - "VaR tells you the door; ES tells you what's behind it." ES is coherent and tail-sensitive; FRTB uses **97.5% ES**. - "The Gaussian copula has zero tail dependence — that's *why* it missed correlated defaults, not the correlation number." - "Sklar lets me fix fat tails and dependence separately." Marginals and copula are independent choices. - "Historical sim can't imagine a scenario worse than its worst day — so I stress test." - "More paths cut simulation error, never model error." - "Backtesting sets the Basel multiplier: Green k=3.0, Red k=4.0 — model quality *is* capital." - "SR 11-7: independent validation and effective challenge — someone competent and empowered must be able to say no." - "Procyclicality and endogeneity: models measure low risk in booms and, because everyone uses them, *create* the correlated selling they later measure."

If asked "why did risk models fail in 2008?": thin-tailed marginals (Normal) + zero-tail-dependence dependence (Gaussian copula) + backward-looking calibration on a benign window + procyclical leverage + endogenous forced-selling — a failure of every layer at once, and a governance failure in not challenging the assumptions.

Q&A — Risk Models and Measurement

Practice bank for Chapter 11. Every question is followed by a full answer. Constants used throughout: Normal quantiles $z_{0.99}=2.326$, $z_{0.975}=1.960$, $z_{0.95}=1.645$; $\phi(2.326)=0.0267$, $\phi(1.960)=0.0584$; χ^2_1 critical values 3.84 (5%) and 6.63 (1%).

Section A — Concept Check

A1. A risk model has three separable layers. Name them and say what "lives" in each.

(1) **Marginal distributions** — the distribution of each individual risk factor; this is where *fat tails* live. (2) **Dependence structure** — how the factors co-move; this is where *correlation, copulas, and tail dependence* live. (3) **Aggregation and measurement** — combining layers 1 and 2 into a portfolio loss distribution and reading off a measure like VaR or ES; this is where *simulation and scenario analysis* live. Each layer can fail independently, so a model can get the marginals right and the dependence catastrophically wrong.

A2. What single conceptual result lets us model the marginals and the dependence separately, and why is that both liberating and dangerous?

Sklar's theorem: any joint distribution $F(x_1,\dots,x_n)=C\big(F_1(x_1),\dots,F_n(x_n)\big)$, where the F_i are marginals and C is a copula carrying *all* the dependence. It is *liberating* because you can fix fat tails without touching correlations and vice versa. It is *dangerous* because you can glue fat-tailed marginals onto a benign, no-joint-extremes dependence structure and fool yourself into thinking the model is conservative when it is structurally blind to joint crashes.

A3. Why is the tail — not the mean or standard deviation — "the whole game" in risk modelling?

Risk measures (VaR, ES) read the far left of the loss distribution: the 1% or 0.1% worst outcomes. The centre, where returns spend most of their time, almost never bankrupts anyone. So modelling effort is *asymmetric*: getting the tail shape right matters far more than fitting the body. A model can fit 99% of the data beautifully and still be worthless because it misprices the 1% that matters.

A4. Why does the Normal distribution keep getting used even though returns are not Normal?

Analytical convenience: it has one scale parameter (σ), it is closed under linear combinations (a portfolio of jointly-Normal assets is Normal, giving VaR a clean closed form), and correlation *fully* describes its dependence. But real returns are **leptokurtic** (excess kurtosis > 0 — fat-tailed, sharp-peaked) and show **volatility clustering** (calm and stormy periods bunch, so returns are not i.i.d. over time). Observed "5-sigma" days occur orders of magnitude more often than a Normal permits.

A5. VaR is "not coherent." Which axiom does it violate, and what fixes it?

VaR can violate **sub-additivity** — a portfolio's VaR can exceed the sum of its parts' VaRs, i.e. it can *penalise diversification*, which is incoherent. **Expected Shortfall (ES)**, the average loss in the worst $(1-\alpha)$ tail, is sub-additive (coherent) and tail-sensitive; it also captures how bad losses are *beyond* the quantile, which VaR ignores. Basel's FRTB replaced 99% VaR with **97.5% ES** for exactly these reasons.

A6. "A higher correlation input will fix my model's tail risk." True or false, and why?

False — if the model uses a **Gaussian copula**. The Gaussian copula has **zero tail dependence** at any correlation below 1: the probability of joint extreme losses vanishes in the tail no matter how high ρ is. To represent "everything crashes together" you must change the *copula family* (to Student-t or Clayton), not just raise the correlation number. This is the structural flaw behind the Gaussian-copula CDO failures.

A7. Why is historical simulation not truly "assumption-free"?

It drops the Normality assumption but replaces it with an equally strong one: **the chosen historical window represents the future.** It cannot produce any scenario worse than the worst day in its window, it weights stale data equally with fresh data, and a single crash in the window can dominate the tail. The assumption moved from "the distribution is Normal" to "the past repeats" — it did not disappear. This is why it must be complemented with stress testing.

A8. Distinguish the three ways to build a portfolio loss distribution in one line each.

(a) **Parametric/variance–covariance**: assume factors jointly Normal, compute $\sigma_P=\sqrt{w^\top\Sigma w}$, read $\text{VaR}=z_\alpha\sigma_P V$ — fast but wrong for options and fat tails. (b) **Historical simulation**: apply the last N days of actual factor moves to today's book and read the empirical quantile — captures real fat tails and dependence but bounded by history. (c) **Monte Carlo**: specify marginals + copula, draw M scenarios, fully revalue — handles non-linearity and arbitrary dependence but is costly and only as good as the assumed model.

A9. What does reverse stress testing ask that ordinary stress testing does not?

Ordinary stress testing asks "what do we lose in *this specific bad scenario*?" Reverse stress testing inverts it: "**what scenario would render us insolvent?**" — you solve for the state of the world that breaks the firm, then assess how plausible it is. It is valuable precisely because it surfaces vulnerabilities that the analyst would never have thought to test for directly.

A10. Why does a backtest failure directly cost a bank money?

Under Basel's Internal Models Approach, the market-risk capital charge is scaled by a multiplier k tied to the number of VaR exceptions over 250 days. **Green zone** (0–4 breaches) gives $k=3.0$; **Red zone** (10+) gives $k=4.0$. A model that breaches too often is pushed toward higher k, mechanically raising required capital by up to a third. So model quality *is* capital — backtesting is not academic.

Section B — Numerical / Applied (with full solutions)

B1. Normal VaR and the ES calibration check. A trading book worth **V = $50m** has daily σ = 2%, mean ≈ 0. Find (a) the 99% one-day VaR, (b) the 97.5% one-day ES, and confirm Basel's calibration claim that 97.5% ES ≈ 99% VaR under Normality.

Solution. (a) $\text{VaR}{99\%}=z{0.99}\,\sigma V = 2.326\times0.02\times50\text{m}=2.326\times1\text{m}=\mathbf{\$2.326m}$. (b) Normal ES: $\text{ES}\alpha=\sigma\,\dfrac{\phi(z\alpha)}{1-\alpha}V$. At α = 0.975, $\phi(1.960)=0.0584$: $$\text{ES}_{97.5\%}=0.02\times\frac{0.0584}{0.025}\times50\text{m}=0.02\times2.338\times50\text{m}=\mathbf{\$2.338m}.$$ Check: $2.338m \approx 2.326m$ — within 0.5%. ✓ Basel calibrated 97.5% ES to sit right on top of 99% VaR *for Normal returns*; the two diverge only once tails fatten (see B2).

B2. Fat-tailed VaR with a Student-t. Same book (V = $50m, σ = 2%). Daily returns are better described by a Student-t with **ν = 6** degrees of freedom. The one-tailed 99% t-quantile is $t_{0.99,6}=3.143$. Find the fat-tailed 99% VaR and compare with the Normal result.

Solution. The raw t must be **scaled so its variance equals σ²**. A t with ν d.f. has variance $\frac{\nu}{\nu-2}$, so the scale factor is $\sqrt{\frac{\nu-2}{\nu}}=\sqrt{4/6}=\sqrt{0.6667}=0.8165$. Effective 99% multiplier $=3.143\times0.8165=2.566$. $$\text{VaR}_{99\%}^{t}=2.566\times0.02\times50\text{m}=\mathbf{\$2.566m}.$$ Comparison: $2.566m$ vs the Normal $2.326m$ — about **10% more capital** at the *same* volatility and confidence. Both models agree on σ = 2%; the Normal is wrong only about *tail shape*, and the entire $0.24m gap lives in the 1% tail. That gap is the price of honestly representing fat tails.

B3. Matching kurtosis to degrees of freedom. Empirically, this desk's daily returns show **excess kurtosis of 3** (i.e. kurtosis = 6). Using the Student-t kurtosis formula, what ν reproduces this?

Solution. For ν > 4, t-kurtosis $=3+\dfrac{6}{\nu-4}$. Set equal to 6: $$3+\frac{6}{\nu-4}=6 \;\Rightarrow\; \frac{6}{\nu-4}=3 \;\Rightarrow\; \nu-4=2 \;\Rightarrow\; \boxed{\nu=6}.$$ Interpretation: the same ν = 6 used in B2 is not arbitrary — it is pinned by the observed tail heaviness. Lower ν = fatter tails; as ν → ∞ the t collapses to the Normal (excess kurtosis → 0).

B4. Historical-simulation VaR and ES as order statistics. A desk has **500 days** of P&L. The six worst daily P&Ls (in $m) are: **−9.2, −7.4, −6.1, −5.5, −4.9, −4.3**. Find the 99% VaR and 99% ES, and comment on the tail.

Solution. The 99% VaR is the loss at the $(1-0.99)\times500 = 5\text{th}$ worst outcome. - **99% VaR** = 5th-worst loss = **$4.9m** (the least-bad of the worst 1%). - **99% ES** = average of the losses *at least as bad as VaR*, i.e. the worst 5: $$\text{ES}_{99\%}=\frac{9.2+7.4+6.1+5.5+4.9}{5}=\frac{33.1}{5}=\mathbf{\$6.62m}.$$ Comment: ES ($6.62m) > VaR ($4.9m) always, because ES averages the tail VaR merely bounds. The ratio 6.62/4.9 = **1.35**, well above the ≈1.15 you would get under Normality — a quantitative fingerprint that this empirical tail is **fatter than Normal**, obtained with no distributional assumption. (Note the 6th-worst value, −4.3, is irrelevant to a 99% measure on 500 days; only the worst 5 enter.)

B5. Kupiec proportion-of-failures backtest. A 99% one-day VaR model produced **8 exceptions in T = 250 days**. Test it with Kupiec's unconditional-coverage statistic and reconcile with Basel's traffic light.

Solution. Expected exceptions $=0.01\times250=2.5$; observed $\hat p=8/250=0.032$. With $p=0.01$: $$LR_{uc}=-2\ln!\left[\frac{(0.99)^{242}(0.01)^{8}}{(0.968)^{242}(0.032)^{8}}\right].$$ Log-likelihoods: - Under $p=0.01$: $242\ln(0.99)+8\ln(0.01)=242(-0.010050)+8(-4.60517)=-2.432-36.841=-39.273$. - Under $\hat p=0.032$: $242\ln(0.968)+8\ln(0.032)=242(-0.032520)+8(-3.44202)=-7.870-27.536=-35.406$. $$LR_{uc}=-2\big(-39.273-(-35.406)\big)=-2(-3.867)=\mathbf{7.73}.$$ Decision: $7.73 > 6.63$, so we **reject the model at the 1% level** (χ^2_1). But Basel's traffic light puts 8 exceptions in the **Yellow zone** (5–9), meaning *increased scrutiny and a scaled multiplier* rather than outright rejection. The lesson: the pure statistical test and the supervisory rule can disagree at the margin — Kupiec says "reject," Basel says "watch closely and add capital." Both agree the model is under strain.

B6. Basel capital impact. Continue B5. Suppose the bank's 10-day 99% VaR is **$40m** and, before any add-ons, capital = $k \times$ VaR. Compare the charge in the Green zone ($k=3.0$) with a Red-zone outcome ($k=4.0$).

Solution. - Green: $3.0\times40\text{m}=\$120\text{m}$. - Red: $4.0\times40\text{m}=\$160\text{m}$. Difference = **$40m of extra capital (+33%)** purely because the model breached too often. This is the concrete mechanism by which "model quality is capital": the same VaR number attracts a one-third larger charge when backtesting fails.

B7. Gaussian copula rank correlation. Two factors are joined by a Gaussian copula with linear correlation **$\rho = 0.7$**. Compute Kendall's τ using $\tau=\frac{2}{\pi}\arcsin(\rho)$, and state the tail dependence.

Solution. $\arcsin(0.7)=0.7754$ rad. $\tau=\frac{2}{\pi}(0.7754)=0.6366\times0.7754=\mathbf{0.494}.$ So a strong linear correlation of 0.7 corresponds to a rank concordance of ~0.49. Crucially, the **lower tail dependence $\lambda_L=0$** for the Gaussian copula at any $\rho < 1$: despite $\rho = 0.7$ and $\tau = 0.49$, the model assigns *vanishing* probability to joint extreme losses. Swapping to a Student-t copula (same ρ, finite ν) would give $\lambda_L>0$ and restore joint-crash risk — the number that matters for a credit portfolio.

Section C — Interview-Style (with model answers)

C1. "Walk me through why risk models failed in 2008."

Model answer: It was a failure of every layer at once. **Marginals** were thin-tailed (Normal), so extreme moves were treated as near-impossible. **Dependence** used the Gaussian copula, which has *zero tail dependence* and structurally cannot represent correlated defaults. **Calibration** was backward-looking on a benign window, so the models had never "seen" a crisis. Then two feedback effects amplified it: **procyclicality** (low measured risk justified high leverage) and **endogeneity** (everyone used the same models, hit limits together, and was forced into correlated selling — the models *created* the correlation they later measured). And wrapping it all, a **governance** failure: no effective challenge to the assumptions. The lesson is that fat tails and tail dependence are not refinements — they are the phenomenon.

C2. "A trader says his desk's VaR is $2m, so that's the most he can lose in a day. Correct him."

Model answer: That is the single most common VaR error. VaR is a **quantile, not a maximum**. "99% VaR of $2m" means that on the worst 1% of days he loses *at least* $2m — the loss beyond that threshold is unbounded and is exactly what VaR is silent about. To know how bad it gets when the threshold breaks, use **Expected Shortfall** — the average of the tail beyond VaR, always larger. VaR tells you the door; ES tells you what's behind it. And both assume normal markets, so neither covers a gap or liquidity hole — that's what stress testing is for.

C3. "You have three VaR methods. How do you choose?"

Model answer: I choose by the two questions that actually differentiate them. First, *is the book linear and are the factors close to Normal?* If yes, **parametric variance–covariance** is fast and adequate. If the book has options or the factors are fat-tailed, parametric breaks, so I move on. Second, *do I trust my historical window to represent the future?* If the window is rich and relevant, **historical simulation** gives me real fat tails and real dependence with no distributional assumption. If I distrust the window — because it's too calm, too short, or the world has changed — I use **Monte Carlo** with explicitly chosen marginals and a copula, which lets me inject fat tails and tail dependence deliberately. Whichever I pick, I *always* complement it with stress and reverse-stress tests, because those probe the tail without relying on the very assumptions that fail in the tail.

C4. "Sell me on Expected Shortfall over VaR — then tell me its catch."

Model answer: ES wins on two counts. It is **coherent** — specifically sub-additive — so it never penalises diversification, whereas VaR can. And it is **tail-sensitive**: it averages the whole tail rather than reading a single point, so it responds to *how* fat the tail is, not just where the quantile sits. That's why FRTB moved to 97.5% ES. The catch is **backtesting**: ES is not "elicitable" in the simple way a quantile is, so it's harder to validate against realised outcomes. In practice regulators police an ES-based charge using VaR-based backtesting — an accepted awkwardness. So ES is the better *measure* but the harder thing to *check*, and a mature framework holds both facts at once.

C5. "How does your model risk governance actually stop a bad model from shipping?"

Model answer: Through the three lines of defence and the doctrine of **effective challenge** from SR 11-7. The **first line** — developers and model owners — build the model and self-test it. The **second line** — independent validation and risk — must be competent, empowered, and separate enough to say *no*: they check conceptual soundness (is the theory and math right?), run ongoing monitoring (backtesting, benchmarking against alternatives, sensitivity analysis), and do outcomes analysis. The **third line** — internal audit — assures the board that the process itself works. The point of effective challenge is that a validator who is junior, captured, or under-resourced is worthless; the person questioning the model must genuinely be able to block it. Everything sits on a model inventory tiered by materiality, with documented assumptions and limitations. Governance answers a different question from the math: the math says what the risk *is if the model is right*; governance says *whether to believe the model at all*.

Section D — MCQs (with reasoning)

D1. The Gaussian copula's defining weakness for credit risk is that it has: (a) negative correlation; (b) zero tail dependence; (c) infinite variance; (d) non-uniform marginals.

Answer: (b). Its lower tail dependence $\lambda_L=0$ for any $\rho < 1$, so it cannot represent joint extreme losses ("everyone defaults together") regardless of the correlation input. (d) is definitionally wrong — every copula has uniform marginals. (a) and (c) are unrelated to the copula's tail behaviour.

D2. Under Normal returns, which pairing is calibrated to give roughly equal risk numbers? (a) 99% VaR and 99% ES; (b) 95% VaR and 99% ES; (c) 99% VaR and 97.5% ES; (d) 97.5% VaR and 99% ES.

Answer: (c). As shown in B1, 97.5% ES ≈ 99% VaR under Normality — the basis for FRTB's switch. (a) is wrong because ES > VaR at the *same* confidence always. (b) and (d) mismatch the calibration levels.

D3. A Student-t distribution with $\nu = 5$ compared with the Normal has: (a) thinner tails and lower VaR; (b) fatter tails and higher VaR; (c) identical tails; (d) fatter tails but lower VaR.

Answer: (b). Lower ν means fatter tails (kurtosis $=3+\frac{6}{\nu-4}=9$ at $\nu = 5$, vs 3 for Normal), pushing the tail quantile — and hence VaR — higher once variance is matched. (d) is internally contradictory; (a) and (c) reverse or deny the fat-tail fact.

D4. Historical simulation VaR cannot produce a loss worse than: (a) the mean loss; (b) the parametric VaR; (c) the worst observation in its window; (d) the ES.

Answer: (c). It replays actual past moves, so its maximum possible loss is the worst day in the sample window — the core limitation that makes stress testing necessary. The other options are unrelated bounds.

D5. A 99% one-day VaR model shows 12 exceptions in 250 days. Basel places this in the: (a) Green zone, $k=3.0$; (b) Yellow zone; (c) Red zone, $k=4.0$; (d) unclassified.

Answer: (c). 10 or more exceptions is the Red zone (model rejected/overhaul), multiplier $k=4.0$. Green is 0–4, Yellow is 5–9. Expected breaches for a true model are only ~2.5, so 12 is a decisive failure.

D6. Kendall's τ and Spearman's ρ are preferred over Pearson correlation because they: (a) are always larger; (b) depend only on the copula and survive non-linear transforms; (c) require Normality; (d) measure only linear dependence.

Answer: (b). Rank correlations depend on the ranks (the copula), not the marginals, so they survive monotone non-linear transforms and don't assume Normality. (d) describes Pearson correlation; (c) is false and (a) is not generally true.

D7. In Monte Carlo risk estimation, increasing the number of paths reduces: (a) model error; (b) simulation (sampling) error only; (c) both equally; (d) neither.

Answer: (b). More paths shrink *sampling* error but do nothing to *model* error — a billion paths of the wrong marginals/copula is "precisely wrong." Choosing the model right is a separate problem from running it long enough.

D8. Extreme Value Theory's key advantage over historical simulation is that it can: (a) avoid all assumptions; (b) estimate quantiles beyond the range of observed data; (c) eliminate model risk; (d) guarantee sub-additivity.

Answer: (b). Fitting a Generalised Pareto Distribution to exceedances over a high threshold lets EVT extrapolate a principled tail *beyond* the worst observed loss — what historical simulation cannot do. It still carries assumptions and model risk, so (a) and (c) overclaim; (d) confuses EVT with the coherence of ES.

Chapter 12 — Basel Accords and Capital Regulation

1. The Problem / Need

A bank is a strange, fragile machine. It takes in deposits that are payable on demand (a liability that can walk out the door tomorrow) and uses them to fund loans that will not be repaid for years (an illiquid, risky asset). It does this while holding only a thin sliver of its own money — its **capital** — against a mountain of borrowed money. A typical bank funds roughly 90-95% of its assets with debt and deposits, and only 5-10% with shareholders' equity. In leverage terms, that is 10x to 20x. A corporate manufacturer running at 2x leverage would be considered aggressive; a bank at 15x is considered normal.

This structure creates three chronic problems that no single bank can solve on its own:

Problem 1 — The bank can become insolvent from small losses. If a bank funds €100 of assets with €6 of equity and €94 of deposits, then a loss of just 6% on its asset book wipes out the entire equity cushion. The bank is now insolvent: its assets are worth less than what it owes depositors. Because banks take concentrated exposure to credit, market, and operational risk, a 6% loss is not a tail event — it is a bad recession.

Problem 2 — Runs and contagion. Depositors and short-term creditors know they are first-come-first-served. If they suspect a bank is weak, the rational move is to withdraw *first*, before the money runs out. This makes the fear self-fulfilling: a solvent-but-illiquid bank can be destroyed by a run. Worse, banks are interconnected — they lend to each other, clear payments through each other, and hold similar assets. One failure transmits losses and panic to others. This is **systemic risk**: the risk that the failure of one institution cascades through the whole financial system.

Problem 3 — Moral hazard and the safety net. Because bank failures are so damaging, governments provide a safety net: deposit insurance and a lender of last resort (the central bank). But the safety net creates a perverse incentive. If depositors are insured, they stop monitoring the bank's risk-taking, and shareholders — who keep the upside and offload the downside to taxpayers — are tempted to gamble. "Heads I win, tails the taxpayer loses." Left alone, banks will hold *too little* capital, because capital is expensive for shareholders but the cost of failure is socialised.

The market cannot fix these problems by itself. So the state fixes them with **regulation**, and the central instrument of that regulation is a **minimum capital requirement**: a rule that forces every bank to fund a defined minimum fraction of its (risk-adjusted) assets with genuine loss-absorbing equity. This chapter is about how that requirement is defined, measured, and enforced — the story of the **Basel Accords**.

2. The Core Idea

Capital regulation rests on one deceptively simple idea:

> **A bank must hold loss-absorbing capital in proportion to the risk it takes — not in proportion to the size of its balance sheet.**

Two banks can both hold €100 of assets. One holds €100 of short-term government bonds; the other holds €100 of unsecured loans to speculative-grade companies. They are not equally risky, and it would be absurd to require them to hold the same capital. So Basel does not measure requirements against *raw* assets. It measures them against **Risk-Weighted Assets (RWA)** — assets scaled by a risk weight that reflects how likely they are to lose value.

The master equation of the entire framework is the **Capital Adequacy Ratio (CAR)**, also called the Capital-to-Risk-weighted-Assets Ratio (CRAR):

$$ \text{CAR} = \frac{\text{Regulatory Capital}}{\text{Risk-Weighted Assets (RWA)}} \geq \text{Minimum \%} $$

Everything else in Basel is an elaboration of this fraction:

- The **numerator** (capital) is defined precisely and split into quality tiers — the best, most loss-absorbing capital (CET1) at the top, weaker forms below.
- The **denominator** (RWA) is defined precisely — how you turn a portfolio of loans, bonds, trading positions, and operational exposures into a single risk-adjusted number.
- The **minimum ratio** and a stack of **buffers** on top set how much is enough.

Basel then bolts on two ideas that a pure risk-weighted ratio misses:

1. A **leverage ratio** — a simple, non-risk-weighted backstop (capital / total exposure), to catch the case where the risk weights themselves are wrong or gamed.
2. **Liquidity ratios** (LCR and NSFR) — because a bank can be perfectly solvent and *still* die from a run. Capital protects against insolvency; liquidity rules protect against illiquidity. They are different diseases needing different medicine.

That is the whole architecture. The rest of this chapter fills it in.

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3. Why / How It Works

Why risk-weight instead of using raw assets?

Because raw assets punish safe banks and reward reckless ones. A flat "hold 8% of total assets" rule would force a bank full of Treasury bills to hold the same capital as a bank full of junk loans. Rational banks would then *shed the safe assets* (which earn a low return but cost the same capital) and pile into risky ones (high return, same capital). The rule would actively encourage risk-taking. Risk weighting aligns the capital charge with the actual probability and severity of loss, so capital is deployed where losses are likely to come from.

Why tier the capital?

Not all "capital" absorbs losses equally. Common equity absorbs losses immediately and continuously — if the bank loses money, shareholders' equity falls first and there is no obligation to ever pay it back. That is the gold standard: it is loss-absorbing *on a going-concern basis* (while the bank is still alive). Other instruments — certain preferred shares, subordinated debt — only absorb losses in a wind-down (*gone-concern*), or absorb them reluctantly. Basel therefore ranks capital by quality and demands that the bulk of the requirement be met with the highest-quality form. The 2008 crisis proved the point: banks that looked well-capitalised on total-capital measures were revealed to have very little *real* equity underneath, and it was the equity — not the subordinated debt — that the market cared about.

How does this make the system safer? The four channels

1. **Loss absorption.** More equity means the bank can absorb larger losses before becoming insolvent, so fewer banks fail in a given downturn.
2. **Skin in the game.** When shareholders have more of their own money at stake, they internalise more of the downside, which curbs the moral-hazard gamble. Capital is the antidote to "heads I win, tails you lose."
3. **Confidence and run-prevention.** A visibly well-capitalised bank is one that creditors do not need to run from. Capital ratios are public signals; strong ratios dampen the incentive to flee.
4. **Systemic resilience.** Buffers that can be drawn down in stress (rather than triggering instant failure) let banks keep lending through a downturn, and surcharges on systemically important banks force the most dangerous institutions to be the strongest. This reduces contagion.

The key trade-off (why not just require 50% capital?)

Capital is not free from the bank's perspective. Equity investors demand a higher return than depositors, so funding with more equity raises the bank's cost of funds and can shrink lending (this is contested — the Modigliani-Miller view says leverage should not change total funding cost, but frictions like the tax deductibility of debt and the safety-net subsidy make equity privately expensive to banks). Regulation therefore lives on a trade-off: too little capital and the system is fragile; too much and credit becomes scarce and expensive. Basel's minimums are the negotiated answer to *where on that curve to sit*.

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4. Full Content — The Framework, Formulas, and Methods

4.1 A short history: Basel I, II, III

The **Basel Committee on Banking Supervision (BCBS)**, hosted at the Bank for International Settlements (BIS) in Basel, Switzerland, sets these standards. It has no legal power — it issues *standards* that member jurisdictions then write into their own law (in the EU via the CRR/CRD, in the US via the federal banking agencies, in India via RBI). Its output came in three great waves.

Basel I (1988). The first international capital standard. Its innovation was the risk-weighted asset concept, but crudely: it sorted assets into a handful of buckets with fixed weights (0% for OECD government debt, 20% for banks, 50% for residential mortgages, 100% for corporate loans and everything else). It required **total capital ≥ 8% of RWA**, split into Tier 1 (≥4%) and Tier 2. It covered essentially only credit risk. It was simple and got capital onto the global agenda, but the coarse buckets invited gaming: a loan to a shaky corporate and a loan to a blue-chip both weighed 100%, so banks kept the risky, high-yield version — a distortion called **regulatory arbitrage**.

Basel II (2004). A far more risk-sensitive framework, built on **three pillars**:

- **Pillar 1 — Minimum capital requirements.** Kept the 8% ratio but made RWA far more granular. Introduced explicit charges for **credit, market, and operational risk**. Crucially, allowed banks to use either a **Standardised Approach** (regulator-set weights, now tied to external credit ratings) or **Internal Ratings-Based (IRB)** approaches, where sophisticated banks estimate their own risk parameters (PD, LGD, EAD — see Chapter 8/9 on credit risk) and feed them into supervisory formulas.
- **Pillar 2 — Supervisory review.** Supervisors assess risks not captured in Pillar 1 (e.g. interest-rate risk in the banking book, concentration risk) and can require *more* capital bank-by-bank. Banks run an internal

capital adequacy assessment process (ICAAP).

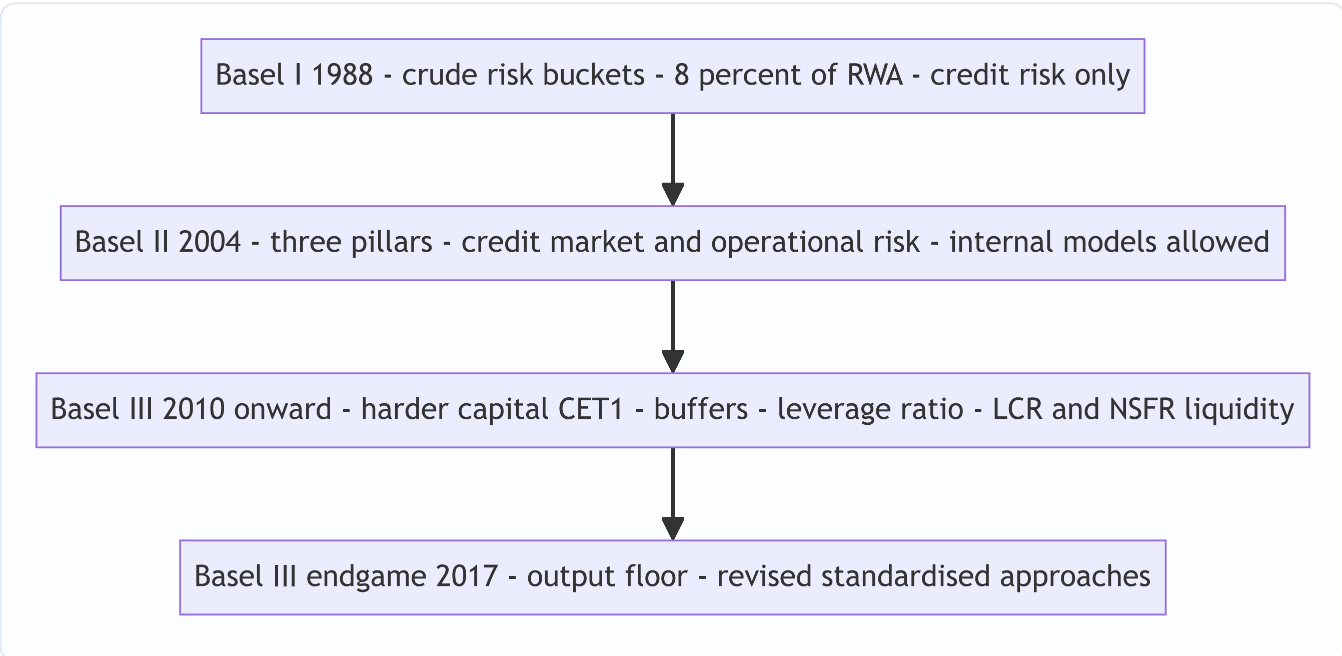
- **Pillar 3 — Market discipline.** Mandatory public disclosure of risk and capital, so the market can price and monitor bank risk.

Basel II's weakness was exposed almost immediately by the 2007-09 crisis: it was **procyclical** (risk weights fell in good times and spiked in bad, amplifying the cycle), it leaned too heavily on external ratings and banks' own optimistic models, it had **no liquidity standard at all**, and its definition of capital was too soft — too much of the "Tier 1" was not genuine common equity.

Basel III (2010 onward, phased in through the 2010s, with a final "Basel III endgame" / Basel 3.1 package agreed in 2017 and implementing into the 2020s). A direct response to the crisis. It did not throw out Basel II's architecture; it hardened every part of it:

- **Better capital.** Redefined capital to put **Common Equity Tier 1 (CET1)** at the centre, with a much higher CET1 minimum and stricter rules on what counts.
- **More capital via buffers.** Added a **capital conservation buffer**, a **countercyclical buffer**, and **surcharges for systemically important banks (G-SIBs/D-SIBs)** on top of the minimums.
- **A leverage ratio.** A non-risk-weighted backstop to the risk-weighted ratios.
- **Liquidity standards — new.** The **Liquidity Coverage Ratio (LCR)** for short-term (30-day) resilience and the **Net Stable Funding Ratio (NSFR)** for structural (1-year) funding stability.
- **The "endgame" reforms (2017).** Constrained internal models with an **output floor** (RWA from internal models cannot fall below 72.5% of the standardised RWA), revised the standardised approaches for credit and operational risk, and replaced the old operational-risk approaches with a single Standardised Measurement Approach.

The diagram below shows how the three accords build on one another.



4.2 The numerator — regulatory capital and its tiers

Regulatory capital is stacked in tiers by loss-absorbing quality. Under Basel III:

Total Capital = Tier 1 + Tier 2, where **Tier 1 = CET1 + Additional Tier 1 (AT1)**.

Tier	Component	What it is	When it absorbs losses
Tier 1	**CET1 (Common Equity Tier 1)**	Common shares, retained earnings, disclosed reserves, share premium, minus regulatory deductions (goodwill, deferred tax assets, etc.)	Continuously, while the bank is a going concern — the highest quality
Tier 1	**AT1 (Additional Tier 1)**	Perpetual instruments with no maturity, discretionary coupons, that convert to equity or write down when CET1 falls below a trigger (e.g. contingent convertibles / CoCos)	Going concern, at the trigger point
Tier 2	**Tier 2**	Subordinated debt (min ~5-year maturity), certain loan-loss provisions	Only in a gone-concern wind-down — absorbs losses after Tier 1 is exhausted

The **regulatory minimums** under Basel III (Pillar 1), as fractions of RWA:

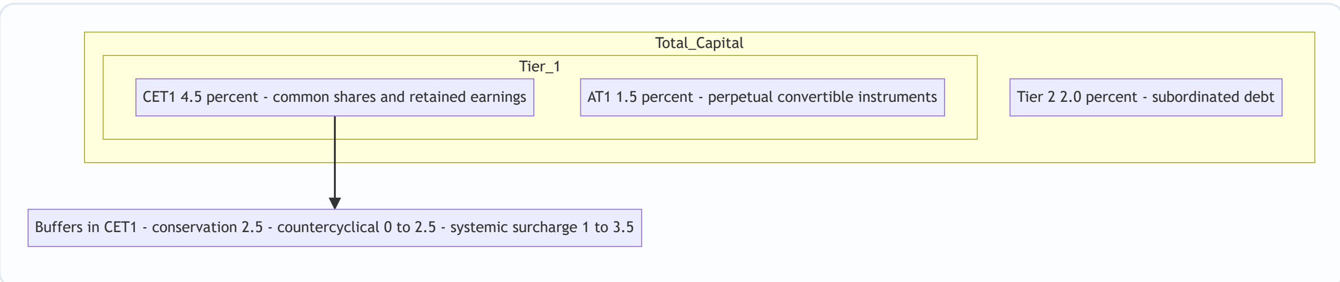
Requirement	Minimum
CET1	**4.5%** of RWA
Tier 1 (CET1 + AT1)	**6.0%** of RWA
Total Capital (Tier 1 + Tier 2)	**8.0%** of RWA

On top of these hard minimums sit the **buffers**, which must be met with **CET1**:

Buffer	Size	Purpose
Capital Conservation Buffer (CCB)	**2.5%**	A drawable cushion above the minimum; dipping into it restricts dividends/bonuses rather than causing failure
Countercyclical Buffer (CCyB)	**0-2.5%**	Set by national regulators; raised in credit booms, released in busts to keep credit flowing
G-SIB / D-SIB surcharge	**1-3.5%**	Extra CET1 for globally/domestically systemic banks — the bigger and more connected, the more

So a normal (non-systemic) bank must hold **CET1 ≥ 4.5% + 2.5% = 7.0%** of RWA in ordinary times, and **Total Capital ≥ 8.0% + 2.5% = 10.5%**. A G-SIB with a 2% surcharge in a period with a 1% countercyclical buffer needs CET1 of 4.5% + 2.5% + 1% + 2% = **10.0%**.

The diagram below shows the capital stack from the highest-quality core out to the buffers.



4.3 The denominator — Risk-Weighted Assets (RWA)

RWA is the sum of risk-weighted exposures across the three risk types:

$$ \text{RWA} = \text{RWA}_{\text{credit}} + \text{RWA}_{\text{market}} + \text{RWA}_{\text{operational}} $$

Credit RWA. Under the Standardised Approach, each exposure is multiplied by a regulatory risk weight (broadly tied to counterparty type and external rating). Representative weights:

Exposure	Typical risk weight
Cash, central government debt (highest-rated)	0%
Highly-rated bank/PSE exposures	20%
Residential mortgages (prudent LTV)	35-50%
Corporate loans (rating-dependent)	20-150%
Unrated / speculative corporate	100%
Past-due / high-risk	150%

$$ \text{RWA}_{\text{credit}} = \sum_i (\text{Exposure}_i \times \text{Risk Weight}_i) $$

Under the **IRB approach**, the bank estimates Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD), and a supervisory formula converts them into a risk weight capturing *unexpected* loss (expected loss is covered by provisions, not capital). The **output floor** then caps how much benefit IRB can give: modelled RWA ≥ 72.5% × standardised RWA.

Market RWA covers the trading book — losses from moves in interest rates, FX, equity, commodity prices. Computed via a standardised sensitivities-based approach or an internal models approach (post-crisis, based on **Expected Shortfall** rather than VaR — see Chapter 6).

Operational RWA covers losses from failed processes, fraud, legal, and systems risk. Basel III replaced the old menu with a single **Standardised Measurement Approach** that scales a Business Indicator (a proxy for size) by an internal loss multiplier.

Because Pillar 1 charges are expressed as capital = 8% of RWA, a common shortcut converts a capital charge *K* back into RWA by multiplying by 12.5 (the reciprocal of 8%):

$$ \text{RWA from a capital charge } K = K \times 12.5 $$

4.4 The leverage ratio — the non-risk-weighted backstop

Risk weighting has a fatal vulnerability: if the *weights* are wrong (or gamed), the risk-weighted ratio lies. Before 2008, banks loaded up on assets that carried tiny risk weights (AAA-rated structured products, sovereign debt) but turned out to be dangerous. Their risk-weighted ratios looked fine right up until they failed. The leverage ratio is the crude, un-foolable check:

$$ \text{Leverage Ratio} = \frac{\text{Tier 1 Capital}}{\text{Total Exposure Measure}} \geq 3\% $$

The **Total Exposure Measure** is *not* risk-weighted — it is on-balance-sheet assets plus derivative exposures, securities-financing exposures, and off-balance-sheet items, with no risk weights applied. A €1 Treasury and a €1 junk loan count the same €1. This is a feature, not a bug: it means a bank cannot escape the leverage limit by claiming its assets are low-risk. The minimum is **3%** (implying a maximum leverage of ~33x), with an additional buffer (typically half the G-SIB surcharge) for the largest banks. The leverage ratio binds when the risk-weighted ratio is very generous; the two together mean a bank must satisfy *both* a risk-sensitive and a risk-blind constraint.

4.5 Liquidity ratios — LCR and NSFR

Capital ratios answer "is the bank solvent?" They say nothing about "can the bank meet cash outflows tomorrow?" A bank can be solvent and still fail if depositors and creditors flee faster than it can raise cash (Northern Rock, 2007). Basel III introduced two liquidity standards.

Liquidity Coverage Ratio (LCR) — short-term, 30-day survival. The bank must hold enough **High-Quality Liquid Assets (HQLA)** — cash and assets that can be sold quickly with little loss (central bank reserves, top-rated government bonds) — to cover 30 days of *stressed* net cash outflows:

$$ \text{LCR} = \frac{\text{Stock of HQLA}}{\text{Total Net Cash Outflows over 30 days (stressed)}} \geq 100\% $$

Net outflows = expected outflows (deposits running off, credit lines drawn — each category multiplied by a stressed run-off rate) minus capped expected inflows. HQLA is split into **Level 1** (0% haircut — cash, reserves, top sovereigns) and **Level 2** (haircuts of 15-50%, and capped at 40% of the total). The intent: survive a month-long acute stress without central-bank help, buying time for an orderly response.

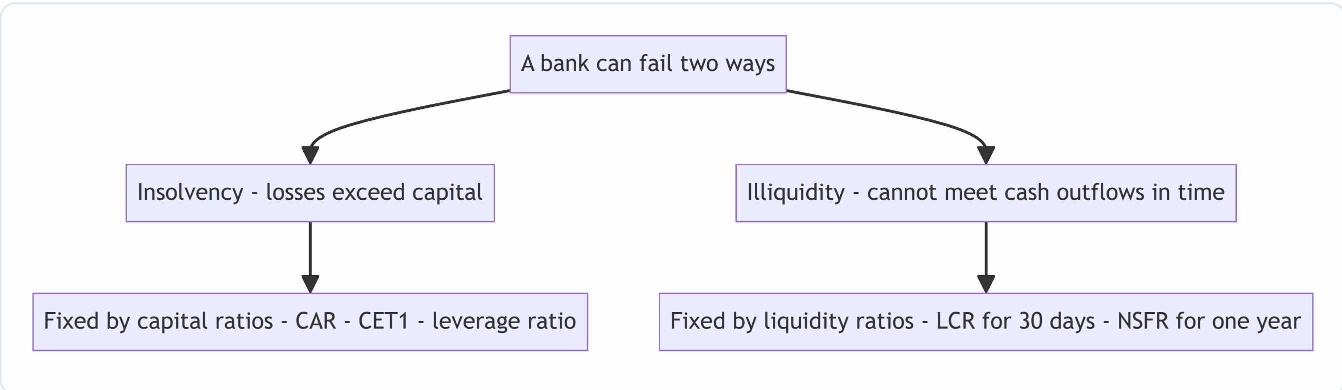
Net Stable Funding Ratio (NSFR) — structural, 1-year stability. LCR is a 30-day snapshot; NSFR addresses the *structural* mismatch — funding illiquid long-term assets with flighty short-term money. It requires that **Available Stable Funding (ASF)** cover **Required Stable Funding (RSF)** over a one-year horizon:

$$ \text{NSFR} = \frac{\text{Available Stable Funding (ASF)}}{\text{Required Stable Funding (RSF)}} \geq 100\% $$

- **ASF** = liabilities and capital weighted by how "sticky" they are (equity and long-term debt get high ASF factors near 100%; overnight wholesale funding gets ~0%).
- **RSF** = assets weighted by how much stable funding they need (cash needs ~0%; long-term illiquid loans need ~85-100%).

NSFR forces the bank to fund long, illiquid assets with genuinely stable funding, curbing the classic "borrow short, lend long" fragility over the medium term. LCR and NSFR are complements: LCR is the sprint (survive the acute run), NSFR is the marathon (be structurally sound over a year).

The diagram below separates the two diseases a bank can die from and the tools that address each.



5. Worked Examples

Example 1 — Computing RWA and the full capital ratio stack

Alpha Bank has the following exposures and capital. Compute credit RWA, total RWA, and check every Basel III ratio.

Assets (exposure at par):

Asset	Exposure (€m)	Risk weight	RWA (€m)
Cash & central bank reserves	500	0%	0
Government bonds (top-rated)	1,000	0%	0
Interbank / bank exposures	500	20%	100
Residential mortgages	2,000	35%	700
Corporate loans (unrated)	3,000	100%	3,000
High-risk / past-due loans	200	150%	300
Total assets	**7,200**		**Credit RWA = 4,100**

Add **market RWA = 300** and **operational RWA = 600**.

$$ \text{Total RWA} = 4{,}100 + 300 + 600 = 5{,}000 \text{ €m} $$

Capital: CET1 = €400m, AT1 = €50m, Tier 2 = €120m.

- Tier 1 = CET1 + AT1 = 400 + 50 = **€450m**
- Total Capital = Tier 1 + Tier 2 = 450 + 120 = **€570m**

Compute the ratios:

$$ \text{CET1 ratio} = \frac{400}{5{,}000} = 8.0\% $$
$$ \text{Tier 1 ratio} = \frac{450}{5{,}000} = 9.0\% $$
$$ \text{Total Capital ratio (CAR)} = \frac{570}{5{,}000} = 11.4\% $$

Compare to requirements (assume a standard non-systemic bank: CET1 min 4.5% + conservation buffer 2.5% = 7.0%; Tier 1 min 6.0%; Total min 8.0% + buffer 2.5% = 10.5%):

Ratio	Bank	Minimum incl. buffer	Pass?	Headroom
CET1	8.0%	7.0%	✅	+1.0%
Tier 1	9.0%	7.5% (6.0 + 2.5 buffer)*	✅	+1.5%
Total	11.4%	10.5%	✅	+0.9%

The 2.5% conservation buffer is met with CET1 and sits above all three minimums.

Self-check / reconciliation. The three capital measures must nest: CET1 (400) ≤ Tier 1 (450) ≤ Total (570). ✅ The ratios rise accordingly: 8.0% ≤ 9.0% ≤ 11.4%. ✅ Convert Total Capital ratio back to a capital number: 11.4% × 5,000 = 570 = Total Capital. ✅ The bank passes every requirement, but its Total-capital headroom is thin at +0.9% — a mild shock could push it into the conservation buffer and trigger dividend restrictions. Note the numerators are unchanged across the three ratios only in the denominator sense; it is the *numerator* that shrinks as we demand higher quality (400 vs 450 vs 570) against the *same* RWA of 5,000.

Example 2 — A shock, and how capital absorbs it

Suppose Alpha Bank's corporate loan book suffers a **€350m loss** (a portion defaults with low recovery). Losses hit CET1 first (retained earnings fall). Trace the impact.

- New CET1 = 400 − 350 = **€50m**
- New Tier 1 = 50 + 50 = **€100m**
- New Total Capital = 100 + 120 = **€220m**

RWA also changes: the €350m of loans that defaulted move from 100% weight to being written off (removed) or to 150% for the past-due remainder — but to keep the arithmetic clean, assume the defaulted exposures are written off entirely, removing €350m of exposure that carried €350m of RWA. New Total RWA = 5,000 − 350 = **€4,650m**.

$$ \text{New CET1 ratio} = \frac{50}{4{,}650} = 1.08\% $$ $$ \text{New Total ratio} = \frac{220}{4{,}650} = 4.73\% $$

Interpretation. The bank is still technically *solvent* (capital > 0), but it has crashed through every minimum: CET1 of 1.08% is far below the 4.5% hard floor, and Total of 4.73% is below the 8% floor. In practice the supervisor would now intervene — force a capital raise, halt all distributions, possibly resolve the bank. This is exactly what capital is *for*: it absorbed €350m of loss so that **depositors lost nothing** (the €7,200m of deposits/liabilities are untouched — losses were borne by shareholders, whose CET1 fell from 400 to 50). Had the bank held only the €400m of CET1 with a leverage-style thin cushion and no buffers, a slightly larger loss would have wiped equity out entirely and hit depositors.

Reconciliation. Loss of 350 fell entirely on CET1 (400 → 50): change of −350. ✅ Tier 1 and Total each fell by the same 350 (AT1 and Tier 2 untouched because CET1 was not fully exhausted): 450 → 100 and 570 → 220. ✅ Capital remained positive, so no tier below CET1 was written down — consistent with the going-concern loss ordering (CET1 absorbs first). ✅

Example 3 — Leverage ratio and LCR checks

Using Alpha Bank's *pre-shock* figures. The leverage exposure measure is not risk-weighted; take total on- and off-balance-sheet exposure = **€7,500m** (the €7,200m of assets plus €300m of off-balance-sheet commitments).

$$ \text{Leverage Ratio} = \frac{\text{Tier 1}}{\text{Total Exposure}} = \frac{450}{7{,}500} = 6.0\% \geq 3\% \;\; ✅ $$

The bank has comfortable leverage headroom (6.0% vs 3.0% minimum), implying leverage of ~16.7x. Note that leverage (6.0%) exceeds the minimum by more than the risk-weighted CET1 ratio does — here the **risk-weighted ratio is the binding constraint**, not leverage. That is typical of a bank with genuinely risky assets. For a bank stuffed with 0%-weighted government bonds, the reverse holds: the risk-weighted ratio looks huge but leverage bites — which is precisely why the backstop exists.

LCR check. Suppose Alpha holds **HQLA = €1,400m** (its €500m reserves + €900m of its top-rated government bonds, all Level 1 with 0% haircut). In the 30-day stress scenario: - Expected outflows: retail deposits €4,000m × 5% run-off = €200m; wholesale funding €1,500m × 40% run-off = €600m; undrawn credit lines €500m × 10% = €50m. **Total outflows = €850m.** - Expected inflows: €300m of contractual inflows, but capped at 75% of outflows = min(300, 0.75 × 850 = 637.5) = **€300m.** - Net cash outflows = 850 − 300 = **€550m.**

$$ \text{LCR} = \frac{1{,}400}{550} = 254\% \geq 100\% \;\; ✅ $$

Reconciliation. Outflows 850 = 200 + 600 + 50. ✅ Inflow cap: 0.75 × 850 = 637.5 ≥ 300, so full 300 counts. ✅ Net 550 = 850 − 300. ✅ LCR 254% means Alpha holds ~2.5x the HQLA needed to survive a 30-day stressed run — a strong liquidity position, driven by its large government-bond book. Observe the interplay: those same government bonds carry a 0% *credit* risk weight (so they contribute nothing to RWA / capital requirements) yet are the backbone of the LCR. This shows why the two regimes are separate — an asset can be capital-cheap and liquidity-rich at the same time, which is exactly why banks crowd into sovereigns.

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6. Connections

- **Chapter 8/9 (Credit Risk, PD/LGD/EAD).** The IRB approach to credit RWA is built directly on the PD, LGD, and EAD parameters. Regulatory capital covers *unexpected* loss; expected loss (PD × LGD × EAD) is covered by provisions, not capital — the two must not double-count.
- **Chapter 6 (Market Risk / VaR & Expected Shortfall).** Market RWA is computed from these measures; post-crisis Basel switched the internal-models charge from 99% VaR to 97.5% Expected Shortfall for better tail capture.
- **Chapter on Operational Risk.** Feeds operational RWA via the Standardised Measurement Approach.
- **Liquidity Risk chapter.** LCR and NSFR are the regulatory encoding of the liquidity concepts (funding vs market liquidity, run dynamics).
- **Stress testing / ICAAP.** Pillar 2 requires banks to hold capital against risks beyond Pillar 1; supervisory stress tests (CCAR, EBA) translate hypothetical shocks into required capital — a forward-looking complement to the static ratios.
- **Systemic risk & macroprudential policy.** The countercyclical buffer and G-SIB surcharges are *macroprudential* tools — aimed at the system, not the individual bank.
- **Resolution / bail-in (TLAC/MREL).** Beyond going-concern capital, the largest banks must hold Total Loss-Absorbing Capacity so they can be recapitalised in resolution without taxpayer money — the "gone-concern" complement to Basel capital.

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7. Key Terms

- **BCBS / BIS** — the Basel Committee (standard-setter) hosted at the Bank for International Settlements; issues standards, not law.
- **Risk-Weighted Assets (RWA)** — assets scaled by risk weights; the denominator of the capital ratio; sum of credit, market, and operational RWA.
- **Capital Adequacy Ratio (CAR / CRAR)** — regulatory capital ÷ RWA; minimum 8% total.
- **CET1 (Common Equity Tier 1)** — highest-quality, going-concern capital: common shares + retained earnings minus deductions; minimum 4.5% of RWA.
- **AT1 (Additional Tier 1)** — perpetual instruments (e.g. CoCos) that convert or write down at a trigger; going-concern.
- **Tier 2** — subordinated debt and certain provisions; gone-concern, absorbs loss after Tier 1.
- **Capital Conservation Buffer** — 2.5% CET1 cushion above the minimum; breaching it restricts distributions.

- **Countercyclical Buffer (CCyB)** — 0-2.5% CET1, raised in booms and released in busts.
- **G-SIB / D-SIB surcharge** — extra CET1 (1-3.5%) for systemically important banks.
- **Leverage Ratio** — Tier 1 ÷ total (non-risk-weighted) exposure; minimum 3%; the risk-blind backstop.
- **LCR (Liquidity Coverage Ratio)** — HQLA ÷ 30-day stressed net outflows ≥ 100%.
- **HQLA** — high-quality liquid assets (Level 1: cash, top sovereigns; Level 2: haircut, capped).
- **NSFR (Net Stable Funding Ratio)** — Available ÷ Required Stable Funding over 1 year ≥ 100%.
- **Output floor** — modelled (IRB) RWA cannot fall below 72.5% of standardised RWA.
- **Three Pillars** — (1) minimum capital, (2) supervisory review, (3) market discipline.
- **Regulatory arbitrage** — exploiting crude or mismatched rules to hold risk at low capital cost.
- **Procyclicality** — the tendency of risk-based rules to loosen in booms and tighten in busts, amplifying the cycle.

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8. Common Confusions

"Capital is cash the bank keeps in a vault." No. Capital is a *funding source* on the liability/equity side — it is how much of the bank is funded by shareholders rather than creditors. It is not a pot of money set aside; it is the difference between assets and liabilities. Liquidity (HQLA) is the asset-side buffer; capital is the equity-side buffer. Conflating them is the single most common error.

"A higher tier number is better." The opposite. **Tier 1 is better than Tier 2**, and within Tier 1, CET1 is the best. The tiers rank *downward* in quality. "Common Equity Tier 1" is the top of the pile.

"CAR of 8% means the bank holds 8% of its assets as capital." No — 8% of its *risk-weighted* assets. Because RWA is usually much smaller than total assets (many assets carry weights below 100%), the capital as a fraction of *total* assets is lower than 8%. That gap is exactly why the leverage ratio (which uses total, unweighted exposure) was added.

"The leverage ratio and the capital ratio measure the same thing." They use the same numerator idea (capital) but different denominators: the CAR uses risk-weighted assets, the leverage ratio uses raw exposure. The leverage ratio deliberately ignores risk weights so that a bank cannot game it by claiming low risk. They are designed to bind in different situations.

"LCR and NSFR are both about the same liquidity." Different horizons. LCR is a 30-day acute-stress survival test; NSFR is a 1-year structural funding test. A bank can pass one and fail the other.

"Expected loss should be covered by capital." No — expected loss is covered by *provisions* (loan-loss reserves), priced into the loan. Regulatory capital covers *unexpected* loss (the deviation above expectation). Double-counting the expected loss inflates the requirement.

"Basel is law." Basel standards are *not* directly binding. They become enforceable only when a jurisdiction transposes them into local regulation (EU CRR/CRD, US rules, RBI norms), often with local deviations and timelines.

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9. Recap

Banks are uniquely fragile: highly leveraged, funded by runnable deposits, interconnected, and backed by a public safety net that invites moral hazard. Left alone they hold too little capital. Regulation fixes this by forcing a minimum ratio of loss-absorbing capital to risk. The master equation is **CAR = Capital / RWA**, where the denominator is *risk-weighted* (so safe and risky assets are treated differently) and the numerator is *tiered by quality* (CET1 best, then AT1, then Tier 2).

The Basel Accords evolved this idea: **Basel I** introduced crude risk buckets and the 8% ratio; **Basel II** made RWA risk-sensitive via three pillars and internal models; **Basel III** — the crisis response — hardened capital around CET1, added conservation/countercyclical/systemic **buffers**, introduced a non-risk-weighted **leverage ratio** backstop, and added the **LCR** and **NSFR** liquidity standards, later constraining models with an **output floor**.

Capital protects against *insolvency*; liquidity ratios protect against *illiquidity* — two distinct failure modes needing distinct tools. The worked examples showed how RWA and the CET1/Tier 1/Total ratios are computed, how a loss is absorbed by CET1 first (protecting depositors), and how the leverage and LCR checks reconcile. The system is safer because banks now absorb more loss, have more skin in the game, inspire more creditor confidence, and — for the most dangerous institutions — carry the largest cushions.

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10. Quick Reference / Interview Points

The formulas — memorise cold:

Metric	Formula	Minimum
CET1 ratio	CET1 / RWA	4.5% (+2.5% buffer = 7.0%)
Tier 1 ratio	(CET1 + AT1) / RWA	6.0%
Total CAR	(Tier 1 + Tier 2) / RWA	8.0% (+2.5% = 10.5%)
Leverage ratio	Tier 1 / Total Exposure (unweighted)	3.0%
LCR	HQLA / 30-day stressed net outflows	100%
NSFR	Available Stable Funding / Required Stable Funding	100%
RWA ↔ capital charge	RWA = Capital charge × 12.5	—
Output floor	IRB RWA ≥ 72.5% × Standardised RWA	—

Talking points for interviews:

- *Why risk-weight?* Flat asset-based rules reward risk-taking; risk weighting aligns capital with actual loss potential.
- *Why the leverage ratio if you already have the CAR?* Risk weights can be wrong or gamed (2008 AAA structured products); the leverage ratio is the risk-blind backstop that binds when weights are too generous.
- *CET1 vs Tier 2 in one line:* CET1 absorbs losses while the bank is alive (going concern); Tier 2 only in wind-down (gone concern).
- *Capital vs liquidity:* solvency vs cash-timing — different diseases, different cures (CAR/leverage vs LCR/NSFR).
- *What did Basel III fix from Basel II?* Softer capital → CET1-centric; procyclicality → countercyclical buffer; no liquidity rules → LCR + NSFR; over-reliance on models → output floor + revised standardised approaches.

- *How does capital protect depositors?* Losses fall on equity (CET1) first; only if equity is fully exhausted do creditors/depositors take a hit — so the bigger the CET1 cushion, the safer the deposit.
- *Buffers vs minimums:* breaching a *minimum* threatens the bank's licence/triggers resolution; dipping into the *conservation buffer* only restricts dividends and bonuses — buffers are designed to be usable in stress so banks keep lending.
- *Systemic angle:* G-SIB surcharges and the countercyclical buffer are macroprudential — they target the system, forcing the most dangerous banks to be the strongest and leaning against the credit cycle.

Q&A — Basel Accords and Capital Regulation

A practice bank for the Basel Accords chapter. Work each question before reading the answer. Numerical answers are self-checked against the master identities **CAR = Capital / RWA**, **RWA = credit + market + operational**, and the tier nesting **CET1 ≤ Tier 1 ≤ Total Capital**.

Section A — Concept-Check (short answer)

A1. State the master equation of capital regulation and name its two parts.

The Capital Adequacy Ratio (CAR / CRAR): **CAR = Regulatory Capital / Risk-Weighted Assets ≥ minimum %.** The numerator is *loss-absorbing capital* (tiered by quality — CET1, then AT1, then Tier 2). The denominator is *risk-weighted assets* — assets scaled by a weight reflecting how likely they are to lose value, not raw balance-sheet size. The whole framework is an elaboration of this fraction.

A2. Why does Basel measure capital against risk-weighted assets rather than raw assets?

A flat "hold 8% of total assets" rule would charge the same capital for a Treasury bill as for a junk loan. Rational banks would then shed safe, low-yield assets (which cost capital but earn little) and pile into risky, high-yield ones (same capital, more return). A raw-asset rule actively *encourages* risk-taking. Risk weighting ties the capital charge to the actual probability and severity of loss, so capital sits where losses are likely to come from.

A3. Why is bank capital tiered by quality?

Not all capital absorbs loss equally. Common equity (CET1) absorbs loss immediately and continuously while the bank is alive (going concern) with no obligation to repay. Subordinated debt (Tier 2) only absorbs loss in a wind-down (gone concern). Basel ranks capital by loss-absorbing quality and demands the bulk of the requirement be met with the highest form. 2008 proved the point: banks that looked well-capitalised on total-capital measures had little real equity underneath, and it was the equity the market cared about.

A4. Note the counter-intuitive tier ordering.

Tiers rank *downward* in quality: **Tier 1 is better than Tier 2**, and within Tier 1, **CET1 is the best**. A higher tier number is worse capital, not better — the common trap.

A5. State the three pillars of Basel II.

Pillar 1 — minimum capital requirements (quantitative charges for credit, market, operational risk). **Pillar 2** — supervisory review (supervisors assess risks missed by Pillar 1, e.g. concentration, interest-rate risk in the banking book, and can require more capital bank-by-bank via ICAAP). **Pillar 3** — market discipline (mandatory public disclosure so the market can price and monitor bank risk).

A6. What are the Basel III Pillar 1 minimum ratios?

CET1 ≥ **4.5%** of RWA; Tier 1 (CET1 + AT1) ≥ **6.0%**; Total Capital (Tier 1 + Tier 2) ≥ **8.0%**. On top sit buffers met with CET1: the **capital conservation buffer (2.5%)**, the **countercyclical buffer (0–2.5%)**, and **G-SIB/D-SIB surcharges (1–3.5%)**.

A7. What does the leverage ratio do that the CAR cannot?

The leverage ratio = **Tier 1 / Total Exposure (unweighted) ≥ 3%**. Its denominator applies *no risk weights* — a €1 Treasury and a €1 junk loan count the same. This is the risk-blind backstop: if the risk weights themselves are wrong or gamed (as with AAA structured products pre-2008), the risk-weighted CAR lies, but the leverage ratio still binds. A bank must satisfy *both* a risk-sensitive and a risk-blind constraint.

A8. Distinguish LCR from NSFR.

Both are liquidity standards but over different horizons. **LCR** (Liquidity Coverage Ratio) = HQLA / 30-day stressed net outflows ≥ 100% — the *sprint*, survive an acute month-long run. **NSFR** (Net Stable Funding Ratio) = Available Stable Funding / Required Stable Funding over one year ≥ 100% — the *marathon*, be structurally sound so illiquid long assets are funded with sticky money. A bank can pass one and fail the other.

A9. Capital ratios and liquidity ratios cure different diseases — which?

Capital protects against **insolvency** (losses exceeding the equity cushion). Liquidity ratios protect against **illiquidity** (unable to meet cash outflows in time, even while solvent). Northern Rock (2007) was solvent but died from a run. Different failure modes, different medicine: CAR/leverage vs LCR/NSFR.

A10. What is the output floor and what problem does it solve?

Under the Basel III endgame (2017), RWA computed with a bank's internal (IRB) models cannot fall below **72.5% of the standardised-approach RWA**. It caps how much capital benefit banks can extract from optimistic internal models — a floor against model-gaming while retaining risk sensitivity.

A11. Why is expected loss NOT covered by regulatory capital?

Expected loss (PD × LGD × EAD) is the predictable average, covered by **provisions** and priced into the loan. Regulatory capital covers **unexpected loss** — the deviation *above* the mean. Treating expected loss as a capital item double-counts and inflates the requirement.

A12. Is Basel law?

No. The BCBS issues *standards*, not legislation. They become binding only when a jurisdiction transposes them into local rules — EU via CRR/CRD, US via the federal banking agencies, India via RBI norms — often with local deviations and timelines.

Section B — Numerical / Applied (full solutions)

B1. Compute credit RWA. A bank holds: cash €400m (0%), top-rated government bonds €800m (0%), interbank exposures €300m (20%), residential mortgages €1,500m (35%), unrated corporate loans €2,000m (100%), past-due loans €100m (150%). Find credit RWA.

Weight each exposure: - Cash: 400 × 0% = 0 - Govt bonds: 800 × 0% = 0 - Interbank: 300 × 20% = 60 - Mortgages: 1,500 × 35% = 525 - Corporate: 2,000 × 100% = 2,000 - Past-due: 100 × 150% = 150

Credit RWA = 0 + 0 + 60 + 525 + 2,000 + 150 = **€2,735m.**

B2. Full ratio stack. Add market RWA = €200m and operational RWA = €400m to B1. Capital: CET1 = €300m, AT1 = €40m, Tier 2 = €90m. Compute total RWA and all three capital ratios.

Total RWA = 2,735 + 200 + 400 = **€3,335m.** Tier 1 = 300 + 40 = €340m. Total Capital = 340 + 90 = €430m.

- CET1 ratio = 300 / 3,335 = **9.00%**
- Tier 1 ratio = 340 / 3,335 = **10.19%**
- Total CAR = 430 / 3,335 = **12.89%**

Self-check: tiers nest, 300 ≤ 340 ≤ 430, and ratios rise accordingly 9.00% ≤ 10.19% ≤ 12.89%. ✅ Against a non-systemic bank's requirements (CET1 7.0% incl. buffer, Tier 1 6.0%, Total 10.5% incl. buffer), the bank passes all three.

B3. RWA from a capital charge. A market-risk model produces a capital charge of K = €24m. Express this as RWA.

RWA = K × 12.5 = 24 × 12.5 = **€300m.** (The 12.5 is the reciprocal of the 8% total-capital ratio, so a €24m charge at 8% implies €300m of RWA.)

Self-check: 8% × 300 = 24. ✅

B4. Loss absorption. The bank in B2 suffers a €260m loss on its corporate book; assume the defaulted exposures (which carried €260m of 100%-weighted RWA) are written off entirely. Trace CET1 and recompute the CET1 ratio.

Losses hit CET1 first: new CET1 = 300 − 260 = **€40m.** New total RWA = 3,335 − 260 = **€3,075m.**

New CET1 ratio = 40 / 3,075 = **1.30%.**

Interpretation: the bank is still solvent (capital > 0) but has crashed through the 4.5% CET1 minimum. Crucially, depositors lost nothing — the entire €260m loss fell on shareholders' equity (CET1 300 → 40). This is exactly what capital is for. AT1 and Tier 2 are untouched because CET1 was not fully exhausted, consistent with the going-concern loss ordering.

B5. Leverage ratio. Using B2's capital, the leverage exposure measure (unweighted on- and off-balance-sheet) is €5,500m. Compute the leverage ratio and state whether it or the CET1 ratio is the binding constraint.

Leverage ratio = Tier 1 / Total Exposure = 340 / 5,500 = **6.18% ≥ 3%.** ✅

Leverage headroom is +3.18% (6.18 − 3.0). CET1 headroom is +2.00% (9.00 − 7.0). The **risk-weighted CET1 constraint is tighter**, typical of a bank with genuinely risky assets. For a bank stuffed with 0%-weighted sovereigns, the reverse holds and leverage bites — which is why the backstop exists.

B6. LCR. A bank holds HQLA = €900m (all Level 1, 0% haircut). Stress outflows: retail deposits €3,000m × 5% run-off; wholesale funding €1,000m × 40%; undrawn lines €400m × 10%. Contractual inflows €250m (cap at 75% of outflows). Compute the LCR.

Outflows = 3,000×5% + 1,000×40% + 400×10% = 150 + 400 + 40 = **€590m.** Inflow cap = 0.75 × 590 = 442.5 ≥ 250, so full €250m counts. Net outflows = 590 − 250 = **€340m.**

LCR = 900 / 340 = **264.7% ≥ 100%.** ✅

Self-check: outflows 590 = 150 + 400 + 40. ✅ Inflows uncapped since 250 < 442.5. ✅ The bank holds ~2.6× the HQLA needed for a 30-day stressed run.

B7. NSFR. Available Stable Funding = €4,200m; Required Stable Funding = €4,000m. Compute NSFR and interpret.

NSFR = ASF / RSF = 4,200 / 4,000 = **105% ≥ 100%.** ✅ The bank's stable funding exceeds what its assets require over a one-year horizon — structurally sound, though the 5% margin is thin, so growth in illiquid long-term lending (high RSF) without matching stable funding could breach it.

B8. Buffer requirement for a G-SIB. A globally systemic bank faces a 2.0% G-SIB surcharge in a period with a 1.0% countercyclical buffer active. What total CET1 (as % of RWA) must it hold?

CET1 minimum 4.5% + conservation buffer 2.5% + countercyclical 1.0% + G-SIB surcharge 2.0% = **10.0% of RWA**, all in CET1.

Section C — Interview-Style (model answers)

C1. Why add a leverage ratio if you already have the CAR?

Because the CAR trusts the risk weights, and risk weights can be wrong or deliberately gamed. Before 2008, banks loaded up on AAA-rated structured products and sovereign debt carrying tiny risk weights; their risk-weighted ratios looked healthy right up until they collapsed. The leverage ratio (Tier 1 / total unweighted exposure ≥ 3%) is the risk-blind backstop — it treats every euro of exposure the same, so a bank cannot escape it by claiming its assets are low-risk. The two constraints bind in different situations: the CAR bites on a genuinely risky book, the leverage ratio bites on a book stuffed with low-weight assets. Together they mean a bank must be safe on both a risk-sensitive and a risk-blind measure.

C2. Explain CET1 versus Tier 2 in one breath, then why it mattered in 2008.

CET1 absorbs losses while the bank is still alive — a going-concern buffer of common equity and retained earnings that falls the instant the bank loses money, with no repayment obligation. Tier 2 (subordinated debt) only absorbs loss in a wind-down, gone-concern. In 2008, many banks met their headline total-capital requirements largely with lower-quality instruments, so on paper they looked fine, but the market recognised that only real equity could actually absorb the mounting losses. Basel III's central fix was to put CET1 at the centre — raising the CET1 minimum and tightening what qualifies — because it is the capital that actually keeps a bank standing.

C3. What exactly did Basel III fix from Basel II?

Four things. First, *soft capital* — Basel III made the framework CET1-centric with stricter definitions. Second, *procyclicality* — the countercyclical buffer leans against the credit cycle, built up in booms, released in busts. Third, *no liquidity standard at all* — it added the LCR (30-day) and NSFR (1-year). Fourth, *over-reliance on internal models and external ratings* — the endgame reforms added an output floor (IRB RWA ≥ 72.5% of standardised) and revised the standardised approaches. It didn't discard Basel II's three-pillar architecture; it hardened every part of it.

C4. A junior analyst says "our CAR is 12%, so we hold 12% of our assets as capital." Correct them.

That's 12% of *risk-weighted* assets, not total assets. Because most assets carry weights below 100% — cash and top sovereigns are 0%, mortgages ~35% — RWA is usually much smaller than the balance sheet. So capital as a fraction of *total* assets is materially lower than 12%. That very gap is why the leverage ratio was introduced: it puts capital over *unweighted* exposure to reveal the true fulcrum the bank is levered on.

C5. How does capital actually protect a depositor?

Through the loss waterfall. When the bank loses money, the loss falls on CET1 (shareholders' equity) first, then AT1, then Tier 2, and only if all of that is exhausted do depositors and senior creditors take a hit. So the larger the CET1 cushion, the more loss the bank can absorb before a depositor loses a cent. In my B4 example, a €260m loss took CET1 from €300m to €40m and depositors were entirely untouched — equity did its job. Capital is the shield standing between asset losses and the deposit base.

C6. Why not just require banks to hold 50% capital and end the fragility debate?

Because capital isn't free from the bank's perspective. Equity investors demand a higher return than depositors, and frictions like the tax-deductibility of debt and the safety-net subsidy make equity privately expensive, so more equity can raise funding costs and shrink lending. (The Modigliani-Miller view argues leverage shouldn't change *total* funding cost, but those frictions break the idealisation in practice.) Regulation therefore sits on a trade-off curve: too little capital and the system is fragile; too much and credit becomes scarce and costly. Basel's minimums are the negotiated answer to where on that curve to sit — not a claim that more is always better.

Section D — MCQs (with reasoning)

D1. Under Basel III, the minimum CET1 ratio (before buffers) is: (a) 2.0% (b) 4.5% (c) 6.0% (d) 8.0%

Answer: (b) 4.5%. 6.0% is the Tier 1 minimum; 8.0% is Total Capital. Adding the 2.5% conservation buffer brings the *effective* CET1 requirement to 7.0%, but the hard minimum is 4.5%.

D2. The leverage ratio differs from the CAR primarily because its denominator: (a) uses Tier 1 capital only (b) applies no risk weights (c) excludes off-balance-sheet items (d) is measured over 30 days

Answer: (b) applies no risk weights. The leverage ratio's exposure measure is unweighted — that risk-blindness is the entire point of a backstop. (a) describes the numerator, not the denominator; (c) is false — it *includes* off-balance-sheet items; (d) confuses it with the LCR.

D3. Expected loss on a loan book is properly covered by: (a) CET1 capital (b) Tier 2 capital (c) provisions (d) the countercyclical buffer

Answer: (c) provisions. Expected loss (PD×LGD×EAD) is the predictable average, provisioned and priced in. Regulatory capital of all tiers covers *unexpected* loss. Charging expected loss to capital double-counts.

D4. Which correctly orders capital from highest to lowest loss-absorbing quality? (a) Tier 2 > AT1 > CET1 (b) AT1 > CET1 > Tier 2 (c) CET1 > AT1 > Tier 2 (d) CET1 > Tier 2 > AT1

Answer: (c) CET1 > AT1 > Tier 2. Tiers rank downward in quality; CET1 (common equity, going-concern) is best, AT1 next (perpetual, converts/writes down at trigger), Tier 2 (subordinated debt, gone-concern) last.

D5. The LCR requires a bank to survive a stressed period of: (a) 7 days (b) 30 days (c) 90 days (d) one year

Answer: (b) 30 days. LCR is the short-term sprint over 30 days; the one-year structural test is the NSFR.

D6. The output floor under the Basel III endgame sets modelled (IRB) RWA at no less than: (a) 50% (b) 60% (c) 72.5% (d) 100% of standardised RWA

Answer: (c) 72.5%. IRB RWA ≥ 72.5% × standardised RWA — a cap on the capital benefit of internal models while keeping risk sensitivity.

D7. To convert a capital charge K into RWA under Basel, you multiply by: (a) 8 (b) 10 (c) 12.5 (d) 0.08

Answer: (c) 12.5. RWA = K × 12.5, the reciprocal of the 8% total-capital ratio. Multiplying by 0.08 would go the wrong direction (RWA → charge).

D8. A bank with a 6% CET1 ratio, 3% conservation buffer requirement, and no other buffers: (a) fully meets its CET1 requirement with headroom (b) meets the minimum but is inside the conservation buffer (c) breaches the hard CET1 minimum (d) must be resolved immediately

Answer: (b) meets the minimum but is inside the conservation buffer. The 4.5% hard minimum is met (6% > 4.5%), but the effective requirement is 4.5% + 2.5% = 7.0% (the standard conservation buffer is 2.5%, not 3%). At 6% the bank sits *inside* the buffer, which restricts dividends and bonuses but does not trigger resolution — buffers are designed to be usable in stress. (The 3% in the question is a distractor; the standard conservation buffer is 2.5%.)

Self-verification note. Every numerical answer was checked against CAR = Capital/RWA and the tier nesting CET1 ≤ Tier 1 ≤ Total. B2 ratios rise monotonically with tier (9.00 ≤ 10.19 ≤ 12.89). B3/B7 reconcile by the 12.5 and 100% identities. B4 confirms losses hit CET1 first and depositors are protected. Minimums used throughout: CET1 4.5%, Tier 1 6.0%, Total 8.0%, conservation buffer 2.5%, leverage 3%, LCR/NSFR 100%, output floor 72.5%.

Chapter 13 — Stress Testing and Scenario Analysis

1. The Problem / The Need

Chapter 12 left us with a number that looks reassuring: "Our 1-day 99% VaR is $10 million." Management reads that as "we should not lose more than $10 million on 99 days out of 100." But three uncomfortable questions immediately follow, and none of them can be answered by VaR:

1. **What about the other day?** VaR is silent about the size of the loss on the 1-in-100 day. It draws a line at the 99th percentile and refuses to look past it. The loss beyond the line could be $12 million or $120 million — VaR gives the same answer either way.
2. **What if tomorrow is not like the last two years?** VaR is estimated from a historical window (say 2 years of daily returns). If that window happens to be calm — no crash, no rate shock, no default wave — then the model has literally never seen a disaster and cannot imagine one. The 2004–2006 pre-crisis VaR models were calibrated on the most placid credit environment in a generation.
3. **What specifically would break us?** VaR answers "how much could we lose?" It does not answer "what has to happen for us to lose it?" A CEO cannot manage a percentile; she can manage a story — "oil to $30, the dollar up 15%, high-yield spreads doubling."

These three gaps — **tail blindness**, **history dependence**, and **lack of narrative** — are exactly what stress testing and scenario analysis exist to fill. Where VaR asks "*How bad is a normal-ish bad day?*", stress testing asks "*How bad could a genuinely bad day be, and what would cause it?*"

The 2007–2009 Global Financial Crisis was the brutal empirical proof. Banks reported "10-day 99% VaR of a few hundred million" and then lost tens of billions. Their models were not lying; they were answering a narrow question honestly. The regulators' response was decisive: stress testing moved from a nice-to-have risk-committee slide to a **legally mandated, capital-binding exercise** (CCAR in the US, EBA/EU-wide stress tests in Europe). Today, a bank's dividend and buyback plans can be blocked by a stress-test result. That is how central stress testing has become.

2. The Core Idea

Stress testing is the deliberate examination of how a portfolio, a book, or an entire institution behaves under severe but plausible conditions that lie outside the range of ordinary day-to-day fluctuations.

Two mental moves define it:

- **We leave the world of probabilities and enter the world of conditions.** VaR says "there is a 1% chance of losing more than X." A stress test says "*if* equities fall 40% and credit spreads triple, we lose Y" — no probability attached to the *if*. We are asking a conditional ("what if") rather than a distributional ("how likely") question. This is liberating: we no longer need to know the odds of a catastrophe to prepare for it.
- **We push the inputs to extremes on purpose.** In normal risk measurement we take the market as given. In stress testing we *impose* a state of the world — a scenario — and revalue everything under it.

A useful way to hold the whole discipline in your head:

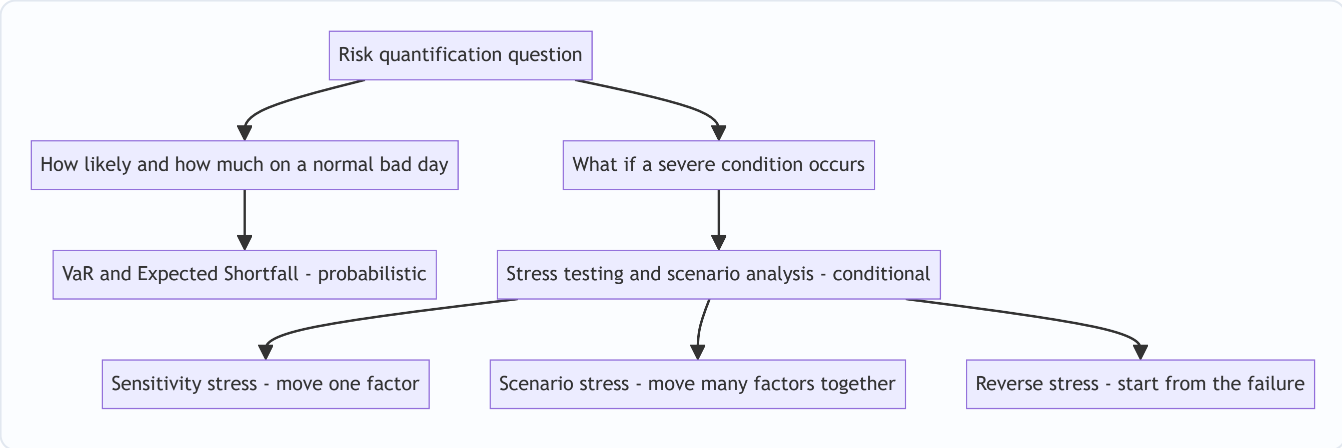

Figure 1: VaR asks how likely; stress testing asks what if. The two are complements, not substitutes.

The relationship to VaR is **complementary, not competitive**. VaR maps the centre and the near-tail of the loss distribution where you have data. Stress testing probes the far tail and the "off-distribution" states where you have no data but plenty of imagination and history. A mature risk function runs both and reconciles the stories they tell.

3. Why / How It Works

3.1 Why VaR misses tail events — the mechanics

VaR's blindness is not a bug that better calibration can fix; it is structural. Three reasons:

(a) It is a quantile, so it discards the tail by construction. The 99% VaR is defined as the loss level with 1% mass beyond it. The *shape* of that 1% — thin or fat, bounded or catastrophic — is thrown away. Two portfolios can have identical VaR and wildly different tails. (Expected Shortfall, the average loss *beyond* VaR, partially repairs this — see Chapter 12 — but even ES is estimated from the same limited data.)

(b) It assumes correlations and volatilities that hold in calm markets. Most VaR engines are calibrated on a recent window. In that window, diversification "works": your long equities and long credit are imperfectly correlated, so the model nets them off. In a crisis, **correlations converge toward 1** — everything risky sells off together as investors flee to cash and Treasuries. The diversification the VaR relied on evaporates precisely when you need it. Stress testing lets you *impose* crisis correlations by hand.

(c) It is anchored to sampled history. If your data window contains no sovereign default, your model assigns near-zero risk to one. The map has no dragon because the cartographer never sailed that sea.

3.2 How stress testing gets around this

Stress testing sidesteps all three problems by **specifying the state of the world exogenously** rather than sampling it:

- You *choose* the shocks (equities −40%, credit spreads +300bp, USD +15%), so you are not limited to what history sampled.
- You *choose* the correlation structure — typically by moving all risk factors in the adverse direction *together*, encoding the crisis "everything falls at once" behaviour.
- You then **fully revalue** the portfolio under those shocks (full repricing of options, bonds, loans), so non-linearities and convexity are captured rather than approximated by a linear sensitivity.

The engine underneath is the same repricing machinery used for P&L; the difference is entirely in the *inputs* you feed it.

3.3 The severity–plausibility trade-off

Every scenario lives on a dial between two failure modes:

- **Too mild** → the test is reassuring theatre; it never binds, it never informs a decision.
- **Too extreme** → "a meteor hits Manhattan" — the result (we lose everything) is true but useless; no one will act on it.

The craft is choosing **"severe but plausible."** Severe enough to hurt, plausible enough that a rational board will authorise capital or hedges in response. Regulators formalise this: the CCAR "severely adverse" scenario is roughly calibrated to a deep recession comparable to 2008, not to the end of the world.

4. Full Content

4.1 The taxonomy of stress tests

There are three structural families, ordered from simplest to most sophisticated:

Type	What you move	Question answered	Strength	Weakness
Sensitivity / factor stress	One risk factor at a time (e.g. rates +100bp)	"How exposed am I to *this* factor?"	Simple, transparent, fast	Ignores that factors move together
Scenario stress	Many factors jointly, in a coherent story	"How do I fare in *this world*?"	Captures correlations and contagion	Harder to build; story-dependent
Reverse stress	Start from the outcome (failure), solve for the cause	"What *breaks* me?"	Finds hidden vulnerabilities	Can be technically hard to invert

We take each in turn.

4.2 Sensitivity (single-factor) stresses

The most basic stress: bump one risk factor by a defined amount, hold everything else constant, revalue. Examples: parallel yield-curve shift of +200bp; equity index −10%; credit spreads +50bp; implied vol +5 vol points; a specific FX rate −20%.

These are the "Greeks made discrete." A DV01 or delta is an *infinitesimal* sensitivity; a sensitivity stress is a *finite, large* move that also captures **convexity** (gamma) — the way exposures accelerate for big moves. For an option book, a −1% equity move captured by delta and a −25% move captured by full revaluation can imply very different hedges, because gamma bites.

Sensitivity stresses are the diagnostic layer: they tell you *which factor* you are most exposed to. But they are unrealistic on their own — in the real world rates, equities and spreads do not move in isolation.

4.3 Scenario analysis — the heart of the discipline

A **scenario** is an internally consistent, simultaneous move of many risk factors that together tell a story. This is where correlation and contagion are modelled. There are two ways to build one.

(a) Historical scenarios. Take an actual past episode and replay its market moves against *today's* portfolio. Classic templates:

- **Black Monday, Oct 1987** — S&P −20% in a day.
- **1998 LTCM / Russia default** — flight to quality, spread blow-out, liquidity evaporation.
- **2008 Lehman week** — equities −30%+, credit spreads triple, funding markets freeze, correlations → 1.
- **2011 Eurozone sovereign crisis** — peripheral spreads explode.
- **March 2020 COVID crash** — fastest 30% equity drawdown in history, oil negative.

You take the *factor moves* from the episode (not the P&L — the P&L was someone else's portfolio) and apply them to your current book.

- **Advantage:** every combination of moves actually happened, so no one can call the scenario implausible. Correlations are real, not assumed.
- **Weakness:** the next crisis rarely rhymes exactly with the last. A book perfectly hedged against 2008 can be wide open to a shock that history has not yet produced. History has "survivorship" and template bias.

(b) Hypothetical scenarios. The risk team *invents* a coherent future that has not happened but could. Examples: "a Middle East conflict spikes oil to $180, the Fed hikes 200bp to fight the resulting inflation, and tech multiples compress 30%." Or a cyber-attack on payment infrastructure, or a disorderly break-up of a currency union.

- **Advantage:** forward-looking; can capture *new* vulnerabilities (a concentration you have built up recently, a novel product, a geopolitical fault line) that no historical episode covers.
- **Weakness:** subjective. Plausibility and internal consistency are matters of judgement, and it is easy to build a scenario that is either internally contradictory (rates up *and* bond prices up) or politically convenient (mild enough not to embarrass anyone).

Good practice runs **both**: historical scenarios for credibility and discipline, hypothetical scenarios for imagination and forward-looking coverage.

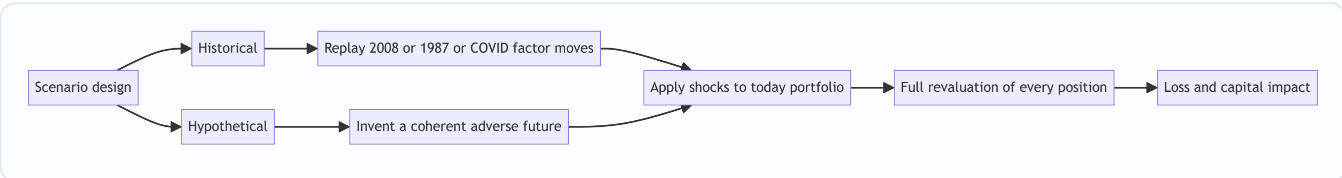

Figure 2: Two routes to a scenario converge on the same revaluation engine.

Building a coherent scenario — the internal-consistency test. A scenario is only useful if the factor moves hang together economically. If you shock equities down hard, you should typically also move credit spreads *wider*, safe-haven rates *lower* (flight to quality), the VIX *up*, and the currencies of risky economies *down*. Bolting on a random combination ("equities crash but high-yield spreads tighten") destroys credibility. Many firms use a **macro model** or **conditional-correlation approach**: specify a few "core" macro shocks (GDP, unemployment, a key rate) and let statistical relationships propagate them to the hundreds of granular pricing factors, so the whole scenario is coherent by construction.

4.4 Regulatory stress testing

After 2009, supervisors institutionalised stress testing as the binding constraint on bank capital. Two flagship regimes:

CCAR / DFAST (United States, Federal Reserve). - **CCAR** = Comprehensive Capital Analysis and Review; **DFAST** = Dodd-Frank Act Stress Test. DFAST is the quantitative loss-projection engine; CCAR adds the *qualitative* assessment of the bank's capital-planning process and its proposed dividends/buybacks. - The Fed publishes **scenarios** each year: *baseline*, *adverse*, and *severely adverse*. The severely adverse scenario is a deep recession — historically things like unemployment rising to ~10%, equities falling ~50%, house prices falling sharply, plus a **global market shock** and, for the largest banks, a **counterparty default** component. - Banks project their balance sheet, revenues, losses, and **capital ratios over a 9-quarter horizon** under each scenario. The binding test: does the bank's **CET1 ratio stay above the minimum** throughout the stress path? - **Teeth:** the result now feeds the **Stress Capital Buffer (SCB)** — the peak-to-trough CET1 decline in the severely adverse scenario is converted directly into a capital add-on the bank must hold. A weak result raises required capital and can force the bank to cut or cancel planned dividends and buybacks. This is the crucial modern point: **the stress test sets capital, not the other way round.**

EBA / EU-wide stress test (European Banking Authority, with the ECB). - Conducted roughly biennially across major EU banks using a **common methodology** and a **common macro scenario** (baseline and adverse) set by the EBA/ESRB, applied over a **3-year horizon**. - Historically a **constrained bottom-up** exercise: banks compute their own results using their models but under strict EBA methodological constraints (e.g. static-balance-sheet assumption, prescribed treatment of specific risks), which makes results comparable across banks. - Results are **published bank-by-bank** — a deliberate transparency/market-discipline mechanism. There is no single automatic pass/fail hurdle in the way CCAR historically had; instead the results feed the **SREP** (Supervisory Review and Evaluation Process) and inform each bank's **Pillar 2 Guidance** capital.

Feature	CCAR / DFAST (US Fed)	EBA / EU-wide (EBA + ECB)
Horizon	9 quarters (~2.25 yrs)	3 years
Scenarios	Baseline, adverse, severely adverse	Baseline, adverse
Approach	Fed also runs own models; supervisor-led	Constrained bottom-up by banks
Balance sheet	Assumptions vary; incl. planned actions	Static balance sheet (traditionally)
Output link	Stress Capital Buffer; approve/block payouts	Feeds SREP and Pillar 2 Guidance
Disclosure	Aggregate + firm results published	Detailed firm-by-firm published

Both regimes force a bank to prove, *before* a crisis, that it can absorb one. That is a profound shift from the pre-2008 world where capital adequacy was assessed against a static snapshot.

4.5 Reverse stress testing

Ordinary stress testing runs **forward**: *scenario → loss*. **Reverse stress testing runs backward**: it *fixes the outcome* — typically "the point at which the business model becomes unviable" (capital breached, funding lost, the firm fails) — and asks **"what set of events would produce this?"**

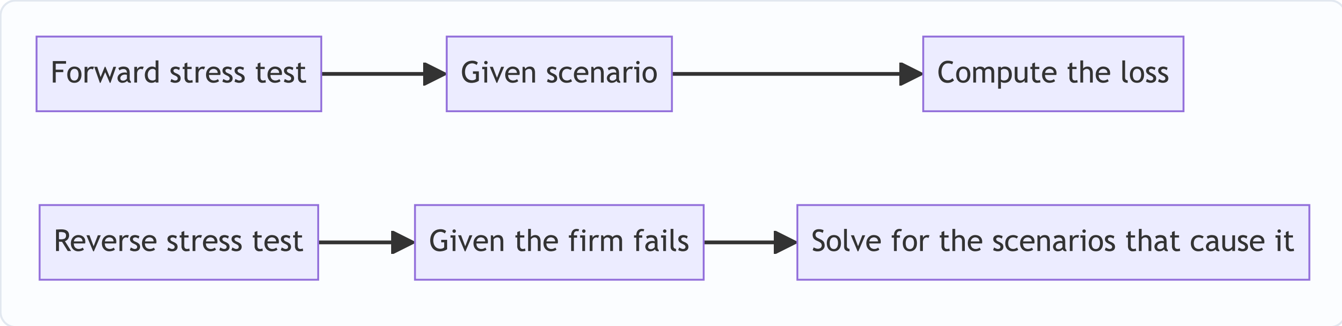

Figure 3: Reverse stress testing inverts the arrow — outcome first, cause second.

Why it is powerful:

- **It removes optimism bias.** Forward tests tend to probe scenarios management already worries about. Reverse tests force you to confront the *specific* combination that kills you — which is often an uncomfortable, unfashionable, or concentrated exposure no one wanted to name.
- **It surfaces hidden concentrations and non-linear cliffs.** The answer might be "we don't die from a broad 30% equity fall, but a 15% fall *combined with* our single largest counterparty defaulting *and* a ratings downgrade triggering collateral calls" — a narrow, correlated, self-reinforcing path that a broad forward scenario would average away.
- **Regulators require it.** Reverse stress testing is an explicit expectation in the UK (PRA), EU, and elsewhere, precisely because it counteracts the tendency to only test comfortable scenarios.

The mechanics can be hard: mathematically you are inverting a many-to-one mapping (many scenarios cause failure), so the output is usually a *family* of break-the-bank scenarios, from which management selects the most plausible and asks "how do we make sure *that* cannot happen, or that we would survive it?"

4.6 From results to decisions — closing the loop

A stress test that produces a number and dies in a slide deck is worthless. The value is in the **feedback into decisions**:

1. **Capital.** The headline use. If the severely adverse scenario would take CET1 below the required minimum, the firm must hold **more capital today** (the CCAR Stress Capital Buffer is exactly this). Stress losses set the size of the buffer.
2. **Risk appetite and limits.** If a scenario shows an outsized loss from, say, a single-sector credit concentration, the risk committee tightens the limit on that sector *before* the scenario materialises.
3. **Hedging and de-risking.** A large tail loss to a specific factor (oil, a currency, a counterparty) is a direct prompt to buy protection or reduce the position.
4. **Contingency and recovery planning.** Reverse-stress results feed **recovery and resolution plans** ("living wills"): if *this* is what would kill us, here are the pre-agreed actions (asset sales, capital raises, business-line exits) we would take.
5. **Capital distribution.** Dividends and buybacks are explicitly gated by stress results in CCAR — a bank cannot return capital it would need in the severely adverse world.

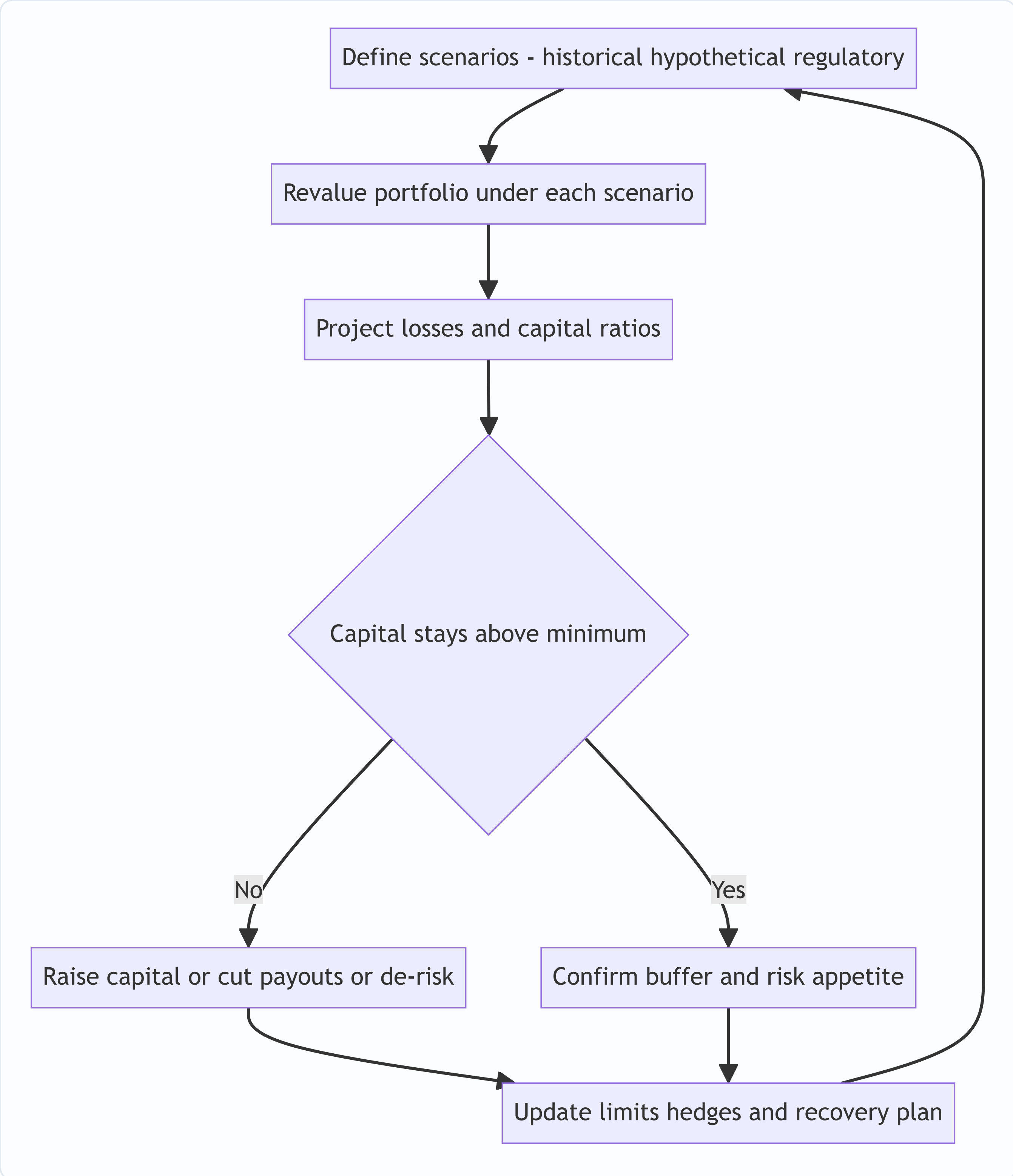

Figure 4: The stress-testing loop — design, revalue, decide, act, and feed back into the next cycle.

4.7 Practical pitfalls

- **Scenario staleness / fighting the last war.** Templates get reused until they no longer bind. Refresh the library; add a genuinely novel hypothetical each cycle.
- **False comfort from static balance sheets.** Assuming the book does not change during a stress ignores that in a real crisis you may be *forced* to sell into a falling market (fire sales), and that clients withdraw funding.

- **Ignoring second-round / feedback effects.** Your de-risking is everyone's de-risking; correlated selling amplifies the shock (the "liquidity spiral"). Simple stress tests miss this.
- **Model risk in the stress engine itself.** The revaluation models are calibrated in normal times; their behaviour in extreme states may itself be unreliable.
- **Correlation naivety.** Applying calm-market correlations to a crisis scenario understates losses — the whole point is that correlations jump. Impose crisis correlations explicitly.

5. Worked / Applied Examples

Example 1 — Sensitivity vs scenario on a simple bank book (numbers reconcile)

A small bank holds two positions:

- **A** — a long government-bond portfolio, market value **$1,000m**, **DV01 = $0.8m per 1bp** (so a +100bp parallel rate rise costs ≈ $80m).
- **B** — a long equity portfolio, market value **$500m**, beta 1.0 to the index.

Single-factor sensitivity stresses: - Rates **+100bp** → bond loss ≈ 100 × $0.8m = **−$80m**. (Equities unchanged in this isolated stress.) - Equities **−20%** → equity loss = 20% × $500m = **−$100m**. (Bonds unchanged.)

Run in isolation, the *worst single* stress is −$100m (the equity shock).

Now a joint scenario — "stagflation shock": the central bank hikes to fight inflation *and* the equity market falls on the growth hit. Suppose the coherent scenario is **rates +100bp AND equities −20%, simultaneously.**

- Bond loss: **−$80m**
- Equity loss: **−$100m**
- **Total scenario loss: −$180m**

The reconciliation and the lesson: the joint scenario loss ($180m) is the *sum* of the two single-factor losses because we imposed both shocks together, whereas each sensitivity stress moved only one factor. A naïve reading of the sensitivity table ("worst case is $100m") **understates the coherent scenario by $80m**. This is exactly the correlation problem: in a stagflation world, rates and equities hurt this book *at the same time*, and a single-factor view cannot see it.

Note the contrast with VaR: a VaR model calibrated on a period where rate rises usually coincided with equity *rallies* (a "risk-on" regime) might have assumed rates and equities are *negatively* correlated and **netted the two losses**, reporting a comforting figure well below $180m. The scenario forces the adverse joint move and exposes the true stagflation vulnerability.

Example 2 — Historical replay: 2008 on a credit book

A desk holds **$2,000m** notional of investment-grade corporate bonds with **spread DV01 (CS01) of $1.5m per 1bp** of spread widening, plus a **−$300m equity hedge overlay** (short index futures, so it *gains* when equities fall).

Apply the **Lehman-week historical scenario**: IG credit spreads **+150bp**, equities **−30%**.

- Credit loss: 150bp × $1.5m/bp = **−$225m**.
- Equity hedge gain: short $300m notional × 30% fall = **+$90m**.

- **Net scenario P&L: −$225m + $90m = −$135m.**

Interpretation. The hedge offsets 40% of the credit loss ($90m of $225m), which is *why* the desk holds it — in the exact historical episode where credit blows out, equities also crash, so the short-equity overlay pays off. This is a **coherent** hedge because the two legs are negatively correlated *in a crisis*. A key insight for interviews: the value of this hedge is invisible in a single-factor credit stress (which would show the full −$225m); only a joint historical scenario reveals that the hedge and the risk are correlated in the right direction. (Caveat: basis risk — the hedge is index equity, the risk is idiosyncratic credit — means the offset is imperfect and could fail in a scenario where credit widens *without* an equity crash.)

Example 3 — Reverse stress test: what breaks the bank?

A bank has **CET1 capital of $10bn** and a stated **minimum requirement of 8% of $80bn risk-weighted assets = $6.4bn**. So it can absorb losses of up to **$10bn − $6.4bn = $3.6bn** before breaching the minimum (ignoring RWA changes for simplicity).

Reverse question: what scenarios generate a **$3.6bn** hit? The team works backward and finds several break-points:

- **Path A — broad recession:** a 25% fall across the loan book's stressed loss rate produces ~$2.5bn of credit losses — *survivable* alone.
- **Path B — concentration cliff:** the bank's **single largest borrower group** (a $4bn exposure to one leveraged sector) defaulting at 60% loss-given-default = **$2.4bn**, *plus* a moderate $1.3bn market loss from the associated risk-off move = **$3.7bn → breach.**

The finding: the bank does *not* die from a broad, diversified recession (Path A); it dies from a **narrow, concentrated** event (Path B) — the failure of one sector to which it has quietly built an outsized exposure. A forward stress test using only broad macro scenarios would have shown "$2.5bn loss, comfortably survivable" and missed this entirely. The reverse test converts an abstract capital buffer ($3.6bn) into a **specific, actionable vulnerability**: reduce the single-name/sector concentration, or buy protection on it, or hold more capital against it. That is the decision the reverse test drives.

6. Connections

- **VaR and Expected Shortfall (Ch. 12).** Stress testing is the tail complement to VaR's body. ES is the bridge — it is VaR reaching *toward* the tail; stress testing reaches *past* the data entirely. In practice firms report VaR, ES, *and* a suite of stress losses side by side.
- **Regulatory capital (Basel).** Basel's **Stressed VaR** (SVaR) requires calibrating market-risk capital to a stress period (2007–2009), directly baking a historical stress into the capital charge. CCAR/EBA sit on top as institution-wide, forward stress tests. Stress testing is now a **capital-setting mechanism**, not just a monitoring tool.
- **Liquidity risk.** Stress scenarios usually include a **funding freeze** (deposit run-off, loss of wholesale funding). The LCR and NSFR ratios are themselves stress-style constructs (survive a 30-day stressed outflow).
- **Credit risk.** Stressed PD/LGD parameters and IFRS 9 / CECL expected-loss provisioning use forward-looking scenarios — the same macro paths often feed both provisioning and stress capital.
- **Model risk (Ch. on model governance).** The stress engine is itself a model; its assumptions (crisis correlations, revaluation behaviour) are a governance concern.

- **Recovery and resolution planning.** Reverse stress tests are a direct input to living wills.

7. Key Terms

- **Stress test** — examination of portfolio/firm behaviour under severe but plausible conditions.
- **Scenario analysis** — joint movement of many risk factors in a coherent story.
- **Sensitivity (factor) stress** — moving one risk factor by a large, defined amount.
- **Historical scenario** — replay of an actual past episode's factor moves against today's book.
- **Hypothetical scenario** — an invented but internally consistent adverse future.
- **Reverse stress test** — fix the failure outcome, solve for the scenarios that cause it.
- **Severe but plausible** — the calibration target: painful enough to matter, credible enough to act on.
- **CCAR / DFAST** — the US Fed's comprehensive capital analysis and Dodd-Frank stress tests.
- **Severely adverse scenario** — the harshest prescribed regulatory macro path (deep-recession calibrated).
- **Stress Capital Buffer (SCB)** — CET1 add-on set by a bank's peak-to-trough decline in the severely adverse scenario.
- **EBA / EU-wide stress test** — the European constrained bottom-up, published, SREP-feeding exercise.
- **Stressed VaR (SVaR)** — VaR calibrated to a historical stress window, part of Basel market-risk capital.
- **Static balance sheet** — the assumption that the book does not change over the stress horizon.
- **Second-round / feedback effects** — amplification from correlated de-risking, fire sales, liquidity spirals.

8. Common Confusions

- **"Stress testing replaces VaR."** No — they answer different questions (conditional vs probabilistic) and are used *together*. VaR maps where you have data; stress testing maps where you do not.
- **"A stress scenario has a probability."** Deliberately not. The power of "what if" is that you need not know the odds. Attaching a spurious probability to a hand-built scenario invites false precision.
- **"Historical scenarios are safe because they are real."** They are *credible* but backward-looking; the next crisis may not rhyme. Real ≠ complete.
- **"Reverse stress testing is just a harsh forward test."** No — the *arrow is reversed*. Forward = scenario→loss; reverse = failure→cause. Reverse tests find vulnerabilities you did not think to test.
- **"More severe is always better."** No — beyond plausibility, severity destroys usefulness (the meteor problem). The target is *severe but plausible*.
- **"Sensitivity stresses and scenarios are the same."** A sensitivity stress moves *one* factor; a scenario moves *many jointly*. Example 1 shows the difference costs $80m of hidden loss.
- **"CCAR and EBA are basically identical."** They differ in horizon, methodology (Fed runs its own models; EBA is constrained bottom-up), and consequence (SCB and payout approval vs SREP/Pillar 2 Guidance).
- **"The stress test result is an output."** Increasingly it is an *input* to capital — the SCB literally sets required capital from the stress loss. The causation now runs stress → capital.

9. Recap

VaR answers "how bad is a normal bad day and how likely?" but is structurally blind to the tail, dependent on sampled history, and offers no narrative a board can act on. **Stress testing fills all three gaps** by imposing severe-but-plausible *conditions* and fully revaluing the book under them. Its three families are **sensitivity stresses** (one factor), **scenario analysis** (many factors jointly — historical replays for credibility, hypotheticals for imagination), and **reverse stress testing** (start from failure, solve for cause). Regulators have made it binding: **CCAR/DFAST** in the US (9-quarter horizon, severely adverse scenario, feeding the Stress Capital Buffer and gating payouts) and the **EBA EU-wide test** (3-year, constrained bottom-up, published, feeding Pillar 2). The discipline only earns its keep when results **feed decisions** — sizing capital buffers, tightening limits, buying hedges, and shaping recovery plans. The examples showed how a joint scenario can hide $80m of loss that single-factor stresses miss, how a crisis-correlated hedge only reveals its value in a joint historical replay, and how a reverse stress test converts an abstract capital buffer into a specific, fixable concentration.

10. Quick-Reference / Interview Points

- **One-liner:** "VaR asks *how likely and how much on a normal bad day*; stress testing asks *what if a severe condition occurs* — they are complements, run together."
- **Why VaR misses tails:** it is a quantile (discards the tail), uses calm-market correlations (which jump to 1 in a crisis), and is anchored to sampled history (no dragon on the map).
- **Three types:** sensitivity (one factor), scenario (many factors — historical vs hypothetical), reverse (outcome→cause).
- **Historical vs hypothetical:** historical = credible but backward-looking; hypothetical = forward-looking but subjective. Run both.
- **Calibration target:** *severe but plausible* — not the meteor.
- **Coherent scenario check:** factor moves must hang together economically (equities down → spreads wider, safe rates lower, vol up, risky FX down).
- **CCAR:** US Fed, 9 quarters, baseline/adverse/severely adverse, CET1 must stay above minimum, sets the **Stress Capital Buffer**, gates dividends/buybacks. DFAST = the quantitative engine; CCAR adds qualitative + capital plan.
- **EBA:** EU, 3 years, baseline/adverse, constrained bottom-up, **published firm-by-firm**, feeds SREP / Pillar 2 Guidance. Traditionally static balance sheet.
- **Reverse stress test:** fix "the firm fails," solve for the scenarios; removes optimism bias, finds hidden concentrations; regulator-required.
- **How results bind:** stress loss → size of capital buffer; scenario exposure → limits and hedges; reverse result → recovery plan. **Stress sets capital, not vice versa.**
- **Basel link:** Stressed VaR bakes a 2007–09 window into market-risk capital.
- **Top pitfalls:** fighting the last war, static-balance-sheet false comfort, ignoring second-round/liquidity-spiral feedback, applying calm-market correlations.
- **Killer example soundbite:** "A single-factor table said worst case $100m; the coherent stagflation scenario (rates +100bp *and* equities −20% together) was $180m — the $80m gap is the correlation VaR and sensitivity views both miss."

Q&A — Stress Testing and Scenario Analysis

Practice bank for Chapter 13. Every question is followed by a full answer. Work each one before reading the solution. Stress testing is deliberately *not* probabilistic in the VaR sense — treat "how bad, if it happens" as the central question throughout.

Section A — Concept Check

A1. Give the one-line definition of stress testing and say how it differs in spirit from VaR.

Stress testing estimates the loss a portfolio or institution would suffer under a specific, severe-but-plausible set of market or economic conditions. VaR answers "how bad, at what probability, under normal conditions"; stress testing answers "how bad *if* this particular nasty thing happens" and deliberately drops the probability question. VaR is distributional and statistical; stress testing is conditional and narrative. They are complements — VaR covers the body of the distribution, stress testing probes the tail VaR is silent about.

A2. Distinguish "scenario analysis" from "sensitivity analysis."

Sensitivity analysis moves **one** risk factor at a time (e.g. rates +100 bp, all else held) to isolate the portfolio's exposure to that single driver. Scenario analysis moves **many** factors together in an internally consistent way (e.g. equities −30%, credit spreads +250 bp, INR −10%, rates −150 bp — a coordinated flight-to-quality). Sensitivity is a partial derivative; scenario analysis is a coherent joint move that respects how factors actually co-move in a crisis.

A3. Why is "severe but plausible" the governing phrase, and what fails if you drop either word?

If a scenario is severe but *implausible* (e.g. equities −99% overnight), the board dismisses the output and no action follows — the test loses credibility. If it is plausible but *not severe*, it tells you nothing you did not already see in VaR — it fails to probe the tail. The value sits in the intersection: bad enough to hurt, believable enough to act on.

A4. Contrast historical scenarios with hypothetical scenarios. Give one strength and one weakness of each.

A **historical** scenario replays an actual episode (2008 GFC, March 2020 COVID crash, 2013 taper tantrum). Strength: it is indisputably plausible — it happened, so no one argues the calibration. Weakness: the next crisis rarely rhymes exactly with the last, so it can miss novel risks. A **hypothetical** scenario is constructed by experts (e.g. a Middle-East oil shock plus a domestic banking freeze). Strength: it can capture forward-looking, never-seen risks and current portfolio-specific vulnerabilities. Weakness: it is subjective and easy to challenge on calibration.

A5. What is reverse stress testing and why is it psychologically valuable?

Reverse stress testing starts from the *outcome* — "what set of events would make us insolvent / breach our capital floor / exhaust our liquidity?" — and works backwards to find the scenarios that produce it. Normal stress testing asks "given this shock, how bad?"; reverse asks "how bad does it have to get before we die, and how could that happen?" It is valuable because it removes the comforting bias of only testing scenarios management already believes are survivable, and it surfaces the specific vulnerabilities that would actually kill the firm.

A6. Name the four broad categories a good stress-testing programme should span.

(1) **Market risk** stresses (prices, rates, vols, spreads, FX). (2) **Credit risk** stresses (rating downgrades, default waves, rising PD/LGD). (3) **Liquidity** stresses (funding runs, margin calls, market illiquidity). (4) **Firm-wide / macro** stresses that combine the above into an integrated recession or systemic scenario. A programme that only stresses market prices misses how a crisis actually compounds across risk types.

A7. Why does correlation "breaking down" matter so much in stress design?

Diversification benefits rest on assumed correlations. In a crisis, correlations tend toward extremes — risky assets fall together (correlations → +1) while the safe-haven leg decouples — so the offsets your VaR relied on vanish exactly when you need them. A credible stress scenario must therefore override normal-period correlations, otherwise it will understate loss by crediting diversification that evaporates under stress.

A8. What is a "second-round" or feedback effect, and why do simple stress tests miss it?

A first-round effect is the direct mark-to-market loss from the shock. Second-round effects are the reactions the shock triggers: forced deleveraging that pushes prices down further, margin calls that drain liquidity, counterparty failures, and fire sales that feed back into prices. Simple stress tests revalue the book once and stop, so they capture only the first round and understate systemic losses, which are dominated by these amplifying feedback loops.

A9. Why can two banks run "the same" 2008 scenario and get very different, both-correct results?

The scenario is a set of factor shocks, but the loss depends on each bank's *own* positions, hedges, and sensitivities. A bank long credit and short volatility suffers very differently from one positioned the opposite way. Stress loss = scenario shocks applied to *this* portfolio — the scenario is shared, the exposure is not.

A10. How does stress testing relate to regulatory capital and to management action?

Regulators (Basel, and supervisory exercises like the Fed's CCAR/DFAST or the RBI's stress-testing guidance) require banks to hold capital and liquidity sufficient to survive prescribed stresses, making stress results a direct input to capital adequacy. Internally, results must trigger *action* — reducing exposures, buying hedges, raising limits on liquidity buffers — or the exercise is theatre. A stress test that never changes a decision has failed regardless of how sophisticated the modelling is.

Section B — Numerical / Applied (with full solutions)

B1. Single-scenario revaluation. A ₹500 crore portfolio holds 60% equities (beta 1.0 to Nifty) and 40% in a 7-year bond (modified duration 6.0). A scenario specifies Nifty −35% and yields +200 bp. Estimate the stressed loss.

Solution. - Equity leg = 0.60 × 500 = ₹300 cr. Loss = 300 × 35% = **₹105 cr**. - Bond leg = 0.40 × 500 = ₹200 cr. Price change ≈ −Duration × Δy = −6.0 × 0.02 = −12%. Loss = 200 × 12% = **₹24 cr**. - Total stressed loss ≈ 105 + 24 = **₹129 cr**, i.e. 25.8% of the portfolio.

Sanity check: both legs lose (equity crash + rising rates hit bonds), so no offset here — consistent with a rate-hike-plus-equity-selloff scenario rather than a flight-to-quality.

B2. Same portfolio, flight-to-quality scenario. Now Nifty −35% but yields *fall* 150 bp (safe-haven rally into bonds). Estimate the loss.

Solution. - Equity leg loss = 300 × 35% = ₹105 cr (unchanged). - Bond leg gain ≈ −6.0 × (−0.015) = +9%. Gain = 200 × 9% = +₹18 cr. - Net loss ≈ 105 − 18 = **₹87 cr** (17.4%).

Teaching point: same equity shock, but the *direction of the rate move* flips the bond leg from a ₹24 cr loss to an ₹18 cr gain — a ₹42 cr swing. This is exactly why scenarios must specify factors jointly and consistently; the correlation assumption is the scenario.

B3. Adding second-order (convexity/gamma) correction. Reconsider B1's bond leg. The bond has convexity 50. Refine the price change for $\Delta y = +200$ bp.

Solution. Price change $\approx -D\cdot\Delta y + \tfrac{1}{2}\cdot C\cdot(\Delta y)^2 = -6.0(0.02) + 0.5(50)(0.02)^2 = -0.12 + 0.5(50)(0.0004) = -0.12 + 0.01 = -0.11$, i.e. −11% (not −12%). Refined bond loss = 200 × 11% = **₹22 cr**; refined total ≈ 105 + 22 = **₹127 cr**.

Point: convexity cushions large moves, so a duration-only stress *overstates* the loss for big yield jumps. For genuinely severe stresses, full revaluation beats greek-based approximation.

B4. Historical scenario scaling. During March 2020 the portfolio's key factor (a credit spread) widened 350 bp over 20 trading days. Your book has spread DV01 (sensitivity to 1 bp) of ₹8 lakh. What is the historical stress loss, and what is the 1-day-equivalent if you want to compare with 1-day VaR?

Solution. - Full-episode loss = 350 bp × ₹8 lakh/bp = **₹2,800 lakh = ₹28 cr**. - To put on a 1-day footing you would *not* simply divide by 20 (that assumes linear time). If comparing to a 1-day VaR under an i.i.d. assumption, scale the *shock* by √: 1-day-equivalent spread move $\approx 350/\sqrt{20} \approx 78$ bp, giving ≈ 78 × 8 = ₹624 lakh ≈ **₹6.2 cr**. But note the honest answer: stress tests are usually reported at the full episode horizon precisely because crisis moves are *not* i.i.d., and the 20-day cumulative figure is the economically meaningful one.

B5. Reverse stress test. A bank has ₹1,200 cr of capital and a policy that it must never fall below ₹300 cr (its regulatory floor plus buffer). Its trading book loses ₹3 cr per 1% equity decline (linear). Ignoring other exposures, what equity crash would breach the floor?

Solution. Allowable loss before breach = 1,200 − 300 = ₹900 cr. Required equity move = 900 / 3 = **300%** decline — impossible, so equities *alone* cannot break the bank. The reverse-stress insight: the killing scenario is not a pure equity crash but a *combination* — equity losses plus credit blow-ups plus a funding freeze acting together. Reverse stress testing forces you to find that combination rather than resting on "no single shock can sink us."

B6. Correlation breakdown, quantified. In normal times two ₹100 cr positions have return correlation +0.3, each with 20% volatility. Under stress, correlation jumps to +0.9. Compare portfolio volatility (in ₹) normal vs stressed. ($\sigma\_p = \sqrt{(w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2\rho w_1 w_2 \sigma_1 \sigma_2)}$, equal weights, absolute ₹ terms with each leg σ = ₹20 cr.)

Solution. Each leg contributes σ = ₹20 cr. - Normal: $\sigma\_p = \sqrt{(20^2 + 20^2 + 2\cdot 0.3\cdot 20\cdot 20)} = \sqrt{(400 + 400 + 240)} = \sqrt{1040}$ = **₹32.2 cr**. - Stressed: $\sigma\_p = \sqrt{(400 + 400 + 2\cdot 0.9\cdot 20\cdot 20)} = \sqrt{(400 + 400 + 720)} = \sqrt{1520}$ = **₹39.0 cr**.

Risk rises ~21% purely from correlation moving 0.3 → 0.9, with *no change in individual volatilities or positions*. This is the diversification benefit evaporating — the single most under-appreciated driver of stress losses.

B7. Aggregating a multi-factor scenario with offsets. A scenario gives four factor P&Ls: equities −₹40 cr, credit −₹25 cr, rates +₹15 cr (flight to quality helps the bond book), FX −₹10 cr. State the total and explain why you must not take the worst single factor.

Solution. Net stress P&L = −40 − 25 + 15 − 10 = **−₹60 cr**. Reporting only the worst factor (equities, −₹40 cr) understates the loss by ₹20 cr because the credit and FX legs pile on; reporting the sum of *only* the losing legs (−75 cr) overstates it by ignoring the ₹15 cr rate hedge. The correct number is the coherent net across all factors moving together. Consistency of the joint move is the whole point.

Section C — Interview-Style (with model answers)

C1. "VaR already gives us a tail number. Why does the firm also spend money on stress testing?"

Model answer: VaR describes the distribution *under normal conditions* and is estimated from recent history, so it is structurally blind to the rare, regime-shifting events that do the real damage — and those are precisely the events that matter for solvency. Stress testing fills three gaps VaR cannot: it probes losses *beyond* the confidence level (the tail VaR ignores), it uses forward-looking or historical crisis scenarios rather than only the recent calm window, and it captures correlation breakdown and feedback effects that a covariance-based VaR assumes away. VaR is for day-to-day limit-setting; stress testing is for survival. You need both.

C2. "Walk me through how you would design a firm-wide stress scenario from scratch."

Model answer: I would start from the firm's *material vulnerabilities* — the largest concentrations and the exposures whose loss would threaten capital or liquidity. Then choose a narrative: anchor on a historical analogue (a 2008-style credit-and-liquidity crisis) or build a hypothetical macro story (a sharp domestic slowdown with capital outflows and INR depreciation). I translate the narrative into a *consistent* set of factor shocks — equity, rates, spreads, FX, defaults — with correlations reflecting crisis behaviour, not calm-period estimates. I apply them by full revaluation, then layer second-round effects: margin calls, funding cost increases, forced sales. Finally I map the loss to capital and liquidity metrics and define the management actions it would trigger. The deliverable is not a number — it is a decision.

C3. "Our stress test showed we survive 2008 comfortably. Are we safe?"

Model answer: Not necessarily, and I would push back gently on the framing. Surviving a *replay* of 2008 tells you that you are prepared for the *last* war. The value of that result is limited by three things: the next crisis will have a different epicentre and different correlations; your current portfolio is not the one that existed in 2008; and a single historical scenario cannot reveal your specific kill-switch. I would want to complement it with hypothetical forward-looking scenarios tailored to today's concentrations and, most importantly, a *reverse* stress test that asks what it would actually take to breach our capital or liquidity floor. Comfort against one historical scenario is a starting point, not a clean bill of health.

C4. "How do you decide the severity of a scenario — where's the line between 'severe but plausible' and 'fantasy'?"

Model answer: I calibrate against evidence. Historical episodes give hard anchors — the actual peak-to-trough moves in 2008, 2013, and 2020 bound what markets have really done. For hypothetical scenarios, I keep individual factor moves within, or modestly beyond, historically observed extremes and — critically — keep the *joint* move coherent, since implausibility usually creeps in through incoherent combinations rather than any single shock. I also let the audience set the bar: the scenario must be severe enough that the board takes the loss seriously, yet grounded enough that they cannot wave it away as impossible. If risk managers and the business can both look at it and say "yes, that could happen," it is in the right zone. Reverse stress testing then deliberately pushes *past* plausibility to find the breaking point, but that is labelled as such.

C5. "What are the main limitations of stress testing itself? Sell me the weaknesses."

Model answer: Four honest ones. First, *scenario selection risk* — you can only stress what you imagine, and the crisis that gets you is usually the one no one scripted. Second, *no probability* — it tells you how bad, not how likely, so it cannot be aggregated the way VaR can, and it is hard to size capital against a scenario with no attached probability. Third, *model and data limits* — feedback effects, liquidity spirals, and contagion are very hard to model, so most tests understate systemic loss. Fourth, *false comfort and gaming* — passing a regulator's prescribed scenario breeds complacency, and firms can position to look good against the published test. The discipline is only as good as the imagination and follow-through behind it.

Section D — Multiple Choice (with reasoning)

D1. The defining feature that separates stress testing from VaR is that stress testing: (a) always produces a larger number (b) attaches no probability to the scenario (c) uses only historical data (d) is required by regulators

Answer: **(b).** Stress testing deliberately answers "how bad if this happens" without assigning a probability. (a) is usually but not necessarily true; (c) is false (scenarios can be hypothetical); (d) is a consequence, not the defining feature.

D2. Moving only interest rates +100 bp while holding all other factors fixed is an example of: (a) scenario analysis (b) reverse stress testing (c) sensitivity analysis (d) Monte Carlo simulation

Answer: **(c).** One factor moved in isolation is sensitivity analysis. Scenario analysis moves multiple factors jointly and consistently.

D3. A reverse stress test begins by specifying: (a) a set of factor shocks (b) the failure outcome, then finds scenarios that cause it (c) the probability distribution of losses (d) the regulatory capital requirement

Answer: **(b).** Reverse stress testing starts from the outcome (e.g. insolvency) and works backwards to the scenarios. (a) describes a conventional forward stress test.

D4. In a crisis scenario, assuming *normal-period* correlations between risky assets will tend to: (a) overstate the stress loss (b) understate the stress loss (c) have no effect on the loss (d) always leave the loss unchanged

Answer: **(b).** In crises, risky-asset correlations rise toward +1, so diversification offsets shrink. Using lower normal-period correlations credits offsets that vanish, understating the loss.

D5. "Second-round effects" in stress testing refer to: (a) running the test a second time for accuracy (b) the direct mark-to-market loss from the shock (c) reactions like margin calls, forced sales, and contagion that amplify the initial loss (d) the second most severe scenario in the suite

Answer: **(c).** Second-round (feedback) effects are the amplifying reactions triggered by the first-round loss; ignoring them understates systemic loss. (b) is the first-round effect.

D6. The phrase "severe but plausible" fails on the "plausible" side when a scenario is: (a) too mild to reveal new risk (b) so extreme the board dismisses it and takes no action (c) taken from an actual historical crisis (d) tailored to the firm's concentrations

Answer: **(b).** Losing plausibility means the scenario is dismissed as fantasy and drives no action. Being too mild (a) is the failure on the "severe" side.

D7. Two banks run the identical 2008 historical scenario and report very different losses. The best explanation is: (a) one of them made a calculation error (b) the scenario was applied inconsistently (c) their portfolios and hedges differ, so the same shocks produce different P&L (d) historical scenarios are unreliable

Answer: **(c).** Stress loss depends on each firm's own exposures. The same shocks acting on different books legitimately give different results — no error required.

D8. Which is the strongest argument for supplementing a historical 2008 scenario with hypothetical scenarios? (a) hypothetical scenarios are always more severe (b) regulators forbid historical scenarios (c) the next crisis may have a different epicentre and correlations that 2008 does not capture (d) hypothetical scenarios are easier to calibrate

Answer: **(c).** Historical replays are backward-looking; hypothetical scenarios add forward-looking, portfolio-specific risks. (a) and (d) are false, and (b) is untrue.

Self-check summary

- **Numbers verified:** B1 (₹129 cr), B2 (₹87 cr — sign flip on the bond leg), B3 (convexity cushions to ₹127 cr), B6 (correlation 0.3→0.9 lifts risk ₹32.2 cr→₹39.0 cr), B7 (net −₹60 cr).
- **Recurring theme:** the danger lives in what the model assumes away — correlation breakdown, second-round feedback, and the scenario you failed to imagine. A stress test is a decision tool, not a number generator.

Chapter 14 — Enterprise Risk Management

1. The Problem / The Need

Imagine a large bank in 2007. Its trading desk is loading up on mortgage-backed securities and reporting record profits. Its treasury is funding those positions with cheap overnight repo. Its retail arm is writing sub-prime mortgages to feed the securitisation machine. Each unit has its own risk manager, its own limits, its own dashboard. Every one of them is inside its limits. Every one of them looks safe.

And then the bank collapses.

How can an institution where *every individual desk is within its risk limits* fail catastrophically? Because nobody was adding it all up. The market-risk team measured price moves. The credit team measured borrower defaults. The liquidity team measured funding gaps. But mortgage default, MBS price collapse, and repo funding drying up are **the same event viewed from three angles** — they are perfectly correlated in a crisis. The bank's *true* exposure was not the sum of three moderate risks; it was one enormous, concentrated bet on US housing, invisible to anyone looking through a single-function lens.

This is the fundamental problem that **Enterprise Risk Management (ERM)** exists to solve. Traditional risk management is **siloed**: each risk type (market, credit, operational, liquidity, reputational, strategic) is owned by a specialist team that manages it in isolation. Siloed risk management fails for four structural reasons:

1. **Correlations are invisible.** Risks that look independent inside silos are often driven by the same underlying factor. Adding up silo exposures understates the aggregate.
2. **Nobody owns the total.** If every silo covers its own patch, the *portfolio* of risks — the thing that actually determines whether the firm survives — has no owner.
3. **Risk is treated as a cost to minimise, not a resource to allocate.** Silos say "reduce my risk." Nobody asks "are we taking the *right* risks in the right amounts to earn our required return?"
4. **Emerging and strategic risks fall through the cracks.** A new competitor, a regulatory shift, a technology disruption — these don't belong to any single silo, so no silo watches for them.

The result is an organisation that is locally optimised and globally fragile. Every part is prudent; the whole is reckless. ERM is the discipline that raises the vantage point from the desk to the boardroom and asks a different question: *not "is each risk controlled?" but "is the firm as a whole taking risk deliberately, in amounts it can survive, in pursuit of its strategy?"*

2. The Core Idea

Enterprise Risk Management is the holistic, integrated, top-down management of the full portfolio of risks facing an organisation, aligned with its strategy and governed from the board.

Unpack the four load-bearing words:

- **Holistic** — every risk type is in scope: financial *and* non-financial, quantifiable *and* judgemental, downside threats *and* the risk of missing upside. Strategic, operational, financial, compliance, reputational, and emerging risks are all on one map.
- **Integrated** — risks are aggregated and viewed together so correlations, concentrations, and offsets become visible. ERM manages the *portfolio*, not the pieces.

- **Top-down** — direction flows from the board's articulation of how much risk the firm is willing to take (its **risk appetite**) down to the limits and controls in each unit. This is the mirror image of siloed, bottom-up risk management.
- **Strategy-aligned** — risk is not a compliance afterthought bolted on after the plan is set. Risk is a lens *on the strategy itself*: which objectives, which bets, which uncertainties could stop us achieving what we set out to do — and are those the risks we *want* to be taking?

The single most important mental shift ERM demands is this: **risk is not just something to be minimised; it is something to be consciously chosen and allocated.** A firm that takes no risk earns no return and eventually dies. The job is not to eliminate risk but to take the *right* risks, in *deliberate* amounts, that the firm is *capable of bearing*, in service of objectives the board has endorsed. ERM turns risk from a defensive, "keep us out of trouble" function into a strategic capability that shapes where the firm competes.

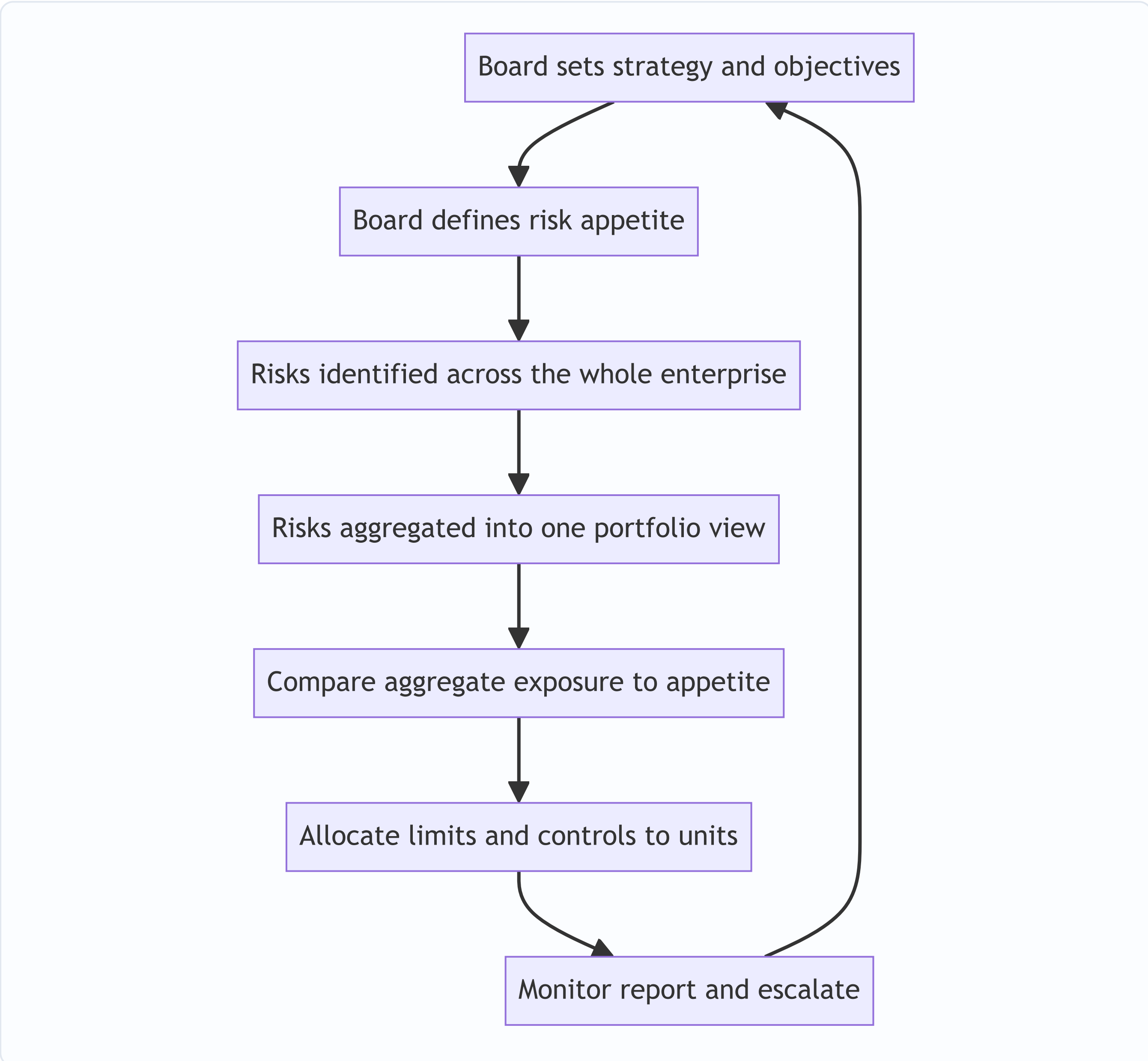

Figure 14.1 — ERM as a closed top-down loop from strategy to monitoring and back.

3. Why / How It Works

Why does aggregating risk change the answer so dramatically? The mathematics of correlation is the engine underneath ERM.

The diversification argument. If two risks are less than perfectly correlated, the risk of holding both together is *less* than the sum of holding each alone. This is the same portfolio insight from investing, applied to the whole firm. A general insurer writing both hurricane cover in Florida and earthquake cover in Japan is safer than the sum of its parts, because a Florida hurricane and a Japanese earthquake almost never happen the same week. Silos miss this benefit — they can't net offsetting exposures they never see together — so a firm managed only in silos will *over-hold* capital against diversifiable risk and, worse, *under-hold* against concentrated risk.

The concentration argument. The mirror image is more dangerous. When risks *are* highly correlated — the 2007 bank's mortgage default, MBS price, and repo funding — the aggregate is far *larger* than any single silo shows. Silos systematically understate concentrated risk because the concentration only appears when you add the pieces up. ERM's aggregation step is precisely what surfaces the hidden bet.

Formally, for two exposures with standard deviations σ_1 and σ_2 and correlation ρ, the combined risk is:

$\sigma\_portfolio = \sqrt{(\sigma_1^2 + \sigma_2^2 + 2 \cdot \rho \cdot \sigma_1 \cdot \sigma_2)}$

When $\rho = 1$ (perfect correlation), this collapses to $\sigma_1 + \sigma_2$ — risks simply add. When $\rho < 1$, the combined risk is strictly *less* than the sum. When ρ is negative, the risks partly cancel. **A silo implicitly assumes $\rho = 1$ with everything it can't see, or $\rho = 0$ by ignoring the rest — and it is almost never right about either.** Only enterprise-level aggregation gets ρ approximately correct.

Why top-down governance is essential. Even with perfect aggregation, someone must *decide* how much total risk is acceptable and *enforce* it downward. Left to themselves, individual units maximise their own performance metrics and collectively overshoot — a classic tragedy of the commons where the shared resource being over-consumed is the firm's risk-bearing capacity (its capital and its survival). The board owns that capacity on behalf of shareholders, so the board must set the ceiling. ERM works because it puts the aggregate ceiling and the strategy in the same hands.

Why culture makes or breaks it. Frameworks, limits, and committees are necessary but not sufficient. Every risk failure autopsy — Barings, Enron, Lehman, the London Whale — reveals that the formal apparatus existed but was ignored, overridden, or gamed. Risk lives in thousands of daily decisions by people the framework never directly touches. If the tacit message from the top is "hit the numbers and don't slow us down," no policy document will stop the next blow-up. ERM works only when the *culture* — the shared, often-unspoken norms about how risk is really treated — reinforces the framework rather than undermining it.

4. Full Content

4.1 The three building blocks of ERM

ERM rests on three interlocking pillars. Miss any one and the structure fails.

Pillar	Question it answers	Key artefacts
Risk governance	Who is accountable for risk, and how is authority delegated?	Board risk committee, three lines of defence, CRO mandate, risk policies
Risk appetite	How much and what kinds of risk are we willing to take?	Risk appetite statement, tolerances, limits cascade
Risk culture	How do people actually behave toward risk day to day?	Tone from the top, incentives, escalation norms, "speak-up" behaviour

4.2 Risk governance and the three lines of defence

Governance is the structure of accountability: who decides, who executes, who challenges, and who assures. The board holds ultimate responsibility for risk — it cannot delegate that — but it delegates *execution*. In most large firms the board operates a dedicated **Board Risk Committee** (separate from the Audit Committee, which focuses on financial reporting and controls) that reviews the risk appetite, the risk profile against it, and major risk decisions.

The workhorse operating model for governance is the **Three Lines of Defence**. It answers the question "who owns risk?" with a clear separation of duties:

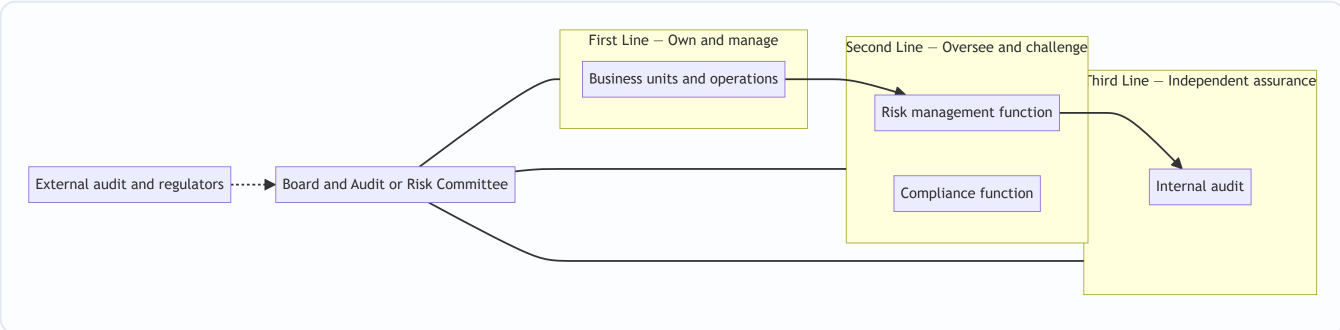

Figure 14.2 — The three lines of defence separating ownership challenge and assurance.

- **First line — the business.** The people who *take* the risk *own* the risk. A trading desk, a lending team, a factory manager — they generate exposures and are the first responsible for identifying, controlling, and managing them within their limits. Risk ownership is *not* outsourced to a risk department; it sits with the business that creates it. This is the single most important governance principle and the one firms most often get wrong.
- **Second line — risk and compliance oversight.** Independent functions that set the framework, aggregate the enterprise view, monitor limits, challenge the first line's decisions, and report to the board. This is where the CRO and the risk-management function live, alongside compliance (which specifically oversees legal and regulatory conformance). The second line does *not* own the risk — it oversees how the first line owns it.
- **Third line — internal audit.** Fully independent assurance that both the first and second lines are working as designed. Internal audit reports functionally to the Audit Committee, not to management, so it can challenge without fear. It provides the board objective evidence that the risk apparatus is real, not theatre.

A frequently added **"fourth line"** comprises external audit and regulators — outside-the-firm assurance. The updated 2020 model from the Institute of Internal Auditors reframes this as the **"Three Lines Model,"** softening the militaristic "defence" language and stressing collaboration and value creation over pure control, but the core separation — ownership, oversight, assurance — remains.

The critical failure mode this model prevents is **the poacher guarding the henhouse**: a business unit marking its own homework, with no independent challenge and no independent assurance. When the second and third lines are weak, captured, or under-resourced, the first line's optimism goes unchecked.

4.3 The risk appetite framework

Risk appetite is the amount and type of risk an organisation is willing to accept in pursuit of its objectives. It is the hinge that connects strategy to day-to-day limits. Without it, "how much risk is too much?" has no answer and every limit is arbitrary.

The framework has a hierarchy, from abstract intent at the top to hard numbers at the bottom:

Level	Term	Nature	Example
1	**Risk capacity**	The *maximum* risk the firm could bear before breaching a hard constraint	Total loss-absorbing capital before insolvency or covenant breach
2	**Risk appetite**	How much risk the firm *chooses* to take, within capacity	"We accept up to 15% of capital at risk over one year"
3	**Risk tolerance**	Acceptable variation around a specific objective	"Credit losses may vary +/- 20% around the plan"
4	**Risk limits**	Hard operational thresholds allocated to units	"Desk X VaR ceiling of 10 million; single-name exposure cap of 50 million"

Appetite is expressed both **qualitatively** ("we have zero appetite for regulatory breaches or safety incidents; moderate appetite for market risk in core products; no appetite for reputational damage") and **quantitatively** (VaR ceilings, economic capital allocations, ratings targets, maximum acceptable loss, liquidity coverage floors). The qualitative statement gives meaning; the quantitative metrics make it enforceable.

The **Risk Appetite Statement (RAS)** is the board-approved document that captures all of this. A well-built RAS is: - **Linked to strategy** — appetite is set *for* the objectives, not in the abstract. - **Cascaded** — the enterprise appetite is broken down into limits for each business line, so a desk's limit is a *slice* of the firm's total appetite, not an isolated number. - **Actionable and monitored** — every appetite statement maps to a metric with a threshold, a green/amber/red status, and an escalation path when breached. - **Forward-looking** — tested against stress scenarios, not just current exposure.

The appetite framework is what makes ERM *top-down* in practice: the board sets the total, and limits cascade down so that the sum of what every unit is allowed to take equals what the firm as a whole is willing to bear.

4.4 Risk culture

If appetite is the framework's *design*, culture is whether the framework is *lived*. **Risk culture is the set of shared values, attitudes, and behaviours that shape how risk is actually treated across the organisation** — especially in the moments no policy anticipates.

The reason culture dominates is empirical: nearly every large risk failure had adequate formal controls that people simply bypassed. Barings had position limits; Nick Leeson hid his losses in an unreconciled error account and no one challenged a "star" trader. The controls existed; the culture — deference to a rainmaker, weak back-office challenge — defeated them.

The observable drivers and markers of a healthy risk culture:

- **Tone from the top.** Senior leaders visibly treat risk as everyone's business, welcome bad news, and act consistently with the stated appetite even when it costs short-term profit. Behaviour, not memos.
- **Incentives aligned with prudent risk-taking.** If bonuses reward short-term revenue with no clawback for later losses, the incentive system is *manufacturing* excessive risk regardless of the stated appetite. Compensation must be risk-adjusted and deferred.
- **Psychological safety and speak-up.** People can raise concerns, challenge decisions, and report near-misses without fear of retaliation. The health of a risk culture is measured less by how confident everyone is and more by how freely bad news travels upward.
- **Accountability and consequences.** Limit breaches and control failures have real consequences, applied consistently regardless of seniority or how much revenue the person generates.
- **Effective challenge.** The second line genuinely challenges the first, and the challenge is respected rather than resented as an obstacle.

Culture is diagnosed through leading indicators — employee surveys, near-miss reporting rates, how limit breaches are handled, whether risk teams are seen as partners or police, turnover in control functions — because by the time it shows up in losses, it is too late.

4.5 The COSO ERM framework

The most widely adopted conceptual framework for ERM is **COSO's "Enterprise Risk Management — Integrating with Strategy and Performance"** (the 2017 update of the original 2004 cube). COSO — the Committee of Sponsoring Organizations of the Treadway Commission — is the same body behind the dominant internal-control framework. The 2017 revision made a decisive shift: it moved ERM away from a control-and-compliance checklist and toward **integration with strategy and the creation and preservation of value.**

The 2017 framework organises ERM into **five interrelated components** supported by **twenty principles**:

#	Component	What it covers
1	**Governance and Culture**	Board oversight, operating structures, defining the desired culture, commitment to core values, attracting and retaining capable people
2	**Strategy and Objective-Setting**	Analysing business context, defining risk appetite, evaluating alternative strategies, formulating objectives aligned with appetite
3	**Performance**	Identifying risk, assessing severity, prioritising, implementing responses, developing a portfolio view of risk
4	**Review and Revision**	Assessing substantial change, reviewing risk and performance, pursuing improvement in ERM
5	**Information, Communication, and Reporting**	Leveraging information systems, communicating risk information, reporting on risk culture and performance

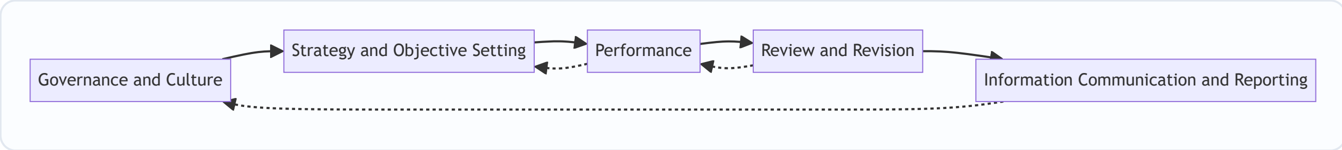

Figure 14.3 — The five components of the COSO 2017 ERM framework as an integrated flow.

The three big messages of COSO 2017 are worth memorising because interviewers probe them:

1. **ERM is not a separate function bolted on — it is woven into strategy-setting and performance management.** Risk consideration begins *when strategy is chosen*, including the risk that the *chosen strategy doesn't align with mission* and the risk of *implications from the strategy* selected. This "strategy risk" framing is the heart of the 2017 update.
2. **The goal is value.** ERM enhances the ability to create, preserve, and realise value — not merely to avoid loss. This reframes risk from purely defensive to genuinely strategic.
3. **Culture and governance come first, literally.** The framework leads with governance and culture because, as section 4.4 argued, everything downstream depends on them.

The **ISO 31000** standard is a lighter, more generic alternative — principles, framework, and a plan-do-check-act process (establish context → risk assessment [identify, analyse, evaluate] → treat → monitor → communicate) — applicable to any organisation of any size, not just corporates. COSO is more detailed and governance-heavy; ISO 31000 is more universal and process-oriented. Many firms borrow from both.

4.6 The Chief Risk Officer (CRO)

ERM needs an owner at the executive table, and that owner is the **Chief Risk Officer**. The CRO is the senior executive accountable for the enterprise-wide risk framework — the person who makes "someone owns the total" a reality.

Core responsibilities: - **Own the ERM framework** — design, implement, and maintain the risk taxonomy, appetite framework, methodologies, and policies. - **Produce the enterprise portfolio view** — aggregate risks across silos into one integrated picture and report the firm's actual profile against appetite to the board. - **Provide independent challenge** — as head of the second line, challenge the business's risk-taking, sign off on major transactions, and say "no" or "not at that size" when appetite would be breached. - **Advise on strategy** — bring the risk lens into strategic decisions early, quantifying the risk-adjusted trade-offs of alternative plans. - **Champion risk culture** — set the tone for the risk function and drive the behavioural norms across the firm.

The CRO's **independence** is structural and non-negotiable. The role rose to prominence after the 2008 crisis precisely because risk heads had too often been buried under revenue-generating executives who could overrule them. Best practice — and, for banks, regulatory requirement — is that the CRO reports directly to the **CEO** *and* has a direct, unfettered line to the **Board Risk Committee**, with the committee (not the CEO alone) controlling the CRO's appointment, removal, and compensation. This dual reporting protects the CRO from being silenced by the very executives whose risk-taking they must challenge.

The essential tension in the role: the CRO must be **close enough to the business to be credible and informed, yet independent enough to challenge it.** A CRO captured by the business is useless; a CRO so detached they don't understand the business is ignored. Walking that line is the craft of the job.

4.7 Why silos fail and ERM adds value — pulled together

The value ERM adds over siloed risk management, summarised:

Dimension	Siloed risk management	Enterprise Risk Management
Scope	One risk type per team	All risks, one portfolio
Correlations	Invisible — assumed away	Explicitly aggregated
Concentrations	Hidden across silos	Surfaced by aggregation
Direction	Bottom-up, local	Top-down from board appetite
Ownership of the total	Nobody	CRO and board
View of risk	Cost to minimise	Resource to allocate to strategy
Strategic/emerging risk	Falls through cracks	Explicitly in scope
Capital	Over-held vs diversifiable, under-held vs concentrated	Optimised against true aggregate

ERM's payoff is both defensive and offensive. **Defensively**, it prevents the "every silo fine, whole firm doomed" failure by making concentrations visible and putting a board-owned ceiling on the total. **Offensively**, it lets a firm *deliberately* take more of the risks it is good at and well-capitalised for — allocating its finite risk-bearing capacity to its highest risk-adjusted-return opportunities — because it can now see and price the whole picture. That is the difference between risk management as a brake and risk management as a steering wheel.

5. Worked / Applied Examples

Example 1 — The correlation that silos miss (a bank)

A bank has two exposures, each measured by its own silo: - **Market-risk silo:** its MBS portfolio has a one-year 99% VaR (loss) of **£100m**. - **Credit-risk silo:** its mortgage loan book has a one-year 99% loss estimate of **£100m**.

Each silo reports "£100m at risk, within my limit." Naively, a manager might think total risk is somewhere between £100m (if fully offsetting) and £200m (if additive), and comfort themselves it's "diversified."

Now apply the aggregation formula. What is the *true* correlation ρ between MBS losses and mortgage-book losses? Both are driven by the **same underlying factor** — US house prices and mortgage defaults. In a housing downturn, mortgages default *and* MBS prices collapse together. Realistically $\rho \approx 0.9$.

$\sigma\_portfolio = \sqrt{(100^2 + 100^2 + 2 \times 0.9 \times 100 \times 100)} = \sqrt{(10{,}000 + 10{,}000 + 18{,}000)} = \sqrt{38{,}000} \approx$ **£195m**

The aggregate risk is **£195m — essentially additive**, almost the full £200m, because the two "different" risks are nearly the same bet. The silos, each seeing £100m, radically understated the firm's true exposure. Had the bank instead diversified into, say, a European corporate loan book with $\rho \approx 0.2$ against its MBS:

$\sigma\_portfolio = \sqrt{(100^2 + 100^2 + 2 \times 0.2 \times 100 \times 100)} = \sqrt{24{,}000} \approx$ **£155m**

Reconciliation and lesson: same £100m per silo in both cases, but the enterprise risk is £195m for the concentrated pair versus £155m for the diversified pair — a £40m difference invisible to any silo. ERM's aggregation step is the *only* place this shows up. The concentrated bank is carrying near-double the risk it thinks it is; only an enterprise view reveals it and forces either a capital increase or a hedge.

Example 2 — Cascading a risk appetite statement (a manufacturer)

"SafeBuild Ltd," a construction-materials group, holds **£500m of loss-absorbing capital** (its risk *capacity*). The board sets a **risk appetite** of "we will not put more than 20% of capital at risk over a one-year horizon at a 95% confidence level" → an enterprise economic-capital budget of **£100m**.

The CRO cascades this £100m across the group's risk types, choosing the allocation to match strategy (SafeBuild wants to grow its overseas operations, so it deliberately funds more market/FX and operational appetite there):

Risk type	Allocated economic capital (appetite)	Illustrative hard limit
Operational (safety, plant)	£40m	Zero appetite for fatalities; max £5m single-event loss
Market / FX	£25m	FX VaR ceiling £8m
Credit (customer receivables)	£20m	Single-customer exposure cap £15m
Strategic / project	£15m	No single project > £30m committed capital
Total	**£100m**	= board appetite

Six months in, the FX desk's VaR reaches £7.5m against its £8m limit — **amber**. Under the framework, this triggers escalation to the CRO and Risk Committee *before* the limit is breached, not after a loss.

Reconciliation and lesson: capacity £500m > appetite £100m > sum of allocated limits £100m. The chain holds: the firm never allocates more appetite than the board authorised, keeps a £400m buffer between appetite and capacity for the truly unexpected, and the desk-level £8m limit is a *deliberate slice* of the enterprise total — not an isolated number. The appetite framework converts one board sentence into an enforceable, monitored limit on a single trading desk.

Example 3 — When culture defeats the framework (a trading loss)

"Meridian Bank" has a textbook ERM apparatus: an independent CRO reporting to a Board Risk Committee, VaR limits on every desk, and daily risk reports. Yet it loses **£1.2bn** on a single credit-derivatives book.

Trace the failure through the framework, not around it: - **First line:** a star trader builds an outsized position. His desk head, whose bonus depends on the desk's revenue, waves it through and pressures the back office to accept the trader's own generous valuations. - **Second line:** the risk function notices the VaR model is being changed in ways that *lower* reported risk. But the risk analysts are junior, poorly paid relative to the desk, and their challenges are dismissed as "not understanding the business." Effective challenge fails. - **Culture:** the tone from the top is "revenue is king." Raising concerns about a rainmaker is career-limiting. Near-misses go unreported.

Every *structural* box was ticked — CRO, limits, reports all existed — yet the firm lost £1.2bn because the **culture** (deference to revenue, weak challenge, no psychological safety) hollowed out the framework.

Reconciliation and lesson: this is the London Whale / Barings pattern in miniature. It demonstrates the chapter's central caveat — governance structure and risk appetite are necessary but *not sufficient*. Without a culture that makes challenge safe and prudent behaviour rewarded, the finest framework on paper is theatre. ERM is 30% framework and 70% behaviour.

6. Connections

- **Portfolio theory (Chapter on diversification/Markowitz):** ERM is portfolio theory applied to the whole firm. The correlation mathematics in Example 1 is identical to two-asset portfolio variance — the "assets" are just risk exposures instead of securities.
- **Value at Risk and Economic Capital:** VaR (market-risk chapters) and economic capital are the quantitative engines that let ERM aggregate and allocate. Risk appetite is typically *denominated* in these units.
- **Operational risk and internal control:** COSO ERM is the sibling of the COSO Internal Control framework; the three lines of defence originated as an internal-control concept and now anchors ERM governance. For CA students, this dovetails directly with auditing's control-environment and governance topics.
- **Basel and Solvency regulation:** Basel's ICAAP (Internal Capital Adequacy Assessment Process) for banks and Solvency II's ORSA (Own Risk and Solvency Assessment) for insurers are *regulatory mandates for ERM* — they force firms to run exactly this appetite-capacity-aggregation loop and prove it to supervisors.
- **Corporate governance:** the Board Risk Committee, CRO independence, and audit-committee relationships tie ERM into the broader governance syllabus — board composition, non-executive oversight, and stewardship.
- **Behavioural finance / agency theory:** risk culture and misaligned incentives are agency problems. The tragedy-of-the-commons framing of why silos overshoot the firm's risk capacity is pure agency economics.
- **Strategic management:** COSO 2017's core message — risk-in-strategy — connects ERM directly to strategy formulation, competitive positioning, and scenario planning.

7. Key Terms

- **Enterprise Risk Management (ERM):** holistic, integrated, top-down, strategy-aligned management of the entire portfolio of an organisation's risks.
- **Silo (siloed risk management):** managing each risk type in isolation, blind to correlations and concentrations across the firm.
- **Risk governance:** the structure of accountability, delegation, oversight, and assurance for risk, anchored by the board.
- **Three Lines of Defence / Three Lines Model:** first line (business — owns risk), second line (risk & compliance — oversees), third line (internal audit — assures).
- **Risk capacity:** the maximum risk a firm *could* bear before hitting a hard constraint (insolvency, covenant breach).
- **Risk appetite:** the amount and type of risk a firm *chooses* to take in pursuit of objectives; less than capacity.
- **Risk tolerance:** acceptable variation around a specific objective or metric.
- **Risk limit:** a hard operational threshold allocated to a unit; a slice of enterprise appetite.

- **Risk Appetite Statement (RAS):** board-approved document articulating appetite qualitatively and quantitatively.
- **Risk culture:** shared values, attitudes, and behaviours that shape how risk is actually treated day to day.
- **Effective challenge:** the second line genuinely questioning and, when needed, overriding first-line risk decisions.
- **COSO ERM (2017):** the leading ERM framework — five components, twenty principles, integrating risk with strategy and performance.
- **ISO 31000:** a generic, universal risk-management standard (principles, framework, process).
- **Chief Risk Officer (CRO):** senior executive accountable for the enterprise risk framework, head of the second line, with independent board access.
- **Portfolio view of risk:** the aggregated, correlation-aware picture of all risks together — the deliverable ERM exists to produce.
- **Economic capital:** capital a firm calculates it needs to absorb unexpected losses to a chosen confidence level; a common currency for appetite.

8. Common Confusions

- **"ERM means minimising or eliminating risk."** No. ERM is about taking the *right* risks *deliberately* and in amounts the firm can bear. A firm with zero risk earns zero return. The goal is optimisation and conscious allocation, not minimisation.
- **"Risk appetite and risk capacity are the same."** No. Capacity is the *maximum you could* bear (a constraint); appetite is *how much you choose* to take (a decision), and it sits *below* capacity, leaving a buffer.
- **"Risk tolerance = risk appetite."** Related but distinct. Appetite is the firm-wide willingness to take risk overall; tolerance is the acceptable variation around a *specific* objective or metric. Appetite is strategic and broad; tolerance is operational and narrow.
- **"The three lines of defence means three teams that all do risk."** No — it's a *separation of duties*. The first line *owns and takes* the risk (it is not a control function); the second *oversees and challenges*; the third *independently assures*. Blurring them (e.g., letting risk teams take positions, or internal audit report to management) breaks the model.
- **"The second line owns the risk."** No. The *business* (first line) owns the risk it creates. The risk function oversees; it does not take the exposure or absolve the business of ownership. This is the most common governance error.
- **"COSO ERM and COSO Internal Control are the same thing."** They are siblings from the same body but distinct. Internal Control is about reliable reporting, compliance, and operational controls; ERM is broader — strategy, appetite, and the full risk portfolio. ERM subsumes control concepts but goes well beyond them.
- **"ERM is a software system / a report."** ERM is a *management discipline and culture*, not a dashboard. Buying a GRC tool does not give you ERM any more than buying a treadmill makes you fit. Example 3 shows a firm with all the tooling that still failed on culture.
- **"The CRO is responsible for all the firm's risk."** No. The *business* is responsible for taking and managing its risk; the CRO is responsible for the *framework*, the *aggregate view*, and *independent*

challenge. Making the CRO the sole owner of risk lets the business abdicate — the opposite of the intended model.

- **"More capital always means safer."** Not if it's held against the wrong (diversifiable) risks while concentrated risks go uncovered. ERM's point is to hold the *right* capital against the *true aggregate*, which is why silo-based capital allocation is both wasteful and dangerous.

9. Recap

Siloed risk management fails because it manages each risk type in isolation and so cannot see correlations, cannot see concentrations, gives nobody ownership of the total, treats risk as a cost to minimise, and lets strategic and emerging risks fall through the cracks. The 2007-style failure — every desk within limits, the whole firm doomed — is the signature symptom.

ERM answers this by managing risk **holistically, integrated, top-down, and aligned with strategy**. It rests on three pillars: **governance** (who is accountable — anchored by the board, the three lines of defence, and an independent CRO), **risk appetite** (how much risk we choose to take — cascaded from board-set capacity through appetite and tolerance down to unit limits), and **culture** (whether the framework is actually lived — the behaviours, incentives, and speak-up norms that make or break everything else).

The mathematics of correlation is the engine: aggregating exposures reveals that concentrated risks add up to far more than any silo shows, while genuinely diversified risks add up to less — knowledge that lets a firm allocate its finite risk-bearing capacity deliberately. The **COSO 2017 framework** codifies this in five components (Governance & Culture; Strategy & Objective-Setting; Performance; Review & Revision; Information, Communication & Reporting), reframing ERM from compliance chore to strategic, value-creating discipline. The **CRO** owns the framework and the portfolio view, challenges the business independently, and reports to both CEO and board. And the enduring caveat — proven by every blow-up from Barings to the London Whale — is that structure and appetite are necessary but never sufficient: **without a healthy risk culture, the finest framework is theatre.**

10. Quick-Reference / Interview Points

- **One-line definition:** ERM is the holistic, integrated, top-down management of an organisation's entire risk portfolio, aligned with strategy and governed from the board.
- **Why silos fail (four reasons):** invisible correlations; hidden concentrations; nobody owns the total; strategic/emerging risks fall through the cracks. Signature: every desk within limits, whole firm collapses.
- **The value ERM adds:** defensively, surfaces concentrations and caps the total; offensively, lets the firm allocate finite risk capacity to its best risk-adjusted opportunities.
- **Three lines of defence:** 1st = business (owns/takes risk), 2nd = risk & compliance (oversees/challenges), 3rd = internal audit (independent assurance). Business owns the risk — *not* the risk department.
- **Appetite hierarchy:** capacity (max you *could* bear) > appetite (how much you *choose*) > tolerance (variation around an objective) > limits (hard unit thresholds). Cascade the top down into the bottom.
- **Risk Appetite Statement:** board-approved, strategy-linked, qualitative + quantitative, cascaded, monitored with green/amber/red escalation.
- **COSO ERM 2017 — five components:** Governance & Culture; Strategy & Objective-Setting; Performance; Review & Revision; Information, Communication & Reporting. Big message: integrate risk with strategy and value creation, not compliance.

- **COSO vs ISO 31000:** COSO is detailed, governance-heavy, corporate-focused; ISO 31000 is generic, principles-based, universal. Basel ICAAP (banks) and Solvency II ORSA (insurers) are regulatory ERM mandates.
- **CRO essentials:** owns the framework and portfolio view; independent challenge; dual reporting to CEO *and* Board Risk Committee; committee controls appointment/removal/pay. Tension: close enough to be credible, independent enough to challenge.
- **Risk culture markers:** tone from the top, risk-aligned incentives with clawback/deferral, psychological safety to speak up, consistent accountability, effective challenge. Diagnosed via leading indicators (near-miss reporting, breach handling, survey data).
- **Correlation formula (be ready to compute):** $\sigma_{portfolio} = \sqrt{(\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2)}$. $\rho \to 1$ means risks add (concentration); $\rho < 1$ means diversification benefit. Silos implicitly get ρ wrong.
- **Killer interview soundbite:** "ERM turns risk management from a brake into a steering wheel — from minimising risk in silos to deliberately allocating the firm's risk-bearing capacity to its strategy. But it's 30% framework and 70% culture: Barings and the London Whale had the controls and still blew up."
- **If asked 'why did ERM rise after 2008?':** the crisis exposed that firms were fatally concentrated in correlated risks their silos couldn't see, and that risk heads had no independent voice. ERM's board-owned appetite, aggregation, and independent CRO are the direct institutional response.

Q&A — Enterprise Risk Management

A practice bank for the Enterprise Risk Management chapter. Work each question before reading the answer. Numerical answers are self-checked against the two-exposure aggregation identity $\sigma_p = \sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2}$ and the appetite chain capacity ≥ appetite ≥ Σ limits.

Section A — Concept-Check (short answer)

A1. Define Enterprise Risk Management in one sentence, and identify the four load-bearing words.

ERM is the **holistic, integrated, top-down management of an organisation's entire portfolio of risks, aligned with its strategy and governed from the board.** The four load-bearing words: **holistic** (every risk type — financial and non-financial, downside and missed-upside — is in scope), **integrated** (risks are aggregated so correlations and concentrations become visible), **top-down** (direction flows from board-set appetite down to unit limits), and **strategy-aligned** (risk is a lens on the strategy itself, not a compliance afterthought).

A2. Why does siloed risk management fail? Give the four structural reasons and the signature symptom.

Silos fail because: (1) **correlations are invisible** — risks driven by the same factor look independent inside separate silos; (2) **nobody owns the total** — each silo covers its patch, so the risk *portfolio* has no owner; (3) **risk is treated as a cost to minimise, not a resource to allocate**; and (4) **strategic and emerging risks fall through the cracks** because they belong to no single silo. The signature symptom: **every desk is within its limits, yet the whole firm collapses** (the 2007-style bank).

A3. Distinguish risk capacity, risk appetite, risk tolerance, and risk limits.

They form a top-down hierarchy. **Risk capacity** is the *maximum* risk the firm could bear before a hard constraint (insolvency, covenant breach) — a constraint, not a choice. **Risk appetite** is how much risk the firm *chooses* to take within that capacity — a board decision. **Risk tolerance** is the acceptable *variation* around a specific objective or metric ("credit losses may vary ±20% around plan"). **Risk limits** are the hard operational thresholds allocated to individual units (a desk's VaR ceiling). Capacity > appetite > tolerance > limits.

A4. What are the three pillars ERM rests on?

Risk governance (who is accountable — the board, the three lines of defence, the CRO mandate), **risk appetite** (how much and what kinds of risk we choose to take — the RAS and limit cascade), and **risk culture** (how people actually behave toward risk day to day — tone, incentives, escalation norms). Miss any one and the structure fails.

A5. State the three lines of defence, and name the single most common governance error.

First line — the business: the people who take the risk own it (identify, control, manage within limits). **Second line — risk & compliance**: independent oversight that sets the framework, aggregates the enterprise view, and challenges the first line. **Third line — internal audit**: independent assurance to the board that lines one and two work. The most common error: believing the **second line (the risk department) owns the risk**. It does not — the *business* owns the risk it creates; the risk function only oversees.

A6. Why must the CRO have dual reporting lines, and to whom?

The CRO reports directly to the **CEO** *and* has an unfettered line to the **Board Risk Committee**, with the committee (not the CEO alone) controlling the CRO's appointment, removal, and pay. Dual reporting exists because pre-2008, risk heads were buried under revenue-generating executives who could overrule or silence them. The direct board line protects the CRO from being muzzled by the very people whose risk-taking they must challenge.

A7. Name the five components of the COSO 2017 ERM framework in order.

(1) **Governance and Culture**; (2) **Strategy and Objective-Setting**; (3) **Performance**; (4) **Review and Revision**; (5) **Information, Communication, and Reporting**. The framework deliberately leads with governance and culture because everything downstream depends on them.

A8. What is the single biggest message of the COSO 2017 update versus the 2004 cube?

The 2017 revision moved ERM **away from a control-and-compliance checklist toward integration with strategy and the creation and preservation of value.** Risk consideration now begins *when strategy is chosen* — including the risk that the chosen strategy misaligns with the mission and the risk of implications flowing from the strategy. ERM became a value-creating strategic discipline, not a defensive compliance chore.

A9. Give the two-exposure aggregation formula and explain what ρ = 1 versus ρ < 1 means for a firm.

$\sigma\_portfolio = \sqrt{\sigma_1^2 + \sigma_2^2 + 2 \cdot \rho \cdot \sigma_1 \cdot \sigma_2}$. When **$\rho = 1$** (perfect correlation), it collapses to $\sigma_1 + \sigma_2$ — risks simply add, the signature of dangerous **concentration**. When **$\rho < 1$**, combined risk is strictly less than the sum — the **diversification** benefit. A silo implicitly assumes $\rho = 1$ with what it can see and $\rho = 0$ with what it cannot, and is almost never right about either; only enterprise aggregation gets ρ approximately correct.

A10. Why is culture said to be "70% of ERM," and how is it diagnosed before losses appear?

Because nearly every large risk failure — Barings, Enron, Lehman, the London Whale — had adequate formal controls that people bypassed, overrode, or gamed. Structure and appetite are necessary but never sufficient; risk lives in thousands of daily decisions the framework never touches. Culture is diagnosed through **leading indicators** — near-miss reporting rates, how limit breaches are handled, employee survey data, turnover in control functions, whether risk teams are seen as partners or police — because by the time it shows up in losses, it is too late.

Section B — Numerical / Applied (full solutions)

B1. The correlation silos miss. A bank's market-risk silo reports a one-year 99% VaR of £120m on its MBS book; its credit silo reports £120m on its mortgage loan book. Both are driven by US house prices, so the true correlation is $\rho = 0.9$. Compute the true aggregate and compare to what each silo sees.

- $\sigma\_p = \sqrt{120^2 + 120^2 + 2 \times 0.9 \times 120 \times 120}$
- $= \sqrt{14{,}400 + 14{,}400 + 25{,}920} = \sqrt{54{,}720} \approx$ **£234m**.

Self-check. Each silo sees £120m; the naive "diversified" instinct hopes the total sits well below £240m. But at $\rho = 0.9$ the aggregate is £234m — **essentially additive**, because the two "different" risks are nearly the same bet on housing. The concentration only appears when you add the pieces up, which no silo does. ✓

B2. The diversification benefit. The same bank instead pairs its £120m MBS book with a £120m European corporate loan book at $\rho = 0.2$. Compute the aggregate and the capital saved versus B1.

- $\sigma_p = \sqrt{(120^2 + 120^2 + 2 \times 0.2 \times 120 \times 120)} = \sqrt{(14{,}400 + 14{,}400 + 5{,}760)} = \sqrt{34{,}560} \approx$ **£186m**.
- Saving versus the concentrated pair = 234 − 186 = **£48m** of risk.

Self-check. Same £120m per silo in both cases, but genuine diversification ($\rho = 0.2$) cuts the aggregate from £234m to £186m — a £48m difference invisible to any silo. This is ERM's offensive payoff: seeing correlations lets the firm deliberately hold the *diversifying* book and free up capital. ✓

B3. Negative correlation — partial hedge. An insurer writes Florida hurricane cover with σ = £80m and buys a reinsurance-linked position with σ = £30m that pays out in the same storms, giving $\rho = -0.6$ against the hurricane book. Compute the net risk.

- $\sigma_p = \sqrt{(80^2 + 30^2 + 2 \times (-0.6) \times 80 \times 30)} = \sqrt{(6{,}400 + 900 - 2{,}880)} = \sqrt{4{,}420} \approx$ **£66.5m**.

Self-check. The net £66.5m is *below* the standalone hurricane risk of £80m — the negatively-correlated position partly cancels it, which is the whole point of a hedge. A silo looking only at the £80m book, or naively adding £80m + £30m = £110m, would both mis-state the true £66.5m. Negative ρ subtracts. ✓

B4. Cascading a risk appetite statement. "BuildCo" holds £600m of loss-absorbing capital. The board sets appetite at "no more than 20% of capital at risk over one year." The CRO must allocate the resulting economic-capital budget across four risk types. State the budget and check a proposed allocation of Operational £45m, Market/FX £30m, Credit £25m, Strategic £20m.

- Enterprise appetite = 20% × £600m = **£120m**.
- Proposed allocation sum = 45 + 30 + 25 + 20 = **£120m** — exactly equals appetite. ✓
- Buffer to capacity = 600 − 120 = **£480m** held against the truly unexpected.

Self-check. The chain holds: capacity £600m > appetite £120m = Σ limits £120m. The firm never allocates more appetite than the board authorised, and each unit limit is a deliberate *slice* of the enterprise total, not an isolated number. Had the allocation summed to, say, £140m, the CRO would be authorising £20m more risk than the board permits — a framework breach. ✓

B5. Amber-trigger monitoring. BuildCo's FX desk has a hard VaR limit of £10m and an amber-escalation trigger set at 80% of limit. Its VaR climbs to £8.4m. What status is it, and what must happen?

- Amber threshold = 80% × £10m = £8.0m. Current £8.4m > £8.0m but < £10m limit.
- Status = **amber** (breach of the escalation trigger, not yet the hard limit).
- Action: escalate to the CRO and Risk Committee *now*, **before** the £10m limit is breached — the point of amber is to act ahead of a loss, not after one.

Self-check. A well-built RAS maps every appetite statement to a metric with a green/amber/red threshold and an escalation path. £8.4m sits in the amber band (£8m–£10m), so the framework fires early. If the desk hit £10.2m it would be red — a hard breach requiring position reduction. ✓

B6. Silo capital versus true aggregate. Three silos each report £50m of standalone risk (σ = 50 each). Under the naive silo approach they are simply summed. Compute (a) the silo sum, (b) the true aggregate if all three are pairwise correlated at $\rho = 0.3$, and comment on over- or under-holding.

- (a) Silo sum = 50 + 50 + 50 = **£150m**.
- (b) For three equal exposures: $\sigma_p = \sqrt{(3 \times 50^2 + 2 \times 0.3 \times 3 \text{ pairs} \times 50 \times 50)} = \sqrt{(3 \times 2{,}500 + 6 \times 0.3 \times 2{,}500)} = \sqrt{(7{,}500 + 4{,}500)} = \sqrt{12{,}000} \approx$ **£109.5m**.

Self-check. The true diversified aggregate (£109.5m) is well below the silo sum (£150m), so a firm that simply adds silo numbers **over-holds capital against diversifiable risk** by ~£40m. The mirror danger (from

B1) is that when ρ is near 1 the silo sum *understates* the true aggregate. Silos are wrong in both directions; only aggregation gets it right. Formula note: n equal exposures give $\sigma_p = \sigma\cdot\sqrt{n + n(n-1)\rho}$; here $50\cdot\sqrt{3 + 6\times0.3} = 50\cdot\sqrt{4.8} = 50 \times 2.19 \approx 109.5$. ✓

Section C — Interview-Style (model answers)

C1. "Every desk was within its limits, so how did the bank fail?"

Because nobody was adding it all up. Each silo measured one angle of what was, in reality, a single concentrated bet — the market team measured MBS price moves, the credit team measured mortgage defaults, the treasury measured repo funding, and in a housing crisis those three are the *same event* seen three ways, correlated near 1. The firm's true exposure was not three moderate risks that diversify away; it was one enormous position on US housing, invisible to anyone looking through a single-function lens. The lesson: **local prudence does not aggregate to global safety.** ERM's aggregation step exists precisely to surface the hidden concentration that every individual limit misses.

C2. "Is ERM just about minimising risk?"

No — that is the most common misconception. A firm that takes no risk earns no return and eventually dies. ERM's central shift is that **risk is a resource to be consciously chosen and allocated, not merely a cost to minimise.** The job is to take the *right* risks — the ones the firm is good at and well-capitalised for — in deliberate amounts it can survive, in service of board-endorsed objectives. ERM turns risk management from a brake into a steering wheel: defensively it caps the total and surfaces concentrations; offensively it lets the firm point its finite risk-bearing capacity at its highest risk-adjusted-return opportunities.

C3. "Walk me through the three lines of defence and tell me who owns the risk."

The first line is the business — trading desks, lending teams, plant managers — the people who *take* the risk and therefore *own* it; they identify, control, and manage exposures within their limits. The second line is risk and compliance: independent functions that set the framework, aggregate the enterprise view, monitor limits, and provide *effective challenge* to the first line — but they do **not** own the risk. The third line is internal audit, which reports to the Audit Committee rather than management and gives the board independent assurance that lines one and two actually work. Who owns the risk? **The business — always.** The failure mode this prevents is the poacher guarding the henhouse: a unit marking its own homework with no independent challenge or assurance.

C4. "Distinguish risk appetite from risk capacity, and why the gap matters."

Capacity is the *maximum* the firm could bear before hitting a hard constraint — the loss-absorbing capital before insolvency or a covenant breach. Appetite is *how much the firm chooses to take*, deliberately set below capacity by the board. The gap between them is the buffer for the truly unexpected — the stress event, the model error, the correlation that spikes to 1 in a crisis. A firm that sets appetite equal to capacity has no margin for being wrong, and it will eventually be wrong. Concretely: if a firm has £600m of capacity and sets appetite at £120m, the £480m gap is what keeps a bad year from becoming a fatal one. Appetite is a choice; capacity is a limit; the distance between them is survival.

C5. "A firm has a CRO, VaR limits on every desk, and daily risk reports — yet it loses a billion on one book. What went wrong?"

Structure was present but culture was absent — the 30%-framework-70%-behaviour problem. Trace it through the framework: a star trader builds an outsized position and his revenue-incentivised desk head waves it through; risk analysts notice the VaR model being tweaked to *lower* reported risk but are junior, underpaid, and dismissed as "not understanding the business," so effective challenge fails; and the tone from the top is "revenue is king," making challenge of a rainmaker career-limiting. Every structural box was ticked, yet the firm blew up because the culture hollowed the framework out — the Barings and London Whale pattern. **The finest framework on paper is theatre without a culture that makes challenge safe and prudent behaviour rewarded.**

C6. "Why did ERM and the modern CRO rise to prominence after 2008?"

The crisis exposed two gaps ERM directly answers. First, firms were catastrophically concentrated in correlated risks their silos could not see — mortgage default, MBS prices, and repo funding were one bet no single-function team aggregated; ERM's board-owned appetite and enterprise portfolio view are the response. Second, risk heads had been buried beneath revenue-generating executives who could overrule them, so warnings never reached the board; the modern independent CRO with dual reporting is the structural fix. Regulators then hard-wired the loop into rules — Basel's ICAAP for banks and Solvency II's ORSA for insurers are, in effect, *mandatory ERM*, forcing firms to run the appetite-capacity-aggregation loop and prove it to supervisors.

Section D — MCQs (with reasoning)

D1. Which word does NOT describe the core nature of ERM? A) Holistic B) Integrated C) Bottom-up D) Strategy-aligned

Answer: C. ERM is **top-down** — direction flows from board-set appetite down to unit limits. "Bottom-up" describes the *siloed* model ERM replaces, where each unit optimises locally and the firm overshoots collectively. A, B, and D are three of the four load-bearing words.

D2. In the three lines of defence, who owns the risk? A) The risk management function B) Internal audit C) The business that takes it D) The Chief Risk Officer

Answer: C. The first line — the business that creates the exposure — owns the risk. The risk function (A) and CRO (D) form the second line, which *oversees and challenges* but does not own. Internal audit (B) is the third line, providing independent assurance. Believing the risk department owns the risk is the single most common governance error.

D3. Which correctly orders the risk appetite hierarchy from broadest to narrowest? A) Limits > tolerance > appetite > capacity B) Capacity > appetite > tolerance > limits C) Appetite > capacity > limits > tolerance D) Tolerance > limits > appetite > capacity

Answer: B. Capacity (the maximum the firm *could* bear) sits at the top as a hard constraint; appetite (how much it *chooses*) sits below it; tolerance is the acceptable variation around a specific objective; and limits are the hard unit-level thresholds at the bottom. The board sets the top and it cascades down.

D4. Two £100m exposures are perfectly correlated ($\rho = 1$). Their aggregate risk is: A) £100m B) £141m C) £200m D) £0

Answer: C. At $\rho = 1$ the formula $\sqrt{(100^2 + 100^2 + 2\times1\times100\times100)} = \sqrt{40{,}000}$ = £200m — risks simply add. This is the concentration case. Option B (£141m) is the $\rho = 0$ diversified answer $\sqrt{20{,}000}$; a silo that assumes independence when the true ρ is 1 would dangerously understate the risk.

D5. The COSO 2017 ERM framework leads with which component? A) Performance B) Information, Communication and Reporting C) Governance and Culture D) Review and Revision

Answer: C. COSO deliberately places **Governance and Culture** first because everything downstream — strategy, performance, review, reporting — depends on it. The five components in order are Governance & Culture; Strategy & Objective-Setting; Performance; Review & Revision; Information, Communication & Reporting.

D6. The biggest shift in COSO's 2017 ERM update was to: A) Add more detailed control checklists B) Integrate ERM with strategy and value creation C) Replace the CRO with a committee D) Mandate a specific VaR model

Answer: B. The 2017 revision reframed ERM from a control-and-compliance checklist toward integration with strategy and the creation and preservation of value — the "strategy risk" framing. It did the opposite of A. C and D are not part of COSO at all; COSO is a conceptual framework, not a modelling mandate.

D7. Which statement about the Chief Risk Officer is correct? A) The CRO owns all of the firm's risk B) The CRO reports only to the CEO C) The CRO owns the framework and the portfolio view, and reports to both CEO and Board Risk Committee D) The CRO sits in the first line of defence

Answer: C. The CRO owns the ERM *framework* and the aggregate portfolio view and provides independent challenge, with dual reporting to the CEO and the Board Risk Committee. The *business* owns the risk, so A is wrong (that abdication is the trap). B removes the board independence that protects the role. D is wrong — the CRO heads the *second* line.

D8. A firm with a complete ERM framework — independent CRO, desk VaR limits, daily reports — still suffers a huge rogue-trading loss. The most likely root cause is: A) The VaR model was miscalibrated B) A weak risk culture defeated the framework C) The firm held too little capital D) The appetite statement was not board-approved

Answer: B. When the entire structural apparatus exists yet the firm blows up, the failure is almost always cultural — deference to revenue, weak effective challenge, no psychological safety to report near-misses (the Barings / London Whale pattern). The chapter's central caveat: structure and appetite are necessary but not sufficient. ERM is roughly 30% framework and 70% culture.

D9. ISO 31000 differs from COSO ERM chiefly in that it is: A) Bank-specific and capital-focused B) Generic, principles-based, and universal to any organisation C) A regulatory capital requirement D) Concerned only with internal financial controls

Answer: B. ISO 31000 is a lighter, generic standard (principles, framework, and a plan-do-check-act process) applicable to any organisation of any size. COSO is more detailed and governance-heavy. Neither is itself a capital rule — that would be Basel's ICAAP or Solvency II's ORSA (C confuses the two). D describes COSO *Internal Control*, not ISO 31000.

D10. A silo that ignores every risk outside its own patch is implicitly assuming, for those unseen risks, a correlation of: A) $\rho = 0$ B) $\rho = 1$ C) $\rho = -1$ D) $\rho = 0.5$

Answer: A. By ignoring other risks entirely, a silo behaves as if they are independent of its own — $\rho = 0$, capturing a full diversification benefit it has not verified. Conversely, when it can *see* an adjacent risk it often lumps it in additively ($\rho = 1$). It is almost never right about either; only enterprise aggregation estimates the true ρ.

Recap of key relationships (self-verify against these)

- **ERM definition:** holistic, integrated, top-down, strategy-aligned management of the whole risk portfolio, governed from the board.
- **Why silos fail:** invisible correlations; hidden concentrations; nobody owns the total; strategic/emerging risks fall through the cracks. Signature: every desk within limits, whole firm collapses.
- **Three lines:** 1st = business (owns/takes risk); 2nd = risk & compliance (oversees/challenges); 3rd = internal audit (independent assurance). Business owns the risk.
- **Appetite chain:** capacity (max you *could* bear) ≥ appetite (how much you *choose*) ≥ tolerance (variation around an objective) ≥ limits (hard unit thresholds); Σ limits ≤ appetite.
- **Aggregation:** $\sigma_p = \sqrt{(\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2)}$; $\rho \to 1$ risks add (concentration), $\rho<1$ diversification benefit, $\rho<0$ partial hedge. For n equal exposures: $\sigma_p = \sigma \cdot \sqrt{(n + n(n-1)\rho)}$.
- **COSO 2017 — five components:** Governance & Culture; Strategy & Objective-Setting; Performance; Review & Revision; Information, Communication & Reporting. Message: integrate risk with strategy and value.
- **CRO:** owns framework + portfolio view; independent challenge; dual reporting to CEO and Board Risk Committee. Culture is 70% of ERM — structure alone is theatre.

Chapter 15 — Using Derivatives to Manage Risk

1. The Problem / The Need

Every operating business and every investing institution carries risks it never chose and does not want. A wheat mill wants to make flour, not to speculate on wheat prices — yet the moment it agrees to sell flour at a fixed price for the next six months while buying wheat at whatever the market charges, it has become an accidental commodity speculator. An Indian software exporter wants to write code, not to bet on the rupee — yet an invoice denominated in dollars and payable in ninety days is a directional currency position whether the CFO likes it or not. A bank wants to lend, not to gamble on the direction of interest rates — yet funding a ten-year fixed-rate mortgage with overnight deposits is one of the largest interest-rate bets in the economy.

These are *residual* risks. They arise as an unavoidable by-product of doing legitimate business. The firm has three choices for each of them:

1. **Accept** the risk — do nothing, and let earnings swing with the market.
2. **Avoid** the risk operationally — refuse the dollar contract, don't lend long, don't buy inventory in advance. This usually means refusing the business itself.
3. **Transfer or offset** the risk — keep the underlying business but lay off the unwanted price exposure to someone willing to hold it.

The third route is the province of **derivatives**. A derivative is a contract whose value *derives* from the price of something else — an interest rate, an exchange rate, a share price, a commodity, or the creditworthiness of a borrower. Because a derivative's payoff can be engineered to move in the exact opposite direction to an existing exposure, it lets a firm surgically remove a specific price risk while keeping everything else about its business intact. That surgical precision — decoupling *what you do* from *what price risk you bear* — is the core reason derivatives exist and the reason they are the primary tools of modern risk management.

The problem this chapter solves is therefore practical and repeated daily in every treasury and trading desk: **given a specific unwanted exposure, which derivative neutralises it, how much of it do you need, and how do you make sure the hedge keeps working over time without quietly turning into a new speculation of its own?**

2. The Core Idea

The core idea is **offset**. If you own an asset whose value falls when some market variable moves against you, you construct a derivative position whose value *rises* by an equal amount when that same variable moves. Combine the two and the net position is insensitive — fully or partly — to that variable. The gain on one leg cancels the loss on the other.

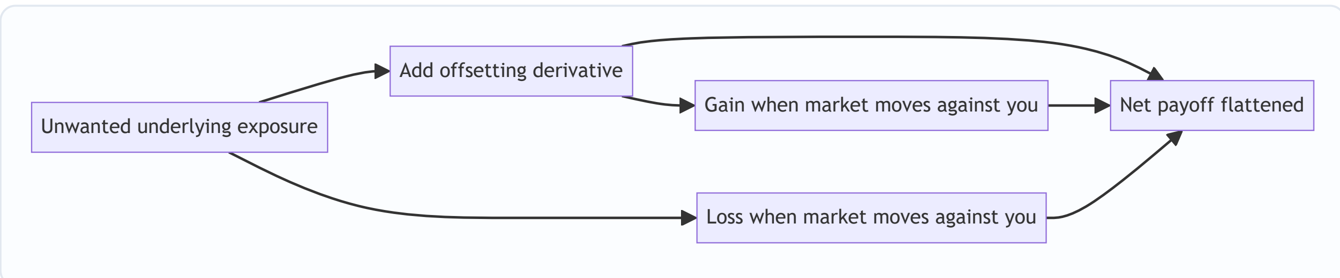

Figure 15.1 — A hedge pairs an exposure with an opposite-signed derivative so the combined payoff is stabilised.

Two things make this idea powerful and worth understanding deeply.

First, a derivative requires little or no capital to establish a large offsetting position. A forward or a swap typically costs nothing to enter; a futures contract needs only a margin deposit that is a small fraction of the notional; an option costs a premium far smaller than the value it controls. This **leverage** is what lets a firm hedge a crore of exposure without tying up a crore of cash — and it is also, as Part 3 and Part 8 explain, the exact mechanism by which derivatives can *create* catastrophic risk when used to speculate rather than to offset.

Second, the offset can be shaped. A **forward, future or swap** locks in a price and removes both downside *and* upside symmetrically — you are protected if the market moves against you but you forgo the gain if it moves in your favour. An **option** removes the downside while preserving the upside, in exchange for an upfront premium. The choice between the two is one of the central design decisions in any hedge and reflects whether the firm wants certainty (lock it in) or protection-with-participation (buy insurance).

3. Why / How It Works

A hedge works because of a single mechanical fact: the derivative and the exposure share the same **underlying** driver. When two contracts reference the same variable, their values move together (or in controlled opposition), and by holding them in opposite directions you engineer cancellation.

Consider the exporter with a USD 1 million receivable due in 90 days. Its home-currency value falls if the rupee strengthens (fewer rupees per dollar). The firm sells USD 1 million forward at today's agreed rate. If in 90 days the rupee has strengthened, the receivable converts to fewer rupees — a loss — but the forward now lets the firm sell dollars above the spot rate — an offsetting gain. If instead the rupee weakens, the receivable is worth more but the forward loses exactly that extra amount. Either way the firm banks the forward rate. The *reason* it works is that both the receivable and the forward settle against the same USD/INR rate on the same date.

Three principles govern whether the offset is clean or leaky:

Matching. The hedge cancels the exposure only to the extent the derivative's terms mirror the exposure's terms — same underlying, same amount (notional), same maturity/timing, and ideally the same reference rate or grade. Mismatch on any dimension leaves a residual, called **basis risk**: the risk that the price of what you hedged with and the price of what you actually own do not move one-for-one.

Payoff symmetry versus asymmetry. Linear instruments (forwards, futures, swaps) produce a symmetric offset — the hedge line is a mirror image of the exposure line, so the sum is flat. Options produce a kinked payoff — they cut off one tail while leaving the other open. This is why a linear hedge *removes* volatility whereas an option *reshapes* it.

Hedge ratio. You rarely offset one-for-one by notional. The right quantity is the number of derivative units whose value change matches the exposure's value change for a given move in the underlying. For a bond portfolio you scale by relative duration/DV01; for an equity portfolio you scale by beta; for an option-based hedge you scale by delta. Getting the ratio wrong is the difference between a hedge that neutralises the exposure and one that over- or under-shoots.

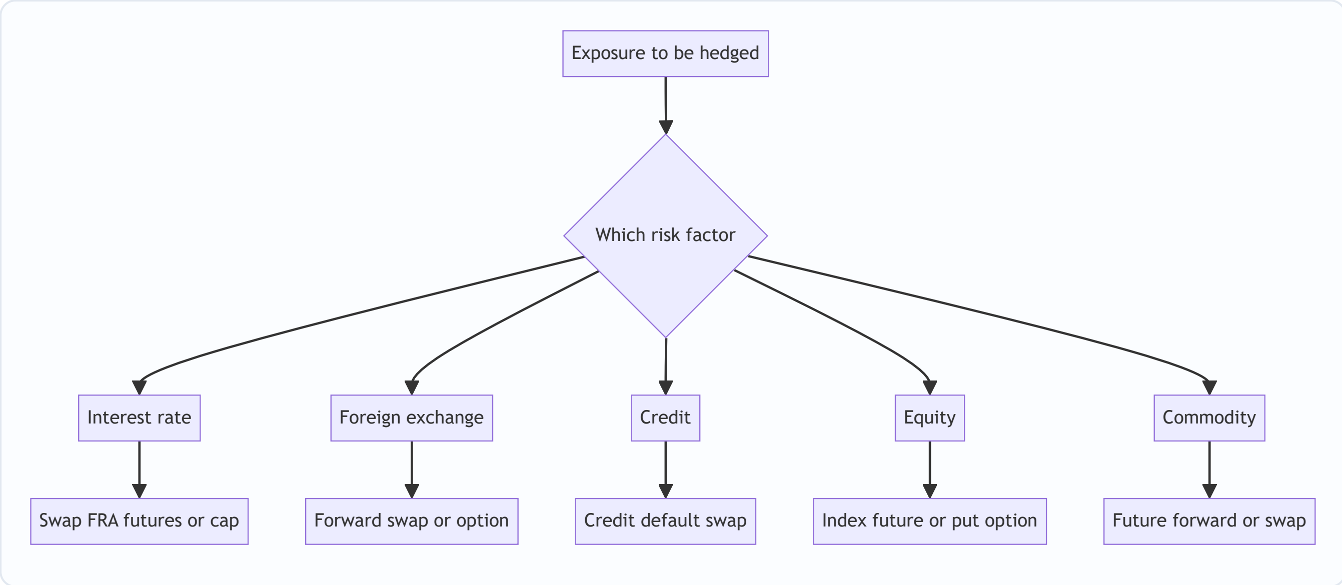

Figure 15.2 — The exposure's dominant risk factor selects the instrument family.

4. Full Content

4.1 The four building-block instruments

Almost every hedge is assembled from four primitives. Understanding these four — and precisely how their payoff profiles differ — lets you reason about any structured product.

Instrument	Obligation	Cost to enter	Payoff shape	Removes downside	Keeps upside
Forward	Both parties must transact at agreed price on a future date	Zero	Linear symmetric	Yes	No
Future	Same as forward but exchange-traded and margined daily	Margin only	Linear symmetric	Yes	No
Swap	Exchange one stream of cash flows for another over time	Zero	Linear symmetric	Yes	No
Option	Buyer has the right not the obligation to transact	Premium paid upfront	Non-linear kinked	Yes (if bought)	Yes

A **forward** is a bilateral, customisable contract to buy or sell an asset at a set price on a set date. It is precise and flexible but carries counterparty credit risk and is illiquid.

A **future** is a standardised forward traded on an exchange, marked to market daily through a clearing house that eliminates counterparty risk via margining. Standardisation buys liquidity but sacrifices exact matching (fixed contract sizes and delivery dates), which introduces basis risk.

A **swap** is a series of forwards bundled into one contract — an agreement to exchange cash flow streams. The interest-rate swap (fixed for floating) is the single most-used risk-management instrument in the world; currency swaps, and to a degree commodity swaps, extend the same idea.

An **option** confers a right without an obligation. A **call** is the right to buy; a **put** is the right to sell. The buyer pays a **premium** and thereby caps their loss at that premium while retaining unlimited favourable participation. The seller (writer) collects the premium and takes on the obligation — and potentially large losses. Options are insurance: you pay a fee to transfer the bad tail while keeping the good one.

4.2 Hedging interest-rate risk

Interest-rate risk is the sensitivity of earnings or asset values to changes in interest rates. A borrower on a floating-rate loan suffers when rates rise; a bond investor suffers when rates rise (prices fall); a bank with a duration mismatch suffers when the yield curve moves.

Instruments and their use:

- **Interest-rate swap (IRS).** A firm with floating-rate debt that fears rising rates enters a *pay-fixed, receive-floating* swap. The floating leg received cancels the floating interest paid on the loan, leaving the firm paying a net fixed rate. It has synthetically converted floating debt to fixed without renegotiating the loan.
- **Forward rate agreement (FRA).** Locks a single future borrowing rate for one period — a one-shot swaplet, useful for a known future funding need.
- **Interest-rate futures / bond futures.** Used to adjust portfolio duration quickly and liquidly. Selling bond futures shortens duration and protects a bond portfolio against rising yields.
- **Caps, floors and collars.** A **cap** is a series of interest-rate options that pays out when a floating rate exceeds a strike — insurance against rising rates while keeping the benefit of falling rates. A **floor** protects a lender against falling rates. A **collar** (buy a cap, sell a floor) finances the cap premium by giving up some downside benefit, often at zero net premium.

The instrument choice mirrors Part 2's certainty-versus-protection trade-off: a swap *locks* the rate; a cap *insures* it.

4.3 Hedging foreign-exchange risk

FX risk comes in three flavours: **transaction** exposure (a known future foreign-currency cash flow, like the exporter's receivable), **translation** exposure (the accounting value of foreign subsidiaries), and **economic** exposure (the effect of exchange rates on future competitiveness).

- **FX forward.** The workhorse for transaction exposure. Lock the conversion rate for a known amount on a known date. Zero cost, exact matching, but symmetric — no upside participation.
- **Currency futures.** Exchange-traded equivalent; liquid and margined but standardised, so basis risk on amount and date.
- **Currency swap.** Exchanges principal and interest in one currency for another over time. Ideal for a firm with foreign-currency debt or long-dated foreign cash-flow streams — it hedges both the interest and the principal re-exchange.
- **Currency options.** A firm bidding on a foreign contract it may not win faces *contingent* exposure — a forward would over-hedge because the underlying cash flow might not materialise. A put option on the foreign currency protects the rate if the bid wins while costing only the premium if it does not. This is the classic case where optionality is worth its price.

4.4 Hedging credit risk

Credit risk is the risk that a borrower or bond issuer defaults or deteriorates. The dedicated derivative is the **credit default swap (CDS)**.

A CDS is insurance on a reference entity's debt. The **protection buyer** pays a periodic premium (the *spread*, in basis points on notional); the **protection seller** pays out if a defined **credit event** (default, bankruptcy, restructuring) occurs, typically compensating for the loss of face value. A bank holding a large corporate loan can buy CDS protection on that borrower: if the borrower defaults, the loan loss is offset by the CDS payout. The bank keeps the client relationship and the loan on its books but sheds the default risk.

CDS also let investors *take* credit risk synthetically (by selling protection) or express negative views (by buying protection without owning the bond). That flexibility — the same feature that makes CDS a precise hedge — is exactly what turned them into an instrument of speculation and systemic contagion in 2008, the paradox Part 8 develops.

4.5 Hedging equity risk

Equity risk is exposure to falling share prices, at the level of a single stock or a whole portfolio.

- **Index futures.** A fund manager expecting a market decline, or wanting to reduce exposure without selling holdings (avoiding transaction costs and tax events), sells index futures. The futures gain offsets the portfolio's fall. The correct number of contracts is scaled by the portfolio's **beta** — a high-beta portfolio needs proportionally more index futures to hedge.
- **Protective put.** Buying a put on the index (or a stock) sets a floor under the portfolio value while retaining upside — insurance against a crash, at the cost of the premium.
- **Collar.** Buy a put and sell a call to fund it — protect against a fall while capping the gain, often at zero net cost. Widely used by concentrated holders (e.g. a founder with restricted stock) to protect wealth cheaply.

4.6 Hedging commodity risk

Commodity risk affects producers (who fear falling prices) and consumers (who fear rising prices).

- **Producer** — an oil producer or a farmer sells futures/forwards to lock in a selling price, protecting against a price collapse.
- **Consumer** — an airline (jet fuel), a food manufacturer (wheat, sugar), or a power utility (coal, gas) buys futures/forwards or swaps to lock in input costs.
- **Options** let a consumer cap input cost while benefiting if prices fall (buy a call), and let a producer set a price floor while keeping upside (buy a put).
- **Commodity swaps** exchange a floating commodity price for a fixed price over many periods — an airline can fix its fuel cost for two years without physically pre-buying fuel.

4.7 Matching the hedge to the exposure

The single most important design skill is *matching*. A hedge is only as good as the fidelity between the derivative and the exposure across five dimensions:

1. **Underlying** — hedge jet fuel exposure with a jet-fuel or closely correlated crude/heating-oil contract, not an unrelated commodity.
2. **Direction** — long exposure needs a short hedge and vice versa.
3. **Amount (notional)** — sized by the hedge ratio (duration, beta, delta), not naively one-for-one.
4. **Timing / maturity** — the derivative should expire when the exposure crystallises; a maturity mismatch forces rolling and creates roll risk.

5. **Reference / grade** — the index or grade underlying the derivative should match the priced exposure to minimise basis risk.

Any residual after matching is **basis risk** — the irreducible gap when a perfect instrument does not exist and a proxy must be used. A jet-fuel hedger using crude futures accepts the risk that the *crack spread* between crude and jet fuel moves. Basis risk is often unavoidable and is itself something to be measured and monitored, not ignored.

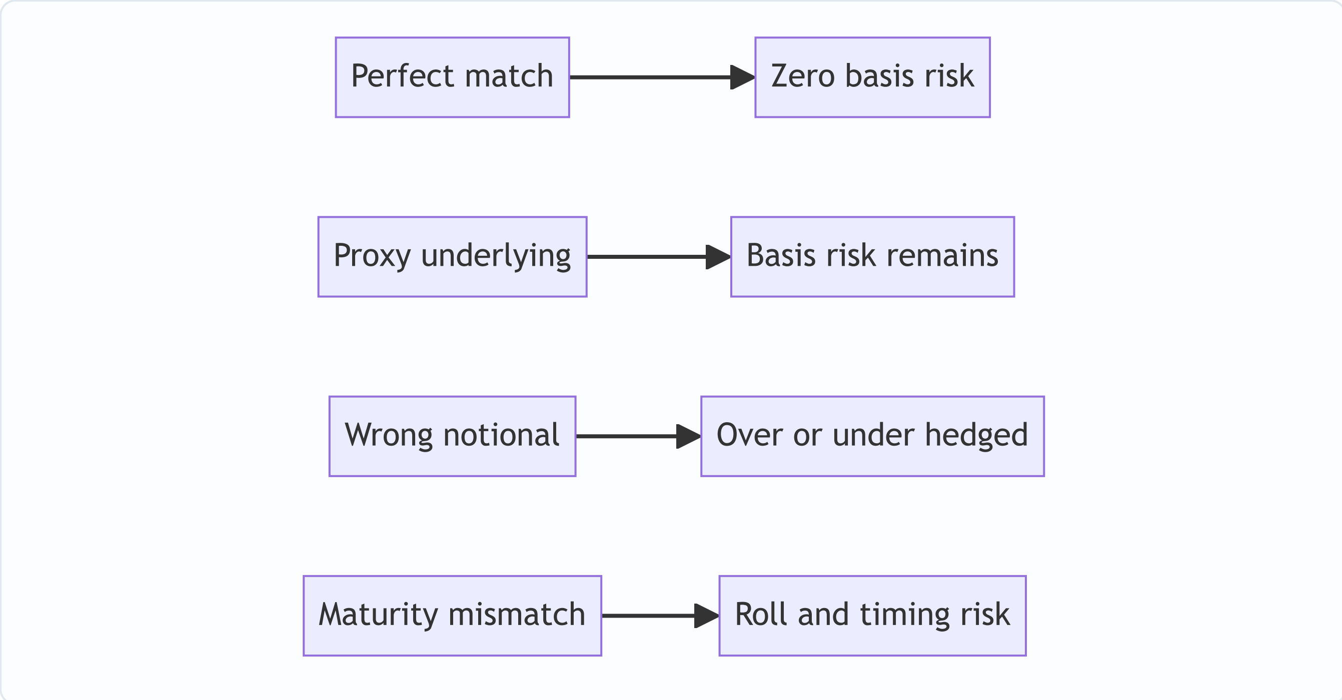

Figure 15.3 — Every dimension of mismatch between hedge and exposure leaves a specific residual risk.

4.8 The hedge-ratio mathematics

The hedge ratio answers "how many units?" and differs by risk factor:

- **Interest rate:** number of futures = (portfolio DV01) / (futures DV01), or scale by duration ratio. If a bond portfolio has a modified duration of 6 and a value of ₹100 crore, its DV01 is roughly ₹60,000 per basis point; you sell enough bond futures to match that sensitivity.
- **Equity:** number of index futures = (portfolio value × portfolio beta) / (index level × futures multiplier). Beta scaling is essential — a beta of 1.2 means the portfolio moves 20 percent more than the index, so you need 20 percent more futures than a naive value match implies.
- **Options:** delta hedge — the number of option contracts is scaled by delta (the sensitivity of option price to underlying). Because delta changes as the market moves (gamma), an option hedge must be **rebalanced dynamically**, unlike a static forward hedge.

4.9 Accounting and effectiveness

For a hedge to receive **hedge accounting** treatment (so the derivative's mark-to-market gains and losses are recognised alongside the hedged item rather than dumped into P&L each period), standards such as **IFRS 9** and **Ind AS 109** require the firm to *document* the hedge relationship at inception and to demonstrate it is **effective** — that the derivative and the exposure actually offset within a required tolerance. This is not mere paperwork; it forces the discipline of matching and ongoing monitoring, which is why it sits at the heart of Part 4.8 and Part 9.

5. Worked / Applied Examples

Example 1 — FX forward hedge for an exporter (self-verifying)

An Indian IT firm will receive **USD 1,000,000** in 90 days. Spot is ₹83.00/USD. The 90-day forward rate is **₹83.50/USD** (rupee at a forward discount, reflecting higher INR interest rates). The firm sells USD 1,000,000 forward at ₹83.50.

Locked-in receipt: 1,000,000 × 83.50 = **₹8,35,00,000**, regardless of where spot lands.

Scenario A — rupee strengthens to ₹80.00 at maturity. - Convert receivable at spot: 1,000,000 × 80.00 = ₹8,00,00,000. - Forward settlement gain: firm sells at 83.50, market is 80.00 → gain of (83.50 − 80.00) × 1,000,000 = ₹35,00,000. - **Total: 8,00,00,000 + 35,00,000 = ₹8,35,00,000.** ✓

Scenario B — rupee weakens to ₹86.00 at maturity. - Convert receivable at spot: 1,000,000 × 86.00 = ₹8,60,00,000. - Forward settlement loss: sells at 83.50, market is 86.00 → loss of (86.00 − 83.50) × 1,000,000 = ₹25,00,000. - **Total: 8,60,00,000 − 25,00,000 = ₹8,35,00,000.** ✓

Both scenarios reconcile to the locked ₹8.35 crore. The forward has removed *all* variability — including the ₹25 lakh windfall the firm forgoes in Scenario B. That forgone upside is the price of certainty and the reason a firm expecting favourable moves might instead buy a **USD put option**: it would keep the Scenario B windfall and only sacrifice the premium.

Example 2 — Interest-rate swap converting floating to fixed

A company has a **₹100 crore** loan at **MIBOR + 2.00%**, reset annually, and fears rising rates. It enters a 5-year **pay-fixed, receive-floating** swap: it pays **7.00%** fixed and receives **MIBOR** on ₹100 crore notional.

Net annual cost = loan interest − swap received + swap paid = (MIBOR + 2.00%) − MIBOR + 7.00% = **9.00% fixed**, independent of MIBOR.

Check at MIBOR = 6%: loan pays 8% (₹8 cr), swap receives 6% (₹6 cr), swap pays 7% (₹7 cr) → net 8 − 6 + 7 = **9% = ₹9 cr.** ✓ *Check at MIBOR = 9%:* loan pays 11% (₹11 cr), swap receives 9% (₹9 cr), swap pays 7% (₹7 cr) → net 11 − 9 + 7 = **9% = ₹9 cr.** ✓

The floating leg cancels and the firm has locked a 9% all-in cost. If rates fall it will regret the lock (it would have paid less floating) — the symmetric trade-off again.

Example 3 — Equity portfolio hedge with index futures (beta-scaled)

A fund holds **₹50 crore** of equities with **beta 1.25** against the Nifty. Nifty is at **20,000**; one futures contract is **50 units** (contract value = 50 × 20,000 = ₹10,00,000).

Number of contracts to fully hedge = (portfolio value × beta) / contract value = (50,00,00,000 × 1.25) / 10,00,000 = 62,50,00,000 / 10,00,000 = **625 contracts sold.**

Suppose the Nifty falls 10% to 18,000. - Portfolio, beta 1.25, falls ≈ 12.5% → loss ≈ 50 cr × 0.125 = **₹6.25 cr.** - Futures gain: index fell 2,000 points × 50 units × 625 contracts = 2,000 × 50 × 625 = **₹6.25 cr.** ✓

The hedge offsets the loss almost exactly. Note that had the manager ignored beta and sold only 500 contracts (naive value match), the futures gain would have been ₹5 cr against a ₹6.25 cr loss — a ₹1.25 cr shortfall. **Beta scaling is what makes the offset clean.**

Example 4 — Protective put versus forward (shape matters)

Return to the exporter but suppose it buys a **USD put** struck at ₹83.00 for a premium of **₹1.00/USD** (₹10,00,000 total) instead of selling forward.

Rupee strengthens to ₹80.00: exercise the put, sell dollars at 83.00 → receive 8,30,00,000, less ₹10,00,000 premium = **₹8,20,00,000**. (Forward would have given 8,35,00,000 — the option's floor is lower by the premium.)

Rupee weakens to ₹86.00: let the put lapse, convert at spot 86.00 = 8,60,00,000, less ₹10,00,000 premium = **₹8,50,00,000**. (Forward would have given only 8,35,00,000 — the option keeps ₹15,00,000 of upside net of premium.)

The option is *worse* when the market moves against you (floor 8.20 cr vs forward's 8.35 cr) but *better* when it moves in your favour (8.50 cr vs 8.35 cr). This is the essence of the linear-versus-optional choice: forwards buy certainty, options buy protection while renting upside.

6. Connections

To the rest of risk management. Derivatives are the *execution* layer of the risk-transfer decision framed in the risk-response chapters (accept, avoid, mitigate, transfer). They presuppose that the exposure has first been **identified and measured** — you cannot size a hedge ratio without knowing your DV01, beta, or delta, which ties this chapter directly to the measurement chapters on duration, VaR and sensitivity analysis.

To market-risk measurement. The residual left after hedging (basis risk, roll risk, un-hedged gamma) is precisely what feeds a portfolio's **Value at Risk**. A well-hedged book has low VaR; a hedge that has drifted out of effectiveness quietly raises VaR — which is why monitoring links this chapter to VaR and stress-testing.

To credit-risk management. The CDS section connects to counterparty risk: a derivative used to hedge introduces a *new* counterparty exposure (the swap or CDS dealer might default), which is why over-the-counter derivatives are increasingly **centrally cleared** and collateralised. Hedging one risk can import another.

To governance and control. The paradox in Part 8 connects to operational risk and internal control — Barings, LTCM, and 2008 were failures of *oversight of derivative positions*, not of the instruments themselves.

To accounting. Hedge effectiveness under Ind AS 109 / IFRS 9 links financial-reporting treatment to the economic quality of the hedge — good matching earns favourable accounting; sloppy matching produces earnings volatility.

7. Key Terms

- **Derivative** — a contract whose value derives from an underlying price, rate or event.
- **Underlying** — the reference asset or variable (rate, FX rate, index, commodity, credit) a derivative is written on.
- **Notional** — the face amount on which a derivative's payments are calculated; usually not exchanged (except currency swaps).
- **Forward / Future** — obligation to transact at a set price on a future date; OTC (forward) or exchange-traded and margined (future).

- **Swap** — exchange of two cash-flow streams over time; the interest-rate swap (fixed-for-floating) is the archetype.
- **Option / Call / Put / Premium / Strike** — right without obligation to buy (call) or sell (put) at the strike price; the buyer pays a premium.
- **Hedge ratio** — the quantity of derivative per unit of exposure that equalises value sensitivity; scaled by duration, beta or delta.
- **Basis risk** — residual risk when the hedge instrument and the exposure do not move perfectly together.
- **Roll risk** — the risk incurred when a shorter-dated hedge must be rolled forward to cover a longer exposure.
- **Delta / Gamma** — an option's sensitivity to the underlying (delta) and the rate of change of that sensitivity (gamma), driving dynamic rebalancing.
- **CDS (Credit Default Swap)** — a contract paying out on a reference entity's credit event in exchange for a periodic spread.
- **Cap / Floor / Collar** — interest-rate (or price) option structures that set a ceiling, a floor, or a range.
- **Hedge effectiveness** — the degree to which the derivative's value changes offset the hedged item's, required for hedge accounting.
- **Speculation** — using a derivative to *take* a directional position rather than offset an existing one.

8. Common Confusions

"Hedging eliminates risk." No — hedging *transfers* the targeted risk and leaves behind basis risk, roll risk, and counterparty risk. It swaps a large, unwanted price risk for a smaller set of residual risks. A "perfect hedge" is rare.

"A hedge that loses money was a bad hedge." The most common and most damaging confusion. If the exposure gained what the derivative lost, the hedge did its job — it produced *certainty*, which is the whole point. Judging a hedge by the standalone P&L of the derivative leg, ignoring the offsetting exposure, is exactly the error that leads managers to abandon hedges at the worst moment. A hedge is evaluated on the *combined* position.

"Forwards and options are interchangeable." They are not. Forwards remove upside and downside symmetrically at zero cost; options remove only the downside but cost a premium. Choosing between them is a deliberate decision about whether you want certainty or asymmetric protection.

"Derivatives are inherently dangerous / are inherently safe." Both wrong. The identical instrument is a hedge or a bomb depending on whether it *offsets* or *adds to* an existing exposure. A pay-fixed swap on top of floating debt is a hedge; the same swap with no underlying loan is a naked bet on rates.

"More hedging is always better." Over-hedging (a hedge ratio above 1, or hedging exposures you do not have) converts a hedger into a speculator in the opposite direction. The goal is to *match* the exposure, not to maximise the derivative position.

"Margin and premium are the cost of the hedge." Margin is a returnable performance deposit, not a cost; an option premium *is* a sunk cost. Confusing the two mis-estimates the economics of the hedge.

9. Recap

Derivatives are the primary, precision tools of risk management because they let a firm keep its underlying business while surgically removing a specific price risk. The mechanism is **offset**: pair an exposure with a derivative on the same underlying, sized by the correct **hedge ratio**, so their value changes cancel. Four building blocks do almost all the work — **forwards, futures, swaps** (linear, symmetric, certainty-creating, near-zero cost) and **options** (non-linear, asymmetric, protection-with-upside, premium-costing).

Each risk factor has its natural instruments: **swaps, FRAs, futures, caps and collars** for interest rates; **forwards, currency swaps and options** for FX; **credit default swaps** for credit; **index futures and protective puts** for equity; **futures, swaps and options** for commodities. The quality of any hedge depends on **matching** across underlying, direction, notional, timing and grade — and whatever cannot be matched perfectly remains as **basis risk**.

The defining paradox is that the same leverage and flexibility that make derivatives ideal for reducing risk make them ideal for *creating* risk when used to speculate rather than offset — which is why documentation, hedge-ratio discipline, effectiveness testing and continuous monitoring are inseparable from hedging itself. A hedge is not "set and forget": beta drifts, delta moves, correlations break, and positions roll — so effectiveness must be watched over the life of the exposure.

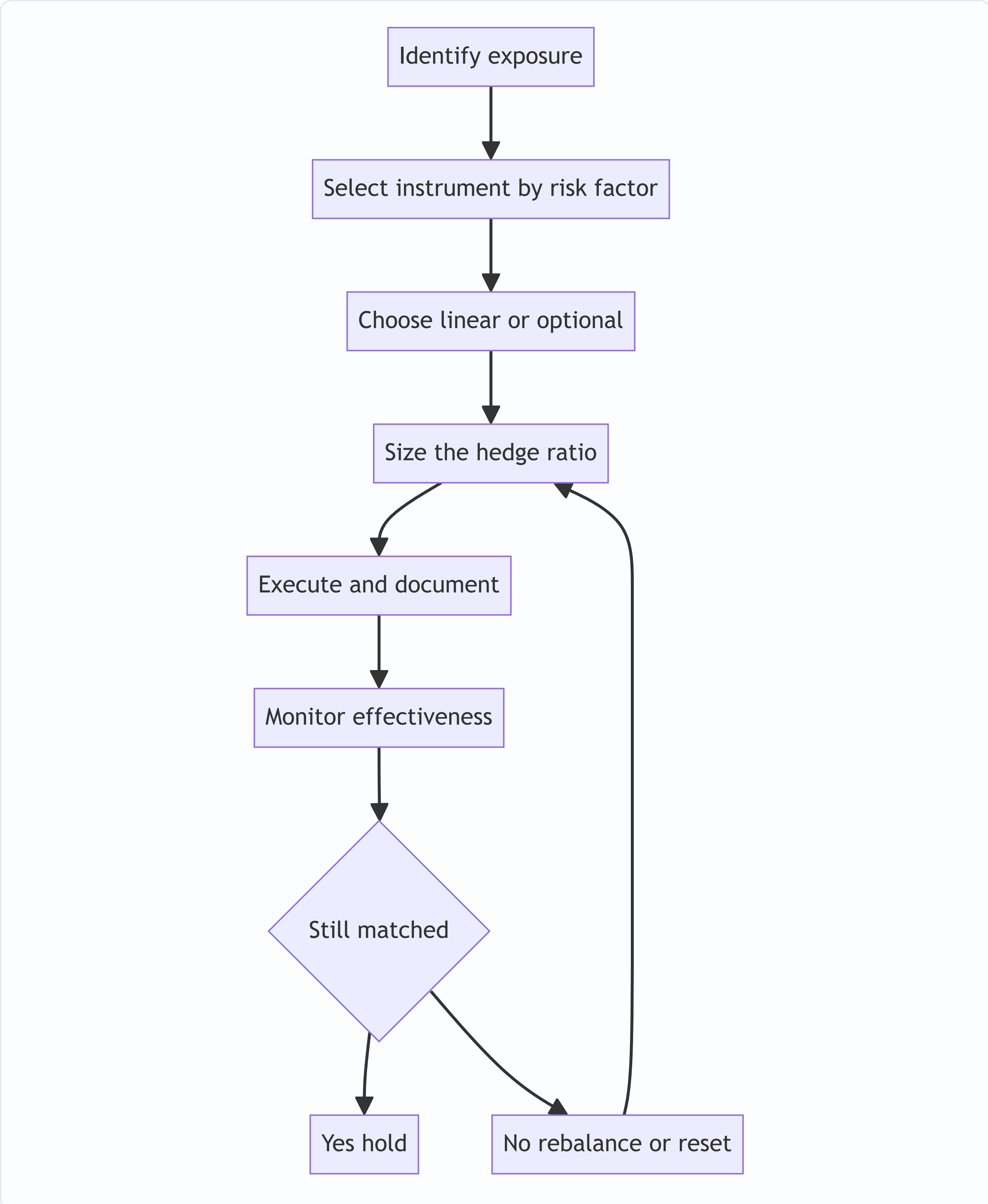

Figure 15.4 — Hedging is a continuous loop not a one-off trade with monitoring feeding back into re-sizing.

10. Quick-Reference / Interview Points

One-line definition to lead with: "A derivative hedges by creating a payoff that moves opposite to an existing exposure on the same underlying, so gains on one leg cancel losses on the other."

Instrument-to-risk cheat sheet:

Risk	Lock it in (linear)	Insure it (option)
Interest rate	Swap, FRA, bond future	Cap, floor, collar
Foreign exchange	Forward, currency swap	Currency option
Credit	(buy) CDS protection	—
Equity	Index future	Protective put, collar
Commodity	Future, forward, commodity swap	Commodity call/put

Interview-ready talking points:

- *The paradox in one sentence:* the leverage that lets you hedge a large exposure with little capital is the same leverage that turns a speculative position into a catastrophic loss — the instrument is neutral; the presence or absence of an offsetting exposure decides whether it hedges or gambles.
- *Judge a hedge on the combined position, never the derivative alone.* A hedge that "lost money" while the exposure gained is a hedge that worked.
- *Hedge ratio drivers:* duration/DV01 for rates, beta for equity, delta for options — and delta hedges must be rebalanced because of gamma.
- *Forward vs option:* forward = zero cost, symmetric, removes upside and downside; option = premium cost, asymmetric, keeps upside. Use options for *contingent* exposures (uncertain bids, uncertain volumes) where a forward would over-hedge.
- *Basis risk* is the residual when you cannot match perfectly (jet fuel hedged with crude); measure and monitor it — it does not vanish.
- *Hedging imports counterparty risk*, mitigated by exchange clearing, margining and collateral (CSAs); OTC hedges add a credit exposure to the dealer.
- *Effectiveness and accounting:* IFRS 9 / Ind AS 109 require documented, effective hedge relationships for hedge accounting — the economic discipline and the accounting treatment reinforce each other.
- *Classic cautionary cases:* Barings (Leeson's un-hedged futures), LTCM (leverage plus correlation breakdown), Metallgesellschaft (roll/basis and margin-funding mismatch on a long-dated oil hedge), and 2008 CDS on subprime — each a failure of *use and oversight*, not of the instruments.

The single most important sentence: match the hedge to the exposure and monitor that the match holds — a derivative reduces risk only for as long as it keeps offsetting the thing it was bought to offset.

Q&A — Using Derivatives to Manage Risk

Practice bank for Chapter 15 (Using Derivatives to Manage Risk). Every question is followed by a full answer. Attempt each before reading the solution. Numerical answers are reconciled at least two independent ways where possible. Conventions: long = agree to buy / benefits when price rises; a "hedge" offsets an existing exposure, "speculation" creates a new one.

Section A — Concept Check

A1. In one sentence, what is a derivative, and what is the single feature that makes it useful for risk management? A derivative is a contract whose value is *derived* from an underlying variable (a price, rate, index or event) rather than from any intrinsic worth of its own. The feature that makes it a risk tool is **leverage without ownership**: it lets you take on, or shed, exposure to the underlying with little or no upfront capital, so you can offset an existing exposure precisely instead of buying or selling the physical asset.

A2. State the one-line difference between a forward, a future, and an option. A **forward** is a private, customised agreement to trade at a set price on a future date — both sides are obligated. A **future** is the same economics but exchange-traded, standardised, and daily margined. An **option** gives the holder the *right but not the obligation* to trade — obligation is one-sided (the writer's), and the holder pays a premium for that asymmetry.

A3. Why does hedging with a forward or future eliminate downside but *also* remove upside, while an option does not? A forward/future locks a price: whatever the market does, you transact at the agreed level, so a favourable move is surrendered along with the unfavourable one — the payoff is *symmetric* and linear. An option only pays out on one side; you keep the good outcome and cap the bad one, because you can simply let the option lapse. That asymmetry is exactly what the premium buys.

A4. Explain the distinction between hedging, speculation, and arbitrage using derivatives. Hedging uses a derivative to *offset* a pre-existing exposure, reducing net risk. **Speculation** uses a derivative to *take* a directional view with no offsetting position, increasing risk in exchange for expected return. **Arbitrage** exploits a price discrepancy between a derivative and its underlying (or between two derivatives) to lock a riskless profit. The same instrument serves all three; intent and existing exposure decide which it is.

A5. Define the hedge ratio and explain why it is rarely exactly 1. The hedge ratio is the number of units of the hedging instrument per unit of exposure being hedged. It departs from 1 because the hedge and the exposure differ in sensitivity: the futures contract may cover a different quantity (contract-size mismatch), the underlying may differ (cross-hedge), or the price relationship may be less than one-for-one (a beta below/above 1 for equities, a duration mismatch for bonds). The minimum-variance hedge ratio is $h^* = \rho\,\sigma_S/\sigma_F$.

A6. What is basis risk, and why does it survive even a "perfect-sized" futures hedge? Basis = spot price − futures price. Basis risk is the risk that this gap moves unpredictably before the hedge is unwound. It survives because a futures hedge fixes the *futures* price you transact at, not the *spot* you actually face; unless spot and futures converge exactly at your horizon (same asset, same date, same location/grade), the residual basis changes your net outcome. It converts absolute price risk into the much smaller — but non-zero — risk of the basis.

A7. A firm buys a put option on an asset it owns. What insurance analogy applies, and what are the "premium" and "deductible"? It is a **protective put** — portfolio insurance. The option **premium** is literally the insurance premium; the gap between the current price and the strike is the **deductible** (the loss you bear before protection kicks in). Below the strike you are made whole (less the premium); above it you keep all the upside and have simply spent the premium.

A8. Why might a treasurer prefer a collar to a plain protective put? A collar buys a protective put *and* sells a call to fund it, so the premium is reduced or zeroed. In return the firm gives up upside above the call strike. A treasurer who wants cheap downside protection and is willing to cap gains — a common trade-off when the priority is certainty, not speculation — prefers the collar's lower (often zero) cost.

A9. What does an interest-rate swap let a borrower do, in one sentence, and what risk does it convert? A plain-vanilla interest-rate swap lets a borrower exchange floating-rate cash flows for fixed (or vice versa) on a notional principal, converting **interest-rate (cash-flow) risk** — the uncertainty of future floating payments — into certainty, without refinancing the underlying loan.

A10. Why is the notional principal in a swap "notional" — and what actually gets exchanged? The notional is never exchanged; it is only the reference figure used to *calculate* the interest payments. What changes hands is the **net difference** between the fixed and floating legs on each settlement date. That is why a large swap can carry huge notional but modest actual cash flows and credit exposure.

A11. How does a currency forward hedge a foreign-currency receivable, and what is the cost of the hedge? An exporter expecting foreign currency in three months sells that currency forward today, locking the domestic-currency amount it will receive regardless of spot moves. The "cost" is not a premium (forwards have zero initial value) but the **forward points** — the difference between spot and forward rates, set by the interest-rate differential (covered interest parity). You may end up worse than the eventual spot; that surrendered upside is the price of certainty.

A12. What is a credit default swap (CDS), and how does a lender use it to manage risk? A CDS is a contract in which the protection buyer pays a periodic premium (the spread) and receives a payout if a reference entity defaults. A lender holding a bond or loan buys CDS protection to **transfer the credit risk** to the seller while keeping the asset — effectively insurance against default — for the cost of the ongoing spread.

A13. Distinguish exchange-traded from OTC derivatives on three axes. Customisation: OTC is bespoke, exchange-traded is standardised. **Counterparty risk**: OTC faces the counterparty directly (mitigated by CSAs/collateral); exchange-traded is novated to a central clearing house that guarantees performance. **Liquidity/transparency**: exchange contracts are liquid and price-transparent; OTC can be illiquid and opaque. The trade-off is precision (OTC) versus safety and liquidity (exchange).

A14. Why can a derivative hedge *increase* risk if mismatched or over-sized? If the notional exceeds the exposure, the "hedge" becomes a net directional bet in the opposite direction — over-hedging speculates. If the underlying, maturity, or sensitivity is wrong (a bad cross-hedge), the derivative may move independently of the exposure and add variance rather than remove it. A hedge only reduces risk when its payoff is genuinely negatively correlated with, and correctly scaled to, the exposure.

A15. State the four principal option Greeks and what each measures in one phrase. Delta — sensitivity of option value to the underlying price. **Gamma** — rate of change of delta (curvature). **Vega** — sensitivity to implied volatility. **Theta** — sensitivity to the passage of time (decay). Rho (rate sensitivity) is the common fifth.

Section B — Numerical / Applied

B1 — Futures hedge and basis risk

Setup. A refiner will buy 1,000,000 litres of crude in three months. Spot today is ₹60/litre; the 3-month future is ₹62/litre (one contract = 100,000 litres). It hedges by going long 10 futures. At delivery, spot = ₹70 and the future it sells to close = ₹71.

Step 1 — Physical cost. Buys 1,000,000 × ₹70 = ₹7,00,00,000. **Step 2 — Futures gain.** Long at 62, close at 71 → gain ₹9/litre × 1,000,000 = ₹90,00,000. **Step 3 — Net effective cost.** ₹7,00,00,000 − ₹90,00,000 = ₹6,10,00,000 → **₹61.00/litre. Cross-check via basis.** Effective price = futures entry (₹62) + change in basis. Initial basis (spot − fut) = 60 − 62 = −2; final basis = 70 − 71 = −1. Basis rose by ₹1. Effective price = 62 + (−1) = **₹61**, matching. Had basis been unchanged at −2, the firm would have locked exactly ₹62; the ₹1 improvement is pure basis risk — it helped here, but could equally have hurt.

B2 — Minimum-variance hedge ratio and contract count

Setup. A fund holds ₹50 crore of equity with portfolio beta 1.20 relative to the Nifty. Nifty future = 20,000 index points; lot size = 50; so one contract notionally covers 20,000 × 50 = ₹10,00,000.

Step 1 — Beta-adjusted exposure. Effective exposure to hedge = ₹50 crore × 1.20 = ₹60 crore. **Step 2 — Number of contracts.** N = 60,00,00,000 ÷ 10,00,000 = **600 contracts, sold short. Check.** If Nifty falls 10%, portfolio (β 1.2) falls ≈ ₹50cr × 12% = ₹6 crore. Futures: index drops 2,000 pts × 50 × 600 = ₹6 crore gain. Offsetting — the hedge neutralises systematic risk. Residual (idiosyncratic) risk remains, which is why a single-index hedge is never perfect for a specific portfolio.

B3 — Protective put payoff

Setup. An investor owns a share at ₹100 and buys a 3-month put, strike ₹95, premium ₹3.

Case price = ₹80. Stock loss = 100 − 80 = ₹20. Put pays 95 − 80 = ₹15. Net position value = 80 + 15 − 3 = **₹92.** Maximum loss is fixed: 100 − 95 + 3 = ₹8, whatever the price below 95. **Case price = ₹120.** Put expires worthless. Net = 120 − 3 = **₹117.** Upside kept, less the ₹3 premium. **Break-even on the upside** = 100 + 3 = ₹103. Interpretation: the put converts an open-ended downside into a capped ₹8 loss, in exchange for a ₹3 drag on gains — textbook insurance economics.

B4 — Zero-cost collar

Setup. Same share at ₹100. Buy put strike 95 for ₹3; sell call strike 110 for ₹3. Net premium = 0.

Below 95: protected — floor value 95 (net of zero premium). **Between 95 and 110:** you ride the stock one-for-one. **Above 110:** call is exercised against you; upside capped at 110. So the outcome is bounded to the corridor **[95, 110] at zero cost.** The price of free protection is the surrendered gains above 110 — a fair trade only if you value certainty over the tail upside.

B5 — Interest-rate swap converting floating to fixed

Setup. A firm has a ₹100 crore loan at floating MIBOR + 1.5%. It enters a swap to **pay fixed 7%, receive MIBOR** on ₹100 crore notional.

Net cost after swap. Loan pays (MIBOR + 1.5%); swap: receives MIBOR, pays 7%. Combined = −(MIBOR + 1.5%) + MIBOR − 7% = **−8.5% fixed.** MIBOR cancels. **Check at MIBOR = 6%:** loan = 7.5%, swap net = pay

7% − receive 6% = pay 1% → total 8.5%. At MIBOR = 9%: loan = 10.5%, swap net = pay 7% − receive 9% = receive 2% → total 8.5%. Rate risk is fully removed; the firm now pays a certain 8.5% regardless of MIBOR.

B6 — Currency forward on a receivable

Setup. An Indian exporter will receive \$1,000,000 in 90 days. Spot = ₹83.00/\$; 90-day forward = ₹83.60/\$. It sells the dollars forward.

Locked receipt = 1,000,000 × 83.60 = **₹8,36,00,000**, guaranteed. **If spot at settlement = ₹81:** unhedged would have got ₹8,10,00,000 → hedge saved ₹26,00,000. **If spot = ₹85:** unhedged would have got ₹8,50,00,000 → hedge cost ₹14,00,000 of upside. The forward removes all FX uncertainty; whether it "wins" ex post is irrelevant to its risk-management purpose — it delivered certainty, which was the objective.

B7 — Delta hedging an option book

Setup. A desk is short 100 call contracts (100 shares each = 10,000 shares) with delta 0.40.

Step 1 — Net delta. Short calls → position delta = −0.40 × 10,000 = **−4,000.** The book loses if the stock rises. **Step 2 — Hedge.** Buy 4,000 shares to bring net delta to zero. **Step 3 — Why it is not "done".** As the stock rises, gamma raises the call's delta toward, say, 0.55, so position delta becomes −5,500 while the hedge still holds only 4,000 shares → net −1,500; the desk must buy 1,500 more shares. Delta-neutral is a one-instant state; gamma forces continuous re-hedging, and each re-hedge is "buy high / sell low," which is the cost that the option premium (theta) compensates.

Section C — Interview-Style

C1. "Your CFO says derivatives are dangerous and wants to ban them. How do you respond?" I'd separate the instrument from its use. Derivatives are dangerous only when used to *take* risk without an offsetting exposure, or when sized wrongly. Used as intended — a forward on a known receivable, a swap on a floating loan, a put on a held asset — they *reduce* the firm's risk and smooth cash flows, which lowers the cost of capital and protects covenants. The famous blow-ups (Barings, Amaranth, LTCM) were speculation or leverage failures, not hedging failures. The right response is a policy: hedge only identified exposures, cap notionals to underlying quantities, mark to market daily, and separate the dealing desk from settlement. Banning hedging would leave the firm *more* exposed, not less.

C2. "When would you hedge with options rather than futures, and what do you pay for that choice?" Options when the exposure is *uncertain* or when I want to keep upside. If a bid I've tendered might not be won, a forward would over-hedge and create a naked position if the bid fails — an option lets me walk away. Options also suit asymmetric views: protect the downside, keep the rally. The cost is the premium — a certain cash outflow versus a forward's zero upfront cost. So the decision is: pay a known premium for optionality and retained upside (options), or lock a price for free but surrender the upside (futures/forwards). Contingent exposures and convex payoffs argue for options; certain, symmetric exposures argue for forwards.

C3. "Walk me through the risks that *remain* after you put on a 'perfect' futures hedge." Four residuals. **Basis risk** — spot and futures may not converge at my horizon. **Rollover risk** — if my exposure outlasts the liquid contract, I must roll and face uncertain roll costs. **Margin/liquidity risk** — daily variation margin can demand large cash even when the hedge is "working," which sank Metallgesellschaft. **Correlation/quantity risk** — in a cross-hedge or when the hedged quantity itself is uncertain (a farmer's crop yield), the offset is imperfect. A hedge doesn't make risk vanish; it *transforms* price risk into these smaller, more manageable residuals.

C4. "A trader shows a huge notional swap position but says credit exposure is small. Is that possible?" Yes, and it's the norm. Notional is only the reference for computing payments — it is never exchanged. The actual credit exposure on a swap is the *replacement cost* if the counterparty defaults, i.e. the current mark-to-market of the net future cash flows, which is a small fraction of notional and can be near zero at inception. Add potential future exposure for how it might grow, then net across trades and subtract collateral held under the CSA. So a ₹1,000 crore notional swap might carry only a few crore of current exposure. The caveat: that exposure is *live* and grows with rate moves, so it must be monitored, not dismissed.

C5. "How do you decide the hedge ratio for a bond portfolio using interest-rate futures?" I match *dollar duration*, not face value. Compute the portfolio's DV01 (or PV01), compute the DV01 of one futures contract via the cheapest-to-deliver bond's DV01 divided by its conversion factor, then N = portfolio DV01 ÷ futures DV01. If I want to *reduce* duration to a target rather than zero it, I hedge the DV01 gap. I'd then stress the hedge for a non-parallel curve shift, because a duration hedge only neutralises parallel moves — twists and butterflies leave residual risk that may need key-rate or multiple-maturity futures.

C6. "Explain to a non-finance board member why a zero-cost collar isn't actually free." Nothing is free; "zero-cost" only means no cash changes hands upfront. We buy downside protection and pay for it by *selling away* our upside above a ceiling. If the asset soars, we forgo those gains — that surrendered profit is the real cost, paid in the future and only if things go well. So it's the right tool when we value certainty and don't need the home-run outcome, but the board should understand we've traded the top of the range for protection on the bottom.

Section D — MCQs (with reasoning)

D1. A wheat farmer expecting a harvest in three months is most naturally hedged by: (a) buying wheat futures (b) selling wheat futures (c) buying a wheat call (d) selling a wheat put **Answer: (b).** The farmer is *long* physical wheat (will own it) and fears a price *fall*. Selling futures locks the sale price. Buying futures (a) would double the long exposure — speculation. A call (c) protects against a rise, the wrong direction. Selling a put (d) adds downside risk for premium — the opposite of a hedge.

D2. Basis risk in a futures hedge is best described as the risk that: (a) the futures exchange defaults (b) the spot–futures gap changes before the hedge is closed (c) volatility rises (d) the option expires worthless **Answer: (b).** Basis = spot − futures; basis risk is uncertainty in that gap at the horizon. (a) is clearing/counterparty risk, essentially removed by the CCP. (c) is vega, an options concept. (d) applies to options, not futures.

D3. The notional principal of an interest-rate swap: (a) is exchanged at maturity (b) is exchanged at inception and maturity (c) is never exchanged and only scales the interest calculation (d) equals the credit exposure **Answer: (c).** Only net interest differences change hands; the notional is a reference figure. (a)/(b) describe cross-currency principal exchange, not a single-currency IRS. (d) is false — credit exposure is the mark-to-market replacement cost, far smaller than notional.

D4. Which position has a symmetric (linear) payoff? (a) long call (b) long put (c) long forward (d) protective put **Answer: (c).** A forward gains and loses one-for-one with the underlying — symmetric. Calls (a), puts (b), and the protective put (d) all have kinked, asymmetric payoffs because of the option's one-sided exercise.

D5. A firm sells a call against stock it owns (a covered call). Its main risk is: (a) unlimited loss if the stock falls (b) forgone gains if the stock rises above the strike (c) margin calls on the stock (d) losing the premium **Answer: (b).** The covered call keeps the premium and full downside exposure of the stock but caps upside at the strike; the real "cost" is the surrendered rally. (a) is wrong — the loss is bounded by the stock going to zero, cushioned by the premium. (d) is wrong — the writer *receives* the premium.

D6. Over-hedging (notional greater than exposure) causes the hedge to: (a) remain a perfect hedge (b) become a net speculative position in the opposite direction (c) eliminate basis risk (d) reduce margin requirements **Answer: (b).** Beyond the exposure, the extra notional is unmatched and creates fresh directional risk opposite to the original — speculation. It neither removes basis risk (c) nor lowers margin (d).

D7. A lender wanting to keep a loan on its books but shed the default risk should: (a) sell the loan (b) buy CDS protection on the borrower (c) sell CDS protection (d) buy a call on the borrower's stock **Answer: (b).** Buying CDS transfers default risk to the seller while the lender retains the asset and relationship. Selling the loan (a) removes the asset entirely; selling CDS (c) *adds* credit risk; an equity call (d) is a directional bet, not credit protection.

D8. Delta-hedging a short option position requires continuous re-hedging chiefly because of: (a) theta (b) rho (c) gamma (d) the risk-free rate **Answer: (c).** Gamma makes delta change as the underlying moves, so a static share hedge drifts out of neutrality and must be rebalanced. Theta (a) is time decay — the P&L cost, not the trigger for re-hedging. Rho (b) and the rate (d) are second-order for equity option hedging.

D9. Compared with an OTC forward, an exchange-traded future primarily reduces: (a) basis risk (b) counterparty (default) risk (c) market risk (d) the need to hedge **Answer: (b).** Central clearing and daily margining substitute the clearing house's guarantee for direct counterparty exposure. Standardisation can actually *increase* basis risk (a) relative to a bespoke forward; market risk (c) is unchanged; the need to hedge (d) is unaffected.

D10. A treasurer buys a floor on a floating-rate *asset* (an investment earning MIBOR). This protects against: (a) rates rising (b) rates falling below the floor strike (c) the counterparty defaulting (d) currency moves **Answer: (b).** A floor pays when the reference rate drops below the strike, protecting the income on a floating-rate *asset* from falling rates. Rising rates (a) would be a *cap*, which protects a floating-rate *borrower*. (c)/(d) are unrelated risk types.

End of practice bank. If you can explain why each hedge removes risk — which exposure it offsets, in which direction, and what residual (basis, gamma, roll, margin) survives — you understand the chapter, not just the formulas.

Chapter 16 — Model Risk

1. The Problem / The Need

Walk the floor of any modern financial institution and you will find that almost nothing is decided by human judgement alone. A trading desk prices a five-year interest-rate swap with a curve-stripping model. A retail bank approves a home loan using a credit-scoring model. The treasury reports a firm-wide 99% Value at Risk from a market-risk model. The finance team books the fair value of an illiquid structured note from a valuation model. Regulatory capital itself — the cushion that decides whether the bank is deemed safe — is computed by internal models blessed under Basel. The entire machinery of risk, pricing, and capital runs on models.

A **model** is any quantitative method that takes inputs, applies theories and assumptions, and produces an output used to make a decision. That output arrives on a screen as a crisp number: "fair value ₹9.84 crore", "PD 1.7%", "1-day VaR ₹46.6 lakh", "capital requirement ₹212 crore". The number *looks* like a fact. It has decimals. It came from mathematics and a computer, so it carries an aura of objectivity.

But every one of those numbers is the end of a long chain of choices: which theory to use, which assumptions to make, which data to feed in, how to code the mathematics, and how to interpret the result. Break any link in that chain and the number is wrong — yet it will still be displayed to the same number of decimal places, with the same air of authority. A trader will hedge against a wrong price. A committee will lend against a wrong PD. A board will believe it is safe because a wrong VaR said so. The danger is not that the model is uncertain; it is that a **confidently wrong number is trusted as if it were right.**

This is the problem **model risk** addresses: the risk of adverse consequences — financial loss, bad decisions, reputational or regulatory damage — from **decisions based on models that are incorrect or misused.** It is the risk hiding inside every other risk measure in this book, because all of those measures are themselves model outputs.

The core motivation: a model output is an opinion dressed as a fact, and the risk is that we forget the difference.

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2. The Core Idea

The US Federal Reserve's supervisory guidance **SR 11-7** (2011) — the single most important document on this topic and a near-certain interview reference — defines model risk with beautiful economy as:

> *"The potential for adverse consequences from decisions based on incorrect or misused model outputs and reports."*

Unpack that sentence and you find the whole subject. Model risk has exactly **two roots**:

1. **The model is fundamentally wrong** — it produces inaccurate outputs even when used exactly as intended. The map does not match the territory.
2. **The model is used wrong** — it may be perfectly sound for its designed purpose, but it is applied to the wrong problem, fed the wrong inputs, or its output is misinterpreted.

A model can be right and still cause a disaster (misuse). A model can be used impeccably and still cause a disaster (it was wrong). Both are model risk.

The founding intellectual anchor is the statistician **George Box's** aphorism: **"All models are wrong, but some are useful."** A model is by definition a *simplification* of reality. It throws away detail to become tractable. The Black-Scholes option model assumes constant volatility, no jumps, and continuous trading — all false. A credit-scoring model compresses a human being's financial life into a few dozen variables. This simplification is not a bug; it is the entire point. A map that reproduced every detail of the territory would be useless.

So the goal of model risk management is **not** to build a "correct" model — that is impossible. The goal is to:

- **know how the model is wrong** (where its assumptions bite),
- **quantify the consequences** of being wrong, and
- **prevent the model from being trusted beyond the range where it is useful.**

Model risk is therefore fundamentally a discipline of **humility and boundaries**. Every model has a domain of validity — a set of conditions under which its simplifications are harmless — and the entire craft is knowing where that domain ends.

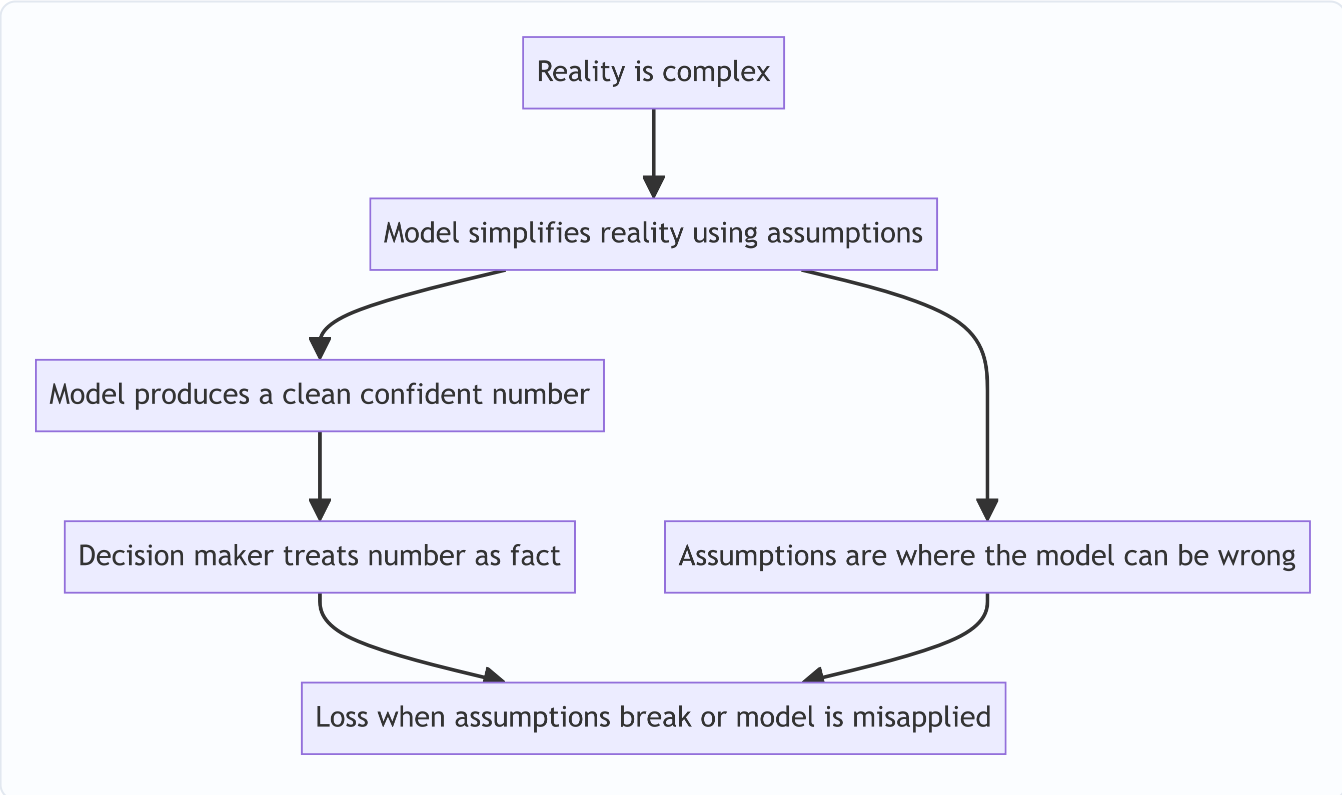

Figure 16.1 — Model risk is born the moment a simplified assumption is trusted as if it were reality.

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3. Why / How It Works — Where Model Risk Comes From

To manage model risk you must know where it enters. It is useful to trace a model's life as a pipeline and see that **each stage introduces its own kind of error.** SR 11-7 groups these into two broad categories — *the model itself* and *the use of the model* — but in practice risk analysts point to four concrete **sources**.

Source 1 — Bad assumptions (specification error)

The model's underlying theory or structure does not match how the world actually behaves. This is the deepest and most dangerous source because it is invisible in the code and the data — the mathematics can be flawless and the answer still wrong.

Classic examples: - Assuming asset returns are **normally distributed** when real returns have **fat tails** (extreme moves are far more common than the bell curve predicts). This single assumption underlies most VaR failures. - Assuming **correlations are constant**, when in a crisis "all correlations go to one" — diversification vanishes exactly when you need it. - Assuming **continuous liquid markets** with no gaps, when real prices jump and markets freeze. - Assuming the **future resembles the past** — that a relationship estimated on historical data will keep holding.

Source 2 — Bad data (input and calibration error)

Even a perfectly specified model is only as good as what you feed it. "Garbage in, garbage out." Data problems include: - **Poor quality**: errors, gaps, stale prices, wrong sign conventions. - **Insufficient history**: a model calibrated only on the calm 2003–2006 period never "saw" a housing crash, so it assigns it near-zero probability. - **Unrepresentative sample**: data drawn from conditions unlike those the model will face (a **regime change** the data does not contain). - **Proxying**: an illiquid instrument has no price history, so a proxy is used — and the proxy behaves differently in stress.

Source 3 — Bad implementation (coding and technical error)

The theory is sound and the data is clean, but the model is built wrong. This is the most mundane and most common source, and it is pure operational risk wearing a quant costume: - Programming bugs — an off-by-one error, a wrong sign, a mis-transcribed formula. - Numerical issues — approximation errors, non-convergence, rounding. - The **spreadsheet risk** that famously produced the "London Whale" and the Reinhart-Rogoff error: a formula that averaged the wrong range of cells. - Wrong linking of systems — the model receives yesterday's rates, or notionals in the wrong currency.

Source 4 — Misuse (application error)

The model is correct and correctly built, but it is used outside the boundary where it is valid: - **Wrong context**: a model calibrated for liquid large-cap equities applied to illiquid small-caps or a new product it was never designed for. - **Wrong inputs at runtime**: reasonable model, unreasonable scenario fed in. - **Misinterpreting the output**: treating a 99% VaR as the *maximum possible* loss rather than a *threshold that is breached 1 day in 100*; treating a credit rating as a guarantee rather than an opinion. - **Ignoring the limitations** the model builders explicitly documented — using the number past its stated shelf life or outside its stated range.

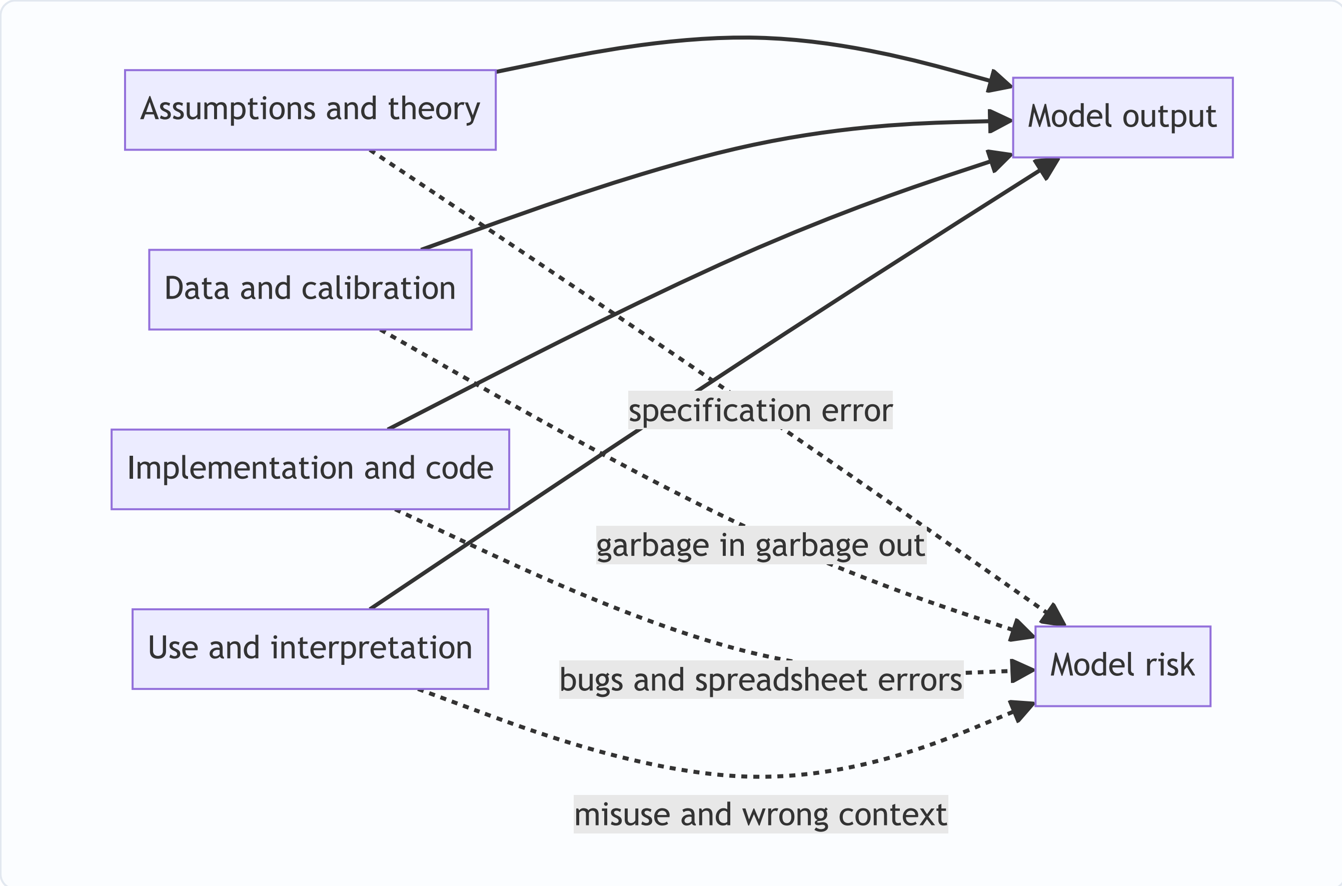

Figure 16.2 — Four entry points for model risk across a model's life. A failure at any single stage corrupts the final number.

The reason this framing matters for an interview: when asked "what could go wrong with this model?" a strong candidate does not give a vague answer. They walk the four sources — assumptions, data, implementation, use — and name a concrete failure at each.

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4. Full Content — Famous Failures, Validation, and Governance

4.1 Why famous failures are the best teacher

Model risk is abstract until you see money vanish. Each landmark failure isolates one of the four sources, and together they form the canon every risk professional is expected to know.

Long-Term Capital Management (LTCM), 1998 — when assumptions break

LTCM was a hedge fund run by star traders and two Nobel laureates (Myron Scholes and Robert Merton, of Black-Scholes fame). Its strategy was **relative-value arbitrage**: identify two securities whose prices had diverged slightly from a historical relationship, bet that they would **converge**, and lever the tiny spread up enormously — leverage reached roughly **25-to-1** on the balance sheet and far higher through derivatives.

The models said these convergence trades were nearly riskless because, historically, the spreads *always* narrowed. Two model assumptions were fatal:

1. **Normal-ish, stable volatility and correlations** estimated from recent calm data. The models assigned a vanishingly small probability to a simultaneous, correlated blow-out across many unrelated markets.

2. **Continuous liquid markets** in which you can always trade out of a position near the last price.

In August 1998, Russia defaulted on its domestic debt. Panicked investors fled to safety everywhere at once. Spreads that were "supposed" to converge instead **widened violently and in unison** — correlations the model treated as low jumped toward one. Because LTCM was levered 25-to-1, losses that the model called a once-in-several-billion-years event wiped out most of its ~$4.6 billion capital in weeks. The Federal Reserve organised a $3.6 billion bailout by a consortium of banks to prevent a systemic cascade.

Lesson: the model was internally elegant but rested on assumptions (stable correlations, permanent liquidity, thin tails) that held in normal times and shattered in stress. High leverage turned a model error into an extinction event. This is the archetypal **bad-assumptions** failure.

The 2008 crisis — ratings models and the Gaussian copula

The 2008 financial crisis was, at its quantitative heart, a **model-risk catastrophe** built from mortgage-backed securities (MBS) and collateralised debt obligations (CDOs). Two model failures interlocked.

(a) The rating-agency default models. Agencies rated tranches of mortgage securities AAA — the same grade as sovereign debt — based on models fed almost entirely on data from a period of **continuously rising US house prices.** The models effectively assumed national house prices could not fall significantly *at the same time* across the whole country, because in the data they never had. When national prices fell together, the "diversification" across geographies that made pools look safe evaporated.

(b) The Gaussian copula and default correlation. The workhorse for pricing CDO tranches was David Li's **Gaussian copula** model, which reduced the fearsomely complex question "how likely are many mortgages to default together?" to a single **correlation parameter**, calibrated conveniently from credit-default-swap spreads rather than actual joint-default history. The model dramatically **understated tail correlation** — the tendency of defaults to cluster in a downturn. Senior tranches sold as near-riskless were, in truth, highly exposed to exactly the correlated-default scenario the model waved away.

Layer on **VaR** at the banks holding these assets: VaR models, calibrated on the placid pre-crisis years, reported small daily risk numbers for portfolios that were in fact catastrophically exposed. The models could not see a risk that was absent from their input window.

Lesson: this was **bad assumptions plus bad data plus misuse** compounding. Assumption: defaults are weakly correlated and normally behaved. Data: calibration windows that never contained a national housing bust. Misuse: a AAA label, meant as an opinion under stated assumptions, was traded as a fact — and everyone relied on the same handful of models, so the errors were perfectly correlated across the system.

The VaR critique — a threshold, not a ceiling

Both episodes indict **VaR** specifically. VaR answers "what loss will I not exceed 99% of the time?" It says **nothing about how bad the other 1% is.** Institutions managed to the number and forgot the tail. As one memorable line from the crisis put it, VaR was "an airbag that works all the time, except when you have a car accident." The failure was as much **misuse** (mistaking a threshold for a maximum) as it was the model. This is precisely why regulators later shifted the trading book toward **Expected Shortfall**, which averages the losses *beyond* the VaR threshold (Chapter 5).

Honourable mentions

- **JPMorgan "London Whale" (2012):** a revised VaR model with a **spreadsheet error** (a formula divided by a sum instead of an average, roughly halving reported volatility) helped hide a build-up of credit derivatives that lost ~$6.2 billion. A textbook **implementation** failure.
- **Reinhart-Rogoff (2013):** an influential austerity-economics paper whose headline result partly rested on an Excel range that **omitted five countries** — a spreadsheet coding error with global policy consequences.

Failure	Year	Dominant source	One-line lesson
LTCM	1998	Bad assumptions	Stable correlations and permanent liquidity are stress-time fictions; leverage magnifies model error
Ratings / CDO models	2008	Assumptions + data	A model cannot price a risk absent from its calibration window
Gaussian copula	2008	Assumptions	Compressing joint default into one correlation number understated tail clustering
VaR reliance	2008	Misuse	A 99% threshold is not a maximum; it is silent about the tail
London Whale	2012	Implementation	A single spreadsheet formula error can hide billions in risk

4.2 The defence — model validation

If all models are wrong, the institutional response is not to stop using models but to **challenge them systematically.** This is **model validation**: an ongoing, independent set of activities to verify that a model works as intended and to establish where it does not.

SR 11-7 frames sound validation around three components:

(1) Evaluation of conceptual soundness. Does the model's design and theory make sense for its purpose? Are the assumptions reasonable and documented? Is there support in the literature and evidence? This is where a validator attacks Source 1 (assumptions) — asking, for instance, "you assume normality; have you tested the tails?"

(2) Ongoing monitoring, including process verification and benchmarking. Confirm the model is implemented correctly and still performing as conditions change. **Benchmarking** compares the model's outputs against an alternative model or a challenger approach — if two independent methods disagree materially, something is wrong. This attacks Source 3 (implementation).

(3) Outcomes analysis, especially back-testing. Compare model predictions against **actual realised outcomes.** For a 99% 1-day VaR, you count exceptions: over 250 trading days you *expect* about 2.5 breaches. If you see 12, the model is understating risk and fails back-testing (Basel's "traffic-light" test formalises exactly this). Back-testing attacks the whole chain by confronting the model with reality.

Two more validation pillars sit alongside these:

- **Benchmarking / challenger models** — never rely on a single model; a materially different second model is a permanent reality check.
- **Sensitivity and stress testing** — deliberately push inputs to extreme values and to historical stress scenarios to find where the model breaks. If small input changes cause wild output swings, the model is fragile.

A cardinal principle: validation must be **effectively independent.** The people who challenge the model cannot be the people who built it and whose bonuses depend on it being approved. Independence and competent challenge are the heart of the discipline.

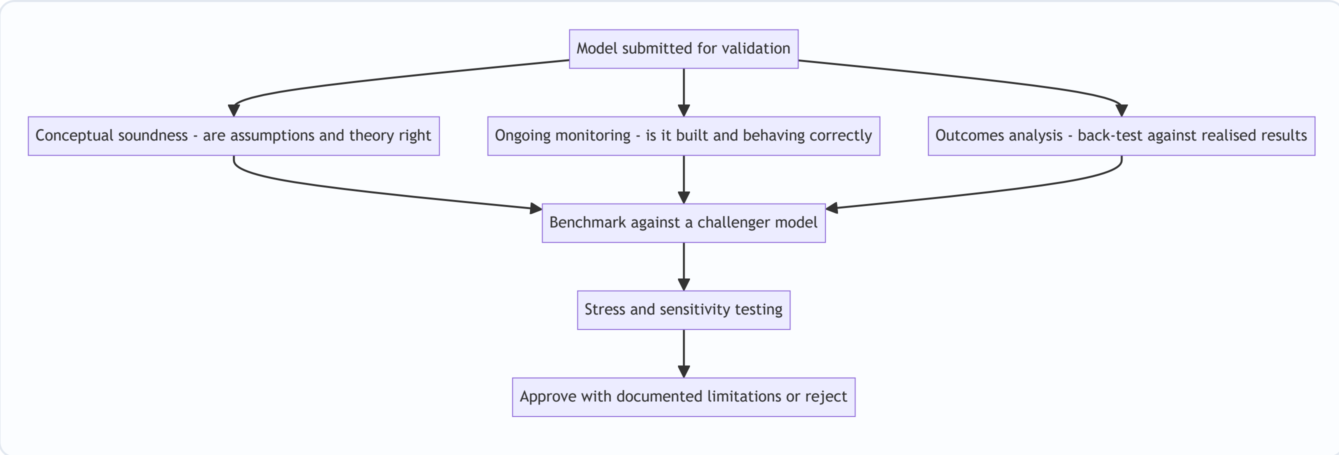

Figure 16.3 — The three pillars of validation under SR 11-7 plus benchmarking and stress testing.

4.3 The framework — model risk governance and SR 11-7

Validation is a technical activity; **governance** is the organisational structure that makes sure validation actually happens, is independent, and has teeth. SR 11-7 is the reference framework and rests on a few load-bearing ideas.

Model risk is managed like any other risk — identify, measure, monitor, control. The guidance insists model risk be treated as a first-class risk, not an afterthought buried in operational risk.

Two guiding principles run through the document:

1. **"Effective challenge"** — critical, competent, independent review by parties who have the incentive, competence, and standing to push back and, if necessary, to say no.
2. **A model is only as good as its use** — governance must control not just how models are built but how they are used, because misuse (Source 4) is half the risk.

A firm-wide model inventory. You cannot manage what you have not catalogued. Every model is registered with its owner, purpose, assumptions, limitations, validation status, and a **materiality tier** (a model driving billions in capital gets far more scrutiny than one that formats a report).

The "three lines of defence." This is the organisational skeleton — expect to be asked to draw it:

Line	Who	Role in model risk
First line	Model **owners / developers** (the business)	Build, document, and use models correctly; own the risk day-to-day
Second line	Independent **model risk / validation** function	Challenge, validate, maintain the inventory, set policy
Third line	**Internal audit**	Assess whether the whole framework is designed and operating effectively

Documentation and the "no black box" rule. Models must be documented well enough that a knowledgeable third party can understand them and reproduce results. A model no one can explain cannot be validated — and an unvalidated model must not drive material decisions. This principle collides directly with modern **machine-learning / AI models**, whose opacity ("explainability" problem) is the frontier of model risk today.

Vendor and AI models. Buying a model from a vendor does not outsource the risk. The firm remains responsible for validating it, understanding its limitations, and having contingency if the vendor changes or fails. The same logic now extends to AI/ML: bias in training data, drift as the world changes, and the inability to explain a decision are all model risk in a new guise.

4.4 Managing model risk in practice

Pulling it together, an institution manages model risk through a continuous loop:

- **Inventory and tier** every model by materiality.
- **Validate** independently before first use, and **re-validate** periodically and after any material change.
- **Document** assumptions, limitations, and the approved range of use.
- **Monitor and back-test** continuously; track exceptions.
- **Set boundaries** — hard limits on where and how a model may be used, plus **model reserves / valuation adjustments** (a capital or P&L buffer held explicitly against model uncertainty for hard-to-value positions).
- **Apply conservatism** — where a model is uncertain, err toward the more prudent number rather than the more flattering one.
- **Keep human oversight** — the "sceptical eye": a number that violates common sense should trigger investigation, not blind trust.
- **Retire** models that no longer fit the environment.

The philosophical bottom line — the sentence to carry into any risk role — is this: **every quantitative number is the output of a model, and every model embeds assumptions that can fail; therefore no number deserves blind trust. The professional's job is not to worship the number but to know where it breaks.**

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5. Worked / Applied Examples

Example 1 — VaR back-testing: catching an understated risk model

A bank reports a **1-day 99% VaR of ₹50 lakh** for its trading book. At 99% confidence the model implicitly claims that daily losses will exceed ₹50 lakh only **1% of the time.** Over **250 trading days**, the expected number of "exceptions" (days the actual loss beat the VaR) is:

$$\text{Expected exceptions} = 250 \times (1 - 0.99) = 250 \times 0.01 = 2.5 \text{ days.}$$

Now suppose that over the year the desk actually recorded **11 days** with a loss above ₹50 lakh. Is that just bad luck, or is the model broken?

Under the model, exceptions follow a binomial distribution with n = 250, p = 0.01, so mean 2.5 and standard deviation:

$$\sqrt{n p (1-p)} = \sqrt{250 \times 0.01 \times 0.99} = \sqrt{2.475} \approx 1.57.$$

Eleven exceptions sits roughly $(11 - 2.5)/1.57 \approx 5.4$ standard deviations above expectation. The probability of 10 or more exceptions when the true rate is 1% is minuscule (far below 0.1%). Under **Basel's traffic-light back-testing**, 5–9 exceptions land in the amber "yellow" zone (model suspect, capital multiplier increased); **10 or more is the red zone** — the model is rejected as understating risk.

Interpretation and reconciliation. The realised breach rate is $11/250 = 4.4\%$, more than four times the promised 1%. The model is systematically underestimating risk — most likely a **bad-data** problem (calibrated on a calm window that omitted recent volatility) or a **bad-assumption** problem (thin-tailed distribution missing fat tails). The response: the model is flagged, capital is penalised via a higher multiplier, and re-calibration or re-specification is required. This is outcomes analysis — Section 4.2's third pillar — catching model risk before it costs real money.

Example 2 — Misuse: a good model applied to the wrong instrument

A quant desk has a well-built, well-validated **Black-Scholes** pricer for **liquid, large-cap European equity options.** It has passed conceptual review and back-tests cleanly *for that product*.

A structuring desk now uses the same pricer to value a **long-dated, deep out-of-the-money option on an illiquid small-cap stock.** The model returns a clean price of **₹2.10 per option**, and the desk books a large position at that value.

Every one of Black-Scholes' assumptions is now violated in a way that matters: - **Constant volatility** — small-caps exhibit a pronounced **volatility skew**; deep-OTM options trade at far higher implied vols than the at-the-money vol the desk plugged in. Using the ATM vol *underprices* the option. - **Continuous liquid trading / costless hedging** — the underlying barely trades, so the delta-hedging that justifies the formula is impossible; the real position carries large unhedgeable risk. - **No jumps** — small-caps gap on news, exactly the tail the model ignores.

The model is not broken. It is **misused** — applied outside its domain of validity (Source 4). When the desk later tries to unwind, the market price is **₹5.80**, and the position shows a large loss versus the booked value. A validator's boundary — "*approved for liquid large-cap European options only*" — would have blocked this. The lesson reconciles cleanly: the same number (₹2.10) that was *correct* for one instrument was *dangerously wrong* for another, and nothing on the screen told the user which case they were in.

Example 3 — Implementation error: the spreadsheet that halved risk

A risk team migrates its VaR calculation to a new spreadsheet-based model. The intended formula scales position risk by the **average** of a series of daily volatility estimates. Due to a copy-paste slip, one cell computes the **sum** of two volatilities divided by their **count-minus-one**, and another divides by the wrong range — the net effect is that reported volatility comes out roughly **half** of the true value.

Because VaR scales linearly with volatility, a true 1-day VaR of **₹100 crore** is reported as **₹52 crore.** Traders, seeing "plenty of room under the limit," add risk. The book grows to a genuine VaR near **₹180 crore** while the report still shows a comfortable **~₹94 crore.** When markets move, the realised loss dwarfs anything the risk report ever suggested.

This is the mechanism behind the real **London Whale** loss (~$6.2 billion): a mundane **implementation error** (Source 3), invisible to anyone reading only the output, silently disabled the risk control that was supposed to prevent exactly this build-up. The defences that would have caught it — independent revalidation of the new model, **benchmarking** the new spreadsheet against the old system (they would have disagreed by ~2x), and process verification — are precisely the SR 11-7 activities of Section 4.2. The numbers reconcile: halving the volatility input roughly halves the reported VaR, which is why the reported ₹52 crore looked safe while the true ₹100 crore was not.

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6. Connections

- **Value at Risk (Ch. 4) and Expected Shortfall (Ch. 5):** VaR is the canonical model-risk case study — its "threshold not ceiling" silence and its calibration-window blindness drove 2008 losses, and the regulatory move to Expected Shortfall under FRTB was a direct response to VaR's tail-blindness.
- **Risk models and measurement (Ch. 11):** every measurement technique in that chapter is a model and therefore a source of model risk; this chapter is the meta-layer that sits above all of them.
- **Operational risk (Ch. 8):** implementation errors, spreadsheet bugs, and misuse are operational-risk events. Model risk overlaps heavily with op risk but is elevated to its own discipline because of its systemic reach.
- **Credit risk (Ch. 6) and counterparty risk (Ch. 7):** PD, LGD, and CVA are all model outputs; the 2008 rating and copula failures were credit-model risk at systemic scale.
- **Basel and regulation (Ch. 12):** internal-models approaches (IMA, IRB) let banks compute capital from their own models — which is precisely why regulators demand model validation, back-testing, and the SR 11-7-style governance covered here.
- **Liquidity risk (Ch. 9):** the "continuous liquid markets" assumption that broke LTCM and 2008 links model risk to liquidity risk — models routinely assume away the liquidity holes that define crises.
- **Behavioural angle:** model risk is amplified by **automation bias** and **overconfidence** — the human tendency to trust a number more because a machine produced it.

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7. Key Terms

- **Model:** a quantitative method that transforms inputs into an output used for a decision (pricing, risk, capital, lending).
- **Model risk:** the potential for adverse consequences from decisions based on incorrect or misused model outputs (SR 11-7 definition).
- **Specification error:** the model's theory/assumptions do not match reality (Source 1).
- **Calibration:** setting a model's parameters from data; poor or unrepresentative data causes calibration error.
- **Model validation:** independent activities verifying a model works as intended and mapping where it does not.
- **Conceptual soundness:** whether the model's design, theory, and assumptions are appropriate for its purpose.
- **Back-testing:** comparing model predictions to realised outcomes (e.g., counting VaR exceptions).
- **Benchmarking / challenger model:** comparing a model's output to an alternative model as a reality check.
- **Effective challenge:** critical, competent, independent review with the standing to reject a model (SR 11-7's core principle).
- **Three lines of defence:** owners (1st), independent validation (2nd), internal audit (3rd).

- **Model inventory:** the firm-wide catalogue of all models, their owners, uses, and validation status.
- **Materiality tiering:** ranking models by potential impact to allocate scrutiny.
- **Model reserve / valuation adjustment:** a buffer held explicitly against model/valuation uncertainty.
- **SR 11-7:** the US Federal Reserve / OCC 2011 supervisory guidance on model risk management — the field's reference framework.
- **Gaussian copula:** the 2008-era model that compressed joint-default risk into a single correlation parameter and understated tail clustering.
- **Explainability (AI/ML):** the ability to understand why a model produced an output — the frontier challenge for machine-learning models.

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8. Common Confusions

"A model is wrong" vs "a model is useless." Box's point is that *all* models are wrong — the useful ones are wrong in known, bounded, tolerable ways. Model risk management does not chase a correct model; it maps the boundaries of a useful one.

Model risk is not the same as market/credit risk. Market risk is the risk that prices move against you. Model risk is the risk that your *measurement* of market (or credit, or any) risk is itself wrong. It is a risk *about* your risk numbers — a meta-risk.

Model risk ≠ model error only. Roughly half of model risk is **misuse** — a perfectly sound model applied to the wrong problem or misinterpreted. Candidates who define model risk as "the model is buggy" miss the SR 11-7 emphasis on *use*.

A more complex model is not a lower-risk model. Complexity can *increase* model risk: more assumptions, more parameters to calibrate wrongly, more code to break, and less transparency. The Gaussian copula was sophisticated and catastrophically wrong. Simplicity and understandability are risk-reducing virtues.

VaR is not the maximum loss. The single most common misuse in the book: 99% VaR is a *threshold* breached ~1 day in 100. It is deliberately silent about how bad the breach is. Treating it as a worst case is a misuse, not a model flaw.

Validation is not a one-time sign-off. It is ongoing. A model validated in 2006 on calm data was "valid" and still failed in 2008 because the *world* changed (regime change) — hence periodic re-validation and continuous monitoring.

Passing back-tests does not prove a model is right. Back-testing on a benign period only confirms the model fits benign periods. Absence of exceptions in calm markets is exactly the false comfort that preceded both LTCM and 2008.

Buying a vendor model does not transfer the risk. The institution remains fully responsible for validating and understanding any model it relies on, however it was sourced.

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9. Recap

Every number that drives a financial decision — a price, a PD, a VaR, a capital charge — is the output of a **model**, and a model is a **simplification of reality** that is, in George Box's words, "wrong but useful." **Model risk** (SR 11-7: *adverse consequences from incorrect or misused model outputs*) is the risk that we trust that simplified number beyond where it holds. It enters through **four sources**: bad **assumptions** (specification error), bad **data** (calibration error), bad **implementation** (coding/spreadsheet error), and **misuse** (wrong context or misinterpretation).

The canon of failures maps to these sources: **LTCM (1998)** — stable-correlation and permanent-liquidity assumptions shattered, magnified by 25-to-1 leverage; the **2008 crisis** — ratings and Gaussian-copula models calibrated on data with no housing bust, understating tail default correlation, with **VaR** misread as a ceiling; the **London Whale (2012)** — a spreadsheet error that halved reported risk. The defence is **independent validation** (conceptual soundness, ongoing monitoring/benchmarking, outcomes analysis/back-testing) plus **stress testing**, all wrapped in **governance** — SR 11-7's "effective challenge," a firm-wide **model inventory**, **materiality tiering**, and the **three lines of defence**. Managed well, model risk becomes a discipline of **humility and boundaries**: know how your model is wrong, quantify the cost of being wrong, and never let a confident number outrun its domain of validity.

The one-sentence takeaway: a model output is an opinion wearing the costume of a fact — the professional's job is to keep a sceptical eye and know exactly where it breaks.

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10. Quick-Reference / Interview Points

- **Definition (memorise, SR 11-7):** model risk = "*the potential for adverse consequences from decisions based on incorrect or misused model outputs and reports.*"
- **The philosophy:** George Box — "*All models are wrong, but some are useful.*" Goal is not correctness but knowing the boundaries of usefulness.
- **Four sources (be ready to name and give an example of each):** bad **assumptions**, bad **data**, bad **implementation**, **misuse**. Two SR 11-7 roots: the model is *wrong*, or the model is *used wrong*.
- **LTCM (1998):** convergence arbitrage, ~25:1 leverage; assumed stable correlations and permanent liquidity; Russian default made correlations spike to one; ~$4.6bn lost, Fed-organised bailout. *Source: bad assumptions + leverage.*
- **2008:** rating models calibrated on ever-rising house prices; **Gaussian copula** compressed joint default into one correlation and understated tail clustering; VaR read as a ceiling. *Source: assumptions + data + misuse, all correlated across the system.*
- **London Whale (2012):** spreadsheet error roughly halved reported volatility/VaR; ~$6.2bn loss. *Source: implementation.*
- **VaR critique:** a 99% VaR is a *threshold breached ~1 day in 100*, not a maximum; silent about the tail → motivated the shift to **Expected Shortfall** (FRTB).
- **Validation (SR 11-7 three pillars):** (1) conceptual soundness, (2) ongoing monitoring + benchmarking, (3) outcomes analysis / back-testing. Plus stress & sensitivity testing. Must be **independent**.
- **Back-testing math:** expected VaR exceptions = $n(1-c)$; for 250 days at 99%, expect 2.5; Basel traffic light — green ≤4, yellow 5–9, red ≥10.
- **Governance: effective challenge**; firm-wide **model inventory**; **materiality tiering**; **three lines of defence** (owners → independent validation → internal audit); "no black box" documentation rule.
- **Managing it:** validate before use and re-validate periodically; document assumptions and approved use range; monitor and back-test; hold **model reserves/valuation adjustments**; apply conservatism; keep human oversight.

- **Modern frontier: AI/ML model risk** — opacity/explainability, data bias, drift; vendor models do not transfer responsibility.
- **The closing line:** *every quant number is a model output with embedded assumptions that can fail — so no number deserves blind trust; the job is to know where it breaks.*

Q&A — Model Risk

Practice bank for the Model Risk chapter. Every question is followed by a full answer. Reference facts used throughout: the two SR 11-7 sources of model risk (fundamental errors; incorrect or inappropriate use); the three components of a validation framework (conceptual soundness, ongoing monitoring, outcomes analysis); the three lines of defence; and the FRTB distinction between modellable and non-modellable risk factors.

Section A — Concept Check

A1. Define model risk precisely, and name its two root sources.

Model risk is the potential for adverse consequences — financial loss, bad decisions, or reputational and regulatory damage — arising from **decisions based on incorrect or misused model outputs**. Following the Federal Reserve's SR 11-7, it has two root sources: (1) **the model may have fundamental errors** — wrong theory, wrong mathematics, coding bugs, or poor data — so it produces inaccurate outputs even when used as intended; and (2) **the model may be used incorrectly or inappropriately** — applied outside the conditions it was built for, fed the wrong inputs, or its results misunderstood. The crucial point is that a *perfectly correct* model still carries model risk through the second channel.

A2. What is a "model" for governance purposes, and why does the definition matter?

SR 11-7 defines a model as a quantitative method that applies statistical, economic, financial, or mathematical **theory, techniques, and assumptions** to process input data into quantitative estimates. The definition matters because it sets the **perimeter of the model inventory**: anything meeting it must be inventoried, tiered by materiality, and validated. Firms that define "model" too narrowly leave spreadsheets, vendor tools, and end-user computing applications ungoverned — which is exactly where uncontrolled model risk accumulates.

A3. "All models are wrong, but some are useful." How does this shape the risk manager's job?

The George Box aphorism reframes the goal. A model is a **deliberate simplification** of reality, so the question is never "is it correct?" (nothing is) but "**is it fit for its intended purpose, and do we understand where it breaks?**" The risk manager's job is therefore not to eliminate error but to (a) document the model's assumptions and limitations, (b) quantify the error it introduces, and (c) ensure it is only used where those errors are tolerable. Managing model risk is managing the *gap between the map and the territory*.

A4. Distinguish model risk from parameter (estimation) risk.

Parameter risk is uncertainty in the *inputs* to a chosen model — the estimated volatility, correlation, or default probability could be wrong because they are estimated from finite, noisy data. **Model risk** is broader: it includes parameter risk but also **structural risk** — the possibility that the entire model *form* is wrong (e.g. using a Gaussian copula when tail dependence exists, or Black–Scholes when volatility is stochastic). You can have perfect parameters in the wrong model. Structural error is the more dangerous of the two because it cannot be fixed by better estimation.

A5. What are the three pillars of an effective model validation framework?

(1) **Evaluation of conceptual soundness** — is the theory defensible, the methodology appropriate, and are the assumptions and limitations documented? This includes *developmental evidence* review. (2) **Ongoing**

monitoring — process verification and benchmarking to confirm the model still works as markets and portfolios change; this catches degradation over time. (3) **Outcomes analysis** — comparing model outputs to actual realised results, the most direct of which is **backtesting**. A validation missing any pillar is incomplete: soundness without monitoring goes stale, and monitoring without outcomes analysis never confronts reality.

A6. What does "effective challenge" mean, and what three things must a challenger have?

Effective challenge is the *critical analysis by objective, informed parties who can identify model limitations and produce appropriate changes*. For it to be real, the challenger must have three things: **competence** (the technical skill to understand the model), **influence** (the standing and authority to force change), and **incentive/independence** (organisational separation so they are willing to say no). Strip away any one — a validator who is skilled but junior, or independent but under-resourced — and challenge becomes a rubber stamp. Effective challenge is the human mechanism that actually stops a bad model from shipping.

A7. How do the "three lines of defence" allocate responsibility for model risk?

First line — model owners, developers, and users — own the risk: they build, use, and self-test the model and are accountable for it. **Second line** — independent model validation / risk management — provides effective challenge, sets standards, maintains the inventory, and can block deployment. **Third line** — internal audit — does not re-validate models but assures the board that the *framework itself* is designed and operating effectively. The separation prevents the people who benefit from a model's approval from being the only ones checking it.

A8. Why is a model inventory tiered by materiality rather than validated uniformly?

Validation is costly and scarce; not every model threatens the firm equally. **Tiering by materiality** — based on the size of exposure, complexity, and reliance placed on the model — lets the firm concentrate deep, frequent validation on high-tier models (e.g. the regulatory capital or trading VaR engine) while applying proportionate, lighter scrutiny to low-tier ones. Treating a small pricing spreadsheet like the capital model wastes resources; treating the capital model like a spreadsheet is negligent. Tiering aligns validation intensity with the harm a failure would cause.

A9. What is "model overlay" or a management adjustment, and why is it a double-edged sword?

An overlay is a manual adjustment applied *on top of* a model's raw output — for example, adding to a loss-provision estimate because management judges the model has not captured a new risk (as widely done for COVID-19 and post-pandemic inflation under IFRS 9 / CECL). It is **necessary** because it corrects for known model blind spots the model cannot yet see. It is **dangerous** because it can become an unauditable channel for smoothing results or hiding a broken model. Good practice makes overlays explicit, documented, governed, and *temporary* — an overlay that never expires is a signal the underlying model needs rebuilding.

A10. What are non-modellable risk factors (NMRFs) under FRTB, and why does the distinction exist?

Under the Fundamental Review of the Trading Book, a risk factor is **modellable** only if it has enough *real, observable* price observations (broadly, at least 24 per year with no 90-day gap). Factors failing this test are **non-modellable** and attract a separate, punitive **stressed-expected-shortfall capital add-on** computed via stress scenarios rather than the internal ES model. The distinction exists because model risk is highest exactly where **data is thin** — you cannot reliably calibrate or backtest a factor you rarely observe — so the regulator forces conservative, non-model treatment there rather than trusting a poorly-supported model.

Section B — Numerical / Applied (with full solutions)

B1. Sizing model risk from a pricing-model gap. A desk holds a book of exotic options. Model A (a simple Black–Scholes proxy) values the book at **$100.0m**. Model B (a stochastic-volatility model the validators consider more accurate) values it at **$97.4m**. (a) What is the model-risk provision (the "prudent valuation" reserve) implied by the disagreement? (b) If the book has been marked at Model A's value, what is the P&L impact of switching to B?

Solution. (a) The model-uncertainty reserve is the difference between the marked value and the more conservative defensible value: $100.0\text{m} - 97.4\text{m} = \mathbf{\$2.6m}$. This is held as an **Additional Valuation Adjustment (AVA)** for model risk so the book is not carried at an optimistic mark. (b) Switching to Model B writes the book down by the same **$2.6m loss** (a 2.6% haircut). The number is not a market move — nothing in the world changed — it is *pure model risk crystallising*. This is why unreserved model uncertainty is a hidden short position in your own valuation.

B2. Backtesting exceptions and the traffic-light multiplier. A bank's 99% one-day VaR model recorded **7 exceptions over 250 trading days**. (a) How many exceptions are expected for a correct model? (b) Which Basel traffic-light zone is this, and what happens to the capital multiplier? (c) If the 10-day 99% VaR is $30m, quantify the capital difference between the resulting multiplier and the green-zone baseline of $k=3.0$.

Solution. (a) Expected exceptions $= (1-0.99)\times 250 = 0.01\times250 = \mathbf{2.5}$. (b) 7 exceptions falls in the **Yellow zone** (5–9 breaches). Basel raises the multiplier from the green-zone $k=3.0$ by a yellow-zone add-on; for 7 exceptions $k = 3.65$. (c) Baseline charge $= 3.0\times30\text{m} = \$90\text{m}$. Yellow charge $= 3.65\times30\text{m} = \$109.5\text{m}$. Extra capital $= \mathbf{\$19.5m\ (+21.7\%)}$. A model that under-forecasts risk is punished with a mechanically larger capital charge — model quality converts directly into capital.

B3. Kupiec proportion-of-failures test on the same result. Test the B2 model (7 exceptions in 250 days, target $p=0.01$) with Kupiec's unconditional-coverage statistic against the χ^2_1 critical values 3.84 (5%) and 6.63 (1%).

Solution. Observed rate $\hat p = 7/250 = 0.028$. The likelihood-ratio statistic is $$LR_{uc} = -2\ln!\left[\frac{(1-p)^{T-x}p^{x}}{(1-\hat p)^{T-x}\hat p^{x}}\right].$$ - Under $p=0.01$: $243\ln(0.99)+7\ln(0.01) = 243(-0.010050)+7(-4.60517) = -2.442 - 32.236 = -34.678$. - Under $\hat p=0.028$: $243\ln(0.972)+7\ln(0.028) = 243(-0.028399)+7(-3.575551) = -6.901 - 25.029 = -31.930$. $$LR_{uc} = -2\big(-34.678 - (-31.930)\big) = -2(-2.748) = \mathbf{5.50}.$$ Decision: $3.84 < 5.50 < 6.63$, so we **reject at 5% but not at 1%**. Consistent with the Yellow-zone verdict of B2 — statistically the model is under strain but not decisively broken. The test and the supervisory rule agree: watch it closely.

B4. Expected number of firms breaching by chance. A regulator oversees **100 banks**, each running a *genuinely correct* 99% VaR model, over 250 days. (a) What is the expected number of exceptions per bank, and its standard deviation? (b) Roughly how many of the 100 banks would you expect to land in the Yellow zone (≥5 exceptions) purely by chance?

Solution. (a) Exceptions $\sim \text{Binomial}(250, 0.01)$. Mean $= 250\times0.01 = 2.5$; SD $= \sqrt{250\times0.01\times0.99} = \sqrt{2.475} = \mathbf{1.573}$. (b) $P(X\ge 5)$ for a Poisson/Binomial with mean 2.5 is about **0.11** (using Poisson: $P(X\le4)\approx0.891$, so $P(X\ge5)\approx0.109$). Across 100 correct models, that is roughly $\mathbf{11\ banks}$ in the Yellow zone by pure luck. The lesson for model risk governance: backtesting exceptions are noisy, so a single Yellow result is *not* proof of a bad model — you must weigh it against the base rate of false alarms before condemning a model.

B5. Benchmarking two challenger models. During validation, the champion model estimates portfolio 99% VaR at **\$12.0m**. Two independent challenger models produce **\$14.5m** and **\$15.2m**. (a) What does the benchmark spread reveal? (b) If realised outcomes over the next quarter breach the \$12.0m level far more often than 1% of the time, what is the validator's conclusion?

Solution. (a) Both challengers sit **20–27% above** the champion ($14.5/12.0 = 1.21$; $15.2/12.0 = 1.27$). Consistent disagreement in the *same direction* is a red flag that the champion may be **systematically under-stating risk** — benchmarking has surfaced a possible structural bias that a single model in isolation would never reveal. (b) Outcomes analysis (backtesting) then *confirms* what benchmarking suspected: excessive breaches mean the champion under-forecasts the tail. The validator escalates — the champion should be recalibrated or replaced, and in the interim a conservative overlay (or use of the challenger level) applied. This is the two pillars working together: benchmarking flags, outcomes analysis proves.

B6. Vendor-model input error. A firm licenses a third-party credit model. The vendor's default model was calibrated on data with an average PD of 2%, but the firm's actual portfolio has an average PD of 5%. Expected loss per \$100m of exposure is $EL = PD \times LGD \times EAD$ with LGD = 40%. (a) What EL does the mis-applied model report? (b) What is the true EL, and what is the model-risk understatement?

Solution. (a) Using the vendor's calibration PD: $EL = 0.02\times0.40\times100\text{m} = \mathbf{\$0.8m}$. (b) Using the firm's true PD: $EL = 0.05\times0.40\times100\text{m} = \mathbf{\$2.0m}$. Understatement $= 2.0 - 0.8 = \mathbf{\$1.2m}$, i.e. the model reports only **40%** of the true expected loss. Nothing is wrong with the vendor's mathematics — this is SR 11-7's *second* source of model risk: a correct model **used outside its calibration domain**. The fix is not a better model but a re-calibration to the firm's own portfolio, and a control that flags when input distributions drift from the calibration set.

B7. Materiality tiering by exposure-weighted error. Three models each carry an estimated valuation-error rate of 3%, on exposures of \$2,000m (Model X), \$150m (Model Y), and \$8m (Model Z). Rank them for validation priority by dollar error at risk.

Solution. Dollar error $=$ exposure $\times$ error rate: - Model X: $2{,}000\text{m}\times0.03 = \mathbf{\$60m}$. - Model Y: $150\text{m}\times0.03 = \mathbf{\$4.5m}$. - Model Z: $8\text{m}\times0.03 = \mathbf{\$0.24m}$. Priority order **X ≫ Y ≫ Z**. Although all three share the *same* error *rate*, the materiality — the harm a failure causes — spans a factor of 250. Tiering directs the scarce validation budget to Model X, where an identical modelling flaw does 250 times the damage. This is why materiality, not model complexity alone, drives the validation calendar.

Section C — Interview-Style (with model answers)

C1. "What is model risk, and why is it not just 'the risk the model is wrong'?"

Model answer: Model risk is the risk of loss or bad decisions from relying on model outputs — but the naive version, "the model is wrong," misses half of it. SR 11-7 splits it into two sources: **fundamental errors** in the model (bad theory, bugs, poor data) *and* **incorrect or inappropriate use** — running the model outside its intended domain, feeding it wrong inputs, or misreading its results. The second source means a *flawless* model still carries model risk. So managing it is not only about building better models; it is about controlling assumptions, use, inputs, and interpretation across the model's whole life. The mindset is: every model is a simplification with a domain of validity, and risk lives at the edge of that domain.

C2. "Walk me through how you would validate a new pricing model before it goes live."

Model answer: I work the three pillars. First, **conceptual soundness**: I review the developmental evidence — is the theory appropriate for this product, are the assumptions and limitations documented, does the math check out, and is the data representative? Second, before it can go live I run **process verification and benchmarking**: I re-implement or compare against an independent challenger model and known analytical cases to confirm it does what it claims. Third, I set up **outcomes analysis** — a backtesting and P&L-attribution plan so that once live, model output is continuously compared to realised results. Throughout, the deliverable is not a yes/no but a documented statement of the model's *limitations and safe operating range*, plus any conditions or overlays required for approval. And critically, I must have the independence and authority to withhold approval — validation without the power to say no is theatre.

C3. "Give me a real example where model risk caused a disaster, and name the specific failure."

Model answer: Two canonical ones. **LTCM (1998)**: models assumed relationships would mean-revert and treated correlations as stable; when Russia defaulted, correlations went to one and liquidity vanished — the model's *assumption* of stable, diversifiable relationships was the structural error, amplified by leverage. **The 2008 Gaussian-copula CDO losses**: the copula had *zero tail dependence*, so it structurally could not represent correlated defaults; raising the correlation input could never fix a model that was blind to joint crashes by construction. The common thread is **structural model risk** — not a bad parameter but the wrong model *form* — combined with a governance failure to challenge the assumption. More recent and mundane: the 2012 "London Whale," where a spreadsheet VaR model with a copy-paste and a divide-by-sum-instead-of-average error understated risk. Disasters come from both exotic structural flaws and boring operational ones.

C4. "How do you manage the model risk you cannot eliminate?"

Model answer: You can never drive model risk to zero, so you *bound and reserve* it. First, **quantify** it — through benchmark spreads, sensitivity analysis, and comparison of alternative model outputs, so you have a dollar range for the uncertainty. Second, **reserve** against it — hold a model-uncertainty valuation adjustment (an AVA under prudent valuation) so the book is not marked at an optimistic single point. Third, **constrain use** — restrict the model to its validated domain with hard controls that flag inputs drifting outside calibration. Fourth, **overlay** with documented, temporary management adjustments where the model has a known blind spot. And fifth, **monitor** continuously so degradation is caught early. The philosophy is the same as market risk: you don't eliminate the exposure, you measure it, limit it, reserve for it, and watch it.

C5. "Your validation team keeps signing off models that later fail. Diagnose the governance problem."

Model answer: That is almost always a breakdown of **effective challenge**, and I'd check its three ingredients. **Competence** — are the validators technically able to understand the models, or are they out of their depth on the complex ones? **Influence** — do they actually have the standing and authority to block a model, or can the business override them? **Independence and incentive** — are they organisationally separate and free from pressure, or are they captured by the desk whose revenue depends on approval? A rubber-stamp pattern usually means at least one is missing — commonly influence: validators who *can* identify limitations but can't force change. I'd also check the **three lines of defence** are intact and that internal audit is assuring the *framework*, not just individual models. And I'd look at incentives — if validators are rewarded for throughput rather than for catching problems, they will optimise for sign-offs.

Section D — MCQs (with reasoning)

D1. According to SR 11-7, the two sources of model risk are: (a) market risk and credit risk; (b) fundamental errors and incorrect/inappropriate use; (c) parameter risk and liquidity risk; (d) coding bugs and data errors.

Answer: (b). The model may have fundamental errors, *or* it may be used incorrectly — the second source means even a correct model carries risk. (d) names only sub-types of the first source; (a) and (c) name other risk categories entirely.

D2. The three components of a sound model validation framework are: (a) pricing, hedging, and reporting; (b) first, second, and third lines of defence; (c) conceptual soundness, ongoing monitoring, and outcomes analysis; (d) VaR, ES, and stress testing.

Answer: (c). These are SR 11-7's three validation pillars. (b) is the *governance* structure, not the validation framework; (a) and (d) are model uses and measures, not validation components.

D3. "Effective challenge" of a model requires the challenger to have: (a) a PhD; (b) competence, influence, and incentive/independence; (c) veto power only; (d) access to the source code only.

Answer: (b). All three together — skill to understand, standing to force change, and independence/incentive to say no. Any one alone (e.g. veto power without competence) is insufficient. (a) and (d) are neither necessary nor sufficient.

D4. A model produces a perfect valuation but is applied to a portfolio outside its calibration range. This is: (a) not model risk, since the model is correct; (b) parameter risk only; (c) model risk from inappropriate use; (d) market risk.

Answer: (c). This is SR 11-7's second source — correct model, wrong use. (a) is the classic mistake: a correct model *still* carries model risk through misuse. B6 quantifies exactly this case.

D5. Under Basel's backtesting traffic light, a 99% VaR model with 10+ exceptions in 250 days lands in the: (a) Green zone, $k=3.0$; (b) Yellow zone; (c) Red zone, model rejected; (d) unclassified.

Answer: (c). Ten or more exceptions is the Red zone (multiplier $k=4.0$, model overhaul). Green is 0–4, Yellow is 5–9. Expected breaches for a correct model are only ~2.5.

D6. Non-modellable risk factors (NMRFs) under FRTB arise mainly because: (a) the factor is too volatile; (b) there are too few real price observations to calibrate reliably; (c) the factor is illiquid to trade; (d) the regulator dislikes internal models.

Answer: (b). Modellability turns on having enough *observable* prices (broadly ≥24/year, no 90-day gap). Sparse data means the factor cannot be reliably calibrated or backtested, so it gets a conservative stressed-ES add-on. Volatility and tradability are related but not the defining test.

D7. Benchmarking a champion model against independent challenger models primarily tests: (a) whether the model is coded correctly; (b) whether the model's outputs are reasonable relative to credible alternatives; (c) realised P&L; (d) regulatory capital.

Answer: (b). Benchmarking is a form of *ongoing monitoring* that flags systematic bias when alternatives consistently disagree (see B5). Comparing to realised results is *outcomes analysis* (c), a different pillar; (a) is process verification.

D8. A management overlay on a model's output should ideally be: (a) permanent and undocumented; (b) explicit, documented, governed, and temporary; (c) applied silently to smooth earnings; (d) larger than the model output.

Answer: (b). Overlays correct known model blind spots but must be transparent and time-limited, or they become an unauditable channel for manipulation. An overlay that never expires signals the underlying model needs rebuilding. (a) and (c) describe exactly the abuse to avoid.